CAUTION DO NOT LOOSEN THE BOLTS MORE THAN ONE-HALF OF A TURN. IF YOU DO NOT OBEY THIS INSTRUCTION, DAMAGE TO THE ENGINE MOUNTS OR THE STRUT CAN OCCUR.

CAUTION BE CAREFUL WHEN YOU LOOSEN OR TIGHTEN THE BOLTS WITH TORQUE EQUIPMENT ON THE AFT SIDE OF THE AFT ENGINE MOUNT. IF YOU DO NOT OBEY, DAMAGE TO THE HANGER FITTING OF THE AFT ENGINE MOUNT CAN OCCUR.

(c) Loosen the bolts [493] and bolts [514] for the forward mount and aft mount one half turn as follows:

1) Make sure that the forward dynamometer assembly [415] and aft dynamometer assembly [454] do not have more than the load limits.

2) Make sure that there is no clearance between the engine mounts again.

NOTE: The entire weight of the engine is now supported by the bootstrap hoists.

(d) Remove the forward mount bolts [493] that attach the forward engine strut mount to the forward engine mount [492] as follows (Figure 411):

1) Remove the forward mount bolts [493] and washers [494] from the barrel nuts [495] and barrel nut retainers [496].

(e) Remove the aft mount bolts [514] that attach the aft engine strut mount to the aft engine mount [513] as follows (Figure 412):

1) Remove the aft mount bolts [514] and washers [515] from the barrel nuts [511] and barrel nut retainers [512].

SUBTASK 71-00-02-860-019-G00

(10) Move the engine stand, SPL-14726, into its position from the front of the power plant as follows:

(a) Use the engine stand center red mark line to center the engine stand under the engine.

(b) Move the two forward stand arms outwards.

(c) If it is necessary, adjust the height of the stand cradle by the stand hydraulic jack that is located on the left side of the engine stand.

(d) If it is necessary, adjust the wheel positions.

SUBTASK 71-00-02-860-022-G00

WARNING LET ONLY APPROVED PERSONS OPERATE THE LEVER HOISTS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT WORKSTANDS AND PERSONNEL ARE AWAY FROM THE ENGINE BEFORE YOU LOWER IT. THE ENGINE CAN MOVE RELATIVE TO THE STRUT WHEN THE SHEAR PINS ARE NOT IN THE UPPER ENGINE MOUNTS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(11) Lower the engine to the engine stand as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Slowly lower the forward lever hoists [413] and aft lever hoist [453] equally until the shear pins are clear of the upper engine mounts.

NOTE: Binding of the aft engine mount can occur if the forward hoists are lowered too far ahead of the aft hoist.

CAUTION

MONITOR THE THRUST LINKS WHEN YOU LOWER AND TURN THE ENGINE. MAKE SURE IT DOES NOT INTERFERE WITH THE PRECOOLER OR DAMAGE TO THE PRECOOLER CAN OCCUR.

(b) Lower the forward lever hoists [413] and aft lever hoist [453] equally until the engine is approximately 4 in. (10 cm) - 6 in. (15 cm) above the engine stand.

(c) Adjust the lever hoists [413] and aft lever hoist [453] to align the engine with the engine stand as follows:

1) If it is necessary, adjust the height of the cradle, wheels, and engine stand positions.

2) Align the engine stand forward adapters [472] to the forward engine stand support arms.

3) Secure with the adapter pins [474] and retainers [475].

NOTE: Insert the pins from the rear.

4) Raise the aft engine stand support columns from the stowed position.

5) Attach the engine stand aft adapters [476] to the TCF lifting bracket adapter, SPL-15012 as follows:

a) Align the engine stand aft adapters [476] to the aft engine stand support columns.

WARNING

MAKE SURE THAT YOU CORRECTLY ATTACH SUPPORT COLUMNS ON THE ENGINE STAND. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONNEL AND DAMAGE TO THE ENGINE CAN OCCUR.

b) Secure with the adapter pin [477], axle [479], and retainer [478].

c) Make sure that the axle [479] is installed from the rear (View C, Figure 410).

(d) Remove the forward lever hoists [413] with the forward dynamometer assembly [415], the aft lever hoist [453] with the aft dynamometer assembly [454], shackles [414], and shackle [451] from the engine attached brackets and from the bootstrap equipment (View C, Figure 409).

SUBTASK 71-00-02-860-024-G00

(12) Carefully pull the engine stand with the engine attached, forward until it is clear of the strut.

(a) Make sure that you monitor the clearance between the engine and other components while you pull the engine stand forward.

SUBTASK 71-00-02-860-065-G00

(13) If you lower the engine stand cradle and casters, do these steps to put the engine stand in a position to taxi the removed engine:

(a) Use the stand hydraulic device to lift the cradle until engine stand casters off the ground.

(b) Install the four pins in the cradle to activate the cradle damping.

(c) If it is necessary, put the wheel casters on the engine transportation stand to 3.15 in. (80.01 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Lower the engine stand with the stand hydraulic device.

(e) Make sure that the engine stand hydraulic actuators are fully retracted.

SUBTASK 71-00-02-480-003-G00

(14) If it is necessary, install the transportation brace, SPL-14557, on the forward and aft engine mount for shipping.

NOTE: Boeing recommends using the transportation brace, SPL-14557, for engine transportation with the forward and aft engine mounts installed. The tool is designed to prevent the movement and damage of the engine mounts and the engine core case when installed on the engine. If the engine mount stabilization equipment is not used, the engine mounts can be removed to prevent the mounts from moving during transportation. Alternatively, commercially available packaging materials can be used in lieu of transportation brace, SPL-14557, as long as they serve the same purpose.

SUBTASK 71-00-02-020-001-G00

(15) If it is necessary, remove the forward and aft engine mount (TASK 71-21-01-000-801-G00 and TASK 71-21-02-000-801-G00).

SUBTASK 71-00-02-860-025-G00

(16) Put the covers on the open ends of the ducts on the engine and strut.

SUBTASK 71-00-02-080-001-G00

(17) If it is necessary, remove the bootstrap equipment as follows:

(a) Remove the two pins [447] and two retainers [443] that attach the aft outboard arm assembly [448] (View C, Figure 408).

(b) Remove the aft outboard arm assembly [448] from the aft torque beam assembly [446].

(c) Remove the retainer [443] and pin [441] from the outboard aft support beam assembly [442] (View A, Figure 408).

(d) Remove the outboard aft support beam assembly [442] from the outboard forward strut bracket.

(e) Remove the two retainers [443] from the two pins [441] that attach the outboard aft cross beam assembly [444] (View A, Figure 408).

(f) Remove the outboard aft cross beam assembly [444] from the outboard forward strut two brackets.

(g) Remove the four retainers [443] from the two pins [447] and two pins [445] (View B, Figure 408).

(h) Remove the two pins [447] and two pins [445] that attach the aft torque beam assembly [446] to the outboard aft support beam assembly [442] and outboard aft cross beam assembly [444].

(i) Remove the pin [420] and retainer [421] that join the forward outboard arm assembly [409] and forward inboard arm assembly [416] to the top of the forward support arm assembly [406] (View C, Figure 409 and View E, Figure 409).

(j) Remove the two pins [423] and two retainers [422] that attach the forward outboard arm assembly [409] and forward inboard arm assembly [416] to the forward center beam [401] (View C, Figure 409 and View E, Figure 409).

(k) Remove the forward outboard arm assembly [409] and forward inboard arm assembly [416] from the forward center beam [401].

(l) Remove the pin [408] and retainer [407] that attach the forward support arm assembly [406] to the forward center beam [401] (View B, Figure 409).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) Remove the pin [418] and retainer [419] that attach the other end of the forward support arm assembly [406] to the strut fan cowl support beam (View D, Figure 409).

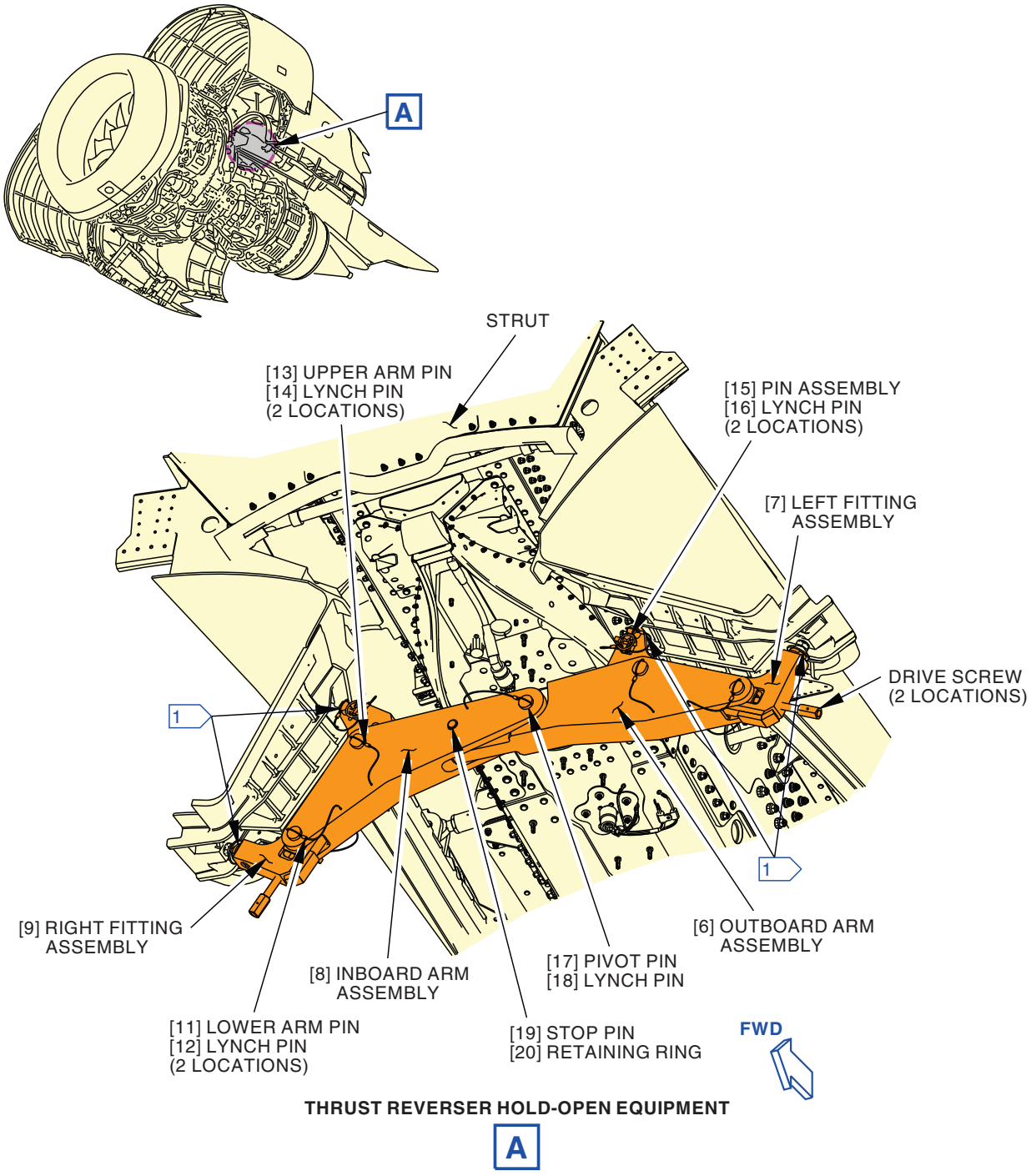
(n) Remove the forward support arm assembly [406] from top of the forward center beam [401].

(o) Remove the two pins [403] and two retainers [402] that attach the forward center beam [401] to the forward mount brackets (View A, Figure 409).

(p) Remove the forward center beam [401] from top of the forward mount brackets.

——— END OF TASK ———

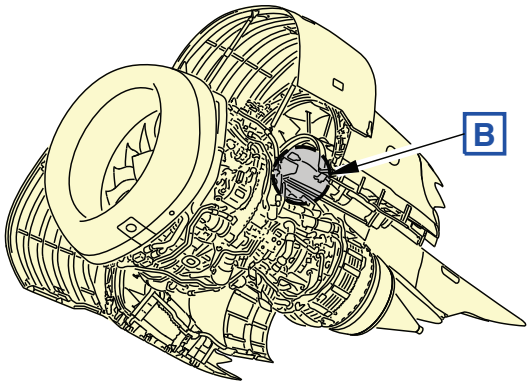
ECCN 9E991 BOEING PROPRIETARY - See title page for details

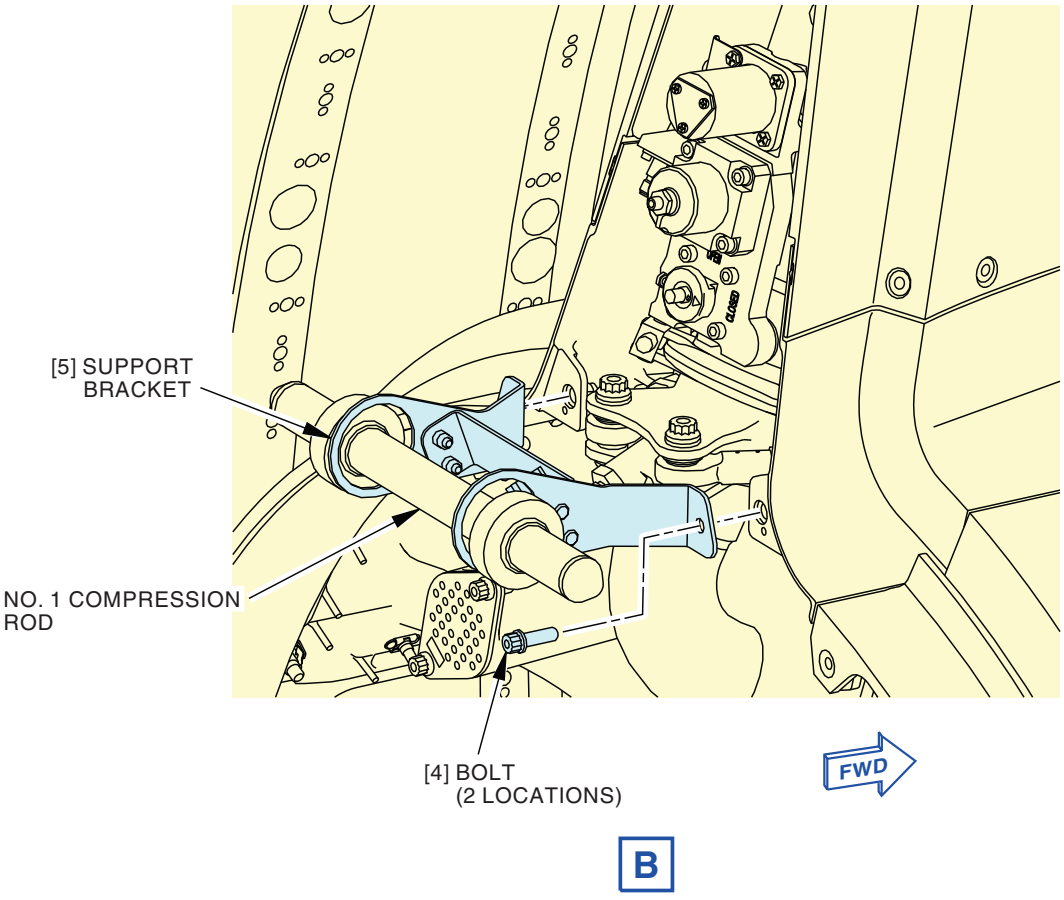


2433567 S0000562680_V2

Thrust Reverser Hold-Open Equipment Installation
Figure 401/71-00-02-990-815-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

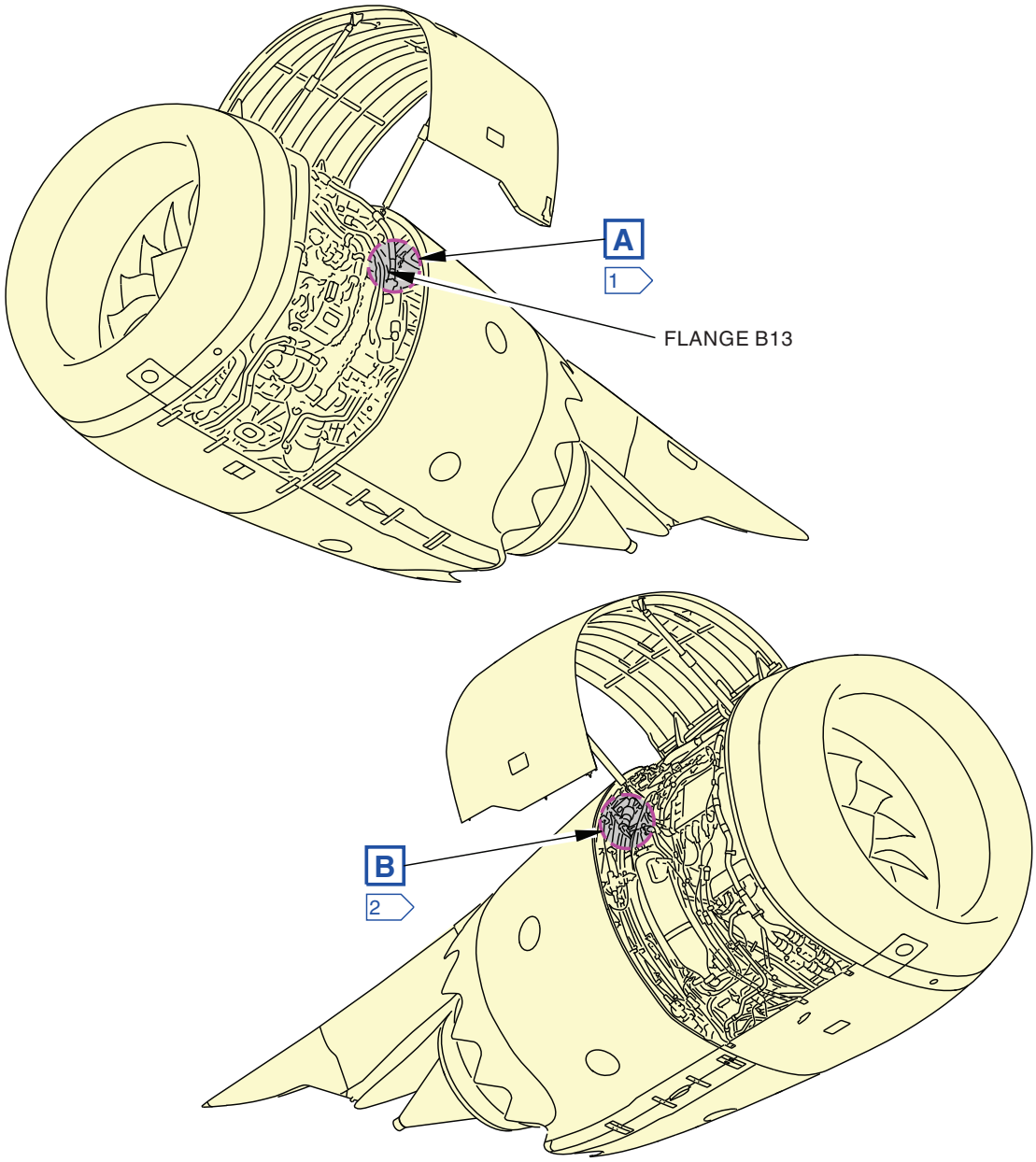




2935851 S0000711688_V1

Thrust Reverser Hold-Open Equipment Installation
Figure 401/71-00-02-990-815-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

THE MDOS ACTUATORS ARE SPECIFIC FOR THE INBOARD AND OUTBOARD SIDE OF THE ENGINE.

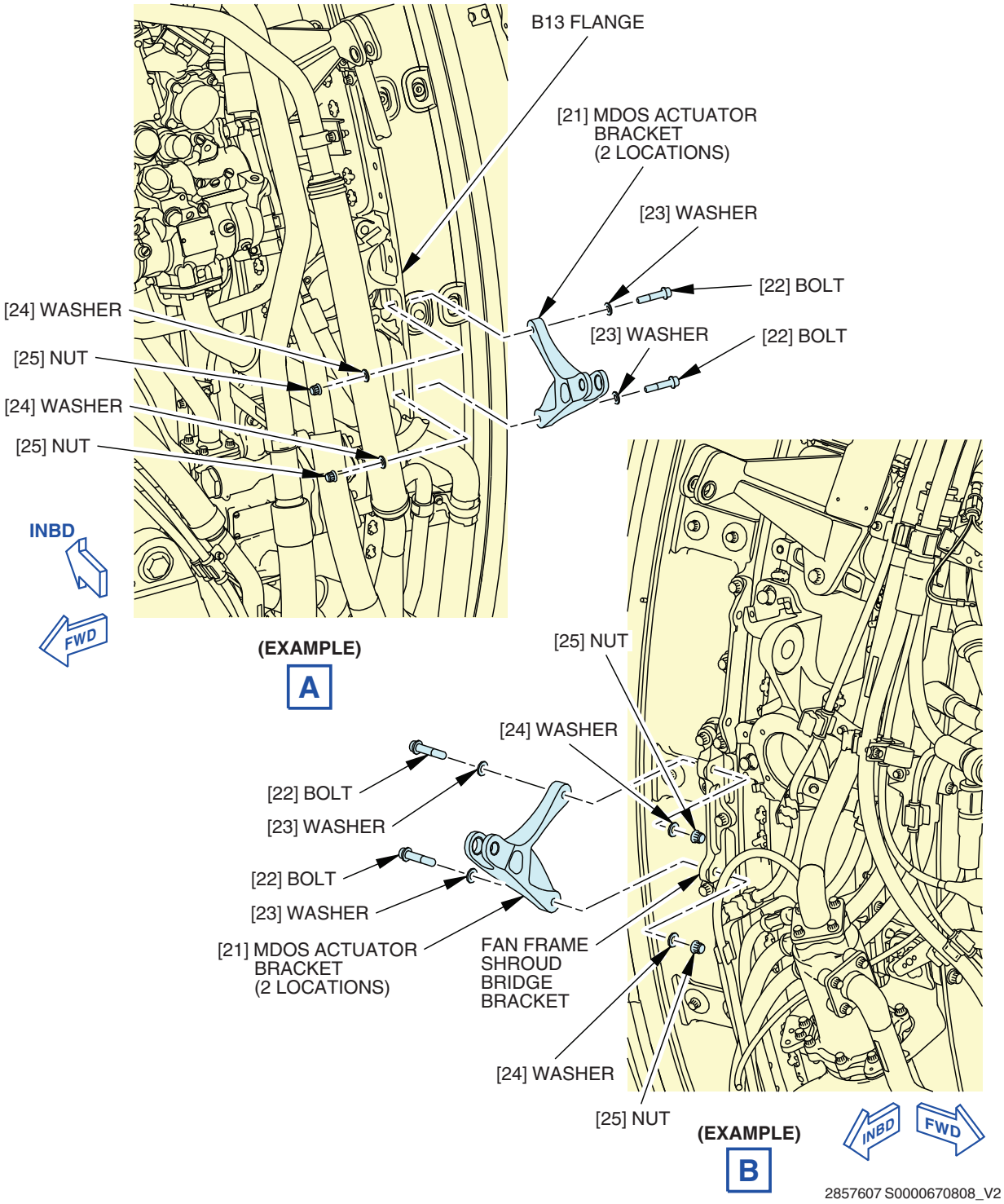
1 MDOS ACTUATOR BRACKETS INSTALL ON THE AFT SIDE OF THE FLANGE B13 INBOARD AND OUTBOARD SIDE OF THE ENGINE.

2 MDOS ACTUATOR BRACKETS INSTALL ON THE FAN FRAME SHROUD BRIDGE BRACKETS. UPPER BRACKET FOR THE INBOARD ACTUATOR AND LOWER BRACKET FOR THE OUTBOARD ACTUATOR.

2857378 S0000670950_V1

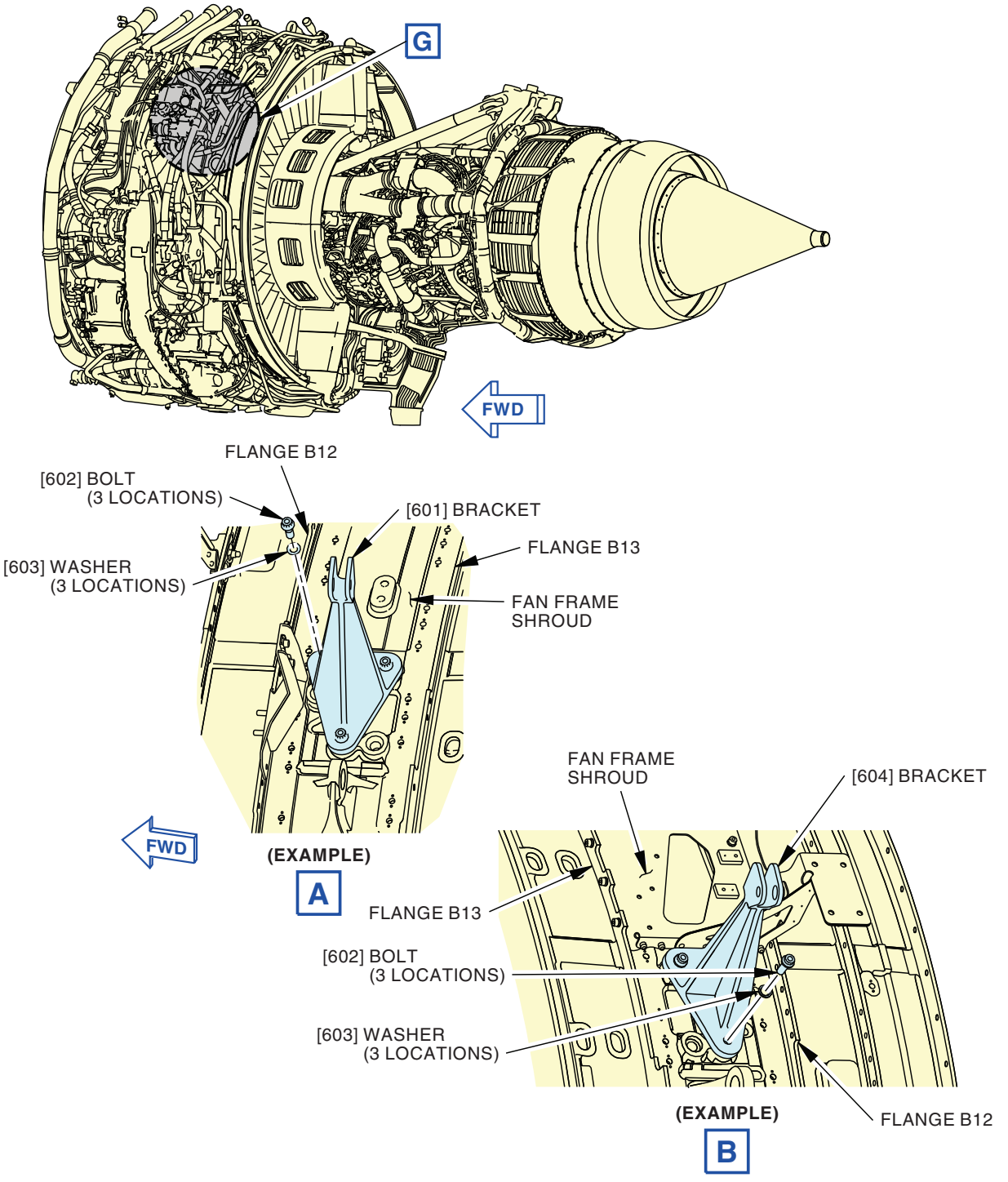
MDOS Engine Bracket Installation
Figure 402/71-00-02-990-819-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



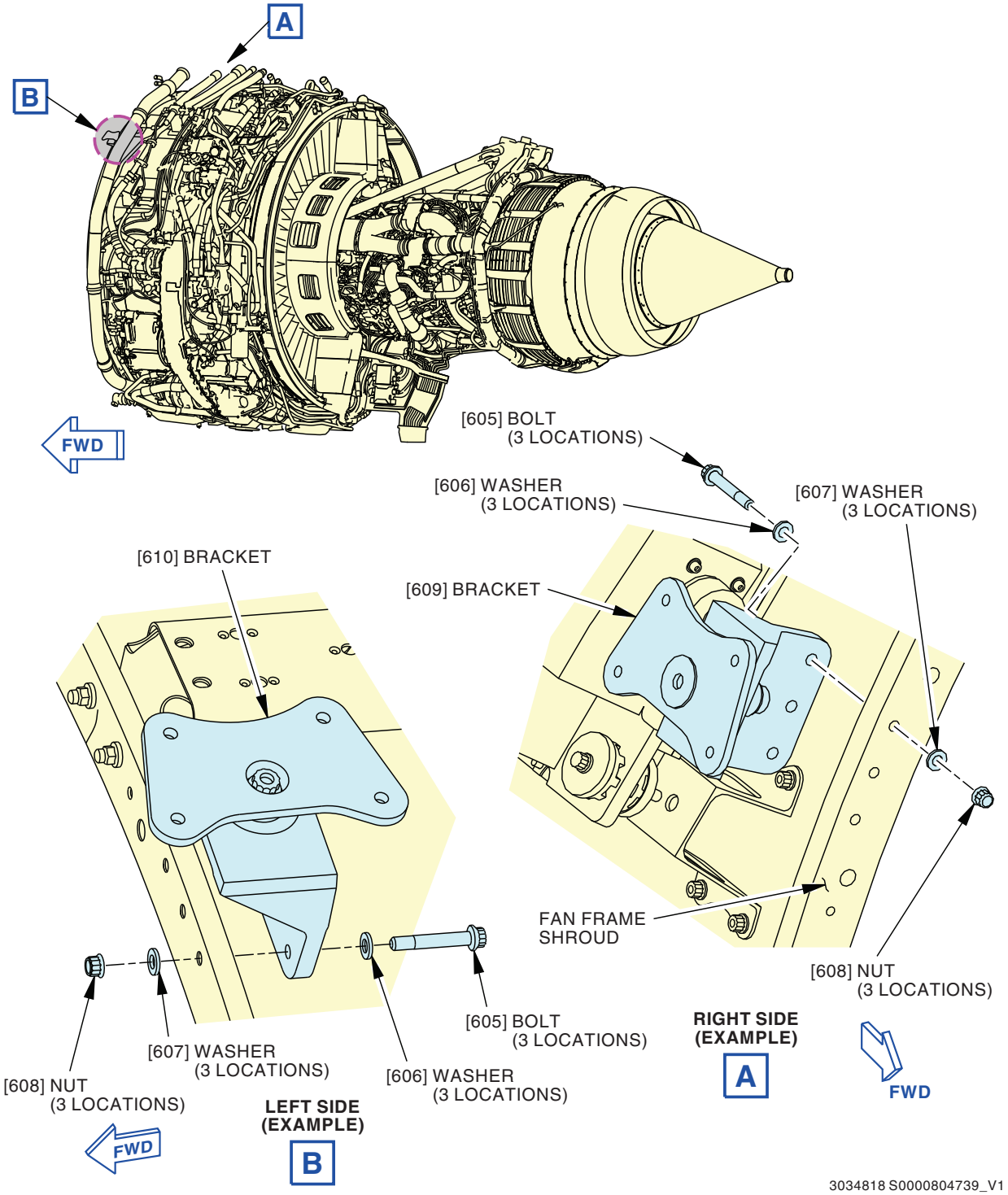
MDOS Engine Bracket Installation
Figure 402/71-00-02-990-819-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



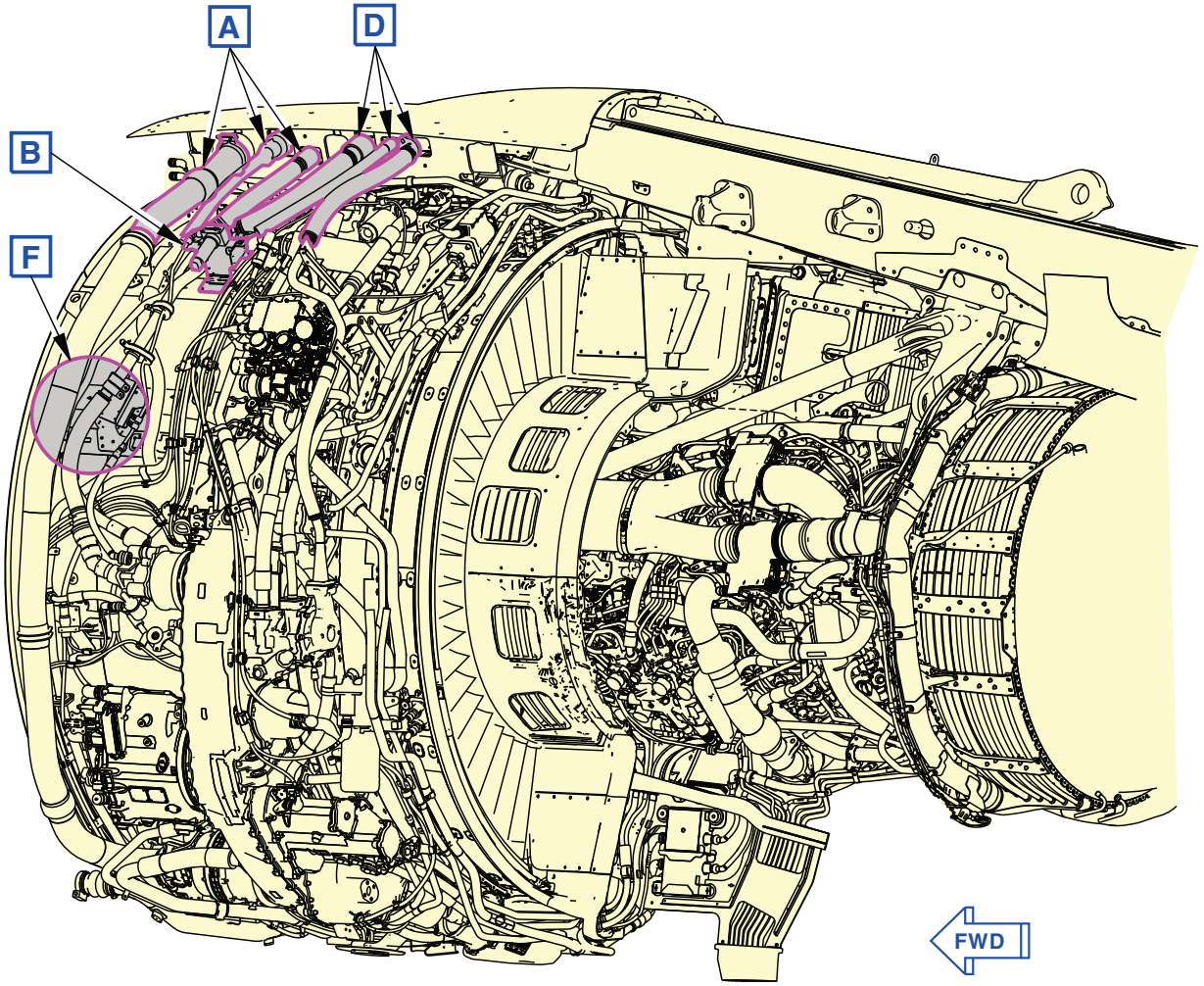
EBU Brackets Installation
Figure 403/71-00-02-990-818-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EBU Brackets Installation
Figure 403/71-00-02-990-818-G00 (Sheet 2 of 2)

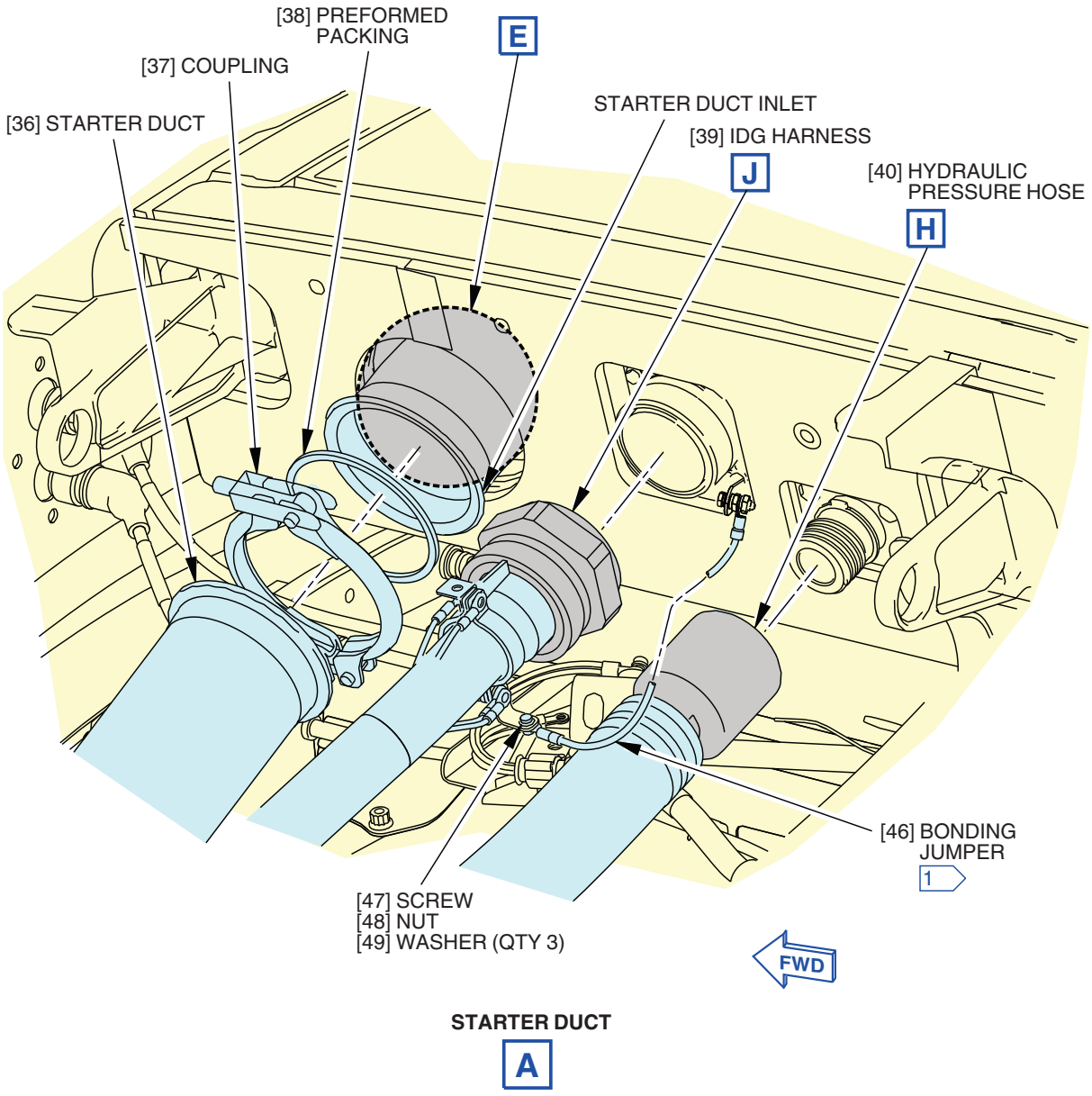
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354177 S00061514929_V4

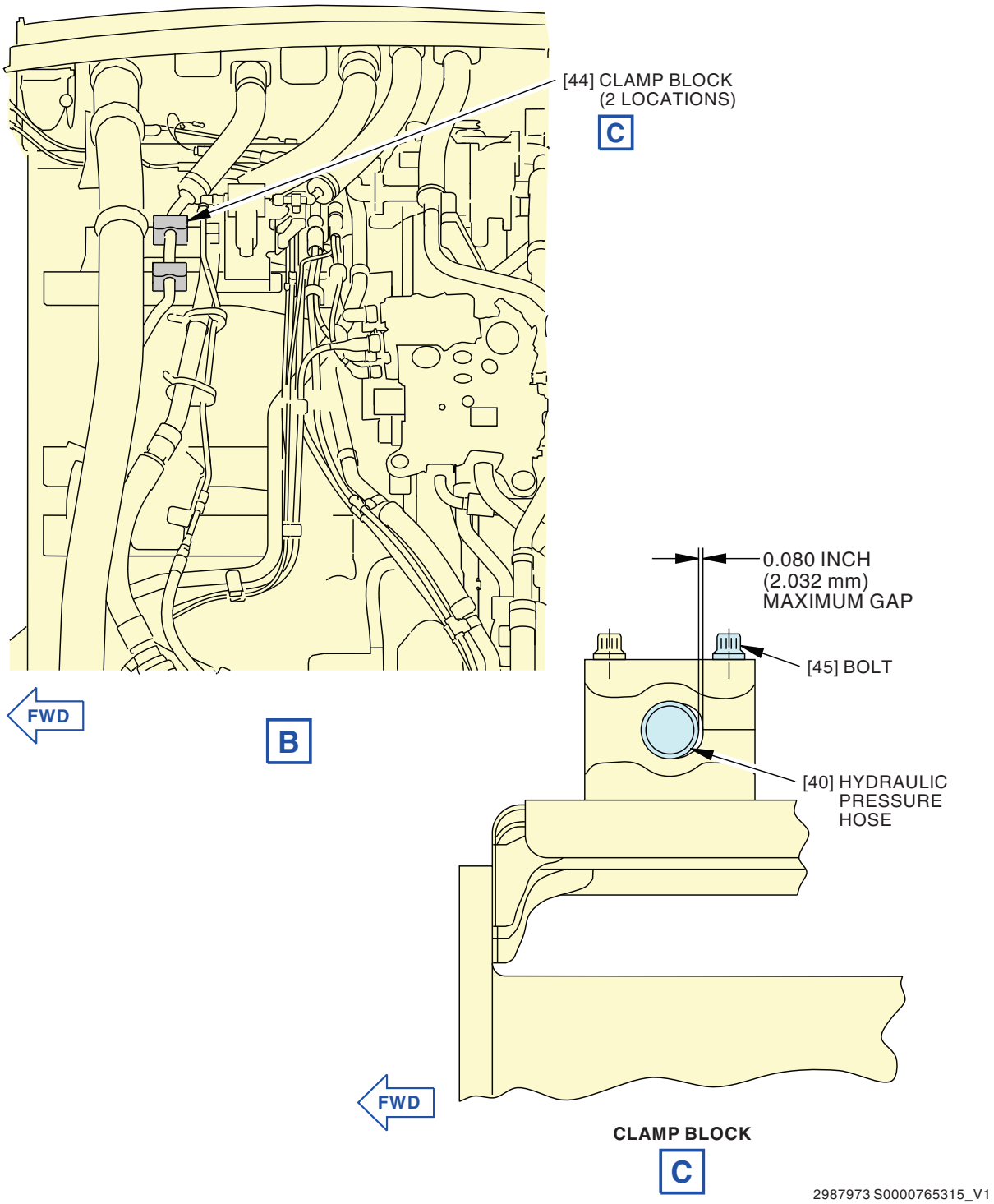
Left Side of Engine Installation
Figure 404/71-00-02-990-802-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



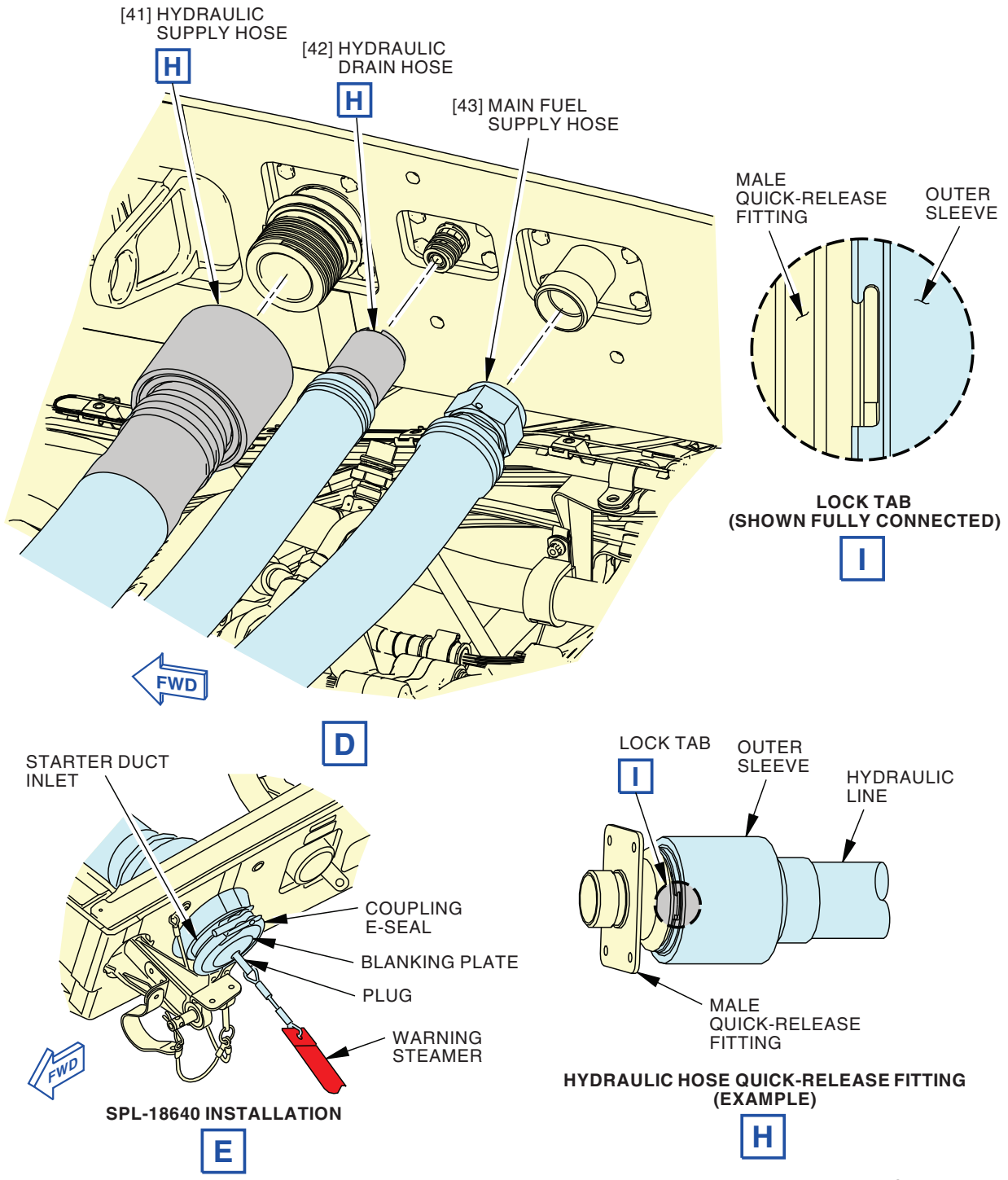
Left Side of Engine Installation
Figure 404/71-00-02-990-802-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



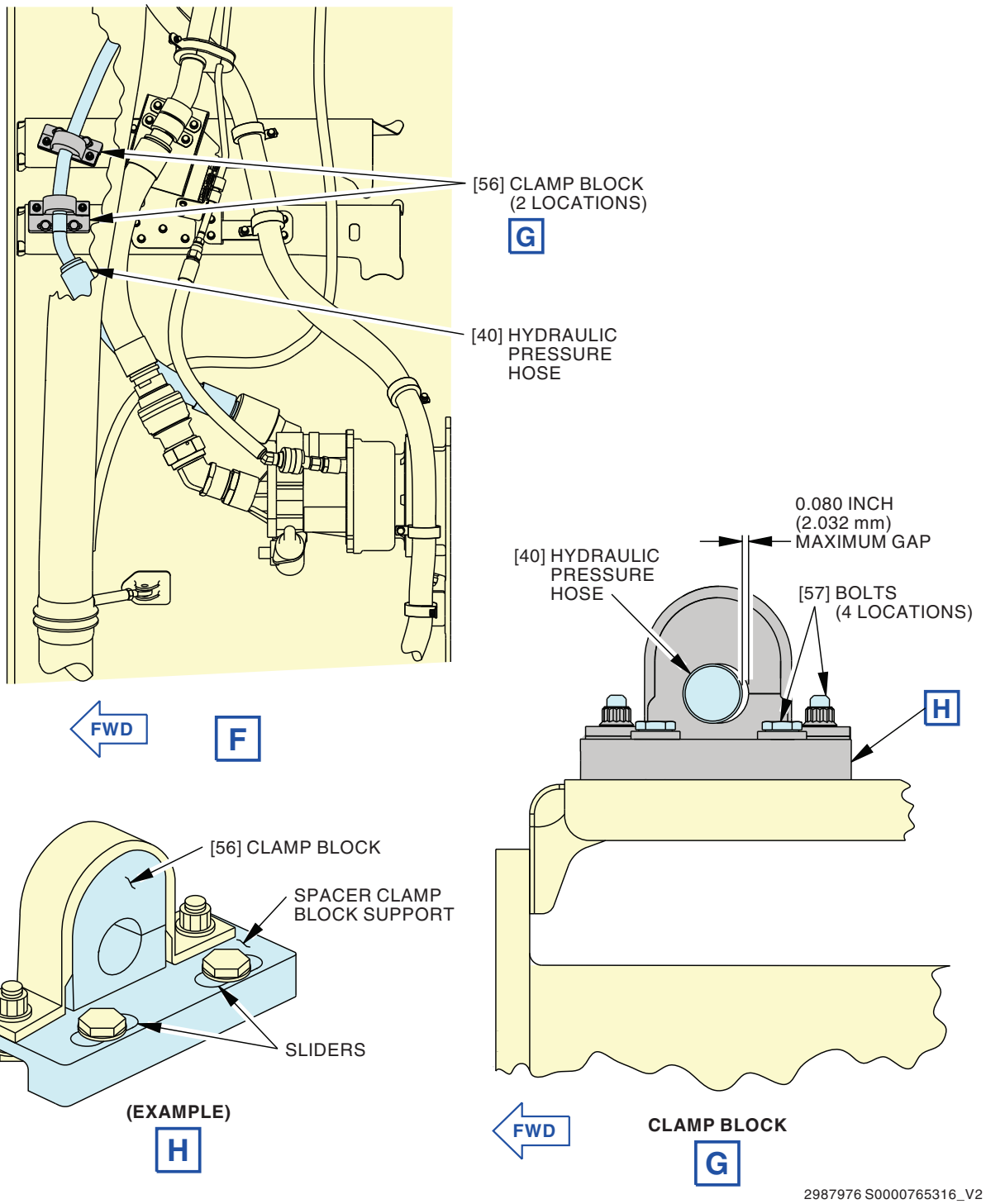
Left Side of Engine Installation
Figure 404/71-00-02-990-802-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



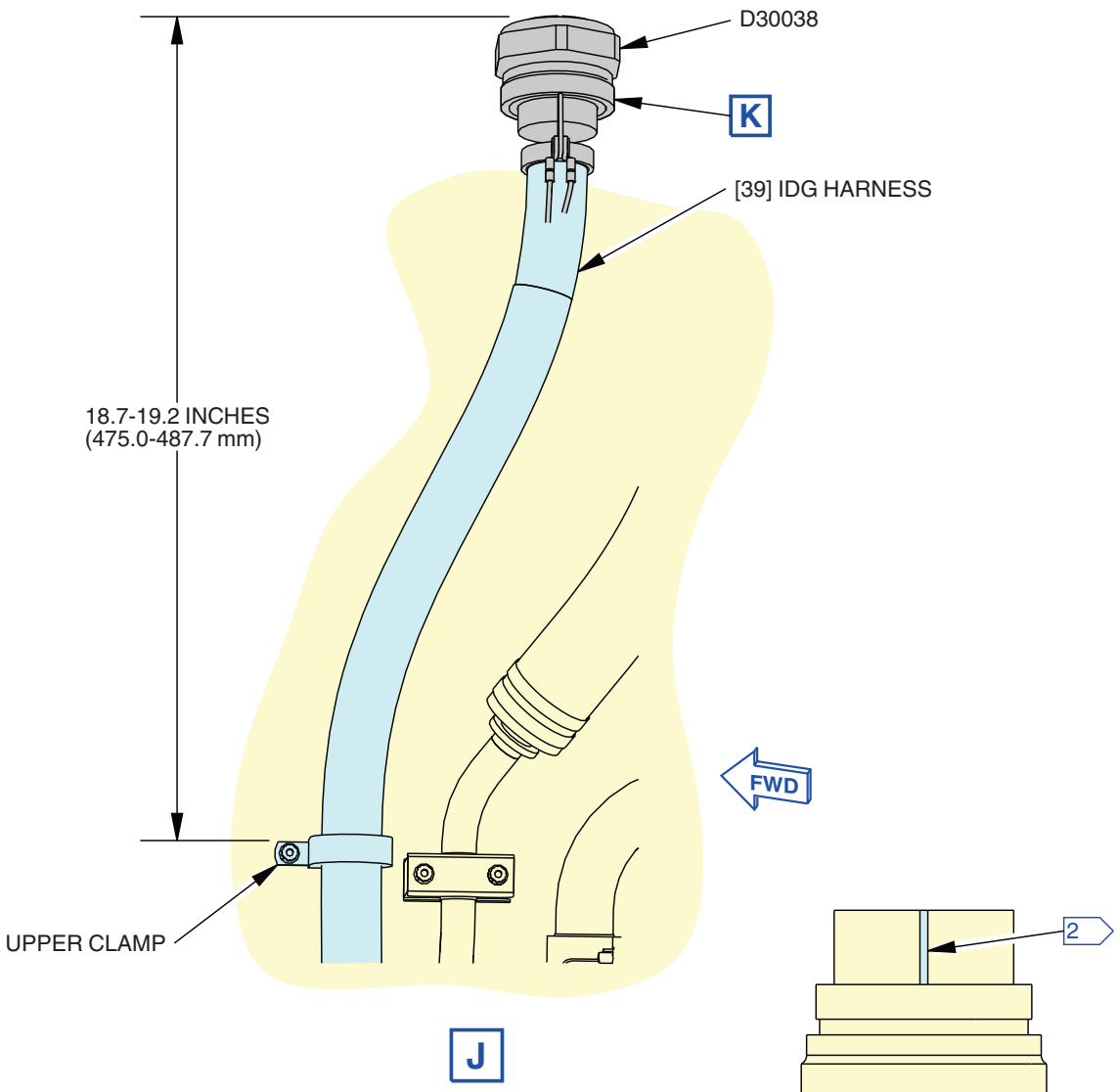
Left Side of Engine Installation
Figure 404/71-00-02-990-802-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left Side of Engine Installation
Figure 404/71-00-02-990-802-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

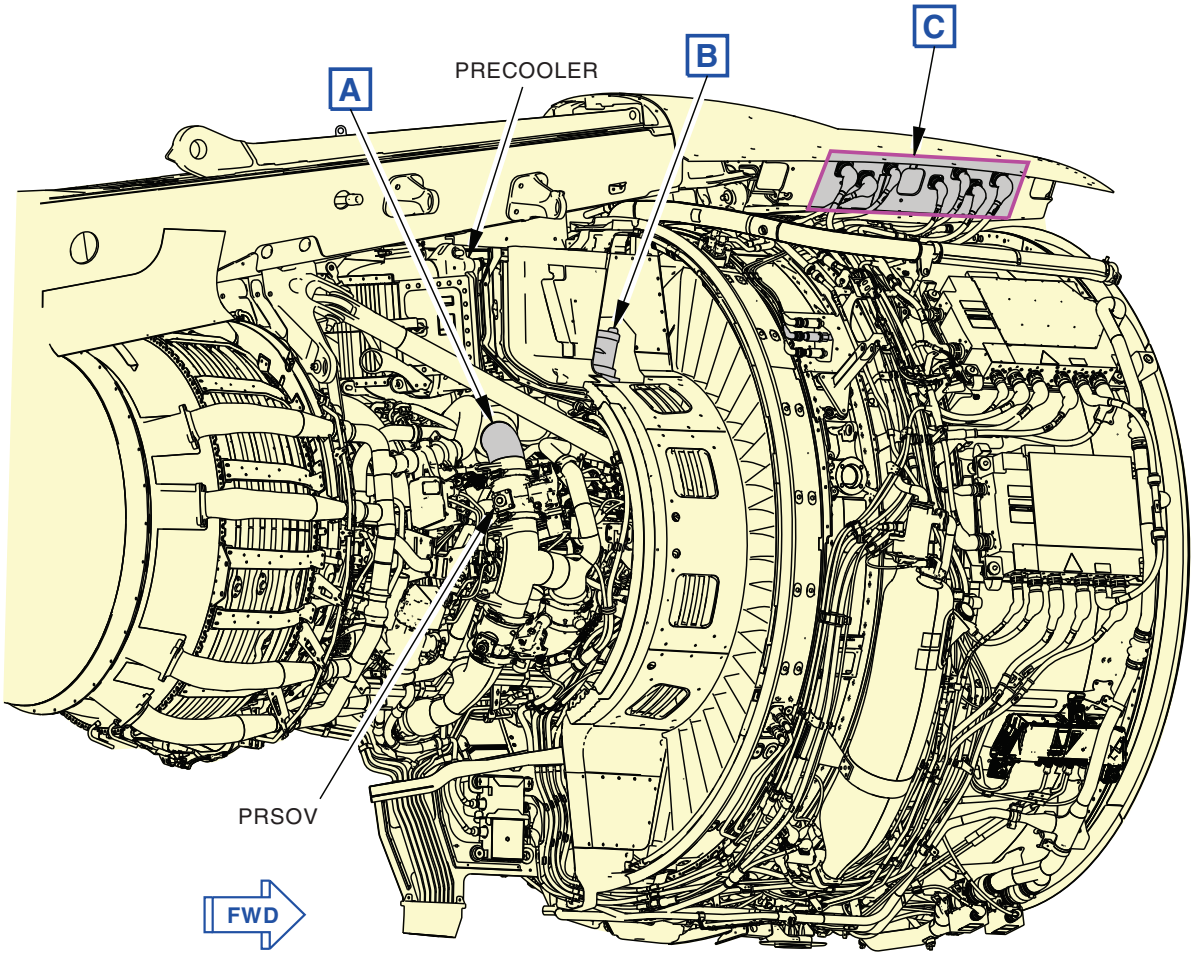


2 THE WIRE BUNDLE MASTER KEYWAY (NOTCH) MUST BE IN THE UPRIGHT POSITION (12 O'CLOCK POSITION) ±10 DEGREES AFTER WIRE HARNESS IS INSTALLED ON ENGINE.

3085789 S0000839311_V2

Left Side of Engine Installation
Figure 404/71-00-02-990-802-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

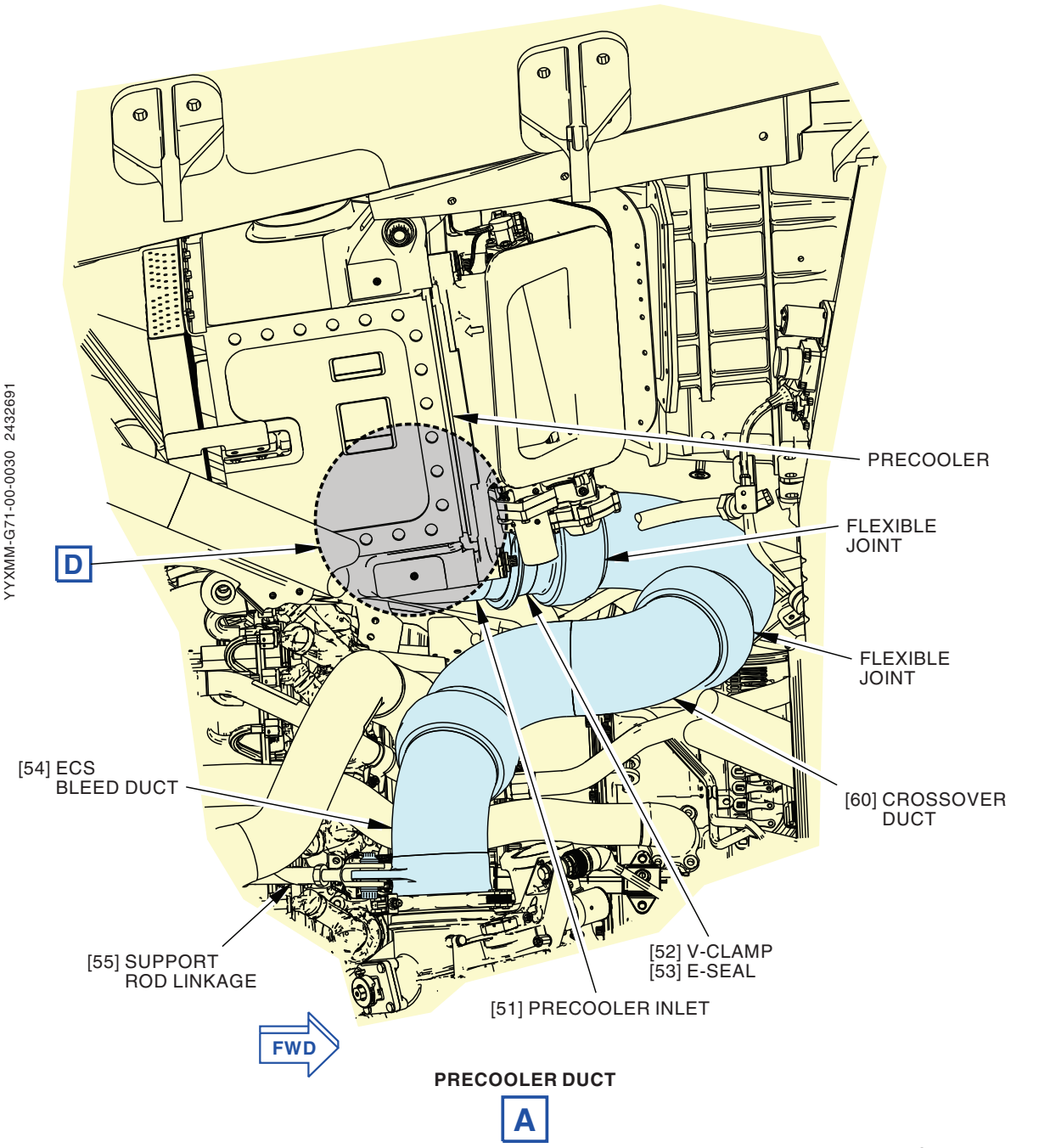


(TYPICAL)

2354184 S00061514936_V3

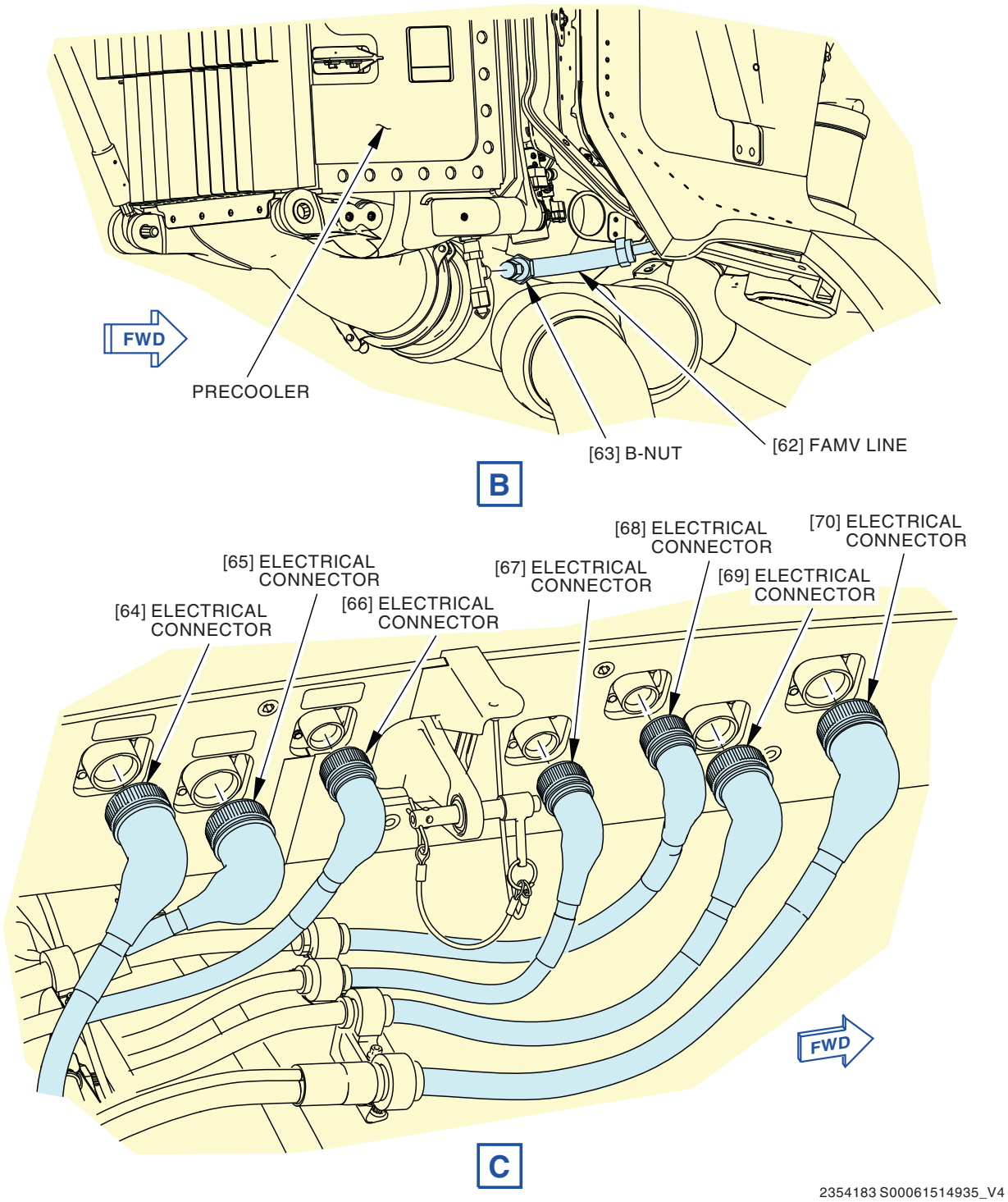
Right Side of Engine Installation
Figure 405/71-00-02-990-805-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



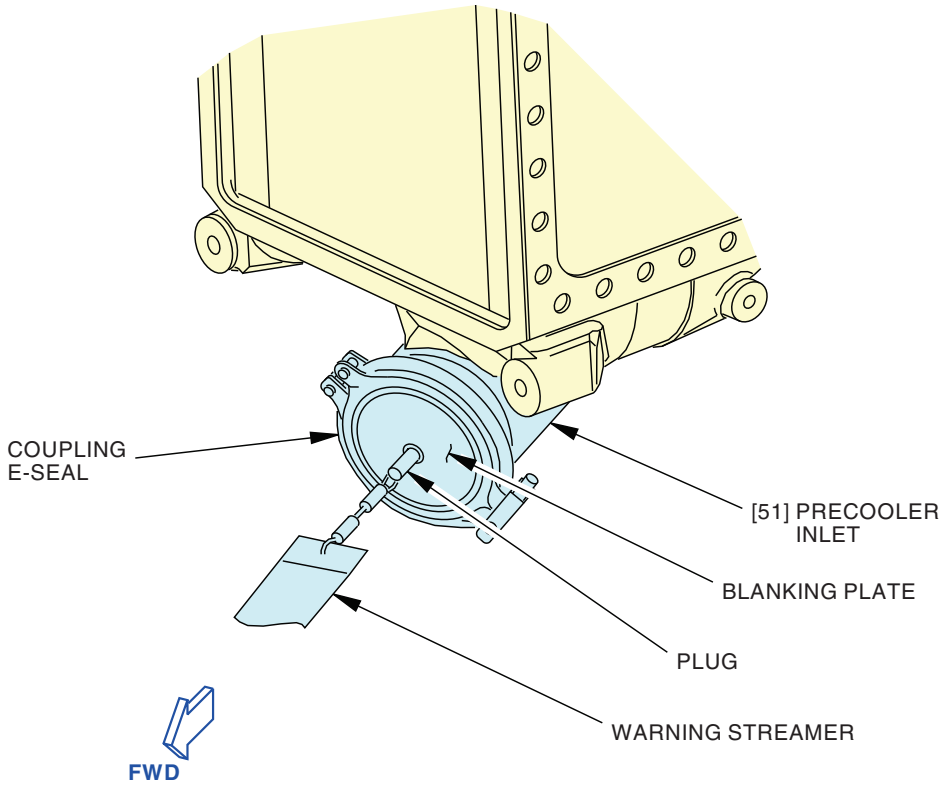
Right Side of Engine Installation
Figure 405/71-00-02-990-805-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right Side of Engine Installation
Figure 405/71-00-02-990-805-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



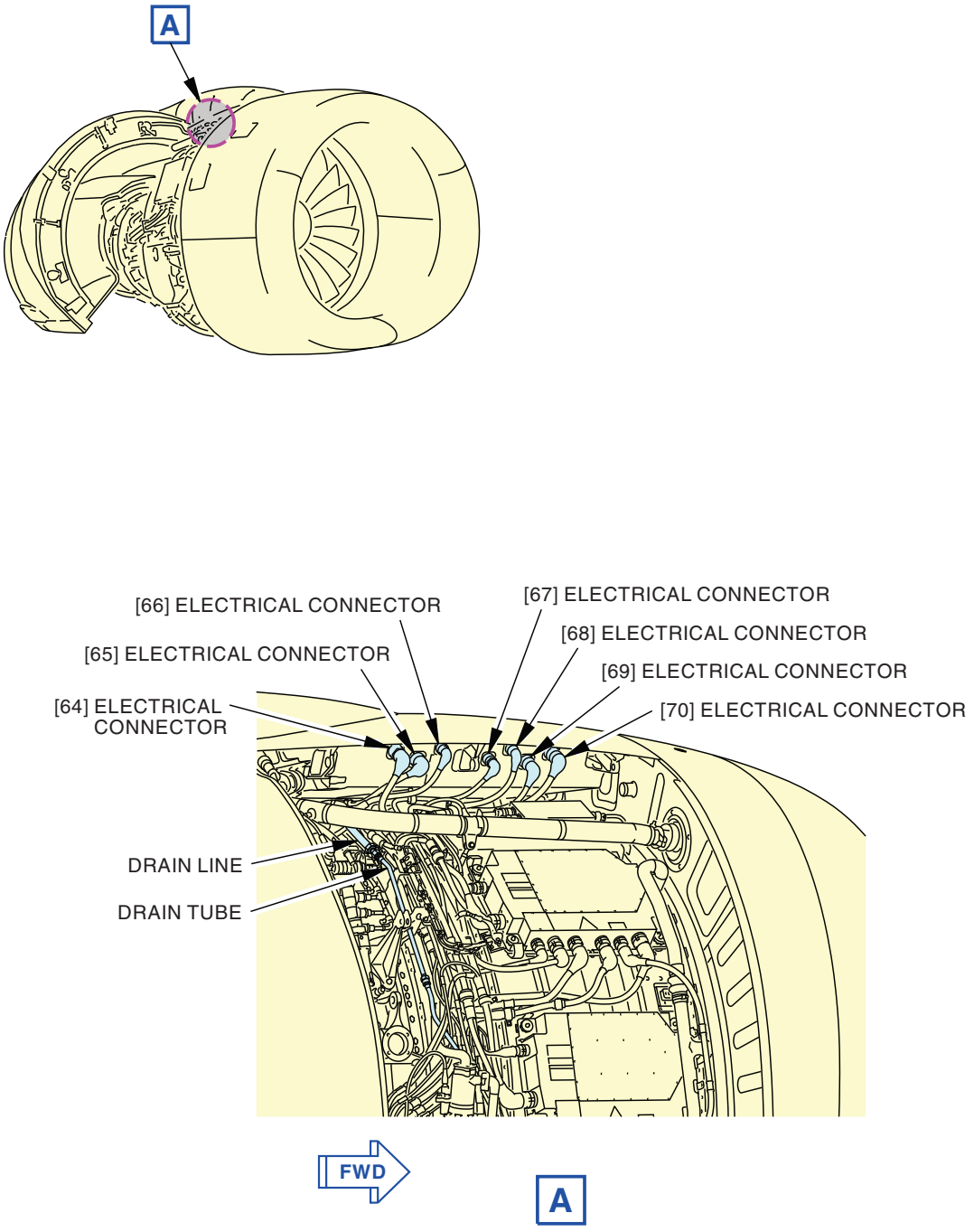
SPL-18258 INSTALLATION

D

2913921 S0000699687_V1

Right Side of Engine Installation
Figure 405/71-00-02-990-805-G00 (Sheet 4 of 4)

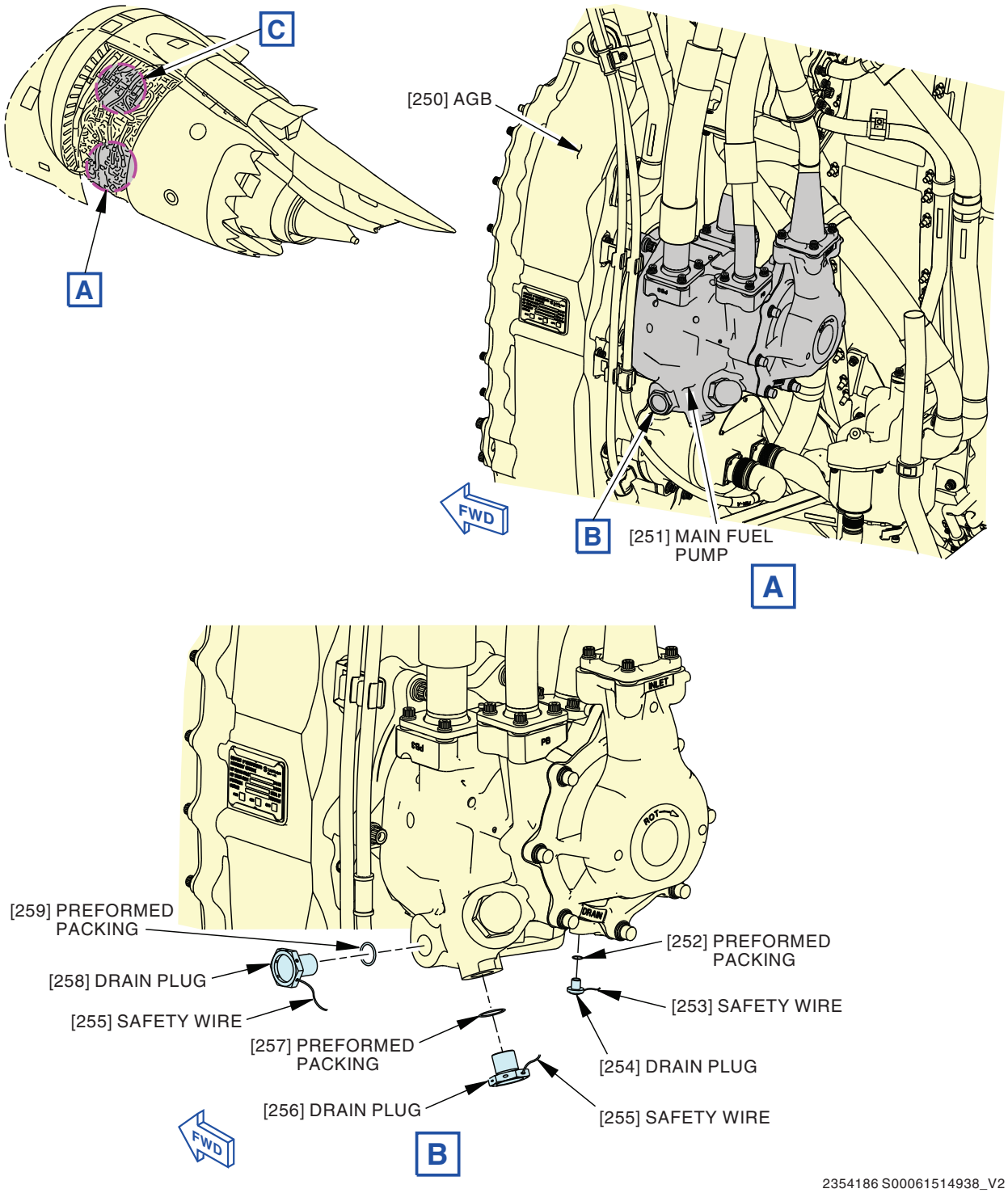
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3012378 S0000786011_V1

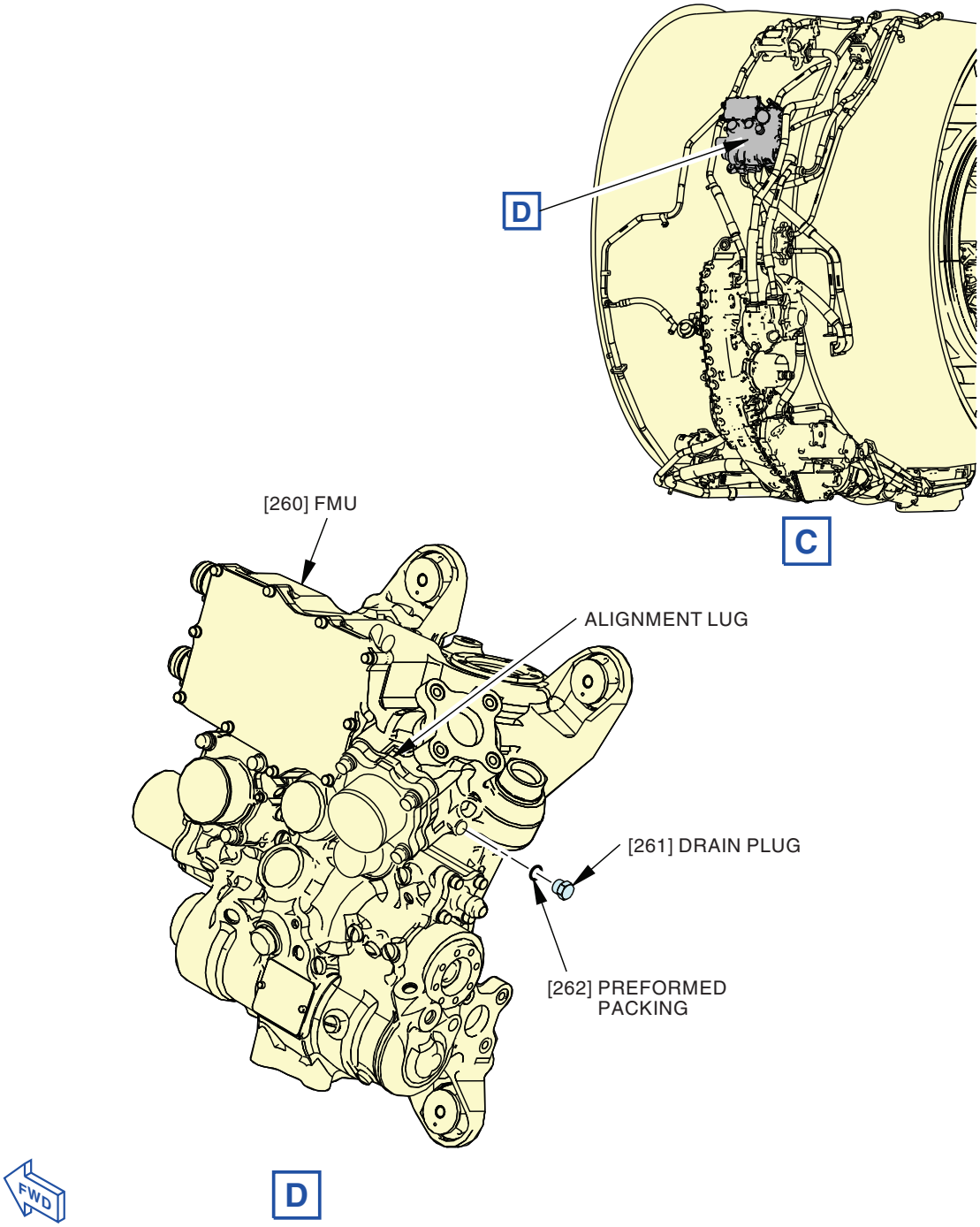
Fan Case Disconnects on the Right Side
Figure 406/71-00-02-990-820-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



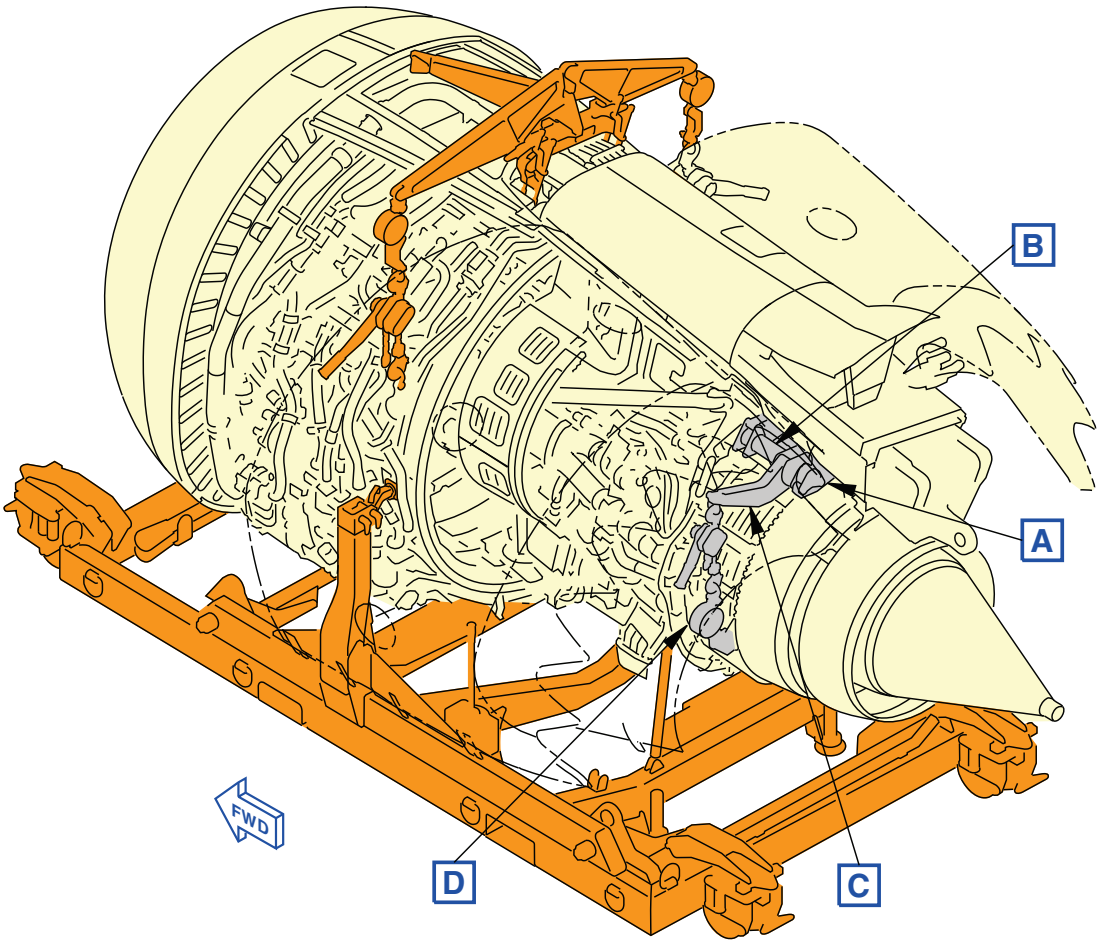
Engine Fuel Drain Plug Installation
Figure 407/71-00-02-990-806-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fuel Drain Plug Installation
Figure 407/71-00-02-990-806-G00 (Sheet 2 of 2)

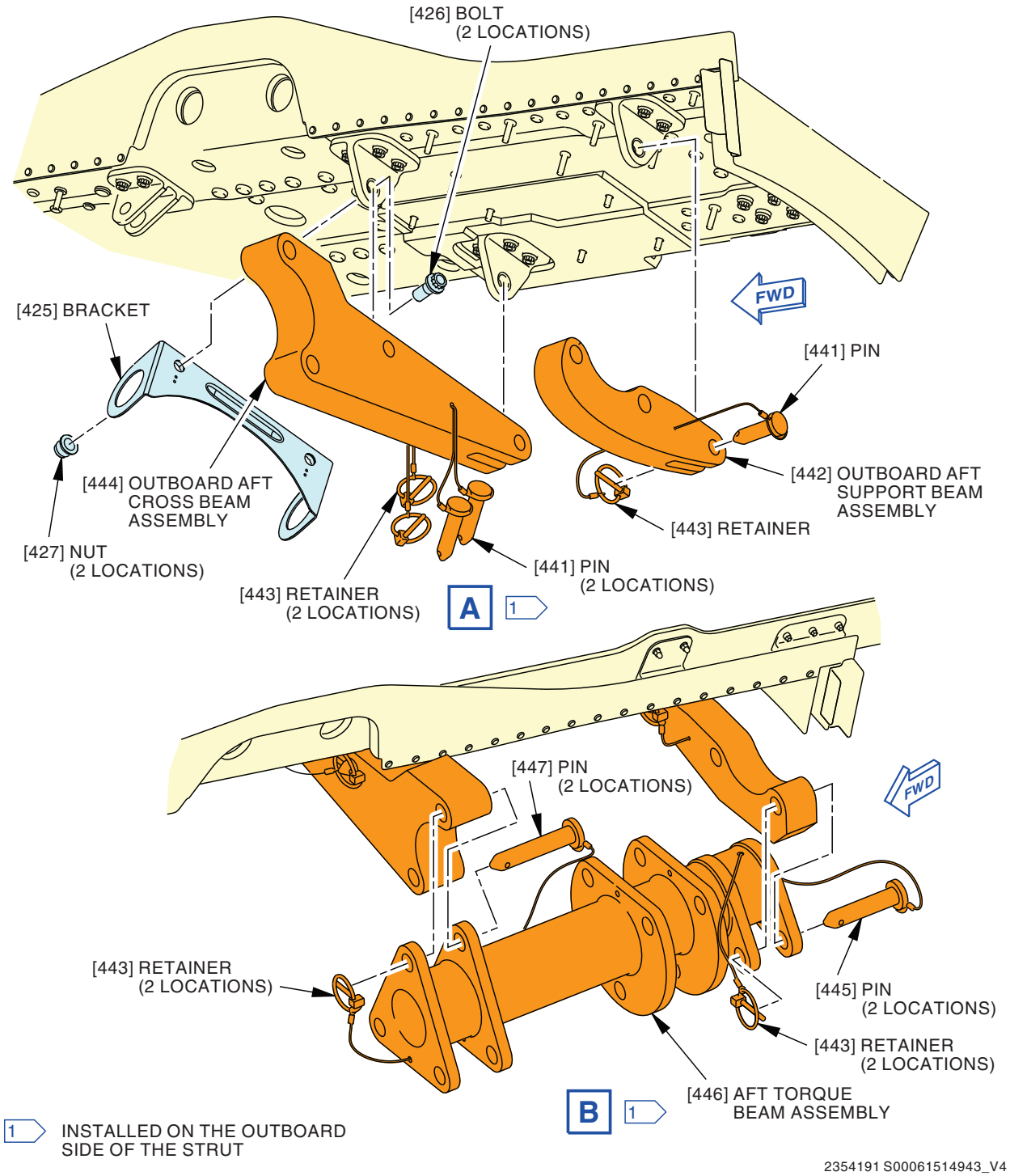
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354187 S00061514939_V2

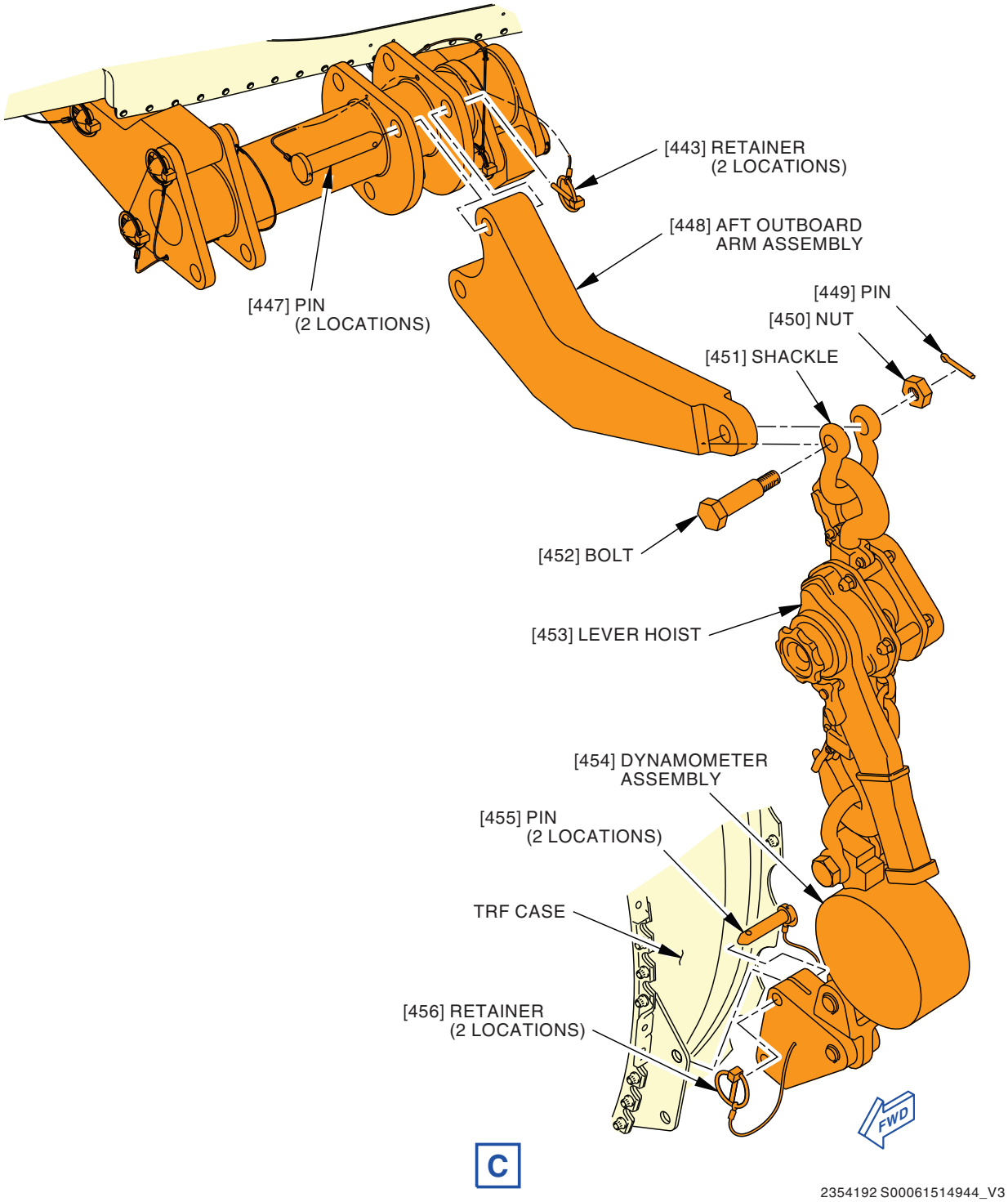
Aft Bootstrap Equipment Installation
Figure 408/71-00-02-990-809-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



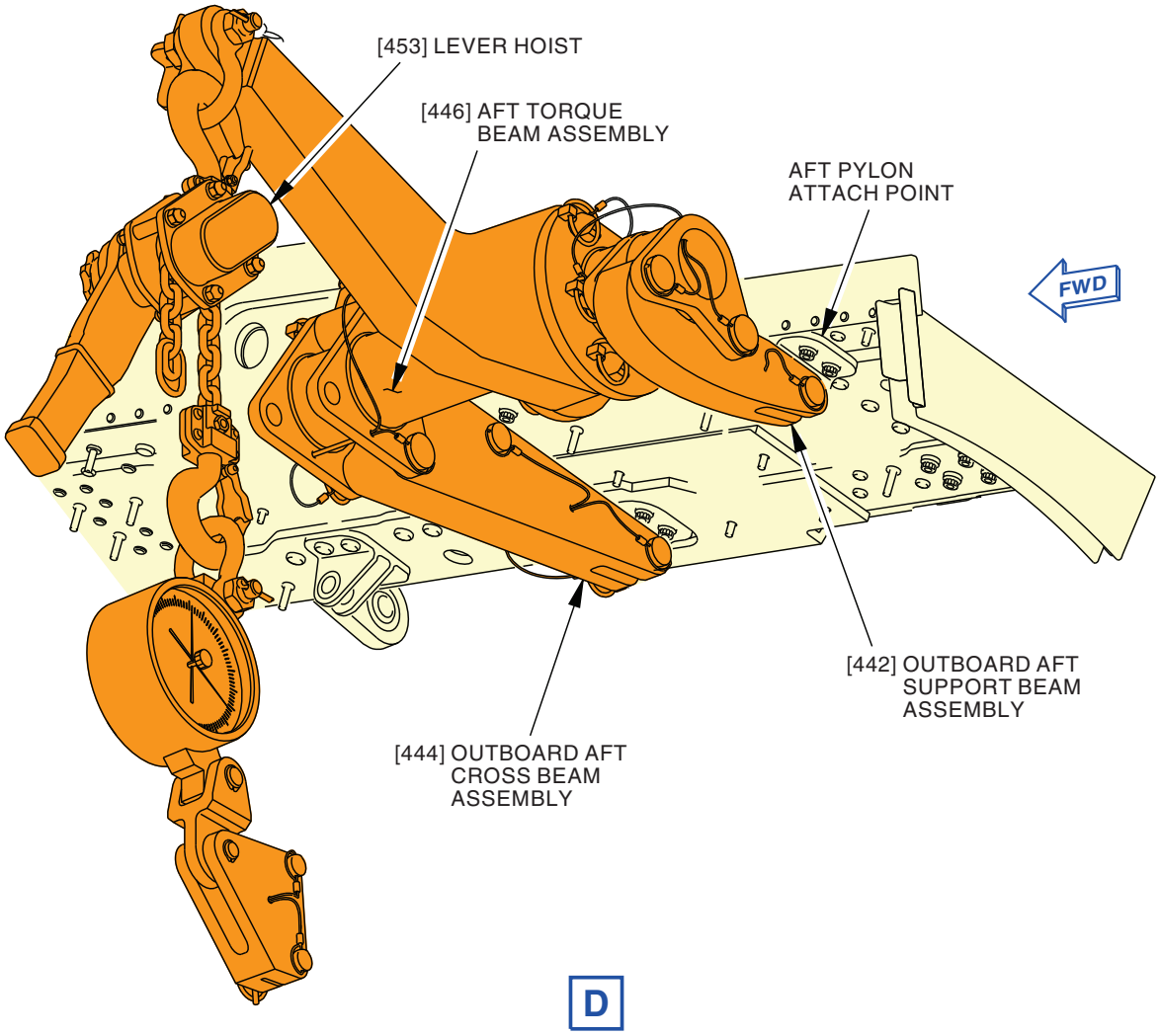
Aft Bootstrap Equipment Installation
Figure 408/71-00-02-990-809-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Bootstrap Equipment Installation
Figure 408/71-00-02-990-809-G00 (Sheet 3 of 4)

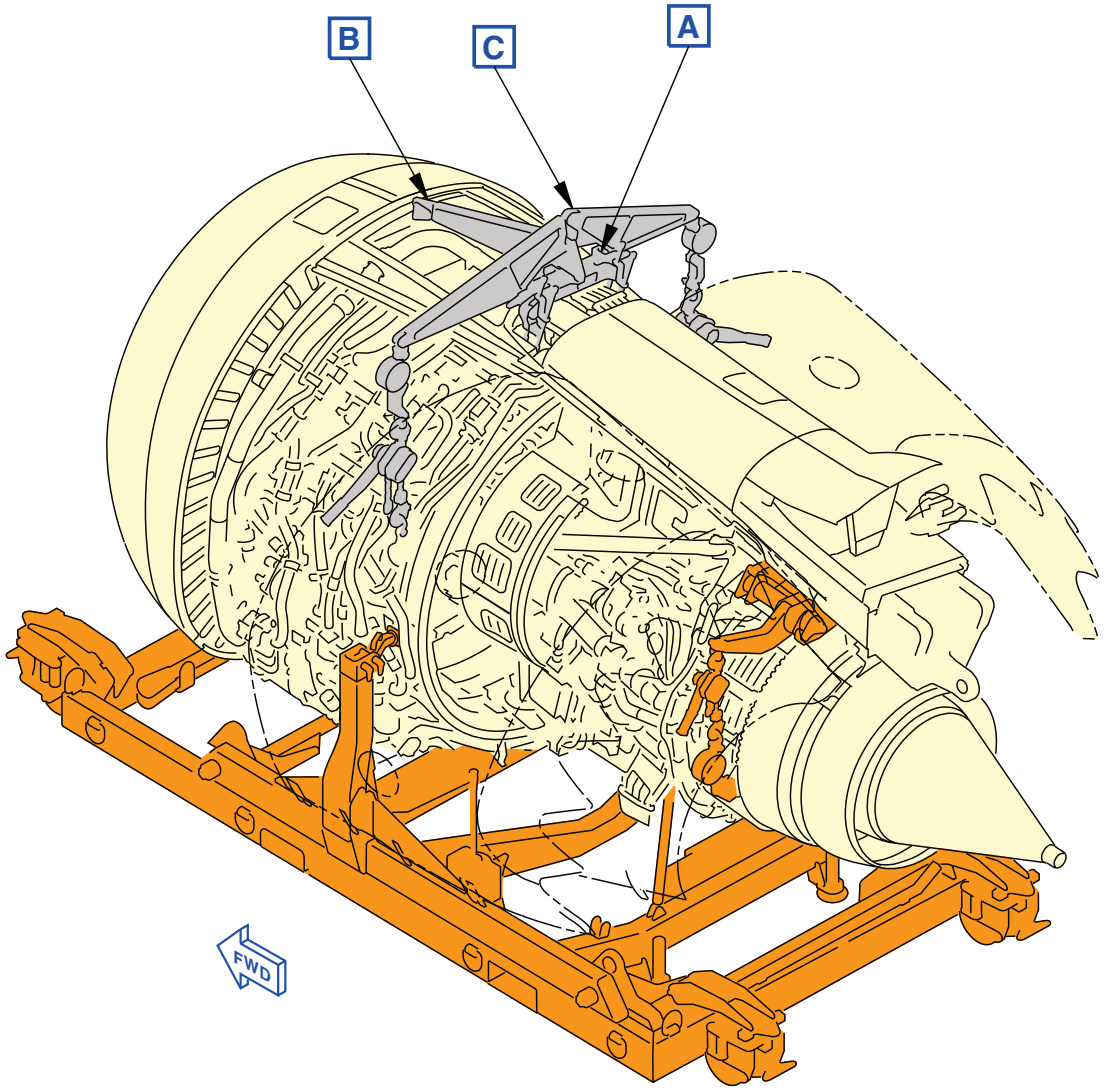
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354190 S00061514942_V2

Aft Bootstrap Equipment Installation
Figure 408/71-00-02-990-809-G00 (Sheet 4 of 4)

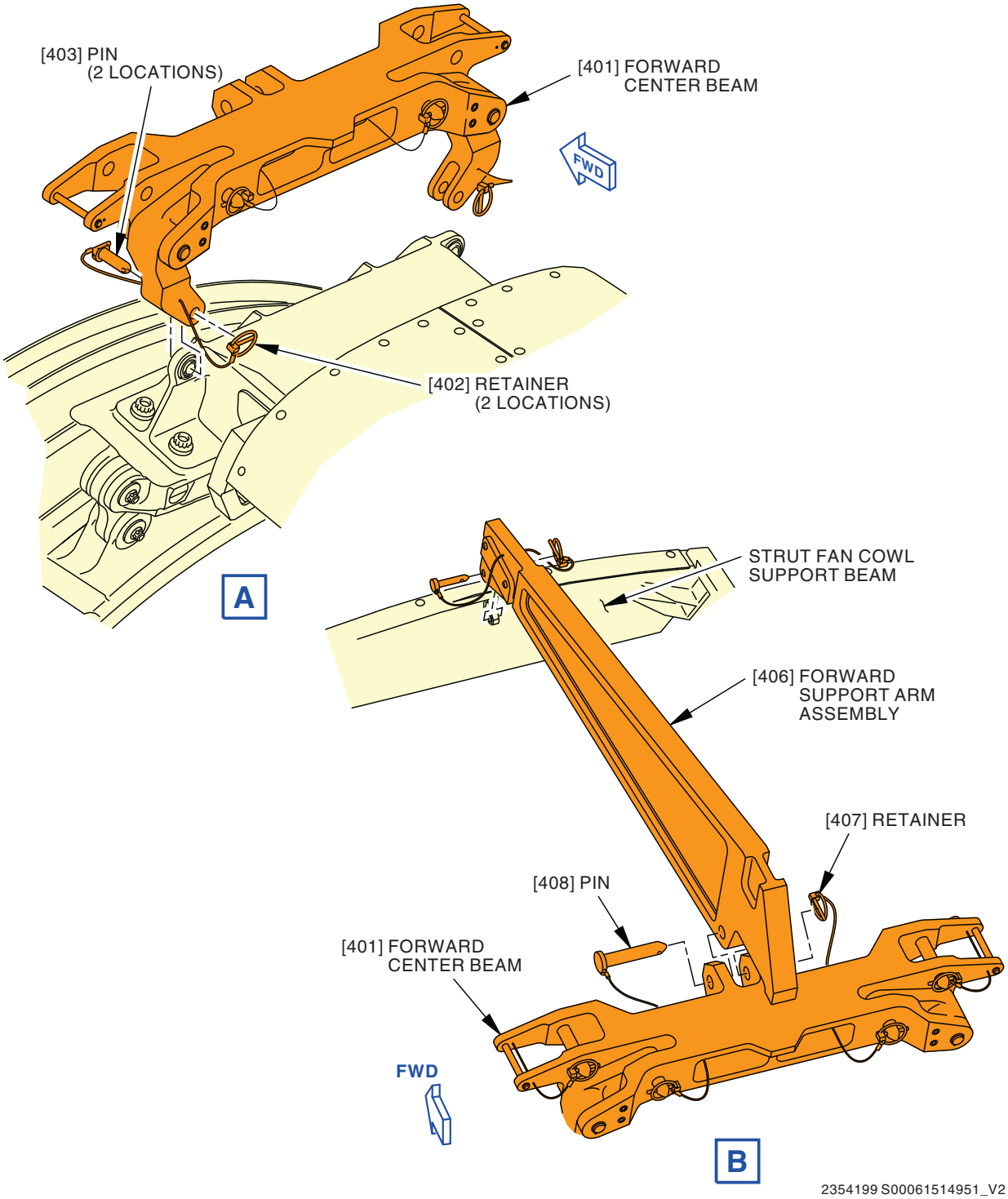
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2432304 S0000563253_V2

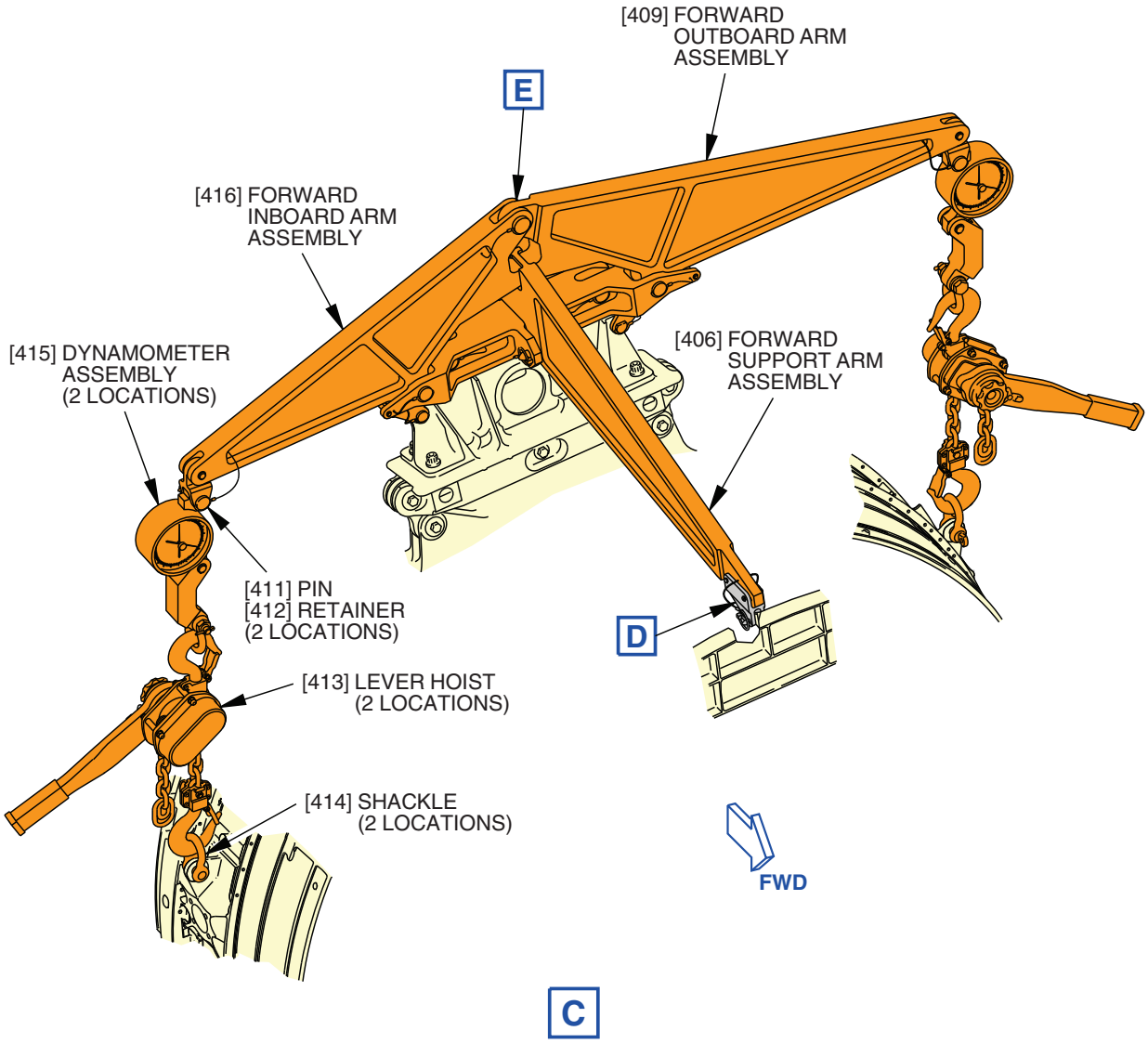
Forward Bootstrap Equipment Installation
Figure 409/71-00-02-990-811-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Bootstrap Equipment Installation
Figure 409/71-00-02-990-811-G00 (Sheet 2 of 6)

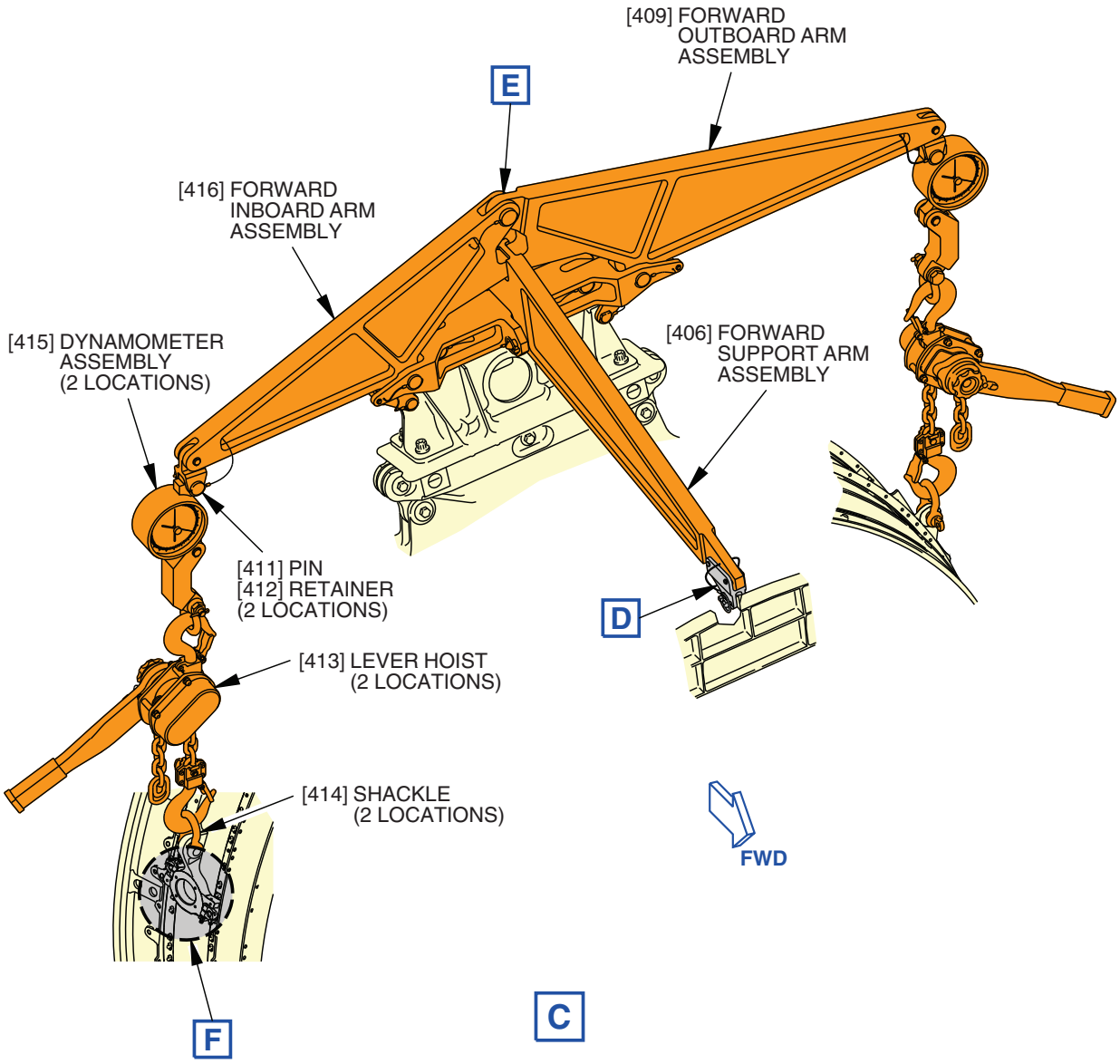
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354195 S00061514947_V4

Forward Bootstrap Equipment Installation
Figure 409/71-00-02-990-811-G00 (Sheet 3 of 6)

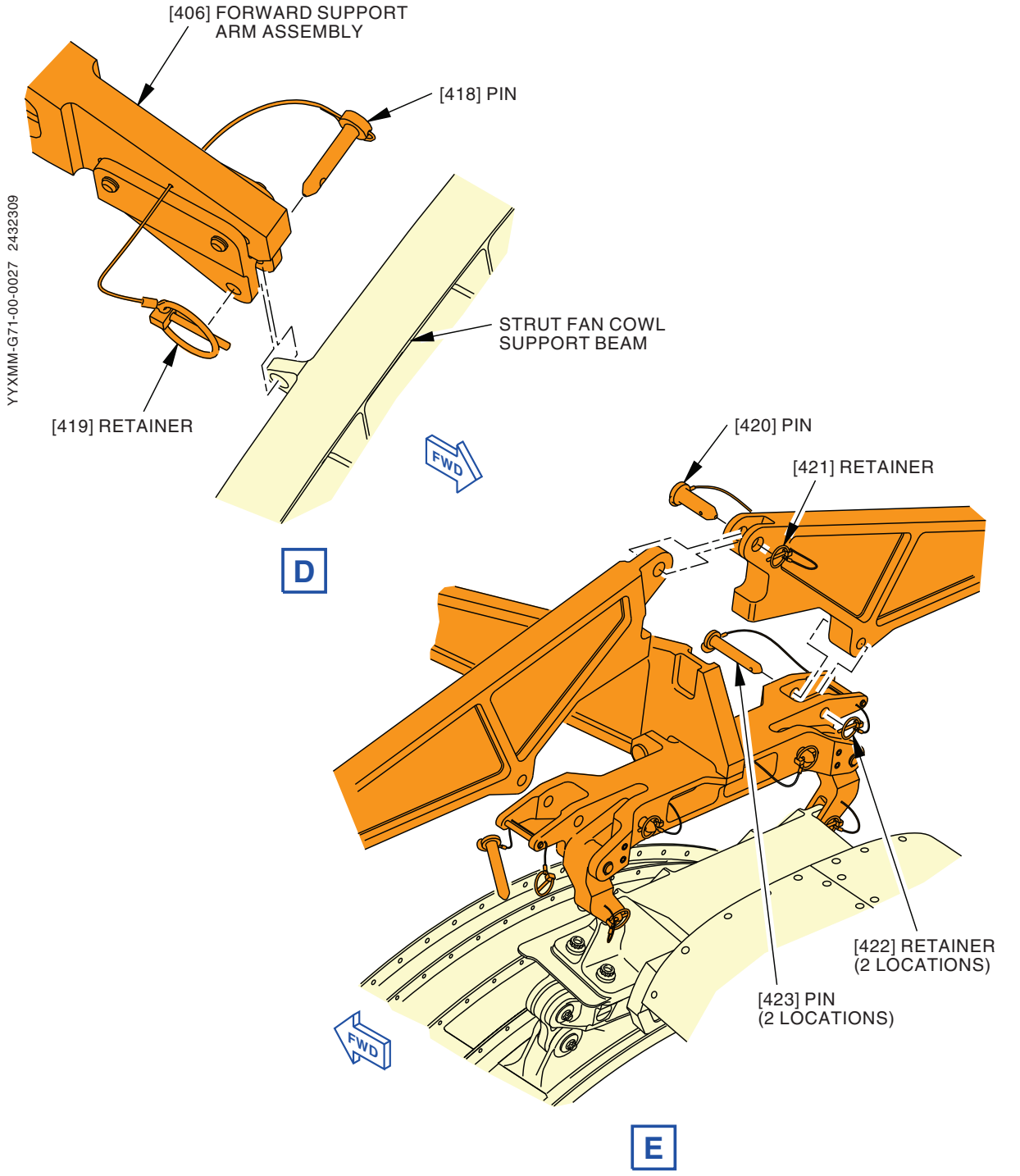
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2872175 S0000678476_V3

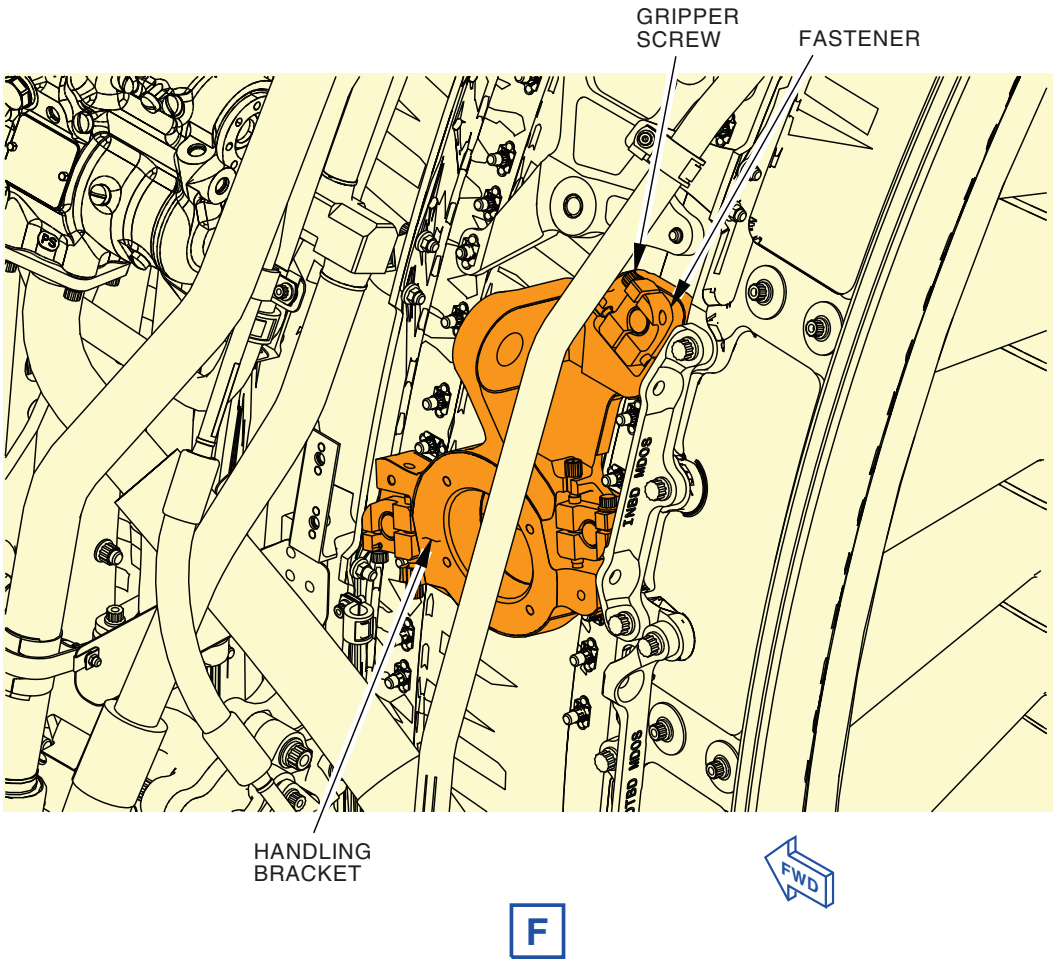
Forward Bootstrap Equipment Installation
Figure 409/71-00-02-990-811-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Bootstrap Equipment Installation
Figure 409/71-00-02-990-811-G00 (Sheet 5 of 6)

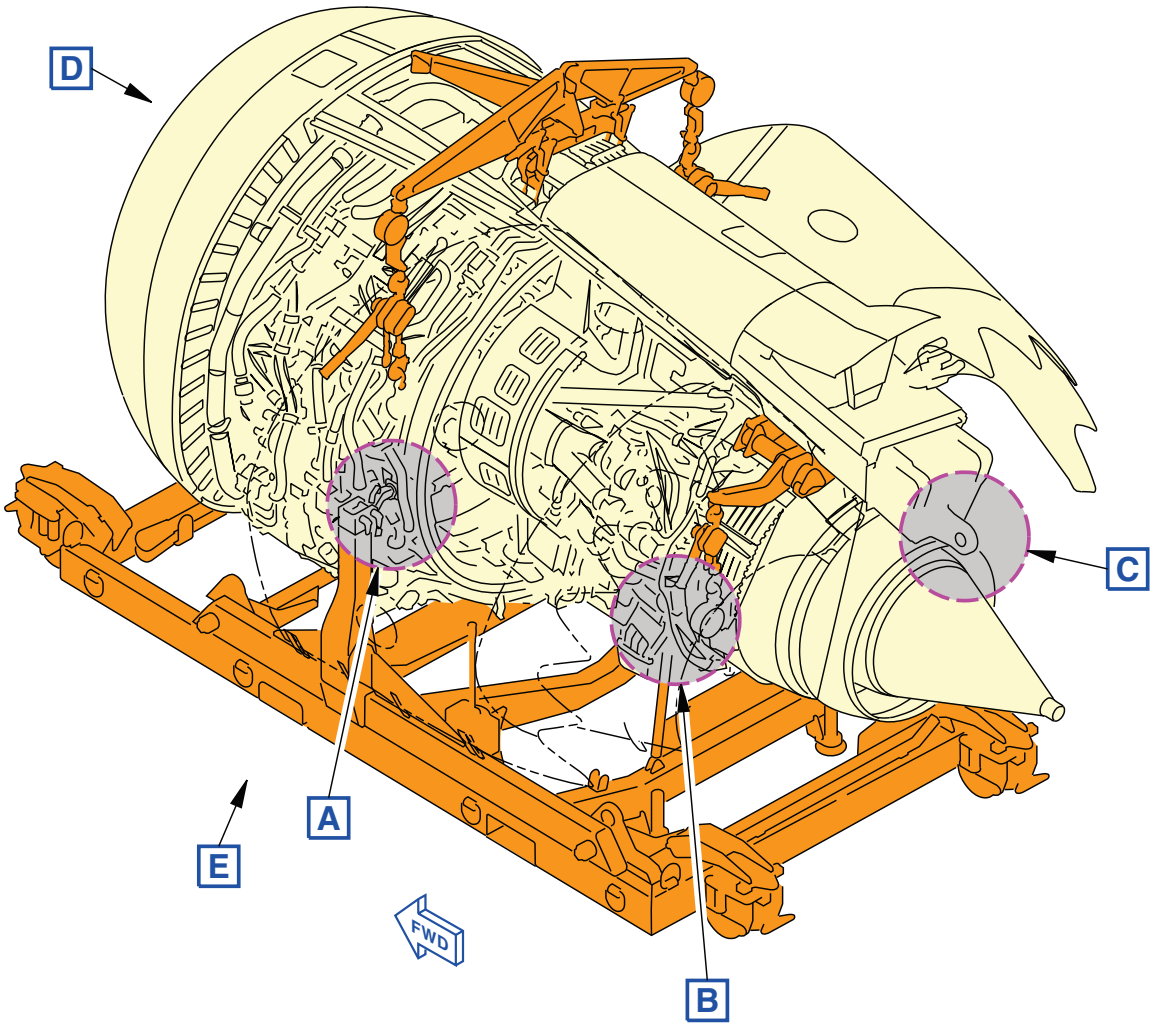
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-XXXXXX-B-F0301-XXXXX-A-001-02
2872189 S0000678477_V1

Forward Bootstrap Equipment Installation
Figure 409/71-00-02-990-811-G00 (Sheet 6 of 6)

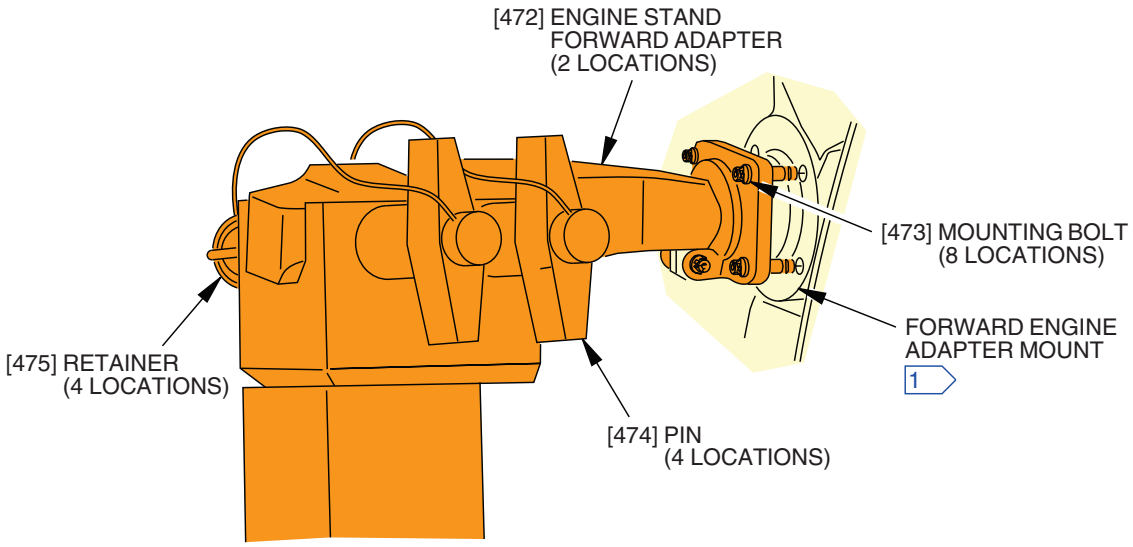
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B710002D5882800001A00101
2354201 S00061514953_V4

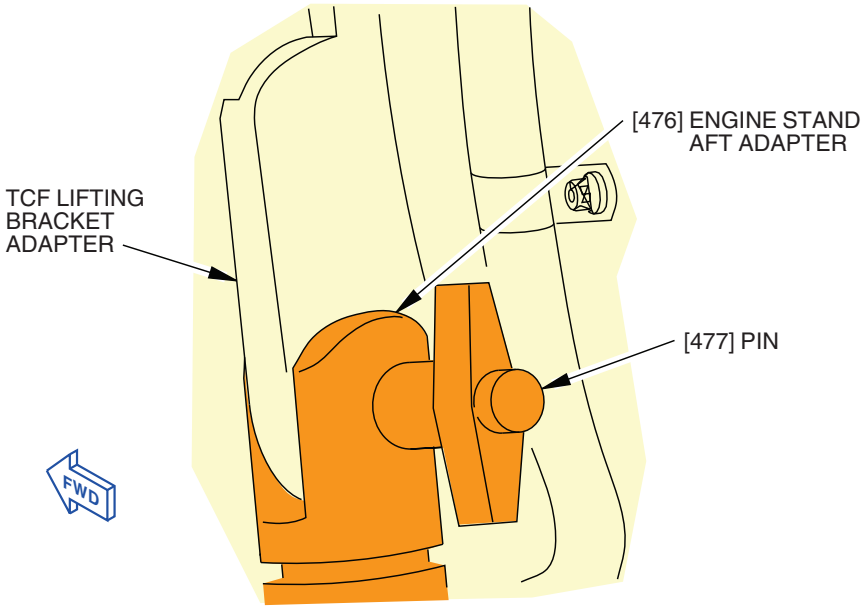
Engine and Transportation Stand Installation
Figure 410/71-00-02-990-812-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A



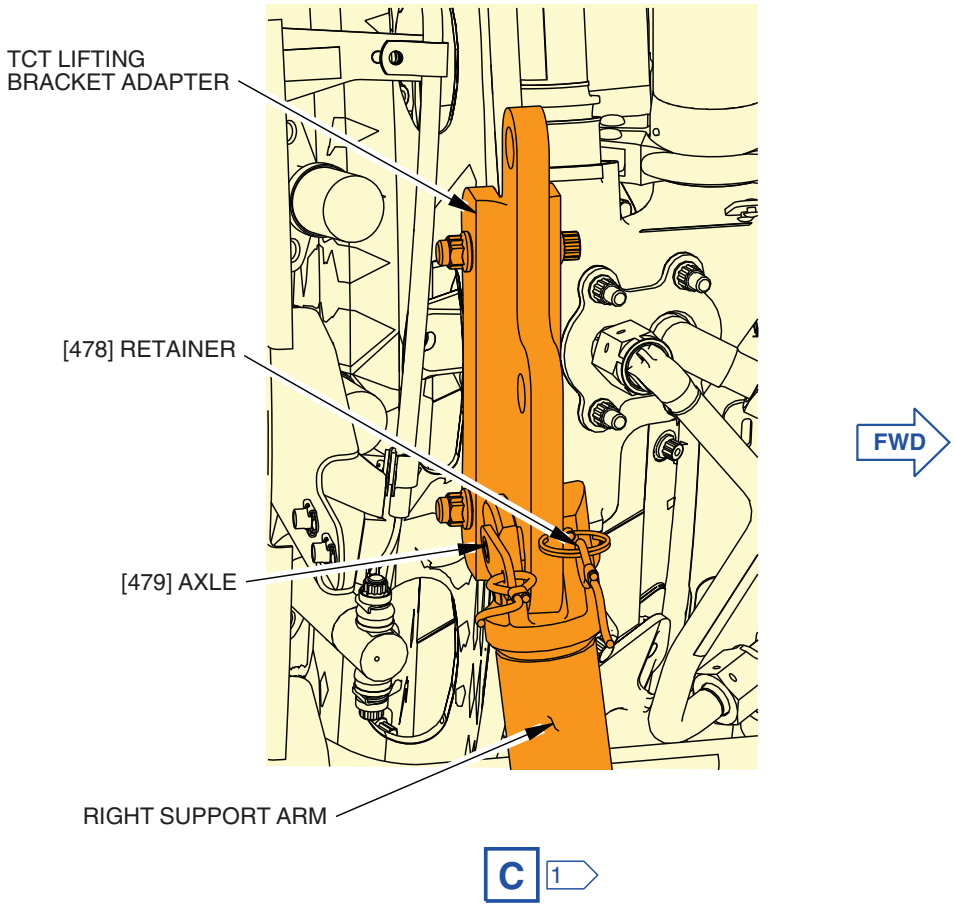
B

1 THIS IS THE HANDLING BRACKET FOR POST LEAP-1B SB 72-0181.

ICNLEAP1B710002D5882800002A00101
2354202 S00061514954_V6

Engine and Transportation Stand Installation
Figure 410/71-00-02-990-812-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

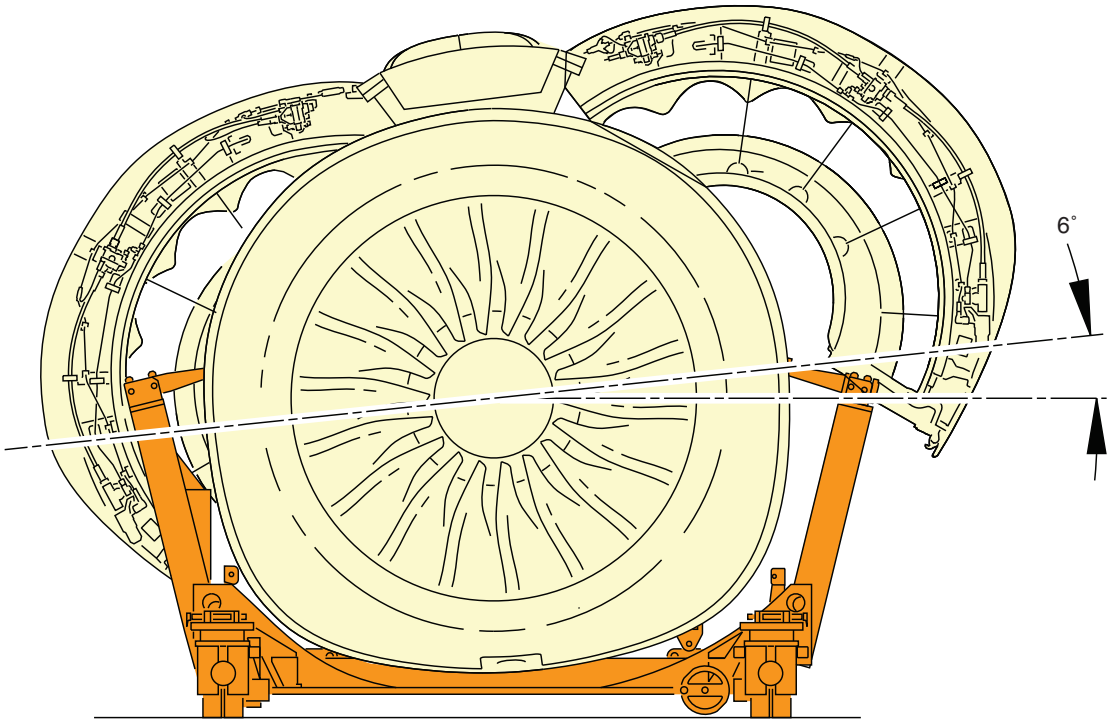


1 THE AXLE [479] IS ENGAGED FROM THE REAR.
THE POSITION OF THE RETAINER [478] IS DIFFERENT AS PER ENGINE STAND VERSION 956A8600G03 OR 956A8600G04

ICNLEAP1B710002D5882800003A00101
3099463 S0000845641_V1

Engine and Transportation Stand Installation
Figure 410/71-00-02-990-812-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

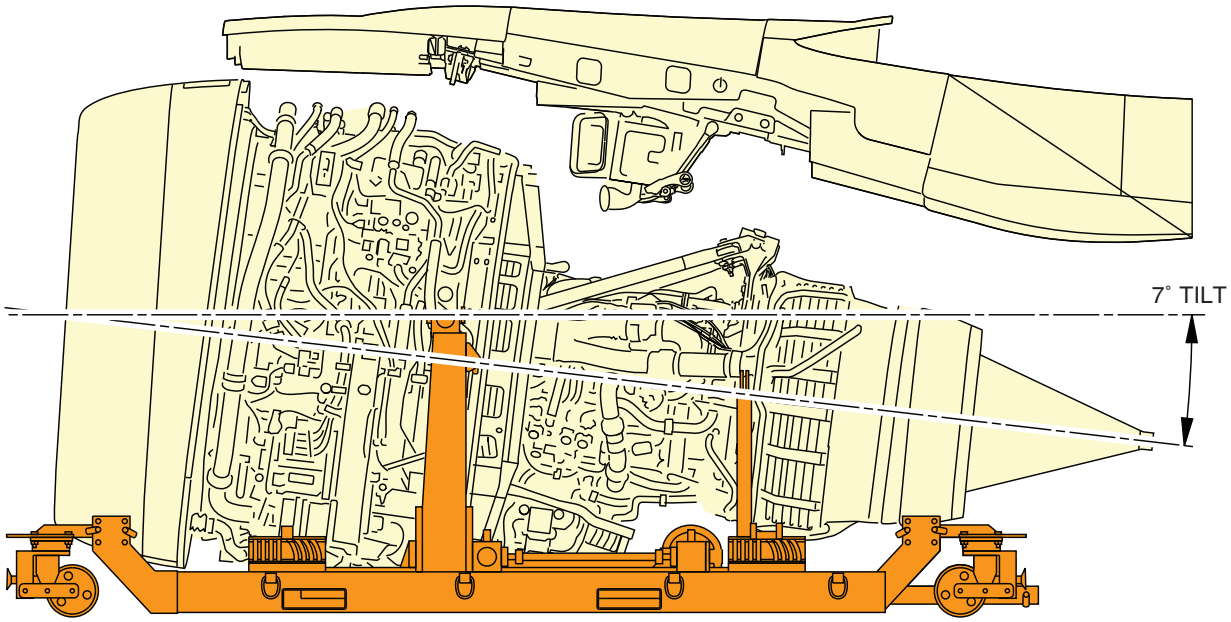


D

ICNLEAP1B710002D5882800004A00101
2354203 S00061514955_V3

Engine and Transportation Stand Installation
Figure 410/71-00-02-990-812-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

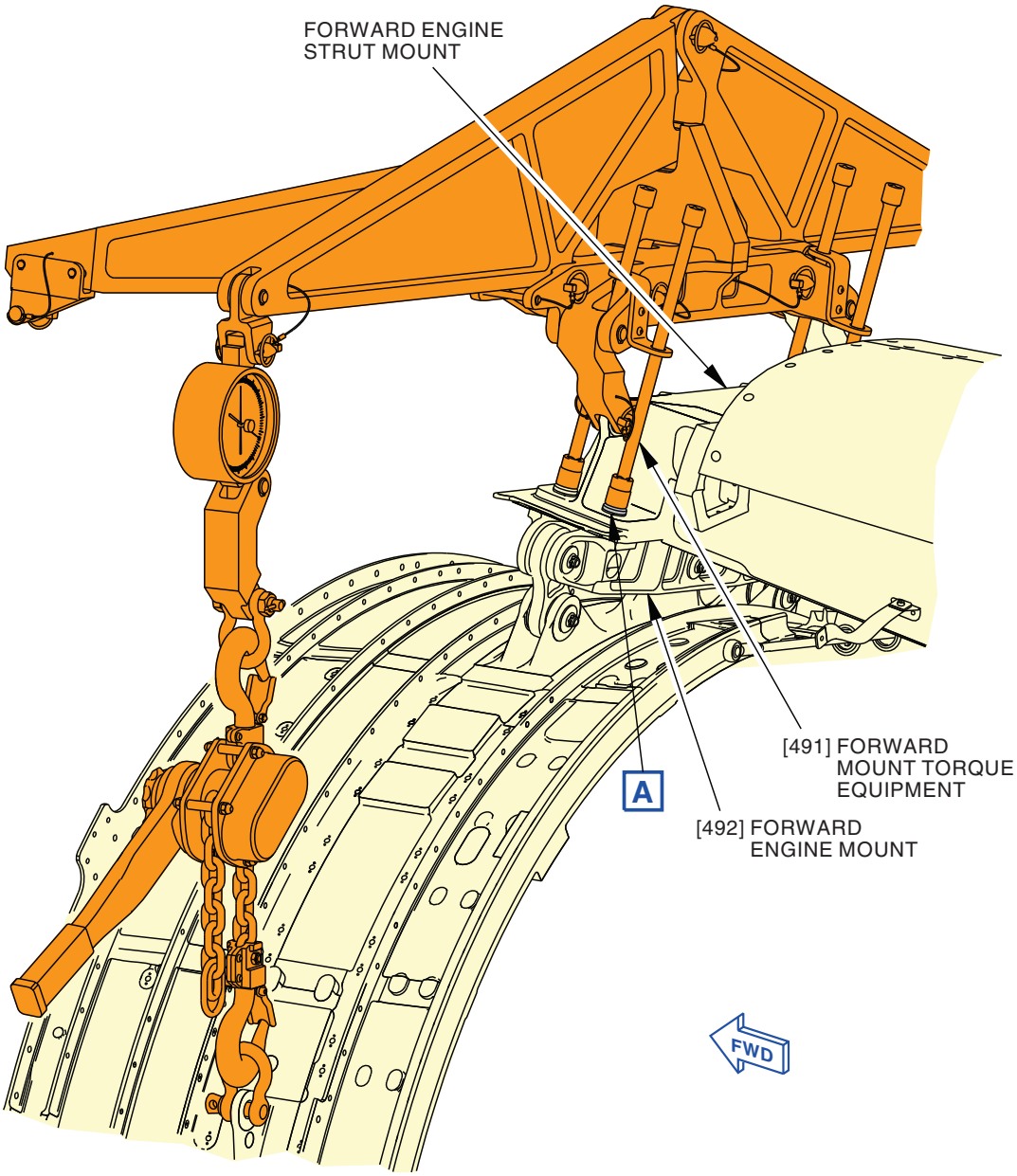


E

ICNLEAP1B710002D5882800005A00101
2434798 S0000564623_V3

Engine and Transportation Stand Installation
Figure 410/71-00-02-990-812-G00 (Sheet 5 of 5)

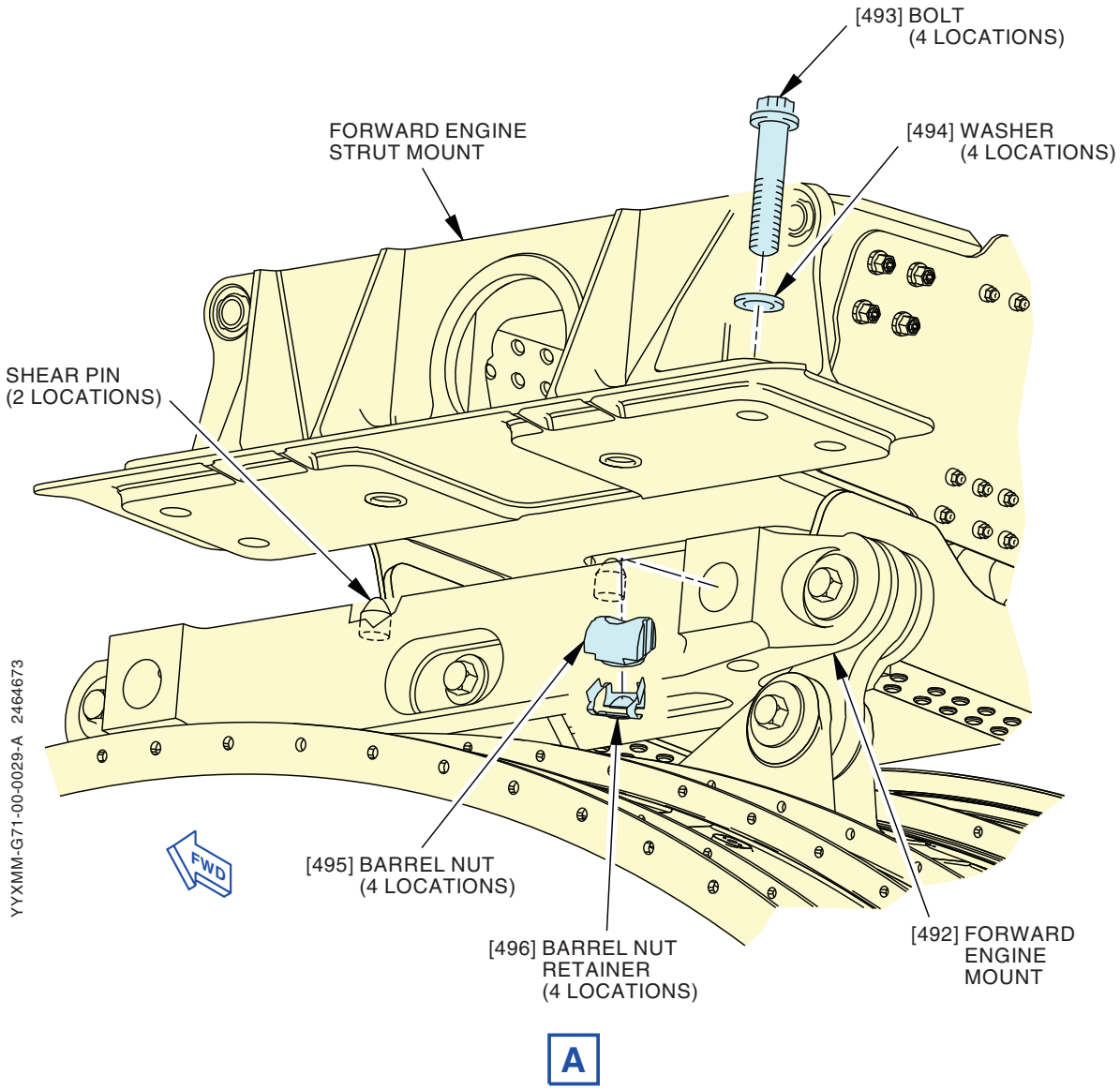
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354198 S00061514950_V3

Forward Engine Mount Installation
Figure 411/71-00-02-990-813-G00 (Sheet 1 of 2)

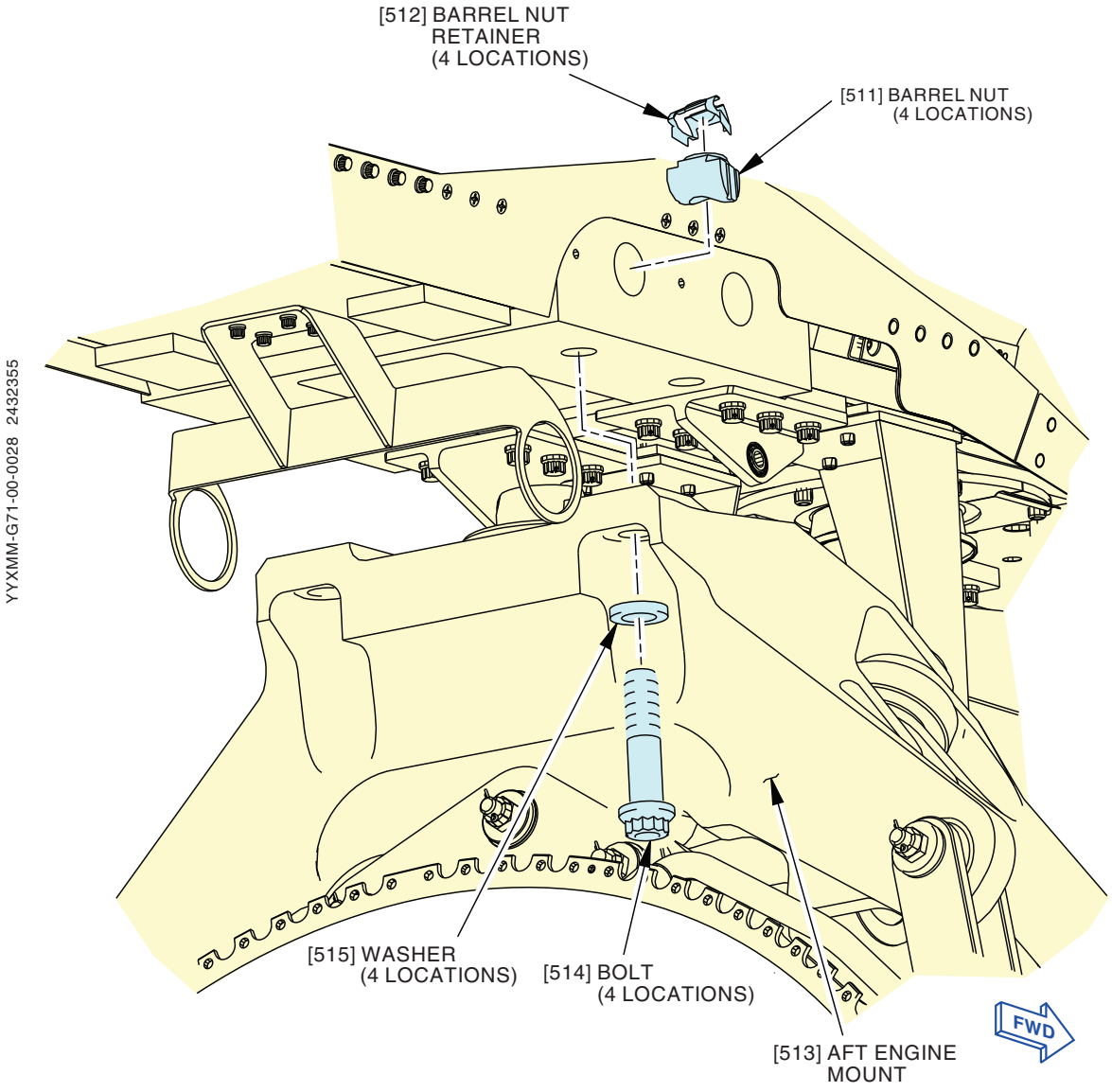
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354204 S00061514956_V7

Forward Engine Mount Installation
Figure 411/71-00-02-990-813-G00 (Sheet 2 of 2)

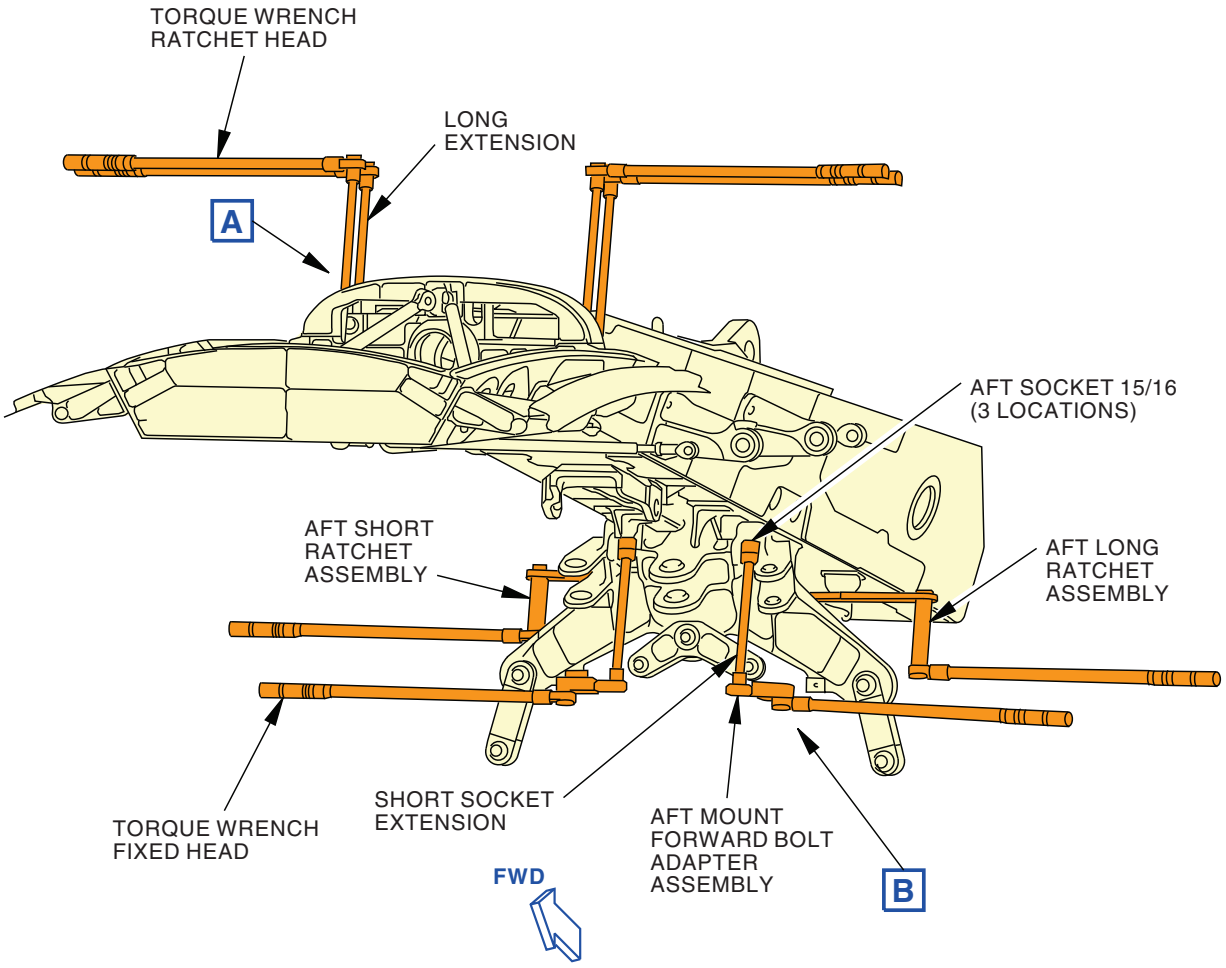
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354205 S00061514957_V5

Aft Engine Mount Installation
Figure 412/71-00-02-990-814-G00

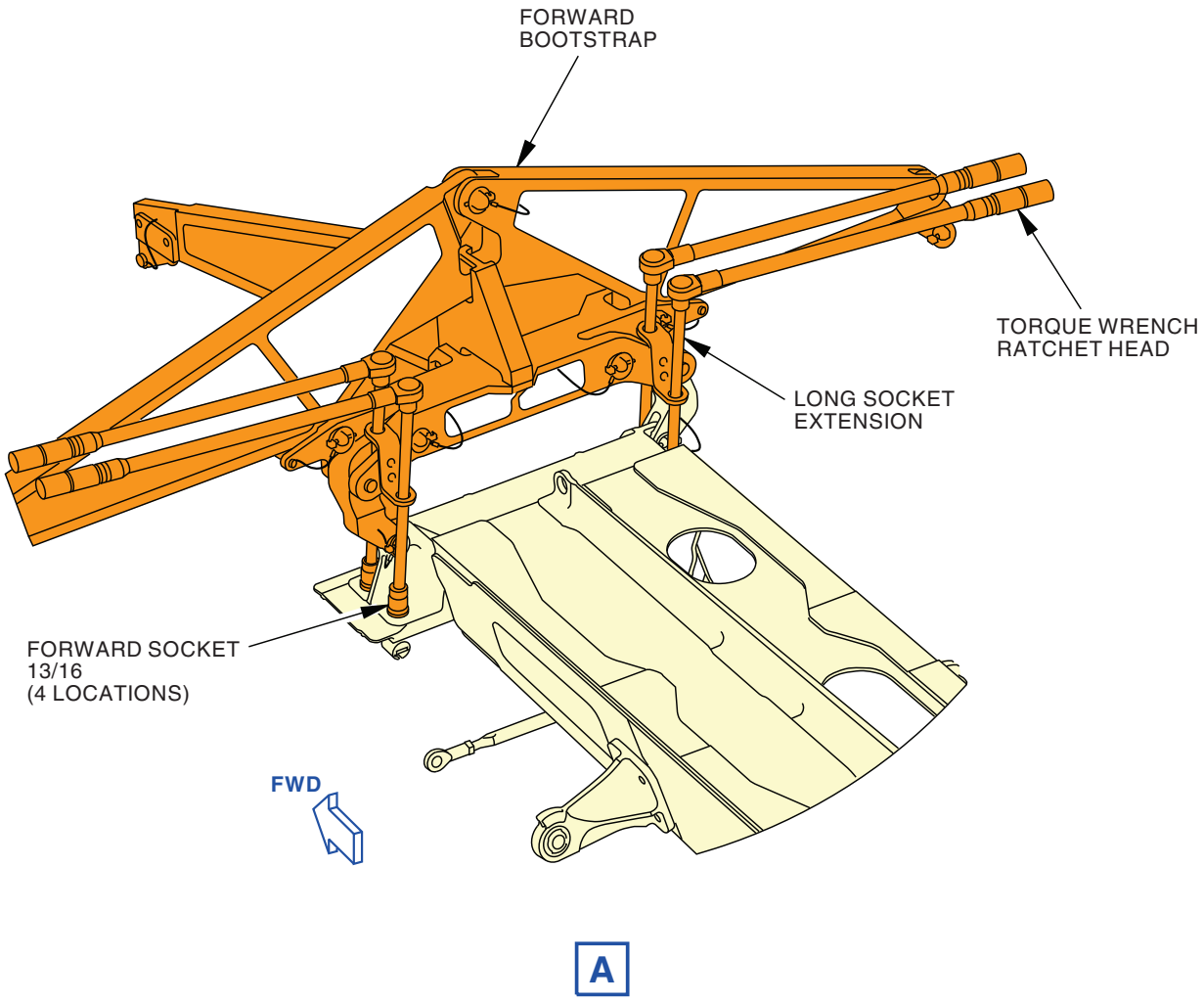
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2453743 S0000569587_V2

Engine Mount Bolts Torque Equipment Installation (Torque Wrench with Ratchet Head)
Figure 413/71-00-02-990-817-G00 (Sheet 1 of 4)

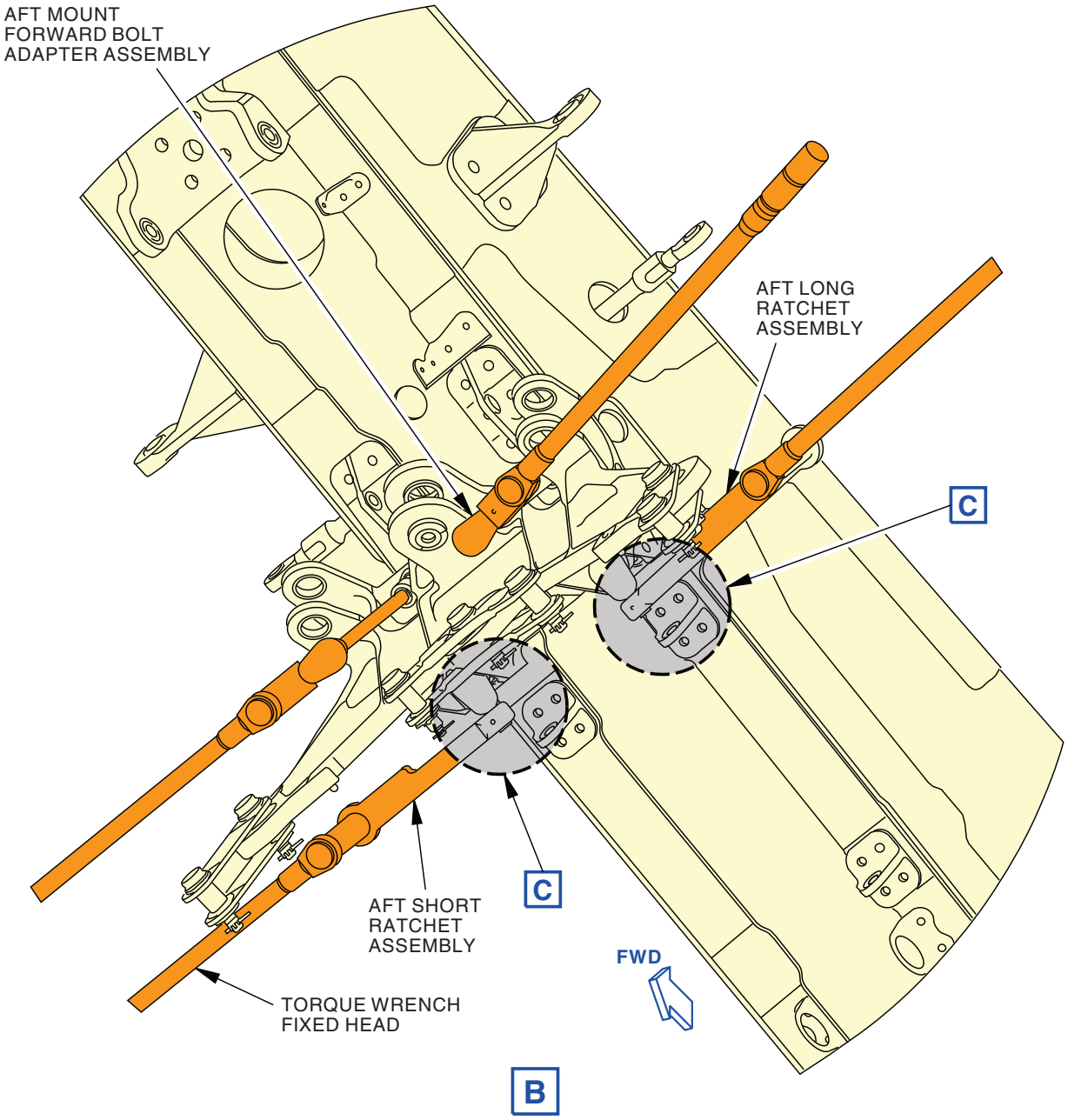
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2453744 S0000568639_V2

Engine Mount Bolts Torque Equipment Installation (Torque Wrench with Ratchet Head)
Figure 413/71-00-02-990-817-G00 (Sheet 2 of 4)

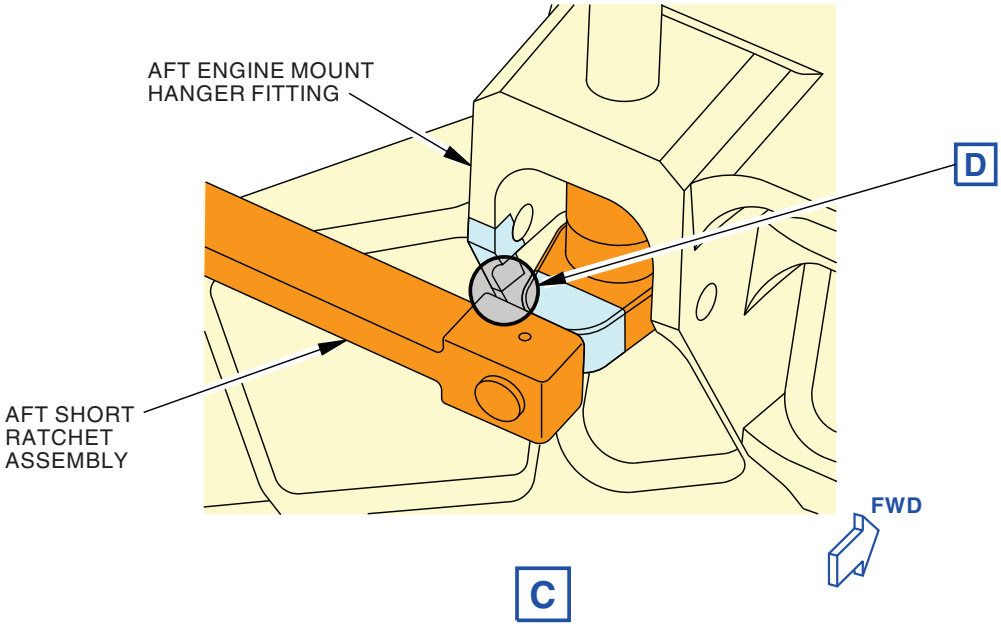
ECCN 9E991 BOEING PROPRIETARY - See title page for details

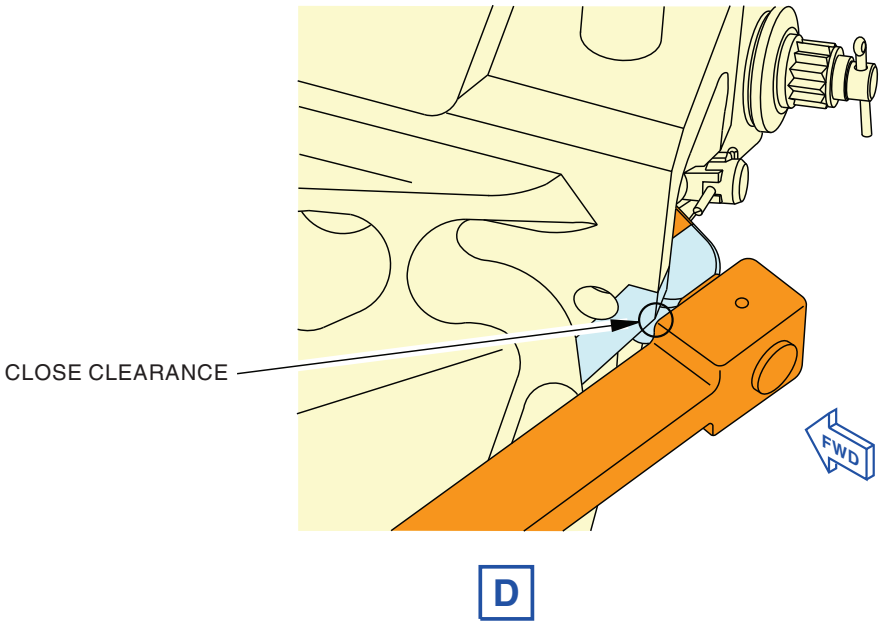


2453768 S0000569588_V2

Engine Mount Bolts Torque Equipment Installation (Torque Wrench with Ratchet Head)
Figure 413/71-00-02-990-817-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



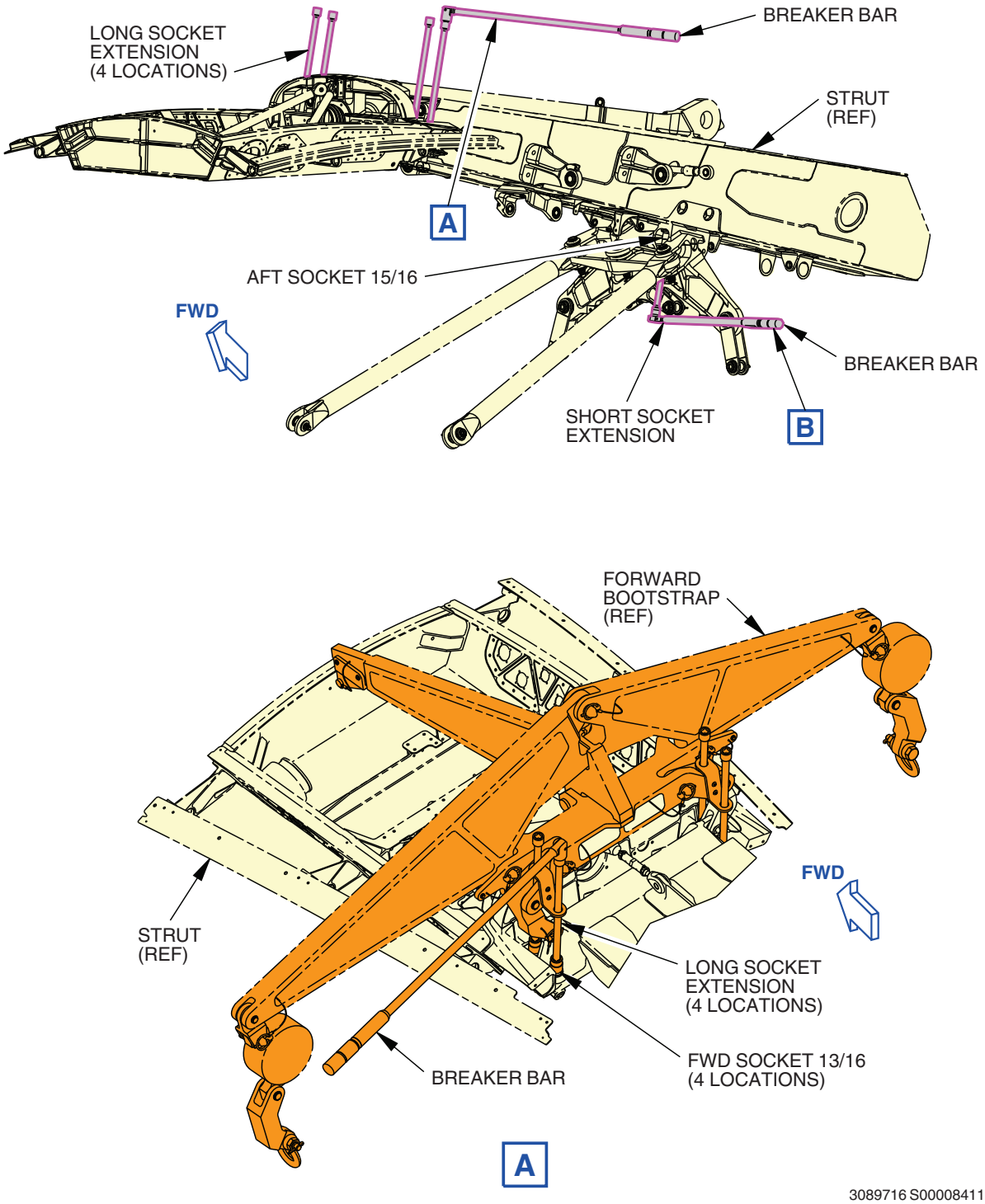


TEFLON TAPE OR ANOTHER
APPROPRIATE MATERIAL

3010861 S0000784803_V1

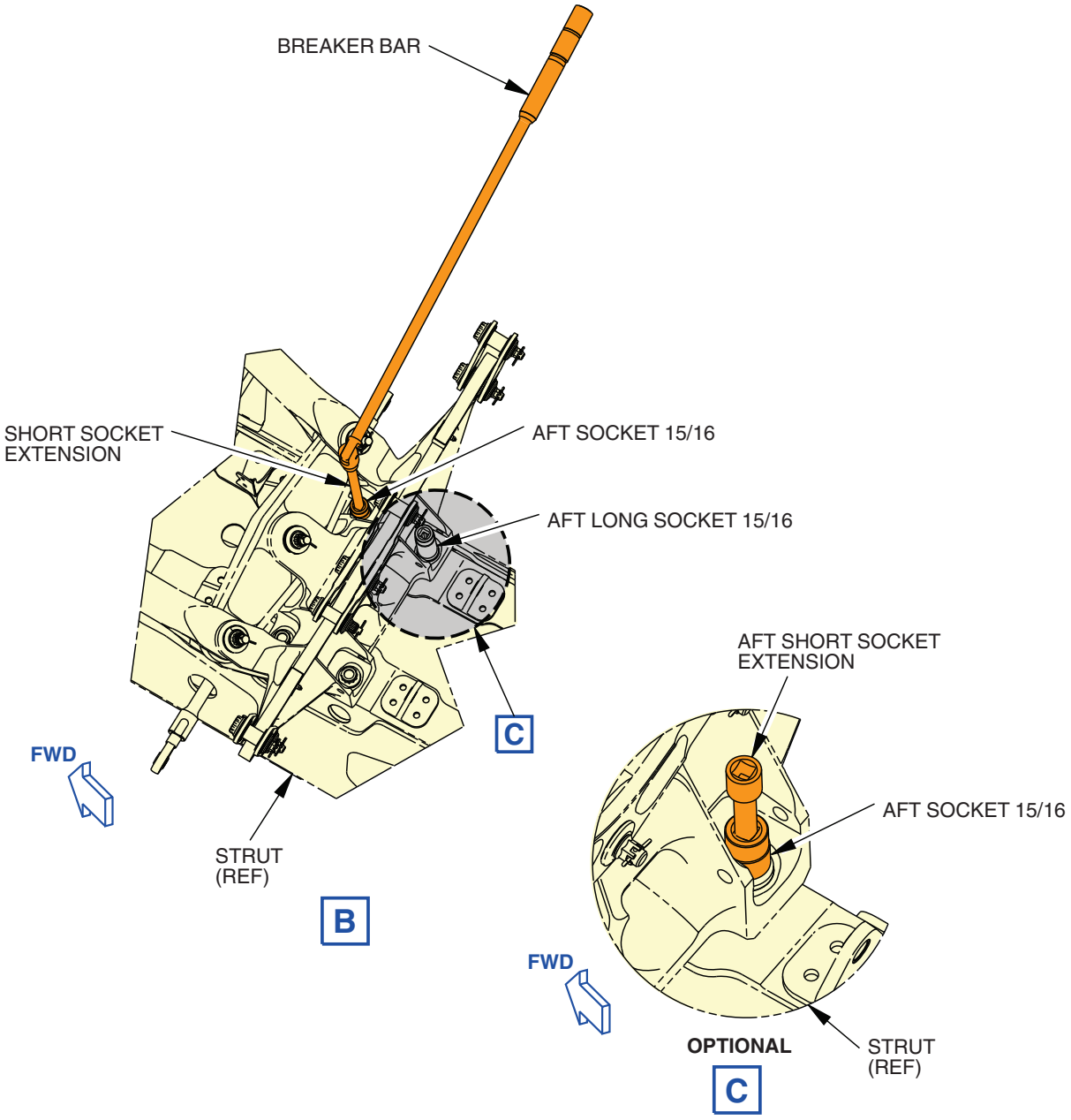
Engine Mount Bolts Torque Equipment Installation (Torque Wrench with Ratchet Head)
Figure 413/71-00-02-990-817-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Mount Bolts Torque Equipment Installation (Torque Wrench with Breaker Bar)
Figure 414/71-00-02-990-822-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TORQUE EQUIPMENT - ENGINE MOUNT BOLTS, CFM LEAP-1B

NOTE:

USE BREAKER BAR TO BREAK LOOSE THE ENGINE MOUNT BOLTS.

3089717 S0000841122_V1

Engine Mount Bolts Torque Equipment Installation (Torque Wrench with Breaker Bar)
Figure 414/71-00-02-990-822-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-00-02-400-801-G00

3. Power Plant - Installation

(Figure 401, Figure 404, Figure 405, Figure 406, Figure 407, Figure 408, Figure 409, Figure 410, Figure 411, Figure 412, Figure 413, and Figure 414)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to install the power plant.

NOTE: The total acoustic panel blockage for the engine installation must be recalculated when components are changed per SRM SUBJECT 54-00-01 .

(2) The power plant weighs approximately 7300 lb (3311 kg). Include the weight of the ground support equipment and calculate the load.

(3) Install the engine with the inlet cowl or without the inlet cowl installed. Without the inlet cowl installed, the aft loads will be slightly higher than the engine with the inlet cowl installed.

B. References

Reference	Title
08-20-02-580-801	Level the Airplane With a Plumb Bob and Inclinometers (P/B 201)
08-20-03-580-801	Level the Airplane With a Attitude Gage (P/B 201)
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
12-13-21-600-801	IDG Servicing (Oil Fill) (P/B 301)
12-15-31-610-803	Main Landing Gear Strut Servicing, Airplane on Jacks (P/B 301)
20-40-11-910-801	Static Grounding (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)
54-05-03-210-805	INTERNAL - GENERAL VISUAL: LEFT STRUT FORWARD AND AFT ENGINE MOUNT TO STRUT SHEAR PINS (P/B 201)
54-05-03-210-806	INTERNAL - GENERAL VISUAL: RIGHT STRUT FORWARD AND AFT ENGINE MOUNT TO STRUT SHEAR PINS (P/B 201)
54-05-03-210-807	EXTERNAL - GENERAL VISUAL: LEFT STRUT AFT ENGINE MOUNT ASSEMBLY (P/B 201)
54-05-03-210-808	EXTERNAL - GENERAL VISUAL: RIGHT STRUT AFT ENGINE MOUNT ASSEMBLY (P/B 201)
54-52-01-410-801	Forward Fairing Installation (P/B 401)
54-52-07-200-801	Leaf Spring Seal - Inspection/Check (P/B 601)
54-54-00-200-801	Forward Strut Fireseal Inspection (P/B 201)
54-54-01-200-801	Strut Thermal Barriers Inspection (P/B 601)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-700-810-G00	Test No. 34 - Menu Mode Hardware Code Entry Special Functions Test (P/B 501)
71-00-00-700-813-G00	Test No. 37 - Engine Serial Number Entry Special Functions Test (P/B 501)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)

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(Continued)

Reference	Title
71-00-03-630-802-G00	Depreservation of an Engine on Wing (P/B 201)
71-11-01-000-802-G00	Inlet Ground Cooling Duct Cap and Cover Removal (P/B 201)
71-11-01-400-801-G00	Inlet Cowl Installation (P/B 401)
71-11-01-400-802-G00	Inlet Ground Cooling Duct Cap and Cover Installation (P/B 201)
71-11-04-400-801-G00	Fan Cowl Panels Installation (P/B 401)
71-11-11-400-801-G00	Fan Cowl Hold Open Rod Installation (P/B 201)
71-12-01-400-801-G00	Fan Cowl SDOS Actuators Installation (P/B 401)
71-21-00-200-801-G00	Forward Engine Mount Visual Inspection (P/B 601)
71-21-00-200-802-G00	Aft Engine Mount Visual Inspection (P/B 601)
71-21-03-200-801-G00	Engine Mount Bolts and Nuts Inspection (P/B 601)
71-71-01-400-801-G00	Drain Mast Installation (P/B 401)
73-21-00-470-801-G00	EEC System Configuration Data Functions Check (P/B 201)
73-21-00-470-803-G00	EEC Onboard Software Installation (P/B 201)
73-21-05-970-801-G00	Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT (P/B 201)
73-21-09-700-801-G00	Data Entry Plug Configuration Check (P/B 601)
78-11-00-210-802-G00	Primary Nozzle Assembly and Primary Plug Assembly Inspection (P/B 601)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-01-280-801-G00	Thrust Reverser Rigging Check (P/B 601)
78-31-01-400-801-G00	Thrust Reverser Installation (P/B 401)
78-31-08-400-801-G00	Thrust Reverser Opening Actuator Installation (P/B 401)
78-31-13-400-801-G00	Insulation Blankets - Installation (P/B 401)
78-31-13-400-802-G00	Insulation Blankets - Installation (P/B 401)
80-11-01-610-802-G00	Engine Starter Servicing (Oil Fill) (P/B 301)
SOPM 20-30-03	General Cleaning Procedures
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-60-01	Cleaning of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
SPL-14548	Engine Anti-Ice Duct Cover/Support, LEAP-1B Part #: C71037-1 Supplier: 81205
SPL-14550	Equipment - Hold Open, Thrust Reverser Cowl, LEAP-1B Engine Part #: C78033-40 Supplier: 81205 Opt Part #: C78033-1 Supplier: 81205
SPL-14557	Stabilizer - Engine Mount, CFM LEAP-1B Part #: C71036-1 Supplier: 81205
SPL-14637	Torque Equipment - Engine Mount Bolt Part #: C71040-34 Supplier: 81205 Opt Part #: C71040-1 Supplier: 81205 Opt Part #: C71040-30 Supplier: 81205
SPL-14726	Engine Stand - LEAP-1B Part #: 956A8600G04 Supplier: 58828 Opt Part #: 956A8600G03 Supplier: 58828
SPL-15012	Bracket, Lifting - Frame, Turbine Center (TCF) Part #: 956A8667G04 Supplier: 58828
SPL-17099	Handling Brackets - Engine Fan Case Part #: 956A8018G02 Supplier: 58828
SPL-18258	Blanking Plate - PRSOV Part #: C36007-11 Supplier: 81205 Opt Part #: C36007-1 Supplier: 81205
SPL-18640	Blanking Plates - Starter Air Duct & Precooler, CFM Engine Series Part #: C36007-12 Supplier: 81205
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE
STD-3906	Mallet - Rubber

D. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00010	Compound - Thread Antiseize, High Temperature	MIL-PRF-907
D50011	Grease - Perfluoropolyether - Christo-lube MCG111	
G51905	Putty/Lacquer - Tamper-proof, Hydraulic Fluid Resistant, Yellow	BMS8-45 Type III

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
38	Preformed packing	71-40-51-05-015	PTF ALL
53	E-seal	71-40-51-05-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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G. Access Panels

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Installation

SUBTASK 71-00-02-860-082-G00

(1) Make sure that the applicable engine circuit breakers are open.

(2) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L
C	4	C00154	ENGINE 2 START VALVE
C	18	C00348	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 2
C	19	C00347	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 1
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	6	C04035	AIR COND BLEED AIR CONTROL PRIMARY LEFT
B	6	C04036	AIR COND BLEED AIR CONTROL PRIMARY RIGHT
D	5	C04037	AIR COND BLEED AIR CONTROL BACKUP LEFT
D	6	C04038	AIR COND BLEED AIR CONTROL BACKUP RIGHT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 71-00-02-860-026-G00

(3) Make sure that the airplane is statically grounded (TASK 20-40-11-910-801).

SUBTASK 71-00-02-860-061-G00

(4) Make sure that the airplane is in a level position (TASK 08-20-03-580-801 or TASK 08-20-02-580-801).

NOTE: Inflate the Main Landing Gear (MLG) and Nose Landing Gear (NLG) struts when necessary. The airplane must be level (no more than ±0.5° in the pitch axis and ±1.0° about the roll axis).

SUBTASK 71-00-02-860-083-G00

(5) Make sure that the weight and balance Center of Gravity (CG) is in the forward section or install a tail jack.

SUBTASK 71-00-02-860-029-G00

(6) If the thrust reversers remain installed, make sure that the thrust reverser hold-open tools are properly installed (TASK 71-00-02-000-801-G00).

SUBTASK 71-00-02-020-014-G00

(7) If the hold open equipment, SPL-14550 (P/N C78033-1), is used, remove the No. 1 compression rod and bracket from the engine being installed as follows:

NOTE: It is not necessary to remove the No. 1 compression rod and bracket if the hold open equipment, SPL-14550 (P/N C78033-40), is used.

(a) Remove the rubber grommet from the inboard side of the No. 1 compression rod.

(b) Move the No. 1 compression rod in the outboard direction to remove the compression rod.

(c) Keep the compression rod and grommet for installation.

(d) Remove the two bolts [4] that attach the No. 1 compression rod support bracket [5].

(e) Keep the No. 1 compression rod support bracket [5] and two bolts [4] for installation.

SUBTASK 71-00-02-860-030-G00

(8) If the bolts [493], bolts [514], barrel nuts [495], and barrel nuts [511] are not new, do the inspection as follows (Figure 411 (Sheet 2), Figure 412 (Sheet 1)):

(a) Examine the bolts [493], bolts [514], barrel nuts [495], and barrel nuts [511] with a dye penetrant inspection (TASK 71-21-03-200-801-G00).

(b) Make sure that the bolts [493] and bolts [514] have the correct length.

NOTE: The length of the bolt is on the part number for the bolt in the Aircraft Illustrated Parts Catalog (AIPC).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-00-02-860-031-G00

(9) Examine the forward engine mount [492], aft engine mount [513], forward and aft strut mounts, and assembly for the conditions as follows:

(a) Discoloration, cracks, corrosion, or signs of worn areas.

(b) Loose fasteners, damage thrust link clevis bushings, worn bearings, or signs of failures in the adjacent structure.

(c) Do the inspections of the components and areas as follows:

1) Do this task: Forward Engine Mount Visual Inspection, TASK 71-21-00-200-801-G00.

2) Do this task: Aft Engine Mount Visual Inspection, TASK 71-21-00-200-802-G00.

3) Do this task: INTERNAL - GENERAL VISUAL: LEFT STRUT FORWARD AND AFT ENGINE MOUNT TO STRUT SHEAR PINS, TASK 54-05-03-210-805 or INTERNAL - GENERAL VISUAL: RIGHT STRUT FORWARD AND AFT ENGINE MOUNT TO STRUT SHEAR PINS, TASK 54-05-03-210-806.

4) Do this task: EXTERNAL - GENERAL VISUAL: LEFT STRUT AFT ENGINE MOUNT ASSEMBLY, TASK 54-05-03-210-807 or EXTERNAL - GENERAL VISUAL: RIGHT STRUT AFT ENGINE MOUNT ASSEMBLY, TASK 54-05-03-210-808.

5) Do this task: Primary Nozzle Assembly and Primary Plug Assembly Inspection, TASK 78-11-00-210-802-G00.

SUBTASK 71-00-02-200-001-G00

(10) Make sure that the fireseal is serviceable (TASK 54-54-00-200-801).

SUBTASK 71-00-02-200-002-G00

(11) Make sure that the strut thermal barriers are serviceable (TASK 54-54-01-200-801).

SUBTASK 71-00-02-200-003-G00

(12) Make sure that the leaf spring seals are serviceable, do this task: Leaf Spring Seal - Inspection/Check, TASK 54-52-07-200-801.

SUBTASK 71-00-02-860-032-G00

(13) Prepare the forward engine mount [492] and aft engine mount [513] for the installation of the engine as follows:

(a) Make sure that the surfaces of the forward engine mount [492] and aft engine mount [513] (that touch) are free of oil, grease, and other unwanted material.

(b) Apply Pure Nickel Special compound, D00006, to the shear pins on the forward engine mount [492] and aft engine mount [513].

SUBTASK 71-00-02-860-033-G00

CAUTION

STOP THE TORQUE PROCEDURE WHEN YOU CAN SEE A FULL BOLT THREAD ABOVE THE END OF THE NUT. IF THE BOLT GOES TOO FAR THROUGH THE INSTALLED NUT DURING THE SELF-LOCKING TORQUE CHECK, DAMAGE TO THE NUT RETAINER CAN OCCUR.

(14) Do a self-locking torque check of the barrel nuts [495] in the forward engine mount [492] and barrel nuts [511] in the aft engine mount [513] as follows:

(a) Apply Pure Nickel Special compound, D00006, to the shank and to the threads of the bolts [493] and bolts [514].

(b) Install the bolts [493] and bolts [514] in the (captive) nuts to do a check of the self-locking torque as follows:

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1) Make sure that the forward mount barrel nuts [495] self-locking torque is between 50.0 in-lb (5.6 N·m) - 400 in-lb (45 N·m).

2) Make sure that the aft mount barrel nuts [511] self-locking torque is between 70.0 in-lb (7.9 N·m) - 600 in-lb (68 N·m).

3) If the self-locking torque on the barrel nuts [495] and barrel nuts [511] is not within the limits specified above, replace the barrel nuts [495] and barrel nuts [511].

SUBTASK 71-00-02-860-035-G00

(15) If it is necessary, install the bootstrap equipment (TASK 71-00-02-000-801-G00).

I. Power Plant Installation

SUBTASK 71-00-02-860-037-G00

(1) Push the engine stand, SPL-14726, with the engine rearward from front of the strut until the engine stand is aligned under the strut.

SUBTASK 71-00-02-910-001-G00

(2) Lock the engine stand wheels.

SUBTASK 71-00-02-020-007-G00

(3) If it is installed, remove the transportation brace, SPL-14557, from the engine mounts.

SUBTASK 71-00-02-010-016-G00

(4) Attach the bootstrap equipment to the engine as follows:

(a) Refer to the power plant removal task for the bootstrap equipment installation (TASK 71-00-02-000-801-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(b) Install the lever hoists [413] and dynamometer assembly [415] on the forward engine adapter mounts with the two shackles [414] (Figure 409).

NOTE: If it is necessary, use an extra shackle to attach the lever hoists to the forward engine adapter mounts.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(c) Install the lever hoists [413] and dynamometer assembly [415] on the engine fan case handling bracket, SPL-17099, with the two shackles [414] (Figure 409).

NOTE: If it is necessary, use an extra shackle to attach the lever hoists to the handling brackets.

PTF ALL

(d) Install the lever hoist [453] and dynamometer assembly [454] on the TRF case bracket with the two pins [455] and two retainers [456] (Figure 409).

(e) Take up the slack in the hoist chains.

SUBTASK 71-00-02-010-017-G00

WARNING

MONITOR THE DYNAMOMETERS ON THE FORWARD AND THE AFT HOIST. MAKE SURE THAT THE LOADS ARE WITHIN THE LIMITS. IF THE LOADS ARE MORE THAN THE LIMITS, THE ENGINE CAN FALL. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Disconnect the engine from the engine stand as follows (View A and View B, Figure 410):

(a) Remove the adapter pins [474] and retainers [475].

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(b) Disconnect the engine stand forward adapters [472] from the forward engine stand support arms.

(c) Remove the adapter pin [477], axle [479], and retainer [478].

(d) Disconnect the engine stand aft adapters [476] from the TCF lifting bracket adapters, SPL-15012.

(e) Move the two forward stand arms outwards.

(f) Make sure that the aft engine stand support columns are clear from the TCF lifting bracket adapters, SPL-15012.

(g) If the inlet cowl is installed on the engine, make sure that you do not exceed these dynamometer values:

1) 3180 lb (1442 kg) on the forward outboard dynamometer

2) 4100 lb (1860 kg) on the forward inboard dynamometer

3) 2500 lb (1134 kg) on the aft dynamometer.

(h) If the inlet cowl is not installed on the engine, make sure that you do not exceed these dynamometer values:

1) 3000 lb (1361 kg) on the forward outboard dynamometer

2) 3500 lb (1588 kg) on the forward inboard dynamometer

3) 2500 lb (1134 kg) on the aft dynamometer.

SUBTASK 71-00-02-860-038-G00

WARNING LET ONLY APPROVED PERSONS OPERATE THE LEVER HOISTS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

(6) Operate the forward lever hoists [413] and aft lever hoist [453] equally to lift the engine up to the strut as follows (View C, Figure 409 and View D, Figure 408):

(a) Make sure that the loads on all three dynamometers are within limits and are continuously monitored when the lever hoists are operating.

NOTE: Operation of one hoist can affect the loads at the other two hoist locations.

(b) Make sure that the engine is lifted slowly until it clears the engine stand.

(c) Use the forward lever hoists [413] and aft lever hoist [453] to change the roll of the engine to align the engine mount platform to the strut alignment.

(d) Use the aft lever hoist [453] to change the pitch of the engine.

(e) Lift the engine until there is approximately 6.0 in. (15.2 cm) between the engine mounts and strut mounts as follows:

1) Make sure that you monitor the clearance between the engine and thrust reversers while you lift the engine.

CAUTION MONITOR THE THRUST LINKS, THE PRECOOLER, AND THE HINGE BEAM FLANGE WHEN THE ENGINE MOVES OR TURNS. DO NOT MOVE OR TURN THE ENGINE TOO QUICKLY OR THE ENGINE CAN HIT AND CAUSE DAMAGE TO THESE COMPONENTS.

2) Make sure that the engine is clear from the thrust links, pre-cooler, and forward thrust reversers.

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(f) Operate the forward lever hoists [413] and aft lever hoist [453] until the clearance is equal and mounts are aligned.

(g) Operate the forward lever hoists [413] and the aft lever hoist [453] until you have a clearance of 0.50 in. (1.27 cm) between the engine mounts and the strut mounts.

1) Align both the forward and aft shear pins with the strut mounts.

2) Make sure that the forward and aft engine mounts are level.

NOTE: If the angular position of the forward or aft engine mount is not aligned correctly, it will cause binding of the shear pins and lock-up.

WARNING LET ONLY APPROVED PERSONS OPERATE THE LEVER HOISTS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MONITOR THE DYNAMOMETERS ON THE FORWARD AND THE AFT HOIST. MAKE SURE THAT THE LOADS ARE WITHIN THE LIMITS. IF THE LOADS ARE MORE THAN THE LIMITS, THE ENGINE CAN FALL. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(h) Lift the engine until you see no clearance between the forward engine mount and strut mount, and between the aft engine mount and strut mount.

NOTE: There is no specific load value that must be applied. The only limits are that the loads must be more than the applied engine load values, and that the strut mating surfaces and engine mounts stay together.

(i) If the inlet cowl is installed on the engine, make sure that you do not exceed these dynamometer values:

1) 3180 lb (1442 kg) on the forward outboard dynamometer

2) 4100 lb (1860 kg) on the forward inboard dynamometer

3) 2500 lb (1134 kg) on the aft dynamometer.

(j) If the inlet cowl is not installed on the engine, make sure that you do not exceed these dynamometer values:

1) 3000 lb (1361 kg) on the forward outboard dynamometer

2) 3500 lb (1588 kg) on the forward inboard dynamometer

3) 2500 lb (1134 kg) on the aft dynamometer.

SUBTASK 71-00-02-410-002-G00

(7) Install the forward engine mount [492] as follows (Figure 411):

(a) Make sure that the barrel nut retainers [496], bolts [493], washers [494], and barrel nuts [495] are not bent, cracks, have corrosion, or signs of worn areas.

(b) Lubricate the four bolts [493] and exterior cylindrical section of the barrel nuts [495] and barrel nut retainers [496] with Pure Nickel Special compound, D00006.

NOTE: Do not apply compound to the barrel nuts thread locking feature.

(c) Install the four bolts [493], four washers [494], four barrel nuts [495], and four barrel nut retainers [496] that attach the forward engine strut mount to the forward engine mount [492].

1) Install the countersunk side of the washer next to the bolt head.

(d) Tighten the forward mount bolts [493] as follows:

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1) Tighten to 50% of final torque each bolt [493] to 989.0 in-lb (111.7 N·m) - 1161.0 in-lb (131.2 N·m) or 82.4 ft-lb (111.7 N·m) - 96.8 ft-lb (131.2 N·m).

2) Tighten the bolts [493] to a final torque of 1978.0 in-lb (223.5 N·m) - 2322.0 in-lb (262.3 N·m) or 164.8 ft-lb (223.5 N·m) - 193.5 ft-lb (262.3 N·m).

3) Make sure that the threads of the barrel nuts [495] are fully engaged and end of the forward mount bolts [493] chamfer extends out of the nuts.

4) Make sure that the mating connection of the forward mount bolts [493] is free of oil, grease, and other unwanted material (SOPM 20-30-03).

5) Apply tamper-proof putty/lacquer, G51905, to the mating connection of the forward mount bolts [493].

SUBTASK 71-00-02-420-006-G00

(8) Install the aft engine mount [513] as follows (Figure 412):

(a) Make sure that the barrel nut retainers [512], bolts [514], washers [515], and barrel nuts [511] have no cracks, corrosion, or signs of worn areas.

(b) Lubricate the bolts [514] and exterior cylindrical section of the barrel nuts [511] and barrel nut retainers [512] with Pure Nickel Special compound, D00006.

NOTE: Do not apply compound to the barrel nuts thread locking feature.

(c) Install the four bolts [514], four washers [515], four barrel nuts [511], and four barrel nut retainers [512] that attach the aft engine strut mount to the aft engine mount [513].

1) Install the countersunk side of the washer next to the bolt head.

(d) Tighten the aft mount bolts [514] as follows:

1) If the torque equipment, SPL-14637, is used, do this step (Figure 413 or Figure 414):

NOTE: The indicated torque depends on the length of the torque wrench. The indicated torque must be calculated in accordance with the tool placard.

a) Make sure that Teflon tape or another appropriate material is applied to the wrench/ratchet head and aft engine mount fitting.

CAUTION

BE CAREFUL WHEN YOU LOOSEN OR TIGHTEN THE BOLTS WITH TORQUE EQUIPMENT ON THE AFT SIDE OF THE AFT ENGINE MOUNT. IF YOU DO NOT OBEY, DAMAGE TO THE HANGER FITTING OF THE AFT ENGINE MOUNT CAN OCCUR.

2) Tighten to 50% of final torque each bolt [514] to 1564.0 in-lb (176.7 N·m) - 1836.0 in-lb (207.4 N·m) or 130.3 ft-lb (176.7 N·m) - 153.0 ft-lb (207.4 N·m).

3) Tighten the bolts [514] to a final torque 3128.0 in-lb (353.4 N·m) - 3672.0 in-lb (414.9 N·m) or 260.7 ft-lb (353.4 N·m) - 306.0 ft-lb (414.9 N·m).

4) Make sure that the threads of the barrel nuts [511] are fully engaged and end of the aft mount bolts [514] chamfer extends out of the nuts.

5) Make sure that the mating connection of the aft mount bolts [514] is free of oil, grease, and other unwanted material (SOPM 20-30-03).

6) Apply tamper-proof putty/lacquer, G51905, to the mating connection of the aft mount bolts [514].

7) If it is necessary remove applied Teflon tape or another appropriate material from the wrench/ratchet head and aft engine mount fitting.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-00-02-860-063-G00

(9) Move the engine stand forward out from below the engine and put the engine stand in the taxi position if the cradle and casters were lowered.

(a) Use the stand hydraulic device to lift the cradle until engine stand casters off the ground.

(b) Install the four pins in the cradle to activate the cradle damping.

(c) If it is necessary, put the wheel casters on the engine transportation stand to 3.15 in. (80 mm).

(d) Lower the engine stand with the stand hydraulic device.

(e) Make sure that the engine stand hydraulic actuators are fully retracted.

(f) Make sure that other parts of the engine transportation stand are in the stow positions.

SUBTASK 71-00-02-010-018-G00

(10) Remove the bootstrap equipment as follows:

(a) Remove the forward lever hoists [413] with the forward dynamometer assembly [415] and shackles [414] from the engine attached brackets and from bootstrap equipment (View C, Figure 409).

(b) Remove the pin [420] and retainer [421] that join the forward outboard arm assembly [409] and forward inboard arm assembly [416] to the top of the forward support arm assembly [406] (Views C and E, Figure 409).

(c) Remove the two pins [423] and two retainers [422] that attach the forward outboard arm assembly [409] and forward inboard arm assembly [416] to the forward center beam [401] (Views B, C, and E, Figure 409).

(d) Remove the forward outboard arm assembly [409] and forward inboard arm assembly [416] from the forward center beam [401].

(e) Remove the pin [408] and retainer [407] that attach the forward support arm assembly [406] to the forward center beam [401] (View B, Figure 409).

(f) Remove the pin [418] and retainer [419] that attach the other end of the forward support arm assembly [406] to the fan cowl support beam (View D, Figure 409).

(g) Remove the forward support arm assembly [406] from top of the forward center beam [401].

(h) Remove the two pins [403] and two retainers [402] that attach the forward center beam [401] to the forward mount brackets (View A, Figure 409).

(i) Remove the forward center beam [401] from top of the forward mount brackets.

(j) Remove the pin [455] and retainer [456] from the dynamometer assembly [454].

(k) Remove the aft lever hoist [453] with the aft dynamometer assembly [454] and shackle [451] from the engine attached brackets and from bootstrap equipment (View C, Figure 408).

(l) Remove the two pins [447] and two retainers [443] that attach the aft outboard arm assembly [448] to the aft torque beam assembly [446] (View C, Figure 408).

(m) Remove the four retainers [443], two pins [447], and two pins [445] from the aft torque beam assembly [446] (View B, Figure 408).

(n) Remove the aft torque beam assembly [446] from the outboard aft cross beam assembly [444] and outboard aft support beam assembly [442].

(o) Remove the two pins [441] and two retainers [443] that attach the outboard aft cross beam assembly [444] to the outboard forward strut two brackets (View A, Figure 408).

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(p) Remove the pin [441] and retainer [443] that attach the outboard aft support beam assembly [442] to the outboard forward strut bracket (View A, Figure 408).

SUBTASK 71-00-02-080-008-G00

(11) Remove the handling brackets, SPL-17099, from the engine.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

NOTE: The handling brackets are included on the fan frame shroud of the engine.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

NOTE: The handling brackets, SPL-17099, are tools, not installed on the fan frame shroud.

(a) For each handling brackets, SPL-17099, do these steps:

1) Fully loosen the two gripper screws and long gripper screw with the screw head.

NOTE: Gripper screws are captive screws.

2) Fully loosen the three double-threaded screws with the three screw heads.

3) Make sure that the three fasteners of the handling mount are fully opened.

4) Remove the handling mount from the fan frame shroud.

5) Store the handling mount in the engine stand, SPL-14726, storage box.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 71-00-02-080-005-G00

(12) Remove the engine stand forward adapters [472] from the forward engine adapter mounts (Figure 410).

(a) Remove the eight adapter mounting bolts [473] that attach the engine stand forward adapters [472] to the forward engine adapter mounts.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 71-00-02-860-081-G00

(13) Remove the engine stand forward adapters [472] from the handling brackets, SPL-17099 (Figure 410).

(a) Remove the eight adapter mounting bolts [473] that attach the engine stand forward adapters [472] to the handling brackets, SPL-17099.

PTF ALL

SUBTASK 71-00-02-080-006-G00

(14) Remove the TCF lifting bracket adapters, SPL-15012, from the turbine center frame.

SUBTASK 71-00-02-860-084-G00

(15) If it is necessary, do this task: Inlet Cowl Installation, TASK 71-11-01-400-801-G00.

(a) If it is installed, remove the duct support, SPL-14548.

J. Fan Case Connections on the Left Side

SUBTASK 71-00-02-210-003-G00

(1) Do a check of the installed engine data plate.

NOTE: The steps that follow relates to the interchangeability between engines that have the Electronic Engine Control (EEC) blower hardware and those that do not.

NOTE: The LEAP-1BXXGXX engine model designation is identified on the engine data plate, that is located on the RH side of the engine fan case.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0010 OR PRE LEAP-1B SERVICE BULLETIN 75-0011

(a) If the engine installed is PRE-SB 75-0010 and PRE-SB 75-0011, do these steps:

NOTE: LEAP-1BXXG05 engines were delivered from new engine production PRE-SB 75-0010 and PRE-SB 75-0011. Check engine Service Bulletin records to determine compliance to SB 75-0010 and SB 75-0011.

1) Make sure that the ground cooling duct is on the inlet cowl.

2) If there is a cap and cover on the inlet ground cooling duct, remove them (TASK 71-11-01-000-802-G00).

3) Make sure that the EEC blower duct is connected to the inlet ground cooling duct.

a) If it is necessary to connect the EEC blower duct to the inlet ground cooling duct, do the applicable steps in this task: Inlet Cowl Installation, TASK 71-11-01-400-801-G00.

4) If it is installed, remove the INOP label from the EEC blower circuit breakers.

NOTE: Refer to Boeing service bulletin 737-71-1891 to remove INOP labels, remove collars, and close the EEC blower circuit breakers.

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 75-0010 OR POST LEAP-1B SB 75-0011

(b) If the engine installed is POST-SB 75-0010 or POST-SB 75-0011, do these steps:

NOTE: LEAP-1BXXG06 and later engines were delivered POST-SB 75-0010 from new engine production. For LEAP-1BXXG05 engines, check engine Service Bulletin records to determine compliance to SB 75-0010 and SB 75-0011.

1) If there is an inlet ground cooling duct, make sure that you install the cap and cover (TASK 71-11-01-400-802-G00).

2) Make sure that the EEC blower circuit breakers are opened and labeled INOP.

NOTE: Refer to Boeing service bulletin 737-71-1891 to open the EEC blower circuit breaker, install collars, and install INOP labels.

PTF ALL

SUBTASK 71-00-02-010-020-G00

(2) Connect the main fuel supply hose [43] on the left side of the fan cowl support beam (Figure 404).

(a) Remove the protective caps from the main fuel supply hose [43] and on the fitting in the fan cowl support beam.

(b) Lubricate the B-nut of the main fuel supply hose [43] with Christo-lube MCG111 grease, D50011.

(c) Connect the main fuel supply hose [43] to the fan cowl support beam fitting.

(d) Tighten the B-nut to 1140.0 in-lb (128.8 N·m) - 1260.0 in-lb (142.4 N·m).

NOTE: Use two wrenches to tighten the B-nut. Use one wrench to hold the hose and the other wrench to tighten the B-nut.

SUBTASK 71-00-02-410-007-G00

CAUTION

BE CAREFUL WITH THE HYDRAULIC HOSE. DO NOT BEND OR PUT THE HOSE INTO A COIL TOO TIGHTLY. YOU CAN CAUSE A BLOCKAGE OF THE HOSE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE A FAILURE OF THE ENGINE-DRIVEN PUMP (EDP).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE A WRENCH AND A BACK-UP WRENCH WHEN YOU TIGHTEN OR LOOSEN THE B-NUT FOR THE HYDRAULIC LINE. GALLING DAMAGE TO THE LINE OR HYDRAULIC FITTING CAN OCCUR IF YOU DO NOT USE A BACK-UP WRENCH.

CAUTION

DO NOT LET HYDRAULIC FLUID LEAKAGE COLLECT. IF YOU DO NOT MAKE THE AREA CLEAN, DAMAGE CAN OCCUR FROM THE HYDRAULIC FLUID.

(3) Connect the hydraulic pressure hose [40], hydraulic supply hose [41], and hydraulic case drain hose [42] to the left side of the fan cowl support beam as follows (Figure 404):

(a) Remove the protective caps from the hydraulic pressure hose [40], hydraulic supply hose [41], hydraulic case drain hose [42], and fan cowl support beam connectors.

(b) Align and connect the hydraulic pressure hose [40] end to the quick disconnect on the support beam and engage the mating threads.

NOTE: The hydraulic hoses are equipped with self-locking quick disconnect fittings.

(c) Hand tighten until the hydraulic pressure hose [40] end until the outer sleeve snaps forward.

NOTE: You will hear an audible click sound.

(d) Make sure that the outer sleeve snaps fully forward to prevent the axial movement of the sleeve.

CAUTION

MAKE SURE THAT YOU FULLY CONNECT THE COUPLING. IF YOU DO NOT, YOU CAN CAUSE DAMAGE TO THE HYDRAULIC PUMP.

(e) Make sure that the hydraulic pressure hose [40] quick-release fitting is properly connected, do the steps that follow:

CAUTION

MAKE SURE THAT YOU TURN THE OUTER SLEEVE. DO NOT PULL ON IT (AXIAL FORCE). IF YOU PULL ON THE OUTER SLEEVE (AXIALLY), THE LOCKING FEATURE CAN ACCIDENTALLY PULL OUT FROM THE TAB.

1) Without pulling (axial force), rotate the outer sleeve counterclockwise to make sure that it is fully engaged in the locking feature (with the lock tab on the male fitting) (View I, Figure 404).

NOTE: If the locking feature is engaged, the sleeve will not rotate counterclockwise.

2) If the outer sleeve rotates counterclockwise or pulls away from the male quick-release fitting, repeat the steps above to connect the hydraulic hose.

NOTE: The quick-release fitting is not properly installed.

(f) Align and connect the hydraulic supply hose [41] end to the quick disconnect on the support beam and engage the mating threads.

(g) Hand tighten until the hydraulic supply hose [41] end until the outer sleeve snaps forward.

NOTE: You will hear an audible click sound.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Make sure that the outer sleeve snaps fully forward to prevent the axial movement of the sleeve.

CAUTION

MAKE SURE THAT YOU FULLY CONNECT THE COUPLING. IF YOU DO NOT, YOU CAN CAUSE DAMAGE TO THE HYDRAULIC PUMP.

(i) Make sure that the hydraulic supply hose [41] quick-release fitting is properly connected, do the steps that follow:

CAUTION

MAKE SURE THAT YOU TURN THE OUTER SLEEVE. DO NOT PULL ON IT (AXIAL FORCE). IF YOU PULL ON THE OUTER SLEEVE (AXIALLY), THE LOCKING FEATURE CAN ACCIDENTALLY PULL OUT FROM THE TAB.

1) Without pulling (axial force), rotate the outer sleeve counterclockwise to make sure that it is fully engaged in the locking feature (with the lock tab on the male fitting) (View I, Figure 404).

NOTE: If the locking feature is engaged, the sleeve will not rotate counterclockwise.

2) If the outer sleeve rotates counterclockwise or pulls away from the male quick-release fitting, repeat the steps above to connect the hydraulic hose.

NOTE: The quick-release fitting is not properly installed.

(j) Align and connect the hydraulic case drain hose [42] end to the quick disconnect on the support beam and engage the mating threads.

(k) Hand tighten until the hydraulic case drain hose [42] end until the outer sleeve snaps forward.

NOTE: You will hear an audible click sound.

(l) Make sure that the outer sleeve snaps fully forward to prevent the axial movement of the sleeve.

CAUTION

MAKE SURE THAT YOU FULLY CONNECT THE COUPLING. IF YOU DO NOT, YOU CAN CAUSE DAMAGE TO THE HYDRAULIC PUMP.

(m) Make sure that the hydraulic case drain hose [42] quick-release fitting is properly connected, do the steps that follow:

CAUTION

MAKE SURE THAT YOU TURN THE OUTER SLEEVE. DO NOT PULL ON IT (AXIAL FORCE). IF YOU PULL ON THE OUTER SLEEVE (AXIALLY), THE LOCKING FEATURE CAN ACCIDENTALLY PULL OUT FROM THE TAB.

1) Without pulling (axial force), rotate the outer sleeve counterclockwise to make sure that it is fully engaged in the locking feature (with the lock tab on the male fitting) (View I, Figure 404).

NOTE: If the locking feature is engaged, the sleeve will not rotate counterclockwise.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If the outer sleeve rotates counterclockwise or pulls away from the male quick-release fitting, repeat the steps above to connect the hydraulic hose.

NOTE: The quick-release fitting is not properly installed.

SUBTASK 71-00-02-100-001-G00

(4) If silicone/titanium dioxide (white powder) is found on the clamp block [44], clean the clamp block [44] hardware and the surrounding surfaces, such as brackets and the engine fan case.

(a) Make sure to remove all silicone/titanium dioxide (white powder).

NOTE: Refer to SOPM 20-30-03 as an accepted procedure.

SUBTASK 71-00-02-220-001-G00

(5) Make sure that the gap between the outer circumference of the hydraulic pressure hose [40] and each clamp block [44] is less than 0.080 in. (2.032 mm).

(a) Tighten the bolt [45] to 80.0 ±4.0 in-lb (9.0 ±0.5 N·m) on each clamp block [44].

SUBTASK 71-00-02-220-002-G00

CAUTION

MAKE SURE TO MEASURE THE CLEARANCE BETWEEN THE PRESSURE HOSE AND THE CLAMP BLOCK. IF THE CLEARANCE IS NOT IN THE LIMITS, THERE CAN BE A PRELOAD CONDITION. THIS CAN CAUSE A LOSS OF THE HYDRAULIC SYSTEM. IF YOU DO NOT OBEY THE INSTRUCTIONS, DAMAGE TO THE ENGINE-DRIVEN PUMP (EDP) CAN OCCUR.

(6) Make sure that the gap between the outer circumference of the hydraulic pressure hose [40] and each clamp block [56] is less than 0.080 in. (2.032 mm).

NOTE: The spacer block clamp support has sliders that allow to adjust the clamp block to remove preload.

(a) Tighten the bolts [57] to 75 ±2 in-lb (8.47 ±0.23 N·m) on each clamp block [56].

SUBTASK 71-00-02-410-008-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(7) Connect the IDG power feeder harness [39] as follows (Figure 404):

(a) Remove the protective caps from the IDG power feeder harness [39] electrical connector and fan cowl support beam receptacle.

(b) Use the solvent cleaning method to prepare the connector threads and mating keyways (SWPM 20-60-01, Section 2).

(c) Make sure that the distance between top of the connector and upper end of clamp is 18.7 in. (475.0 mm) - 19.2 in. (487.7 mm) (View J, Figure 404).

(d) Align the connector key/keyway and mate the connector (View K, Figure 404).

1) Make sure not to twist the harness wires during routing.

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2) Make sure that the wire bundle connector master keyway (notch) is in the upright position (12 o'clock) +/- 10 degrees after wire harness is installed on engine, when in a relaxed state, prior to strut disconnect hook up.

a) If the connector master keyway (notch) does not meet the upright position (12 o'clock) +/- 10 degrees requirement, adjust the wire harness before the wire harness is connected to the strut disconnect.

NOTE: The wire harness connector master keyway (notch) must be at the upright position without the need to twist the connector.

(e) Rotate the coupling ring by hand in the clockwise direction until the coupling ring stops.

(f) Make sure that the indicator band on the engaging end of the fan cowl support beam receptacle is fully covered.

(g) Tighten the connector to completely mate with the fan cowl support beam receptacle to a torque-tool set-point of 485 ±14 in-lb (54.8 ±1.6 N·m) to make sure of metal-to-metal bottoming.

PTF ALL POST SB 737-24-1233

(h) Do these steps to connect the bonding jumper [46] to the IDG power feeder harness [39] clip:

1) Use the cleaning Procedure 1 to manually clean the bonding jumper [46] terminal and IDG power feeder harness [39] clip (SWPM 20-20-00).

2) Use the cleaning Procedure 5 to clean the fasteners (SWPM 20-20-00).

3) Install the screw [47], washers [49], and nut [48] that attach the bonding jumper [46] to the IDG power feeder harness [39] clip.

NOTE: Use the clip that is near to the fan cowl support beam bracket.

4) Measure the electrical bonding resistance between IDG power feeder harness [39] clip and fan cowl support beam bracket (SWPM 20-20-00).

a) Use an intrinsically safe approved bonding meter, COM-1550, to measure the bonding resistance.

b) Make sure that the resistance is 0.001 ohm (1.0 milliohm) or less.

PTF ALL

SUBTASK 71-00-02-020-006-G00

(8) If the blanking plate, SPL-18640, is installed, do these steps (View E, Figure 404):

(a) Remove the coupling from the starter duct inlet.

(b) Remove the blanking plate, SPL-18640.

(c) Remove the E-seal.

(d) Examine the duct opening for unwanted material.

SUBTASK 71-00-02-410-009-G00

(9) Connect the starter duct [36] to the starter duct inlet as follows (Figure 404):

(a) Remove the protective caps from the starter duct [36] and starter duct inlet.

(b) Install the coupling [37] and new preformed packing [38] that attach the starter duct [36] to the starter duct inlet.

(c) Tighten the coupling [37] to 92 ±2 in-lb (10.4 ±0.2 N·m).

(d) Use the rubber mallet, STD-3906, and lightly tap the outer edge of the coupling [37].

(e) Tighten the coupling [37] again to 92 ±2 in-lb (10.4 ±0.2 N·m).

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K. Fan Case Connections on the Right Side

SUBTASK 71-00-02-410-006-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Use the teflon-jawed pliers, STD-664, to connect the electrical connectors on the right side of the raceway as follows (TASK 70-00-01-910-803-G00):

(a) Remove the protective covers from the electrical connectors and raceway electrical receptacles.

(b) Connect the electrical connector [64], electrical connector [65], and electrical connector [66] to the raceway electrical receptacles (Table 401).

(c) Connect the electrical connector [67], electrical connector [68], electrical connector [69], and electrical connector [70] to the raceway electrical receptacles (Table 401).

Table 401/71-00-02-993-801-G00

ITEMS	ENGINE	LEFT STRUT		RIGHT STRUT	
Electrical Connectors	PLUG	WIRE BUNDLE	RECEPTACLE	WIRE BUNDLE	RECEPTACLE
[64]	DP41B	286A1168	D30260	286A1268	D30460
[65]	DP100	286A1164	D30234	286A1264	D30434
[66]	DP300	286A1166	D30220	286A1266	D30420
[67]	DP21B	286A1166	D30256	286A1266	D30456
[68]	DP200	286A1170	D30252	286A1270	D30452
[69]	DP21A	286A1170	D30212	286A1270	D30412
[70]	DP41A	286A1172	D30224	286A1272	D30424

(d) Tighten the electrical connectors (TASK 70-00-01-910-803-G00).

SUBTASK 71-00-02-410-010-G00

(2) Connect the strut drain line to the right side of the strut fitting (Figure 406).

L. Engine Core Connections

SUBTASK 71-00-02-400-001-G00

(1) Do this task: Drain Mast Installation, TASK 71-71-01-400-801-G00.

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SUBTASK 71-00-02-410-005-G00

CAUTION

BE CAREFUL WITH THE HYDRAULIC HOSE. DO NOT BEND OR PUT THE HOSE INTO A COIL TOO TIGHTLY. YOU CAN CAUSE A BLOCKAGE OF THE HOSE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE A FAILURE OF THE ENGINE-DRIVEN PUMP (EDP).

CAUTION

USE A WRENCH AND A BACK-UP WRENCH WHEN YOU TIGHTEN OR LOOSEN THE B-NUT FOR THE HYDRAULIC LINE. GALLING DAMAGE TO THE LINE OR HYDRAULIC FITTING CAN OCCUR IF YOU DO NOT USE A BACK-UP WRENCH.

(2) Connect the FAMV line [62] under the precooler from the right side of the fan cowl support beam as follows (Figure 405):

(a) Remove the protective caps from the FAMV line [62] and from the hose fitting.

(b) Apply Pure Nickel Special compound, D00006 (preferred), or compound, D00010 (alternate), to all male threads before connection the FAMV line [62].

(c) Connect the B-nut [63] on the FAMV line [62].

(d) Tighten the B-nut [63] to 256.0 in-lb (28.9 N·m) - 284.0 in-lb (32.1 N·m).

SUBTASK 71-00-02-020-005-G00

(3) If the blanking plate, SPL-18258, is installed, do these steps (View D, Figure 405):

(a) Remove the coupling from the precooler inlet [51].

(b) Remove the blanking plate, SPL-18258.

(c) Remove the E-seal.

(d) Examine the inlet opening for unwanted material.

SUBTASK 71-00-02-420-002-G00

(4) Connect the crossover duct [60] on the precooler inlet [51] as follows (Figure 405):

(a) Remove the protective caps from the crossover duct [60] and precooler inlet [51].

(b) Align the crossover duct [60] to the precooler inlet [51].

(c) Install the V-clamp [52] and new E-seal [53] on the crossover duct [60].

(d) Tighten the V-clamp [52] to 115.0 in-lb (13.0 N·m) - 125.0 in-lb (14.1 N·m).

(e) Use the rubber mallet, STD-3906, and lightly tap the outer edge of the V-clamp [52].

(f) Tighten the V-clamp [52] again to 115.0 in-lb (13.0 N·m) - 125.0 in-lb (14.1 N·m).

SUBTASK 71-00-02-420-005-G00

(5) Install the thrust reverser No. 2 and No. 3 compression rods in the support brackets as follows (Figure 408):

(a) Apply Pure Nickel Special compound, D00006, to the threads and shank of the bolts [426].

(b) Install the No. 3 compression rod bracket [425] to the lugs.

(c) Install the two bolts [426] and two nuts [427].

(d) Tighten the bolts [426] to 500 in-lb (56.5 N·m) - 700 in-lb (79.1 N·m).

(e) Install the No. 2 and No. 3 compression rods in the support brackets.

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M. Put the Airplane Back to Its Usual Condition

SUBTASK 71-00-02-410-004-G00

(1) Install the thrust reverser opening actuator (Manual Door Opening System (MDOS)) brackets on both the inboard and outboard sides of the engine fan case as follows:

(a) Install each MDOS actuator bracket [21] with two bolts [22], two washers [23], two washers [24], and two nuts [25].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(b) Install the inboard and outboard MDOS actuator brackets [21] on the aft side of flange B13 on the fan case.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(c) Install the inboard MDOS actuator bracket [21] on the aft side of the upper engine bridge bracket.

(d) Install the outboard MDOS actuator bracket [21] on the forward side of the lower engine bridge bracket.

PTF ALL

(e) Tighten the head of the two bolts [22] to 121 in-lb (14 N·m) - 151 in-lb (17 N·m).

SUBTASK 71-00-02-480-006-G00

(2) Attach the thrust reverser opening actuators (MDOS actuators) to the fan case brackets on the inboard and outboard sides of the engine (TASK 78-31-08-400-801-G00).

(a) Make sure that the thrust reverser opening actuators (MDOS actuators) are locked and support the weight of the thrust reversers.

SUBTASK 71-00-02-420-003-G00

(3) Install the applicable thumbnail fairing (TASK 54-52-01-410-801).

SUBTASK 71-00-02-080-003-G00

(4) If the thrust reverser remains installed, remove the thrust reverser hold-open equipment [10] from the thrust reversers as follows (Figure 401):

(a) Back out the drive screws to move the pin holes to the lowest position on the left fitting assembly [7] and right fitting assembly [9].

(b) Remove the lynch pin [18] from the pivot pin [17].

(c) Remove the pivot pin [17] to disconnect the inboard arm assembly [8] from the outboard arm assembly [6].

(d) Remove the lynch pin [12] from the lower arm pin [11].

(e) Remove the lower arm pin [11] that attaches the outboard arm assembly [6] from the lower hole of the left fitting assembly [7].

(f) Remove the lynch pin [14] from the upper arm pin [13].

(g) Remove the upper arm pin [13] that attaches the outboard arm assembly [6] from the upper hole of the left fitting assembly [7].

(h) Remove the outboard arm assembly [6] from the left fitting assembly [7] and from the stop pin [19] and retaining ring [20] of the inboard arm assembly [8].

(i) Remove the lynch pin [12] from the lower arm pin [11].

(j) Remove the lower arm pin [11] that attaches the inboard arm assembly [8] from the lower hole of the right fitting assembly [9].

(k) Remove the lynch pin [14] from the upper arm pin [13].

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(l) Remove the upper arm pin [13] that attaches the inboard arm assembly [8] from the upper hole of the right fitting assembly [9].

(m) Remove the inboard arm assembly [8] from the outboard side to the right fitting assembly [9].

(n) Remove the lynch pins [16] from the fitting brackets.

(o) Remove the pin assemblies [15] from the left fitting assembly [7] and right fitting assembly [9].

(p) Remove the left fitting assembly [7] and right fitting assembly [9] from the thrust reverser precooler fittings.

(q) Remove the thrust reverser hold-open equipment [10] from the thrust reversers.

SUBTASK 71-00-02-400-002-G00

(5) If necessary, install the thrust reverser. Do this task: Thrust Reverser Installation, TASK 78-31-01-400-801-G00

SUBTASK 71-00-02-410-015-G00

(6) Install the insulation blankets around the thrust reverser hold-open equipment fittings on thrust reverser upper bifurcation (TASK 78-31-13-400-801-G00 or TASK 78-31-13-400-802-G00).

SUBTASK 71-00-02-420-007-G00

(7) If it is removed, install the thrust reverser No. 1 compression rod in the support bracket as follows (Figure 401):

(a) Install the No. 1 compression rod support bracket [5] with the two bolts [4].

(b) Tighten the two bolts [4] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

(c) Install the No. 1 compression rod in the support bracket [5] from the outboard side.

(d) Install the rubber grommet on the inboard side of the No. 1 compression rod.

SUBTASK 71-00-02-860-041-G00

(8) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 71-00-02-210-001-G00

(9) Do a check of the fuel system for leaks.

CAUTION

DO NOT TURN THE FIRE HANDLE AFTER YOU PULLED IT UP. YOU WILL CAUSE THE FIRE BOTTLES TO RELEASE THEIR CONTENTS IF YOU TURN THE HANDLE.

(a) Put the applicable engine fire switch handle (DISCH LEFT or RIGHT) to its usual position.

1) Remove the DO NOT OPERATE tag from the engine fire switch handle (DISCH LEFT or RIGHT).

(b) For the applicable engine, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L
C	18	C00348	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 2
C	19	C00347	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

(c) Look at the FUEL QUANTITY indicator for the fuel quantity.

(d) Move the applicable L(R) START LEVER switch to the IDLE position.

(e) Pressurize the engine fuel pump system.

(f) Examine the fuel system on the engine for leaks.

NOTE: Look carefully at the connection for the main fuel supply line at the strut.

(g) Move the applicable L(R) START LEVER switch to the CUTOFF position.

SUBTASK 71-00-02-860-059-G00

(10) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	6	C04035	AIR COND BLEED AIR CONTROL PRIMARY LEFT
B	6	C04036	AIR COND BLEED AIR CONTROL PRIMARY RIGHT
D	5	C04037	AIR COND BLEED AIR CONTROL BACKUP LEFT
D	6	C04038	AIR COND BLEED AIR CONTROL BACKUP RIGHT

SUBTASK 71-00-02-860-068-G00

(11) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

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SUBTASK 71-00-02-860-043-G00

(12) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 71-00-02-860-044-G00

(13) For the left engine, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

SUBTASK 71-00-02-860-045-G00

(14) For the right engine, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

SUBTASK 71-00-02-860-046-G00

(15) For the left engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 71-00-02-860-047-G00

(16) For the right engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 71-00-02-410-018-G00

(17) Do these tasks in sequence to safely close the left and right thrust reversers:

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

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1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 71-00-02-410-022-G00

(18) If necessary, install the Engine Build Up (EBU) bracket [601] and bracket [604] used to attach the fan cowl hold open rods to the left and right side of the engine fan case as follows (Figure 403):

(a) Lubricate the threads of the bolts [602] and shanks with Pure Nickel Special compound, D00006.

(b) Align the EBU bracket [604] and bracket [601] on each side of the fan case module.

(c) Install the three bolts [602] and three washers [603] that attach the bracket [601] and bracket [604] to each side of the fan case.

(d) Tighten the bolts [602] to 194 in-lb (21.9 N·m) - 206 in-lb (23.3 N·m).

SUBTASK 71-00-02-410-025-G00

(19) Install the left and right fan cowl hold open rods. Do this task: Fan Cowl Hold Open Rod Installation, TASK 71-11-11-400-801-G00.

SUBTASK 71-00-02-410-023-G00

(20) If necessary, install the EBU bracket [609] and bracket [610] used to attach the fan cowl Spring Door Opening System (SDOS) to the left and right side of the engine fan case as follows(Figure 403):

(a) Lubricate the threads of the bolts [605] and shanks with Pure Nickel Special compound, D00006.

(b) Align the EBU bracket [609] and bracket [610] on each side of the fan case module.

(c) Install the three bolts [605], three washers [606], three washers [607], and three nuts [608] that attach the bracket [609] and bracket [610] to each side of the fan case.

(d) Tighten the nuts [608] to 107 in-lb (12.1 N·m) - 113 in-lb (12.8 N·m).

SUBTASK 71-00-02-410-024-G00

(21) Install the left and right SDOS. Do this task: Fan Cowl SDOS Actuators Installation, TASK 71-12-01-400-801-G00.

SUBTASK 71-00-02-410-017-G00

(22) Install the left and right fan cowl panels (TASK 71-11-04-400-801-G00).

SUBTASK 71-00-02-820-002-G00

(23) If the engine installed is a replacement engine, do a check of the thrust reverser rigging (TASK 78-31-01-280-801-G00).

NOTE: If the removed engine is the same engine installed, the thrust reverser rigging check is not required.

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SUBTASK 71-00-02-610-004-G00

(24) Make sure that the starter is serviced with oil before engine run to ensure initial start lubrication.

NOTE: The starter can be serviced with oil either before it is installed on the gearbox or after installation.

NOTE: The engine starter must have oil in order to be operated. Initial servicing of the engine oil system does not provide enough oil to the starter for the starter to be operated.

(a) If unsure of oil quantity in starter, fill the starter with oil (TASK 80-11-01-610-802-G00).

SUBTASK 71-00-02-630-001-G00

(25) If it is necessary, do the depreservation of the engine (TASK 71-00-03-630-802-G00).

SUBTASK 71-00-02-610-001-G00

(26) If it is necessary, service the engine oil system (TASK 12-13-11-600-801).

SUBTASK 71-00-02-610-002-G00

(27) If it is necessary, service the Integrated Drive Generator (IDG) (TASK 12-13-21-600-801).

SUBTASK 71-00-02-440-002-G00

(28) Pressurize the pneumatic system on the engine (TASK 36-00-00-860-801).

SUBTASK 71-00-02-480-007-G00

WARNING MAKE SURE THAT THE GROUND LOCKS ARE INSTALLED IN ALL THE LANDING GEARS BEFORE YOU PRESSURIZE THE HYDRAULIC SYSTEM. WITHOUT THE GROUND LOCKS, THE LANDING GEARS CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(29) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 71-00-02-860-066-G00

WARNING KEEP PERSONS AND EQUIPMENT AWAY FROM ALL CONTROL SURFACES AND THE NOSE GEAR WHEN HYDRAULIC POWER IS SUPPLIED. THE AILERONS, ELEVATORS, RUDDER, FLAPS, SLATS, SPOILERS, AND THE NOSE GEAR ARE SUPPLIED WITH POWER BY THE HYDRAULIC SYSTEMS. INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR WHEN HYDRAULIC POWER IS SUPPLIED.

CAUTION DO NOT OPERATE THE EDP FOR MORE THAN 2 MINUTES UNLESS THE APPLICABLE FUEL TANK HAS A MINIMUM OF 1675 LBS (760 KG) OF FUEL IN IT. IF YOU OPERATE THE EDP FOR 2 MINUTES WITHOUT FUEL IN THE TANK, LET THE RESERVOIR TEMPERATURE CHANGE TO AMBIENT TEMPERATURE BEFORE YOU CONTINUE WITH THE TEST. IF YOU CONTINUE TO OPERATE THE EDP, THE HYDRAULIC FLUID BECOMES TOO HOT.

(30) Pressurize the hydraulic system (TASK 29-11-00-860-801).

SUBTASK 71-00-02-860-049-G00

(31) If it is necessary, remove the engine strut shock absorbing lanyard personnel equipment.

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SUBTASK 71-00-02-440-004-G00

WARNING

ENSURE LEADING AND TRAILING EDGE FLAPS ARE CLEAR OF PERSONNEL AND OBSTRUCTIONS

(32) Do this task:Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 71-00-02-860-050-G00

(33) Remove the DO NOT OPERATE tags from the thrust levers.

SUBTASK 71-00-02-860-051-G00

(34) Remove the DO NOT OPERATE tag from the flap control lever.

SUBTASK 71-00-02-860-052-G00

(35) If it is necessary, do the shock strut deflation (TASK 12-15-31-610-803).

SUBTASK 71-00-02-860-054-G00

(36) To determine whether the EECs have preloaded credentials, conduct Method 3 in this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: This is only applicable when the ONS OS9.1 EEC is new and needs to generate client credential for the first time.

(a) If EEC client credentials generate with a current date and time, the EECs have preloaded credentials and no further action is needed to generate client credentials.

(b) If EEC client credentials do not generate with a current date and time, the EECs do not have preloaded credentials.

1) If the installed EEC does not have preloaded credentials, generate the EEC client credential by performing Method 1 or Method 2 in this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: If neither Method 1 or 2 can be performed, this task may be deferred. EEC data will not store to the Network File Server (NFS) until the EEC client credential is generated, but the data can be pulled via a Portable Maintenance Access Terminal (PMAT) from the EEC. This is the task: Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT, TASK 73-21-05-970-801-G00

NOTE: Ensure all EEC credentials have a date and time stamp that occur after the date and time of the EEC installation.

SUBTASK 71-00-02-860-079-G00

(37) Do this task: EEC System Configuration Data Functions Check, TASK 73-21-00-470-801-G00.

(a) Make sure that the EEC software part numbers are correct.

NOTE: Refer to the applicable Boeing and CFM Service Bulletin/Letters for EEC software to make sure that the software part numbers are compatible with the software on the opposite engine.

1) If the EEC software part numbers are not compatible with the EEC software on the opposite engine, install the compatible EEC software parts (TASK 73-21-00-470-803-G00).

2) Make sure that the EEC software version matches between the EEC channels for each engine.

3) Make sure that the EEC software version in the EECs match between the engines.

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(b) Make sure that the Engine Serial Number (ESN) on display matches the ESN on the engine data plate.

1) If the ESN on the display does not match the ESN on the engine data plate, key in the correct ESN, do this task: Test No. 37 - Engine Serial Number Entry Special Functions Test, TASK 71-00-00-700-813-G00.

NOTE: The next engine start after one of the tasks that follow can have a longer pre-start motoring time than typical of cold engines. This is because the engine off time has been erased from EEC Non Volatile Memory (NVM).

- Install new EEC software.

- Enter new ESN.

- Install the engine on a new aircraft tail number.

- Install the engine on the opposite engine position on the same aircraft tail number.

- Install a data entry plug with new thrust rating, N1 trim, or engine configuration.

(c) Make sure that the menu mode hardware code is correct.

1) If it is necessary, do this task: Test No. 34 - Menu Mode Hardware Code Entry Special Functions Test, TASK 71-00-00-700-810-G00.

(d) Make sure that the auto-detected hardware configuration code is correct.

(e) Make sure that the engine thrust rating on the display matches the engine thrust rating on the engine data plate.

1) If it is necessary, do this task: Data Entry Plug Configuration Check, TASK 73-21-09-700-801-G00.

(f) Make sure that the engine thrust rating matches the opposite engine's thrust rating.

(g) Make sure that the engine thrust rating is applicable to the airplane model.

NOTE: It is recommended to print or make a record of the EEC system configuration page for your airline's records.

N. Power Plant Installation Test

SUBTASK 71-00-02-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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POWER PLANT (ENGINE PRESERVATION AND DEPRESERVATION) - MAINTENANCE PRACTICES

1. General

A. Use this Maintenance Practice in conjunction with the applicable Prolonged Parking procedures when storing an airplane (PROLONGED PARKING - MAINTENANCE PRACTICES, PAGEBLOCK 10-12-00/201).

B. This procedure contains these tasks:

(1) Inlet Cowl Lipskin Protective Coating Procedure

(2) Preservation of an engine (task selection)

(3) Preservation of an engine on wing for not more than 10 days

(4) Preservation of an engine on wing for not more than 30 Days

(5) Preservation of an engine on wing for not more than 3 months

(6) Preservation of an engine on wing for 30 days to 1 year

(7) Preservation of an engine on wing that has expired preservation

(8) Preservation renewal requirements

(9) Depreservation of an engine on wing.

TASK 71-00-03-390-801-G00

2. Inlet Cowl Lipskin Protective Coating Procedure

(Figure 201)

A. General

(1) This task is to apply a protective coating on the inlet cowl lipskin to prevent corrosion.

NOTE: It is acceptable to leave the inlet cowl lipskin bare if the engine is preserved for 7 days or less.

NOTE: It is recommended checking the lipskins for visual appearance every two weeks for the first two months and reapplying protective coating as necessary. The aircraft operator may adjust their inspection schedule based on the operational environment and use of the aircraft.

(2) The task includes the procedure to remove the protective coating.

(3) These tasks should be performed if the engine (on wing) is preserved for more than 7 days.

(4) There are three methods to preserve the inlet cowl lipskin:

(a) Method 1 - Apply the corrosion preventative compound X20-18 compound, G51625.

to the lipskin.

NOTE: The X20-18 compound, G51625, can be left on the lipskin during engine idle runs if the engine anti-ice system is deactivated.

NOTE: If the engine anti-ice system is to be turned on during engine idle runs, the X20-18 compound, G51625, coating must be removed and reapplied after the engine run.

NOTE: The X20-18 compound, G51625, coating must be removed prior to return to service.

NOTE: Please refer to the supplier instructions for additional information regarding the application and removal of X20-18 compound, G51625.

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(b) Method 2 - Apply the peelable protective coating system, Protectapeel SC-1074B-1 coating, C50013/Protectapeel SC-1090 topcoat, C50014, PPG Z6160 coating, C50411/PPG Z6148 topcoat, C50412 or Gardo Protect 350 coating, C50410.

NOTE: Boeing recommends that peelable protective coatings be applied to the inlet lipskin when airplanes are stored indoors or when using a typical inlet cover that covers the inlet lipskin and nacelles.

NOTE: The consumable materials Protectapeel SC-1074B-1 coating, C50013 and Protectapeel SC-1090 topcoat, C50014 are now discontinued by the manufacturer. The coatings can still be used as a coating system and operators are encouraged to use any existing stock before using the alternative materials PPG Z6160 coating, C50411/PPG Z6148 topcoat, C50412 or Gardo Protect 350 coating, C50410.

NOTE: Protectapeel SC-1074B-1 coating, C50013/Protectapeel SC-1090 topcoat, C50014, PPG Z6160 coating, C50411/PPG Z6148 topcoat, C50412 or Gardo Protect 350 coating, C50410 must be removed prior to engine runs.

(c) Method 3 - Weekly inspection for corrosion and cleaning.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
30-00-00-040-803	DDG 30-21-01A Engine Anti-Ice Valves Closed Inoperative - Preparation (P/B 901)
30-00-00-440-803	DDG 30-21-01A Engine Anti-Ice Valves Closed Inoperative - Restoration (P/B 901)
SOPM 20-30-03	General Cleaning Procedures

C. Tools/Equipment

Reference	Description
STD-1064	Scraper - Phenolic, Hard Resin

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
C50013	Coating - Protective, Strippable, Sprayable - Protectapeel SC-1074B-1 (Use Until Stock Depleted)	
C50014	Topcoat - Protective, Water-Based Coating Protectapeel SC-1090 (Use Until Stock Depleted)	
C50410	Coating - Peelable - Gardo Protect 350	BAC5034-1
C50411	Coating - Basecoat, Peelable, Preservation - PPG Z6160	BAC5034-1
C50412	Topcoat - Water-Borne, Preservation - Z6148	BAC5034-1
G02356	Tape - Saint Gobain Rulon J Tape	
G02418	Water - De-ionized	
G50933	Cloth - Cleaning, Low-Lint Cloth (General Use)	A-A-59323 Type II (Supersedes MIL-C-85043)

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(Continued)

Reference	Description	Specification
G51625	Compound - Corrosion Preventive (AECI 3 X20-18)	

E. Location Zones

Zone	Area
412	Engine 1 - Nose Inlet Cowl
422	Engine 2 - Nose Inlet Cowl

F. Method 1 - Protective Coating Application.

WARNING MAKE SURE THAT YOU READ THE MANUFACTURER'S SAFETY DATA SHEET (MSDS) FOR THE MATERIALS YOU WILL USE, AND WEAR THE PROTECTIVE CLOTHING AND EQUIPMENT IF IT IS SPECIFIED. IF YOU DO NOT OBEY, INJURY TO PERSONS CAN OCCUR.

SUBTASK 71-00-03-100-002-G00

(1) If necessary, remove existing pitting, corrosion and surface oxidation using the applicable Structural Repair Manual (SRM) procedure. Make sure that the minimum thickness requirements per the SRM are met.

SUBTASK 71-00-03-100-003-G00

(2) Do these steps to prepare the lipskin for the application of X20-18 compound, G51625:

(a) Clean the lipskin with XS3 Aircraft Wash and XG5 Hard Surface Cleaner furnished from the X20-18 compound, G51625, vendor prior to application.

NOTE: X20-18 compound, G51625, will not remove corrosion but will act to assist in preventing future corrosion. It is acceptable to perform general solvent cleaning per Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802 as an alternative to the products recommended by Grantize.

(b) Remove any dust, dirt, unwanted material, oil, grease or moisture.

SUBTASK 71-00-03-950-003-G00

(3) Do these steps to mask the lipskin:

(a) Mask the inner and outer barrel assembly and the fan compartment.

(b) Make sure that the sealant common to the lipskin is protected.

(c) It is permitted for about 0.1 in. (2.54 mm) of the lipskin to be unprotected.

(d) Do not apply the protective coating to the acoustic barrel area.

SUBTASK 71-00-03-370-005-G00

(4) Do these steps to prepare for the application of X20-18 compound, G51625, to the lipskin:

(a) Make sure that the lipskin surface temperature is between 67.5 ±22.5°F (19.7 ±12.5°C).

(b) Avoid application of X20-18 compound, G51625, when the ambient humidity is above 90%.

(c) Apply a few drops of the X20-18 compound, G51625, to a soft cotton applicator. Do not saturate the applicator to the point of dripping.

NOTE: If the X20-18 compound, G51625, pours out in a cloudy or milky consistency, do not use it. Discard the remaining contents.

SUBTASK 71-00-03-370-006-G00

(5) Do these steps to apply X20-18 compound, G51625, to the lipskin:

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(a) Apply a thin film of X20-18 compound, G51625, to an area of 2.5 ±0.5 ft^2 (0.2 ±0.0 m^2) in a sweeping motion.

NOTE: Do not allow the X20-18 compound, G51625, to dry. Wipe the treated area immediately with a dry and soft cloth until all wet film is buffed off and no longer visible. Rotate the cloth as necessary when absorbency decreases.

NOTE: The initial application of theX20-18 compound, G51625, may remove remaining amounts of polishing residue if cleaning wasn't completed adequately. If only a small amount; proceed with application. Otherwise, repeat cleaning steps in previous section.

NOTE: Direct exposure to sunlight or warm surface temperatures can result in streaking.

(b) Repeat application process until entire lipskin is treated. The process is complete when three (3) coats are applied.

NOTE: Please allow a minimum of 30 minutes between coat applications.

NOTE: X20-18 compound, G51625, has a 12 hour cure time.

SUBTASK 71-00-03-370-007-G00

(6) It is recommended to repolish and reapply X20-18 compound, G51625, at a cycle of every ninety (90) days or after the surface loses its shine. Whichever occurs first.

NOTE: Boeing recommends a schedule for checking the lipskin visual appearance every two (2) weeks for the first two (2) months and then adjusting the schedule according to operator experience based on environment and use of airplane.

(a) To detect the presence of X20-18 compound, G51625, on bare metal surfaces, perform one of the following:

1) Use Rulon J tape, G02356, fold one end back to make a tab at the end. Firmly press the tape on a subject treated area and suspected untreated area. Pull the tab back slowly at a 90 degree angle to the surface. Noticeable ease in resistance will take place on a treated surface compared to a non-treated surface.

2) Spray clean de-ionized water, G02418, on the surface. Moisture will bead up on a treated surface when compared to a non-treated surface.

SUBTASK 71-00-03-950-004-G00

(7) Remove the masking material from the inner and outer barrels and the fan compartment.

SUBTASK 71-00-03-040-003-G00

(8) If engine idle runs will be performed throughout aircraft storage, it is required to deactivate the engine anti-ice system if X20-18 compound, G51625, is still applied to the lipskin. Deactivating the engine anti-ice system will allow the X20-18 compound, G51625, to be left on during engine runs.

NOTE: Make sure that the X20-18 compound, G51625, has cured for 12 hours before performing engine runs.

NOTE: Failure to keep the Engine Anti Ice deactivated during the entire engine run with AECI 3 applied to the lipskin would cause AECI 3 burning, and will result in permanent staining of the lipskin.

(a) Deactivate the engine anti-ice system with the engine anti-ice valve inoperative CLOSED (TASK 30-00-00-040-803).

NOTE: The system can be left deactivated throughout storage. There is no need to do the restoration procedure after each engine run.

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G. Method 2 - Protective Coating Application

WARNING

MAKE SURE THAT YOU READ THE MANUFACTURER'S SAFETY DATA SHEET (MSDS) FOR THE MATERIALS YOU WILL USE, AND WEAR THE PROTECTIVE CLOTHING AND EQUIPMENT IF IT IS SPECIFIED. IF YOU DO NOT OBEY, INJURY TO PERSONS CAN OCCUR.

SUBTASK 71-00-03-100-005-G00

(1) If necessary, remove existing pitting, corrosion and surface oxidation using the applicable SRM procedure. Make sure that the minimum thickness requirements per the SRM are met.

SUBTASK 71-00-03-110-001-G00

(2) Do these steps to clean the lipskin and get a water break free surface:

(a) Remove any dust, dirt, unwanted material, oil, grease or moisture (TASK 20-10-34-110-802).

SUBTASK 71-00-03-950-001-G00

(3) Do these steps to mask the lipskin:

(a) Mask the inner and outer barrel assembly and the fan compartment.

(b) Make sure that the sealant common to the lipskin is protected.

(c) It is permitted for about 0.1 in. (2.54 mm) of the lipskin to be unprotected.

(d) Do not apply the protective coating to the acoustic barrel area.

SUBTASK 71-00-03-370-001-G00

(4) Do these steps to prepare the basecoat Protectapeel SC-1074B-1 coating, C50013:

(a) Let the material get to 75 ±5°F (23.9 ±2.8°C).

(b) Mix the material to a homogeneous consistency.

(c) Standard spray equipment with an atomization pressure of 75 psi (517 kPa) and pot pressure of 5 psi (34.5 kPa) is recommended to get the proper fluid delivery.

SUBTASK 71-00-03-370-002-G00

(5) Do these steps to apply the basecoat Protectapeel SC-1074B-1 coating, C50013, to the lipskin:

(a) Apply a 4 mils (0.004 in.) layer in one application with the spray equipment.

(b) Let this layer dry for 1 hour.

NOTE: You can decrease this time with warm air or infrared lamps. Protect the area from freezing.

(c) Apply a second layer to get a total of 8 mils (0.008 in.) thickness.

SUBTASK 71-00-03-370-003-G00

(6) Do these steps to prepare the Protectapeel SC-1090 topcoat, C50014:

(a) Let the material get to 75 ±5°F (23.9 ±2.8°C).

(b) Mix the material to a homogeneous consistency.

(c) Apply the topcoat in two separate layers using standard spray equipment with an atomization pressure of 75 psi (517 kPa) and pot pressure of 5 psi (34.5 kPa).

SUBTASK 71-00-03-370-004-G00

(7) Do these steps to apply the Protectapeel SC-1090 topcoat, C50014, to the lipskin:

(a) Make sure the basecoat is dry (no gloss).

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(b) The temperature at the time of application and drying should be a minimum of 40 ±5°F (4.4 ±2.8°C). Protect the area from freezing.

(c) Dry apply the first layer of topcoat over the basecoat with a wide fan and the gun held about 18 in. (45.7 cm) from the surface.

1) This discontinuous layer should barely cover the black basecoat to minimize bubbles in the protective coating.

(d) Apply a second layer in a usual spray application to get a thickness of 3 mils (0.003 in.) of dry film.

(e) The basecoat and topcoat will fully dry together to make a rain resistant coating with these conditions:

NOTE: Lower temperatures and thicker protective coating layers will have longer dry times.

1) Two hours of drying time with conditions of 50% relative humidity and 70°F (21.1°C).

2) The lipskin can be dried outside if there is no direct rain and the temperature is about 50°F (10°C).

SUBTASK 71-00-03-370-008-G00

(8) Do these steps to prepare the PPG Z6160 coating, C50411:

NOTE: PPG Z6160 coating, C50411 can be used alone or for added exterior durability it can be over coated with Z6148 preservation topcoat.

(a) Let the PPG Z6160 coating, C50411 get to 70°F (21.11°C) – 80°F (26.67°C).

(b) Fully mix the PPG Z6160 coating, C50411 in the container.

NOTE: If a mechanical paint shaker is used, the recommended mix time is 5 minutes.

(c) Apply the basecoat with standard spray equipment.

NOTE: Use spray equipment with an atomization pressure of 75 psi (517 kPa) and a pot pressure of 5 psi (34 kPa) to make sure that the fluid is delivered correctly.

SUBTASK 71-00-03-370-009-G00

(9) Do these steps to apply the PPG Z6160 coating, C50411 to the lip skin:

(a) Apply a 4 mils (0.004 in.) layer in one application with the spray equipment.

(b) Let the first coat dry for one hour.

NOTE: Drying can be accelerated by the circulation of warm air.

(c) Apply a second 4 mils (0.004 in.) layer to achieve a total coating of 8 mils (0.008 in.) thickness.

(d) Let the second coat dry for one hour.

SUBTASK 71-00-03-370-010-G00

(10) Do these steps to prepare the PPG Z6148 topcoat, C50412:

(a) Let the material get to 70°F (21.11°C) – 80°F (26.67°C).

(b) Fully mix the PPG Z6160 coating, C50411 in the container.

NOTE: If a mechanical paint shaker is used, the recommended mix time is 5 minutes.

(c) Apply the topcoat with standard spray equipment.

NOTE: Use spray equipment with an atomization pressure of 75 psi (517 kPa) and a pot pressure of 5 psi (34 kPa) to make sure that the fluid is delivered correctly.

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SUBTASK 71-00-03-370-011-G00

(11) Do these steps to apply the PPG Z6148 topcoat, C50412 to the lip skin:

(a) Make sure that the basecoat is dry (no gloss).

(b) Apply a 2 mils (0.002 in.) layer in one application with the spray equipment.

(c) Allow to dry for two hours.

NOTE: Full cure is achieved in 16 hours.

SUBTASK 71-00-03-370-012-G00

(12) Do these steps to prepare the Gardo Protect 350 coating, C50410:

(a) Let the material get to 70°F (21.11°C) – 80°F (26.67°C).

(b) Fully mix the Gardo Protect 350 coating, C50410 in the container.

(c) Apply the coating with standard spray equipment.

NOTE: Use spray equipment with an atomization pressure of 75 psi (517 kPa) and a pot pressure of 5 psi (34 kPa) to make sure that the fluid is delivered correctly.

SUBTASK 71-00-03-370-013-G00

(13) Do these steps to apply the Gardo Protect 350 coating, C50410 to the lip skin:

(a) Apply a 2 mils (0.002 in.) - 3 mils (0.003 in.) layer in one application with the spray equipment.

(b) Allow to dry for 15 to 30 minutes.

SUBTASK 71-00-03-950-002-G00

(14) Remove the masking material from the inner and outer barrels and the fan compartment.

H. Method 3 - Weekly Inspection for Corrosion and Cleaning

SUBTASK 71-00-03-100-004-G00

(1) If a protective coating is not available, do the following at least once (1) weekly. Daily inspection and cleaning is preferred.

NOTE: Frequency of inspections will depend on the environmental condition that the airplane is stored in. Inspections and cleaning should be done at a frequency that ensures the lipskin does not have corrosive substance build up or prolonged moisture pooling on the lipskin surface.

(a) With a soft low-lint cleaning cloth, G50933, moisten with alcohol, B00130 (preferred), or with de-ionized water, G02418, clean the nose cowl lipskin (TASK 20-10-34-110-802).

(b) If corrosion is found, remove per the applicable SRM procedure. Make sure that the minimum thickness requirements per SRM are met.

I. Protective Coating Removal

SUBTASK 71-00-03-150-001-G00

(1) Do one of these steps to remove the protective coating from the lipskin:

(a) If X20-18 compound, G51625, was used, perform the following steps:

1) Clean the lipskin with XS3 Aircraft Wash and XG5 Hard Surface Cleaner furnished from the X20-18 compound, G51625, vendor.

NOTE: It is acceptable to perform general solvent cleaning per SOPM 20-30-03 as an alternative to the products recommended by Granitize.

2) Clean the surface until there are no signs of X20-18 compound, G51625, present on the lipskin.

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a) To detect the presence of X20-18 compound, G51625, on bare metal surfaces, perform one of the following:

<1> Use Rulon J tape, G02356, to make a tab at the end. Firmly press the tape on a subject treated area and suspected untreated area. Pull the tab back slowly at a 90 degree angle to the surface. Noticeable ease in resistance will take place on a treated compared to the untreated surface.

<2> Spray clean de-ionized water, G02418, on the surface. Moisture will bead up on a treated surface when compared to a non-treated surface.

3) Do this task if the aircraft will be put back into service: DDG 30-21-01A Engine Anti-Ice Valves Closed Inoperative - Restoration, TASK 30-00-00-440-803.

(b) If Protectapeel SC-1074B-1 coating, C50013/Protectapeel SC-1090 topcoat, C50014, PPG Z6160 coating, C50411/PPG Z6148 topcoat, C50412, or Gardo Protect 350 coating, C50410, was used, perform one of the following steps:

1) Use your hand to peel off the coating or use clean shop air.

a) Refer to the coating manufacturer's recommendations for more details.

2) Do not use hand tools harder than the aluminum lipskin to prevent damage.

CAUTION

SCRIBE MARKS OR CUTS WHICH GO INTO THE METAL SURFACE OF THE LIP SKIN CAN CAUSE DAMAGE TO THE LIP SKIN. THE DAMAGE CAUSES LOSS OF FATIGUE LIFE OF THE METAL.

3) You can use a hard resin phenolic scraper, STD-1064, to help you remove the peelable coating with these conditions:

a) The tool does not go completely through the coating.

b) The tool does not score or mar the lipskin.

c) Use a scraper which has an adjustable depth of cut on the coating.

SUBTASK 71-00-03-110-002-G00

(2) Remove any residue or unwanted material (TASK 20-10-34-110-802).

——— END OF TASK ———

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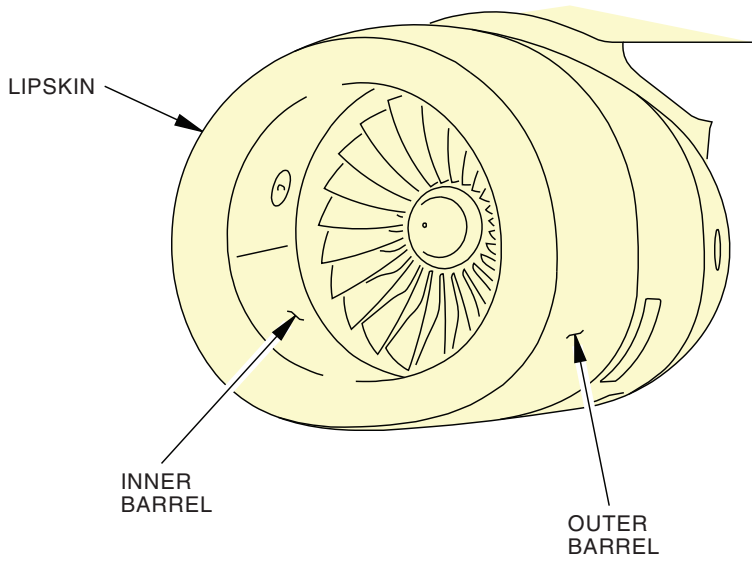

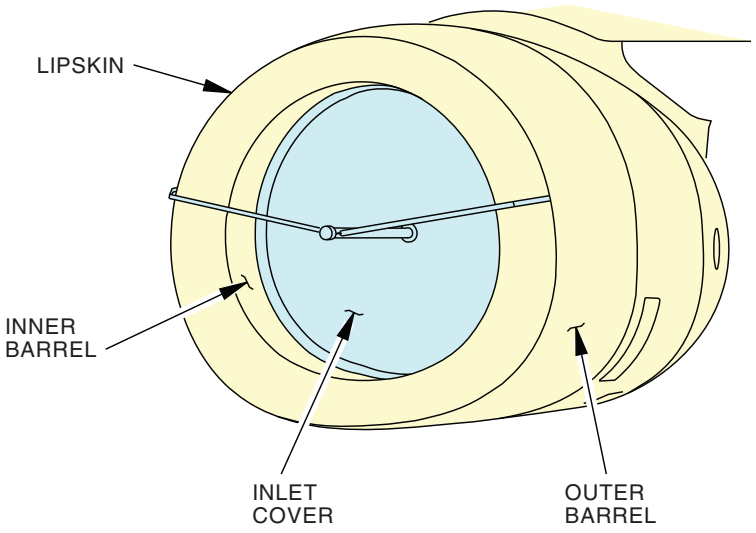

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Inlet Cowl Lipskin Protective Coating
Figure 201/71-00-03-990-820-G00

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TASK 71-00-03-620-801-G00

3. Preservation of an Engine (Task Selection)

A. General

(1) This task gives the instructions for the task selection for engine preservation.

(2) These tasks are recommended for installed engines that are not to be operated. Preservation instructions specific for engines that are not installed (off-wing) are available from the manufacturer.

B. Consumable Materials

Reference	Description	Specification
G02455 [CP2160]	Film, Vapor Barrier, Polyethylene	L-P-512
G02456 [CP2098]	Desiccant - Activated, Bagged, Packaging Use And Static Dehumidification	MIL-D-3464
G51486 [CP2109]	Tape - Plastic, Colored - 3M No. 471	
G51738 [CP2158]	Tape - Masking, Cloth Adhesive - tesa 4651	

C. Preservation of an Engine Task Selection

SUBTASK 71-00-03-620-001-G00

(1) Select the engine preservation procedure per the duration that the engine will be inoperative:

NOTE: Preservation renewal procedures are also covered in this instruction.

NOTE: An engine removed for shop-visit needs to be preserved for the duration the engine will be inoperative, including the shop-visit. Therefore, it is recommended to take into account the expected time engine will spend in shop visit when selecting the task for preservation before engine removal.

(a) Preservation of an engine for not more than 10 days, do this task: Preservation of an Engine on Wing for not more than 10 Days, TASK 71-00-03-600-801-G00.

(b) Preservation of up to 30 days, do this task: Preservation of an Engine on Wing for not more than 30 Days, TASK 71-00-03-620-802-G00.

(c) Preservation of up to 3 months, do this task: Preservation of an Engine on Wing for not more than 3 Months, TASK 71-00-03-620-806-G00.

(d) Preservation for 30 days to 1 year, do this task: Preservation of an Engine on Wing for 30 Days to 1 Year, TASK 71-00-03-630-801-G00.

(e) Preservation of an engine that has expired preservation, do this task: Preservation of an Engine on Wing That Has Expired, TASK 71-00-03-620-808-G00.

(f) Requirements for preservation renewal, do this task: Preservation Renewal Requirements, TASK 71-00-03-620-809-G00.

(g) Engine dry-out and re-lubrication procedure after the engine was shutdown in flight.

(h) Depreservation of an engine to put back in operation, do this task: Depreservation of an Engine on Wing, TASK 71-00-03-630-802-G00.

SUBTASK 71-00-03-620-002-G00

(2) Preservation program guidelines:

(a) The tasks give the instructions for the preservation and de-preservation for operable and non-operable engines.

NOTE: Engines that can be started are considered operable. Engines that cannot be started are considered non-operable.

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(b) The preservation procedures for an operable engine give the minimum recommended steps that are necessary to prevent these conditions:

1) Corrosion

2) Entry of unwanted material into the engine

3) The effects of the climate and weather conditions.

(c) The preservation procedures for a non-operable engine give the minimum recommended steps that are necessary if you cannot start the engine.

(d) The de-preservation procedure puts the installed engine back in operation.

(e) The preservation procedures are different for different periods of non-operational time, different types of preservation procedures and if the power plant is operable or not operable.

(f) Power plant preservation is a flexible program that you can do, and must agree with the applicable weather and storage conditions.

(g) More care is necessary for a program for power plants that are operational in high humidity, in large temperature changes or near a salt water area than for the power plants that are in drier climates or less bad weather conditions.

CAUTION

DO NOT USE CABLE TIES TO ATTACH THE DESICCANT BAGS. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

CAUTION

MAKE SURE THAT THERE IS NO CONTACT BETWEEN DESICCANT BAGS AND LOW PRESSURE TURBINE (LPT) BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(h) In case of engine storage in an environment with a humidity level that is higher than 70%, put in place desiccant bags, G02456 [CP2098], and humidity indicator card (locally manufactured) in the inlet and the exhaust in order to monitor the humidity level.

NOTE: The best condition for engine storage is to keep an environment with less than 70% of humidity.

1) Install desiccant bags, G02456 [CP2098], on Outlet Guide Vane (OGV)/fan frame struts and Turbine Rear Vane (TRV) struts.

a) Use tape to secure desiccant bags, G02456 [CP2098] (with humidity indicator), to prevent movement of desiccant bags, G02456 [CP2098].

2) Protect engine parts with vapor barrier film, G02455 [CP2160], to make sure that there is no contact between engine parts and desiccant bags, G02456 [CP2098], nor between engine parts and tape.

3) Make sure that desiccant bags, G02456 [CP2098], are not damaged.

4) Attach humidity indicator card (locally manufactured) to monitor humidity level during the preservation period.

5) Make sure that desiccant bags, G02456 [CP2098], and humidity indicators are easily accessible.

(i) When you use desiccants, inspect and change them regularly as applicable, taking into account environmental conditions, to make sure that the desiccants give good protection.

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1) If indication of important humidity or saturated desiccant bags is noted during inspection, consider reducing inspection interval and increasing quantity of desiccant bags.

(j) Install a vapor barrier film, G02455 [CP2160], in inlet and exhaust to seal the engine flow path.

NOTE: This will make sure that the desiccants give good protection.

(k) Do a schedule for the preservation programs for power plants that are not operated and also do a schedule for the preservation renewal procedures.

(l) Monitor the schedule regularly to make sure that you do the necessary procedure before the expiration time.

CAUTION

YOU MUST DO ALL THE APPLICABLE PRESERVATION PROCEDURES WHEN YOU PUT AN ENGINE INTO A PERIOD OF NON-OPERATION. IF YOU DO NOT, CORROSION AND GENERAL DETERIORATION OF THE CORE ENGINE AND THE FUEL SYSTEM CAN OCCUR.

(m) Examine the preservation of the power plant as the weather conditions and conditions of power plant protection change, and do the procedures necessary to keep the power plant in an operable condition.

NOTE: You cannot perform engine preservation and put the engines into storage without maintenance. You must do a schedule and apply it.

(n) Close the Variable Bleed Valve (VBV) doors when you do the preservation and store the power plant.

CAUTION

DO NOT USE WHITE, GREY, YELLOW, RED OR GREEN TESA 4651 TAPE (CP2158) OR RED 3M 471 TAPE (CP2109). IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

1) Cover VBV grids with vapor barrier film, G02455 [CP2160], or blue 3M 471 tape, G51486 [CP2109], or blue, brown or black tesa 4651 tape, G51738 [CP2158].

2) Make sure that no unwanted material goes into the core engine inlet through the VBV.

(o) Make sure that you do the engine dry-out and re-lubrication procedure after an engine has had an in-flight shutdown.

(p) Make sure that you do the engine dry-out and re-lubrication procedure when you do a preservation if the engine is not operable for a long time.

(q) Make sure that you remove all tools and clean the work area after you do the preservation procedure.

——— END OF TASK ———

TASK 71-00-03-600-801-G00

4. Preservation of an Engine on Wing for not more than 10 Days

A. General

(1) This task gives the instructions on how to preserve an engine on wing for not more than 10 days.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-00-00-910-803-G00	Start the Engine (Normal Start) (P/B 201)
71-00-00-910-806-G00	Stop the Engine (Usual Engine Stop) (P/B 201)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1517	Cover - Engine Exhaust Part #: 896812 Supplier: SBK11 Part #: AVTB73MLEA1BEPS Supplier: 018PR Part #: C10006-1 Supplier: 81205
SPL-14650	Cover, Engine Inlet - LEAP 1B Part #: 892812 Supplier: SBK11 Part #: AVTB73MLEA1B Supplier: 018PR Part #: B737-104 Supplier: 1LE67 Part #: B737-153 Supplier: 1LE67 Part #: C10005-23 Supplier: 81205 Opt Part #: C10005-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
G00252	Film - Polyethylene Film And Sheeting	ASTM D2103 (Supersedes L-P-512)
G00253	Material - Barrier Materials, Greaseproofed, Waterproof, Flexible, Heat-Sealable	MIL-PRF-121 (Supersedes MIL-B-121)
G00834	Cloth - Lint-free Cotton	
G00920	Tape - Waterproof, Packaging	ASTM D5486
G02455 [CP2160]	Film, Vapor Barrier, Polyethylene	L-P-512
G02456 [CP2098]	Desiccant - Activated, Bagged, Packaging Use And Static Dehumidification	MIL-D-3464

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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F. Preservation of an Engine on Wing for not more than 10 Days

SUBTASK 71-00-03-910-001-G00

(1) If the engine is operable but has not been operated at the idle-power speed for a minimum of three minutes during the last engine operation, do these steps:

NOTE: This step and the two steps that follow are not necessary if the engine was operated at ground idle during the last engine operation. Consider the engine preserved for 10 days from the time of the last engine operation.

(a) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.

NOTE: Keep the engine at stable ground idle for at least 3 minutes.

(b) Dry motor the engine for 2 minutes in order to reduce risk of fuel nozzle coking (TASK 71-00-00-910-808-G00).

(c) In less than 1 hour after engine shutdown, do a check of the oil level on the oil tank sight glass.

(d) If the oil level is above the top of the oil tank sight glass, do the steps as follow:

1) Drain the oil tank down to -3QT sight glass indication (TASK 12-13-11-600-803).

2) Do this task: Start the Engine (Normal Start), TASK 71-00-00-910-803-G00.

3) Operate the engine at stable ground idle for 3 minutes.

4) Do this task: Stop the Engine (Usual Engine Stop), TASK 71-00-00-910-806-G00.

5) Inspect again the oil level on the oil tank sight glass in less than 1 hour after engine shutdown.

6) If the oil level is still above the MAX level, repeat the steps above, until the oil level is visible on the oil tank sight glass.

7) At first opportunity, do the troubleshooting for the cause of oil over-servicing per the applicable IFIM procedure.

a) Until you complete troubleshooting, do not perform oil tank servicing on a cold engine (TASK 12-13-11-600-801).

NOTE: It is necessary to perform only oil tank servicing on warm engine (less than 1 hour after engine shutdown).

SUBTASK 71-00-03-640-001-G00

(2) If the engine is not operable and was shutdown in-flight (IFSD), dry out and lubricate the engine.

NOTE: If the engine is non-operable, consider the engine preserved for 10 days from the time of last operation or the engine dry-out procedure.

SUBTASK 71-00-03-640-005-G00

CAUTION

DO NOT USE CABLE TIES TO ATTACH THE DESICCANT BAGS. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT THERE IS NO CONTACT BETWEEN DESICCANT BAGS AND LOW PRESSURE TURBINE (LPT) BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(3) In case of engine storage in an environment with a humidity level that is higher than 70%, put in place desiccant bags, G02456 [CP2098], and humidity indicator card (locally manufactured) in the inlet and the exhaust in order to monitor the humidity level (TASK 71-00-03-620-801-G00).

NOTE: The best condition for engine storage is to keep an environment with less than 70% of humidity.

(a) Install desiccant bags, G02456 [CP2098], on Outlet Guide Vane (OGV)/fan frame struts and Turbine Rear Vane (TRV) struts.

1) Use tape to secure desiccant bags, G02456 [CP2098] (with humidity indicator), to prevent movement of desiccant bags, G02456 [CP2098].

(b) Protect engine parts with vapor barrier film, G02455 [CP2160], to make sure that there is no contact between engine parts and desiccant bags, G02456 [CP2098], nor between engine parts and tape.

(c) Make sure that desiccant bags, G02456 [CP2098], are not damaged.

(d) Attach humidity indicator card (locally manufactured) to monitor humidity level during the preservation period.

(e) Make sure that desiccant bags, G02456 [CP2098], and humidity indicators are easily accessible.

SUBTASK 71-00-03-480-001-G00

(4) Put the cover, SPL-14650, on the fan cowl entrance.

(a) If the engine inlet cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

1) When using plastic sheeting to cover the inlet, do the steps that follow:

a) Use a lint-free cloth, G00834, and solvent, B00666, to wipe the area where the tape will be used.

b) Cut approximately 70 in. (1.8 m) diameter circle from plastic sheet.

c) Tape the plastic circle on FWD edge of the acoustic panel behind the lipskin.

<1> Apply with at least 3 layers of overlapping tape around the perimeter of the inlet acoustic panel.

<2> Do not place plastic or tape over the lipskin or encase the lipskin with plastic.

d) Apply the tape across the diameter of the circular sheet to reduce the pillowing of the plastic sheeting.

SUBTASK 71-00-03-480-002-G00

(5) Put the exhaust cover, SPL-1517, on the tailpipe exit.

(a) If the engine exhaust cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

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1) Make sure that the drain holes or tubes are open and not covered, when applying plastic sheeting and tape to the exhaust.

——— END OF TASK ———

TASK 71-00-03-620-802-G00

5. Preservation of an Engine on Wing for not more than 30 Days

(Figure 202)

A. General

(1) This task gives the instructions to preserve the engine on wing for not more than 30 days.

(2) If engine was shutdown in flight, the engine must be dried out and re-lubricated within 24 hours.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)
36-11-04-000-801	PRSOV Removal (P/B 401)
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-00-00-910-803-G00	Start the Engine (Normal Start) (P/B 201)
71-00-00-910-806-G00	Stop the Engine (Usual Engine Stop) (P/B 201)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1517	Cover - Engine Exhaust Part #: 896812 Supplier: SBK11 Part #: AVTB73MLEA1BEPS Supplier: 018PR Part #: C10006-1 Supplier: 81205
SPL-14650	Cover, Engine Inlet - LEAP 1B Part #: 892812 Supplier: SBK11 Part #: AVTB73MLEA1B Supplier: 018PR Part #: B737-104 Supplier: 1LE67 Part #: B737-153 Supplier: 1LE67 Part #: C10005-23 Supplier: 81205 Opt Part #: C10005-1 Supplier: 81205
SPL-18258	Blanking Plate - PRSOV Part #: C36007-11 Supplier: 81205 Opt Part #: C36007-1 Supplier: 81205
STD-3906	Mallet - Rubber

D. Consumable Materials

Reference	Description	Specification
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
G00252	Film - Polyethylene Film And Sheeting	ASTM D2103 (Supersedes L-P-512)

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(Continued)

Reference	Description	Specification
G00253	Material - Barrier Materials, Greaseproofed, Waterproof, Flexible, Heat-Sealable	MIL-PRF-121 (Supersedes MIL-B-121)
G00834	Cloth - Lint-free Cotton	
G00920	Tape - Waterproof, Packaging	ASTM D5486
G02304	Tape - Moisture Barrier, Adhesive One-side	BMS8-346
G02455 [CP2160]	Film, Vapor Barrier, Polyethylene	L-P-512
G02456 [CP2098]	Desiccant - Activated, Bagged, Packaging Use And Static Dehumidification	MIL-D-3464
G51486 [CP2109]	Tape - Plastic, Colored - 3M No. 471	
G51738 [CP2158]	Tape - Masking, Cloth Adhesive - tesa 4651	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Preservation of an Engine on Wing for not more than 30 Days

SUBTASK 71-00-03-620-003-G00

CAUTION DO NOT SPRAY PRESERVATIVE OIL OR EQUIVALENT INTO THE ENGINE INLET, CORE COMPRESSOR, TURBINE, OR ENGINE EXHAUST. UNWANTED PARTICLES THAT STAY ON THE BLADES AND VANES WET WITH OIL CAN CAUSE DAMAGE TO THE ENGINE.

(1) If the engine is operable, do these steps:

(a) If the engine was shutdown in flight, dry out and re-lubricate the engine within 24 hours.

(b) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.

NOTE: Keep the engine operational at stable ground idle for 15-20 minutes after the test completes.

NOTE: This step and the steps to operate and shut down the engine are not necessary if the engine was operated at stable ground idle for 15-20 minutes during the last engine operation. Consider the engine preserved for 30 days from the last time of operation.

(c) Dry motor the engine for 2 minutes, in order to reduce risk of fuel nozzle coking (TASK 71-00-00-910-808-G00).

(d) In less than 1 hour after engine shutdown, do a check of the oil level on the oil tank sight glass.

(e) If the oil level is above the top of the oil tank sight glass, do the steps as follow:

1) Drain the oil tank down to -3QT sight glass indication (TASK 12-13-11-600-803).

2) Do this task: Start the Engine (Normal Start), TASK 71-00-00-910-803-G00.

3) Operate the engine at stable ground idle for 3 minutes.

4) Do this task: Stop the Engine (Usual Engine Stop), TASK 71-00-00-910-806-G00.

5) Inspect again the oil level on the oil tank sight glass in less than 1 hour after engine shutdown.

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6) If the oil level is still above the MAX level, repeat the steps above, until the oil level is visible on the oil tank sight glass.

7) At first opportunity, do the troubleshooting for the cause of oil over-servicing per the applicable IFIM procedure.

a) Until you complete troubleshooting, do not perform oil tank servicing on a cold engine (TASK 12-13-11-600-801).

NOTE: It is necessary to perform only oil tank servicing on warm engine (less than 1 hour after engine shutdown).

CAUTION

DO NOT USE CABLE TIES TO ATTACH THE DESICCANT BAGS. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

CAUTION

MAKE SURE THAT THERE IS NO CONTACT BETWEEN DESICCANT BAGS AND LOW PRESSURE TURBINE (LPT) BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(f) In case of engine storage in an environment with a humidity level that is higher than 70%, put in place desiccant bags, G02456 [CP2098], and humidity indicator card (locally manufactured) in the inlet and the exhaust in order to monitor the humidity level (TASK 71-00-03-620-801-G00).

NOTE: The best condition for engine storage is to keep an environment with less than 70% of humidity.

1) Install desiccant bags, G02456 [CP2098], on Outlet Guide Vane (OGV)/fan frame struts and Turbine Rear Vane (TRV) struts.

a) Use tape to secure desiccant bags, G02456 [CP2098] (with humidity indicator), to prevent movement of desiccant bags, G02456 [CP2098].

2) Protect engine parts with vapor barrier film, G02455 [CP2160], to make sure that there is no contact between engine parts and desiccant bags, G02456 [CP2098], nor between engine parts and tape.

3) Make sure that desiccant bags, G02456 [CP2098], are not damaged.

4) Attach humidity indicator card (locally manufactured) to monitor humidity level during the preservation period.

5) Make sure that desiccant bags, G02456 [CP2098], and humidity indicators are easily accessible.

CAUTION

DO NOT USE WHITE, GREY, YELLOW, RED OR GREEN TESA 4651 TAPE (CP2158) OR RED 3M 471 TAPE (CP2109). IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(g) Cover Variable Bleed Valve (VBV) grids with vapor barrier film, G02455 [CP2160], or blue 3M 471 tape, G51486 [CP2109] or blue, brown or black tesa 4651 tape, G51738 [CP2158].

(h) Put the cover, SPL-14650, on the entrance of the fan cowling.

1) If the engine inlet cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

a) When using plastic sheeting to cover the inlet, do the steps that follow:

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<1> Use a lint-free cloth, G00834, and solvent, B00666, to wipe the area where the tape will be used.

<2> Cut approximately 70 in. (1.8 m) diameter circle from plastic sheet.

<3> Tape the plastic circle on FWD edge of the acoustic panel behind the lipskin.

<a> Apply with at least 3 layers of overlapping tape around the perimeter of the inlet acoustic panel.

<b> Do not place plastic or tape over the lipskin or encase the lipskin with plastic.

<4> Apply the tape across the diameter of the circular sheet to reduce the pillowing of the plastic sheeting.

(i) Put the exhaust cover, SPL-1517, on the exit of the tailpipe.

1) If the engine exhaust cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

a) Make sure that the drain holes or tubes are open and not covered, when applying plastic sheeting and tape to the exhaust.

SUBTASK 71-00-03-020-005-G00

(2) If the APU or external air will be used to operate the ECS during the time that the engine will be preserved on wing, it is recommended to remove the PRSOV and store it off of the airplane, do the steps that follow:

(a) Remove the PRSOV, do this task: PRSOV Removal, TASK 36-11-04-000-801.

(b) Use an E-seal and a coupling with a blanking plate, SPL-18258, and install on the crossover duct (Figure 202).

1) When the coupling are in place, tighten the coupling.

a) Tighten the coupling nut to 175 ±5 in-lb (19.8 ±0.6 N·m).

b) Lightly tap the outer surface of the coupling with a rubber mallet, STD-3906.

c) Tighten the coupling nut again to 175 ±5 in-lb (19.8 ±0.6 N·m).

(c) On the mix duct, cover the opening with a protective cover or polyethylene film, G00252, and tape, G00920.

SUBTASK 71-00-03-620-004-G00

(3) If the engine is not operable, do these steps:

(a) If the engine was shutdown in flight, dry out and re-lubricate the engine.

(b) If engine is non-operable, consider the engine preserved for 30 days from the time of last operation or the engine dry-out procedure.

CAUTION DO NOT USE WHITE, GREY, YELLOW, RED OR GREEN TESA 4651 TAPE (CP2158) OR RED 3M 471 TAPE (CP2109). IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(c) Cover VBV grids with vapor barrier film, G02455 [CP2160], or blue 3M 471 tape, G51486 [CP2109] or blue, brown or black tesa 4651 tape, G51738 [CP2158].

1) Make sure that no unwanted material goes into the core engine inlet through the VBV.

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(d) Put a cap or protective cover on all strut, pipes, accessory openings and electrical connections.

1) Make sure that the drain holes or tubes are open and not covered.

CAUTION DO NOT USE CABLE TIES TO ATTACH THE DESICCANT BAGS. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

CAUTION MAKE SURE THAT THERE IS NO CONTACT BETWEEN DESICCANT BAGS AND LOW PRESSURE TURBINE (LPT) BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(e) In case of engine storage in an environment with a humidity level that is higher than 70%, put in place desiccant bags, G02456 [CP2098], and humidity indicator card (locally manufactured) in the inlet and the exhaust in order to monitor the humidity level (TASK 71-00-03-620-801-G00).

NOTE: The best condition for engine storage is to keep an environment with less than 70% of humidity.

1) Install desiccant bags, G02456 [CP2098], on OGV/fan frame struts and TRV struts.

a) Use tape to secure desiccant bags, G02456 [CP2098] (with humidity indicator), to prevent movement of desiccant bags, G02456 [CP2098].

2) Protect engine parts with vapor barrier film, G02455 [CP2160], to make sure that there is no contact between engine parts and desiccant bags, G02456 [CP2098], nor between engine parts and tape.

3) Make sure that desiccant bags, G02456 [CP2098], are not damaged.

4) Attach humidity indicator card (locally manufactured) to monitor humidity level during the preservation period.

5) Make sure that desiccant bags, G02456 [CP2098], and humidity indicators are easily accessible.

(f) Seal front and back openings/exposed areas with vapor barrier film, G02455 [CP2160].

1) Attach the film with moisture barrier tape, G02304.

(g) Put the cover, SPL-14650, on the entrance of the fan cowling.

1) If the engine inlet cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

a) When using plastic sheeting to cover the inlet, do the steps that follow:

<1> Use a lint-free cloth, G00834, and solvent, B00666, to wipe the area where the tape will be used.

<2> Cut approximately 70 in. (1.8 m) diameter circle from plastic sheet.

<3> Tape the plastic circle on FWD edge of the acoustic panel behind the lipskin.

<a> Apply with at least 3 layers of overlapping tape around the perimeter of the inlet acoustic panel.

<b> Do not place plastic or tape over the lipskin or encase the lipskin with plastic.

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<4> Apply the tape across the diameter of the circular sheet to reduce the pillowing of the plastic sheeting.

(h) Put the exhaust cover, SPL-1517, on the exit of the tailpipe.

1) If the engine exhaust cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

a) Make sure that the drain holes or tubes are open and not covered, when applying plastic sheeting and tape to the exhaust.

SUBTASK 71-00-03-160-002-G00

(4) Remove all tools and clean the work area.

——— END OF TASK ———

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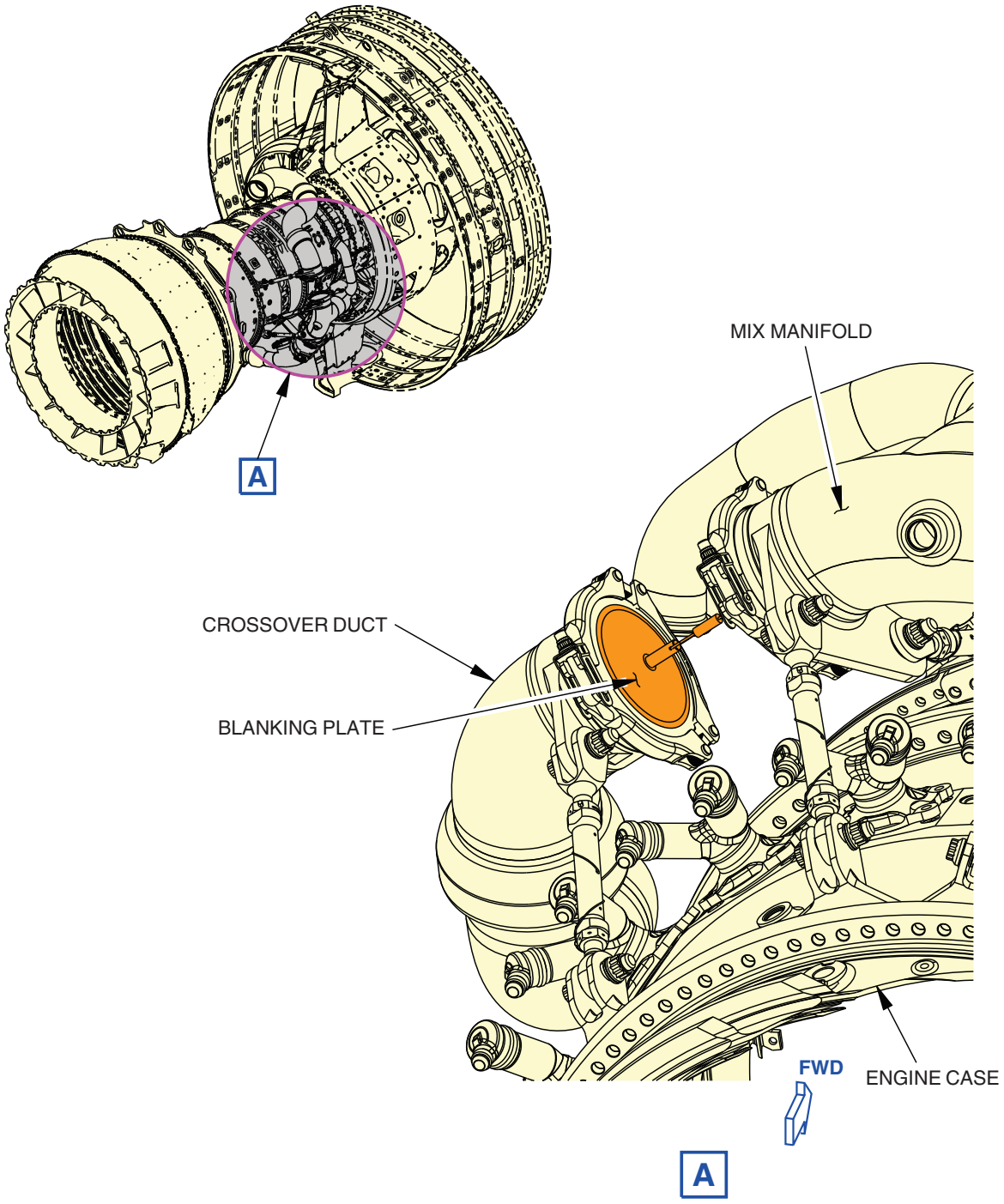

Blanking Plate Installation
Figure 202/71-00-03-990-821-G00

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TASK 71-00-03-620-806-G00

6. Preservation of an Engine on Wing for not more than 3 Months

(Figure 203, Figure 202)

A. General

(1) This task gives the instructions on how to preserve the engine on wing for not more than 3 months.

(2) There are two procedures available for this task:

(a) The first procedure is to perform preservation after the last engine operation (or flight).

(b) The second procedure avoids the idle engine run after end of operations, by servicing the oil tank with preservation mixture before last scheduled flights and not more than one day before end of operations. The last flight or engine run will replace the idle run.

B. References

Reference	Title
12-13-11-170-801	Oil Tank Flushing (P/B 301)
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
36-11-04-000-801	PRSOV Removal (P/B 401)
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
80-11-01-610-802-G00	Engine Starter Servicing (Oil Fill) (P/B 301)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1517	Cover - Engine Exhaust Part #: 896812 Supplier: SBK11 Part #: AVTB73MLEA1BEPS Supplier: 018PR Part #: C10006-1 Supplier: 81205
SPL-14650	Cover, Engine Inlet - LEAP 1B Part #: 892812 Supplier: SBK11 Part #: AVTB73MLEA1B Supplier: 018PR Part #: B737-104 Supplier: 1LE67 Part #: B737-153 Supplier: 1LE67 Part #: C10005-23 Supplier: 81205 Opt Part #: C10005-1 Supplier: 81205
SPL-18258	Blanking Plate - PRSOV Part #: C36007-11 Supplier: 81205 Opt Part #: C36007-1 Supplier: 81205
STD-3906	Mallet - Rubber

D. Consumable Materials

Reference	Description	Specification
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
D00599 [CP2442]	Oil - Engine	

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(Continued)

Reference	Description	Specification
D00662 [CP5075]	Additive - Corrosion Preventive, Brayco 599	
D00664 [CP5067]	Oil - Corrosion Preventive, Lubricative System	MIL-PRF-6085
G00252	Film - Polyethylene Film And Sheeting	ASTM D2103 (Supersedes L-P-512)
G00253	Material - Barrier Materials, Greaseproofed, Waterproof, Flexible, Heat-Sealable	MIL-PRF-121 (Supersedes MIL-B-121)
G00834	Cloth - Lint-free Cotton	
G00920	Tape - Waterproof, Packaging	ASTM D5486
G02455 [CP2160]	Film, Vapor Barrier, Polyethylene	L-P-512
G02456 [CP2098]	Desiccant - Activated, Bagged, Packaging Use And Static Dehumidification	MIL-D-3464
G51486 [CP2109]	Tape - Plastic, Colored - 3M No. 471	
G51738 [CP2158]	Tape - Masking, Cloth Adhesive - tesa 4651	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Prepare for the Preservation

SUBTASK 71-00-03-010-001-G00

(1) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

H. Preservation of an Engine on Wing for not more than 3 Months (At the end of engine operation)

SUBTASK 71-00-03-680-001-G00

WARNING DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING	DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.
WARNING	DO NOT LET THE OIL STAY ON YOUR SKIN. USE THE OIL IN AN AREA WITH GOOD VENTILATION. THE OIL IS POISONOUS AND CAN BE ABSORBED THROUGH YOUR SKIN. THE OIL FUMES CAN IRRITATE YOUR RESPIRATORY TRACT.
CAUTION	DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.
CAUTION	DO NOT SPRAY PRESERVATIVE OIL OR EQUIVALENT INTO THE ENGINE INLET, CORE COMPRESSOR, TURBINE, OR ENGINE EXHAUST. UNWANTED PARTICLES THAT STAY ON THE BLADES AND VANES WET WITH OIL CAN CAUSE DAMAGE TO THE ENGINE.

(1) Prepare the oil tank [1] as follows:

(a) Drain the oil tank [1] (TASK 12-13-11-170-801).

(b) Service the oil tank [1] with a minimum of 5 gal (19 l) mixture of engine oil, D00599 [CP2442], and lubrication system corrosion preventive oil, D00664 [CP5067], as follows (TASK 12-13-11-600-801):

1) Mix the engine oil, D00599 [CP2442], with 5% by volume of lubrication system corrosion preventive oil, D00664 [CP5067].

2) As an alternative, mix the engine oil, D00599 [CP2442], with 7% by volume of corrosion preventive additive, D00662 [CP5075].

(c) If it is necessary, fill the oil tank [1] with engine oil, D00599 [CP2442], after adding the lubrication system corrosion preventive oil, D00664 [CP5067], or corrosion preventive additive, D00662 [CP5075].

SUBTASK 71-00-03-600-004-G00

(2) As an alternative, prepare the oil tank [1] as follows:

(a) Put 1 qt (0.95 l) of lubrication system corrosion preventive oil, D00664 [CP5067], directly into the oil tank [1].

1) If it is necessary, partially drain the oil tank [1] before adding the lubrication system corrosion preventive oil, D00664 [CP5067], to prevent over servicing.

(b) As an alternative, put 1.5 qt (1.4 l) of corrosion preventive additive, D00662 [CP5075], directly into the oil tank.

1) If it is necessary, partially drain the oil tank [1] before adding the corrosion preventive additive, D00662 [CP5075], to prevent over servicing.

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SUBTASK 71-00-03-860-005-G00

(3) Make sure that the starter is serviced with oil before engine run to make sure there is initial start lubrication.

NOTE: The starter can be serviced with oil either before it is installed on the gearbox or after installation.

NOTE: The engine starter must have oil in order to be operated. Initial servicing of the engine oil system does not provide enough oil to the starter for the starter to be operated.

(a) If you are not sure of oil quantity in starter, fill the starter with oil (TASK 80-11-01-610-802-G00).

SUBTASK 71-00-03-700-002-G00

(4) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.

NOTE: Keep the engine operational at stable ground idle for 15-20 minutes after the test completes.

SUBTASK 71-00-03-720-001-G00

(5) Dry motor the engine for 2 minutes, in order to reduce risk of fuel nozzle coking (TASK 71-00-00-910-808-G00).

SUBTASK 71-00-03-020-003-G00

(6) If the Auxiliary Power Unit (APU) or external air will be used to operate the Environmental Control System (ECS) during the time that the engine will be preserved on wing, it is recommended to remove the Pressure Regulating and Shutoff Valve (PRSOV) and store it off the airplane do the steps that follow:

(a) Remove the PRSOV, do this task: PRSOV Removal, TASK 36-11-04-000-801.

(b) Use an E-seal and a coupling with a blanking plate, SPL-18258, and install on the crossover duct (Figure 202).

1) When the coupling are in place, tighten the coupling.

a) Tighten the coupling nut to 175 ±5 in-lb (19.8 ±0.6 N·m).

b) Lightly tap the outer surface of the coupling with a rubber mallet, STD-3906.

c) Tighten the coupling nut again to 175 ±5 in-lb (19.8 ±0.6 N·m).

(c) On the mix duct, cover the opening with a protective cover or polyethylene film, G00252, and tape, G00920.

SUBTASK 71-00-03-620-019-G00

CAUTION

DO NOT USE CABLE TIES TO ATTACH THE DESICCANT BAGS. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION MAKE SURE THAT THERE IS NO CONTACT BETWEEN DESICCANT BAGS AND LOW PRESSURE TURBINE (LPT) BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(7) In case of engine storage in an environment with a humidity level that is higher than 70%, put in place desiccant bags, G02456 [CP2098], and humidity indicator card (locally manufactured) in the inlet and the exhaust in order to monitor the humidity level (TASK 71-00-03-620-801-G00).

NOTE: The best condition for engine storage is to keep an environment with less than 70% of humidity.

(a) Install desiccant bags, G02456 [CP2098], on Outlet Guide Vane (OGV)/fan frame struts and Turbine Rear Vane (TRV) struts.

1) Use tape to secure desiccant bags, G02456 [CP2098] (with humidity indicator), to prevent movement of desiccant bags, G02456 [CP2098].

(b) Protect engine parts with vapor barrier film, G02455 [CP2160], to make sure that there is no contact between engine parts and desiccant bags, G02456 [CP2098], nor between engine parts and tape.

(c) Make sure that desiccant bags, G02456 [CP2098], are not damaged.

(d) Attach humidity indicator card (locally manufactured) to monitor humidity level during the preservation period.

(e) Make sure that desiccant bags, G02456 [CP2098], and humidity indicators are easily accessible.

SUBTASK 71-00-03-620-020-G00

CAUTION DO NOT USE WHITE, GREY, YELLOW, RED OR GREEN TESA 4651 TAPE (CP2158) OR RED 3M 471 TAPE (CP2109). IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(8) Cover Variable Bleed Valve (VBV) grids with vapor barrier film, G02455 [CP2160], or blue 3M 471 tape, G51486 [CP2109], or blue, brown, or black tesa 4651 tape, G51738 [CP2158].

SUBTASK 71-00-03-480-003-G00

(9) Put the cover, SPL-14650, on the entrance of the fan cowling.

(a) If the engine inlet cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

1) When using plastic sheeting to cover the inlet, do the steps that follow:

a) Use a lint-free cloth, G00834, and solvent, B00666, to wipe the area where the tape will be used.

b) Cut approximately 70 in. (1.8 m) diameter circle from plastic sheet.

c) Tape the plastic circle on FWD edge of the acoustic panel behind the lipskin.

<1> Apply with at least 3 layers of overlapping tape around the perimeter of the inlet acoustic panel.

<2> Do not place plastic or tape over the lipskin or encase the lipskin with plastic.

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d) Apply the tape across the diameter of the circular sheet to reduce the pillowing of the plastic sheeting.

SUBTASK 71-00-03-480-004-G00

(10) Put the exhaust cover, SPL-1517, on the engine exhaust.

(a) If the engine exhaust cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

1) Make sure that the drain holes or tubes are open and not covered, when applying plastic sheeting and tape to the exhaust.

SUBTASK 71-00-03-160-003-G00

(11) Remove all tools and clean the work area.

I. Alternate Procedure for Preservation of an Engine on Wing for not more than 3 Months (One day before the end of operations)

NOTE: Use this alternate procedure on the operable power plant before last scheduled flight(s) and not more than one day before end of operations.

SUBTASK 71-00-03-680-002-G00

WARNING	DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.
WARNING	DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.
WARNING	DO NOT LET THE OIL STAY ON YOUR SKIN. USE THE OIL IN AN AREA WITH GOOD VENTILATION. THE OIL IS POISONOUS AND CAN BE ABSORBED THROUGH YOUR SKIN. THE OIL FUMES CAN IRRITATE YOUR RESPIRATORY TRACT.
CAUTION	DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.
CAUTION	DO NOT SPRAY PRESERVATIVE OIL OR EQUIVALENT INTO THE ENGINE INLET, CORE COMPRESSOR, TURBINE, OR ENGINE EXHAUST. UNWANTED PARTICLES THAT STAY ON THE BLADES AND VANES WET WITH OIL CAN CAUSE DAMAGE TO THE ENGINE.
CAUTION	DO NOT USE THE CORROSION PREVENTIVE ADDITIVE CP5075 FOR THIS PROCEDURE. DAMAGE TO THE ENGINE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION DO NOT USE PRESERVATIVE ADDITIVE CP5067 DURING ENGINE OPERATIONS FOR MORE THAN ONE DAY. DAMAGE TO THE ENGINE CAN OCCUR.

(1) Prepare the oil tank [1] as follows:

(a) Drain the oil tank [1] (TASK 12-13-11-170-801).

(b) Service the oil tank [1] with a mixture of engine oil, D00599 [CP2442], and lubrication system corrosion preventive oil, D00664 [CP5067], as follows (TASK 12-13-11-600-801):

CAUTION DO NOT USE THE CORROSION PREVENTIVE ADDITIVE CP5075 FOR THIS PROCEDURE. DAMAGE TO THE ENGINE CAN OCCUR.

1) Mix the engine oil, D00599 [CP2442], with 5% by volume of lubrication system corrosion preventive oil, D00664 [CP5067].

a) Do not use corrosion preventive additive, D00662 [CP5075].

(c) If it is necessary, fill the oil tank [1] with engine oil, D00599 [CP2442], after adding the lubrication system corrosion preventive oil, D00664 [CP5067].

SUBTASK 71-00-03-600-003-G00

(2) As an alternative, prepare the oil tank [1] as follows:

CAUTION DO NOT USE THE CORROSION PREVENTIVE ADDITIVE CP5075 FOR THIS PROCEDURE. DAMAGE TO THE ENGINE CAN OCCUR.

(a) Put 1 qt (0.95 l) of lubrication system corrosion preventive oil, D00664 [CP5067], directly into the oil tank [1].

1) Do not use corrosion preventive additive, D00662 [CP5075].

2) If it is necessary, partially drain the oil tank [1] before adding the lubrication system corrosion preventive oil, D00664 [CP5067], to prevent over servicing.

SUBTASK 71-00-03-860-006-G00

(3) Make sure that the starter is serviced with oil before engine run to make sure there is initial start lubrication.

NOTE: The starter can be serviced with oil either before it is installed on the gearbox or after installation.

NOTE: The engine starter must have oil in order to be operated. Initial servicing of the engine oil system does not provide enough oil to the starter for the starter to be operated.

(a) If you are not sure of oil quantity in starter, fill the starter with oil (TASK 80-11-01-610-802-G00).

SUBTASK 71-00-03-020-004-G00

(4) If the APU or external air will be used to operate the ECS during the time that the engine will be preserved on wing, it is recommended to remove the PRSOV and store it off of the airplane. Do the steps that follow:

(a) Remove the PRSOV, do this task: PRSOV Removal, TASK 36-11-04-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Use an E-seal and a coupling with a blanking plate, SPL-18258, and install on the crossover duct (Figure 202).

1) When the coupling are in place, tighten the coupling.

a) Tighten the coupling nut to 175 ±5 in-lb (19.8 ±0.6 N·m).

b) Lightly tap the outer surface of the coupling with a rubber mallet, STD-3906.

c) Tighten the coupling nut again to 175 ±5 in-lb (19.8 ±0.6 N·m).

(c) On the mix duct, cover the opening with a protective cover or polyethylene film, G00252, and tape, G00920.

SUBTASK 71-00-03-160-004-G00

(5) Remove all tools and clean the work area.

SUBTASK 71-00-03-620-017-G00

CAUTION

DO NOT USE CABLE TIES TO ATTACH THE DESICCANT BAGS. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

CAUTION

MAKE SURE THAT THERE IS NO CONTACT BETWEEN DESICCANT BAGS AND LOW PRESSURE TURBINE (LPT) BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(6) In case of engine storage in an environment with a humidity level that is higher than 70%, put in place desiccant bags, G02456 [CP2098], and humidity indicator card (locally manufactured) in the inlet and the exhaust in order to monitor the humidity level (TASK 71-00-03-620-801-G00).

NOTE: The best condition for engine storage is to keep an environment with less than 70% of humidity.

(a) Install desiccant bags, G02456 [CP2098], on OGV/fan frame struts and TRV struts.

1) Use tape to secure desiccant bags, G02456 [CP2098] (with humidity indicator), to prevent movement of desiccant bags, G02456 [CP2098].

(b) Protect engine parts with vapor barrier film, G02455 [CP2160], to make sure that there is no contact between engine parts and desiccant bags, G02456 [CP2098], nor between engine parts and tape.

(c) Make sure that desiccant bags, G02456 [CP2098], are not damaged.

(d) Attach humidity indicator card (locally manufactured) to monitor humidity level during the preservation period.

(e) Make sure that desiccant bags, G02456 [CP2098], and humidity indicators are easily accessible.

SUBTASK 71-00-03-620-018-G00

CAUTION

DO NOT USE WHITE, GREY, YELLOW, RED OR GREEN TESA 4651 TAPE (CP2158) OR RED 3M 471 TAPE (CP2109). IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(7) Cover VBV grids with vapor barrier film, G02455 [CP2160], or blue 3M 471 tape, G51486 [CP2109], or blue, brown, or black tesa 4651 tape, G51738 [CP2158].

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SUBTASK 71-00-03-480-008-G00

(8) After the last flight, put the covers on the engine.

(a) Put the cover, SPL-14650, on the entrance of the fan cowling.

1) If the engine inlet cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

a) When using plastic sheeting to cover the inlet, do the steps that follow:

<1> Use a lint-free cloth, G00834, and solvent, B00666, to wipe the area where the tape will be used.

<2> Cut approximately 70 in. (1.8 m) diameter circle from plastic sheet.

<3> Tape the plastic circle on FWD edge of the acoustic panel behind the lipskin.

<a> Apply with at least 3 layers of overlapping tape around the perimeter of the inlet acoustic panel.

<b> Do not place plastic or tape over the lipskin or encase the lipskin with plastic.

<4> Apply the tape across the diameter of the circular sheet to reduce the pillowing of the plastic sheeting.

(b) Put the exhaust cover, SPL-1517 on the engine exhaust.

1) If the engine exhaust cover is not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

a) Make sure that the drain holes or tubes are open and not covered, when applying plastic sheeting and tape to the exhaust.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 71-00-03-410-001-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

END OF TASK

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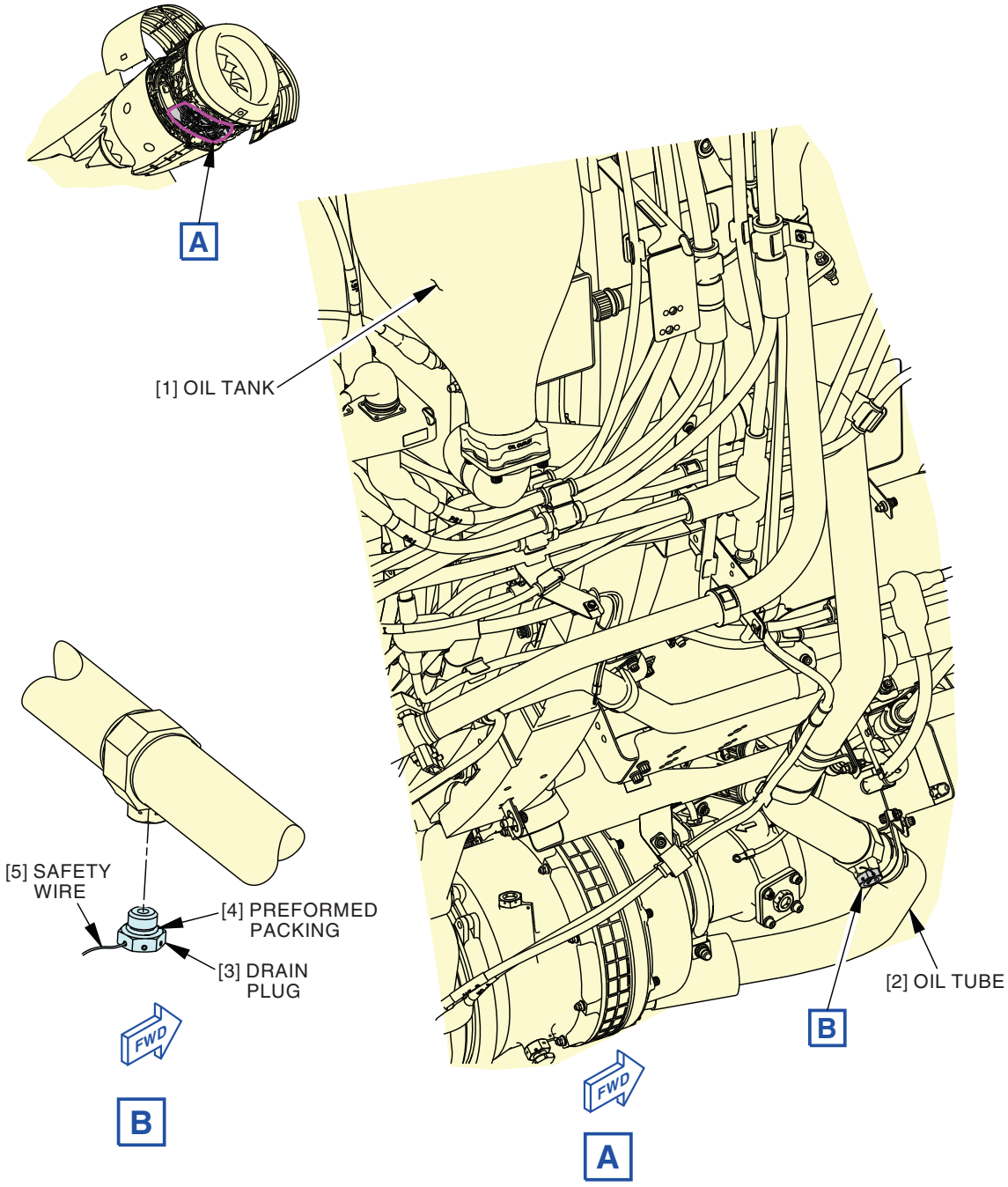

Preservation Of An Engine On Wing For Not More Than Three Months
Figure 203/71-00-03-990-818-G00

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TASK 71-00-03-630-801-G00

7. Preservation of an Engine on Wing for 30 Days to 1 Year

(Figure 202)

A. General

(1) This task gives the instructions to preserve the engine for 30 days to one year while the engine is installed on wing.

(2) This task gives the instructions for an operable engine with “Menu Mode” procedure in the flight deck.

B. References

Reference	Title
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)
36-11-04-000-801	PRSOV Removal (P/B 401)
71-00-00-700-801-G00	Test No. 25 - Fuel System Preservation Special Functions Test (P/B 501)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-01-000-801-G00	Main Fuel Pump Removal (P/B 401)
73-11-07-400-801-G00	Main Fuel Supply Hose Installation (P/B 401)
75-23-02-000-801-G00	Left Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)
75-23-02-000-802-G00	Right Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)
75-23-02-980-801-G00	Left or Right Modulated Turbine Cooling (MTC) Valve Manual Operation (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
80-11-01-610-802-G00	Engine Starter Servicing (Oil Fill) (P/B 301)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1517	Cover - Engine Exhaust Part #: 896812 Supplier: SBK11 Part #: AVTB73MLEA1BEPS Supplier: 018PR Part #: C10006-1 Supplier: 81205
SPL-14650	Cover, Engine Inlet - LEAP 1B Part #: 892812 Supplier: SBK11 Part #: AVTB73MLEA1B Supplier: 018PR Part #: B737-104 Supplier: 1LE67 Part #: B737-153 Supplier: 1LE67 Part #: C10005-23 Supplier: 81205 Opt Part #: C10005-1 Supplier: 81205

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(Continued)

Reference	Description
SPL-14906	Kit - Engine Preservation (On or Off-Wing) Part #: 956A1602P01 Supplier: 58828
SPL-18258	Blanking Plate - PRSOV Part #: C36007-11 Supplier: 81205 Opt Part #: C36007-1 Supplier: 81205
SPL-21586	Kit - Engine Preservation (On Wing) Part #: 956A1602P02 Supplier: 58828
STD-3906	Mallet - Rubber
STD-6638	Air Source - Compressed, Clean, Filtered, and Dry, 0 - 100 psi
STD-6646	Container - Fuel Resistant

D. Consumable Materials

Reference	Description	Specification
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
D00599 [CP2442]	Oil - Engine	
D00623 [CP5066]	Oil - Fuel System, Corrosion Preventive	MIL-PRF-6081, Grade 1010
D00662 [CP5075]	Additive - Corrosion Preventive, Brayco 599	
D00664 [CP5067]	Oil - Corrosion Preventive, Lubricative System	MIL-PRF-6085
G00252	Film - Polyethylene Film And Sheeting	ASTM D2103 (Supersedes L-P-512)
G00253	Material - Barrier Materials, Greaseproofed, Waterproof, Flexible, Heat-Sealable	MIL-PRF-121 (Supersedes MIL-B-121)
G00834	Cloth - Lint-free Cotton	
G00920	Tape - Waterproof, Packaging	ASTM D5486
G02455 [CP2160]	Film, Vapor Barrier, Polyethylene	L-P-512
G02456 [CP2098]	Desiccant - Activated, Bagged, Packaging Use And Static Dehumidification	MIL-D-3464
G51486 [CP2109]	Tape - Plastic, Colored - 3M No. 471	
G51738 [CP2158]	Tape - Masking, Cloth Adhesive - tesa 4651	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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G. Prepare for the Preservation

SUBTASK 71-00-03-840-001-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Preservation of an Engine on Wing for 30 Days to 1 Year

SUBTASK 71-00-03-600-001-G00

(1) Preservation of operable engines with “Menu Mode” in the flight deck.

NOTE: If the engine was shutdown in flight, dry out and re-lubricate the engine within 24 hours.

WARNING DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING	DO NOT LET THE OIL STAY ON YOUR SKIN. USE THE OIL IN AN AREA WITH GOOD VENTILATION. THE OIL IS POISONOUS AND CAN BE ABSORBED THROUGH YOUR SKIN. THE OIL FUMES CAN IRRITATE YOUR RESPIRATORY TRACT.

WARNING	DO NOT USE A SILICONE-BASED OIL WHEN YOU CONNECT THE OIL CART TO FLUSH THE OIL. KEEP CONTAMINATION OUT OF THE FUEL SYSTEM WHEN YOU CONNECT THE CART. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO THE ENGINE COMPONENTS.

CAUTION	DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION	DO NOT SPRAY PRESERVATIVE OIL OR EQUIVALENT INTO THE ENGINE INLET, CORE COMPRESSOR, TURBINE, OR ENGINE EXHAUST. UNWANTED PARTICLES THAT STAY ON THE BLADES AND VANES WET WITH OIL CAN CAUSE DAMAGE TO THE ENGINE.

(a) Preserve the engine oil system:

1) Drain the oil tank.

NOTE: Refer to Engine Oil Change Servicing, TASK 12-13-11-600-803.

2) Replenish the oil tank up to the -3QT mark on the oil tank sight glass with a mixture of

a) Engine oil, D00599 [CP2442], and 7% (by volume) corrosion preventative additive, D00662 [CP5075] or

b) Engine oil, D00599 [CP2442], and 5% (by volume) lubrication system corrosion preventative oil, D00664 [CP5067].

3) Make sure that the starter is serviced with oil before engine run to make sure there is initial start lubrication.

NOTE: The starter can be serviced with oil either before it is installed on the gearbox or after installation.

NOTE: The engine starter must have oil in order to be operated. Initial servicing of the engine oil system does not provide enough oil to the starter for the starter to be operated.

a) If you are not sure of oil quantity in starter, fill the starter with oil (TASK 80-11-01-610-802-G00).

(b) Preserve the engine fuel system:

1) Make sure to do this check of the circuit breakers before you do the wet motor procedure to prevent fuel leakage from the spar valve.

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a) For engine 1, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1

b) For engine 2 open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2

2) Make sure that the position of the circuit breaker is not changed until the fuel system preservation is complete or fuel leakage from spar valve could occur.

3) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

4) Drain the main fuel pump.

a) Refer to Main Fuel Pump Removal, TASK 73-11-01-000-801-G00 for the steps to drain the fuel pump.

5) Disconnect the main fuel supply hose from the main fuel pump.

a) Refer to Main Fuel Pump Removal, TASK 73-11-01-000-801-G00 for the steps to disconnect the main fuel supply hose from the main fuel pump.

b) Retain the gasket.

6) Connect the preservation kit, SPL-21586, or preservation kit, SPL-14906:

a) Fill the reservoir with oil, D00623 [CP5066].

b) Connect the reservoir to the main fuel pump using the adapter.

<1> Tighten the four bolts to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

7) Put the main fuel pump hose cap on the main fuel supply hose using the retained gasket.

8) Pressurize the preservation kit, SPL-21586, or preservation kit, SPL-14906, reservoir:

a) Close the main supply valve on the reservoir.

b) Do one of the steps that follow:

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<1> Raise the tank higher than 3.0 ft (0.9 m) above the fuel pump inlet, or

<2> Pressurize the tank between 20 psi (138 kPa) and 25 psi (172 kPa) using a clean, dry, and regulated air source, STD-6638.

9) Put a fuel resistant container, STD-6646, below the engine drain to collect the fuel.

10) Open the main supply valve on the preservation kit, SPL-21586, or preservation kit, SPL-14906, reservoir.

11) Open the supply valve of the main fuel pump adapter.

12) Do this task: Test No. 25 - Fuel System Preservation Special Functions Test, TASK 71-00-00-700-801-G00.

NOTE: During this step, engine will be motored and fuel system preservation fluid mist will appear at the engine exhaust.

13) Remove the remaining fuel and oil from the flow path:

a) Let the fuel drain for 15 minutes or until no fluid drips.

b) Make sure that there is a minimum of 3.7 gal (14.0 l) remains in the tank.

c) Make sure that the main supply valves on the preservation kit, SPL-21586, or preservation kit, SPL-14906, reservoir are open.

CAUTION

DO NOT APPLY BOOST PUMP PRESSURE TO THE FUEL PUMP INLET WHEN YOU DRY-MOTOR THE ENGINE WITH THE DISCONNECTED FUEL LINE. FUEL WILL SPILL ON THE ENGINE AND CAUSE DAMAGE TO THE EQUIPMENT.

d) Dry motor the engine for two minutes to remove remaining fuel and oil from the flow path (TASK 71-00-00-910-808-G00).

14) Close the main supply valve of the main fuel pump adapter.

15) Rotate the fan rotor through a minimum of two complete turns if it did not turn during the dry motor.

16) Depressurize the preservation kit, SPL-21586, or preservation kit, SPL-14906, and drain all residual fluid.

17) Remove the main fuel pump adapter from the fuel pump inlet.

18) Reconnect the aircraft main fuel supply hose to the main fuel pump inlet port (TASK 73-11-07-400-801-G00).

(c) Install vapor barrier film, G02455 [CP2160], on all open pipes and struts.

CAUTION

DO NOT USE WHITE, GREY, YELLOW, RED OR GREEN TESA 4651 TAPE (CP2158) OR RED 3M 471 TAPE (CP2109). IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(d) Cover Variable Bleed Valve (VBV) grids with vapor barrier film, G02455 [CP2160], or blue 3M 471 tape, G51486 [CP2109], or blue, brown, or black tesa 4651 tape, G51738 [CP2158].

1) Make sure that no unwanted material goes into the core engine inlet through the VBV.

(e) Install protective covers or caps on all open ports, tubes, hoses, and connectors.

NOTE: Drain holes or tubes remain open and should not be covered.

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SUBTASK 71-00-03-020-002-G00

(2) If the Auxiliary Power Unit (APU) or external air will be used to operate the Environmental Control System (ECS) during the time that the engine will be preserved on wing, it is recommended to remove the Pressure Regulating and Shutoff Valve (PRSOV) and store it off of the airplane. Do the steps that follow:

(a) Remove the PRSOV, do this task: PRSOV Removal, TASK 36-11-04-000-801.

(b) Use an E-seal and a coupling with a blanking plate, SPL-18258, and install on the crossover duct (Figure 202).

1) When the coupling are in place, tighten the coupling.

a) Tighten the coupling nut to 175 ±5 in-lb (19.8 ±0.6 N·m).

b) Lightly tap the outer surface of the coupling with a rubber mallet, STD-3906.

c) Tighten the coupling nut again to 175 ±5 in-lb (19.8 ±0.6 N·m).

(c) On the mix duct, cover the opening with a protective cover or polyethylene film, G00252, and tape, G00920.

SUBTASK 71-00-03-550-001-G00

CAUTION

THE MTC SYSTEM OPERATES AT HIGH-POWER OPERATION DURING FLIGHT. IF INACTIVE, DAMAGE TO THE MTC VALVE CAN OCCUR.

(3) If the engine is stored outdoors during the time the engine will be preserved on wing, it is recommended to protect the Modulated Turbine Cooling (MTC) valve by one of the two actions as follow:

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124

NOTE: The left side MTC valve and actuator (aft looking forward) are removed.

PTF ALL

(a) Do these tasks: Left Modulated Turbine Cooling (MTC) Valve Removal, TASK 75-23-02-000-801-G00, and Right Modulated Turbine Cooling (MTC) Valve Removal, TASK 75-23-02-000-802-G00.

1) Store the MTC valve.

(b) Actuate the MTC valve 5 to 20 times every 5 month, do this task: Left or Right Modulated Turbine Cooling (MTC) Valve Manual Operation, TASK 75-23-02-980-801-G00.

I. Job Close-up

SUBTASK 71-00-03-840-002-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 71-00-03-480-009-G00

CAUTION

DO NOT USE CABLE TIES TO ATTACH THE DESICCANT BAGS. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

CAUTION

MAKE SURE THAT THERE IS NO CONTACT BETWEEN DESICCANT BAGS AND LOW PRESSURE TURBINE (LPT) BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(2) In case of engine storage in an environment with a humidity level that is higher than 70%, put in place desiccant bags, G02456 [CP2098], and humidity indicator card (locally manufactured) in the inlet and the exhaust in order to monitor the humidity level (TASK 71-00-03-620-801-G00).

NOTE: The best condition for engine storage is to keep an environment with less than 70% of humidity.

(a) Install desiccant bags, G02456 [CP2098], on Outlet Guide Vane (OGV)/fan frame struts and Turbine Rear Vane (TRV) struts.

1) Use tape to secure desiccant bags, G02456 [CP2098] (with humidity indicator), to prevent movement of desiccant bags, G02456 [CP2098].

(b) Protect engine parts with vapor barrier film, G02455 [CP2160], to make sure that there is no contact between engine parts and desiccant bags, G02456 [CP2098], nor between engine parts and tape.

(c) Make sure that desiccant bags, G02456 [CP2098], are not damaged.

(d) Attach humidity indicator card (locally manufactured) to monitor humidity level during the preservation period.

NOTE: Humidity indicator cards and desiccant bags must be inspected at regular intervals in order to make sure that they are still active. If indication of important humidity or saturated desiccant bags is noted during inspection, consider reducing inspection interval and increasing quantity of desiccant bags. As an example, start monitoring of the desiccant bags as follow in order to determine the effective interval for desiccant bags replacement. A suggested schedule is as follows:

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<1> Second day after beginning of storage period.

<2> Fourteenth day after beginning of storage period.

<3> Every three months after beginning of storage period.

(e) Make sure that desiccant bags, G02456 [CP2098], and humidity indicators are easily accessible.

SUBTASK 71-00-03-480-010-G00

(3) Cover the inlet and exhaust of the engine.

(a) Seal the inlet and exhaust of the engine with vapor barrier film, G02455 [CP2160].

(b) Install the cover, SPL-14650, on the engine inlet and the exhaust cover, SPL-1517, on the rear of the engine.

1) If the engine inlet and exhaust covers are not available, use barrier material, G00253, polyethylene film, G00252, or equivalent 6 MIL or thicker material, and tape, G00920.

NOTE: The drain holes or tubes should not be capped or covered when applying plastic sheeting and tape to the exhaust.

a) When using plastic sheeting to cover the inlet, do the steps that follow:

<1> Use lint-free cloth, G00834, and solvent, B00666, to wipe the area where the tape will be used.

<2> Cut approximately 70 in. (1.8 m) diameter circle from plastic sheet.

<3> Tape the plastic circle on FWD edge of the acoustic panel behind the lipskin. Apply with at least 3 layers of overlapping tape around the perimeter of the inlet acoustic panel.

NOTE: Do not place plastic or tape over the lipskin or encase the lipskin with plastic.

<4> Apply the tape across the diameter of the circular sheet to reduce the pillowing of the plastic sheeting.

(c) Put a tag on the engine to indicate the engine is preserved.

——— END OF TASK ———

TASK 71-00-03-620-808-G00

8. Preservation of an Engine on Wing That Has Expired

A. General

(1) This task gives the instructions on what to do with an engine on wing with an expired preservation period.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Preservation of an Engine on Wing That Has Expired

SUBTASK 71-00-03-600-002-G00

(1) If the engine preservation period is expired, contact your CFMI representative for instruction.

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(a) Provide the following parameters:

NOTE: An engineering analysis must be used to analyze the situation applicable to each special case.

1) LEAP engine model and Engine Serial Number
2) Engine TSN and CSN
3) Date of last preservation task performed and reference of the task
    a) In case of multiple renewal procedure performed, provide complete history and dates
4) Date of next expected preservation or run
5) Total period of improper preservation
    a) Detail if maximum extended period allowed in Preservation of an Engine on Wing That Has Expired, TASK 71-00-03-620-808-G00, has been reached
6) Confirm that the last preservation task was performed properly, with the following steps done, as applicable:
    a) Oil system preserved
    b) Fuel system preserved
    c) VBVs were closed and/or sealed with a vapor barrier film or alternative grid protection
    d) All open tubes, hoses and electrical connections were capped and plugged
    e) Desiccant bags and humidity indicators were installed and monitoring history
    f) Engine inlet and exhausts were sealed with vapor barrier film
    g) Engine covers were installed (whole engine or inlet and exhaust only)
7) Desiccant bags replacement history and humidity indicators monitoring history
8) Engine storage locations and conditions with the information that follows:
    a) Inside storage or outside storage
    b) Temperature variation and humidity level encountered during out of operation period
    c) Confirm exposure to windmilling if engine stored outside without cover
9) Date when the engine will be available for preservation (the latest possible, in order to know how long the period without preservation renewal will be)
10) In particular case of an engine which had an IFSD, confirmation that dry-out and re-lubrication were performed within 24 hours after IFSD
11) If the engine is going to shop visit.

——— END OF TASK ———

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TASK 71-00-03-620-809-G00

9. Preservation Renewal Requirements

A. General

(1) This task provides the requirements for the preservation renewal.

(2) Before engine operation, refer to Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00 for lipskin coating requirements during engine operation.

NOTE: It may be necessary to apply protective coating to the inlet cowl lipskin after engine operation if a renewal is performed.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Preservation Renewal Requirements

SUBTASK 71-00-03-620-007-G00

(1) Preservation renewal for a period up to 10 days (Engine On-Wing).

(a) Operable engine:

1) Do the preservation for a period up to 10 days (Preservation of an Engine on Wing for not more than 10 Days, TASK 71-00-03-600-801-G00).

a) There is no limit on the number of times you can do this procedure.

2) For engine that has preservation period expired, do this task: Preservation of an Engine on Wing That Has Expired, TASK 71-00-03-620-808-G00.

(b) If applicable, apply protective coating to the inlet cowl lipskin. Do this task: Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00.

(c) Non-operable engine:

1) No renewal is allowed, the engine must be serviced and operable before another preservation procedure is done.

SUBTASK 71-00-03-620-008-G00

(2) Preservation renewal of an engine for a period up to 30 days.

(a) Operable engine:

1) Two renewals are permitted.

2) Do the preservation for a period up to 30 days (Preservation of an Engine on Wing for not more than 30 Days, TASK 71-00-03-620-802-G00).

3) If the engine will not be operated for more than the subsequent 30 days, do one of the tasks that follow:

- Preservation of an Engine on Wing for not more than 3 Months, TASK 71-00-03-620-806-G00
- Preservation of an Engine on Wing for 30 Days to 1 Year, TASK 71-00-03-630-801-G00

4) For engine that has preservation period expired, do this task: Preservation of an Engine on Wing That Has Expired, TASK 71-00-03-620-808-G00.

5) If applicable, apply protective coating to the inlet cowl lipskin. Do this task: Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00.

(b) Non-operable engine:

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1) No renewal is allowed, the engine must be serviced and operable before another preservation procedure is done.

SUBTASK 71-00-03-620-009-G00

(3) Preservation renewal of an engine for a period up to 90 days.

(a) Operable engine:

1) One renewal is permitted.

2) Do the preservation for a period up to 90 days (Preservation of an Engine on Wing for not more than 3 Months, TASK 71-00-03-620-806-G00).

NOTE: it is not necessary to drain and service the oil tank to do the renewal procedure.

3) For engine that has preservation period expired, do this task: Preservation of an Engine on Wing That Has Expired, TASK 71-00-03-620-808-G00.

4) If the engine will not be operated for more than the subsequent three months, the engine must then be preserved per the 30 days to 1 year procedure (Preservation of an Engine on Wing for 30 Days to 1 Year, TASK 71-00-03-630-801-G00).

5) If applicable, apply protective coating to the inlet cowl lipskin, do this task: Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00.

(b) Non-operable engine:

1) No renewal is allowed, the engine must be serviced and operable before another preservation procedure is done.

SUBTASK 71-00-03-620-010-G00

(4) Preservation renewal of an engine for a period of 30 days up to 1 year.

(a) Operable engine:

1) Do the preservation renewal procedure again of the engine for a period of 30 days up to 1 year (Preservation of an Engine on Wing for 30 Days to 1 Year, TASK 71-00-03-630-801-G00).

NOTE: There is no limit to the number of times you can do this procedure.

2) For engine that has preservation period expired, do this task: Preservation of an Engine on Wing That Has Expired, TASK 71-00-03-620-808-G00.

3) If applicable, apply protective coating to the inlet cowl lipskin, do this task: Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00.

(b) Non-operable engine:

1) No renewal is allowed, the engine must be serviced and operable before another preservation procedure is done.

——— END OF TASK ———

TASK 71-00-03-630-802-G00

10. Depreservation of an Engine on Wing

(Figure 202)

A. General

(1) This task gives the instructions on how to return the engine that is on wing to service from preservation.

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B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)
36-11-04-400-801	PRSOV Installation (P/B 401)
71-00-00-700-807-G00	Test No. 31 - Wet Motor Ground Test (P/B 501)
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-00-00-710-802-G00	Test No. 7 - Replacement Engine (Pretested Engine) Test (P/B 501)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)
71-00-00-910-809-G00	Wet Motor Procedure (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
75-23-02-400-801-G00	Left Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
75-23-02-400-802-G00	Right Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
80-11-01-610-802-G00	Engine Starter Servicing (Oil Fill) (P/B 301)

C. Consumable Materials

Reference	Description	Specification
G02456 [CP2098]	Desiccant - Activated, Bagged, Packaging Use And Static Dehumidification	MIL-D-3464

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Depreservation

SUBTASK 71-00-03-840-003-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Depreservation of an Engine on Wing

SUBTASK 71-00-03-630-001-G00

(1) Depreserve the engine that has been preserved on wing for a period of up to 30 days as follows:

(a) Make sure that desiccant bags, G02456 [CP2098] (with humidity indicator), are not damaged.

(b) Remove all moisture barrier material, seals, caps, covers, desiccant bags, as applicable, from the engine.

(c) Make sure that there are no foreign materials.

CAUTION MAKE SURE THAT YOU FOLLOW THE LIPSKIN COATING PROCEDURES BEFORE AN ENGINE OPERATION. IF YOU DO NOT FOLLOW INSTRUCTIONS, THE COATING CAN MELT AND GO INTO THE ENGINE. THIS CAN CAUSE ENGINE DAMAGE.

(d) If not already done, remove the lipskin protective coating from the inlet cowl lipskin (Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00).

SUBTASK 71-00-03-420-002-G00

(2) If the PRSOV was removed and stored off the airplane do the following steps:

(a) Remove the coupling, blanking plate, and E-seal from the crossover duct (Figure 202).

(b) On the mix duct, remove the protective cover.

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(c) Inspect the duct openings for unwanted material.

(d) Install the PRSOV, do this task: PRSOV Installation, TASK 36-11-04-400-801.

SUBTASK 71-00-03-420-003-G00

(3) If the Modulated Turbine Cooling (MTC) valve was removed and stored off the airplane, install the MTC valve:

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124

NOTE: The left side MTC valve and actuator (aft looking forward) are removed.

PTF ALL

(a) Do these tasks: Left Modulated Turbine Cooling (MTC) Valve Installation, TASK 75-23-02-400-801-G00 and Right Modulated Turbine Cooling (MTC) Valve Installation, TASK 75-23-02-400-802-G00.

SUBTASK 71-00-03-630-003-G00

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. USE THE OIL IN AN AREA WITH GOOD VENTILATION. THE OIL IS POISONOUS AND CAN BE ABSORBED THROUGH YOUR SKIN. THE OIL FUMES CAN IRRITATE YOUR RESPIRATORY TRACT.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Depreserve the engine that has been preserved on wing for a period of up to three months as follows:

(a) Make sure that desiccant bags, G02456 [CP2098] (with humidity indicator), are not damaged.

(b) Remove all moisture barrier material, seals, caps, covers and desiccant bags, as applicable, from the engine.

(c) Make sure that there are no foreign materials.

CAUTION MAKE SURE THAT YOU FOLLOW THE LIPSKIN COATING PROCEDURES BEFORE AN ENGINE OPERATION. IF YOU DO NOT FOLLOW INSTRUCTIONS, THE COATING CAN MELT AND GO INTO THE ENGINE. THIS CAN CAUSE ENGINE DAMAGE.

(d) If not already done, remove the lipskin protective coating from the inlet cowl lipskin (Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00).

(e) Connect the fuel supply to the engine.

(f) If applicable, reconnect the oil supply and scavenge lines.

(g) Make sure that you drain the oil tank and the accessory gearbox (TASK 12-13-11-600-803).

(h) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

(i) Service the starter oil (TASK 80-11-01-610-802-G00).

NOTE: The engine starter must have oil to operate. The initial servicing of the engine oil system does not provide enough oil to the starter for operation.

(j) Do this task: Wet Motor Procedure, TASK 71-00-00-910-809-G00.

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1) As an alternative, do this task: Test No. 31 - Wet Motor Ground Test, TASK 71-00-00-700-807-G00.

NOTE: This test has a much higher fuel flow rate than the wet motor task. This will de-preserve the engine in less time.

(k) Examine the oil level (TASK 12-13-11-600-801).

(l) Dry motor the engine to remove the fuel from the combustion chamber (TASK 71-00-00-910-808-G00).

NOTE: If it is necessary, do the dry motor several times.

(m) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.

NOTE: Keep the engine at stable ground idle for at least 3 minutes.

SUBTASK 71-00-03-630-002-G00

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. USE THE OIL IN AN AREA WITH GOOD VENTILATION. THE OIL IS POISONOUS AND CAN BE ABSORBED THROUGH YOUR SKIN. THE OIL FUMES CAN IRRITATE YOUR RESPIRATORY TRACT.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(5) Depreserve the engine that has been preserved on wing for a period from 30 days to not more than one year as follows:

(a) Make sure that desiccant bags, G02456 [CP2098] (with humidity indicator), are not damaged.

(b) Remove all moisture barrier material, seals, caps, covers and desiccant bags, as applicable, from the engine.

(c) Make sure that there are no foreign materials.

CAUTION MAKE SURE THAT YOU FOLLOW THE LIPSKIN COATING PROCEDURES BEFORE AN ENGINE OPERATION. IF YOU DO NOT FOLLOW INSTRUCTIONS, THE COATING CAN MELT AND GO INTO THE ENGINE. THIS CAN CAUSE ENGINE DAMAGE.

(d) If not already done, remove the lipskin protective coating from the inlet cowl lipskin (Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00).

(e) If applicable, connect the fuel supply to the engine.

(f) If applicable, reconnect the oil supply and scavenge lines.

(g) Make sure that you drain the oil tank and the accessory gearbox (TASK 12-13-11-600-803).

(h) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

(i) Service the starter oil (TASK 80-11-01-610-802-G00).

NOTE: The engine starter must have oil to operate. The initial servicing of the engine oil system does not provide enough oil to the starter for operation.

(j) Do this task: Wet Motor Procedure, TASK 71-00-00-910-809-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) As an alternative, do this task: Test No. 31 - Wet Motor Ground Test, TASK 71-00-00-700-807-G00.

NOTE: This test has a much higher fuel flow rate than the wet motor task. This will de-preserve the engine in less time.

2) Do the wet motoring with the fuel sprays in view at the exhaust for at least 30 seconds.

(k) Examine the oil level (TASK 12-13-11-600-801).

(l) Dry motor the engine to remove the fuel from the combustion chamber (TASK 71-00-00-910-808-G00).

NOTE: If it is necessary, do the dry motor several times.

(m) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.

NOTE: Keep the engine at stable ground idle for at least 3 minutes.

SUBTASK 71-00-03-860-004-G00

(6) If the engine has been preserved on wing for a period from 90 days to not more than one year, do this task: Test No. 7 - Replacement Engine (Pretested Engine) Test, TASK 71-00-00-710-802-G00.

NOTE: If the engine that is being tested was preserved on wing do not erase faults.

SUBTASK 71-00-03-160-005-G00

(7) Clean the work area and clear the tools and items used to do the task.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-00-03-840-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

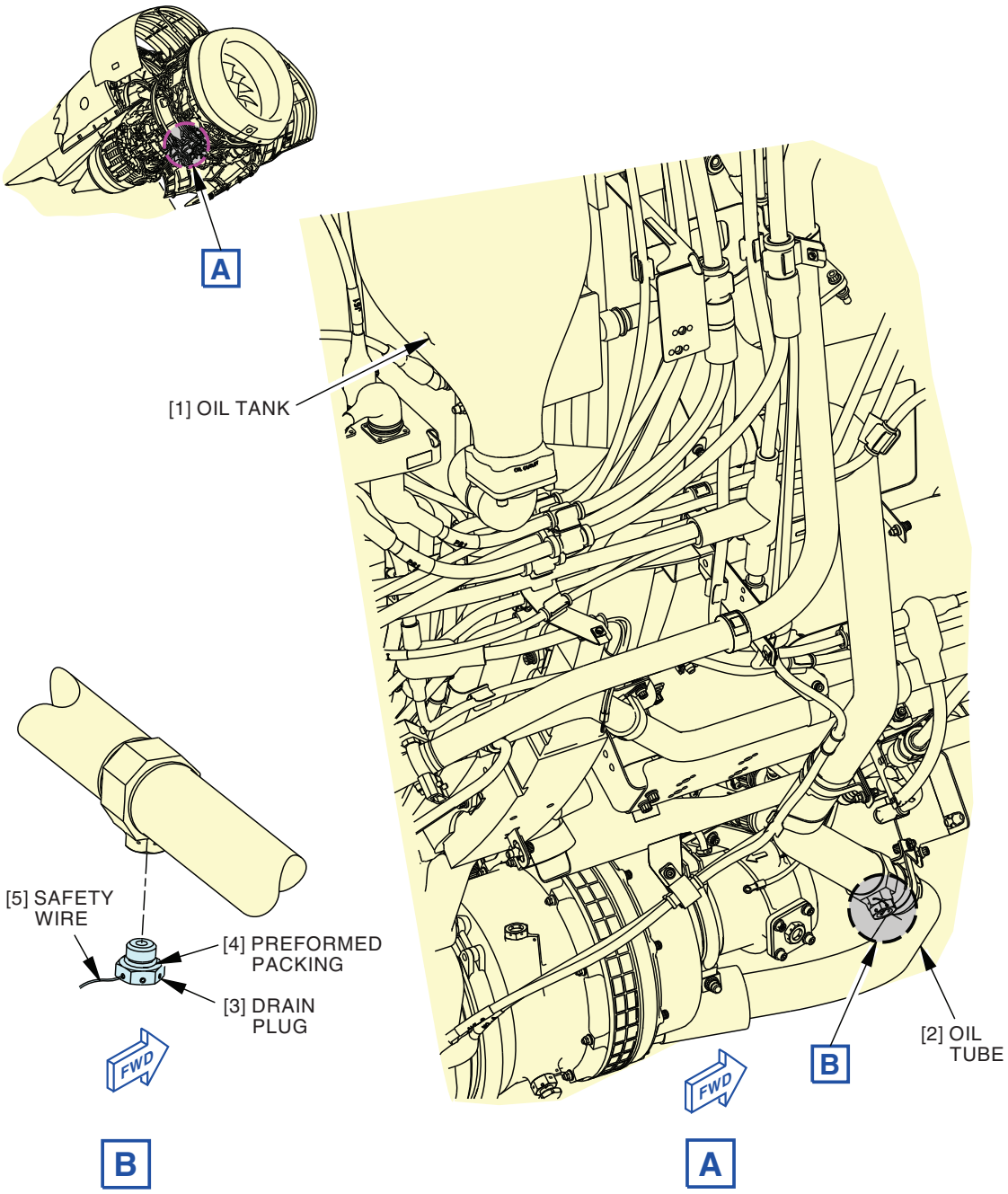
1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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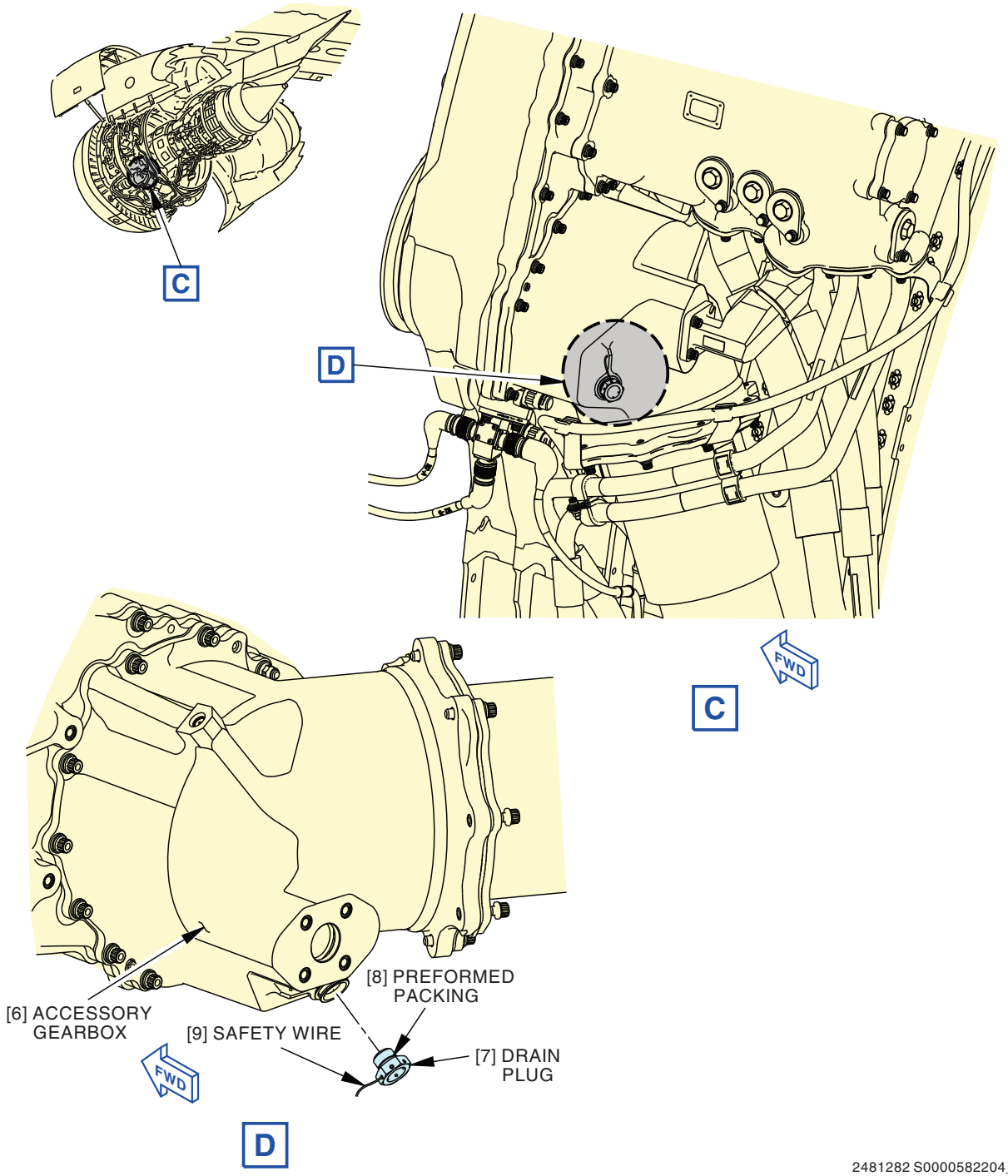
Depreservation of an Engine On Wing
Figure 204/71-00-03-990-819-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Depreservation of an Engine On Wing
Figure 204/71-00-03-990-819-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

POWER PLANT - MAINTENANCE PRACTICES (ENGINE TRANSPORTATION)

1. General

A. This procedure has this task:

(1) The instructions to transport an engine.

TASK 71-00-04-500-801-G01

2. Engine Transportation

(Figure 201, Figure 202, Figure 203, and Figure 204)

A. General

(1) This task gives instructions for the transportation of an engine.

(2) Shipping stands should agree with the CFM specifications. There is a cradle on which the engine is attached and a base on which the cradle is installed through shock absorbers (Figure 201).

(3) Make sure that the shipping stands you use are in good condition and that they could not cause injuries to maintenance personnel or damage to the engines. It is recommended to regularly do an inspection of the engine shipping stands for safe continued operations.

(4) Refer to the documentation supplied by the manufacturer of CFM stands for more instructions.

(5) Make sure that the engine is preserved. If necessary, do the preservation of the engines. Period of up to 30 days or period of 30 to 365 days (TASK 71-00-03-620-802-G00 or TASK 71-00-03-630-801-G00).

(6) An engine can be transported in these configurations:

(a) An engine that has the Quick Engine Change (QEC) equipment items installed and that is ready for aircraft installation.

(b) An engine that has part of the QEC equipment items installed:

1) The QEC engine is without the inlet cowl, exhaust nozzle and plug.

(c) A bare engine without its QEC equipment items installed.

Table 201/71-00-04-993-803-G01 Shipping stand dimensions (base and cradle)

MANUFACTURE CRADLE BASE	HEIGHT WITH ENGINE	WIDTH BARE OR WITH QEC	LENGTH WITH BARE ENGINE	LENGTH WITH Quick Engine Change (QEC)	WEIGHT STAND ONLY	WEIGHT WITH BARE ENGINE	WEIGHT WITH QEC
CFM engine stand, SPL-14726	WITHOUT INLET: 96.4 in. (2449 mm) WITH INLET:101.4 in. (2576 mm)	98.4 in. (2499 mm)	194 in. (4928 mm)	228.3 in. (5799 mm)	5247 lb (2380 kg)	11,352 lb (5149 kg)	12,617 lb (5723 kg)

B. References

Reference	Title
71-00-03-620-802-G00	Preservation of an Engine on Wing for not more than 30 Days (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
71-00-03-630-801-G00	Preservation of an Engine on Wing for 30 Days to 1 Year (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11132	Cover - Engine, Protective (Bare Configuration) Part #: 956A8657G03 Supplier: 58828 Opt Part #: 956A8657G02 Supplier: 58828
SPL-11641	Cover - Engine, Protective (Partial QEC Configuration) Part #: 956A8676G03 Supplier: 58828 Opt Part #: 956A8676G02 Supplier: 58828
SPL-14726	Engine Stand - LEAP-1B Part #: 956A8600G04 Supplier: 58828 Opt Part #: 956A8600G03 Supplier: 58828

D. Engine Transportation Cautions

CAUTION

OBEY ALL THE INSTRUCTIONS AND NECESSARY CONDITIONS GIVEN IN THIS PROCEDURE OTHERWISE ENGINE DAMAGE CAN OCCUR.

CAUTION

FOR ANY DEVIATION FROM THE INSTRUCTIONS GIVEN IN THIS PROCEDURE, CONTACT CFMI FOR ENGINE DISPOSITION. A DEVIATION CAN CAUSE ENGINE DAMAGE.

SUBTASK 71-00-04-760-001-G01

(1) Follow these cautions throughout this procedure, otherwise engine damage can occur.

SUBTASK 71-00-04-860-019-G01

(2) Make sure that the lower cover is installed before the engine on the shipping stand (engine stand, SPL-14726) (Figure 203 and Figure 204).

SUBTASK 71-00-04-860-020-G01

(3) Make sure that the engine drain box is removed (Figure 202).

E. Engine Protection During Shipment

SUBTASK 71-00-04-860-010-G01

CAUTION

INSTALL THE APPLICABLE COVER ON THE ENGINE BEFORE YOU SEND IT. UNWANTED DUST, RAIN OR OBJECTS CAN CAUSE DAMAGE DURING SHIPPING.

(1) Use the engine assy cover [11] (engine protective cover, SPL-11132) or quick engine change storage cover [12] (protective engine cover, SPL-11641) (Figure 203 and Figure 204).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Transportation in the Plant and Access Roads

SUBTASK 71-00-04-860-009-G01

(1) Do these steps to tow the engine in the cradle and base assembly:

(a) Release the four swivel locks of the casters when you tow the engine.

CAUTION THE MAXIMUM SPEED LIMIT TO TOW THE ENGINE, CRADLE, AND BASE ASSEMBLY IS NOT MORE THAN 3 MPH (5 KPH) ON A SMOOTH AND HORIZONTAL SURFACE, OTHERWISE ENGINE DAMAGE CAN OCCUR.

(b) Tow the engine, cradle and base assembly at not more than 3 MPH (5 KPH).

(c) Use spring-type casters or pneumatic tires to tow the engine.

(d) Make sure that the cradle is free of attaching device to be sure that the shock absorbers operate satisfactorily.

(e) Use the caster braking system to prevent movement of the assembly when necessary.

G. Highway Transportation

SUBTASK 71-00-04-210-001-G01

(1) Inform the freight company of engine transportation requirements.

SUBTASK 71-00-04-970-001-G01

CAUTION USE THE "AIR RIDE" (PNEUMATIC) OR THE ALTERNATIVE "HYDRAULIC WITH NITROGEN ACCUMULATORS" (HYDRO-PNEUMATIC) SUSPENSION. THIS YIELDS FILTRATION AT A NATURAL FREQUENCY OF THE ENGINE/STAND SUSPENDED-ASSEMBLY (FROM 7 HZ TO 10 HZ). IF YOU DO NOT FOLLOW THIS CAUTION, IT CAN CAUSE ENGINE DAMAGE AND REPAIRS WILL BE NECESSARY.

CAUTION IF YOU USE ALTERNATIVE “HYDRO-PNEUMATIC” SUSPENSIONS, THE SPEED LIMIT IS 25 MI/H (40 KM/H). IF THE SPEED IS MORE THAN THIS LIMIT, DAMAGE TO THE ENGINE CAN OCCUR.

(2) Requirements for the shipment of the engine on trailer and truck.

(a) A maximum of two engines installed on stand can be transported on a single trailer.

(b) When a single engine is shipped, the trailer must be pneumatically suspended and the engine must be stowed over the rear axle of the trailer.

(c) When two engines are shipped, the tractor and the trailer must be pneumatically suspended.

(d) When engines are shipped on a towed trailer or wagon-bed truck, the axles must be pneumatically suspended.

NOTE: The shipping stands give protection against the shock and vibration loads which could occur during normal transportation.

SUBTASK 71-00-04-210-002-G01

(3) Do a check of the engine/cradle and cradle/base attaching devices.

SUBTASK 71-00-04-410-003-G01

(4) Attach the engine/cradle assembly to the base equipped with shock absorbers.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-00-04-420-007-G01

(5) Install the engine in flight direction in the transportation vehicle.

SUBTASK 71-00-04-860-011-G01

(6) Put the stand base on the truck bed and put the caster wheels and tow bars in "Storage Engine on Stand" position.

SUBTASK 71-00-04-860-012-G01

CAUTION ATTACH ONLY THE BOTTOM OF THE STAND TO THE TRUCK BASE. DO NOT ATTACH THE ENGINE CRADLE ASSEMBLY TO THE TRUCK BED. THIS IS TO BE SURE THAT THE SHOCK ABSORBERS OPERATE SATISFACTORILY.

(7) Use the correct tie-down points to secure the shipping stand base on the truck bed (Figure 201).

(a) Use a minimum of four tie-down straps to attach the base on the trailer.

1) Use six, eight, ten, or twelve tie-down straps when possible.

NOTE: As an alternative, you can use chains through forklift tunnels to attach the base to the truck bed.

CAUTION DO NOT ATTACH THE SHIPPING STAND WITH CHAINS THROUGH FORKLIFT TUNNELS. THIS WILL NOT STOP THE STAND TO MOVE IN THE Y DIRECTION. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

(b) Attach the tie-down straps to the air/truck tie-down rings and pull them along a diagonal across the trailer bed.

1) Attach the straps to the attaching points on the trailer bed.

H. Sea Transportation

SUBTASK 71-00-04-210-003-G01

(1) Inform the freight company of engine transportation requirements.

SUBTASK 71-00-04-860-013-G01

CAUTION IF THE ENGINE IS EXPOSED TO SALTY AIR MORE THAN EIGHT DAYS, INCLUDING LOADING AND UNLOADING FROM SHIP, ENGINE DAMAGE CAN OCCUR.

CAUTION THE ENGINE TO BE MOVED MUST BE INSTALLED ON A TRAILER BED WITH AN ACTIVE SUSPENSION SYSTEM. ENGINE DAMAGE CAN OCCUR IN CASE OF TRANSPORTATION OF AN ENGINE IN A CONTAINER NOT APPROVED.

(2) You can ship the engine on ferry boats in the conditions as follows:

(a) Obey the highway transportation conditions for sea transportation.

(b) The "Air Ride" suspension system of the vehicle must be on (air bottles inflated) during the sea transportation.

(c) Engine exposure to salty air must not be more than eight days, loading and unloading sequences included.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Air Cargo Transportation

SUBTASK 71-00-04-210-004-G01

(1) Inform the freight company of engine transportation requirements.

SUBTASK 71-00-04-860-014-G01

(2) Make sure that caster wheels and tow bars are locked in stow position.

SUBTASK 71-00-04-860-015-G01

(3) Use a forklift truck to install the engine, cradle, and base assembly on a standard air cargo pallet.

SUBTASK 71-00-04-410-004-G01

CAUTION

ATTACH ONLY THE BASE OF THE STAND TO THE AIR CARGO PALLET. DO NOT ATTACH THE ENGINE CRADLE ASSEMBLY TO THE AIR CARGO PALLET. THIS IS TO BE SURE THAT THE SHOCK ABSORBERS OPERATE SATISFACTORILY. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Attach the tie-down straps from the base tie-down points to the pallet fitting points.

SUBTASK 71-00-04-420-008-G01

(5) Tighten and safety all tie-down straps.

J. Rail Transportation

SUBTASK 71-00-04-860-016-G01

(1) It is not recommended that you move an engine by rail.

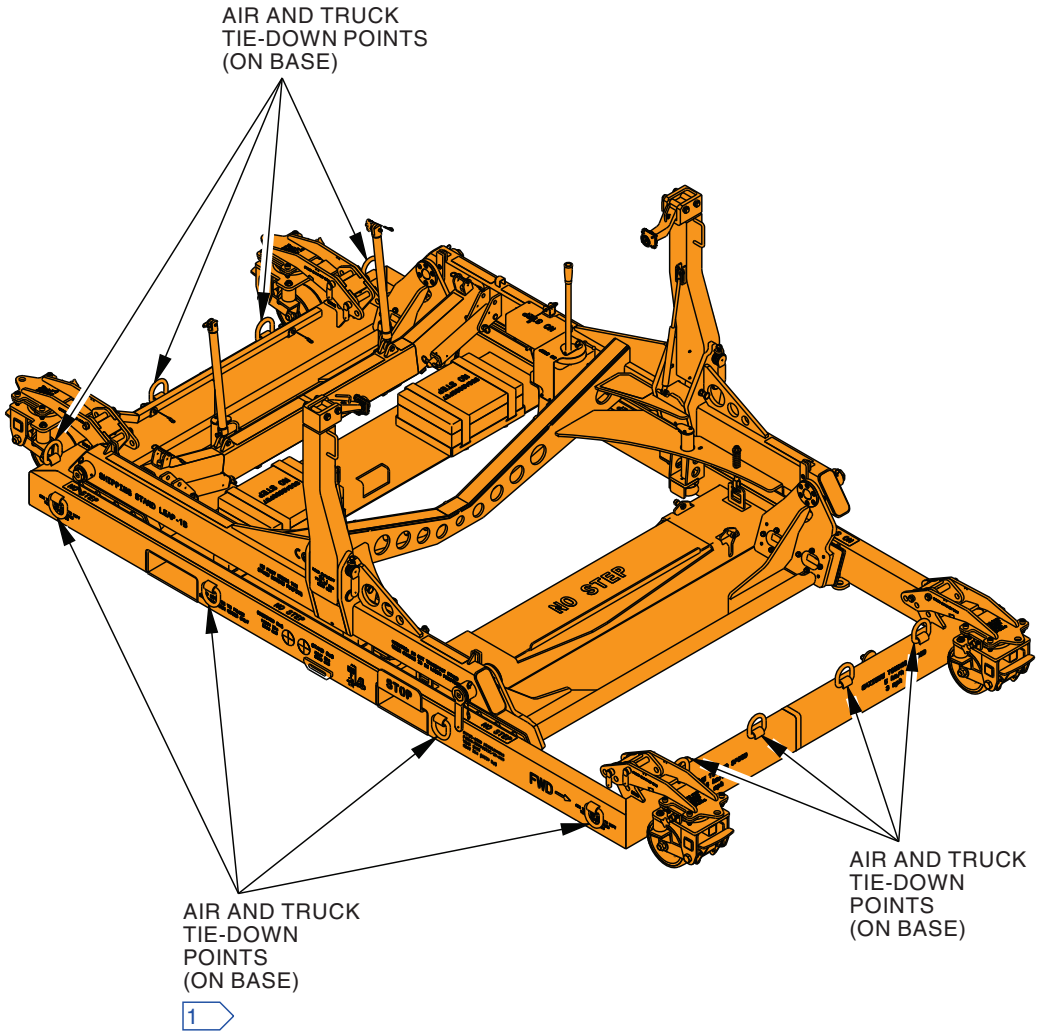
K. Transportation Issue

SUBTASK 71-00-04-860-017-G01

(1) For any transportation issue, contact CFM.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 RIGHT TIE-DOWN POINTS SHOWN, LEFT TIE-DOWN POINTS OPPOSITE.

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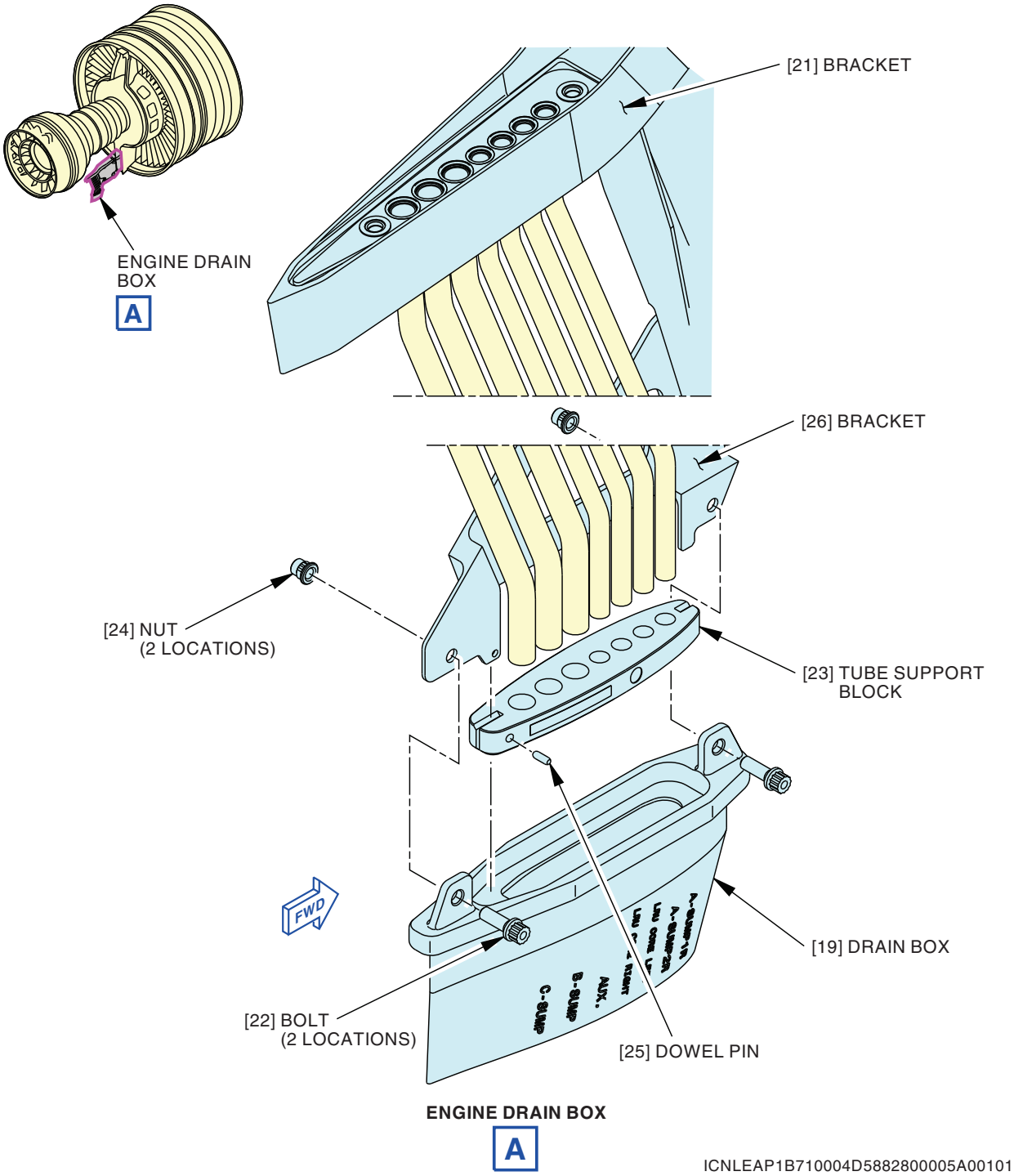
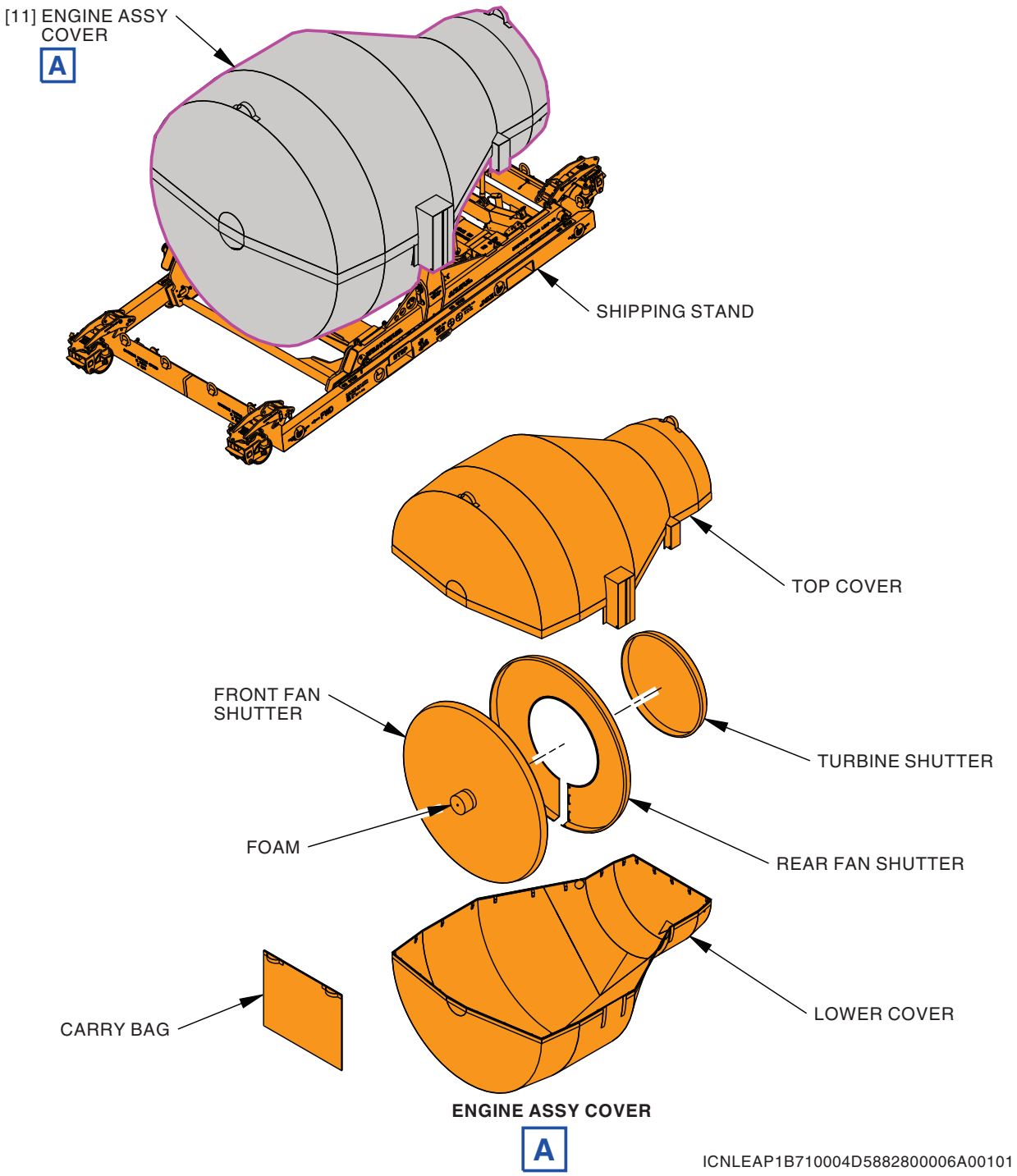
Tie-Down Locations
Figure 201/71-00-04-990-835-G01

ECCN 9E991 BOEING PROPRIETARY - See title page for details



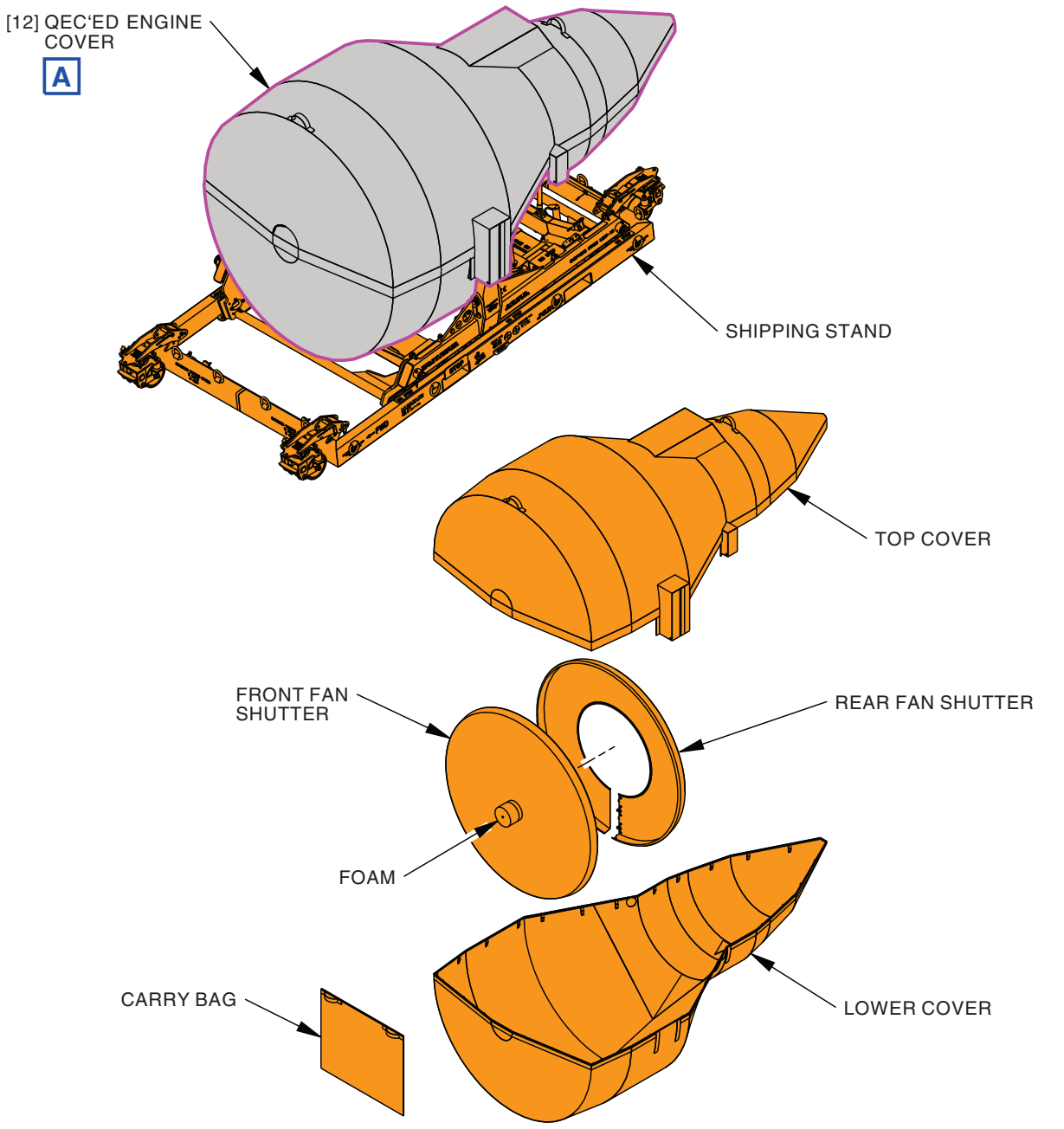
Engine Drain Box
Figure 202/71-00-04-990-836-G01

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Shipping Stand Cover
Figure 203/71-00-04-990-837-G01

ECCN 9E991 BOEING PROPRIETARY - See title page for details



QEC'ED ENGINE COVER
A

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Quick Engine Change Storage Cover
Figure 204/71-00-04-990-838-G01

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE COWLING - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Inlet Cowl Inspection.

TASK 71-11-00-200-801-G00

2. Inlet Cowl Inspection

A. General

(1) This task provides the detailed inspection of the inlet cowl fasteners, the A1 flange, and the inner barrel fasteners.

B. References

Reference	Title
71-11-01-400-801-G00	Inlet Cowl Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 71-11-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-00-010-001-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-00-860-002-G00

(3) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

G. Inlet Cowl Inspection

SUBTASK 71-11-00-211-001-G00

(1) Examine the inlet cowl fasteners for looseness and damage.

(a) If you find the loosen fastener, tighten the fastener to 291 in-lb (32.9 N·m) - 309 in-lb (34.9 N·m).

(b) If you find broken fastener, replace with the new fastener (TASK 71-11-01-400-801-G00).

SUBTASK 71-11-00-220-001-G00

(2) Examine the A1 flange for cracks and damage.

(a) If you find the cracks on the A1 flange, repair or replace the inlet cowl if it is not within the limits to fly.

NOTE: Contact Boeing for inspection interval on cracks if the inlet cowl is serviceable.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-00-860-003-G00

(1) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 71-11-00-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 71-11-00-410-001-G00

(3) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE COWLING - REPAIR

1. General

A. This procedure has this task:

(1) Repair of the engine cowling TV-Chord assembly.

TASK 71-11-00-300-801-G00

2. Engine Cowling - TV-Chord Assembly Repair

(Figure 801 or Figure 802)

A. General

(1) This task is the instruction to repair the TV-Chord assembly rubstrip of the engine cowling.

(2) The TV-Chord rubstrip is located on the aft outer edge of the inlet and interfaces with the fan cowl V-blade. It protects the metallic TV-Chord from damage. If any damage to the metallic TV-Chord is found, blend per the limitations and requirements shown in Reference Not Currently Available.

NOTE: Service history suggests that the worst rubstrip damage typically occurs between the 5 o'clock and 7 o'clock positions. This area is also where damage is most likely to occur on the TV-Chord. Boeing recommends prioritizing repairs to this area of the rubstrip to prevent damage to the metallic TV-Chord.

PTF ALL PRE SB 737-71-1920

(3) If more than 1 in. (25 mm) of TSJ tape, G51451, is hanging loose, trim immediately.

PTF ALL POST SB 737-71-1920

(4) If more than 1 in. (25 mm) of wear strip, G51413, or LH TV-Chord PTFE, or RH TV-Chord PTFE, or LWR TV-Chord PTFE rubstrips is disbonded trim immediately.

PTF ALL PRE SB 737-71-1920

(5) Damage (tear and/or peel) that is less than or equal to 0.375 in. (9.525 mm) wide, less than or equal to 3 in. (76 mm) long, and greater than or equal to 5 in. (127 mm) from the closest adjacent damage may be left as-is after any required trimming per Step (3).

PTF ALL POST SB 737-71-1920

(6) Damage (tear, peel, and/or disbond resulting in exposed TV-Chord substrate) that is less than or equal to 0.375 in. (9.525 mm) wide, less than or equal to 3 in. (76 mm) long, and greater than or equal to 5 in. (127 mm) from the closest adjacent damage may be left as-is after any required trimming per Step (3).

PTF ALL

(7) Perform repairs to damage that exceeds the dimensions provided in Step (4) within 400 flights of finding the damage.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-15157	Heat Gun - Thermogun
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A01057	Adhesive - Hysol EA9394	BMS5-141
A50396	Sealant - Dapco 2200 Primerless Silicone Firewall Sealant	BMS5-63 Type II Class B-1/2
A50407	Adhesive - Epoxy, Two Part, Cures at Room Temperature	BAC5010 Type 94 (BMS5-109 Type II Class 2)
B00068	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002, MIL-E-51454 Type II
B00130	Alcohol - Isopropyl	TT-I-735
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C50240	Primer - Dow Corning PR-1200 Prime Coat Red	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50381	Abrasive - Aluminum Oxide Paper, 180 Grit	
G50629	Tape - Nitto P-422 (Formerly Permacel) PTFE Film Tape	A-A-59474 (Supersedes MIL-I-23594)
G50961	Coating - Corrosion Inhibiting Epoxy Primer - (BAC 377 Yellow Color)	BMS10-11 Type I
G51413	Strip - Wear, P54 Grade 020, Light Duty	
G51451	Tape - Polymer Reinforced, PTFE - TSJ	
G51457	Tape - PTFE Skived Film - 3M 5481	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	LWR PTFE rubstrip	Not Specified	
2	LH PTFE rubsrip	Not Specified	
3	RH PTFE rubsrip	Not Specified	

F. Location Zones

Zone	Area
412	Engine 1 - Nose Inlet Cowl
422	Engine 2 - Nose Inlet Cowl

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

H. Prepare for the Repair

SUBTASK 71-11-00-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-00-010-002-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

PTF ALL PRE SB 737-71-1920

I. Repair of TV-Chord with TSJ tape

SUBTASK 71-11-00-300-001-G00

NOTE: The following procedures are all acceptable for repairing rubstrip damage.

(1) Procedure 1: Teflon tape application

(a) Clean the area to remove any contamination within the damage rubstrip area.

1) Rub the area and approximately 2 in. (51 mm) or more on all sides of the area with a clean cotton wiper, G00034, that is wet with one of the solvents that follow:

a) alcohol, B00068,

b) alcohol, B00130.

2) Wipe dry with a cotton wiper, G00034, before solvent evaporates.

a) If it is necessary repeat steps to completely remove contaminants.

(b) Apply a Teflon backed silicone adhesive tape such as 3M 5481 tape, G51457, or Nitto P-422 tape, G50629, to the damage area.

1) 3M 5481 tape, G51457, or Nitto P-422 tape, G50629, should extend to 0.5 in. (13 mm) - 1 in. (25 mm) beyond the damage area.

NOTE: It is permissible to wrap 3M 5481 tape, G51457, or Nitto P-422 tape, G50629, around the end/edge of the TV-Chord to prevent 3M 5481 tape, G51457, or Nitto P-422 tape, G50629, from peeling.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-71-1920 (Continued)

2) If 3M 5481 tape, G51457, or Nitto P-422 tape, G50629, becomes damaged, carefully remove 3M 5481 tape, G51457, or Nitto P-422 tape, G50629, and reapply with new 3M 5481 tape, G51457, or Nitto P-422 tape, G50629.

SUBTASK 71-11-00-300-004-G00

(2) Procedure 2: Re-bond peeled Polytetraflouroethylene (Teflon) (PTFE) to TV-Chord

(a) Re-bond peeled PTFE to the TV-Chord as follows:

1) Carefully lift the delaminated rubstrip from the end of the TV-Chord up to 12 in. (305 mm).

a) Lift only loose or peeled rubstrip ensuring to minimize amount of peel back.

2) Mask the V-groove in the TV-Chord to facilitate clean-up.

3) Apply Dapco 2200 firewall sealant, A50396, to the mating surfaces of the TV-Chord and rubstrip.

4) Apply the rubstrip to the TV-Chord and allow to cure for 4 hours.

a) Apply pressure to the rubstrip to form a smooth surface.

NOTE: Pressure must be applied within 30 minutes when Dapco 2200 firewall sealant, A50396, is applied to the surfaces.

5) Clean and remove squeeze out.

SUBTASK 71-11-00-300-002-G00

(3) Procedure 3: Bond new piece of PTFE to TV-Chord

(a) Bond new piece of PTFE to the TV-Chord as follows:

1) Carefully trim and remove the damaged section PTFE from the TV-Chord.

a) Make the trim cuts in the fore/aft direction, perpendicular to the aft edge of the TV-Chord.

2) Prepare the primed surface for bonding operation:

a) Rub the area with a clean cotton wiper, G00034, that is wet with one of the solvents that follow:

<1> alcohol, B00068,

<2> alcohol, B00130.

b) Wipe dry with cotton wiper, G00034, before solvent evaporates.

3) Apply thin coat of Dow Corning PR-1200 prime coat red, C50240, by spraying or using a clean brush to the areas that will be bonded.

a) Exercise care to minimize prime coat thickness.

b) Apply just enough primer to wet the surface, resulting in a uniform light pink color after drying.

c) Do not use if solutions turns cloudy or milky.

NOTE: It is acceptable to prime coat the areas 1 in. (25 mm) beyond the bond line.

4) Allow prime coat to air dry for a minimum of 60 minutes and a maximum of 12 hours.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-71-1920 (Continued)

5) Apply one layer of TSJ tape, G51451 (with silicone pressure sensitive adhesive on one side), to the top surface and along the entire length of the TV-Chord flush +0.0 in. (0.0 mm)/ -0.06 in. (2 mm) with the edge (Figure 801).
   a) Material may be butt-spliced with 0.06 in. (2 mm) maximum gaps.
      <1> Do not overlap TSJ tape, G51451, at splice locations.
   b) It is acceptable to slit TSJ tape, G51451, as required to prevent wrinkling or stretching.
      <1> Do not damage the assembly material or finish when trimming or splicing TSJ tape, G51451.
6) Use heat gun, COM-15157, and set to 250 ±10°F (121 ±6°C) to heat area for 10 to 30 seconds dwell time.
7) Squeeze or roll TSJ tape, G51451, onto the part surface while TSJ tape, G51451, is still warm to bond in place.

PTF ALL POST SB 737-71-1920

J. Repair of TV-Chord with P54 Strip or PTFE Rubstrips

NOTE: The following procedures are all acceptable for repairing rubstrip damage.

NOTE: The TV-Chord assemblies (314A6141-71/-72/-73) have specific P54 rubstrip details (314A6144-39/-40/-41) installed. wear strip, G51413, is not contoured to the TV-Chord surface, therefore wear strip, G51413, is only ideal to use in areas that is less than 12 in. (30.5 cm) in length. Assemblies are as follows:

- LH TV-Chord assy: 314A6141-71 + wear strip, G51413: 314A6144-39
- RH TV-Chord assy: 314A6141-72 + wear strip, G51413: 314A6144-40
- LWR TV-Chord assy: 314A6141-73 + wear strip, G51413: 314A6144-41.

SUBTASK 71-11-00-300-005-G00

(1) Procedure 1: PTFE, wear strip, G51413, surface preparation
   (a) Abrade the area to remove any adhesive, A50407, within the area where the damaged rubstrip was removed.
   (b) If base metal of fastener heads is exposed, touch up bared fastener head surfaces with coating, G50961.
      1) Rub the area with a clean cotton wiper, G00034, that is wet with one of the solvents that follow:
         a) alcohol, B00068,
         b) alcohol, B00130.
      2) Wipe dry with cotton wiper, G00034, before solvent evaporates.
      3) Apply liquid coating, G50961.
         a) Allow to dry prior to installing rubstrip.
   (c) If base metal of TV-Chord aft flange, excluding fastener heads, is exposed due to abrasion, apply coating, C00064, and coating, G50961.
      1) Rub the area with a clean cotton wiper, G00034, that is wet with one of the solvents that follow:
         a) alcohol, B00068,

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-71-1920 (Continued)

b) alcohol, B00130.

2) Wipe dry with cotton wiper, G00034, before solvent evaporates.

3) Apply coating, C00064, to the exposed surface.

a) Keep the surface wet until can be seen.

NOTE: The color will come in 2-3 minutes.

CAUTION THE COATING IS SOFT WHEN IT IS INITIALLY APPLIED. USE CARE WHEN THE SURFACE IS FLUSHED AND DRIED. IF YOU DO NOT USE CARE, YOU WILL SCRATCH, OR POSSIBLY REMOVE THE COATING.

4) Use cotton wiper, G00034, wet with water to remove only the excess liquid.

5) Air dry at room temperature.

6) Apply liquid coating, G50961.

a) Allow to dry prior to installing rubstrip.

SUBTASK 71-11-00-300-006-G00

(2) Procedure 2: PTFE, wear strip, G51413, or LWR PTFE rubstrip [1], LH PTFE rubsrip [2], RH PTFE rubsrip [3] application

(a) Bond new piece of PTFE, wear strip, G51413,new LWR PTFE rubstrip [1], new LH PTFE rubsrip [2], new RH PTFE rubsrip [3] to the TV-Chord as follows:

1) Carefully trim and remove the damaged section PTFE from the TV-Chord.

a) Make the trim cuts in the fore/aft direction, perpendicular to the aft edge of the TV-Chord.

2) Prepare the primed surface for bonding operation:

a) Rub the area with a clean cotton wiper, G00034, that is wet with one of the solvents that follow:

<1> alcohol, B00068,

<2> alcohol, B00130.

b) Wipe dry with cotton wiper, G00034, before solvent evaporates.

SUBTASK 71-11-00-300-007-G00

(3) Install wear strip, G51413, or LWR PTFE rubstrip [1], LH PTFE rubsrip [2], RH PTFE rubsrip [3].

(a) If the repair length is greater than or equal to 12 in. (30.5 cm) along the aft flange of the TV-Chord, use the applicable LWR PTFE rubstrip [1], LH PTFE rubsrip [2], RH PTFE rubsrip [3], or wear strip, G51413.

(b) Remove peel ply on the LWR PTFE rubstrip [1], LH PTFE rubsrip [2], RH PTFE rubsrip [3] or wear strip, G51413.

1) Mix part A and part B of adhesive, A50407, or Hysol EA9394 adhesive, A01057, before mixing part A and part B together.

NOTE: The pot life is approximately 40 minutes at 75°F (24°C).

2) Mix part A and part B in the proportions shown below:

a) Part A: 97 - 103

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-71-1920 (Continued)

b) Part B:16.5 - 17.5.

3) Apply a continuous layer of the mixed adhesive to each faying surface.

4) Assemble the parts and apply strong, constant pressure to make sure that all of the mating surfaces touch completely.

5) Bond rubstrip to TV-Chord within 5 minutes of applying adhesive.

6) Remove unwanted adhesive with 180 grit abrasive paper, G50381, before it cures.

a) Use a clean wiper wet with solvent.

b) Do not let solvent get into the bond line.

7) Cure under pressure as shown Figure 803.

(c) Make sure that the wear strip, G51413, is flush +0.00 in. (0.00 mm)/-0.06 in. (2 mm) from the aft edge of TV-Chord surface.

(d) If it is necessary, trim wear strip, G51413, or LWR PTFE rubstrip [1], LH PTFE rubsrip [2], RH PTFE rubsrip [3] to make sure that there is no overlap into the V-groove.

1) Trim wear strip, G51413, or LWR PTFE rubstrip [1], LH PTFE rubsrip [2], RH PTFE rubsrip [3] lengthwise and join smaller segments to meet 5 minutes requirement.

NOTE: Minimum 6 in. (15.2 cm) long repair recommended for durability.

(e) If it necessary, rework 6 in. (15.2 cm) length wear strip, G51413, or LWR PTFE rubstrip [1], LH PTFE rubsrip [2], RH PTFE rubsrip [3] with butt-splicing.

(f) Make sure that the butt spliced wear strips segments do not overlap and have 0.06 in. (2 mm) maximum gaps.

PTF ALL

K. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-00-410-002-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-00-860-008-G00

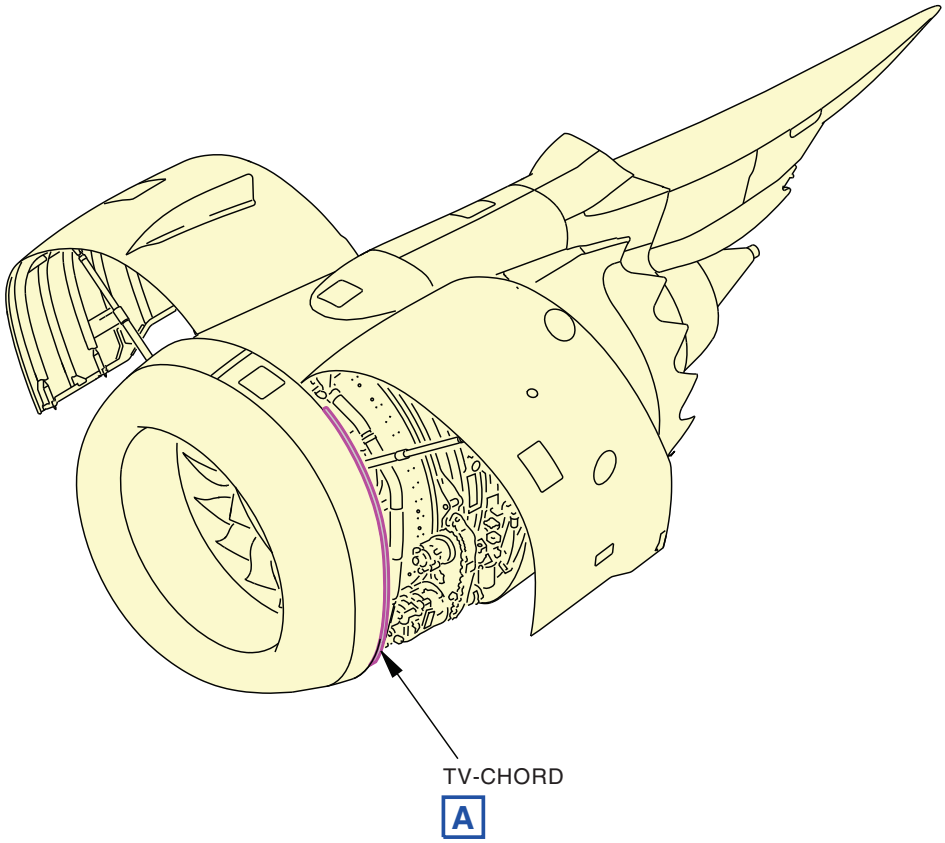
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

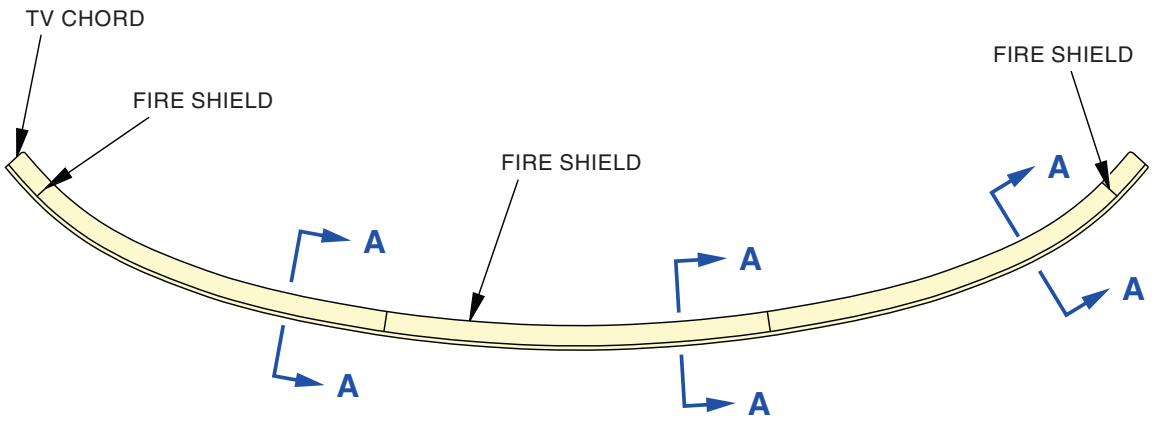
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2809062 S0000646121_V1

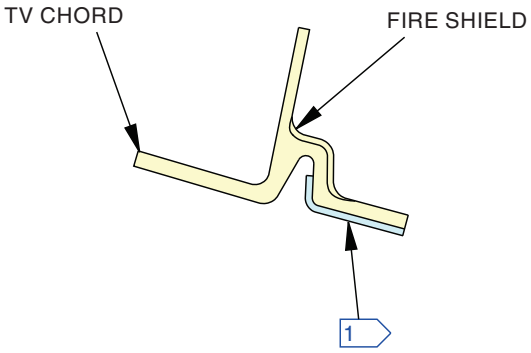
Engine TV-Chord Assembly Repair
Figure 801/71-11-00-990-801-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LOWER CHORD)

A



A-A

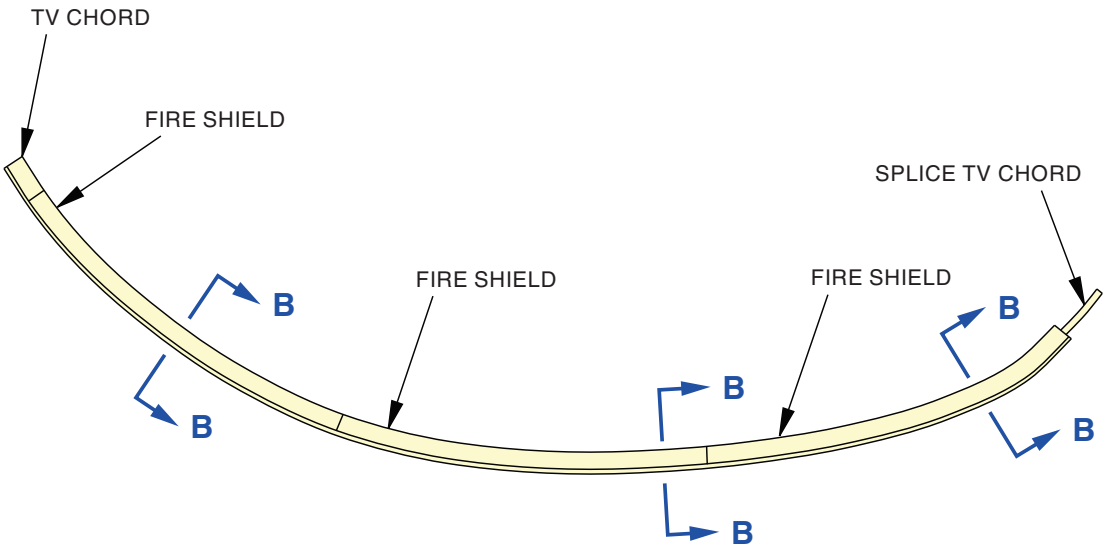
NOTE:

1 APPLY TSJ PTFE TAPE, REFER TO THE TAPE APPLICATION PROCEDURE DETAILS.

2809073 S0000646123_V2

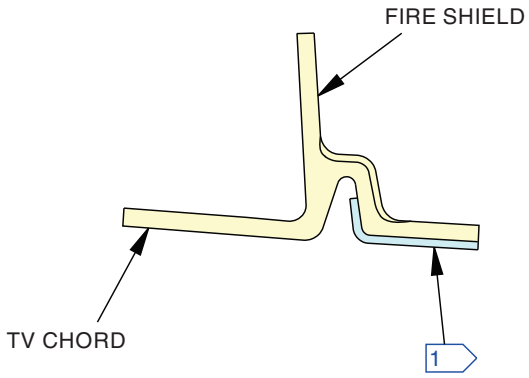
Engine TV-Chord Assembly Repair
Figure 801/71-11-00-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LH TV-CHORD)

A



B-B

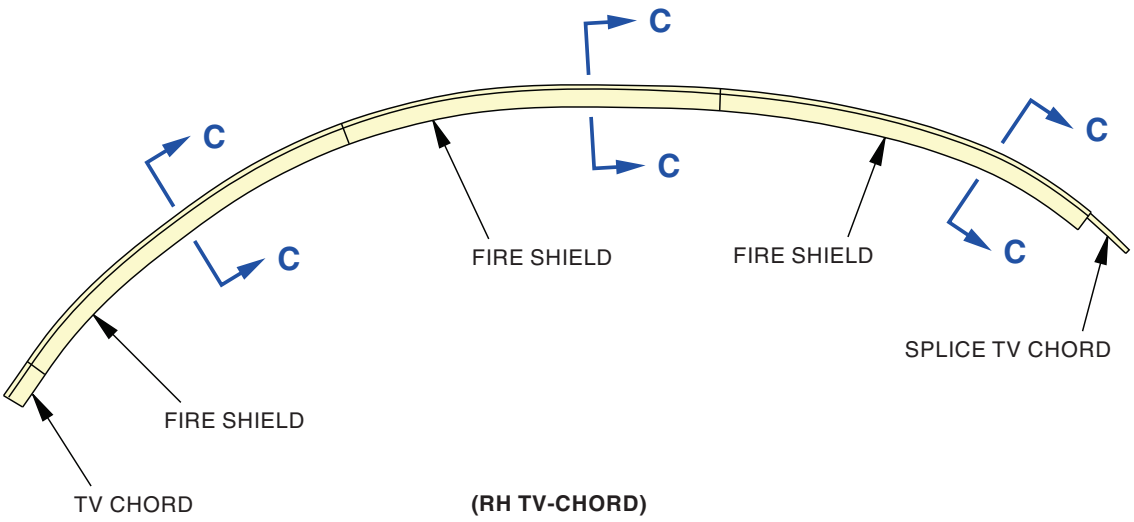
NOTE:

1 APPLY TSJ PTFE TAPE, REFER TO THE TAPE APPLICATION PROCEDURE DETAILS.

2809076 S0000646125_V2

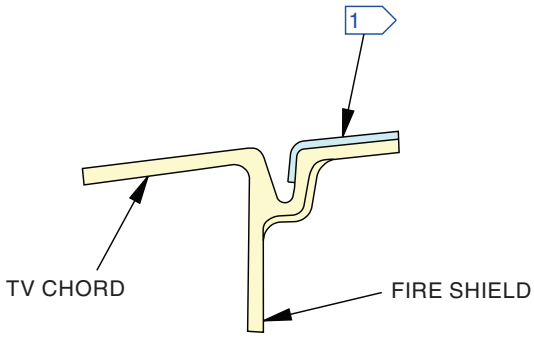
Engine TV-Chord Assembly Repair
Figure 801/71-11-00-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(RH TV-CHORD)

A



C-C

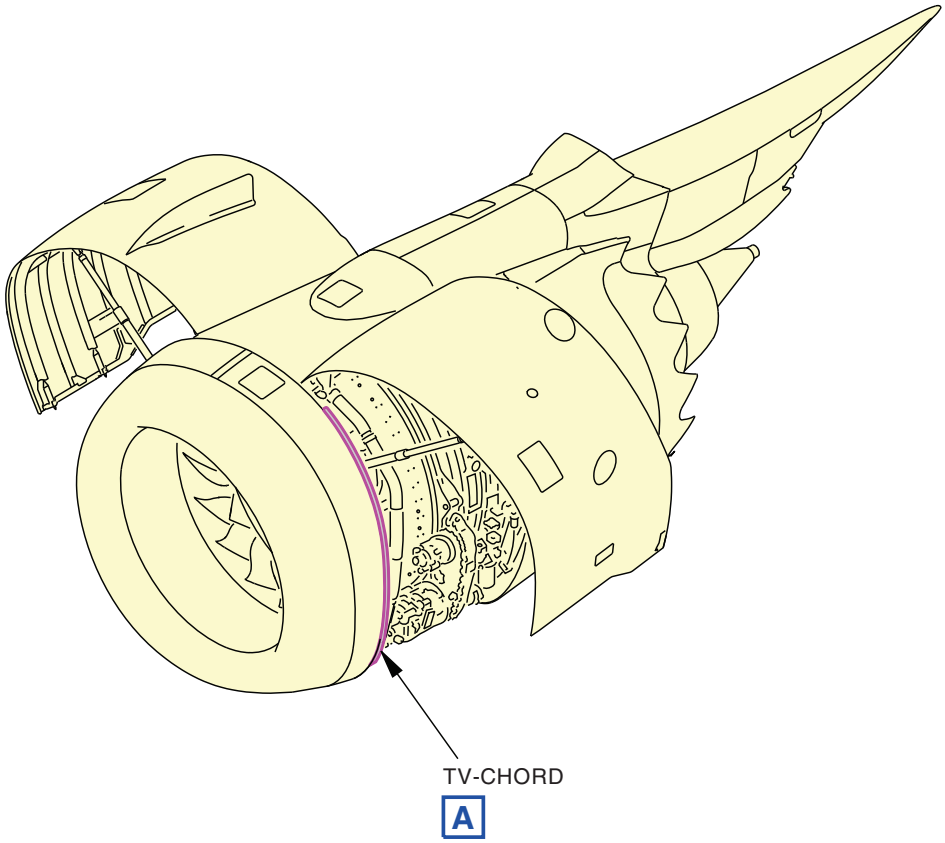
NOTE:

1 APPLY TSJ PTFE TAPE, REFER TO THE TAPE APPLICATION PROCEDURE DETAILS.

2809083 S0000646129_V2

Engine TV-Chord Assembly Repair
Figure 801/71-11-00-990-801-G00 (Sheet 4 of 4)

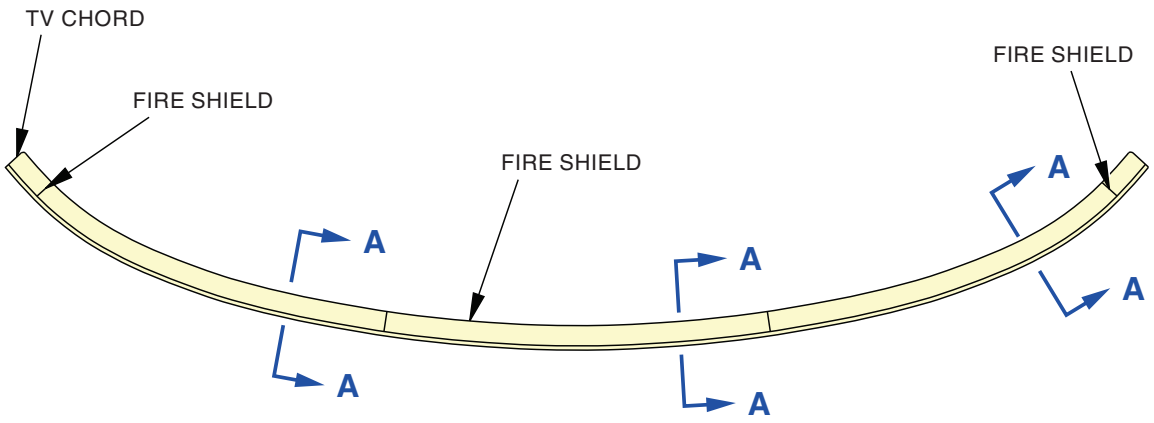
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2809062 S0000646121_V1

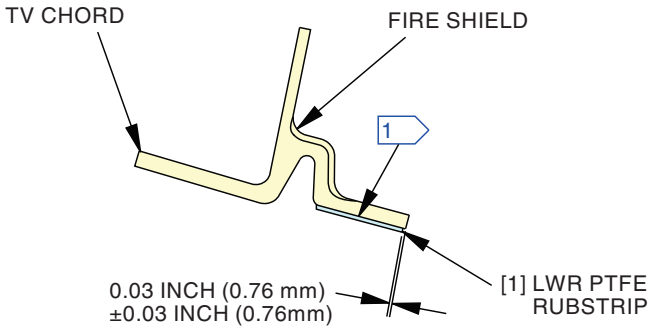
Engine TV-Chord Assembly Repair
Figure 802/71-11-00-990-802-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LOWER CHORD)

A



A-A

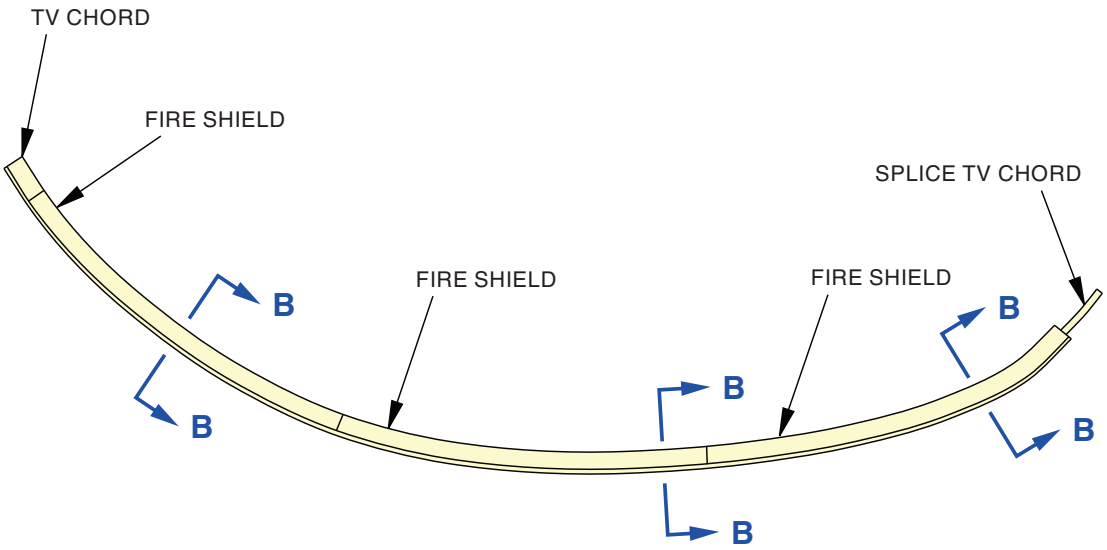
NOTE:

1 APPLY P54 PTFE WEAR STRIP OR LOWER PTFE RUBSTRIP, REFER TO RUBSTRIP APPLICATION DETAILS.

3000055 S0000775901_V1

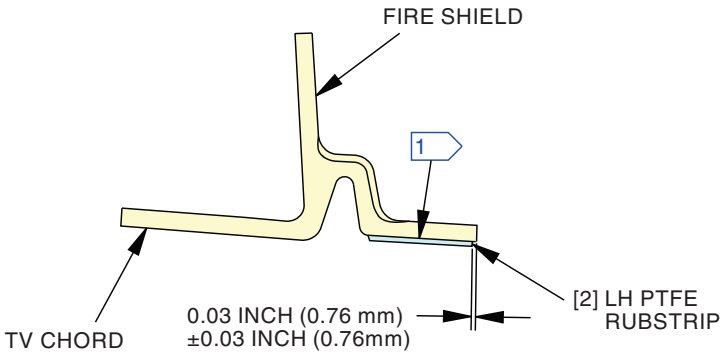
Engine TV-Chord Assembly Repair
Figure 802/71-11-00-990-802-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LH TV-CHORD)

A



B-B

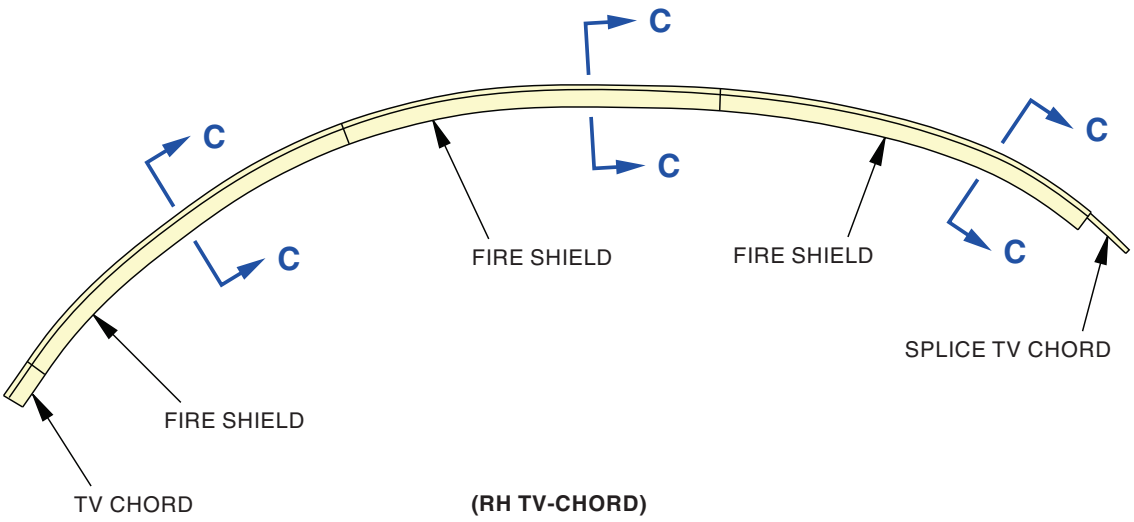
NOTE:

1 APPLY P54 PTFE WEAR STRIP OR
LH PTFE RUBSTRIP,
REFER TO RUBSTRIP APPLICATION DETAILS.

3000056 S0000775902_V1

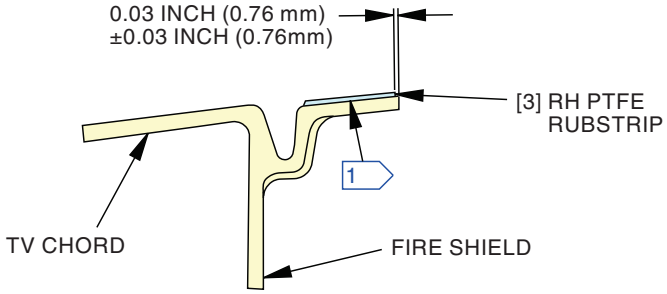
Engine TV-Chord Assembly Repair
Figure 802/71-11-00-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(RH TV-CHORD)

A



C-C

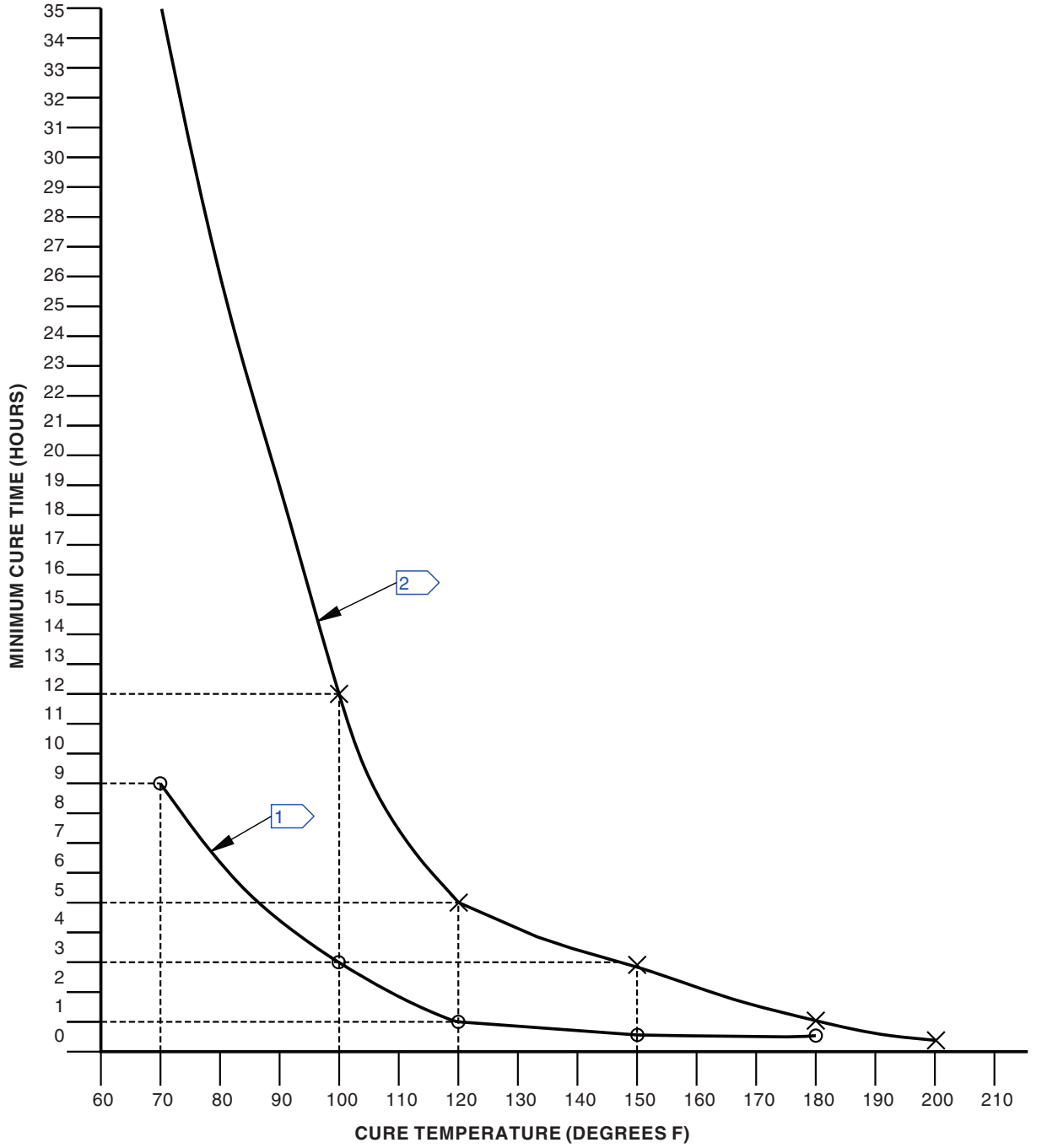
NOTE:

1 APPLY P54 PTFE WEAR STRIP OR
RH PTFE RUBSTRIP,
REFER TO RUBSTRIP APPLICATION DETAILS.

3000058 S0000775903_V1

Engine TV-Chord Assembly Repair
Figure 802/71-11-00-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Adhesive Cure Time and Temperature
Figure 803/71-11-00-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INLET COWL - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Inlet Ground Cooling Duct Cap and Cover Removal

(2) Inlet Ground Cooling Duct Cap and Cover Installation.

TASK 71-11-01-000-802-G00

2. Inlet Ground Cooling Duct Cap and Cover Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove the inlet ground cooling duct cap and cover.

(2) The cap and cover are installed on the inlet ground cooling duct when the inlet is installed on a LEAP-1BXXG06 engine or higher GXX number engine. The LEAP-1BXXG06 engines and higher GXX number engines have the EEC blower system removed or partially removed.

(3) It is necessary to remove the cap and cover from the inlet ground cooling duct when an inlet assembly is installed on a LEAP-1BXXG05 engine with the EEC blower system installed.

B. Tools/Equipment

Reference	Description
STD-765	Scraper - Plastic

C. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Inlet Ground Cooling Duct Cap and Cover Removal

SUBTASK 71-11-01-020-006-G00

(1) Remove the cap [4] from the ground cooling duct [3] on the inlet aft bulkhead as follows:

(a) Remove the hose clamp [5].

(b) Use a knife and plastic scraper, STD-765, to carefully separate the sealant under the clamp bracket [6].

(c) Remove the clamp bracket [6].

(d) Use a knife and plastic scraper, STD-765, to carefully separate the fillet seal around the circumference of the cap [4] and ground cooling duct [3].

(e) Remove the cap [4] from the ground cooling duct [3].

SUBTASK 71-11-01-020-007-G00

(2) Remove the cover [1] from the ground cooling duct [3] on the inlet outer surface as follows:

(a) Remove the eight bolts [2] (P/N BACB30ZE3K5) that attach the cover [1] and the ground cooling duct [3] to the inlet outer surface.

(b) Use a knife and plastic scraper, STD-765, to carefully separate the fay surface seal under the cover [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the cover [1] from the ground cooling duct [3].

(d) Use the sealant, A00767 to wet install the four bolts [2] (P/N BACB30ZE3K5) in the four locations on the aft side of the ground cooling duct [3].

(e) Use the sealant, A00767 to wet install the four bolts [2] (P/N BACB30ZE3K4) in the four locations on the forward side of the ground cooling duct [3].

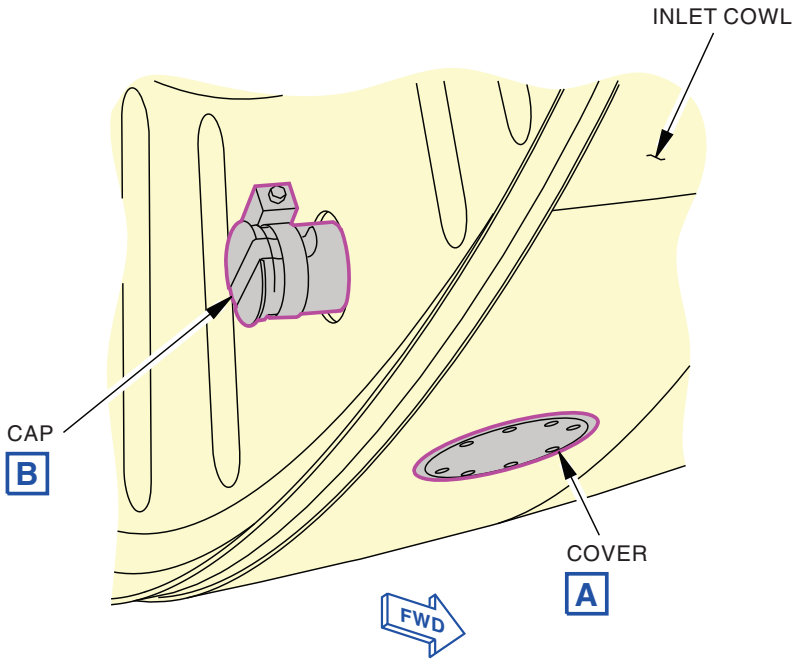
(f) Tighten the bolts [2] to 29.0 in-lb (3.3 N·m) - 31.0 in-lb (3.5 N·m).

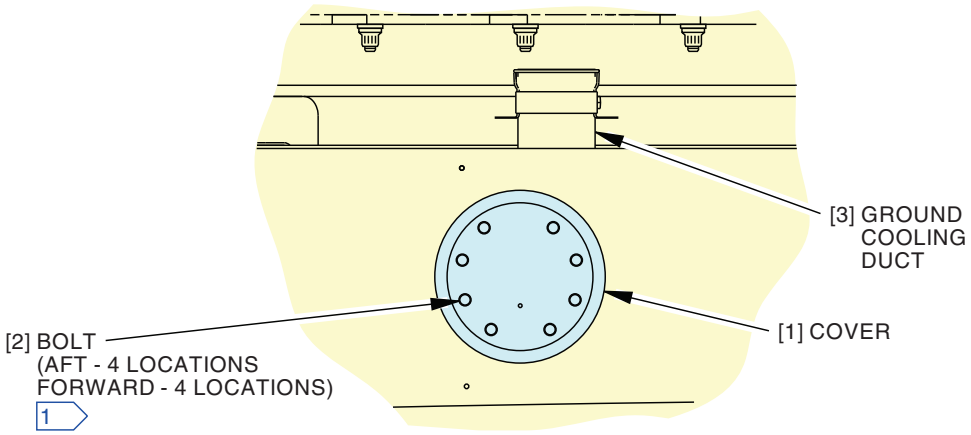
SUBTASK 71-11-01-100-001-G00

(3) Clean the ground cooling duct to remove any remaining sealant from the areas where the cap and the cover were installed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





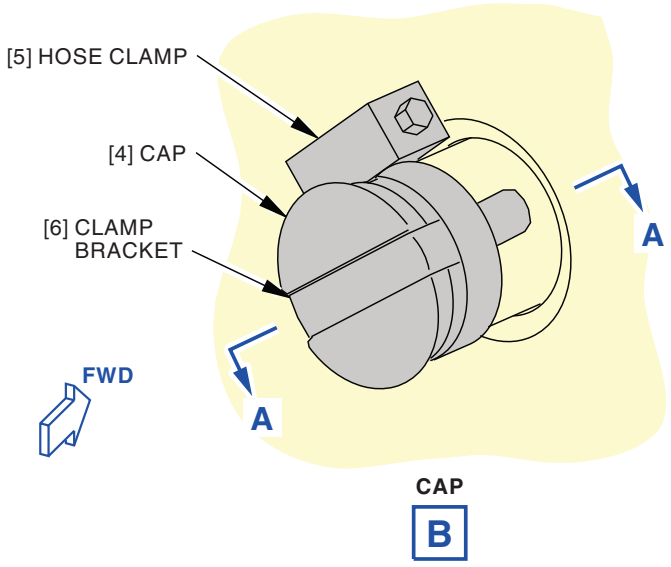
COVER
A

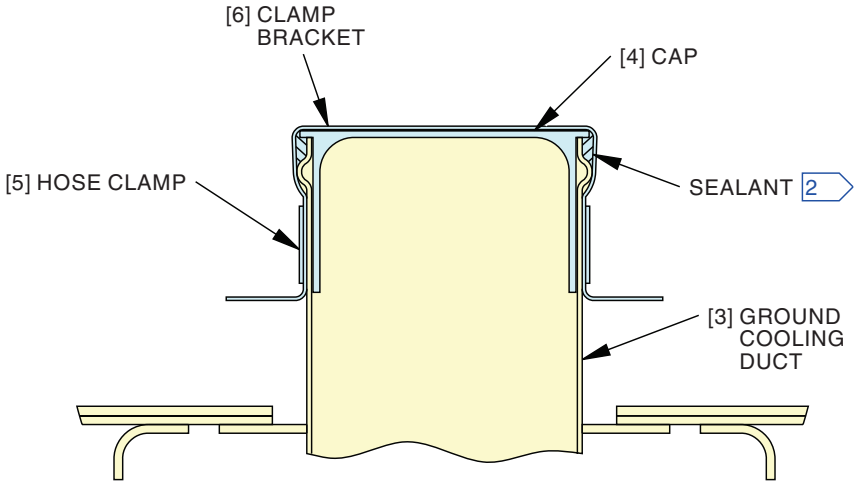
1 FORWARD AND AFT BOLTS BACB30ZE3K5 WITH COVER INSTALLED; WITH COVER REMOVED, AFT BOLTS ARE BACB30ZE3K5 AND FORWARD BOLTS ARE BACB30ZE3K4. ALL BOLTS ARE WET INSTALLED WITH BMS5-45 SEALANT.

2846105 S0000664920_V1

Inlet Ground Cooling Duct Cap and Cover Installation
Figure 201/71-11-01-990-809-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





A-A

2 FILLET AROUND CIRCUMFERENCE UNDER CAP FLANGE. EXTEND SEALANT UNDER CLAMP BRACKET FROM THE CAP FLANGE TO THE DUCT BEAD; MAKE SURE THE SEALANT COMPLETELY FILLS THE SPACE UNDER THE CLAMP BRACKET.

2846119 S0000664921_V1

Inlet Ground Cooling Duct Cap and Cover Installation
Figure 201/71-11-01-990-809-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-01-400-802-G00

3. Inlet Ground Cooling Duct Cap and Cover Installation

(Figure 201)

A. General

(1) This task gives the instructions to install the inlet ground cooling duct cap and cover.

(2) It is necessary to install the cap and cover on the inlet ground cooling duct when an inlet assembly is installed on a LEAP-1BXXG06 engine or higher GXX number engine. The LEAP-1BXXG06 engines and higher GXX number engines have the EEC blower system removed or partially removed.

B. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
A50396	Sealant - Dapco 2200 Primerless Silicone Firewall Sealant	BMS5-63 Type II Class B-1/2

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cover	71-11-01-01-040	PTF ALL
4	Cap	71-11-01-01-035	PTF ALL

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Inlet Ground Cooling Duct Cap and Cover Installation

SUBTASK 71-11-01-010-007-G00

(1) If installed, remove the protective caps from the ends of the inlet ground cooling duct [3].

SUBTASK 71-11-01-420-005-G00

(2) Install the cover [1] over the open end of the ground cooling duct [3] at the inlet outer surface as follows:

(a) Remove the four bolts [2] (P/N BACB30ZE3K5, aft side) and four bolts [2] (P/N BACB30ZE3K4, forward side) that attach the ground cooling duct [3] to the inlet outer surface.

(b) Use the sealant, A00767 to apply a faying surface seal between the cover [1] and the ground cooling duct [3] on the inlet outer surface.

(c) Install the cover [1].

(d) Use the sealant, A00767 to wet install the eight bolts [2] (P/N BACB30ZE3K5) that attach the cover [1] and the ground cooling duct [3] to the inlet outer surface.

(e) Tighten the bolts [2] to 29.0 in-lb (3.3 N·m) - 31.0 in-lb (3.5 N·m).

SUBTASK 71-11-01-420-006-G00

(3) Install the cap [4] on the ground cooling duct [3] at the inlet aft bulkhead as follows:

(a) Install the cap [4] in the ground cooling duct [3].

(b) Use the Dapco 2200 firewall sealant, A50396 to apply a fillet seal under the cap [4] flange around the circumference of the cap [4].

(c) Install the clamp bracket [6] over the cap [4] and the ground cooling duct [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Apply the Dapco 2200 firewall sealant, A50396 under both sides of the clamp bracket [6] from the cap [4] flange to the duct bead.

(e) Make sure that the sealant completely fills the space that extends under the clamp bracket [6].

(f) Install the hose clamp [5].

(g) Tighten the hose clamp [5] to 19.0 in-lb (2.1 N·m) - 21.0 in-lb (2.4 N·m).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INLET COWL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the inlet cowl

(2) An installation of the inlet cowl.

TASK 71-11-01-000-801-G00

2. Inlet Cowl Removal

(Figure 401, Figure 402, Figure 403)

A. General

(1) This task gives the instructions to remove the inlet cowl.

(2) Use the inlet cowl sling, SPL-2062, to remove the inlet cowl.

(a) Use the installation/removal frame equipment, SPL-14544, together with the boom hoist, SPL-2430, as an optional method to remove the inlet cowl.

(b) The optional method is useful where overhead lifting is not possible.

NOTE: This tool combination can also be easily transported as needed.

B. References

Reference	Title
54-52-01-010-801	Forward Fairing Removal (P/B 401)
71-11-01-000-802-G00	Inlet Ground Cooling Duct Cap and Cover Removal (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-21-06-000-801-G00	T12 Sensor Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2062	Sling - Inlet Cowl Part #: B71040-66 Supplier: 81205 Opt Part #: B71040-38 Supplier: 81205 Opt Part #: B71040-39 Supplier: 81205
SPL-2430	Hoist ? Boom, 800 lbs WLL, 250 lbs Load Positioner Compatible Part #: C20002-267 Supplier: 81205
SPL-14544	Installation/Removal Frame Equipment - Inlet Cowl, LEAP-1B Engine Part #: C71033-1 Supplier: 81205
STD-653	Platform - Engine and Strut Access
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-748	Rope - 12 ft (3.6 m) minimum
STD-858	Tag - DO NOT OPERATE
STD-1095	Crane - Lift, 2000 lb Capacity, 30 Foot Height

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 71-11-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-01-010-001-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-01-010-009-G00

(3) Removal the thumbnail fairing (TASK 54-52-01-010-801).

SUBTASK 71-11-01-430-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS OR RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN THE ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(4) Use a teflon-jawed pliers, STD-664, to disconnect the T12-B electrical connector [30] and T12-A electrical connector [31] at the 11:30 o'clock position (TASK 73-21-06-000-801-G00).

(a) Disconnect the T12-B electrical connector [30] and T12-A electrical connector [31] from the T12-B electrical receptacle and T12-A electrical receptacle (View D, Figure 403).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Put the protective covers on the T12-B electrical connector [30], T12-A electrical connector [31], T12-B electrical receptacle, and T12-A electrical receptacle.

SUBTASK 71-11-01-020-001-G00

(5) Disconnect the engine anti-ice duct [2] from the aft inlet cowl [1] at the 1:00 o'clock position (Figure 401).

(a) Remove the coupling [3] and seal [4] that attach the engine anti-ice duct [2] to the inlet cowl [1].

(b) Put the protective covers on the open ends of the engine anti-ice duct [2] and duct connection on the inlet cowl [1].

PTF ALL; LEAP-1B ENGINES WITH THE EEC BLOWER SYSTEM INSTALLED (PRE LEAP-1B SERVICE BULLETIN 75-0010 AND PRE LEAP-1B SERVICE BULLETIN 75-0011)

SUBTASK 71-11-01-020-002-G00

(6) Disconnect the ground cooling duct [37] from the fan compartment ventilation duct [36] at the 7:00 o'clock position of the inlet cowl [1].

(a) Remove the clamp [38] that attaches the fan compartment ventilation duct [36] to the ground cooling duct [37].

(b) Put the protective caps on the open ends of the ground cooling duct [37] and fan compartment ventilation duct [36].

PTF ALL; LEAP-1B ENGINES WITH THE EEC BLOWER SYSTEM REMOVED (POST LEAP-1B SERVICE BULLETIN 75-0010 OR POST LEAP-1B SERVICE BULLETIN 75-0011)

SUBTASK 71-11-01-010-008-G00

(7) If it is necessary, remove the ground cooling duct cap and cover as follows:

NOTE: This step is for inlet cowl with the ground cooling duct installed.

(a) Remove the inlet cap and cover for any of the following steps:

NOTE: The LEAP-1BXXGXX engine model designation is identified on the engine data plate, that is located on the RH side of the engine fan case.

1) If the same inlet is removed and reinstalled to replace the engine, and inlet will be installed on a spare LEAP-1BXXG05 engine.

2) If the inlet will be replaced and spare inlet does not have the cap and cover installed on the ground cooling duct, and spare inlet will be installed on the same LEAP-1BXXG06 engine or higher GXX number engine with the EEC blower system removed.

(b) If it is necessary, remove the cap and cover from the inlet ground cooling duct (TASK 71-11-01-000-802-G00).

PTF ALL

G. Inlet Cowl Removal

SUBTASK 71-11-01-490-001-G00

(1) Assemble the inlet cowl sling, SPL-2062 (Figure 402 (Sheet 1)).

(a) Put the long cable [19] through the two pulleys.

NOTE: The long cable is approximately 24.5 ft (294.0 in.) long in length and has a blue color code identifier.

(b) Connect the master (side) links [18] to the beam with the shackles [16].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Connect the short cable [17] to the beam with the shackles [16].

NOTE: The short cable is approximately 7.2 ft (86.4 in.) long in length and has a white color code identifier.

(d) Connect the short cable [17] to the master link [11] with the lok-A-loys [10].

(e) Connect the dynamometer [12] to the master link [11].

SUBTASK 71-11-01-020-003-G00

(2) Use the engine and strut access platform, STD-653, for two persons that is necessary to remove the inlet cowl when the inlet cowl sling, SPL-2062, is used.

SUBTASK 71-11-01-480-004-G00

(3) If the optional removal tool is used, attach the installation/removal frame equipment, SPL-14544, to the boom hoist, SPL-2430 (Figure 402 (Sheet 2)).

SUBTASK 71-11-01-490-002-G00

(4) Install the inlet cowl sling, SPL-2062, on the inlet cowl [1] (Figure 402 (Sheet 1)).

(a) Install the forward and aft straps [21] with hoist pins [22] to the inlet cowl [1].

(b) Install the chain hoists [20] to the aft straps [21].

(c) With the 30 foot height (2000 lb capacity) lift crane, STD-1095, lift the inlet cowl sling, SPL-2062, and put it over the inlet cowl [1].

(d) Connect the chain hoists [20] to the master (side) links [18].

(e) Connect the long cable [19] to the forward straps [21].

(f) With the 30 foot height (2000 lb capacity) lift crane, STD-1095, and chain hoists [20] tighten the chains and cables.

WARNING GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE THAT THE WEIGHT OF THE INLET COWL AND THE SLING EQUIPMENT IS BALANCED WITH THE CRANE FORCE. IF THE WEIGHT AND FORCE ARE NOT BALANCED, THE COWL CAN MOVE QUICKLY WHEN YOU DISCONNECT THE INLET COWL. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(g) With the 30 foot height (2000 lb capacity) lift crane, STD-1095, lift the inlet cowl and sling up until the dynamometer indicates a preload of approximately 500 lb (227 kg).

NOTE: The inlet cowl weighs approximately 410 lb (186 kg). The sling equipment weighs approximately 100 lb (45 kg).

1) Monitor the load during adjustment.

2) Adjust the 30 foot height (2000 lb capacity) lift crane, STD-1095, and inlet cowl sling, SPL-2062, until all of the strap assemblies are not loose.

SUBTASK 71-11-01-480-005-G00

(5) If the optional removal tool is used, attach the installation/removal frame equipment, SPL-14544, to the inlet cowl [1] (Figure 402 (Sheet 2)).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-01-010-004-G00

WARNING

MAKE SURE THAT THE WEIGHT OF THE INLET COWL AND THE SLING EQUIPMENT IS BALANCED WITH THE CRANE FORCE. IF THE WEIGHT AND FORCE ARE NOT BALANCED, THE COWL CAN MOVE QUICKLY WHEN YOU DISCONNECT THE INLET COWL. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) Remove the inlet cowl [1] (Figure 403).

(a) Remove the 44 bolts [32], 44 spacers [33], 44 washers [35], and 44 nuts [34] that attach the inlet cowl [1] to the fan case flange.

(b) Remove the two brackets [29], four bolts [55], four spacers [33], four washers [35], and four nuts [34] at the 12:00 o'clock position.

NOTE: These brackets were also called the bolt retainer assemblies.

(c) If the inlet cowl [1] will be installed on a different engine, remove the two nuts [41] and two washers [40] from the two alignment pins [39] that install on the fan case flange at the 3:00 and 9:00 o'clock positions.

NOTE: There is no need to remove the alignment pins, washers, and nuts, if the inlet cowl is installed back onto the same engine.

(d) Install a rope - 12 ft (3.6 m) minimum, STD-748, around the inlet cowl [1] to guide the movement of the inlet cowl [1] when the inlet cowl sling, SPL-2062, is used.

(e) Lift the inlet cowl [1] to the ground platform.

NOTE: Ground platform can be sand bags or a wood pallet.

1) Turn and put the inlet cowl [1] on top of the ground platform in an area that is away from the airplane.

2) Use the chain hoists when using the inlet cowl sling, SPL-2062, to turn the inlet cowl [1] to the horizontal position until it is parallel with the ground platform.

3) Slowly lower the inlet cowl [1] until the attach ring rests evenly on the ground platform.

(f) Disconnect the straps [21] from the inlet cowl [1] when the inlet cowl sling, SPL-2062, is used.

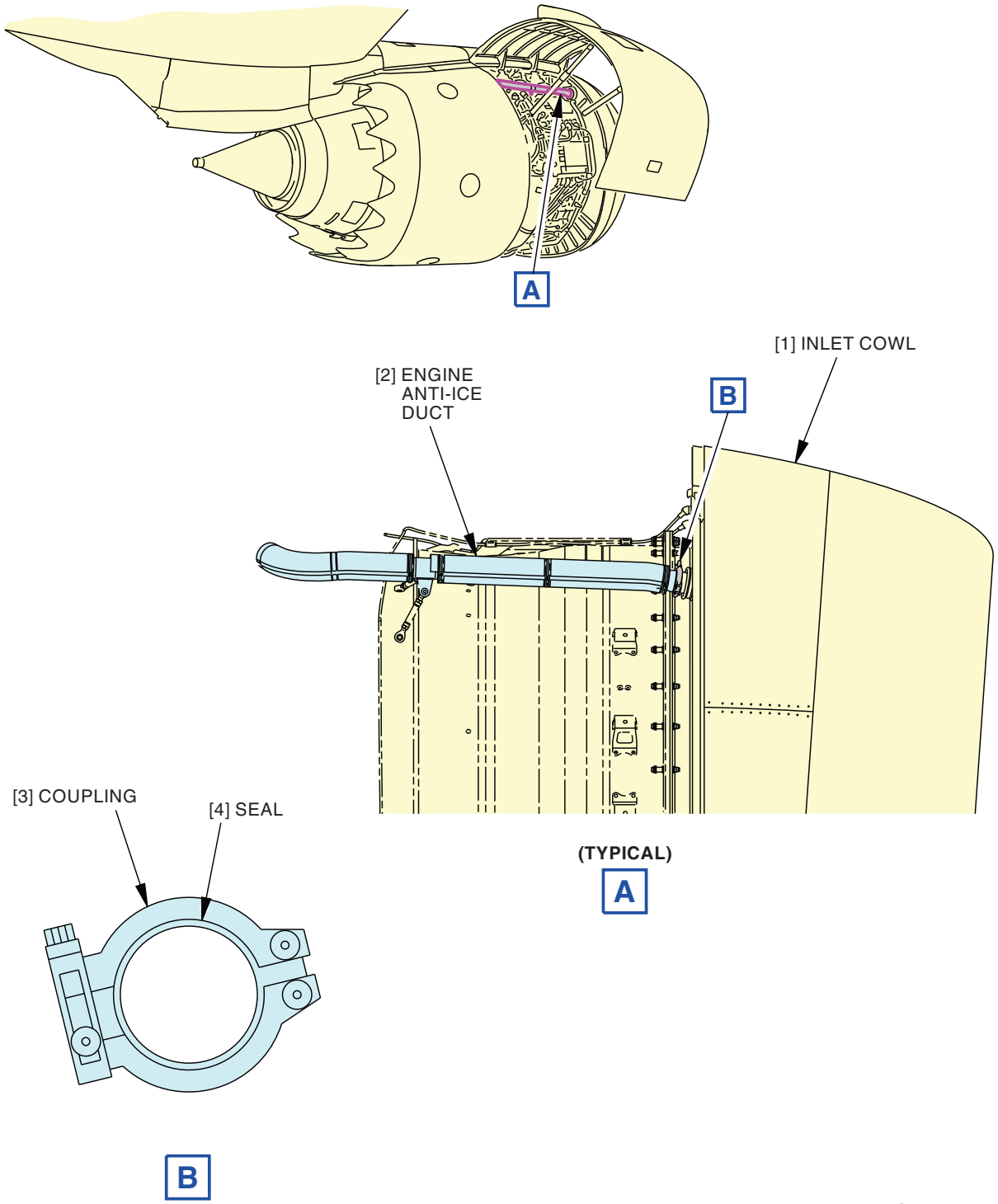
1) Remove the hoist pins [22] that attach the straps [21] to the inlet cowl [1].

(g) Remove the rope - 12 ft (3.6 m) minimum, STD-748, from the inlet cowl [1] when the inlet cowl sling, SPL-2062, is used.

(h) If the optional removal tool is used, disconnect the installation/removal frame equipment, SPL-14544, from the inlet cowl [1].

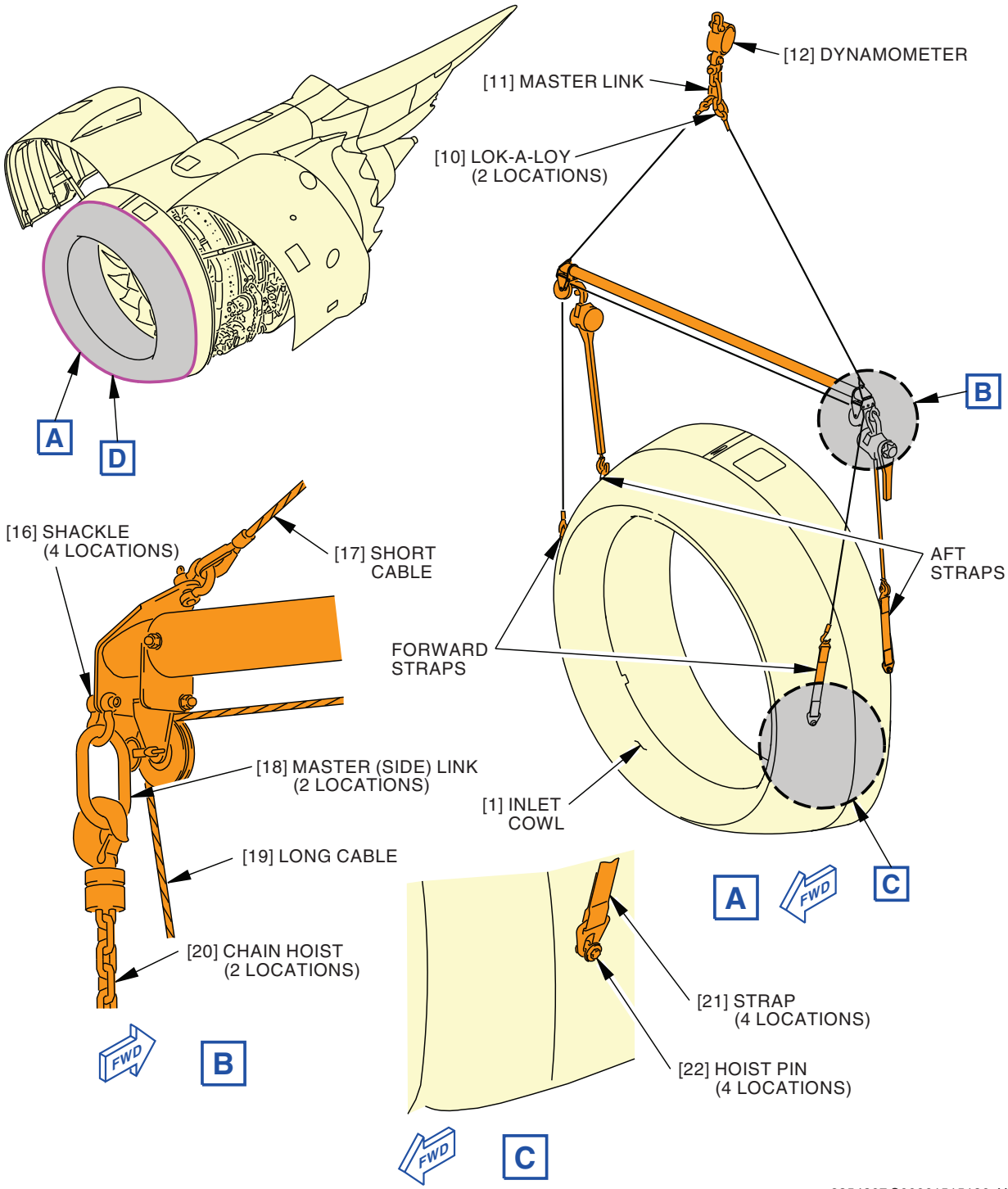
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



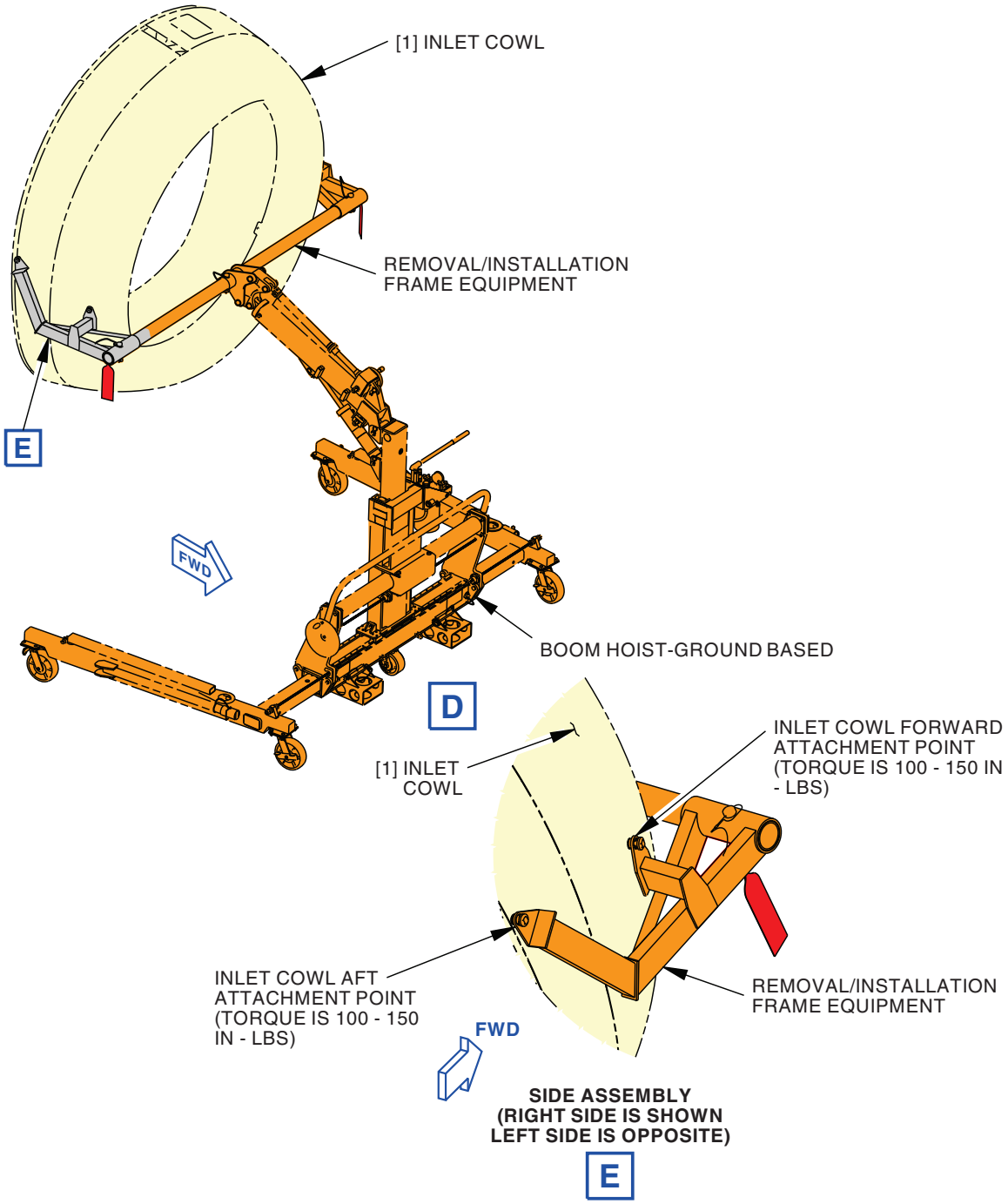
Engine Anti-Ice Duct Installation
Figure 401/71-11-01-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inlet Cowl Tool Installation
Figure 402/71-11-01-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inlet Cowl Tool Installation
Figure 402/71-11-01-990-802-G00 (Sheet 2 of 2)

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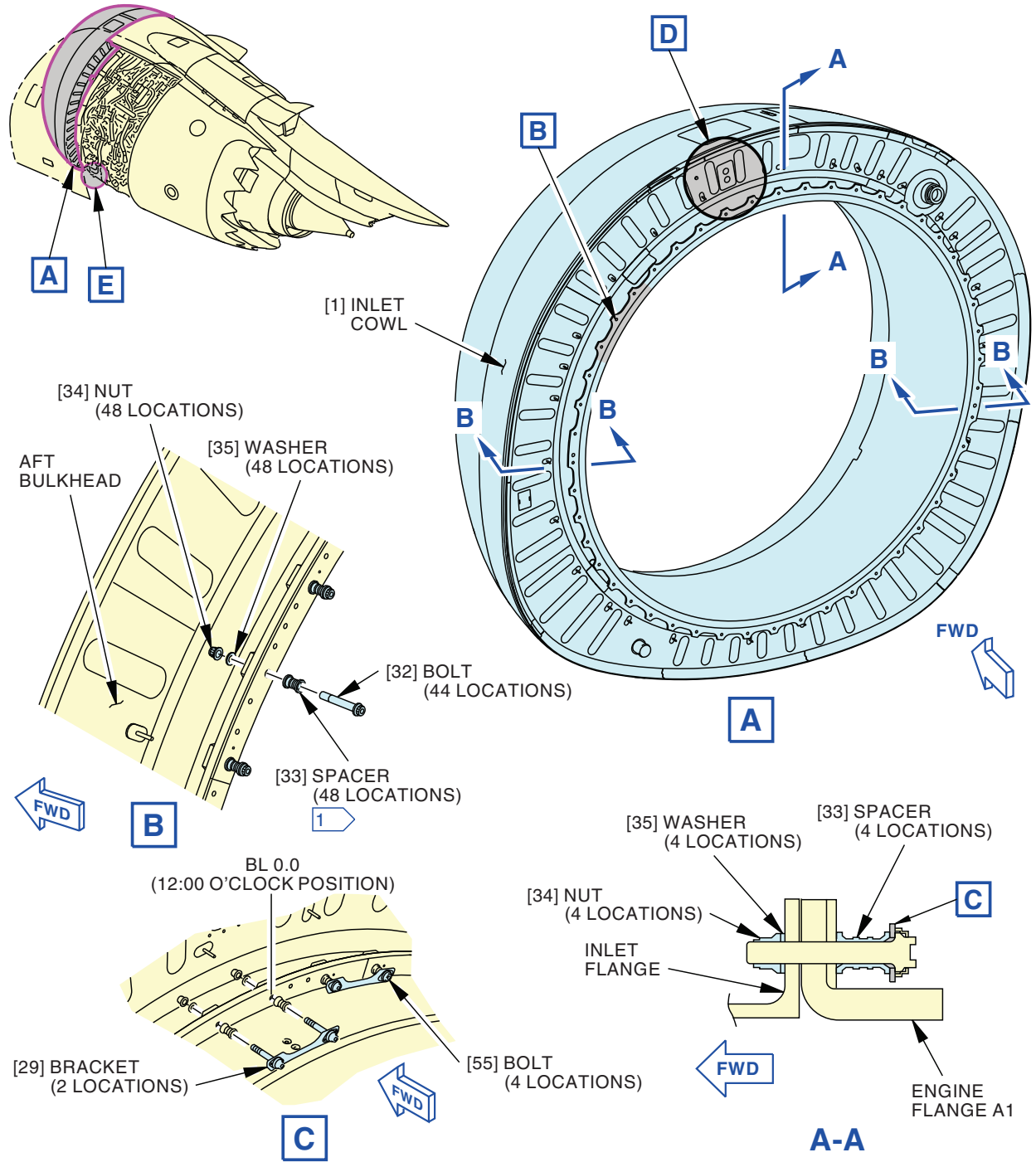

1 INCLUDE 4 LOCATIONS OF BOLT RETAINER ASSEMBLY (BRACKET [29])

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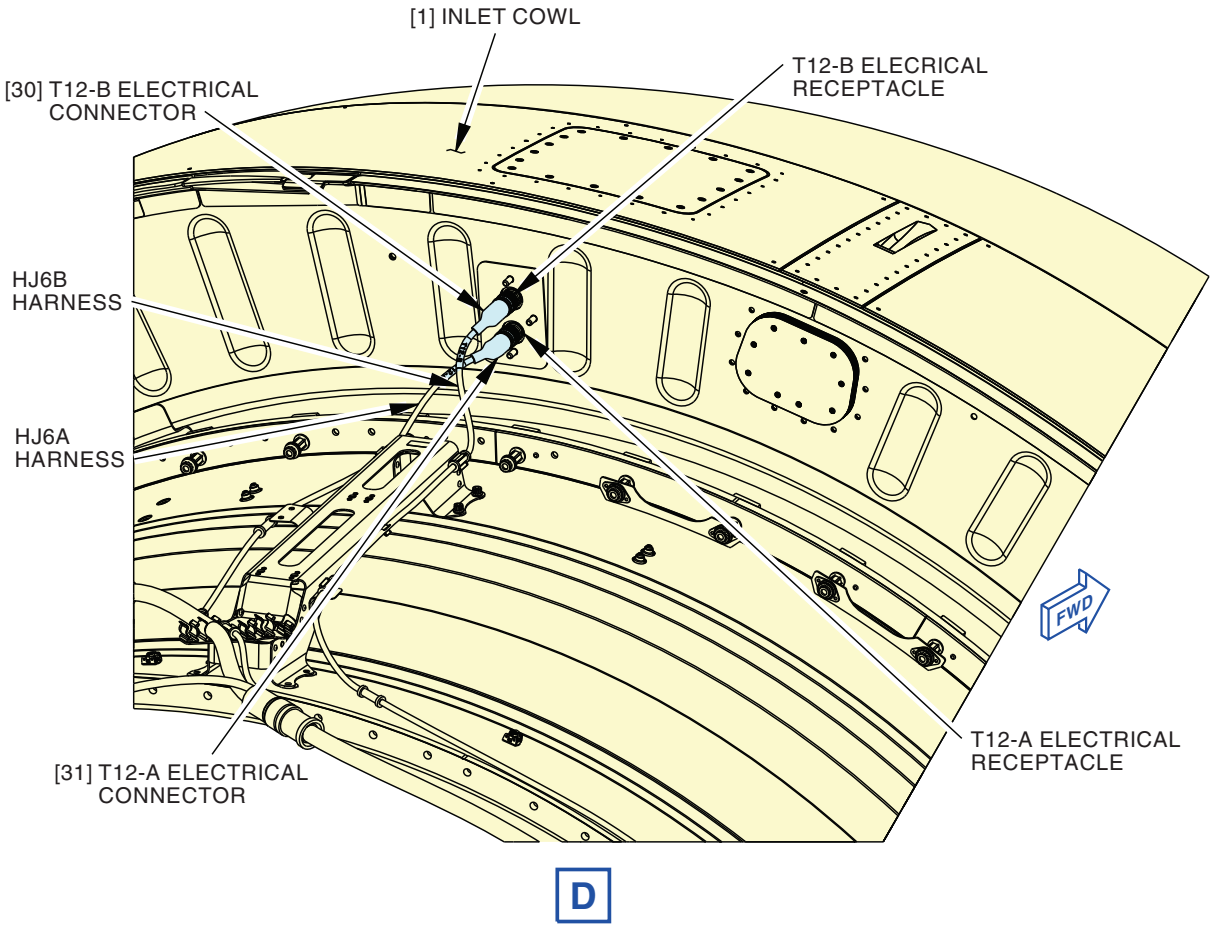
Inlet Cowl Installation
Figure 403/71-11-01-990-803-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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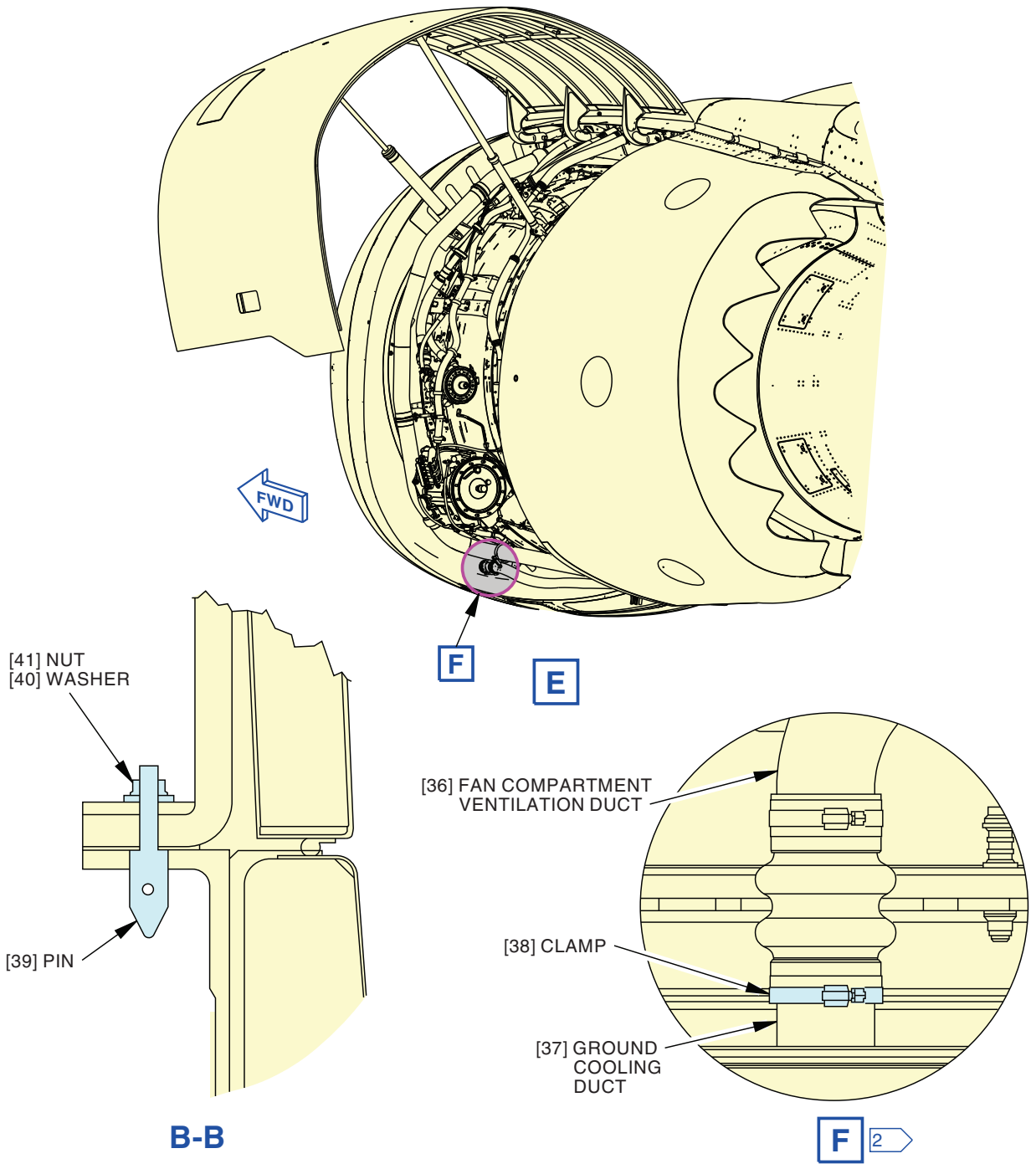
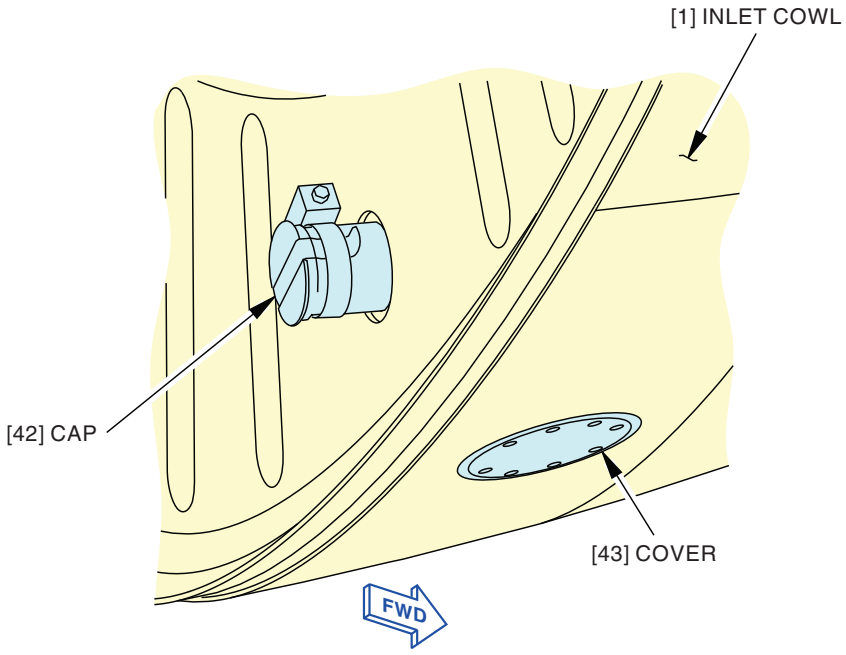
Inlet Cowl Installation
Figure 403/71-11-01-990-803-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inlet Cowl Installation
Figure 403/71-11-01-990-803-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



F 3

3 LEAP-1BXXG05 ENGINES WITH THE EEC BLOWER SYSTEM REMOVED (POST LEAP-1B SERVICE BULLETIN 75-0010 AND POST LEAP-1B SERVICE BULLETIN 75-0011)

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Inlet Cowl Installation
Figure 403/71-11-01-990-803-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-01-400-801-G00

3. Inlet Cowl Installation

(Figure 401, Figure 402, Figure 403)

A. General

(1) This task gives the instructions to install the inlet cowl.

NOTE: The total acoustic panel blockage for the engine installation must be recalculated when components are changed per SRM SUBJECT 54-00-01.

(2) You must use the inlet cowl sling, SPL-2062, to install the inlet cowl.

(a) You may use installation/removal frame equipment, SPL-14544, together with boom hoist, SPL-2430, as an optional method to install the inlet cowl.

(b) The optional method is useful where overhead lifting is not possible. This tool combination can also be easily transported as needed.

PTF ALL; AIRPLANES WITH INLET COWL WITH GROUND COOLING DUCT

(3) If the inlet is to be installed on a LEAP-1BXXG05 engine, the cap and cover must be removed from the ground cooling duct.

(4) If the inlet is to be installed on a LEAP-1BXXG06 engine or higher GXX number engine, the inlet must have the cap and cover installed on the ground cooling duct.

PTF ALL

B. References

Reference	Title
54-52-01-410-801	Forward Fairing Installation (P/B 401)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-00-03-390-801-G00	Inlet Cowl Lipskin Protective Coating Procedure (P/B 201)
71-11-01-000-802-G00	Inlet Ground Cooling Duct Cap and Cover Removal (P/B 201)
71-11-01-200-801-G00	Inlet Cowl Inspection (P/B 601)
71-11-01-211-801-G00	Inlet Cowl Attachment Ring Detailed Inspection (P/B 601)
71-11-01-400-802-G00	Inlet Ground Cooling Duct Cap and Cover Installation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-06-400-801-G00	T12 Sensor Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2062	Sling - Inlet Cowl Part #: B71040-66 Supplier: 81205 Opt Part #: B71040-38 Supplier: 81205 Opt Part #: B71040-39 Supplier: 81205
SPL-2430	Hoist ? Boom, 800 lbs WLL, 250 lbs Load Positioner Compatible Part #: C20002-267 Supplier: 81205
SPL-14544	Installation/Removal Frame Equipment - Inlet Cowl, LEAP-1B Engine Part #: C71033-1 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
STD-583	Mallet - Non-metallic
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-748	Rope - 12 ft (3.6 m) minimum
STD-858	Tag - DO NOT OPERATE
STD-1095	Crane - Lift, 2000 lb Capacity, 30 Foot Height

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Inlet cowl	71-11-01-01-200	PTF ALL
		71-11-01-01-205	PTF ALL
4	Seal	30-21-12-01-355	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Prepare for the Installation

SUBTASK 71-11-01-490-003-G00

(1) If the sling equipment is not assembled, use the sling assemble steps from the removal task to assemble the inlet cowl sling, SPL-2062.

SUBTASK 71-11-01-480-006-G00

(2) Install the inlet cowl sling, SPL-2062, on the inlet cowl [1] (Figure 402 (Sheet 1)).

(a) Install the forward and aft straps [21] with hoist pins [22] to the inlet cowl [1].

(b) Install the chain hoists [20] to the aft straps [21].

(c) With the 30 foot height (2000 lb capacity) lift crane, STD-1095, lift the inlet cowl sling, SPL-2062, and put it over the inlet cowl [1].

(d) Connect the chain hoists [20] to the master (side) links [18]

(e) Connect the long cable [19] to the forward strap [21].

SUBTASK 71-11-01-480-002-G00

(3) If the optional installation tool is used, make sure that the installation/removal frame equipment, SPL-14544, is attached to the boom hoist, SPL-2430.

SUBTASK 71-11-01-480-007-G00

(4) If the optional removal tool is used, attach the installation/removal frame equipment, SPL-14544, to the inlet cowl [1] (Figure 402 (Sheet 2))

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-01-420-003-G00

(5) If the alignment pins [39] were removed or were not installed on the fan case, install the two alignment pins [39], two washers [40], and two nuts [41] on the fan case flange at the 3:00 and 9:00 o'clock positions.

(a) Tighten the two nuts [41] to 52.0 in-lb (5.9 N·m) - 68.0 in-lb (7.7 N·m).

H. Inlet Cowl Installation

SUBTASK 71-11-01-200-001-G00

(1) Examine the inlet cowl (TASK 71-11-01-200-801-G00).

SUBTASK 71-11-01-211-002-G00

(2) Examine the inlet cowl attachment ring (TASK 71-11-01-211-801-G00).

NOTE: Scheduled maintenance requires this inspection when you remove and install the inlet cowl.

SUBTASK 71-11-01-490-004-G00

WARNING

GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

WARNING

MAKE SURE THAT THE WEIGHT OF THE INLET COWL AND THE SLING EQUIPMENT IS BALANCED WITH THE CRANE FORCE. IF THE WEIGHT AND FORCE ARE NOT BALANCED, THE COWL CAN MOVE QUICKLY WHEN YOU DISCONNECT THE INLET COWL. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Lift the inlet cowl [1] with the inlet cowl sling, SPL-2062, from the ground platform.

(a) Adjust the 30 foot height (2000 lb capacity) lift crane, STD-1095, and straps [21] until the straps [21] are not loose.

(b) Install a rope - 12 ft (3.6 m) minimum, STD-748, around the inlet cowl [1] to guide the movement of the inlet cowl [1].

(c) Lift the inlet cowl [1] until you have sufficient clearance to turn the inlet cowl [1].

1) Use the chain hoists to turn the inlet cowl [1] to the vertical position.

SUBTASK 71-11-01-480-003-G00

(4) If the optional installation tool is used, lift the inlet cowl [1] with the installation/removal frame equipment, SPL-14544, and boom hoist, SPL-2430, from the ground platform.

SUBTASK 71-11-01-030-001-G00

(5) Install the inlet cowl [1] as follows:

(a) Position the inlet cowl [1] in front of the fan case flange.

(b) Move the inlet cowl [1] slowly to the fan case flange for alignment.

(c) Remove the rope - 12 ft (3.6 m) minimum, STD-748, from the inlet cowl [1] when the inlet cowl sling, SPL-2062, is used.

(d) Install the 44 bolts [32], 44 spacers [33], 44 washers [35], and 44 nuts [34] that attach the inlet cowl [1] to the fan case flange.

NOTE: The bolts must not be lubricated.

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1) Install the two brackets [29] on the 12:00 o'clock position with the four bolts [55], four spacers [33], four washers [35], and four nuts [34].

NOTE: The bolts must not be lubricated.

2) Snug fit four bolts [32], four spacers [33], four washers [35], and four nuts [34] to 150.0 ±5.0 in-lb (16.9 ±0.6 N·m) in the following sequence: 3:00, 9:00, 6:00 and 12:00 o'clock positions.

3) Snug fit the 40 bolts [32], 40 spacers [33], 40 washers [35], and 40 nuts [34] to 150.0 ±5.0 in-lb (16.9 ±0.6 N·m).

4) Tighten four nuts [34] to 290.0 in-lb (32.8 N·m) - 310.0 in-lb (35.0 N·m) in the following sequence: 3:00, 9:00, 6:00 and 12:00 o'clock positions.

5) Tighten the remaining 44 nuts [34] to 290.0 in-lb (32.8 N·m) - 310.0 in-lb (35.0 N·m).

a) Do a check of the torque at first nut [34], if it is not within the specified range, tighten all nuts [34] again and re-check.

(e) Disconnect the straps [21] from the inlet cowl [1] when the inlet cowl sling, SPL-2062, is used.

1) Remove the hoist pins [22] that attach the straps [21] to the inlet cowl [1].

(f) If the optional installation tool is used, disconnect the installation/removal frame equipment, SPL-14544, from the inlet cowl [1].

SUBTASK 71-11-01-940-001-G00

(6) Move the 30 foot height (2000 lb capacity) lift crane, STD-1095, with the inlet cowl sling, SPL-2062, away from the airplane.

SUBTASK 71-11-01-940-002-G00

(7) If the optional installation tool is used, move the installation/removal frame equipment, SPL-14544, with the boom hoist, SPL-2430, away from the airplane.

SUBTASK 71-11-01-430-002-G00

(8) Connect the engine anti-ice duct [2] at the 1:00 o'clock position.

(a) Remove the protective caps from the ends of the engine anti-ice duct [2].

(b) Examine the new seal [4] and surfaces of the engine anti-ice duct [2] for scratches, cuts, pits, creases, and unwanted material.

(c) Install the seal [4] in the recess of the engine anti-ice duct [2].

1) Align the engine anti-ice duct [2] and duct of the inlet cowl [1].

(d) Install the coupling [3] that attaches the engine anti-ice duct [2] to the inlet cowl [1].

1) Turn the coupling bolt until it is outboard of the fan case.

2) Tighten the coupling [3] to 115.0 in-lb (13.0 N·m) - 125.0 in-lb (14.1 N·m).

3) Lightly hit the outer diameter of the coupling [3] with a non-metallic mallet, STD-583.

4) Tighten the coupling [3] again to 115.0 in-lb (13.0 N·m) - 125.0 in-lb (14.1 N·m).

PTF ALL; LEAP-1B ENGINES WITH THE EEC BLOWER SYSTEM INSTALLED (PRE LEAP-1B SERVICE BULLETIN 75-0010 AND PRE LEAP-1B SERVICE BULLETIN 75-0011)

SUBTASK 71-11-01-420-001-G00

(9) Connect the ground cooling duct [37] to the fan compartment ventilation duct [36] at the 7:00 o'clock position.

NOTE: Do not install inlet cowl without the ground cooling duct, on a LEAP-1BXXG05 engine.

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PTF ALL; LEAP-1B ENGINES WITH THE EEC BLOWER SYSTEM INSTALLED (PRE LEAP-1B SERVICE BULLETIN 75-0010 AND PRE LEAP-1B SERVICE BULLETIN 75-0011) (Continued)

(a) If it is installed, remove the cap and cover from the inlet ground cooling duct [37] (TASK 71-11-01-000-802-G00).

(b) Remove the protective cap from the end of the fan compartment ventilation duct [36].

(c) Install the clamp [38] that attaches the fan compartment ventilation duct [36] to the inlet cowl [1].

1) Tighten the clamp [38] to 19.0 in-lb (2.1 N·m) - 21.0 in-lb (2.4 N·m).

PTF ALL; LEAP-1B ENGINES WITH THE EEC BLOWER SYSTEM REMOVED (POST LEAP-1B SERVICE BULLETIN 75-0010 OR POST LEAP-1B SERVICE BULLETIN 75-0011)

SUBTASK 71-11-01-410-006-G00

(10) Make sure that the cap and cover are installed on the ground cooling duct (TASK 71-11-01-400-802-G00).

NOTE: This step is for inlet cowl with the ground cooling duct installed.

PTF ALL

SUBTASK 71-11-01-430-003-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS OR RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(11) Use a teflon-jawed pliers, STD-664, to connect the T12-B electrical connectors [30] and T12-A electrical connector [31] at the 11:30 o'clock position (TASK 73-21-06-400-801-G00).

(a) Remove the protective caps from the T12-B electrical connector [30], T12-A electrical connector [31], T12-B and T12-A receptacles.

(b) Connect the T12-B electrical connector [30] and T12-A electrical connector [31] to the T12-B and T12-A receptacles (View D, Figure 403).

NOTE: The T12-B and T12-A receptacles masterkey are oriented upwards.

(c) Turn the coupling rings on the electrical connectors 1/8 turn or until the teflon-jawed pliers, STD-664, slippage occurs.

NOTE: Do not over tighten the coupling ring.

SUBTASK 71-11-01-630-001-G00

CAUTION

MAKE SURE THAT YOU FOLLOW THE INLET COWL LIPSKIN PROTECTIVE COATING PROCEDURE BEFORE YOU OPERATE THE ENGINE ANTI ICE SYSTEM. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU FOLLOW THE INLET COWL LIPSKIN PROTECTIVE COATING PROCEDURE BEFORE THE AIRPLANE IS RETURNED TO SERVICE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(12) If necessary, remove the inlet cowl lipskin protective coating (Inlet Cowl Lipskin Protective Coating Procedure, TASK 71-00-03-390-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-01-420-007-G00

(1) Install the thumbnail fairing (TASK 54-52-01-410-801).

SUBTASK 71-11-01-410-002-G00

(2) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-01-860-002-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Inlet Cowl Installation Test

SUBTASK 71-11-01-710-002-G00

(1) Do the applicable test(s) given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INLET COWL - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Inlet Cowl Inspection

(2) Inlet Cowl Attachment Ring Detailed Inspection

TASK 71-11-01-200-801-G00

2. Inlet Cowl Inspection

(Figure 601)

A. General

(1) This task provides the instructions on how to do a visual inspection of the inlet cowl delamination and damage to the areas that follow:

(a) Lip skin

(b) Inner barrel panel

(c) Outer barrel panel

(d) Aft bulkhead

(e) Anti-ice exhaust duct assembly.

B. References

Reference	Title
71-11-01-000-801-G00	Inlet Cowl Removal (P/B 401)
71-11-01-400-801-G00	Inlet Cowl Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 71-11-01-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

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1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-01-010-005-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Inlet Cowl Inspection

SUBTASK 71-11-01-220-001-G00

(1) Examine the inlet cowl lip skin as follows:

(a) Cracks, holes, punctures and loose or missing fasteners

1) If there are cracks, holes, punctures, repair or replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

2) If there are loose or missing fasteners, install and/or tighten the fasteners (TASK 71-11-01-400-801-G00).

(b) Dents, disbond, delamination, nicks, scratches, or gouges

1) For the allowable damage limits for the lip skin, refer to the Structure Repair Manual (SRM).

NOTE: The SRM gives allowable limits for dents, disbond, delamination, nicks, scratches, or gouges of the lipskin.

2) If you find dents, disbond, delamination, nicks, scratches, or gouges, above the allowable limits, replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

SUBTASK 71-11-01-220-002-G00

(2) Examine the inlet cowl inner barrel panel as follows:

(a) Cracks, holes, punctures and loose or missing fasteners

1) If there are cracks, holes, punctures, repair or replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

2) If there are loose or missing fasteners, install and/or tighten the fasteners (TASK 71-11-01-400-801-G00).

(b) Dents, disbond, delamination, nicks, scratches, or gouges

1) For the allowable damage limits for the inner barrel panel, refer to the Structure Repair Manual (SRM).

NOTE: The SRM gives allowable limits for dents, disbond, delamination, nicks, scratches, or gouges of the inner barrel panel.

2) If you find dents, disbond, delamination, nicks, scratches, or gouges, above the allowable limits, replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

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SUBTASK 71-11-01-220-003-G00

(3) Examine the inlet cowl outer barrel panel as follows:

(a) Cracks, holes, punctures and loose or missing fasteners

1) If there are cracks, holes, punctures, repair or replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

2) If there are loose or missing fasteners, install and/or tighten the fasteners (TASK 71-11-01-400-801-G00).

(b) Dents, disbond, delamination, nicks, scratches, or gouges

1) For the allowable damage limits for the outer barrel panel, refer to the Structure Repair Manual (SRM).

NOTE: The SRM gives allowable limits for dents, disbond, delamination, nicks, scratches, or gouges of the outer barrel panel.

2) If you find dents, disbond, delamination, nicks, scratches, or gouges, above the allowable limits, replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

SUBTASK 71-11-01-210-001-G00

(4) Examine the inlet cowl aft bulkhead as follows:

(a) Cracks, holes, punctures, and loose or missing fasteners

1) If there are cracks, holes, punctures, repair or replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

2) If there are loose or missing fasteners, install and/or tighten the fasteners (TASK 71-11-01-400-801-G00).

(b) Dents, nicks, scratches, gouges, or corrosion

1) For the allowable damage limits for the aft bulkhead, refer to the Structure Repair Manual (SRM).

NOTE: The SRM gives allowable limits for dents, nicks, scratches, gouges, or corrosion of the aft bulkhead.

2) If you find dents, nicks, scratches, gouges, or corrosion above the allowable limits, replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

SUBTASK 71-11-01-210-002-G00

(5) Examine the inlet cowl anti-ice exhaust duct assembly as follows:

(a) If the anti-ice exhaust duct assembly is damaged, repair or replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

SUBTASK 71-11-01-220-004-G00

(6) Examine the inlet cowl for delamination as follows:

(a) Delamination of the perforated face sheet is permitted if all the limits that follow are satisfied:

1) The delamination is more than 3.0 inches (76.2 mm) from the edges.

2) Each area of delamination is less than 4.00 inches by 4.00 inches (101.6 x 101.6 mm).

3) The total area of the delamination is less than 50.0 square-inches (1270 square-mm).

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(b) If the delamination is in the permitted limits, do the followed-up maintenance scheduled inspection at the damaged areas.

(c) If the delamination is not in the permitted limits, repair the delamination or replace the inlet cowl (TASK 71-11-01-000-801-G00 and TASK 71-11-01-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-01-410-004-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-01-860-004-G00

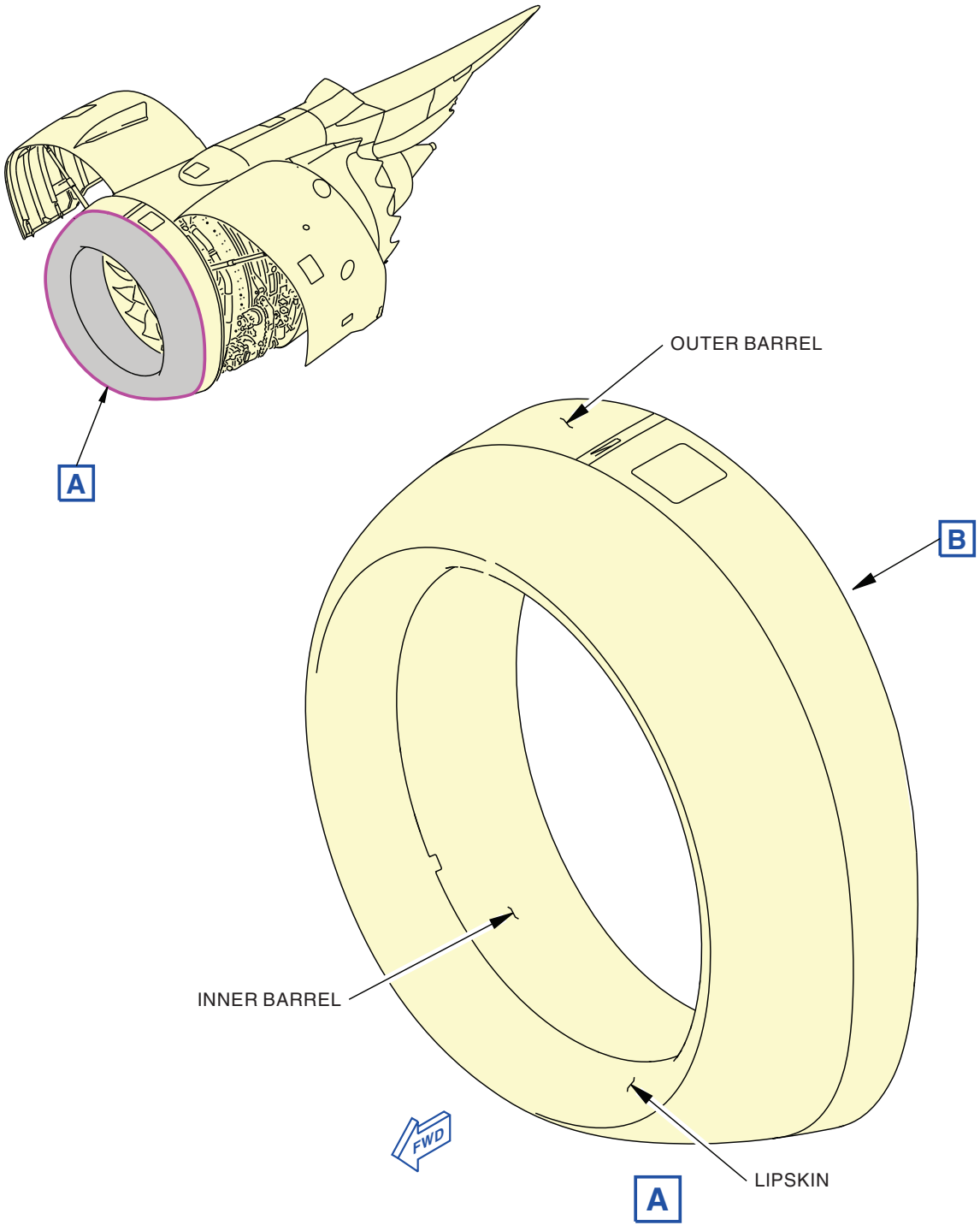
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

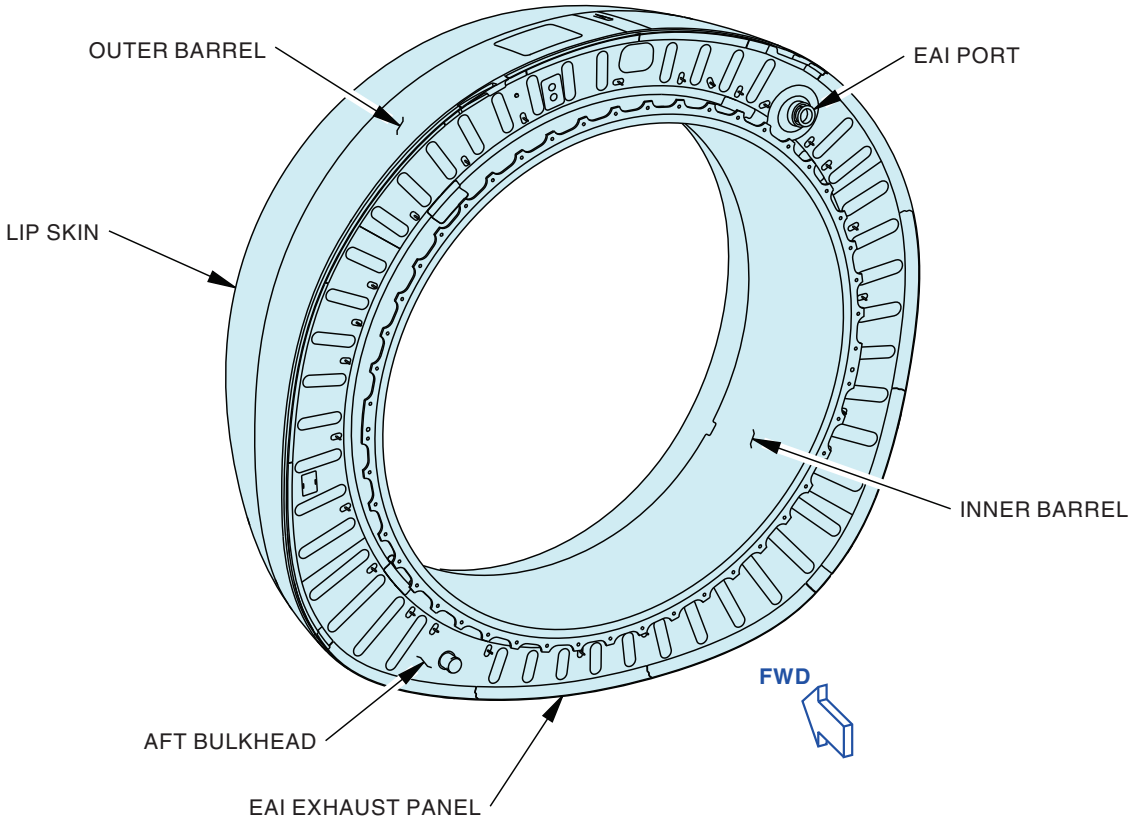
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inlet Cowl Inspection
Figure 601/71-11-01-990-805-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Inlet Cowl Inspection
Figure 601/71-11-01-990-805-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-01-211-801-G00

3. Inlet Cowl Attachment Ring Detailed Inspection

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task provides the instructions on how to do a detailed inspection of the inlet cowl attachment ring.

(2) This inspection requires removal of the inlet cowl.

B. References

Reference	Title
71-11-01-000-801-G00	Inlet Cowl Removal (P/B 401)
71-11-01-400-801-G00	Inlet Cowl Installation (P/B 401)

C. Location Zones

Zone	Area
412	Engine 1 - Nose Inlet Cowl
422	Engine 2 - Nose Inlet Cowl

D. Prepare for the Inspection

SUBTASK 71-11-01-000-001-G00

(1) To remove the inlet cowl, do this task: Inlet Cowl Removal, TASK 71-11-01-000-801-G00.

E. Inlet Cowl Attachment Ring Detailed Inspection

SUBTASK 71-11-01-211-001-G00

(1) Do a detailed inspection of the inlet cowl attachment ring for cracks.

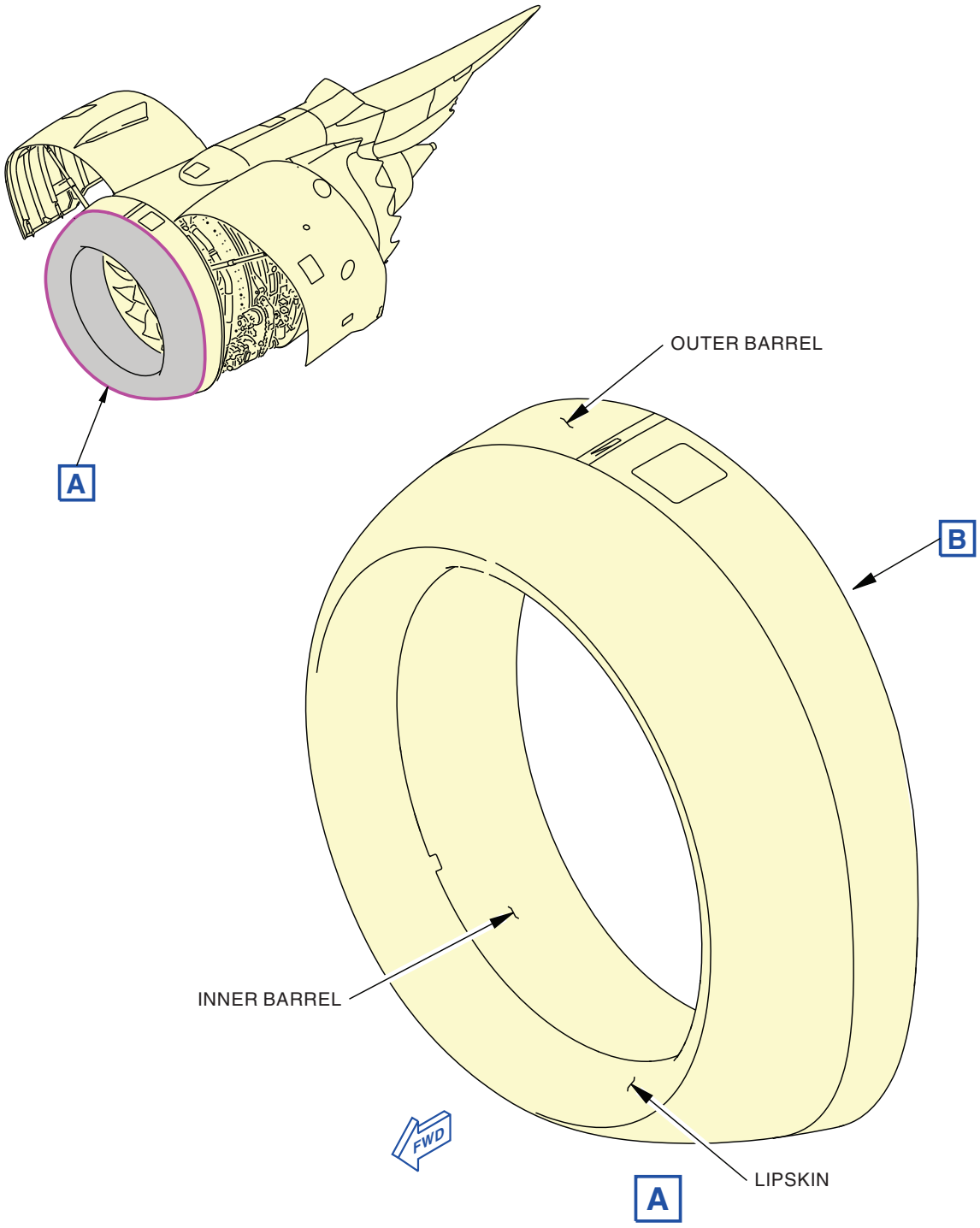
F. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-01-420-002-G00

(1) To install the inlet cowl, do this task: Inlet Cowl Installation, TASK 71-11-01-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inlet Cowl Attachment Ring Detailed Inspection
Figure 602/71-11-01-990-807-G00 (Sheet 1 of 2)

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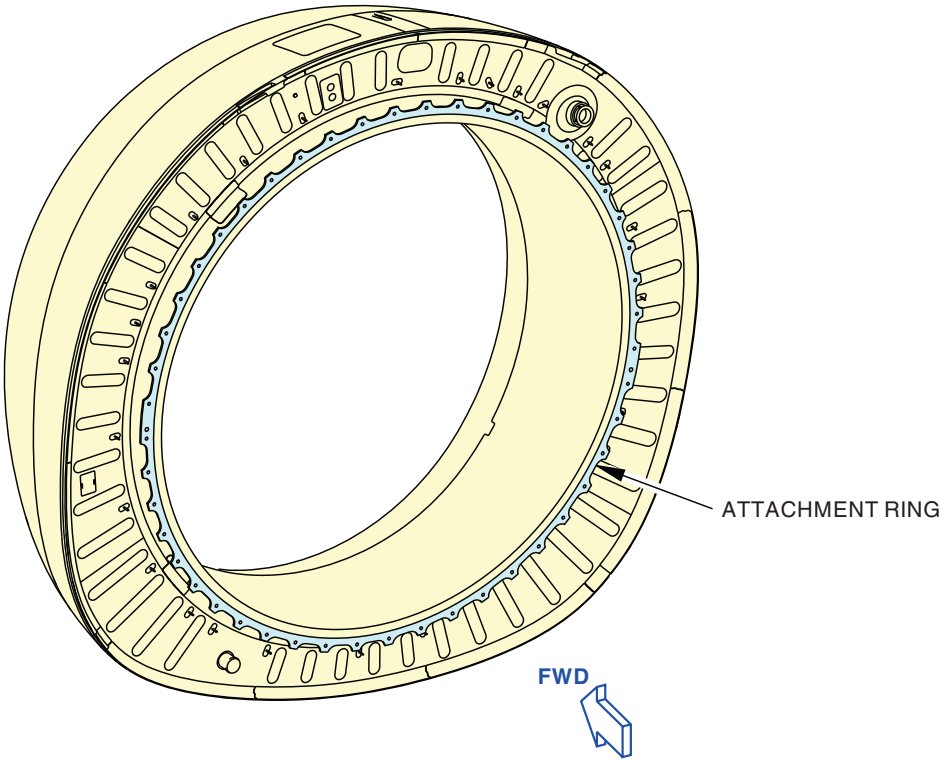

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Inlet Cowl Attachment Ring Detailed Inspection
Figure 602/71-11-01-990-807-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL CHINE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fan cowl chine removal

(2) Fan cowl chine installation.

TASK 71-11-02-000-801-G00

2. Fan Cowl Chine Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the fan cowl chine.

(2) The fan cowl chine is installed on the inboard of the fan cowl panel.

(3) The fan cowl chines are not interchangeable on the fan cowl panel.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-11852	Rapid Desealer System Part #: RDS001 Supplier: 3FM31
STD-653	Platform - Engine and Strut Access
STD-765	Scraper - Plastic
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Removal

SUBTASK 71-11-02-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-02-010-001-G00

(2) Open the applicable fan cowl panel (TASK 71-11-04-010-801-G00).

SUBTASK 71-11-02-490-001-G00

(3) Position the engine and strut access platform, STD-653 to get access to remove the fan cowl chine [1].

G. Fan Cowl Chine Removal

SUBTASK 71-11-02-020-001-G00

(1) Remove the fan cowl chine [1] from the applicable fan cowl panel as follows:

(a) Use a plastic scraper, STD-765 to cut the sealant around the edge of the fan cowl chine [1].

(b) Remove the eight bolts [2], the eight collars [3], and the eight washers [4] that attach the fan cowl chine [1] to the fan cowl self-locking nuts on the fan cowl panel.

(c) Remove the two bolts [5], the two collars [6], and the two washers [4] that attach the fan cowl chine [1] to the fan cowl self-locking nuts on the fan cowl panel.

NOTE: The first upper bolt of the tip of the fan cowl chine and the last bottom bolt of the end of the fan cowl chine.

SUBTASK 71-11-02-160-001-G00

(2) Remove the fay sealant from the fan cowl chine [1] and the skin interface with the rapid desealer, COM-11852.

SUBTASK 71-11-02-410-003-G00

(3) If the fan cowl panel will not install the fan cowl chine [1], do the steps that follow:

(a) Make sure that the surfaces of the fan cowl panel and the fan cowl chine holes are clean.

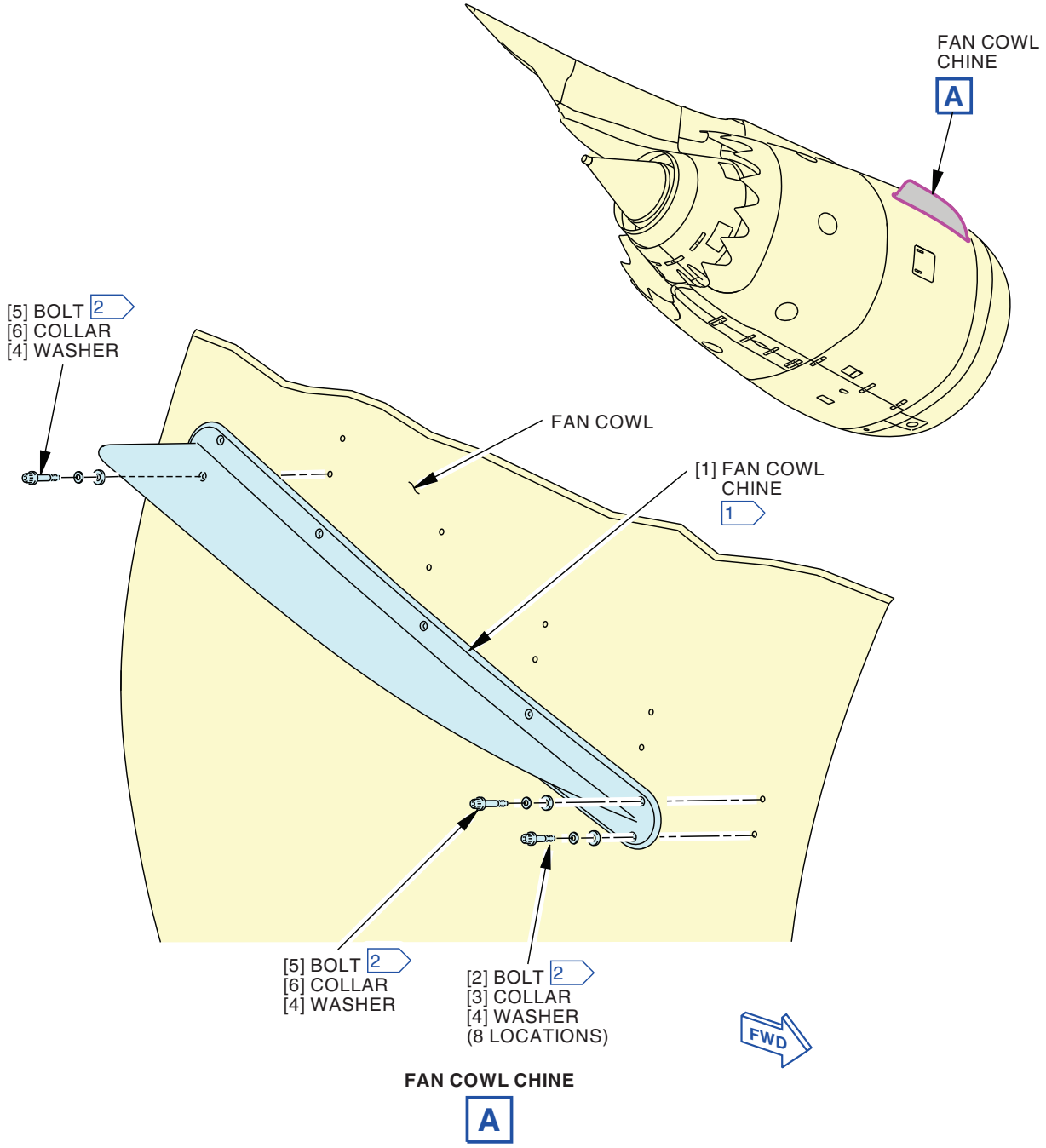
(b) Install the plug fasteners into the open holes on the skin.

(c) Apply the sealant, A50110 to the plug fasteners.

(d) Let the sealant, A50110 dry for 30 minutes.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 THE FAN COWL CHINE IS INSTALLED ON THE INBOARD SIDE OF THE FAN COWL PANEL. PANEL. THE FAN COWL CHINES ARE NOT INTERCHANGEABLE ON THE FAN COWL.

2 DESIGNATED BOND FASTENER RESISTANCE CHECK.

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Fan Cowl Chine Installation
Figure 401/71-11-02-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-02-400-801-G00

3. Fan Cowl Chine Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the fan cowl chine.

(2) The fan cowl chine is installed on the fan cowl on the inboard of the fan cowl panel.

(3) The fan cowl chines are not interchangeable on the fan cowl panel.

B. References

Reference	Title
51-22-00-160-801	Prepare For Sealing (P/B 201)
51-22-00-390-802	Removable Faying (Mated) Surface Seal Application (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fan cowl chine	71-11-02-01-020	PTF ALL
		71-11-02-05-020	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Fan Cowl Chine Installation

SUBTASK 71-11-02-010-002-G00

(1) If the fan cowl panel had the fan cowl chine [1] plug fasteners installed, remove the plug fasteners from the fan cowl chine [1] holes on the skin.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-02-410-001-G00

(2) Do this task: Prepare For Sealing, TASK 51-22-00-160-801.

(a) Make sure that the surfaces of the fan cowl panel and the fan cowl chine are clean.

SUBTASK 71-11-02-420-001-G00

(3) Install the fan cowl chine [1] on the applicable fan cowl panel as follows:

NOTE: The fan cowl chine is installed on the inboard of the fan cowl panel. The fan cowl chines are not interchangeable on the fan cowl panel.

(a) Apply the part agent and the fay sealant to the fan cowl chine [1] and the skin interface (TASK 51-22-00-390-802).

NOTE: Do not remove the existing finish from the fan cowl panel.

(b) Put the fan cowl chine [1] on the surface of the fan cowl panel.

(c) Align the fan cowl chine bolt holes with the fan cowl panel bolt holes.

(d) Install the two bolts [5], the two collars [6], and the two washers [4] that attach the fan cowl chine [1] to the fan cowl self-locking nuts on the fan cowl panel.

NOTE: The first upper bolt of the tip of the fan cowl chine and the last bottom bolt of the end of the fan cowl chine.

NOTE: Install the washer under the collar.

(e) Install the eight bolts [2], the eight collars [3], and the eight washers [4] that attach the fan cowl chine [1] to the fan cowl self-locking nuts on the fan cowl panel.

NOTE: Install the washer under the collar.

(f) Tighten the bolts [2] and the bolts [5] to 87.3 in-lb (9.86 N·m) - 92.7 in-lb (10.47 N·m).

SUBTASK 71-11-02-760-001-G00

(4) Do the designated bond fastener resistance checks:

(a) Use an intrinsically safe approved bonding meter, COM-1550, to do a check of the bonding resistance between the fastener heads (bolts [5] and bolts [2]) and the structure (SWPM 20-20-00).

1) Make sure that the bonding resistance between the fastener head and the structure is 1.0 milliohm (0.001 ohm) or less.

(b) Use an intrinsically safe approved bonding meter, COM-1550, to do a check of the bonding resistance between the collars [3], collars [6], and the structure (SWPM 20-20-00).

1) Make sure that the bonding resistance between the locknut/collar and the structure where the locknut/collar is contacted is 1.5 milliohm (0.0015 ohm) or less.

SUBTASK 71-11-02-390-001-G00

(5) Apply the sealant, A50110 to the fan cowl chine [1] as follows:

(a) Apply the sealant, A50110 all around the fan cowl chine [1] surface interface to the fan cowl panel (TASK 51-22-00-390-806).

(b) Apply the sealant, A50110 to the countersunk areas on the heads of the bolts [2] and the bolts [5].

(c) Let the sealant, A50110 dry for 30 minutes.

SUBTASK 71-11-02-090-001-G00

(6) Move the engine and strut access platform, STD-653 away from the engine.

(a) Make sure that the work area is clean of all tools and other items.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-02-410-002-G00

(1) Close the applicable fan cowl panel (TASK 71-11-04-410-801-G00).

SUBTASK 71-11-02-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL CROSS ENGINE DEBRIS SHIELD - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the fan cowl cross engine debris shield

(2) An installation of the fan cowl cross engine debris shield.

TASK 71-11-03-000-801-G00

2. Fan Cowl Cross Engine Debris Shield Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fan cowl cross engine debris shields.

(2) The fan cowl cross engine debris shields are installed only on the inboard fan cowl panel of engine 1.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 71-11-03-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-03-010-001-G00

(2) Open the applicable right fan cowl panel (TASK 71-11-04-010-801-G00).

SUBTASK 71-11-03-480-001-G00

(3) Position the engine and strut access platform, STD-653 to get access to remove the fan cowl cross engine debris shields.

F. Fan Cowl Cross Engine Debris Shield Removal

SUBTASK 71-11-03-020-001-G00

(1) Remove the forward debris shield [1] from the applicable right fan cowl panel as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the bolt [4] and washer [5] at eight locations.

(b) Remove the bolt [6] and washer [7] at two locations.

(c) Remove the forward debris shield [1].

SUBTASK 71-11-03-020-002-G00

(2) Remove the aft upper debris shield [2] from the applicable right fan cowl panel as follows:

(a) Remove the bolt [8] and washer [9] at eight locations.

(b) Remove the aft upper debris shield [2].

SUBTASK 71-11-03-020-003-G00

(3) Remove the aft lower debris shield [3] from the applicable right fan cowl panel as follows:

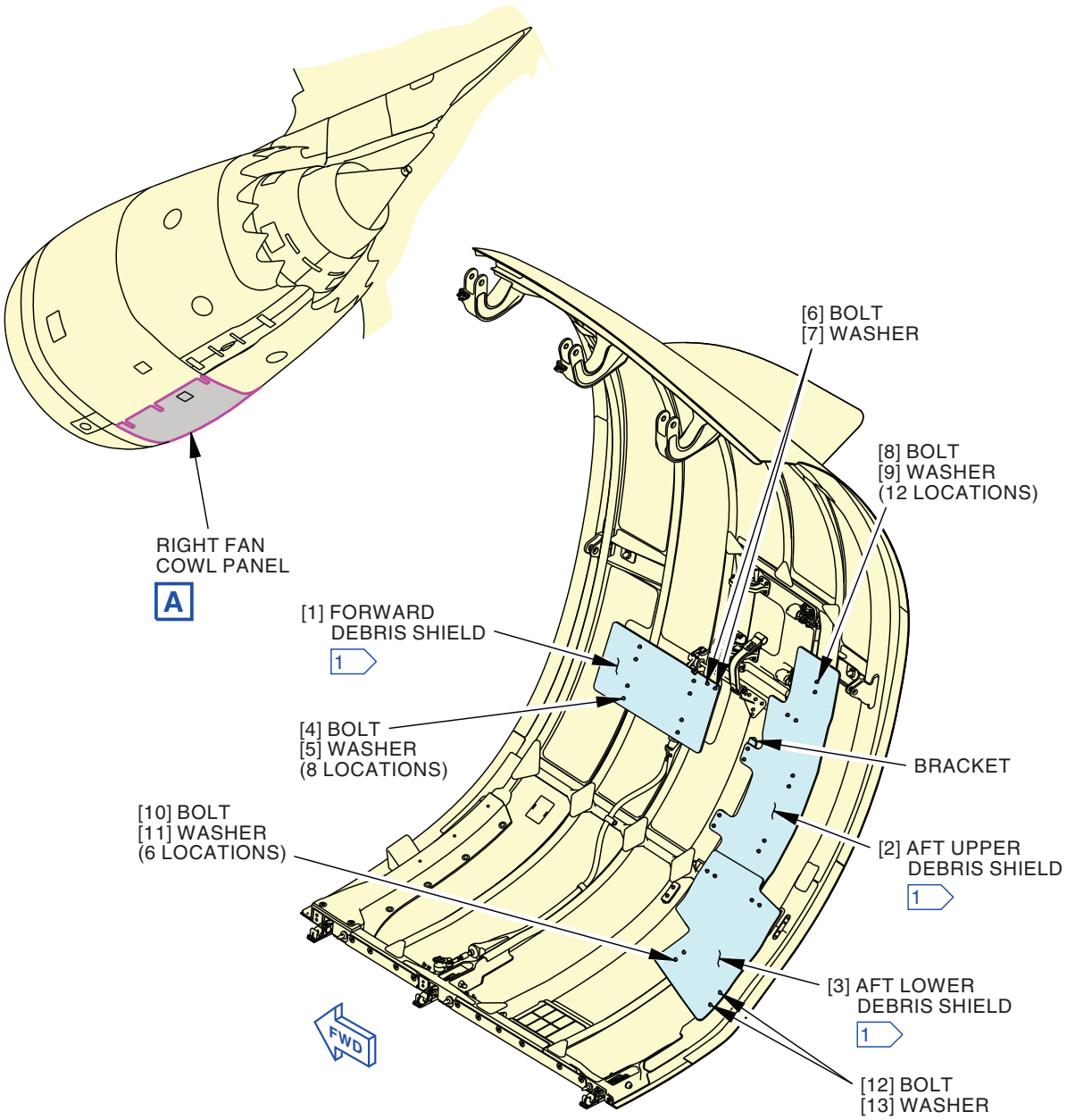
(a) Remove the bolt [12] and washer [11] at eight locations.

(b) Remove the bolt [12] and washer [13] at two locations.

(c) Remove the aft upper debris shield [2].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIGHT FAN COWL PANEL (TYPICAL)

A

1 THE FAN COWL CROSS ENGINE DEBRIS SHIELDS ARE INSTALLED ONLY ON THE INBOARD FAN COWL PANEL OF ENGINE 1.

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Fan Cowl Cross Engine Debris Shield Installation
Figure 401/71-11-03-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-03-400-801-G00

3. Fan Cowl Cross Engine Debris Shield Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fan cowl cross engine debris shields.

(2) The fan cowl cross engine debris shields are installed only on the inboard fan cowl panel of engine 1.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
G50337	Shim - Laminated Polyimide, Square And Rectangular	BACS40V
G50888	Shim - Laminated, Aluminum Alloy, CRES And Titanium, Surface Bonded	BACS40R

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Fan Cowl Cross Engine Debris Shield Installation

SUBTASK 71-11-03-420-001-G00

(1) Install the fan cowl cross engine debris shields on the applicable right fan cowl panel as follows:

(a) If it is necessary, install shim, G50337, or shim, G50888, between the debris shield and the debris shield brackets to meet the required maximum gap of 0.01 in. (0.25 mm) before the fasteners are installed.

1) Remove 0.003 in. (0.08 mm) laminations from the shim as necessary.

2) The maximum thickness of the shims must be 0.05 in. (1.27 mm).

3) Apply primer, C00259, to the shim mating surfaces.

(b) Install the forward debris shield [1] as follows:

1) Hold the forward debris shield [1] in its position on the fan cowl panel.

2) Install the bolt [4] and washer [5] at eight locations.

3) Install the bolt [6] and washer [7] at two locations.

(c) Install the aft upper debris shield [2] as follows:

1) Hold the aft upper debris shield [2] in its position on the fan cowl panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Install the bolt [8] and washer [9] at eight locations.

(d) Install the aft lower debris shield [3] as follows:

1) Hold the aft lower debris shield [3] in its position on the fan cowl panel.

2) Install the bolt [10] and washer [11] at eight locations.

3) Install the bolt [12] and washer [13] at two locations.

SUBTASK 71-11-03-080-001-G00

(2) Move the engine and strut access platform, STD-653, away from the engine.

(a) Make sure that the work area is clean of all tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-03-410-001-G00

(1) Close the applicable right fan cowl panel (TASK 71-11-04-410-801-G00).

SUBTASK 71-11-03-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL PANELS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Open the Fan Cowl Panels (Task Selection)

(2) Open the Fan Cowl Panels (Manual Method)

(3) Open the Fan Cowl Panels (Sling Method)

(4) Close the Fan Cowl Panels (Task Selection)

(5) Close the Fan Cowl Panels (Manual Method)

(6) Close the Fan Cowl Panels (Sling Method).

TASK 71-11-04-010-801-G00

2. Open the Fan Cowl Panels (Selection)

A. General

(1) The purpose of this task is to permit mechanics to select the applicable method to open the fan cowl panels.

(2) This task can be used for the left fan cowl panel, the right fan cowl panel, or both fan cowl panels.

NOTE: When only one of the fan cowl panels is opened, the other fan cowl panel will open slightly.

(3) When you close the fan cowl panels, put the left fan cowl panel into the closed position first.

NOTE: This will make sure that the latches and radial restraint pin engage correctly.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Open the Fan Cowl Panels (Selection)

SUBTASK 71-11-04-010-001-G00

(1) Do one of these tasks to open the applicable fan cowl panel:

(a) Do this task: Open the Fan Cowl Panels (Manual Method), TASK 71-11-04-010-803-G00.

(b) Do this task: Open the Fan Cowl Panels (Sling Method), TASK 71-11-04-010-804-G00.

——— END OF TASK ———

TASK 71-11-04-010-803-G00

3. Open the Fan Cowl Panels (Manual Method)

(Figure 201 and Figure 202 or Figure 203)

A. General

(1) This task gives the instructions to manually open the fan cowl panels.

(2) This task can be used for the left fan cowl panel, the right fan cowl panel, or both fan cowl panels.

NOTE: When only one of the fan cowl panels is opened, the other fan cowl panel will open slightly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) When you close the fan cowl panels, put the left fan cowl panel into the closed position first.

NOTE: This will make sure that the latches and radial restraint pin engage correctly.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-20446	Clamp Set - HOR & SDOS Part #: C71043-1 Supplier: 81205

D. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right

E. Prepare for the Opening

CAUTION DO NOT OPEN THE INBOARD FAN COWL PANEL UNLESS THE LEADING EDGE FLAPS ARE DEACTIVATED IN THE RETRACTED POSITION. IF THERE IS NOT SUFFICIENT CLEARANCE, DAMAGE TO EQUIPMENT CAN OCCUR.

SUBTASK 71-11-04-860-018-G00

(1) When you open the inboard fan cowl panel, do these tasks:

(a) Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804

(b) Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-04-860-024-G00

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL POST SB 737-71-1891 AND PRE SB 737-71-2023			
D	2	C04049	ENGINE 1 EEC FAN BLOWERS
D	5	C04050	ENGINE 2 EEC FAN BLOWERS

PTF ALL

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023			
D	2	C04049	ENGINE 1 EEC FAN BLOWERS (INOP)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023 (Continued)

(Continued)

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL POST SB 737-71-2023			
D	2	C04130	ENGINE 1 RBS BLOWER (INOP)
PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023			
D	5	C04050	ENGINE 2 EEC FAN BLOWERS (INOP)
PTF ALL POST SB 737-71-2023			
D	5	C04131	ENGINE 2 RBS BLOWER (INOP)

PTF ALL

F. Open the Fan Cowl Panels (Manual Method)

SUBTASK 71-11-04-010-004-G00

WARNING DO NOT OPEN THE FAN COWL PANEL DURING AN ENGINE OPERATION. THE FORWARD LATCHES ARE IN THE DANGEROUS AREA ZONE OF THE ENGINE INLET. IF YOU GO INTO THE DANGEROUS AREA, THE ENGINE CAN PULL YOU IN. IF YOU DO NOT OBEY, INJURY OR DEATH CAN OCCUR.

WARNING DO NOT OPEN THE FAN COWL PANELS IN HIGH WINDS OR GUST CONDITIONS THAT ARE MORE THAN 40 KNOTS. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT COULD OCCUR.

(1) Open the fan cowl latches (3 locations) at the 6:00 o'clock position as follows:

NOTE: Start from the aft latch and work forward.

PTF 102-999; PTF 101 POST SB 737-71-1894

(a) For the right fan cowl panel, do the steps that follow:

1) Open the oil tank access door.

2) Pull the fan cowl latch indication system handle out through the oil tank access door opening.

NOTE: Pull the handle out will rotate the blade to uncover the center fan cowl latch.

PTF ALL

(b) Push the triggers to release the latches.

(c) Pull the handles down to release the latches from the keepers.

(d) Make sure that the fan cowl panels are disengaged completely from each other.

(e) Slowly push open the fan cowl panel.

SUBTASK 71-11-04-010-008-G00

(2) Do these steps to open the fan cowl panels:

(a) Manually hold the fan cowl panel and pull it away from the engine.

NOTE: At or near the lower portion of travel, the fan cowl will experience resistance and want to force itself closed. This is due to the internal mechanism of the Spring Door Opening System (SDOS).

1) Open the oil tank or Integrated Drive Generator (IDG) access door as necessary to use the inside door frame as a handle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

WHEN THE FAN COWL IS OPENED, MAKE SURE THAT THE SPRING DOOR OPENING SYSTEM (SDOS) AND HOLD OPEN ROD (HOR) ARE LOCKED. DO NOT WORK UNDERNEATH THE COWL UNTIL YOU VERIFIED THE LOCKING MECHANISMS ARE LOCKED CORRECTLY. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Continue lifting until you reach the full open position.

WARNING

MAKE SURE THAT YOU DO NOT HIT OR LEAN ON THE HOLD-OPEN ROD (HOR) OR THE SPRING DOOR OPENING SYSTEM (SDOS) ACTUATOR WHEN THE FAN COWL IS IN THE OPEN POSITION. THE HOR OR SDOS ACTUATOR CAN UNLOCK AND CAUSE THE FAN COWL TO LOWER. USE THE CLAMP SET SPL-20446 TO DECREASE THE RISK OF A COMPONENT TO UNLOCK. IF THE HOR OR SDOS UNLOCKS, INJURY TO PERSONNEL CAN OCCUR.

(c) Make sure that there is an audible and tactile click from the Hold Open Rod (HOR) and SDOS locking mechanisms when they engage.

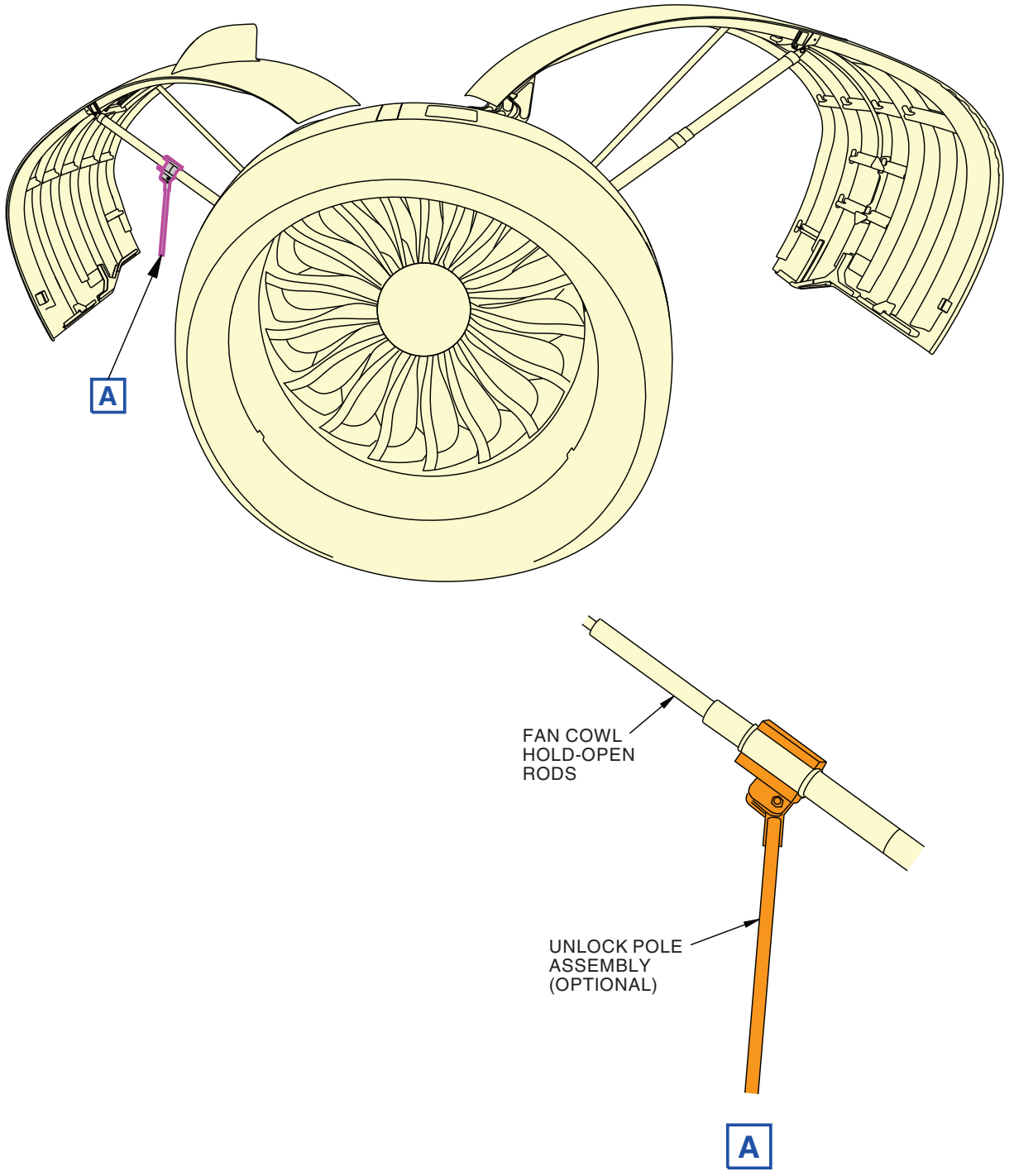
NOTE: It is recommended to shake the fan cowl at the bottom corners in a FWD/AFT motion to make sure that the locking mechanisms are locked correctly.

NOTE: If personnel are working near the HOR or SDOS, clamp, SPL-20446, can be used to prevent an inadvertent unlocking of the HOR or SDOS locking mechanisms.

NOTE: Alternatively, if the steps listed above cannot be completed, the sling method (TASK 71-11-04-010-804-G00) can be used to open the fan cowls. If the sling method is used as a result of the SDOS or HOR not securing properly in this procedure, the fan cowls should be closed using the manual method before using the sling method to open the cowls.

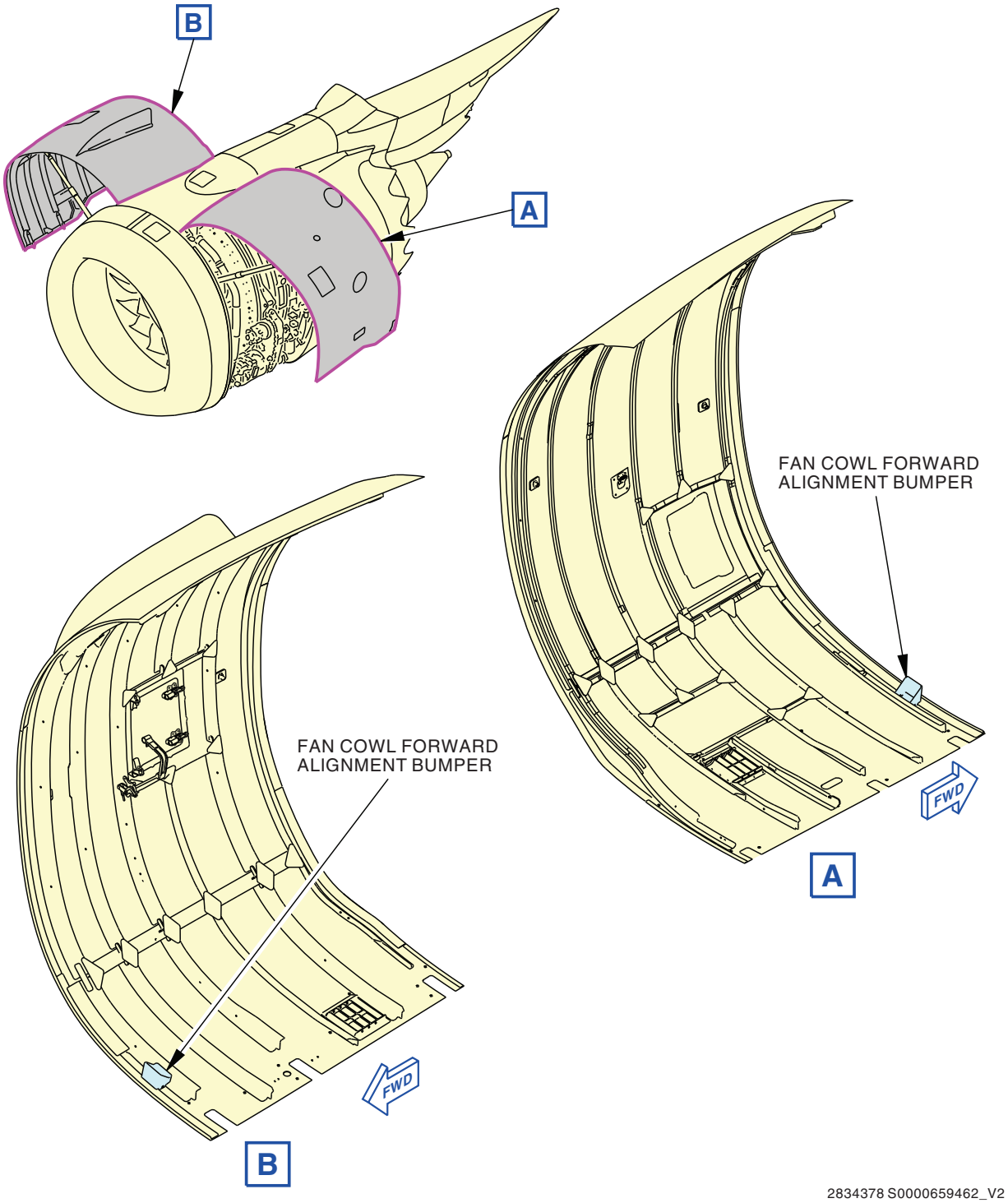
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl Panels Opening and Closing (Manual Method)
Figure 201/71-11-04-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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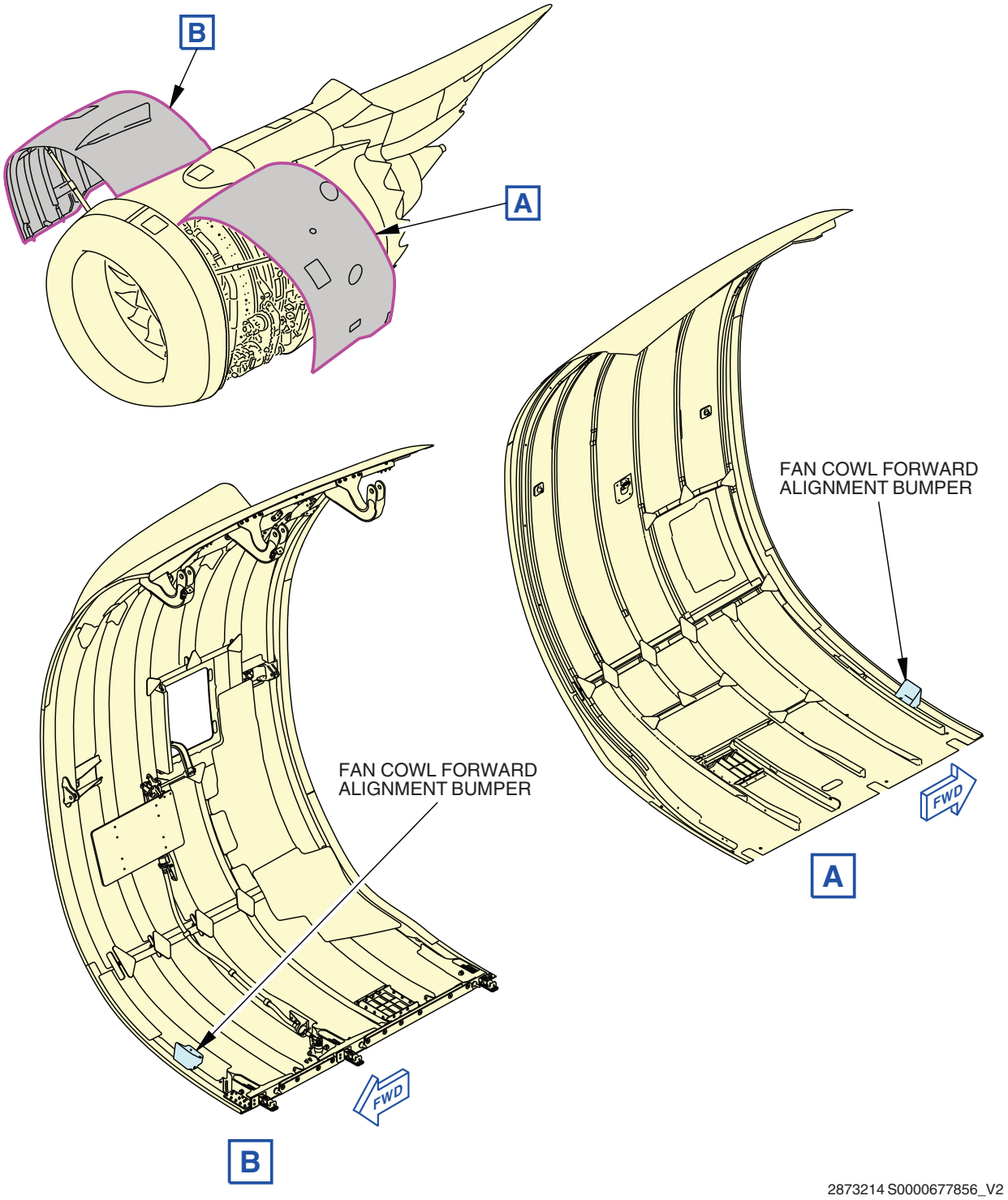
Cowl Panels Forward Alignment Bumpers
Figure 202/71-11-04-990-809-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl Latch Indication Systems
Figure 203/71-11-04-990-811-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-04-010-804-G00

4. Open the Fan Cowl Panels (Sling Method)

(Figure 204 and Figure 202 or Figure 203)

A. General

(1) This task gives the instructions to open the fan cowl panel with the use of the fan cowl sling.

(2) You will need a special fan cowl sling, SPL-14546, and a hoist, STD-467, to do this task.

(3) This task can be used for the left fan cowl panel, the right fan cowl panel, or both fan cowl panels.

NOTE: When only one of the fan cowl panels is opened, the other fan cowl panel will open slightly.

(4) When you close the fan cowl panels, put the left fan cowl panel into the closed position first.

NOTE: This will make sure that the latches and radial restraint pin engage correctly.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14546	Sling Eqpt - Fan Cowl, LEAP-1B Engine Part #: C71031-21 Supplier: 81205 Opt Part #: C71031-1 Supplier: 81205
STD-467	Hoist - 500 lb (226.8 kg)
STD-653	Platform - Engine and Strut Access

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Opening

CAUTION

DO NOT OPEN THE INBOARD FAN COWL PANEL UNLESS THE LEADING EDGE FLAPS ARE DEACTIVATED IN THE RETRACTED POSITION. IF THERE IS NOT SUFFICIENT CLEARANCE, DAMAGE TO EQUIPMENT CAN OCCUR.

SUBTASK 71-11-04-860-021-G00

(1) When you open the inboard fan cowl panel, do these tasks:

(a) Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804

(b) Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-04-490-001-G00

(2) Move the engine and strut access platform, STD-653, next to the fan cowl panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-04-860-025-G00

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL POST SB 737-71-1891 AND PRE SB 737-71-2023			
D	2	C04049	ENGINE 1 EEC FAN BLOWERS
D	5	C04050	ENGINE 2 EEC FAN BLOWERS

PTF ALL

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023			
D	2	C04049	ENGINE 1 EEC FAN BLOWERS (INOP)
PTF ALL POST SB 737-71-2023			
D	2	C04130	ENGINE 1 RBS BLOWER (INOP)
PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023			
D	5	C04050	ENGINE 2 EEC FAN BLOWERS (INOP)
PTF ALL POST SB 737-71-2023			
D	5	C04131	ENGINE 2 RBS BLOWER (INOP)

PTF ALL

F. Open the Fan Cowl Panels (Sling Method)

SUBTASK 71-11-04-010-005-G00

WARNING DO NOT OPEN THE FAN COWL PANEL DURING AN ENGINE OPERATION. THE FORWARD LATCHES ARE IN THE DANGEROUS AREA ZONE OF THE ENGINE INLET. IF YOU GO INTO THE DANGEROUS AREA, THE ENGINE CAN PULL YOU IN. IF YOU DO NOT OBEY, INJURY OR DEATH CAN OCCUR.

WARNING DO NOT OPEN THE FAN COWL PANELS IN HIGH WINDS OR GUST CONDITIONS THAT ARE MORE THAN 40 KNOTS. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT COULD OCCUR.

(1) Open the fan cowl latches (3 locations) at the 6:00 o'clock position as follows:

NOTE: Start from the aft latch and work forward.

PTF 102-999; PTF 101 POST SB 737-71-1894

(a) For the right fan cowl panel, do the steps that follow:

1) Open the oil tank access door.

2) Pull the fan cowl latch indication system handle out through the oil tank access door opening.

NOTE: Pull the handle out will rotate the blade to uncover the center fan cowl latch.

PTF ALL

(b) Push the triggers to release the handles.

(c) Pull the handles down to release the latches from the keepers.

(d) Make sure that the fan cowl panels are disengaged completely from each other.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Slowly push open the fan cowl panel.

SUBTASK 71-11-04-490-002-G00

(2) Attach the fan cowl sling, SPL-14546, to the applicable fan cowl panel as follows:

(a) Install the three bolts [3] that attach the three hoist pins [2] to the fan cowl Ground Support Equipment (GSE) lift points.

1) Tighten the bolts [3] to 40 in-lb (5 N·m) - 80 in-lb (9 N·m).

(b) Engage the hoist, STD-467, with the sling hoist plate.

(c) Put the hoist, STD-467, and the sling hoist plate above the fan cowl panel.

(d) Make sure that the fan cowl sling, SPL-14546, does not touch the surface of the fan cowl panel.

(e) Attach the three sling straps [4] to the sling link [6] with the three Lok-A-Loys [5].

NOTE: The sling straps, the sling link, and the three Lok-A-Loys are part of the fan cowl sling, SPL-14546.

(f) Attach the sling straps [4] to the fan cowl panel.

(g) Attach a dynamometer and a master link on the sling link [6].

SUBTASK 71-11-04-860-003-G00

WARNING

MONITOR THE WEIGHT OF THE LOAD WHILE YOU PUT WEIGHT ON THE SLING. MAKE SURE THAT THE WEIGHT OF THE LOAD IS LESS THAN THE WEIGHT LIMIT. IF THE WEIGHT OF THE LOAD IS MORE THAN THE LIMIT, THE COMPONENT CAN BREAK. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(3) Lift slowly and adjust the hoist, STD-467, to hold the weight of the fan cowl panel.

NOTE: The fan cowl panel weighs approximately 160 lb (73 kg).

(a) Make sure that the fan cowl sling, SPL-14546, does not touch the surface of the fan cowl panel.

(b) Continue lifting until you reach the full open position.

WARNING

WHEN THE FAN COWL IS OPENED, MAKE SURE THAT THE SPRING DOOR OPENING SYSTEM (SDOS) AND HOLD OPEN ROD (HOR) ARE LOCKED. DO NOT WORK UNDERNEATH THE COWL UNTIL YOU VERIFIED THE LOCKING MECHANISMS ARE LOCKED CORRECTLY. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Make sure that there is an audible click from the Hold Open Rod (HOR) and Spring Door Opening System (SDOS) locking mechanisms when they engage.

NOTE: It is recommended to shake the fan cowl at the bottom corners in a FWD/AFT motion to make sure that the locking mechanisms are locked correctly.

SUBTASK 71-11-04-080-001-G00

(4) If the HOR or SDOS will be removed, or if the locking mechanisms on both the HOR and SDOS did not engage properly, the fan cowl must continue to be held open using the hoist, STD-467, and the fan cowl sling, SPL-14546.

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SUBTASK 71-11-04-090-001-G00

(5) If the HOR and SDOS will remain installed and the locking mechanisms on both the HOR and SDOS did engage properly, you can remove the hoist, STD-467, and the fan cowl sling, SPL-14546, if necessary:

(a) Disconnect the three sling straps [4] from the fan cowl panel.

1) Remove the three bolts [3] that attach the three hoist pins [2] to the fan cowl GSE lift points.

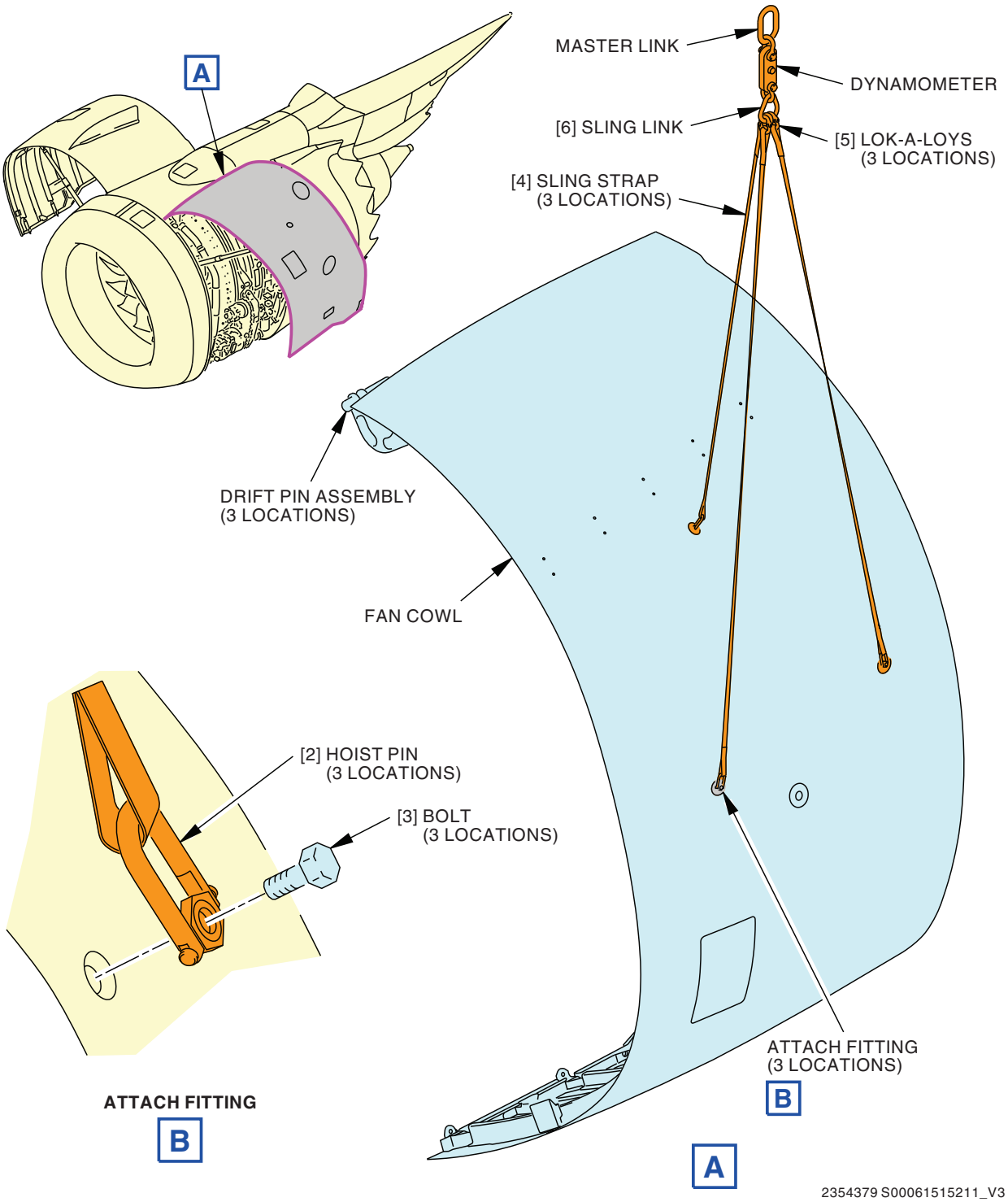
(b) Move the hoist, STD-467, away from the fan cowl panel.

SUBTASK 71-11-04-090-002-G00

(6) Move the engine and strut access platform, STD-653, away from the engine.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl Panel Opening and Closing (Sling Method)
Figure 204/71-11-04-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-04-410-801-G00

5. Close the Fan Cowl Panels (Selection)

A. General

(1) The purpose of this task is to permit mechanics to select the applicable method to close the fan cowl panels.

(2) The procedure can be used for the left fan cowl panel or the right fan cowl panel.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Close the Fan Cowl Panels (Selection)

SUBTASK 71-11-04-410-001-G00

(1) Do one of these tasks to close the applicable fan cowl panel:

(a) Do this task: Close the Fan Cowl Panels (Manual Method), TASK 71-11-04-410-803-G00.

(b) Do this task: Close the Fan Cowl Panels (Sling Method), TASK 71-11-04-410-804-G00.

——— END OF TASK ———

TASK 71-11-04-410-803-G00

6. Close the Fan Cowl Panels (Manual Method)

(Figure 201 and Figure 202 or Figure 203)

A. General

(1) This task gives the instructions to manually close the fan cowl panels.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-05-821-801-G00	Fan Cowl Latch Adjustment (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-10926	Unlock Pole - Fan Cowl Hold Open Rods Part #: C71041-1 Supplier: 81205

D. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Close the Fan Cowl Panels (Manual Method)

SUBTASK 71-11-04-420-002-G00

WARNING DO NOT CLOSE THE FAN COWL PANEL DURING AN ENGINE OPERATION. THE FORWARD LATCHES ARE IN THE HAZARD AREA ZONE OF THE ENGINE INLET. IF YOU GO INTO THE HAZARD AREA, YOU CAN GET PULLED INTO THE ENGINE. IF YOU DO NOT OBEY, DANGEROUS INJURY OR DEATH CAN OCCUR.

WARNING DO NOT TRY TO OPEN OR CLOSE THE FAN COWL PANEL IN HIGH WINDS OR GUST CONDITIONS OF MORE THAN 40 KNOTS. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT COULD OCCUR.

(1) Do these steps to close the fan cowl panels:

(a) Close the left fan cowl panel first.

(b) Lift the fan cowl slightly to relieve the weight from the Spring Door Opening System (SDOS) and Hold Open Rod (HOR) collars.

WARNING HOLD THE FAN COWL PANEL WHILE YOU UNLOCK OR RETRACT THE HOLD-OPEN RODS. IF THE FAN COWL PANEL FALLS, INJURY TO PERSONS AND DAMAGE TO THE ENGINE COMPONENTS CAN OCCUR.

(c) Make sure that the weight of the fan cowl is fully supported while you disengage the SDOS and HOR lock collars.

(d) Use the unlock pole, SPL-10926, to pull the lock collar down to disengage the locking mechanism.

1) As an option, use a ladder as required to disengage the lock collar.

WARNING YOU MUST CONTROL THE WEIGHT AND RATE OF CLOSURE OF THE FAN COWL. FAILURE TO CONTROL THE RATE OF CLOSURE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(e) While you support the fan cowl, lower it into the lower static position as follows:

NOTE: The SDOS will relieve some of the fan cowl weight, but it does not control the rate of closure. The mechanics must control the weight and rate of closure.

1) Open the oil tank or Integrated Drive Generator (IDG) access door as necessary to use the inside door frame as a handle.

2) Control the rate of closure as you lower the fan cowl.

NOTE: Near the last foot of travel for the fan cowl to reach the lower static position, the SDOS will no longer relieve the fan cowl weight. The mechanics must control the full weight and rate of closure of the fan cowl towards the end of travel.

3) Use the forward alignment bumper on the fan cowl to help closing it.

NOTE: The fan cowls naturally swing forward when opening and closing. The alignment bumper on each fan cowl helps to prevent strikes to the inlet cowl and engine components.

(f) Make sure that the forward fan cowl V-blade engages correctly with the inlet TV-chord.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Make sure that the radial restraint pin engages the receiver at the bottom of the engine.

(h) Push to seat the restraint pin.

(i) Repeat the steps above again to close the right-hand fan cowl panel.

(j) Make sure that the shear pins of the right-hand fan cowl panel engage with their respective bushings on the left-hand fan cowl panel.

SUBTASK 71-11-04-860-007-G00

CAUTION

MAKE SURE THAT THE FAN COWLS ARE FULLY CLOSED AND YOU CORRECTLY LATCH THE THREE FAN COWL LATCHES. IF YOU DO NOT FULLY CLOSE AND CORRECTLY LATCH THE THREE FAN COWL LATCHES, DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Close the fan cowl latches (3 locations) at the 6:00 o'clock position as follows:

NOTE: Engage the latch hooks in the specific order.

(a) Engage the latch hook in the center keeper.

(b) Engage the latch hook in the forward keeper.

(c) Engage the latch hook in the aft keeper.

(d) Push the latch handles closed in the same order.

1) If the latch handles do not close smoothly, do this task: Fan Cowl Latch Adjustment, TASK 71-11-05-821-801-G00.

PTF 102-999; PTF 101 POST SB 737-71-1894

(e) For the right fan cowl panel, do the steps that follow:

1) Push the fan cowl latch indication system handle in through the oil tank access door opening.

2) Close the oil tank access door.

3) Make sure that you fully close and secure the oil tank access door.

PTF ALL

(f) If necessary, close the IDG access door.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-04-860-026-G00

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL POST SB 737-71-1891 AND PRE SB 737-71-2023			
D	2	C04049	ENGINE 1 EEC FAN BLOWERS
D	5	C04050	ENGINE 2 EEC FAN BLOWERS

PTF ALL

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023			
D	2	C04049	ENGINE 1 EEC FAN BLOWERS (INOP)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023 (Continued)

(Continued)

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL POST SB 737-71-2023			
D	2	C04130	ENGINE 1 RBS BLOWER (INOP)
PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023			
D	5	C04050	ENGINE 2 EEC FAN BLOWERS (INOP)
PTF ALL POST SB 737-71-2023			
D	5	C04131	ENGINE 2 RBS BLOWER (INOP)

PTF ALL

SUBTASK 71-11-04-440-002-G00

(2) If necessary, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

TASK 71-11-04-410-804-G00

7. Close the Fan Cowl Panels (Sling Method)

(Figure 204 and Figure 202 or Figure 203)

A. General

(1) This task gives the instructions to close the fan cowl panels with the use of the fan cowl sling.

(2) You will need a special fan cowl sling, SPL-14546, and a hoist, STD-467, are necessary to do this task.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-05-821-801-G00	Fan Cowl Latch Adjustment (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-10926	Unlock Pole - Fan Cowl Hold Open Rods Part #: C71041-1 Supplier: 81205
SPL-14546	Sling Eqpt - Fan Cowl, LEAP-1B Engine Part #: C71031-21 Supplier: 81205 Opt Part #: C71031-1 Supplier: 81205
STD-467	Hoist - 500 lb (226.8 kg)
STD-653	Platform - Engine and Strut Access

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Close the Fan Cowl Panels (Sling Method)

WARNING DO NOT CLOSE THE FAN COWL PANEL DURING AN ENGINE OPERATION. THE FORWARD LATCHES ARE IN THE HAZARD AREA ZONE OF THE ENGINE INLET. IF YOU GO INTO THE HAZARD AREA, YOU CAN GET PULLED INTO THE ENGINE. IF YOU DO NOT OBEY, DANGEROUS INJURY OR DEATH CAN OCCUR.

WARNING DO NOT TRY TO OPEN OR CLOSE THE FAN COWL PANEL IN HIGH WINDS OR GUST CONDITIONS OF MORE THAN 40 KNOTS. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT COULD OCCUR.

SUBTASK 71-11-04-490-003-G00

(1) Move an engine and strut access platform, STD-653, next to the fan cowl panel.

SUBTASK 71-11-04-490-004-G00

(2) If it had not done so, attach the fan cowl sling, SPL-14546, to the applicable fan cowl panel as follows:

(a) Install the three bolts [3] that attach the three hoist pins [2] to the fan cowl Ground Support Equipment (GSE) lift points.

1) Tighten the bolts [3] to 40 in-lb (5 N·m) - 80 in-lb (9 N·m).

(b) Engage the hoist, STD-467, with the sling hoist plate.

(c) Put the hoist, STD-467, and the sling hoist plate above the fan cowl panel.

(d) Make sure that the fan cowl sling, SPL-14546, does not touch the surface of the fan cowl panel.

(e) Attach the three sling straps [4] to the sling link [6] with the three Lok-A-Loys [5].

NOTE: The sling straps, the sling link, and the three Lok-A-Loys are part of the fan cowl sling, SPL-14546.

(f) Attach the sling straps [4] to the fan cowl panel.

(g) Attach a dynamometer and a master link on the sling link [6].

SUBTASK 71-11-04-980-002-G00

(3) Close the fan cowl panels as follows:

(a) Close the left fan cowl panel first.

(b) Use the hoist, STD-467, to lift the fan cowl slightly to relieve the weight from the Spring Door Opening System (SDOS) and Hold Open Rod (HOR) collars.

(c) Use the unlock pole, SPL-10926, to pull the lock collar down to disengage the locking mechanism.

1) As an option, use a ladder as required to disengage the lock collar.

WARNING YOU MUST CONTROL THE WEIGHT AND RATE OF CLOSURE OF THE FAN COWL. FAILURE TO CONTROL THE RATE OF CLOSURE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(d) While you support the fan cowl, lower it into the lower static position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Use the forward alignment bumper on the fan cowl to help closing it.

NOTE: The fan cowls naturally swing forward when opening and closing. The alignment bumper on each fan cowl helps to prevent strikes to the inlet cowl and engine components.

(f) Make sure that the forward fan cowl V-blade engages correctly with the inlet TV-chord.

(g) Make sure that the radial restraint pin engages the receiver at the bottom of the engine then push to seat the pin.

(h) Repeat the steps above again to close the right-hand fan cowl panel.

(i) Make sure that the shear pins of the right-hand fan cowl panel engage with their respective bushings on the left-hand fan cowl panel.

SUBTASK 71-11-04-090-003-G00

(4) Remove the fan cowl sling, SPL-14546, as follows:

(a) Remove the three bolts [3] that attach the three hoist pins [2] to the fan cowl GSE lift points.

(b) Move the hoist, STD-467, away from the fan cowl panel.

(c) Disconnect the sling straps [4] from the sling hoist plate.

(d) Remove the sling hoist plate from the hoist, STD-467.

SUBTASK 71-11-04-410-003-G00

CAUTION

MAKE SURE THAT THE FAN COWLS ARE FULLY CLOSED AND YOU CORRECTLY LATCH THE THREE FAN COWL LATCHES. IF YOU DO NOT FULLY CLOSE AND CORRECTLY LATCH THE THREE FAN COWL LATCHES, DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Close the fan cowl latches (3 locations) at the 6:00 o'clock position as follows:

NOTE: Engage the latch hooks in the specific order.

(a) Engage the latch hook in the center keeper.

(b) Engage the latch hook in the forward keeper.

(c) Engage the latch hook in the aft keeper.

(d) Push the latch handles closed in the same order.

1) If the latch handles do not close smoothly, do this task: Fan Cowl Latch Adjustment, TASK 71-11-05-821-801-G00.

PTF 102-999; PTF 101 POST SB 737-71-1894

(e) For the right fan cowl panel, do the steps that follow:

1) Push the fan cowl latch indication system handle in through the oil tank access door opening.

a) Make sure that you do not force the handle back into the fan cowl.

b) Examine the blades if it is not easy to push the handle back.

2) Close the oil tank access door.

3) Make sure that you fully close and secure the oil tank access door.

PTF ALL

(f) If necessary, close the IDG access door.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-04-090-004-G00

(1) Move the engine and strut access platform, STD-653, away from the engine.

SUBTASK 71-11-04-860-027-G00

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL POST SB 737-71-1891 AND PRE SB 737-71-2023			
D	2	C04049	ENGINE 1 EEC FAN BLOWERS
D	5	C04050	ENGINE 2 EEC FAN BLOWERS

PTF ALL

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023			
D	2	C04049	ENGINE 1 EEC FAN BLOWERS (INOP)
PTF ALL POST SB 737-71-2023			
D	2	C04130	ENGINE 1 RBS BLOWER (INOP)
PTF ALL PRE SB 737-71-1891 AND PRE SB 737-71-2023			
D	5	C04050	ENGINE 2 EEC FAN BLOWERS (INOP)
PTF ALL POST SB 737-71-2023			
D	5	C04131	ENGINE 2 RBS BLOWER (INOP)

PTF ALL

SUBTASK 71-11-04-440-003-G00

(3) If necessary, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL PANELS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fan Cowl Panels Removal

(2) Fan Cowl Panels Installation.

TASK 71-11-04-000-801-G00

2. Fan Cowl Panels Removal

(Figure 401, Figure 402, and Figure 403)

A. General

(1) This task gives the instructions to remove the fan cowl panels.

(2) The removal task of the left and right fan cowl panels are the same.

(3) Use a special fan cowl sling, SPL-14546, and hoist, STD-467, to do this task.

(4) Each fan cowl panel weighs approximately 160 lb (73 kg).

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-010-804-G00	Open the Fan Cowl Panels (Sling Method) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14546	Sling Eqpt - Fan Cowl, LEAP-1B Engine Part #: C71031-21 Supplier: 81205 Opt Part #: C71031-1 Supplier: 81205
STD-467	Hoist - 500 lb (226.8 kg)
STD-748	Rope - 12 ft (3.6 m) minimum
STD-858	Tag - DO NOT OPERATE
STD-12395	Load Cell Equipment

D. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right

E. Prepare for the Removal

SUBTASK 71-11-04-860-009-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-04-860-010-G00

CAUTION RETRACT THE LEADING EDGE SLATS, AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THESE PRECAUTIONS, THE FAN COWL PANELS WILL HIT THE SLATS. DAMAGE WILL OCCUR.

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 71-11-04-040-003-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-04-860-019-G00

(4) Make sure that the thrust reverser is closed and latched (TASK 78-31-00-010-802-G00).

SUBTASK 71-11-04-490-005-G00

WARNING DO NOT TRY TO REMOVE OR INSTALL THE FAN COWL PANELS IN HIGH WINDS OR GUST CONDITIONS OF MORE THAN 40 KNOTS. THE WINDS CAN CAUSE THE FAN COWL PANELS TO MOVE SUDDENLY. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Attach the fan cowl sling, SPL-14546, to the applicable fan cowl panel [1] as follows (Figure 401):

(a) Hang the sling hoist plate (master link) onto the hoist, STD-467.

(b) Attach a load cell, STD-12395, to the sling link.

NOTE: The load cell, STD-12395, or equivalent, or similar capacity equipment is customer furnished equipment. The required minimum load cell capacity is 250 lb (113 kg).

(c) Attach the three sling straps [2] to the load cell, STD-12395, with the three Lok-A-Loys and sling link.

NOTE: The sling straps, sling link, and the three Lok-A-Loys are part of the fan cowl sling, SPL-14546.

(d) Set the load cell, STD-12395, to zero.

(e) Put the hoist, STD-467, and assembled fan cowl sling, SPL-14546, above the fan cowl panel [1].

1) Make sure that the fan cowl sling, SPL-14546, does not touch the surface of the fan cowl panel [1].

(f) Install the three hoist pins [3] on the fan cowl panel [1].

1) Install the three bolts [7] the through the three hoist pins [3] in the plug holes.

2) Tighten the bolts [7] to 10 in-lb (1.1 N·m) - 30 in-lb (3.4 N·m).

SUBTASK 71-11-04-010-006-G00

(6) Open the applicable fan cowl panel [1] (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Fan Cowl Panels Removal

SUBTASK 71-11-04-020-001-G00

WARNING GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Remove the fan cowl panel [1] from the engine as follows:

NOTE: Boeing does not recommend the removal of the fan cowl without the use of a crane and sling.

NOTE: The fan cowl panel weighs approximately 160 lb (73 kg).

(a) Attach the fan cowl sling, SPL-14546, to the applicable fan cowl panel [1] (TASK 71-11-04-010-804-G00).

(b) Install the rope - 12 ft (3.6 m) minimum, STD-748, on the latch [20].

NOTE: Use the rope as optional to control the fan cowl panel's position when it is lifted.

WARNING MAKE SURE THAT THE SDOS ACTUATOR IS FULLY EXTENDED AND LOCKED. THE ACTUATOR CONTAINS STORED ENERGY. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY CAN OCCUR.

WARNING BE CAREFUL AFTER YOU REMOVE THE SDOS ACTUATOR. DO NOT POINT THE ACTUATOR END AT A PERSON. THE ACTUATOR CONTAINS ENERGY OF A SPRING. THIS CAN CAUSE INJURY OR DEATH.

CAUTION ONCE THE FAN COWL SDOS ACTUATOR OR THE FAN COWL HOLD-OPEN ROD IS DETACHED FROM THE FAN COWL PANEL FITTING, IT WILL BE FREE TO FALL. CAREFULLY LOWER THE FAN COWL SDOS ACTUATOR OR HOLD-OPEN ROD UNTIL IT RESTS AGAINST THE ENGINE, IN A POSITION THAT DOES NOT ENDANGER PERSONNEL OR COULD CAUSE DAMAGE TO EQUIPMENT.

(c) Disconnect the fan cowl SDOS actuator [12] from the upper fan cowl panel fitting as follows:

1) Make sure that the fan cowl panel [1] is fully open and fan cowl SDOS actuator [12] is locked.

2) Remove the bolt [8], washer [9], bushing [10], washer [11], and nut [13] that attach the fan cowl SDOS actuator [12] to the upper fan cowl bracket fitting.

3) Carefully put the fan cowl SDOS actuator [12] in the space it occupies when the fan cowl panel [1] is closed.

(d) Disconnect the fan cowl hold-open rod [6] from the fan cowl panel fitting as follows:

1) Remove the bolt [14], washer [15], bushing [16], washer [17], and nut [18] that attach the fan cowl hold-open rod [6] to the fan cowl panel fitting.

2) Carefully put the fan cowl hold-open rod [6] in the space it occupies when the fan cowl panel [1] is closed.

(e) Disconnect the bonding jumpers [5] from the brackets on the forward and center hinge assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Remove the screw [21], washer [22], washer [23], and washer [24] that attach the bonding jumper [5] to the bracket.

WARNING

DO NOT FULLY DISENGAGE THE FAN COWL FROM THE STRUT HINGES WHILE A MECHANIC IS UNDER THE COWLING. USE THE DRIFT PINS TO SECURE THE COWLING THEN REMOVE THE DRIFT PINS BY REACHING UNDER THE COWLING. INJURIES TO PERSONS CAN OCCUR.

(f) Disengage the quick-release pins [4] from the forward hinge assembly as follows:

1) Remove the safety pin from the end of the quick-release pin [4].
2) Release the quick-release pin [4] with the button on the end.
3) Remove the quick-release pin [4].
4) Install a drift from the sling kit at the forward hinge.
5) Do the steps above again to disengage the pin from the aft hinge assembly.

(g) Disengage and remove the quick-release pins [4] from the center hinge assembly.

(h) Use the fan cowl sling, SPL-14546, to adjust the position of the fan cowl panel [1] until the forward and aft drift pins can be removed easily.

(i) Make sure that the three sling straps remain tight and balance the weight of the fan cowl panel [1].

(j) Remove the forward and aft drift pins.

(k) Carefully move the fan cowl panel [1] away from the engine with the help of the rope - 12 ft (3.6 m) minimum, STD-748, and/or other personnel.

1) Put the fan cowl panel [1] on a clean, smooth surface.
   a) Rest the fan cowl panel [1] on its hinge fittings and bottom edge.

(l) Remove the rope - 12 ft (3.6 m) minimum, STD-748, from the latch [20] if the fan cowl panel [1] will not be re-installed.

SUBTASK 71-11-04-090-005-G00

(2) Remove the sling straps [2] and hoist pins [3] if the fan cowl panel [1] will not be re-installed.

——— END OF TASK ———

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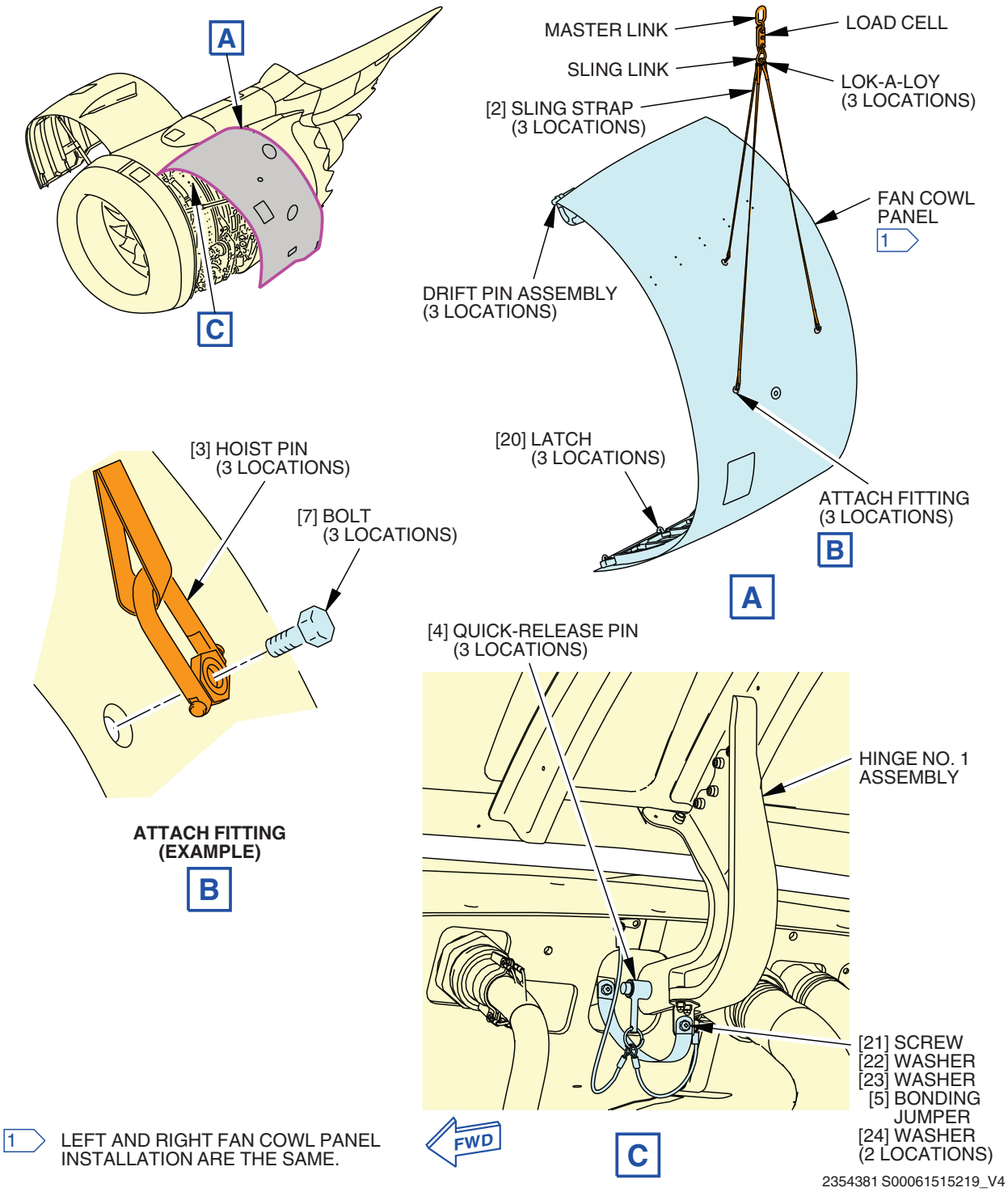
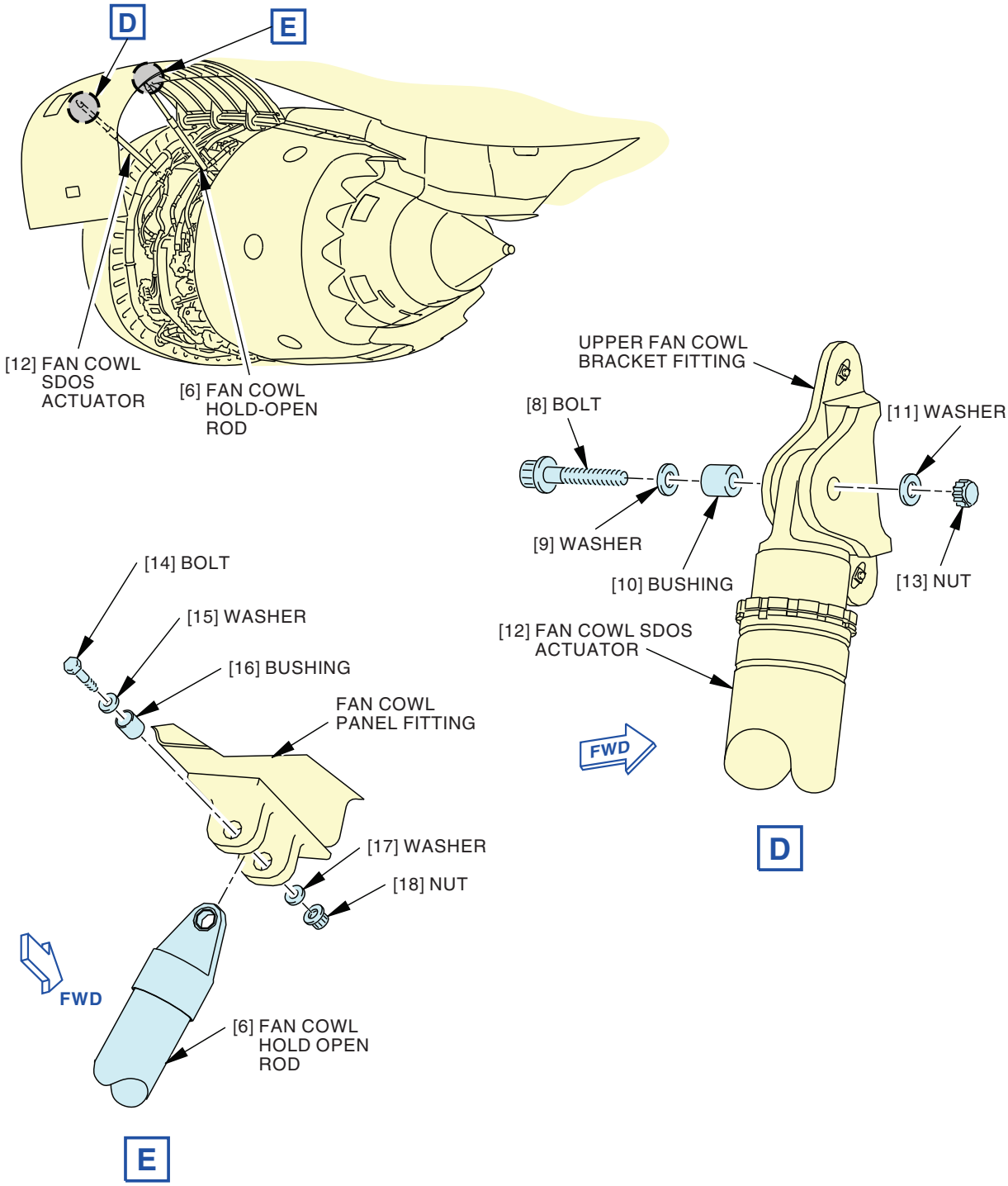


Fan Cowl Panels Installation
Figure 401/71-11-04-990-804-G00 (Sheet 1 of 2)

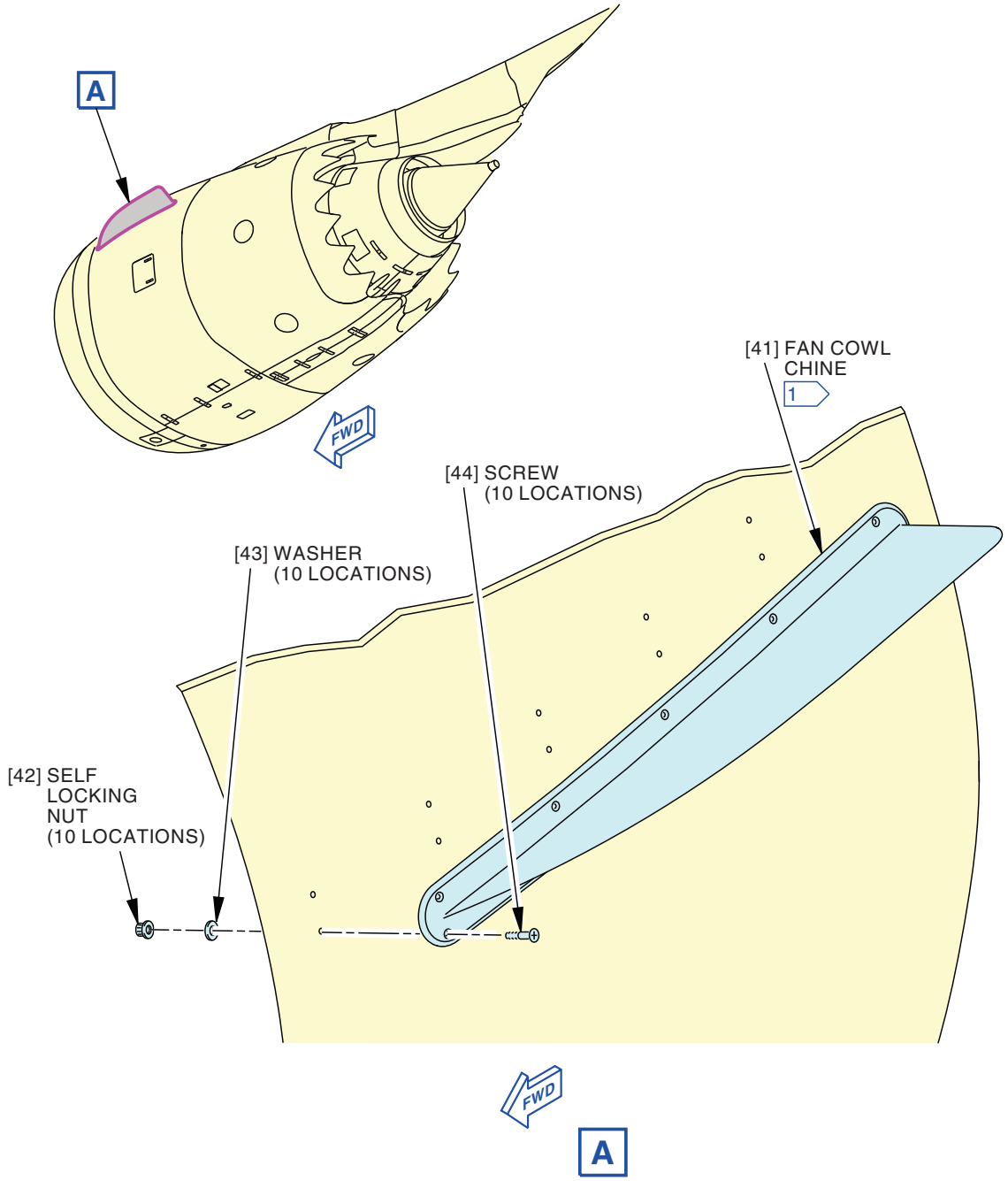
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2570589 S0000616399_V2

Fan Cowl Panels Installation
Figure 401/71-11-04-990-804-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

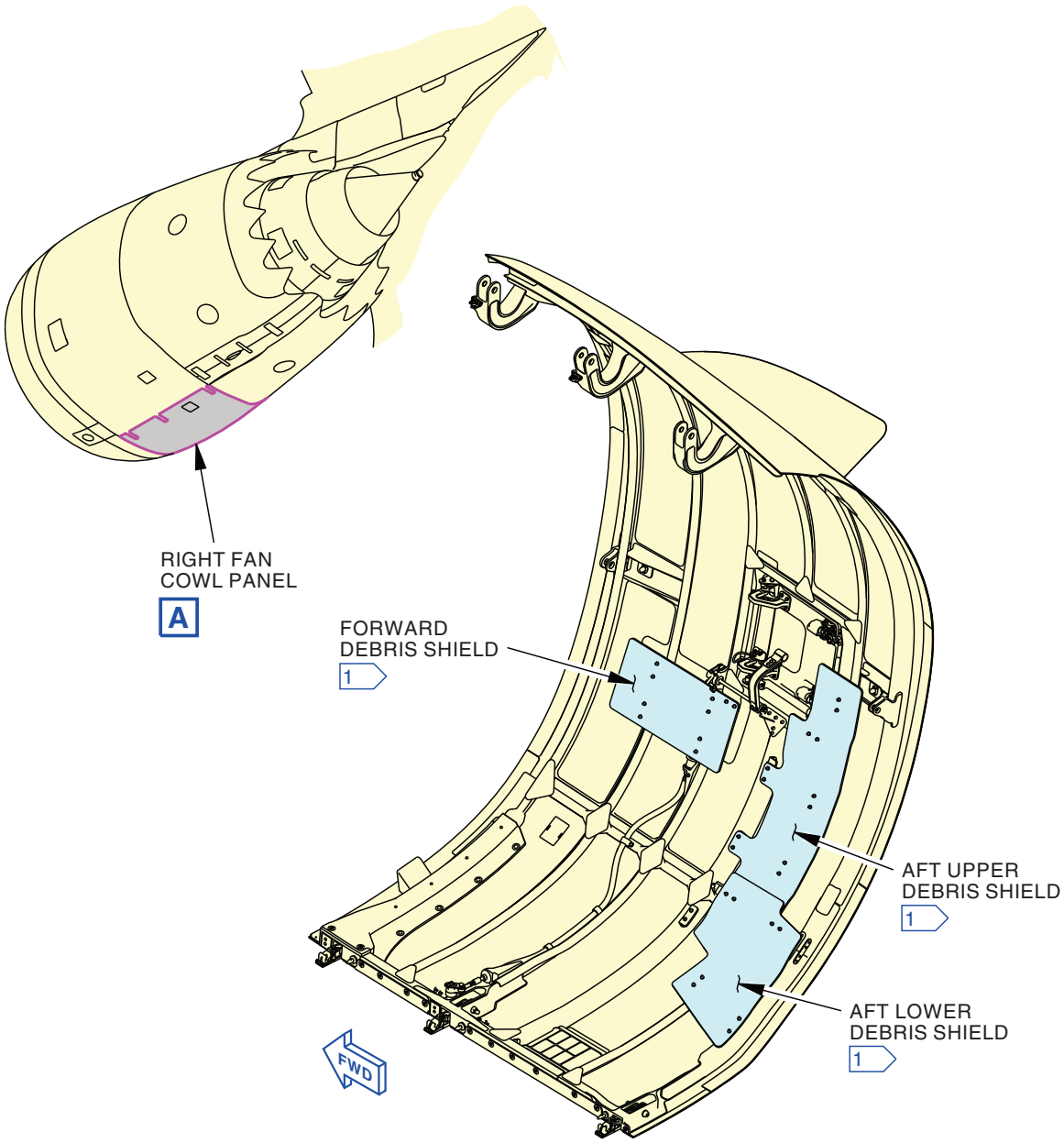


1 THE FAN COWL CHINE IS INSTALLED ON THE INBOARD SIDE OF THE ENGINE. THE FAN COWL CHINES ARE INTERCHANGEABLE ON THE FAN COWL.

2354385 S00061515223_V2

Fan Cowl Chine Provision Installation
Figure 402/71-11-04-990-806-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIGHT FAN COWL PANEL (TYPICAL)

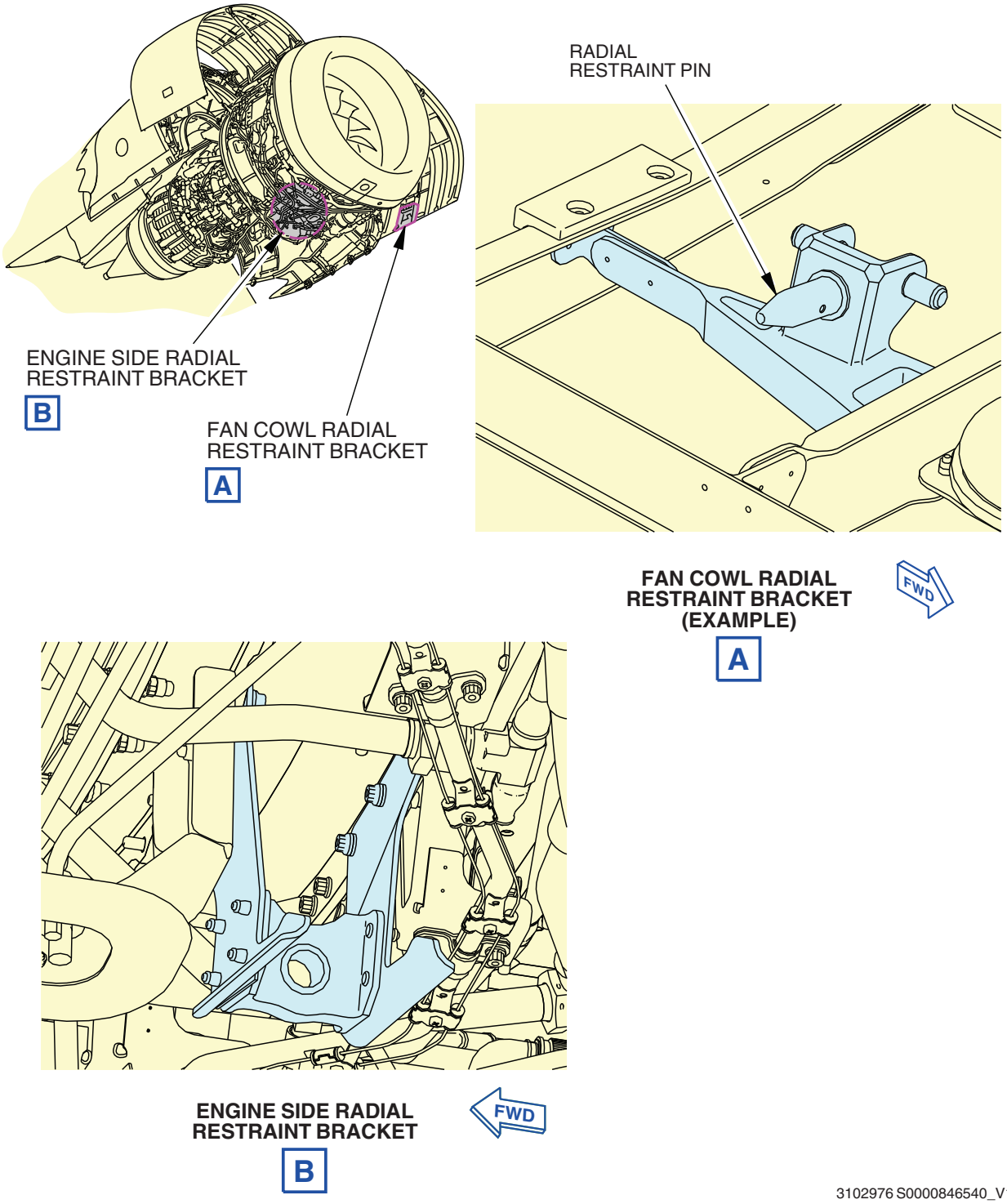
A

1 THE FAN COWL CROSS ENGINE DEBRIS SHIELDS ARE INSTALLED ONLY ON THE INBOARD FAN COWL PANEL OF ENGINE 1.

2848452 S0000666183_V1

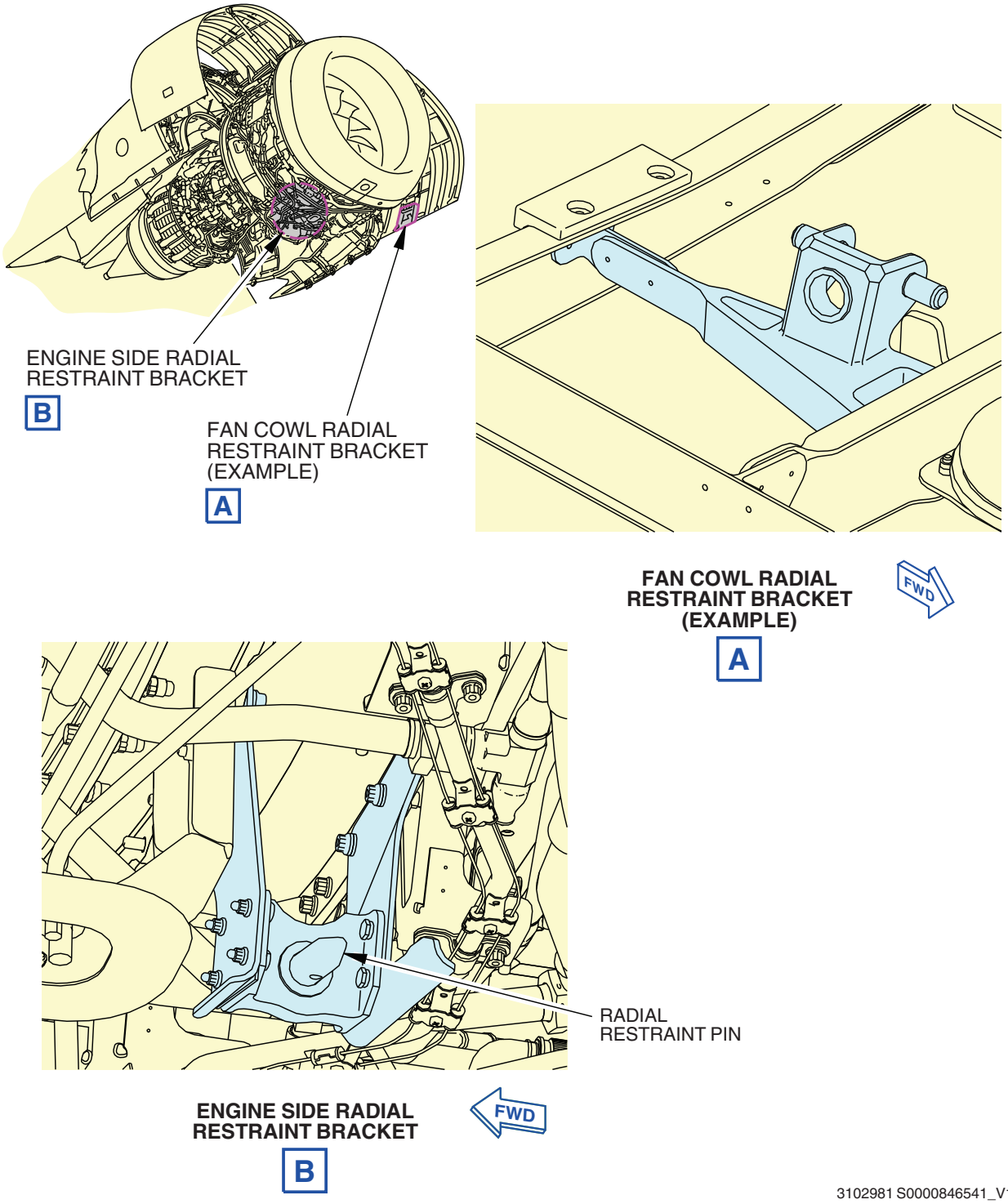
Fan Cowl Cross Engine Debris Shield
Figure 403/71-11-04-990-810-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl and Engine Radial Restraint Pin Configuration
Figure 404/71-11-04-990-813-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl and Engine Radial Restraint Pin Configuration
Figure 404/71-11-04-990-813-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-04-400-801-G00

3. Fan Cowl Panels Installation

(Figure 401, Figure 402, and Figure 403, Figure 404)

A. General

(1) This task gives the instructions to install the fan cowl panels.

(2) The installation procedure of the left and the right fan cowl panels is the same.

(3) You will need a special fan cowl sling, SPL-14546, and hoist, STD-467, to do this task.

(4) Each fan cowl panel weighs approximately 160 lb (73 kg).

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-02-000-801-G00	Fan Cowl Chine Removal (P/B 401)
71-11-02-400-801-G00	Fan Cowl Chine Installation (P/B 401)
71-11-03-000-801-G00	Fan Cowl Cross Engine Debris Shield Removal (P/B 401)
71-11-03-400-801-G00	Fan Cowl Cross Engine Debris Shield Installation (P/B 401)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
71-11-05-821-801-G00	Fan Cowl Latch Adjustment (P/B 501)
71-11-13-000-801-G00	Engine Side Radial Restraint Bracket Removal (P/B 401)
71-11-13-400-801-G00	Engine Side Radial Restraint Bracket Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
SPL-14546	Sling Eqpt - Fan Cowl, LEAP-1B Engine Part #: C71031-21 Supplier: 81205 Opt Part #: C71031-1 Supplier: 81205
STD-467	Hoist - 500 lb (226.8 kg)
STD-748	Rope - 12 ft (3.6 m) minimum
STD-858	Tag - DO NOT OPERATE
STD-12395	Load Cell Equipment

D. Consumable Materials

Reference	Description	Specification
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
D00014	Grease - Molybdenum Disulfide, Low & High Temperature	MIL-G-21164 (NATO G-353)
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fan cowl panel	71-11-04-01-040	PTF ALL
		71-11-04-01-045	PTF ALL
		71-11-04-01-050	PTF ALL
		71-11-04-01-055	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Prepare for the Installation

SUBTASK 71-11-04-860-017-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-04-860-020-G00

(2) Make sure that the thrust reverser is closed and latched (TASK 78-31-00-010-802-G00).

SUBTASK 71-11-04-020-002-G00

WARNING DO NOT TRY TO REMOVE OR INSTALL THE FAN COWL PANELS IN HIGH WINDS OR GUST CONDITIONS OF MORE THAN 40 KNOTS. THE WINDS CAN CAUSE THE FAN COWL PANELS TO MOVE SUDDENLY. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT ALL FAN COWL CROSS ENGINE DEBRIS SHIELDS ARE INSTALLED ON THE INBOARD FAN COWL PANEL OF ENGINE 1. OTHERWISE, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING MAKE SURE THAT THE FAN COWL RADIAL RESTRAINT BRACKET CONFIGURATION IS COMPATIBLE WITH THE ENGINE SIDE RADIAL RESTRAINT BRACKET CONFIGURATION. IF YOU DO NOT OBEY, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT THE FAN COWL CHINE IS INSTALLED ONLY ON THE INBOARD FAN COWL PANEL OF EACH ENGINE. THE INCORRECT INSTALLATION OF THE FAN COWL CHINE CAN CAUSE UNSATISFACTORY AIRPLANE PERFORMANCE.

(3) If the new fan cowl panel [1] is used, do the steps that follow:

NOTE: The fan cowls are interchangeable between engine position, but you must install the chine and Cross Engine Debris (CED) shields in the correct location.

(a) Do a check to make sure that whether the fan cowl panel requires to have the fan cowl chine installed.

NOTE: The fan cowl chine is installed only on the inboard fan cowl panel of each engine.

1) If the fan cowl panel is to be used on the inboard of engine and if the fan cowl chine is not installed, install the fan cowl chine [41] (TASK 71-11-02-400-801-G00).

2) If the fan cowl panel is to be used on the outboard of engine and if the fan cowl chine is installed, remove the fan cowl chine [41] (TASK 71-11-02-000-801-G00).

(b) Do a check to make sure that whether the fan cowl panel requires to have the fan cowl CED shields installed.

NOTE: The fan cowl CED shields are installed only on the inboard fan cowl of engine 1.

1) If the fan cowl panel is to be used on the inboard of engine 1 and if the fan cowl CED shields are not installed, install the fan cowl CED shields (TASK 71-11-03-000-801-G00).

2) If the fan cowl panel is to be used on the outboard of engine 2 and if the fan cowl CED shields are installed, remove the fan cowl CED shields (TASK 71-11-03-400-801-G00).

(c) Do a check to make sure that the fan cowl radial restraint bracket configuration is compatible with the engine radial restraint bracket configuration (Figure 404).

NOTE: A radial restraint pin must be installed in either bracket with a mating bushing installed in the opposite bracket.

1) If it is necessary, replace the compatible engine radial restraint bracket (TASK 71-11-13-000-801-G00 and TASK 71-11-13-400-801-G00).

(d) If the fan cowl sling, SPL-14546, is not attached to the fan cowl panel [1], attach the fan cowl sling, SPL-14546, as follows (Figure 401):

1) Hang the sling hoist plate (master link) onto the hoist, STD-467.

2) Attach a load cell, STD-12395, to the sling link.

NOTE: The load cell, STD-12395, or equivalent or similar capacity equipment is customer furnished equipment. The required minimum load cell capacity is 250 lb (113 kg).

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3) Attach the three sling straps [2] to the load cell, STD-12395, with the three Lok-A-Loys and sling link.

NOTE: The sling straps, sling link, and three Lok-A-Loys are part of the fan cowl sling, SPL-14546.

4) Set the load cell, STD-12395, to zero.

5) Put the hoist, STD-467, and assembled fan cowl sling, SPL-14546, above the fan cowl panel [1].

a) Make sure that the fan cowl sling, SPL-14546, does not touch the surface of the fan cowl panel [1].

WARNING

MONITOR THE WEIGHT OF THE LOAD WHILE YOU PUT WEIGHT ON THE SLING. MAKE SURE THAT THE WEIGHT OF THE LOAD IS LESS THAN THE WEIGHT LIMIT. IF THE WEIGHT OF THE LOAD IS MORE THAN THE LIMIT, THE COMPONENT CAN BREAK. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

6) Adjust the hoist, STD-467, to hold the weight of the fan cowl panel [1].

NOTE: The fan cowl panel weighs approximately 160 lb (73 kg).

7) Install the three hoist pins [3] on the fan cowl panel [1].

a) Install the three bolts [7] through the three hoist pins [3] in the plug holes.

b) Tighten the bolts [7] to 10 in-lb (1.1 N·m) - 30 in-lb (3.4 N·m).

H. Fan Cowl Panels Installation

SUBTASK 71-11-04-410-004-G00

WARNING

GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Install the fan cowl panel [1] on the applicable engine as follows:

NOTE: Boeing does not recommend the installation of the fan cowl without the use of a crane and sling.

(a) If it is installed on the strut, remove the quick-release pins [4] from the hinge assemblies.

NOTE: Let the quick-release pins hang by their lanyards.

(b) Apply grease, D00014, to the quick-release pins [4].

(c) Install a rope - 12 ft (3.6 m) minimum, STD-748, on the latch [20].

NOTE: Use the rope as optional to control the fan cowl panel's position when it is lifted.

(d) Lift the fan cowl panel [1] with the hoist, STD-467.

(e) Adjust the sling strap [2] until the fan cowl panel [1] hangs at approximately 61 degrees (open position).

(f) Align the fan cowl panel [1] hinges with the strut hinge assemblies.

1) Carefully move the fan cowl panel [1] with the help of the rope - 12 ft (3.6 m) minimum, STD-748, and/or other personnel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT FULLY DISENGAGE THE FAN COWL FROM THE STRUT HINGES WHILE A MECHANIC IS UNDER THE COWLING. USE THE DRIFT PINS TO SECURE THE COWLING THEN REMOVE THE DRIFT PINS BY REACHING UNDER THE COWLING. INJURIES TO PERSONS CAN OCCUR.

(g) Install the drift pins from the sling tool kit into the forward and aft hinge assemblies.

(h) Insert the quick-release pins [4] in the center hinge assembly from the forward side.

1) Install the safety pin through the hole in the end of the quick-release pins [4].

(i) Remove the drift pin from the forward hinge assembly and insert the quick-release pins [4] from the aft side.

1) Install the safety pin through the hole in the end of the quick-release pins [4].

(j) Remove the drift pin from the aft hinge assembly and insert the quick-release pins [4] from the forward side.

1) Install the safety pin through the hole in the end of the quick-release pins [4].

(k) Connect the bonding jumpers [5] to the brackets on the forward and center hinge assembly.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the bracket, washer [24], and bonding jumper [5].

2) Install the washer [24], bonding jumper [5], washer [23], washer [22], and screw [21].

3) Tighten the screw [21] to 35 in-lb (4.0 N·m) - 45 in-lb (5.1 N·m).

4) Measure the electrical bonding resistance between the bonding jumper [5] and bracket (SWPM 20-20-00).

a) Use an intrinsically safe approved bonding meter, COM-1550.

b) Make sure that the electrical bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

(l) Connect the fan cowl SDOS actuator [12] to the fan cowl panel fitting as follows:

1) Align the fan cowl SDOS actuator [12] end with the fan cowl panel fitting.

2) Install the bolts [8], washers [9], bushing [10], washer [11], and nut [13] that attach the fan cowl SDOS actuator [12] to the fan cowl panel fitting.

3) Tighten the nut [13] to 310.00 in-lb (35.0 N·m) - 330.00 in-lb (37.3 N·m).

(m) Connect the fan cowl hold-open rod [6] to the fan cowl panel fitting as follows:

1) Align the fan cowl hold-open rod [6] end with the fan cowl panel fitting.

2) Install the bolt [14], washer [15], bushing [16], washer [17], and nut [18] that attach the fan cowl hold-open rods [6] to the fan cowl panel fitting.

3) Tighten the nut [18] to 165 in-lb (18.6 N·m) - 175 in-lb (19.8 N·m).

(n) Remove the rope - 12 ft (3.6 m) minimum, STD-748, from the latch [20].

SUBTASK 71-11-04-490-008-G00

(2) Remove the fan cowl sling, SPL-14546, as follows:

(a) Lower the hoist, STD-467, until there is no load on the fan cowl sling, SPL-14546.

(b) Slowly close the fan cowl panel [1] with the fan cowl sling, SPL-14546, attached.

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1) Make sure that the fan cowl sling, SPL-14546, does not touch the surface of the fan cowl panel [1].

(c) Disconnect the three sling straps [2] from the sling link.

(d) Remove the sling hoist plate (master link) from the hoist, STD-467.

(e) Remove the three hoist pins [3] from the fan cowl panel [1].

1) Remove the three bolts [7] through the three hoist pins [3] in the plug holes.

I. Fan Cowl Panels Installation Test

SUBTASK 71-11-04-610-001-G00

(1) Close the applicable fan cowl panel [1] (TASK 71-11-04-410-801-G00).

SUBTASK 71-11-04-710-001-G00

(2) If the new fan cowl panel is used, do this task: Fan Cowl Latch Adjustment, TASK 71-11-05-821-801-G00.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-04-440-004-G00

(1) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 71-11-04-860-014-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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FAN COWL PANELS - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fan cowl panels inspection.

TASK 71-11-04-200-801-G00

2. Fan Cowl Panels Inspection

A. General

(1) This task gives the instructions to do the visual inspection of the fan cowl panel.

B. References

Reference	Title
71-11-02-000-801-G00	Fan Cowl Chine Removal (P/B 401)
71-11-02-400-801-G00	Fan Cowl Chine Installation (P/B 401)
71-11-04-000-801-G00	Fan Cowl Panels Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-400-801-G00	Fan Cowl Panels Installation (P/B 401)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
71-11-05-000-801-G00	Fan Cowl Latches Removal (P/B 401)
71-11-05-000-802-G00	Fan Cowl Latch Keeper Pin Removal (P/B 401)
71-11-05-400-801-G00	Fan Cowl Latches Installation (P/B 401)
71-11-05-400-802-G00	Fan Cowl Latch Keeper Pin Installation (P/B 401)
71-11-10-000-801-G00	Engine Oil Tank Access Door Removal (P/B 401)
71-11-10-400-801-G00	Engine Oil Tank Access Door Installation (P/B 401)
71-12-01-000-801-G00	Fan Cowl SDOS Actuators Removal (P/B 401)
71-12-01-400-801-G00	Fan Cowl SDOS Actuators Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 71-11-04-860-015-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-04-010-007-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-04-490-009-G00

(3) Put the engine and strut access platform, STD-653 into position.

G. Fan Cowl Panels Inspection

SUBTASK 71-11-04-210-001-G00

(1) Examine the fan cowl panel for damage to the skin with the limits that follow:

(a) Examine the skin of the fan cowl panel for scores, cracks, gouges, nicks, and loose or missing fasteners.

1) If there are cracks and loose or missing fasteners, replace the fan cowl panel (TASK 71-11-04-000-801-G00 and TASK 71-11-04-400-801-G00).

2) If there are scores, cracks, gouges or nicks, refer to the Structural Repair Manual (SRM) for repair instructions.

SUBTASK 71-11-04-210-002-G00

(2) Examine the fan cowl panel for a delamination with the limits that follow:

(a) For the allowable damage limits for the fan cowl panel, refer to the SRM.

NOTE: The SRM gives allowable limits for delamination, dents, nicks, scratches, gouges, or corrosion in the fan cowl panel.

(b) If the delamination is not in the permitted limits, replace the fan cowl panel (TASK 71-11-04-000-801-G00 and TASK 71-11-04-400-801-G00).

SUBTASK 71-11-04-210-003-G00

(3) Examine the edge protection and locators of the fan cowl panel for cracks, gouges, nicks or loose or missing fasteners.

(a) If there are cracks, gouges, nicks, or loose or missing fasteners, replace the fan cowl panel (TASK 71-11-04-000-801-G00 and TASK 71-11-04-400-801-G00).

SUBTASK 71-11-04-210-004-G00

(4) Examine the fan cowl panel hinges for cracks, gouges, nicks or loose or missing fasteners.

(a) If there are cracks, gouges, nicks, or loose or missing fasteners, replace the fan cowl panel (TASK 71-11-04-000-801-G00 and TASK 71-11-04-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-04-210-005-G00

(5) Examine the latch housings of the fan cowl panel for cracks, gouges, nicks or loose or missing fasteners.

(a) If there are cracks, gouges, nicks, or loose or missing fasteners, replace the fan cowl panel latch (TASK 71-11-05-000-801-G00 and TASK 71-11-05-400-801-G00).

SUBTASK 71-11-04-210-006-G00

(6) Examine the latch keepers of the fan cowl panel for cracks, gouges, nicks or loose or missing fasteners.

(a) If there are cracks, gouges, nicks, or loose or missing fasteners replace the latch keepers (TASK 71-11-05-000-802-G00 and TASK 71-11-05-400-802-G00).

SUBTASK 71-11-04-210-007-G00

(7) Examine the oil tank and IDG access doors on the fan cowl panels for cracks, gouges, nicks or loose or missing fasteners.

(a) If there are cracks, gouges, nicks, or loose or missing fasteners, replace the oil tank access door (TASK 71-11-10-000-801-G00 and TASK 71-11-10-400-801-G00).

SUBTASK 71-11-04-210-008-G00

(8) Examine the fan cowl chine for cracks, gouges, nicks or loose or missing fasteners.

(a) If there are cracks, gouges, nicks, or loose or missing fasteners, replace the fan cowl chine (TASK 71-11-02-000-801-G00 and TASK 71-11-02-400-801-G00).

SUBTASK 71-11-04-210-009-G00

(9) Examine the Spring Door Opening System (SDOS) actuators for leaks (TASK 71-12-01-000-801-G00).

SUBTASK 71-11-04-900-001-G00

(10) If there is damage or leaks, replace the SDOS actuator (TASK 71-12-01-000-801-G00 and TASK 71-12-01-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-04-090-006-G00

(1) Remove the engine and strut access platform, STD-653.

NOTE: Make sure that the work area is clean of all tools and items used for the task.

SUBTASK 71-11-04-860-016-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 71-11-04-410-005-G00

(3) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

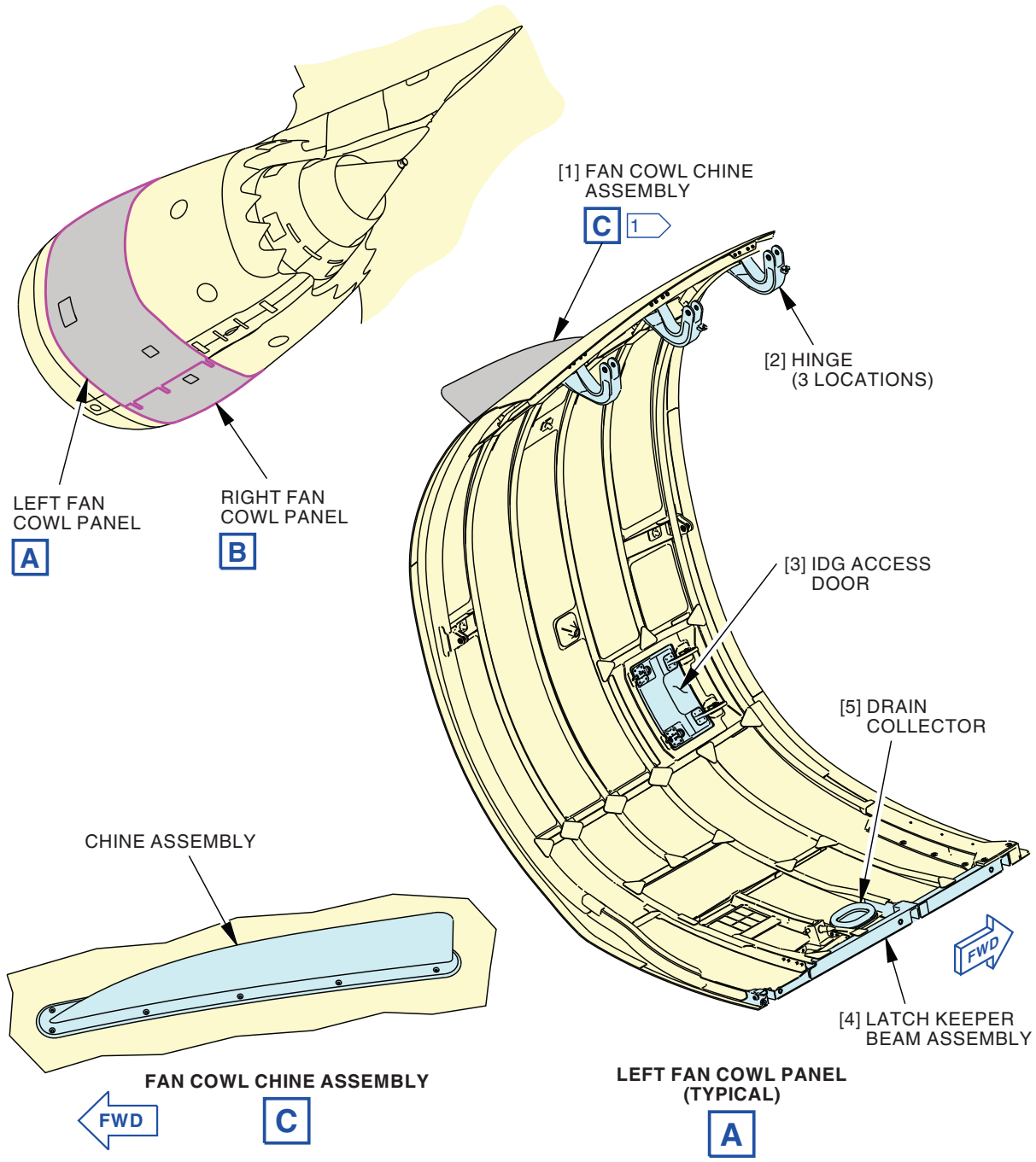
(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



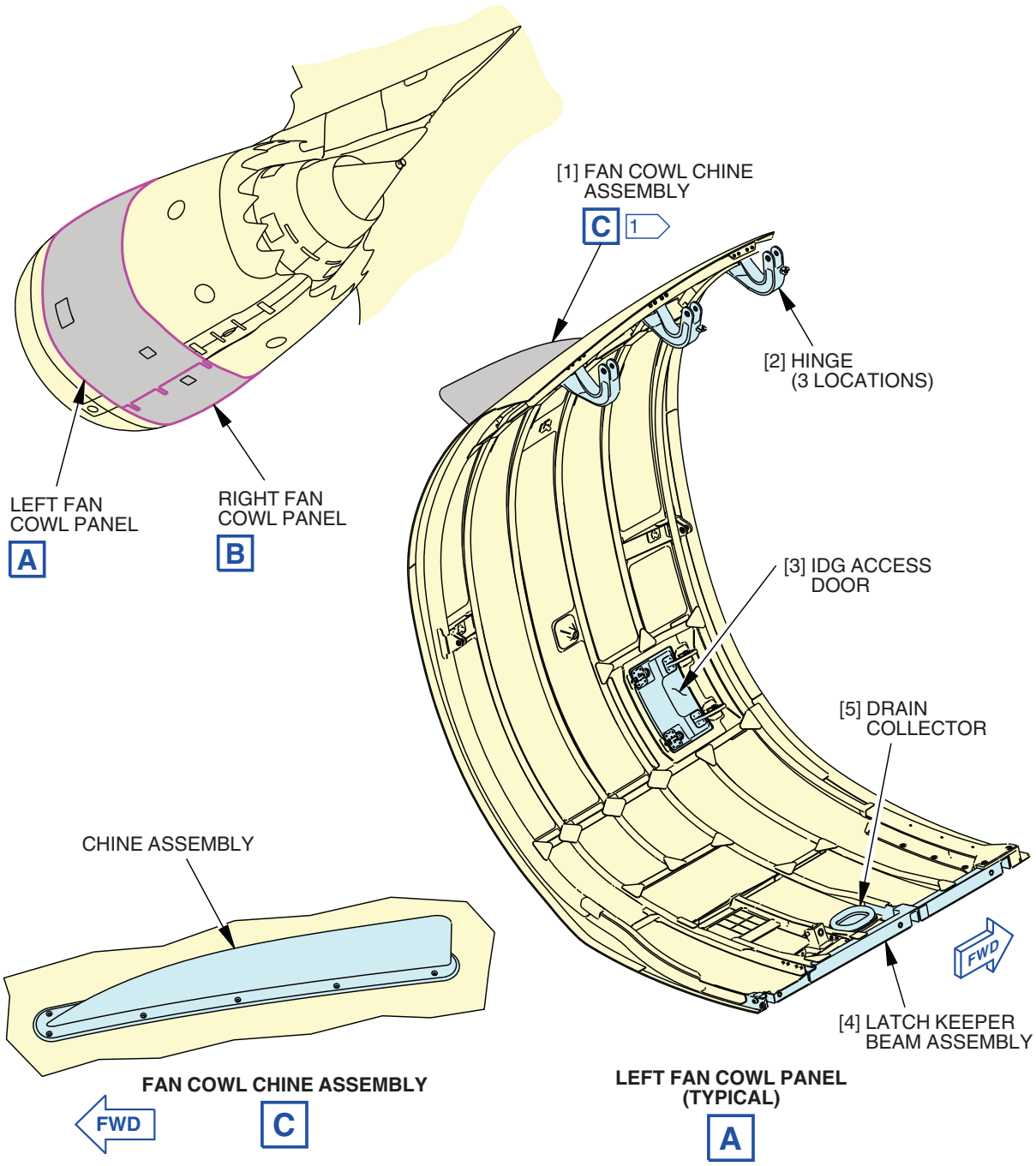
NOTE:

1 CHINE ASSEMBLY INSTALLED ON THE LEFT FAN COWL ON ENGINE 2.

2354386 S00061515227_V2

Fan Cowls Panels Inspection
Figure 601/71-11-04-990-807-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



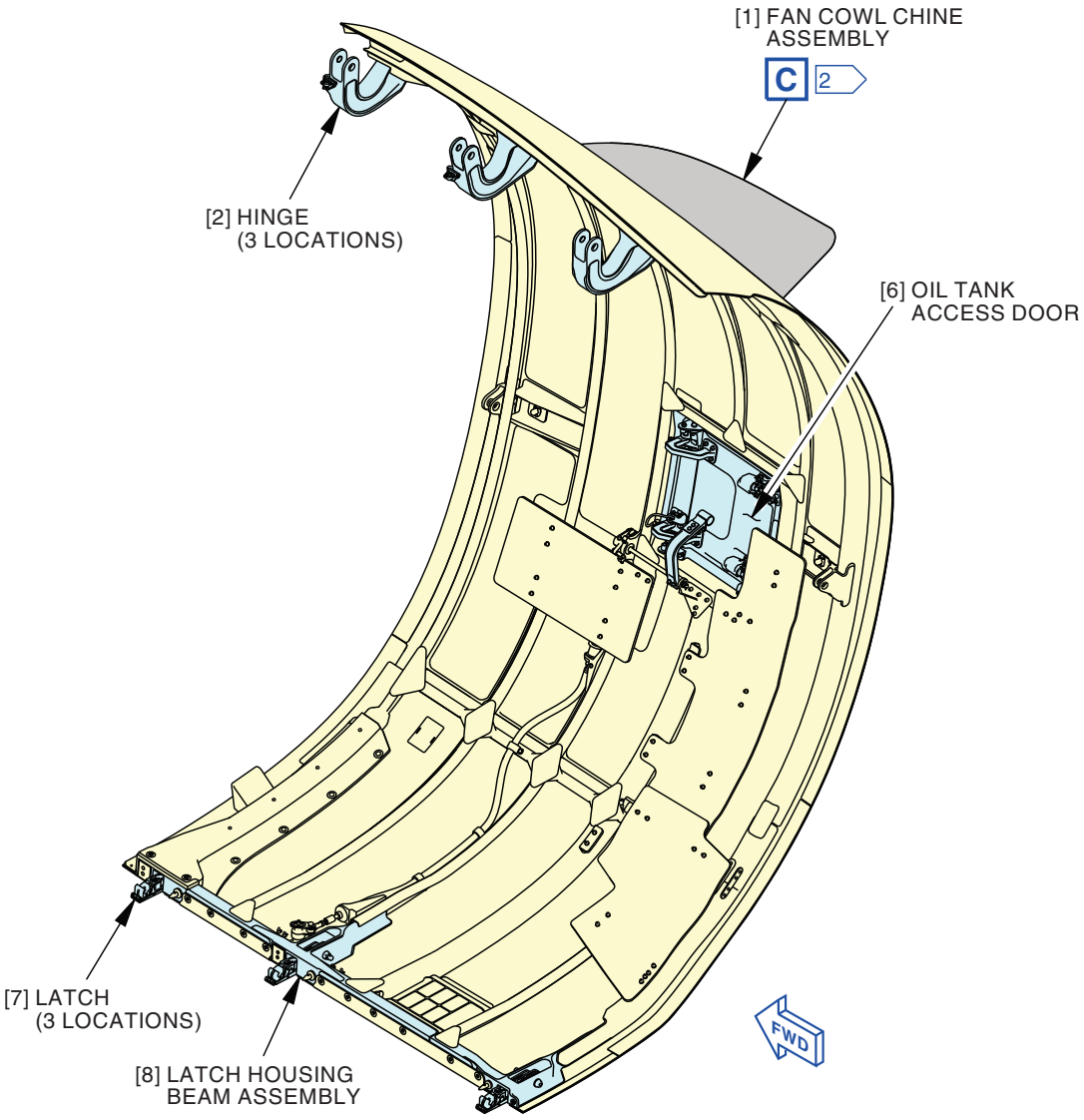
NOTE:

1 CHINE ASSEMBLY INSTALLED ON THE LEFT FAN COWL ON ENGINE 2.

3102983 S0000846868_V1

Fan Cowls Panels Inspection
Figure 601/71-11-04-990-807-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIGHT FAN COWL PANEL (TYPICAL)

B

NOTE:

2 CHINE ASSEMBLY INSTALLED ON THE RIGHT FAN COWL ON ENGINE 1.

2354387 S00061515228_V3

Fan Cowls Panels Inspection
Figure 601/71-11-04-990-807-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL LATCHES AND KEEPERS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the fan cowl panel latches

(2) An installation of the fan cowl panel latches

(3) A removal of the fan cowl panel latch keeper pin

(4) An installation of the fan cowl panel latch keeper pin.

TASK 71-11-05-000-801-G00

2. Fan Cowl Latches Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fan cowl latches.

(2) The fan cowl latch is referred to as the latch.

(3) The latches are installed on the latch housing beams at the 6:00 o'clock position of the right fan cowl panel.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 71-11-05-868-001-G00

CAUTION

RETRACT THE LEADING EDGE SLATS, AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THESE PRECAUTIONS, THE FAN COWL PANELS WILL HIT THE SLATS. DAMAGE WILL OCCUR.

(1) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-05-010-001-G00

(2) Open the applicable right fan cowl panel (TASK 71-11-04-010-801-G00).

E. Fan Cowl Latches Removal

SUBTASK 71-11-05-020-005-G00

(1) Remove the No. 1 latch [1] from the latch housing beam as follows:

(a) Remove the thermal blanket at the No. 1 latch [1] to get access to the bolt [2].

(b) Remove the lockwire, nut [3], washer [4], washer [4A], and bolt [2] that attach the No. 1 latch [1] to the latch housing beam.

1) If it is necessary, remove the screws on the fan cowl structure to remove the bolt [2].

(c) Remove the No. 1 latch [1].

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SUBTASK 71-11-05-020-001-G00

(2) Remove the No. 2 latch [1] from the latch housing beam as follows:

(a) Remove the lockwire, nut [3], washer [4A], washer [4], and bolt [2] that attach the No. 2 latch [1] to the latch housing beam.

(b) Remove the No. 2 latch [1].

SUBTASK 71-11-05-020-002-G00

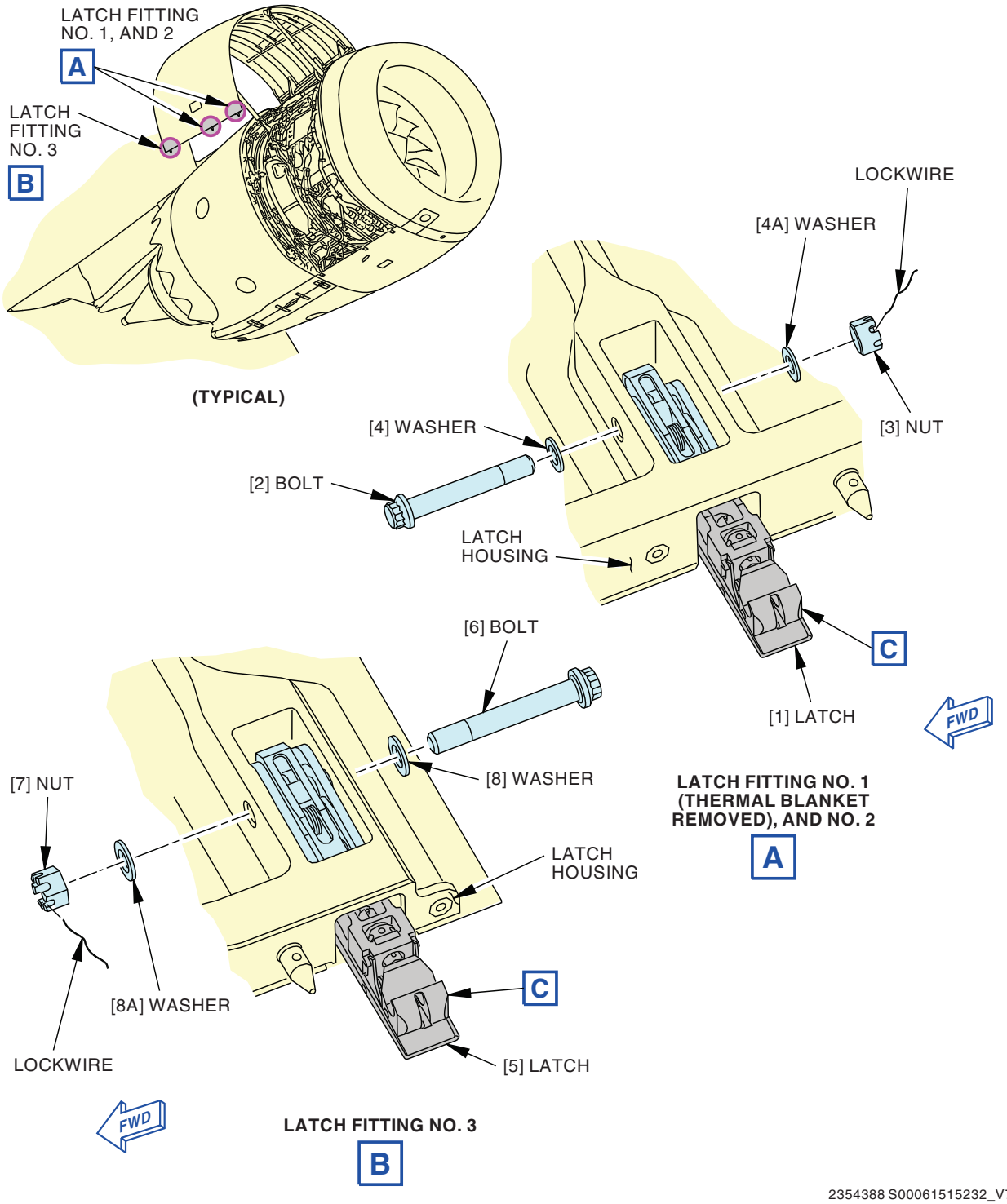
(3) Remove the No. 3 latch [5] from the latch housing beam as follows:

(a) Remove the lockwire, nut [7], washer [8A], washer [8] and bolt [6] that attach the No. 3 latch [5] to the latch housing beam.

(b) Remove the No. 3 latch [5].

——— END OF TASK ———

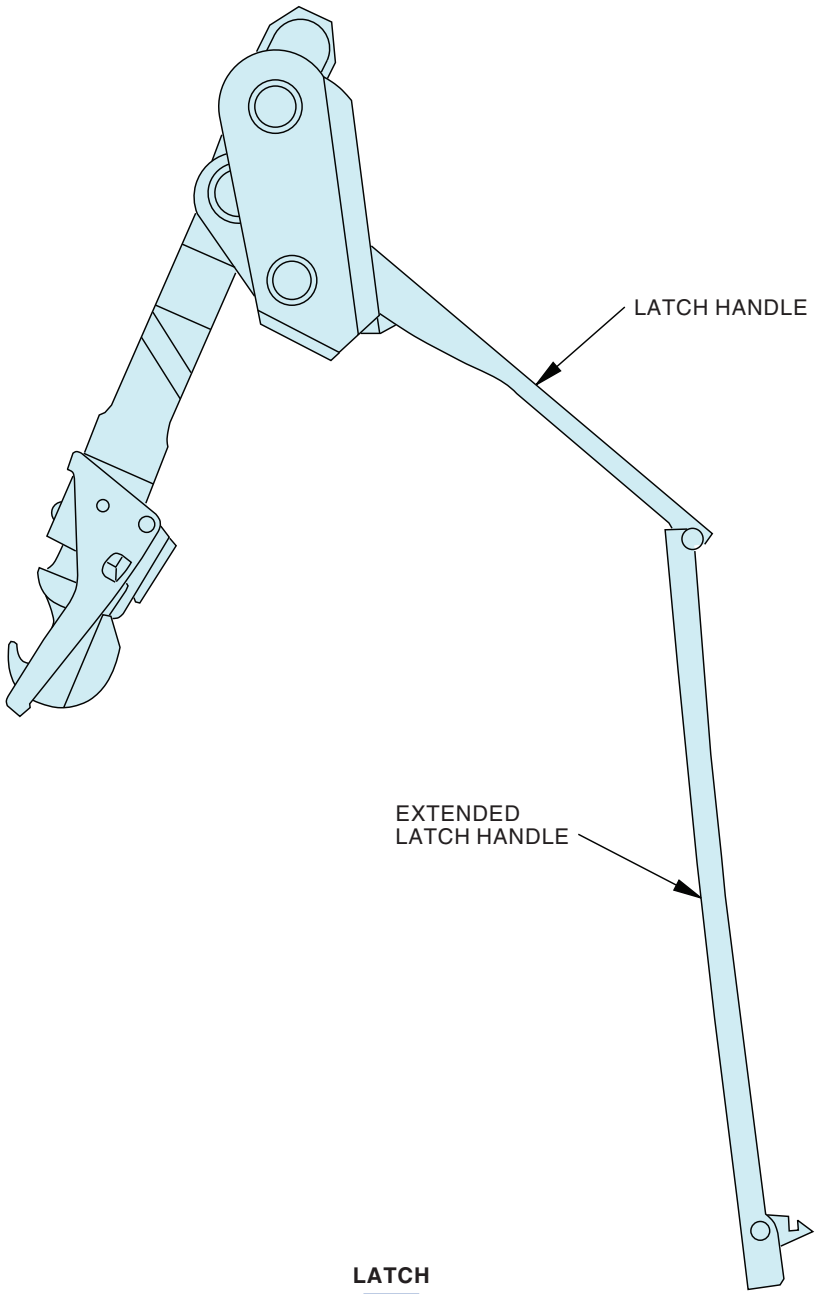
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354388 S00061515232_V7

Fan Cowl Latches Installation
Figure 401/71-11-05-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2483416 S0000571099_V1

Fan Cowl Latches Installation
Figure 401/71-11-05-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-05-400-801-G00

3. Fan Cowl Latches Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fan cowl latches.

(2) The fan cowl latch is referred to as the latch.

(3) You must do an adjustment of the latches after you install the latches.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
71-11-05-821-801-G00	Fan Cowl Latch Adjustment (P/B 501)

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Latch	71-11-05-05-040	PTF ALL
5	Latch	71-11-05-05-040	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Fan Cowl Latches Installation

SUBTASK 71-11-05-420-005-G00

(1) Install the No. 1 latch [1] as follows:

(a) Put the No. 1 latch [1] in its position on the latch housing beam.

(b) Align the bolt hole with the latch holes on the latch housing beam.

(c) Install the bolt [2], washer [4], washer [4A], and nut [3] that attach the No. 1 latch [1] to the latch housing beam.

1) Tighten the nut [3] as required to obtain 0.01 in. (0.25 mm) - 0.04 in. (1.02 mm) axial movement between the bolt head and housing beam assembly.

NOTE: Axial movement can be checked by measuring the gap between the washer and the housing beam assembly with the bolt head in contact with the housing beam assembly or by measuring the gap between the bolt head and the housing beam assembly with the washer/nut in contact with the housing beam assembly.

2) Place a fillet of sealant, A50110, around the bolt shank between washer and housing beam assembly to dampen the washer movement.

3) Lockwire the nut [3].

4) If it is necessary, re-install the screws that were removed from the fan cowl structure.

(d) Install the thermal blanket at the No. 1 latch [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-05-420-001-G00

(2) Install the No. 2 latch [1] as follows:

(a) Put the No. 2 latch [1] in its position on the latch housing beam.

(b) Align the bolt hole with the latch holes on the latch housing beam.

(c) Install the bolt [2], washer [4], washer [4A], and nut [3] that attach the No. 2 latch [1] to the latch housing beam.

1) Tighten the nut [3] as required to obtain 0.01 in. (0.25 mm) - 0.04 in. (1.02 mm) axial movement between the bolt head and housing beam assembly.

NOTE: Axial movement can be checked by measuring the gap between the washer and the housing beam assembly with the bolt head in contact with the housing beam assembly or by measuring the gap between the bolt head and the housing beam assembly with the washer/nut in contact with the housing beam assembly.

2) Place a fillet of sealant, A50110, around the bolt shank between washer and housing beam assembly to dampen the washer movement.

3) Lockwire the nut [3].

SUBTASK 71-11-05-420-002-G00

(3) Install the No. 3 latch [5] as follows:

(a) Put the latch [5] in its position on the latch housing beam.

(b) Align the bolt hole with the latch holes on the latch housing beam.

(c) Install the bolt [6], washer [8], washer [8A], and nut [7] that attach the latch [5] to the latch housing beam.

1) Tighten the nut [7] as required to obtain 0.01 in. (0.25 mm) - 0.04 in. (1.02 mm) axial movement between the bolt head and housing beam assembly.

NOTE: Axial movement can be checked by measuring the gap between the washer and the housing beam assembly with the bolt head in contact with the housing beam assembly or by measuring the gap between the bolt head and the housing beam assembly with the washer/nut in contact with the housing beam assembly.

2) Place a fillet of sealant, A50110, around the bolt shank between washer and housing beam assembly to dampen the washer movement.

3) Lockwire the nut [7].

G. Fan Cowl Latches Installation Test

SUBTASK 71-11-05-860-001-G00

(1) Open and close the fan cowl panels at least three times to see the latches are operated smoothly.

SUBTASK 71-11-05-820-001-G00

(2) Do this task: Fan Cowl Latch Adjustment, TASK 71-11-05-821-801-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-05-410-001-G00

(1) Close the applicable right fan cowl panel (TASK 71-11-04-410-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-05-865-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

END OF TASK

TASK 71-11-05-000-802-G00

4. Fan Cowl Latch Keeper Pin Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the fan cowl latch keeper pin.

(2) The fan cowl latch keeper is referred to as the latch keeper pin.

(3) The latch keeper pins are installed on the keeper beams at the 6:00 o'clock position of the left fan cowl panel.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 71-11-05-868-002-G00

CAUTION RETRACT THE LEADING EDGE SLATS, AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THESE PRECAUTIONS, THE FAN COWL PANELS WILL HIT THE SLATS. DAMAGE WILL OCCUR.

(1) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-05-010-002-G00

(2) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

E. Fan Cowl Latch Keeper Pin Removal

SUBTASK 71-11-05-020-006-G00

(1) Remove the No. 1 latch keeper [14] from the keeper beam as follows:

(a) Remove the thermal blanket at the No. 1 latch keeper [14] to get access to the bolt [15].

(b) Remove the lockwire [18], nut [19], washer [17], washer [16], and bolt [15] that attach the No. 1 latch keeper [14] to the latch keeper beam.

SUBTASK 71-11-05-020-003-G00

(2) Remove the No. 2 latch keeper [14] from the keeper beam as follows:

(a) Remove the lockwire [18], nut [19], washer [17], washer [16], and bolt [15] that attach the No. 2 latch keeper [14] to the latch keeper beam.

SUBTASK 71-11-05-020-004-G00

(3) Remove the No. 3 latch keeper [20] from the keeper beam as follows:

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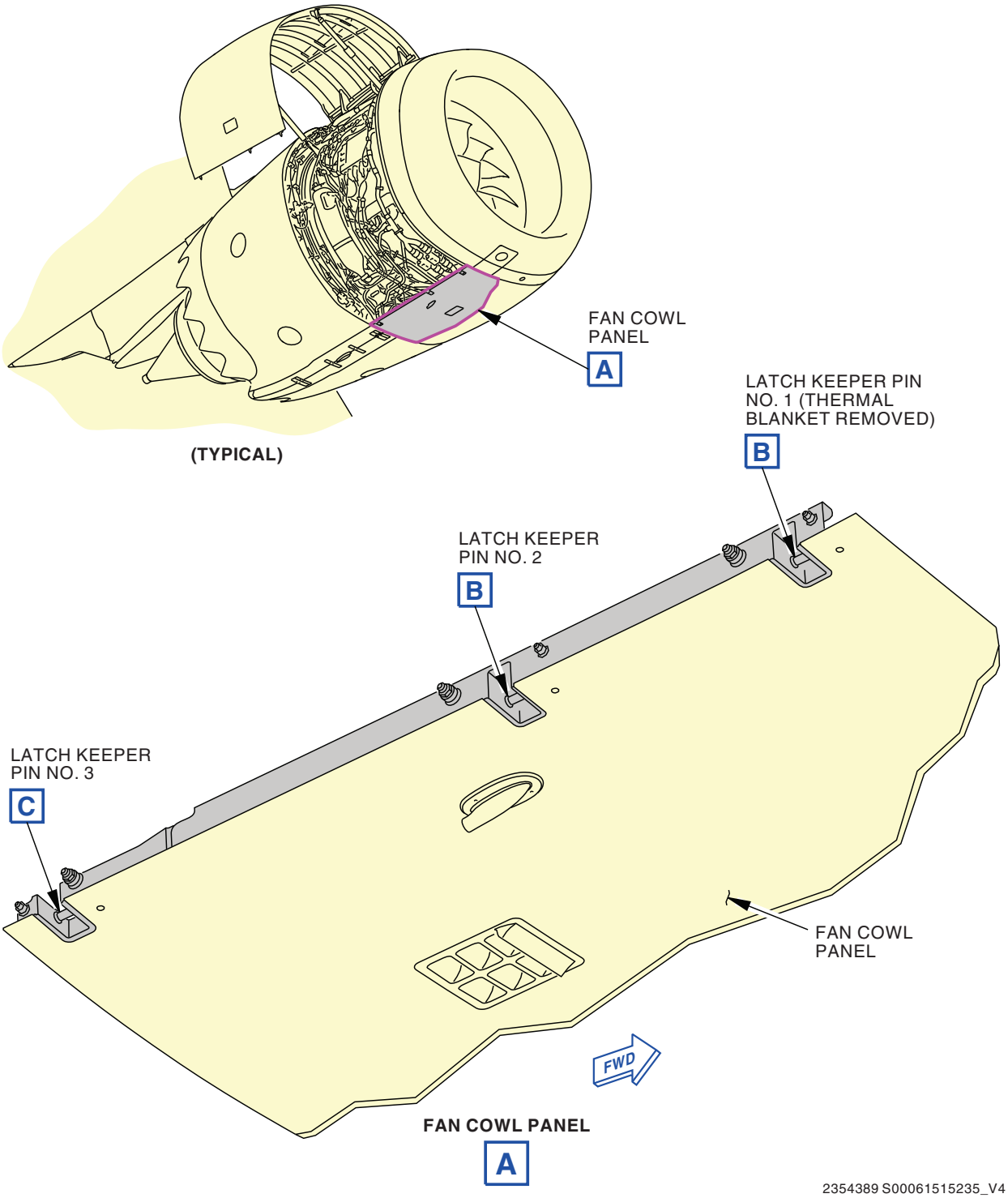
(a) Remove the lockwire [24], nut [25], washer [23], washer [22], and bolt [21] that attach the No. 3 latch keeper [20] to the latch keeper beam.

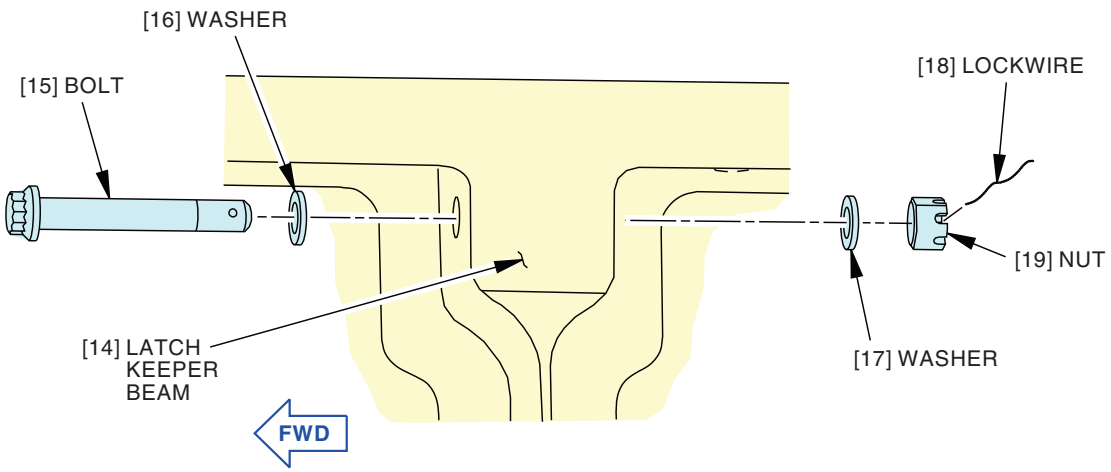
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



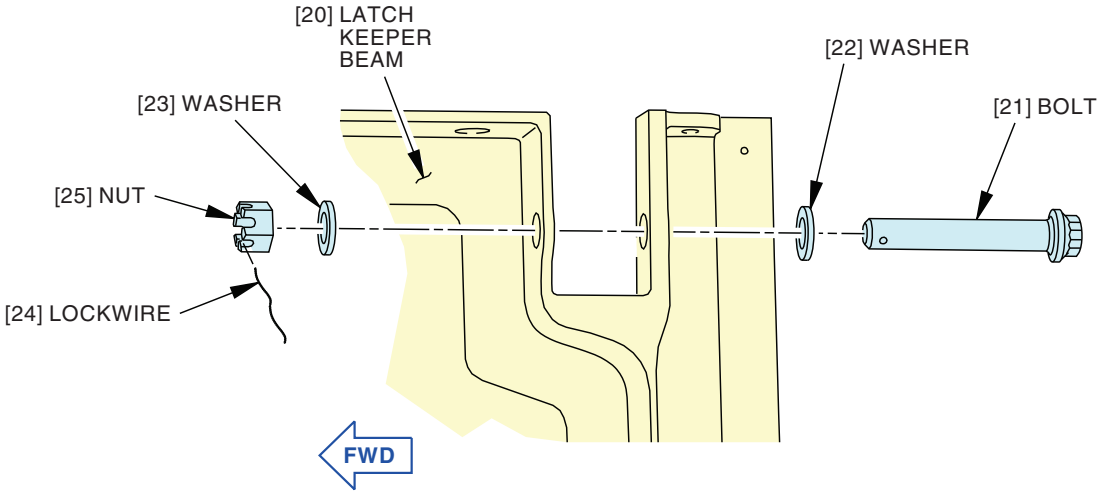
Fan Cowl Latch Keeper Pin Installation
Figure 402/71-11-05-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LATCH KEEPER PIN NO. 1 (THERMAL BLANKET REMOVED), AND NO. 2

B



LATCH KEEPER PIN NO. 3

C

2354390 S00061515236_V5

Fan Cowl Latch Keeper Pin Installation
Figure 402/71-11-05-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-05-400-802-G00

5. Fan Cowl Latch Keeper Pin Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the fan cowl latch keeper pin.

(2) The fan cowl latch keeper pin is referred to as the latch keeper pin.

(3) The latch keeper pins are installed on the keeper beams at the 6:00 o'clock position of the left fan cowl panel.

(4) Do the fan cowl latch keeper adjustment after you install the latch keeper pin.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
71-11-05-821-801-G00	Fan Cowl Latch Adjustment (P/B 501)

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	Latch keeper	71-11-05-01-100	PTF ALL
20	Latch keeper	71-11-05-01-115	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Fan Cowl Latch Keeper Pin Installation

SUBTASK 71-11-05-420-006-G00

(1) Install the No. 1 latch keeper [14] as follows:

(a) Install the bolt [15], washer [16], washer [17], and nut [19] that attach the No. 1 latch keeper [14] to the latch keeper beam.

1) Torque the nut [19] as required to obtain 0.01 in. (0.25 mm) - 0.04 in. (1.02 mm) axial movement between the bolt head and housing beam assembly.

NOTE: Axial movement can be verified by measuring the gap between the washer and the housing beam assembly with the bolt head in contact with the housing beam assembly or by measuring the gap between the bolt head and the housing beam assembly with the washer/nut in contact with the housing beam assembly.

2) Place a fillet of sealant, A50110 around the bolt shank between washer and housing beam assembly to dampen the washer movement.

(b) Install the lockwire [18] on the nut [19].

(c) Install the thermal blanket at the No. 1 latch keeper [14].

SUBTASK 71-11-05-420-003-G00

(2) Install the No. 2 latch keeper [14] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the bolt [15], washer [16], washer [17], and nut [19] that attach the No. 2 latch keeper [14] to the latch keeper beam.

1) Torque the nut [19] as required to obtain 0.01 in. (0.25 mm) - 0.04 in. (1.02 mm) axial movement between the bolt head and housing beam assembly.

NOTE: Axial movement can be verified by measuring the gap between the washer and the housing beam assembly with the bolt head in contact with the housing beam assembly or by measuring the gap between the bolt head and the housing beam assembly with the washer/nut in contact with the housing beam assembly.

2) Place a fillet of sealant, A50110 around the bolt shank between washer and housing beam assembly to dampen the washer movement.

(b) Install the lockwire [18] on the nut [19].

SUBTASK 71-11-05-420-004-G00

(3) Install the No. 3 latch keeper [20] as follows:

(a) Install the bolt [21], washer [22], washer [23], and nut [25] that attach the No. 3 latch keeper [20] to the latch keeper housing beam.

1) Torque the nut [25] as required to obtain 0.01 in. (0.25 mm) - 0.04 in. (1.02 mm) axial movement between the bolt head and housing beam assembly.

NOTE: Axial movement can be verified by measuring the gap between the washer and the housing beam assembly with the bolt head in contact with the housing beam assembly or by measuring the gap between the bolt head and the housing beam assembly with the washer/nut in contact with the housing beam assembly.

2) Place a fillet of sealant, A50110 around the bolt shank between washer and housing beam assembly to dampen the washer movement.

(b) Install the lockwire [24] on the nut [25].

G. Fan Cowl Latch Keeper Pin Installation Test

SUBTASK 71-11-05-860-002-G00

(1) Open and close the fan cowl panels at least three times to see the keepers are engaged correctly.

SUBTASK 71-11-05-820-002-G00

(2) If it is necessary, do this task: Fan Cowl Latch Adjustment, TASK 71-11-05-821-801-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-05-410-002-G00

(1) Close the applicable left fan cowl panel (TASK 71-11-04-410-801-G00).

SUBTASK 71-11-05-865-002-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL LATCHES - ADJUSTMENT/TEST

1. General

A. This procedure has one task:

(1) Fan cowl latch adjustment.

TASK 71-11-05-821-801-G00

2. Fan Cowl Latch Adjustment

(Figure 501)

A. General

(1) This task gives the instructions to adjust the latch to the correct limits if you find the fan cowl panels are not in the aerodynamic smoothness.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-755	Scale - Spring, Push/Pull Type, 0 - 150lbs (0 - 68kg) Capacity

D. Consumable Materials

Reference	Description	Specification
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Prepare for the Adjustment

SUBTASK 71-11-05-865-003-G00

CAUTION

RETRACT THE LEADING EDGE SLATS, AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THESE PRECAUTIONS, THE FAN COWL PANELS WILL HIT THE SLATS. DAMAGE WILL OCCUR.

(1) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-05-040-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-05-010-003-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

H. Fan Cowl Latch Adjustment

SUBTASK 71-11-05-720-001-G00

(1) Make sure that the shims on the latch are not damaged or missing.

(a) If the shims on the latch beam are damaged or missing, do these steps:

1) Remove the bolt (or shear pin), washer, and the nut that attach the shim to the latch beam.

2) Remove and discard the damaged shim(s).

3) Add new shim stock (0.003 in. (0.0762 mm) laminations) to the shim.

4) Apply primer, C00259, to the shim mating surfaces.

5) Install the shim on the latch beam with the bolt (or shear pin), washer, and nut.

NOTE: You can change the grip length (± 2 sizes) of bolt to fit the shim thickness.

a) Make sure that the solid face of the shim is visible.

SUBTASK 71-11-05-410-003-G00

(2) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

SUBTASK 71-11-05-980-002-G00

(3) Make sure that the latches are installed in the correct position.

(a) If the latches are not in the correct position, adjust the latches.

1) Make sure that the end of the latch assembly gap at the latch slot is within 0.1800 ±0.0300 in. (4.5720 ±0.7620 mm).

2) Make sure that the fan cowl panel assembly surface and the latch surface is within 0.000 ±0.060 in. (0.000 ±1.524 mm).

SUBTASK 71-11-05-869-001-G00

(4) Do these steps to adjust the latches:

(a) Start with the middle latch, forward latch, then aft latch.

(b) Apply a force of 15.0 lbf (66.7 N) - 35.0 lbf (155.7 N) with hand pressure.

1) Use a spring scale, STD-755, at the end of the latch handle to close it.

2) Make sure that the scale force is measured between 0.90 in. (22.86 mm) - 1.10 in. (27.94 mm) from the end of the handle.

(c) If the force is not 15.0 lbf (66.7 N) - 35.0 lbf (155.7 N), do these steps to adjust the latches:

1) Keep the latch open to the free travel position.

2) Put a 5/32 in. (4 mm) hexagonal wrench into the holes on top of the hook latch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Turn the wrench in the aft direction to increase the latch tension or the forward direction to decrease the tension.

NOTE: Pushing in the hook latch will lower the resistance when turning the wrench.

(d) Repeat as necessary to make sure that all latches need 15.0 lbf (66.7 N) - 35.0 lbf (155.7 N) to close it.

SUBTASK 71-11-05-820-003-G00

(5) Measure and record the gap at the fan cowl split line at each of the latch location.

SUBTASK 71-11-05-820-004-G00

(6) If the gaps are not in the limits that follow, open the fan cowl panels (TASK 71-11-04-010-801-G00).

(a) If the gap at the fan cowl split line is not 0.04 in. (1.016 mm) - 0.32 in. (8.128 mm), remove or add 0.003 in. (0.0762 mm) laminations as it is necessary to get the correct gap.

1) Make sure to apply the primer, C00259, to the shim mating surfaces.

SUBTASK 71-11-05-410-004-G00

(7) If it is necessary, close the fan cowl panels (TASK 71-11-04-410-801-G00).

SUBTASK 71-11-05-860-003-G00

(8) Repeat all the steps above until the latch force, the gap between the shims and fan cowl panel, and the gap at the fan cowl split line are in the limits.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-05-410-005-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

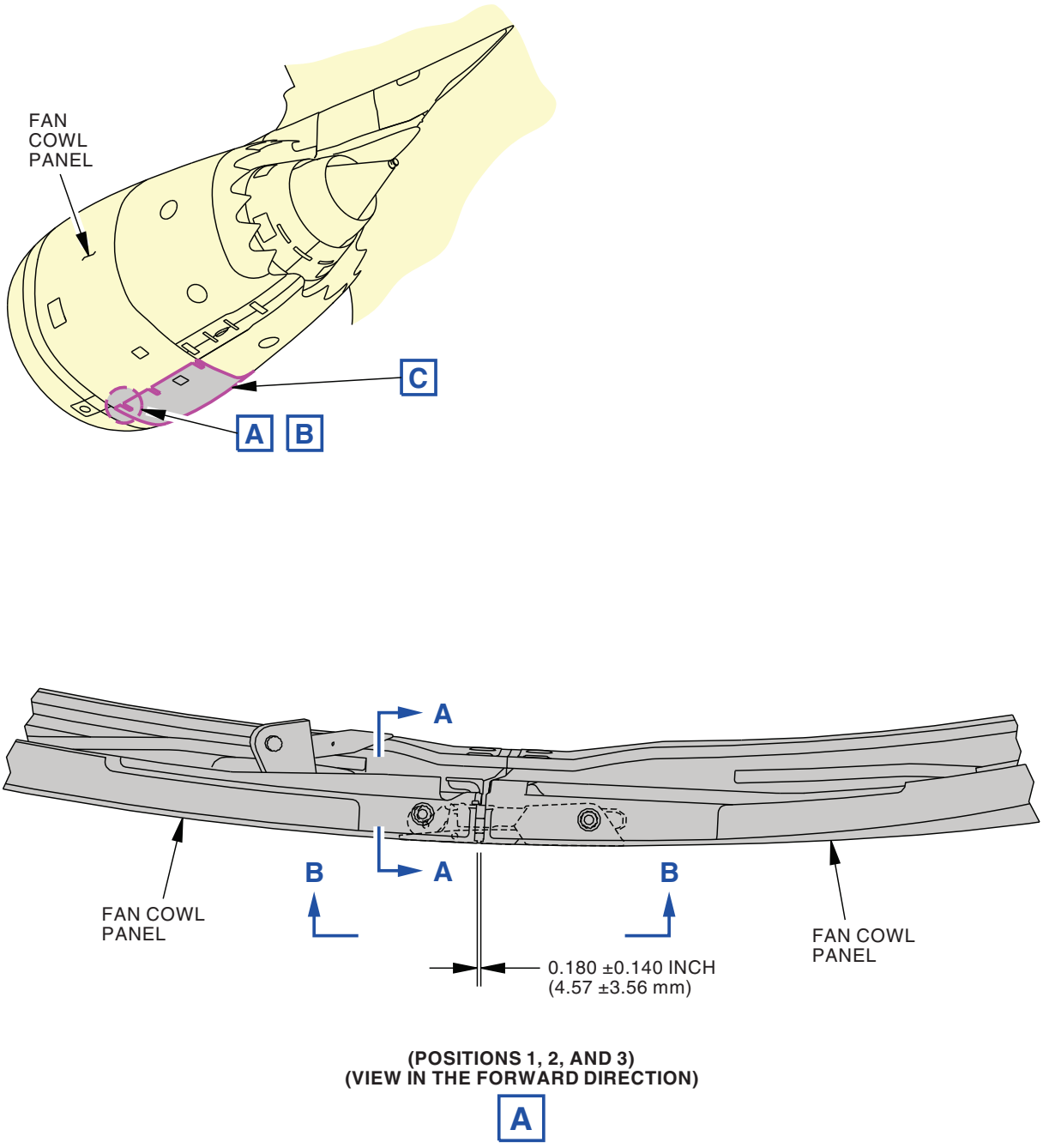
Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-05-410-006-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



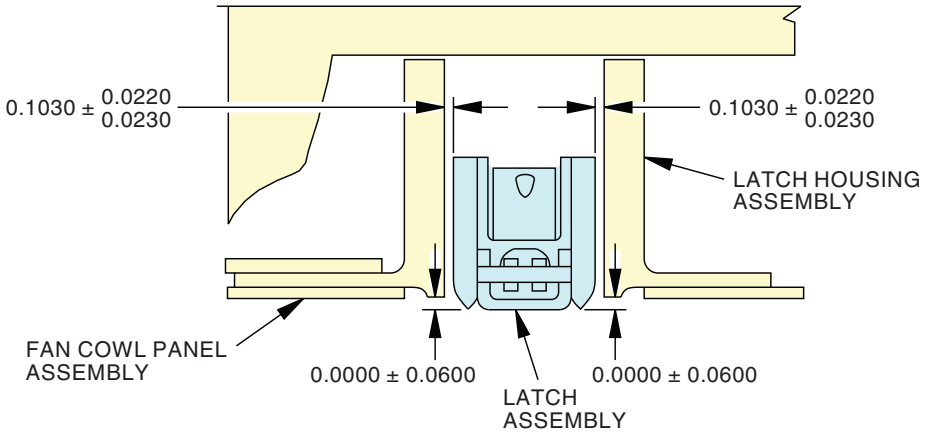
(POSITIONS 1, 2, AND 3)
(VIEW IN THE FORWARD DIRECTION)

A

2354391 S00061515240_V2

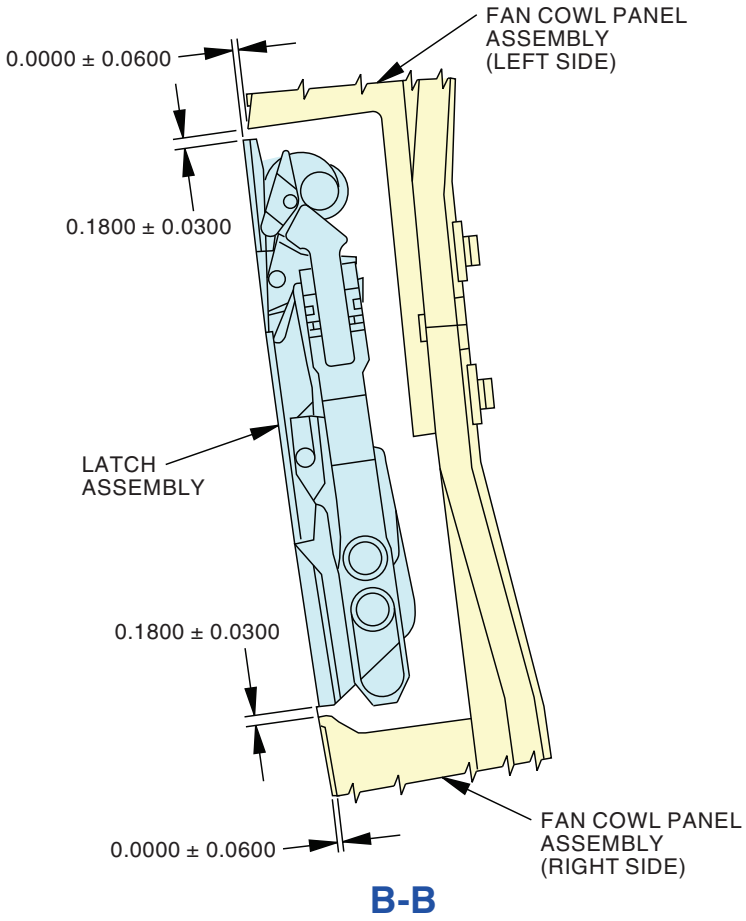
Fan Cowl Latches Adjustment
Figure 501/71-11-05-990-803-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

A-A

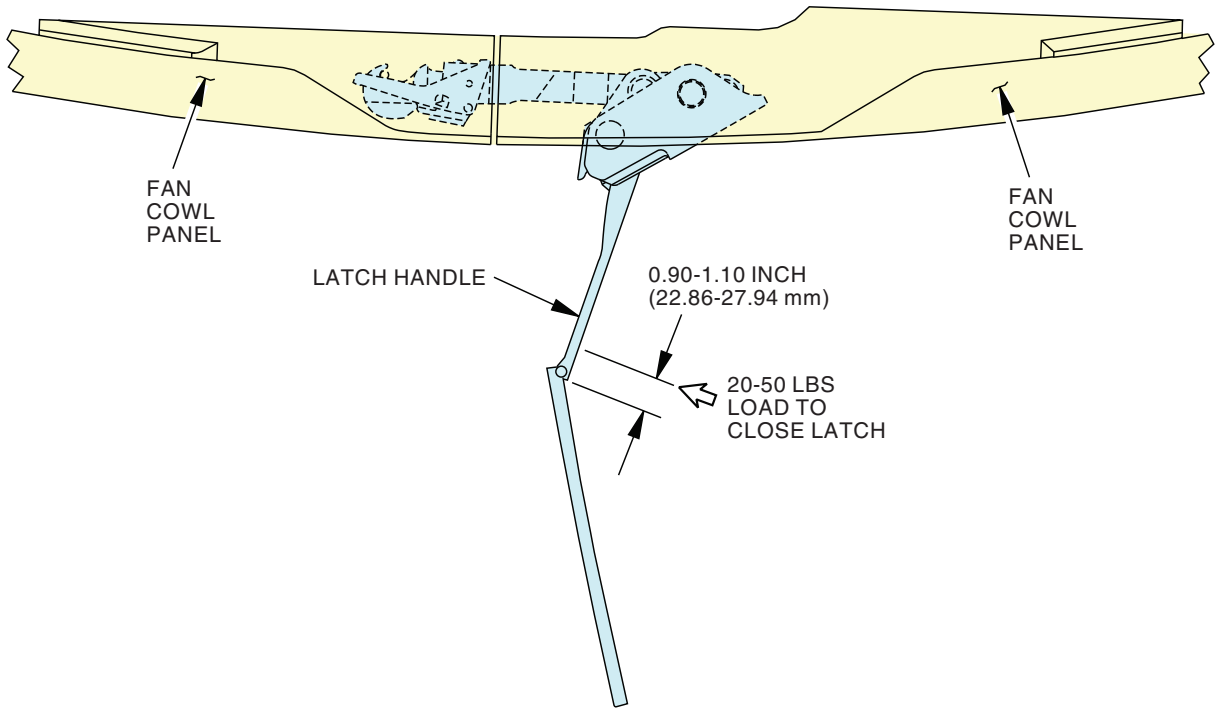


B-B

2354392 S00061515241_V1

Fan Cowl Latches Adjustment
Figure 501/71-11-05-990-803-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



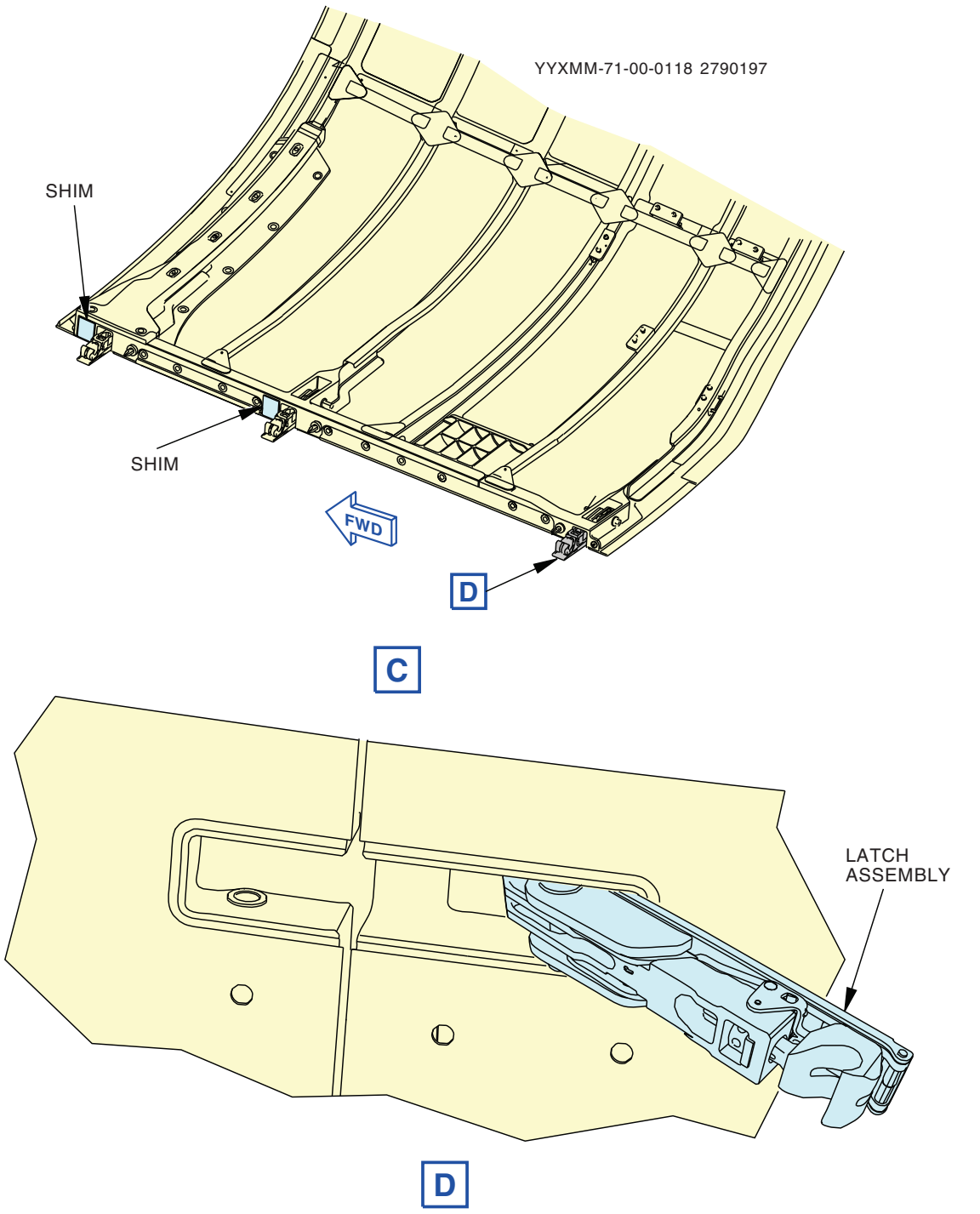
(POSITIONS 1, 2, AND 3)
(VIEW IN THE FORWARD DIRECTION)

B

2354393 S00061515242_V2

Fan Cowl Latches Adjustment
Figure 501/71-11-05-990-803-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl Latches Adjustment
Figure 501/71-11-05-990-803-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL LATCH INDICATION SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Fan Cowl Latch Indication System Removal

(2) Fan Cowl Latch Indication System Installation.

TASK 71-11-06-000-801-G00

2. Fan Cowl Latch Indication System Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove these fan cowl latch indication system components:

(a) Forward fairing

(b) Blade

(c) Aft fairing

(d) Handle

(e) Cable.

(2) The fan cowl latch indication system is installed on the right fan cowl panel.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-765	Scraper - Plastic

D. Location Zones

Zone	Area
411	Engine 1 - Engine
414	Engine 1 - Fan Cowl, Right
421	Engine 2 - Engine
424	Engine 2 - Fan Cowl, Right

E. Prepare for the Removal

SUBTASK 71-11-06-860-001-G00

CAUTION

RETRACT THE LEADING EDGE SLATS, AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THESE PRECAUTIONS, THE FAN COWL PANELS WILL HIT THE SLATS. DAMAGE WILL OCCUR.

(1) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-06-010-001-G00

(2) Open the applicable right fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Fan Cowl Latch Indication System Removal

SUBTASK 71-11-06-020-001-G00

(1) Remove the forward fairing [1] as follow:

(a) Remove the two bolts [2] that attach the forward fairing [1] to the fan cowl.

(b) Use a plastic scraper, STD-765, to carefully separate the fillet seal and remove the forward fairing [1].

(c) Clean the fan cowl surface to remove the remaining fillet seal.

SUBTASK 71-11-06-020-002-G00

(2) Remove the blade [3] as follow:

(a) Remove the bolt [4] that attaches the blade [3] to the fan cowl.

(b) Remove the blade [3].

SUBTASK 71-11-06-020-003-G00

(3) Remove the outer aft fairing [5] as follow:

NOTE: The aft fairing has two parts, an outer aft fairing and an inner aft fairing. The inner aft fairing is bonded to the fan cowl surface with a fay surface seal. It is recommended that the inner aft fairing not be removed.

(a) Remove the two bolts [6] that attach the aft fairing [5] to the fan cowl.

(b) Remove the aft fairing [5].

SUBTASK 71-11-06-020-004-G00

(4) Remove the handle [7] as follow:

(a) Remove the two bolts [8], two nuts [9], two washers [10], and two washers [11] that attach the handle [7] to the handle shaft.

(b) Remove the handle [7].

SUBTASK 71-11-06-020-005-G00

(5) Remove the cable [12] as follow:

(a) Remove the bolt [29], nut [30], washer [31], washer [32], and bushing [33] from the lower crank assembly.

(b) Remove the self-locking nut [13] that attaches the cable [12] to the lower cable rod end.

(c) Remove the loop clamp [14], bolt [15], nut [16], washer [17], and washer [18] at three locations.

(d) Remove the spherical cap [19] and spherical seat [20].

(e) Pull the cable [12] up through the bracket [24].

(f) Remove the washer [21], spring [22], and washer [23].

(g) Remove the lockwire at the jam nut [25].

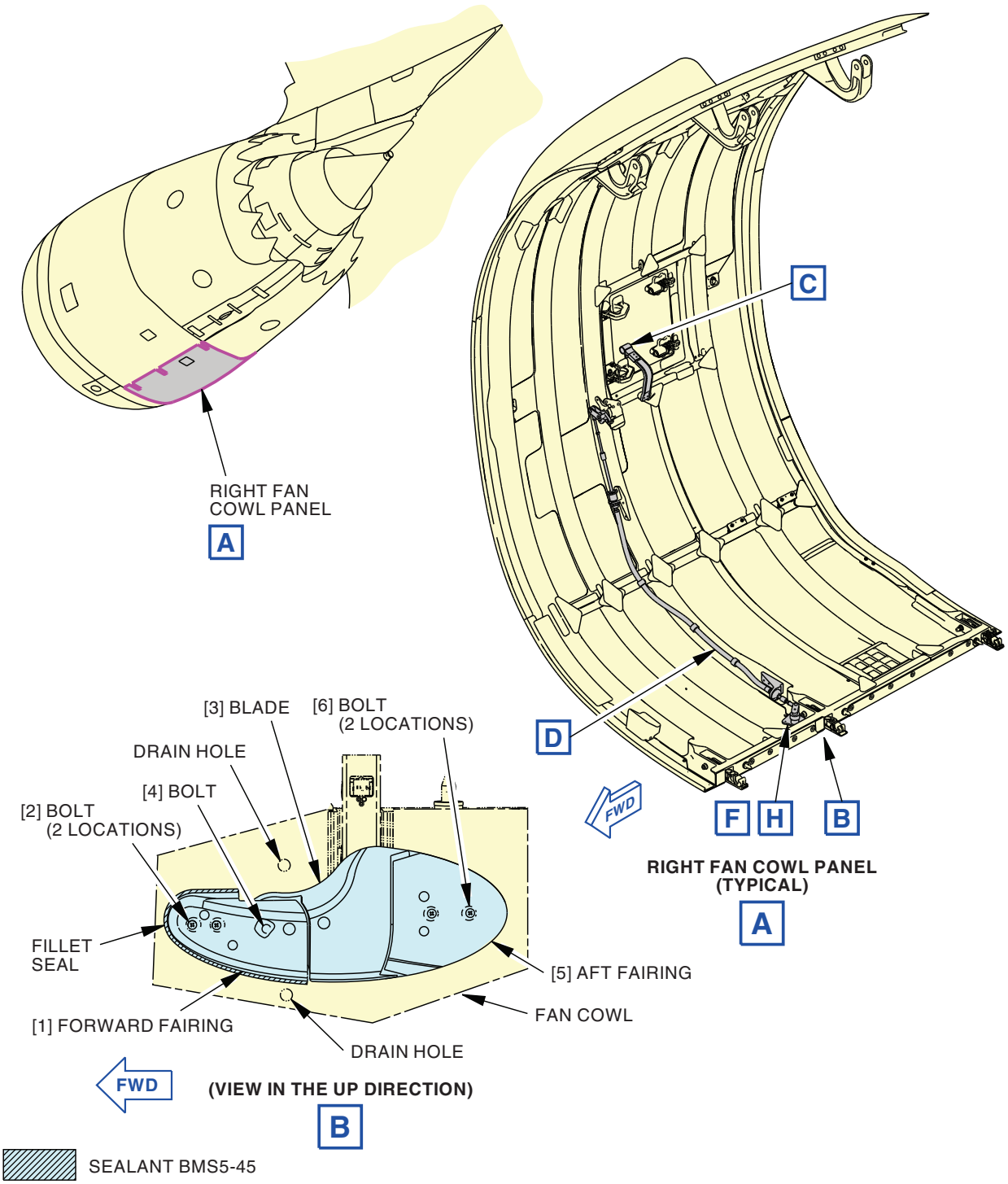
(h) Loosen the jam nut [25].

(i) Remove the bolt [29], nut [30], washer [31], washer [31], and bushing [33] from the upper crank assembly.

(j) Remove the cable [12].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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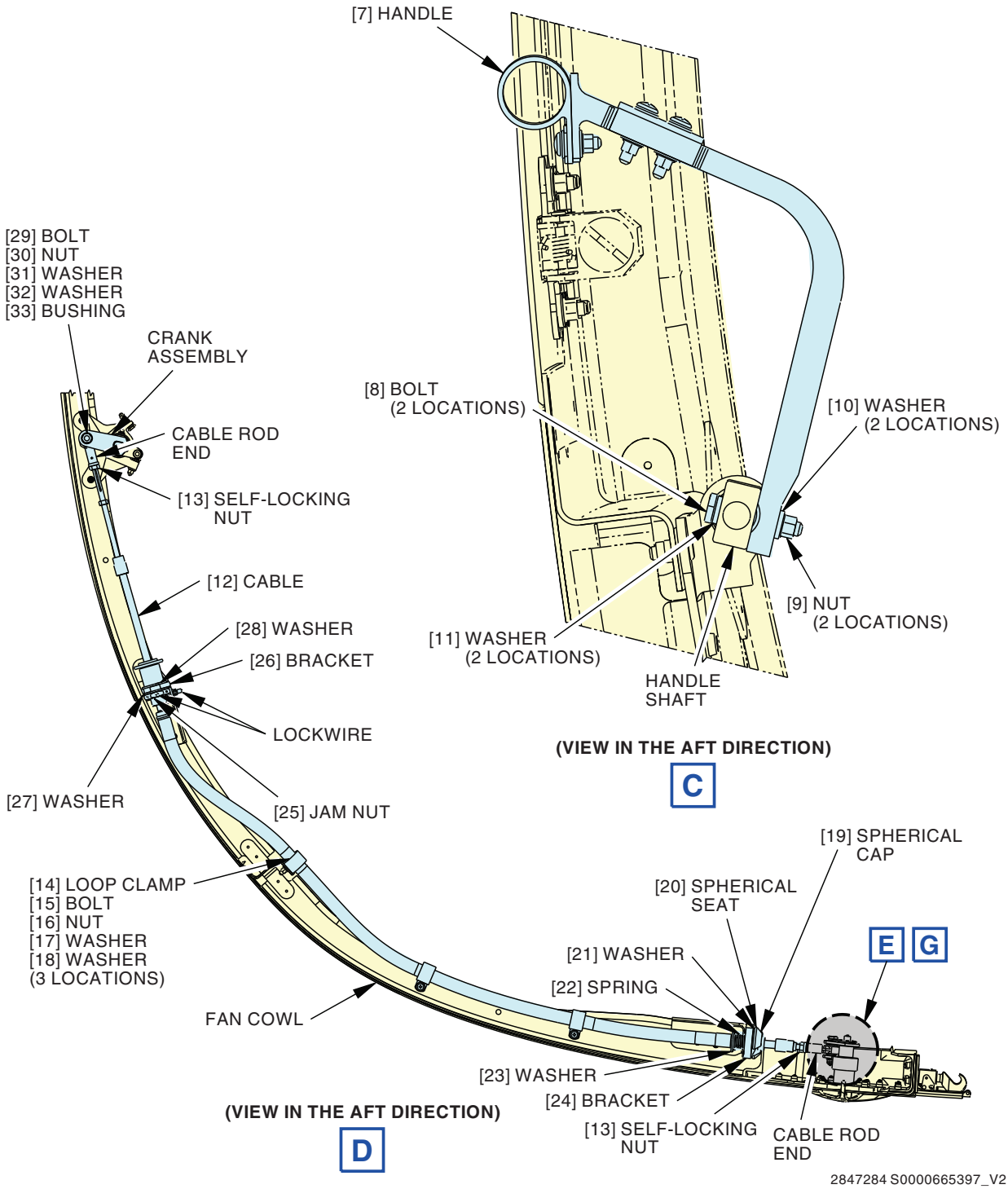
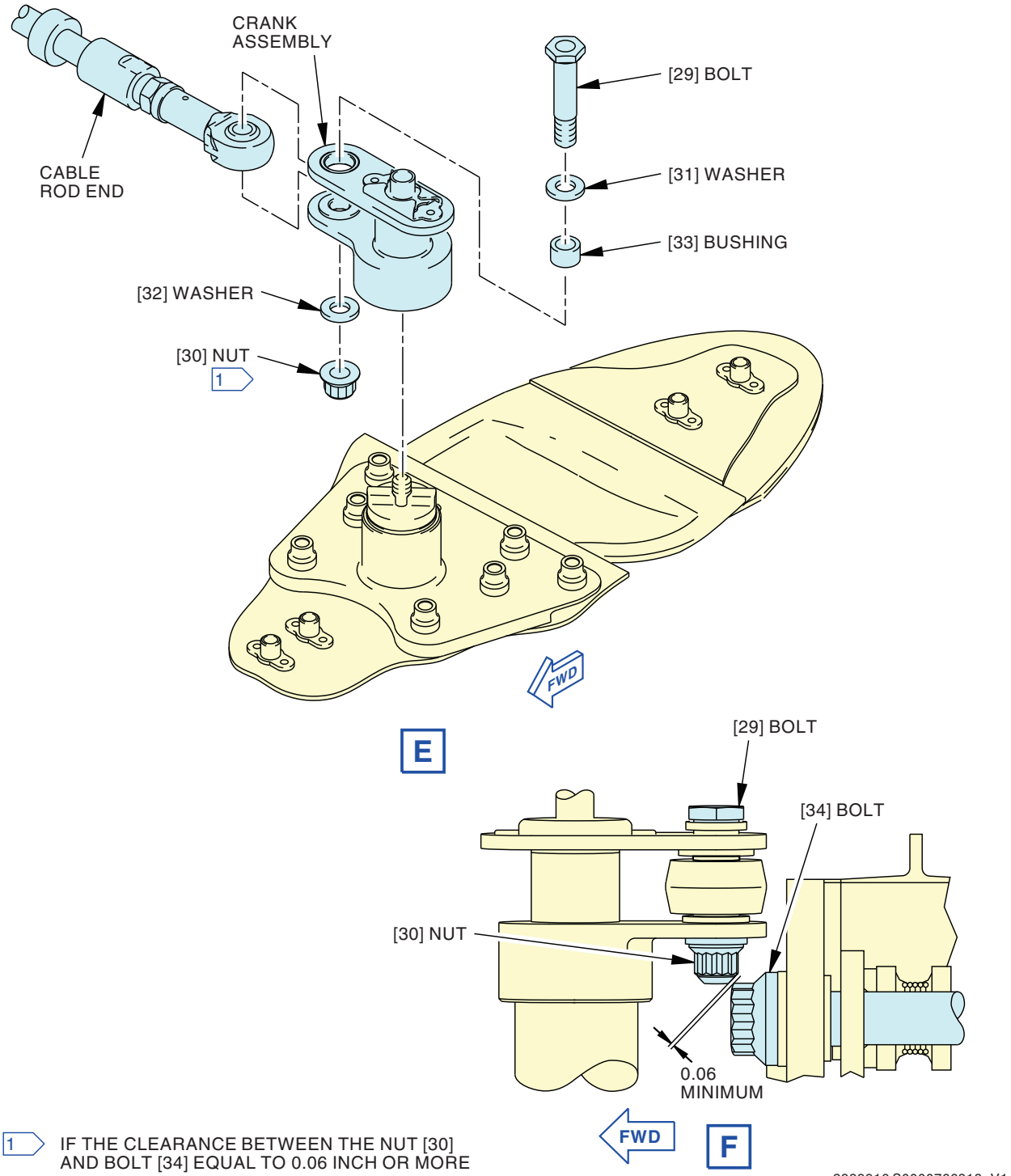
Fan Cowl Latch Indication System Installation
Figure 201/71-11-06-990-801-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



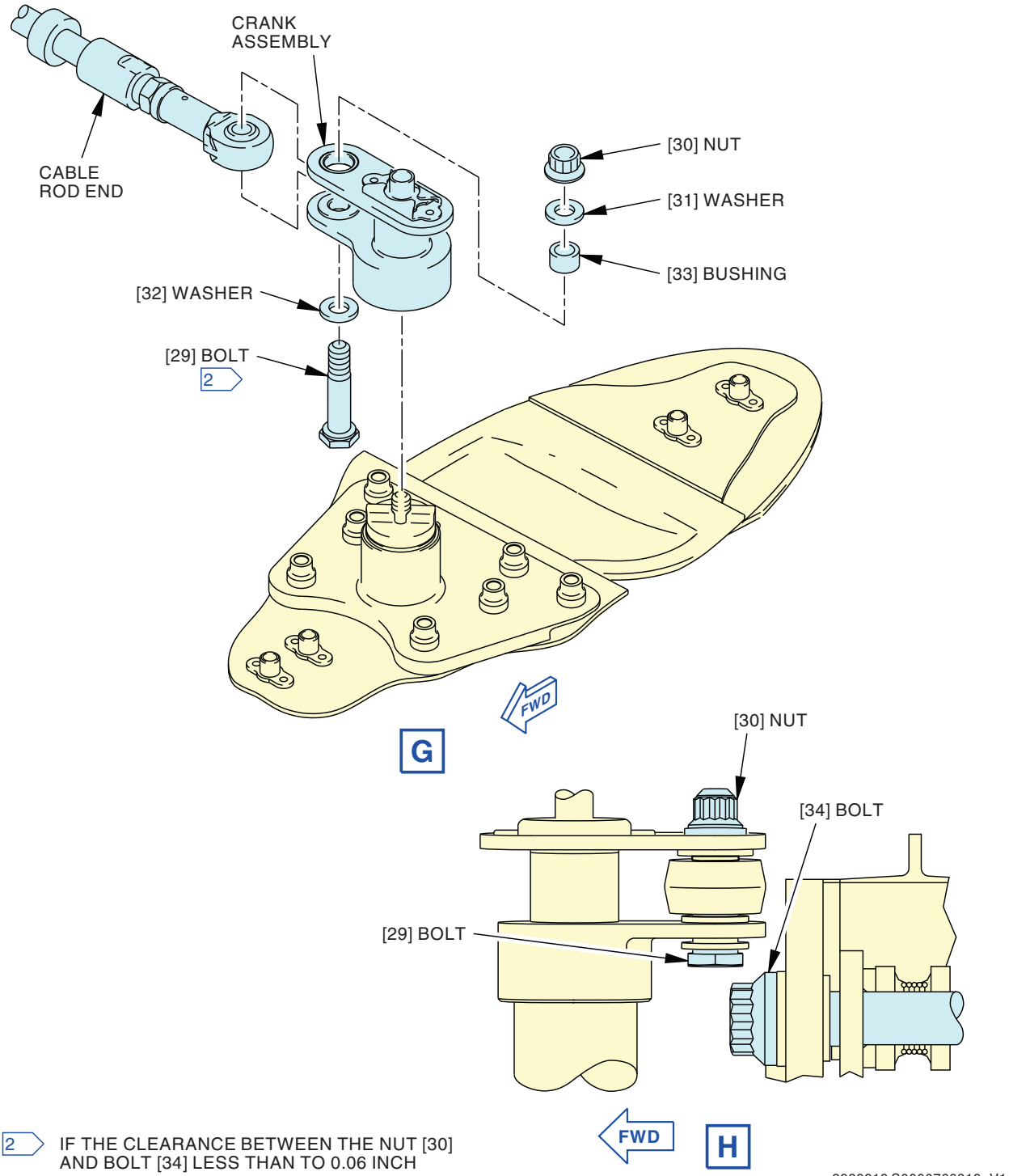
Fan Cowl Latch Indication System Installation
Figure 201/71-11-06-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl Latch Indication System Installation
Figure 201/71-11-06-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl Latch Indication System Installation
Figure 201/71-11-06-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-06-400-801-G00

3. Fan Cowl Latch Indication System Installation

(Figure 201)

A. General

(1) This task gives the instructions to install these fan cowl latch indication system components:

(a) Forward fairing

(b) Blade

(c) Aft fairing

(d) Handle

(e) Cable.

(2) Do an adjustment of the fan cowl latch indication system after the fan cowl latch indication system is installed.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
71-11-06-720-801-G00	Fan Cowl Latch Indication System Rigging Check (P/B 501)

C. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
B00130	Alcohol - Isopropyl	TT-I-735
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
411	Engine 1 - Engine
414	Engine 1 - Fan Cowl, Right
421	Engine 2 - Engine
424	Engine 2 - Fan Cowl, Right

E. Fan Cowl Latch Indication System Installation

SUBTASK 71-11-06-420-001-G00

(1) Install the cable [12] as follow:

(a) Install the bolt [29], nut [30], washer [31], washer [32], and bushing [33] that attach the cable [12] to the upper crank assembly.

NOTE: It is not necessary to do a wet installation if the hole is already primed or if the bolt go through a bushing.

(b) Put the cable [12] into its position on the bracket [26].

(c) Make sure that the washer [27] and washer [28] are in the correct positions.

(d) Tighten the jam nut [25] to 160.0 in-lb (18.1 N·m).

(e) Install MS20995C32 lockwire, G01048, on the jam nut [25].

(f) Put the washer [21], spring [22], and washer [23] in their correct positions on the cable [12].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Put the cable [12] down through the bracket [24].

(h) Install the spherical seat [20] and spherical cap [19].

(i) Install the self-locking nut [13] that attaches the cable [12] to the lower cable rod end.

(j) Install the bolt [29], nut [30], washer [31], washer [31], and bushing [33] that attach the cable [12] to the lower crank assembly.

NOTE: It is not necessary to do a wet installation if the hole is already primed or if the bolt go through a bushing.

(k) Make sure that the clearance between the nut [30] and bolt [34] is equal or more than 0.06 in. (1.52 mm) (View F, Figure 201).

1) If the clearance less than 0.06 in. (1.52 mm), do this step:

a) Change the direction of the bolt [29] and nut [30] (View H, Figure 201).

(l) Install the loop clamp [14], bolt [15], nut [16], washer [17], and washer [18] at three locations.

1) Tighten the fastener head to 35 in-lb (4 N·m).

SUBTASK 71-11-06-420-002-G00

(2) Install the handle [7] as follow:

(a) Install the two bolts [8], two nuts [9], two washers [10], and two washers [11] that attach the handle [7] to the handle shaft.

(b) Tighten the two bolts [8] to 35.0 in-lb (4.0 N·m).

SUBTASK 71-11-06-110-001-G00

(3) Clean the anti-static coatings on the forward fairing [1] and aft fairing [5] with alcohol, B00130.

SUBTASK 71-11-06-420-003-G00

(4) Install the aft fairing [5] as follow:

(a) Use sealant, A00767, to wet install the two bolts [6] that attach the aft fairing [5] to the fan cowl.

NOTE: It is not necessary to do a wet installation if the hole is already primed or if the bolt go through a bushing.

(b) Tighten the two bolts [6] to 18.0 in-lb (2.0 N·m) - 20.0 in-lb (2.3 N·m).

SUBTASK 71-11-06-420-004-G00

(5) Install the blade [3] as follow:

(a) Install the bolt [4] that attaches the blade [3] to the fan cowl.

(b) Tighten the bolt [4] to 18.0 in-lb (2.0 N·m) - 20.0 in-lb (2.3 N·m).

SUBTASK 71-11-06-420-005-G00

(6) Install the forward fairing [1] as follow:

(a) Use sealant, A00767, to wet install the two bolts [2] that attach the forward fairing [1] to the fan cowl.

NOTE: It is not necessary to do a wet installation if the hole is already primed or if the bolt go through a bushing.

(b) Tighten the two bolts [2] to 18.0 in-lb (2.0 N·m) - 20.0 in-lb (2.3 N·m).

(c) Use sealant, A00767, to apply a fillet seal to the external periphery of the forward fairing [1].

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SUBTASK 71-11-06-710-001-G00

(7) Do this task: Fan Cowl Latch Indication System Rigging Check, TASK 71-11-06-720-801-G00.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-06-410-001-G00

(1) Close the applicable right fan cowl panel (TASK 71-11-04-410-801-G00).

SUBTASK 71-11-06-860-002-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL LATCH INDICATION SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Fan Cowl Latch Indication System Rigging Check.

TASK 71-11-06-720-801-G00

2. Fan Cowl Latch Indication System Rigging Check

(Figure 501)

A. General

(1) This task gives the instructions to check the rigging of the fan cowl latch indication system.

(2) The fan cowl latch will need to be held flush with the fan cowl skin to allow cycling of the latch indication blade.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45

D. Location Zones

Zone	Area
411	Engine 1 - Engine
414	Engine 1 - Fan Cowl, Right
421	Engine 2 - Engine
424	Engine 2 - Fan Cowl, Right

E. Fan Cowl Latch Indication System Rigging Check

SUBTASK 71-11-06-040-001-G00

CAUTION

RETRACT THE LEADING EDGE FLAPS, AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THESE PRECAUTIONS, THE FAN COWL PANELS WILL HIT THE FLAPS. DAMAGE WILL OCCUR.

(1) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-06-010-003-G00

(2) Open the right fan cowl panel (TASK 71-11-04-010-801-G00).

SUBTASK 71-11-06-560-001-G00

(3) Hold the center fan cowl latch handle flush with the fan cowl skin.

SUBTASK 71-11-06-860-003-G00

(4) Push the latch indication system handle through the oil tank access door.

SUBTASK 71-11-06-410-002-G00

(5) Make sure that you close and secure the oil tank access door.

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SUBTASK 71-11-06-220-001-G00

(6) Make sure that the bubble seal on the latch indication handle is compressed against the oil tank access door approximately 1/4 to 1/3 of the total compressible distance.

(a) If it is necessary, do these steps to add or remove the washer [4] at the lower stop:

1) Remove the bolt [1], nut [2], washer [3], washer [4], and washer [5].

2) Install the bolt [1], nut [2], washer [3], washer [4], and washer [5].

NOTE: It is not necessary to do a wet installation with the sealant, A00767 if the hole is already primed or if the bolt go through a bushing.

SUBTASK 71-11-06-210-001-G00

(7) With the crank arm against the lower stop, make sure that there is a consistent gap between the blade and the aft fairing.

(a) If it is necessary, loosen and tighten the self-locking nut on the upper cable rod end to adjust the cable.

SUBTASK 71-11-06-010-002-G00

(8) Open the oil tank access door.

SUBTASK 71-11-06-980-001-G00

(9) Pull the fan cowl latch indication system handle out through the oil tank access door opening.

NOTE: Pull the handle out will rotate the blade to uncover the center fan cowl latch.

SUBTASK 71-11-06-210-002-G00

(10) Make sure that the blade clears the fan cowl latch housing and does not contact the forward fairing.

SUBTASK 71-11-06-210-003-G00

(11) While the blade is in the deployed position, make sure that the crank arm touches the upper stop.

(a) If it is necessary, do these steps to add or remove the washer [9] at the upper stop:

1) Remove the bolt [6], nut [7], washer [8], washer [9], and washer [10].

2) Install the bolt [6], nut [7], washer [8], washer [9], and washer [10].

NOTE: It is not necessary to do a wet installation with the sealant, A00767, if the hole is already primed or if the bolt goes through the bushing.

SUBTASK 71-11-06-980-002-G00

(12) Hold the center latch handle flush with the fan cowl skin and do these steps:

(a) Move the handle from the deployed position to the stowed position and back to the deployed position again.

(b) Make sure that the fan cowl latch indication system operates correctly.

SUBTASK 71-11-06-980-003-G00

(13) Pull the fan cowl latch indication system handle out through the oil tank access door opening.

SUBTASK 71-11-06-410-003-G00

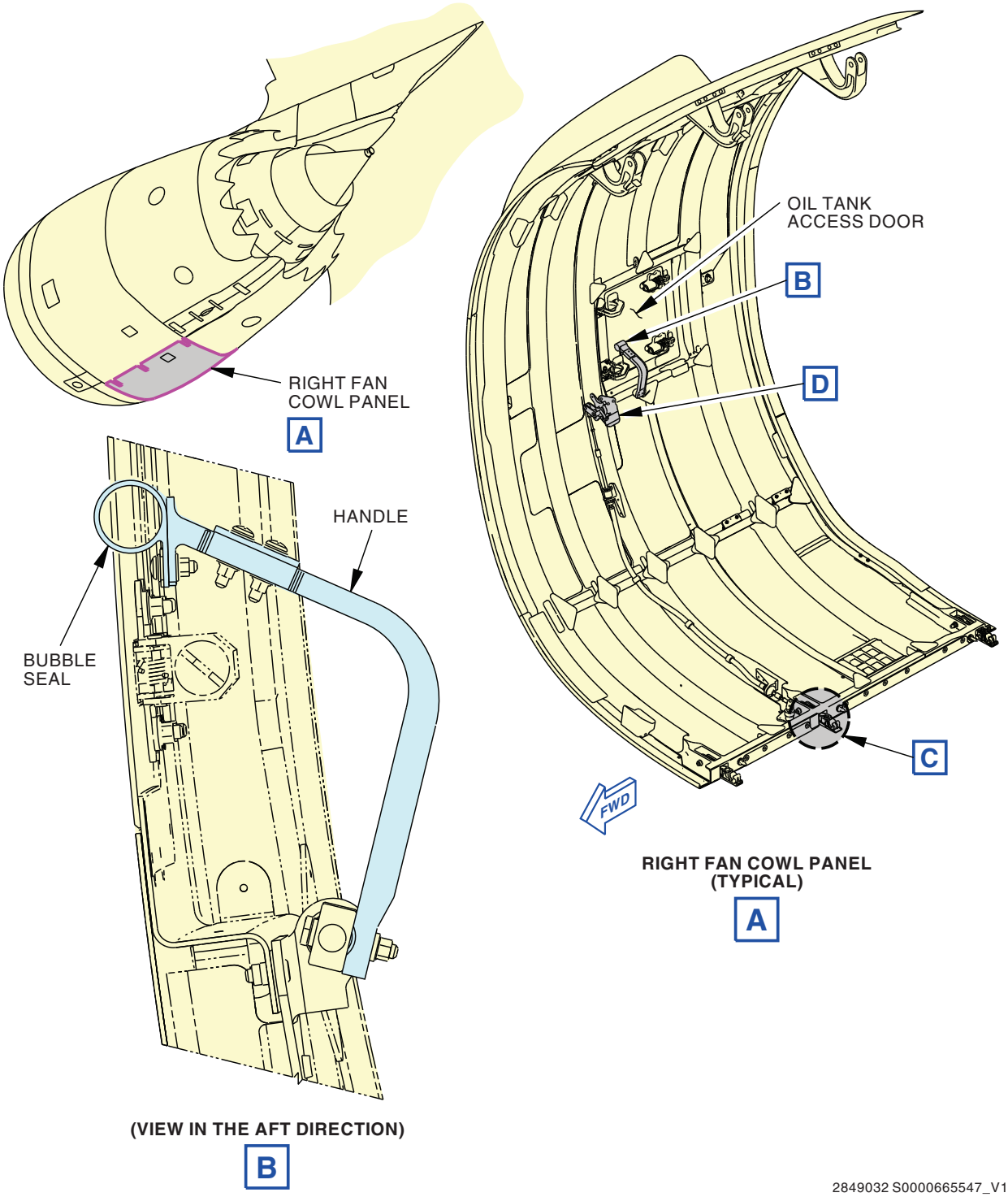
(14) Close the right fan cowl panel (TASK 71-11-04-410-801-G00).

SUBTASK 71-11-06-440-001-G00

(15) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

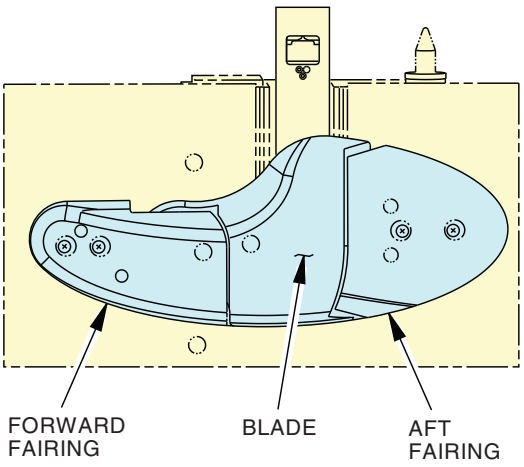
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl Latch Indication System Rigging Check
Figure 501/71-11-06-990-802-G00 (Sheet 1 of 2)

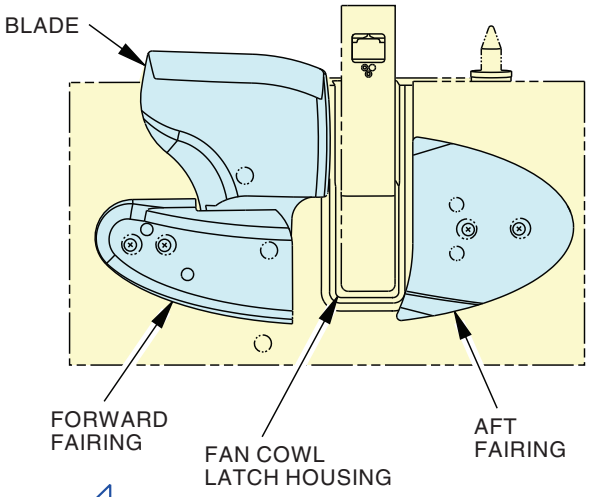
ECCN 9E991 BOEING PROPRIETARY - See title page for details



FWD

STOWED POSITION
(VIEW IN THE UP DIRECTION)

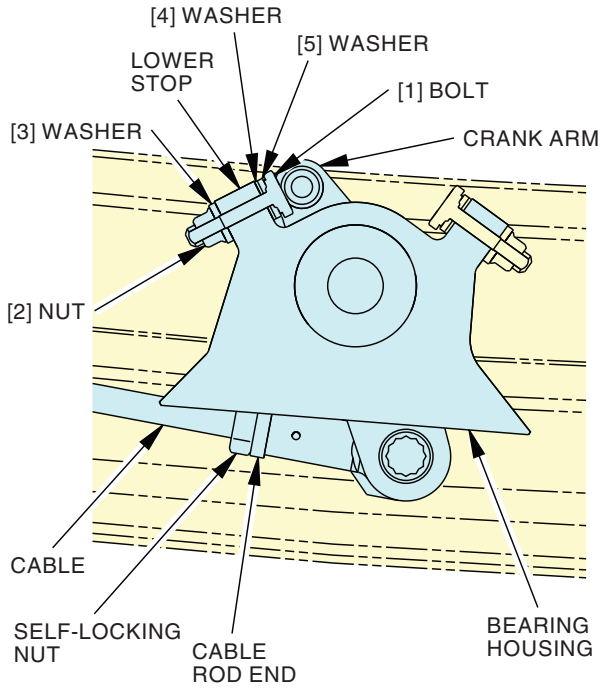
C



FWD

DEPLOYED POSITION
(VIEW IN THE UP DIRECTION)

C



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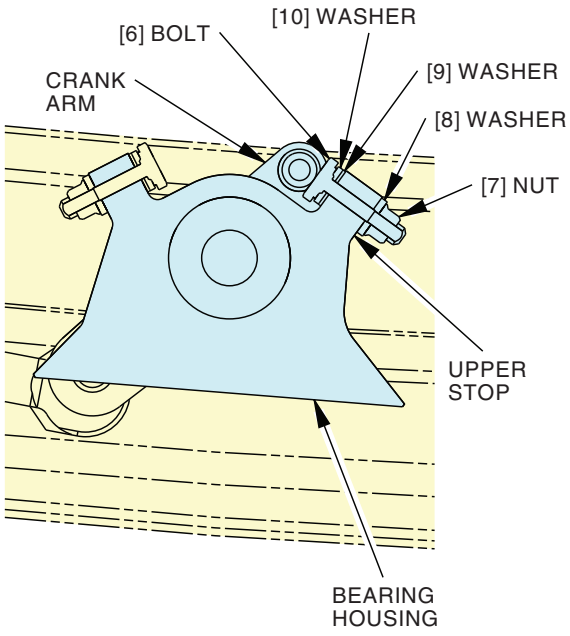

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D

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Fan Cowl Latch Indication System Rigging Check
Figure 501/71-11-06-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

IDG ACCESS DOOR - MAINTENANCE PRACTICES

1. General

A. This procedure has two tasks:

(1) IDG access door removal

(2) IDG access door installation.

TASK 71-11-08-000-801-G00

2. IDG Access Door Removal

(Figure 201)

A. General

(1) This task gives the instructions on how to remove the IDG access door.

(2) The IDG access door is referred to as the access door.

(3) The access door is installed on the left fan cowl panel at the 4:00 o'clock position.

B. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Access Panels

Number	Name/Location
413AL	IDG Access Door, Engine 1
423AL	IDG Access Door, Engine 2

E. Prepare for the Removal

SUBTASK 71-11-08-860-001-G00

(1) Move the engine and strut access platform, STD-653, next to the applicable left fan cowl panel.

F. IDG Access Door Removal

SUBTASK 71-11-08-010-001-G00

(1) Open the access door [1] as follows:

(a) Open the applicable access panels:

Number	Name/Location
413AL	IDG Access Door, Engine 1
423AL	IDG Access Door, Engine 2

(b) Push on the forward part of each latch button to release the latch.

(c) Put the access door [1] in the fully open position.

SUBTASK 71-11-08-869-001-G00

(2) Remove the access door [1] as follows:

(a) Hold the access door [1] open.

(b) Remove the nut [6], two washers [5], washer [3], and bolt [2] that attach the lanyard [4] to the upper hinge fitting.

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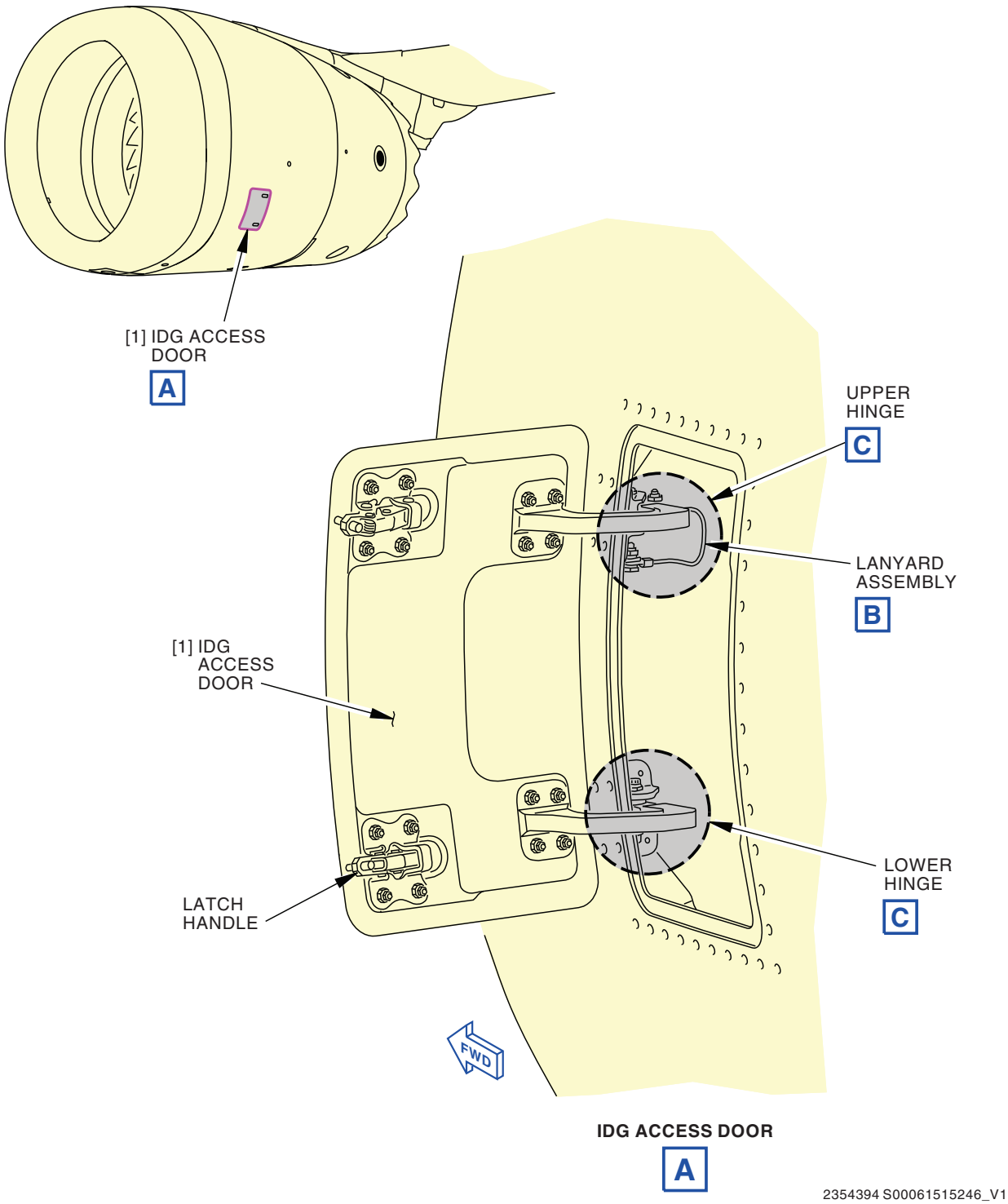
(c) Remove the two nuts [7], two washers [8], two washers [9], and two bolts [10] that attach the upper and lower hinge fittings to the access door [1].

(d) Remove the access door [1] from the fan cowl panel.

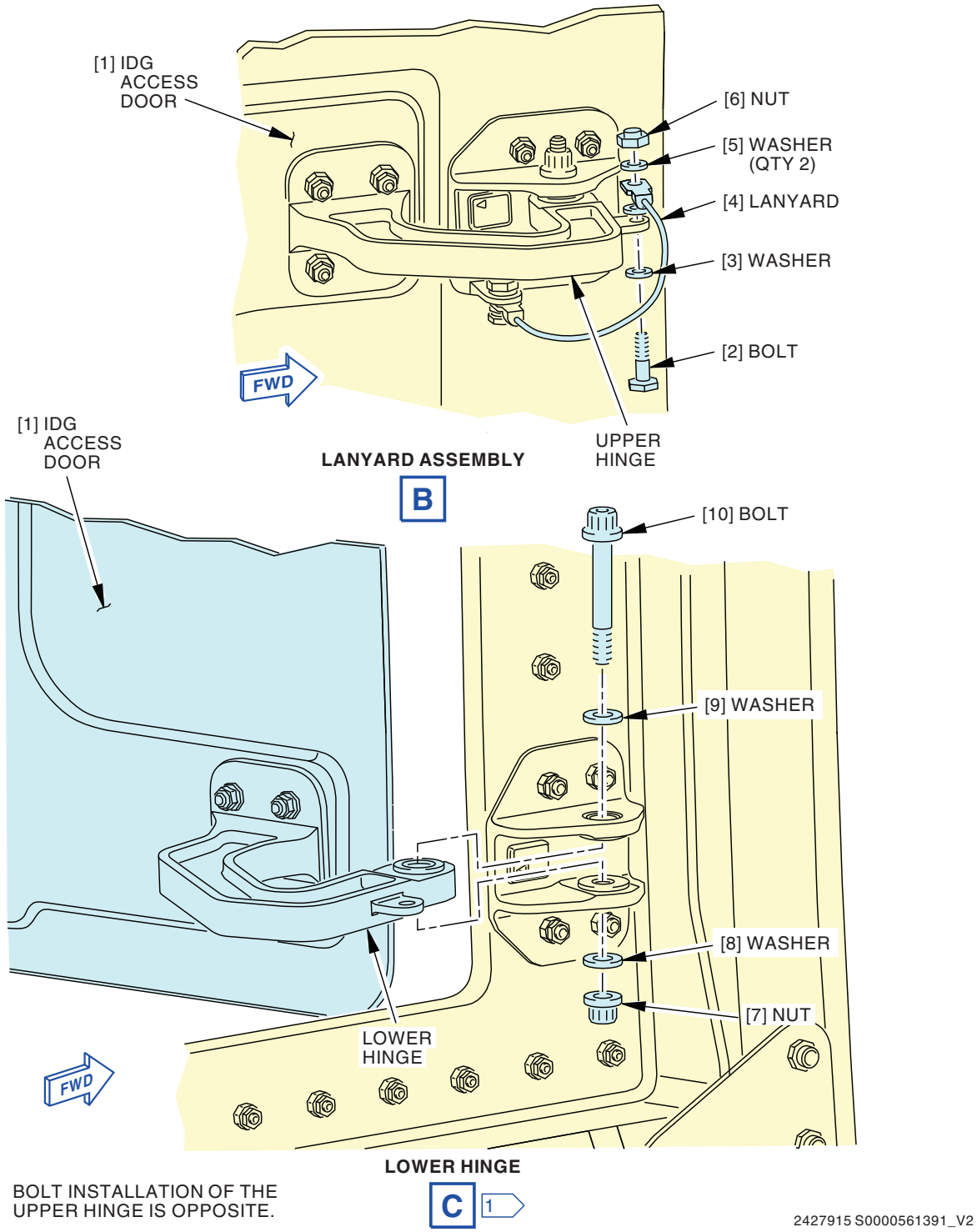
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



IDG Access Door Installation
Figure 201/71-11-08-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



IDG Access Door Installation
Figure 201/71-11-08-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-08-400-801-G00

3. IDG Access Door Installation

(Figure 201)

A. General

(1) This task gives the instructions on how to install the IDG access door.

(2) The IDG access door is referred to as the access door.

(3) The access door is installed on the left fan cowl panel at the 4:00 o'clock position.

B. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access

C. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Access door	71-11-08-01-120	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413AL	IDG Access Door, Engine 1
423AL	IDG Access Door, Engine 2

G. IDG Access Door Installation

SUBTASK 71-11-08-420-001-G00

(1) Install the access door [1] as follows:

(a) Put the access door [1] in its opening position on the fan cowl panel.

(b) Align the hinge fitting bolt holes with the access door bolt holes.

(c) Install the two bolts [10], two washers [9], two washers [8], and two nuts [7] that attach the upper and lower hinge fittings to the access door [1].

NOTE: Do not tighten the bolts at this time.

(d) Install the bolt [2], washer [3], two washers [5], and nut [6] that attach the lanyard [4] to the upper hinge fitting.

1) Tighten the nut to 33 in-lb (3.7 N·m) - 40 in-lb (4.5 N·m).

2) Fillet seal the ground stud and lanyard lug with sealant, A00767.

(e) Close and open the access door [1] to do a check on the clearance of the access door [1].

1) Measure the clearance between the access door [1] and fan cowl panel at the forward edge.

a) Make sure that the clearance is 0.18 ±0.06 in. (4.57 ±1.52 mm).

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2) Measure the clearance between the access door [1] and fan cowl panel at the top, bottom and aft edges.
   a) Make sure that the clearance is 0.07 ±0.06 in. (1.78 ±1.52 mm).
3) Measure the misfair between the access door [1] and fan cowl panel.
   a) Make sure that the misfair is 0.00 ±0.04 in. (0.00 ±1.02 mm).
4) If it is necessary, adjust the hinge fittings.
5) Do a check on the clearance of the access door [1] again to make sure that the access door [1] operates smoothly.

(f) Tighten the nuts [7] to 90 in-lb (10.2 N·m) - 105 in-lb (11.9 N·m).

SUBTASK 71-11-08-410-001-G00

(2) Close the access door [1] as follows:

(a) Make sure that the latches are in the fully open position.

(b) Close these access panels:

Number	Name/Location
413AL	IDG Access Door, Engine 1
423AL	IDG Access Door, Engine 2

1) Press the aft part of the latch level while you push the access door [1] to close.

(c) Make sure that the access door [1] is fully closed.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-08-090-001-G00

(1) Move the engine and strut access platform, STD-653, away from the engine.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE OIL TANK ACCESS DOOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Engine oil tank access door removal

(2) Engine oil tank access door installation.

TASK 71-11-10-000-801-G00

2. Engine Oil Tank Access Door Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the engine oil tank access door.

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Access Panels

Number	Name/Location
414AR	Oil Tank Access / Pressure Relief Door, Engine 1
424AR	Oil Tank Access / Pressure Relief Door, Engine 2

E. Prepare for the Removal

SUBTASK 71-11-10-010-002-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-10-010-001-G00

(2) Open the access door [1] as follows:

(a) Open the applicable access panels:

Number	Name/Location
414AR	Oil Tank Access / Pressure Relief Door, Engine 1
424AR	Oil Tank Access / Pressure Relief Door, Engine 2

(b) Push on the forward button of each latch to release the latch.

(c) Put the access door [1] in the fully open position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Engine Oil Tank Access Door Removal

SUBTASK 71-11-10-869-001-G00

(1) Remove the access door [1] as follows:

(a) Hold the access door [1] open.

(b) Remove the nut [6], the washers [5], the washer [3], and the bolt [2] that attach the lanyard [4] to the lower hinge fitting.

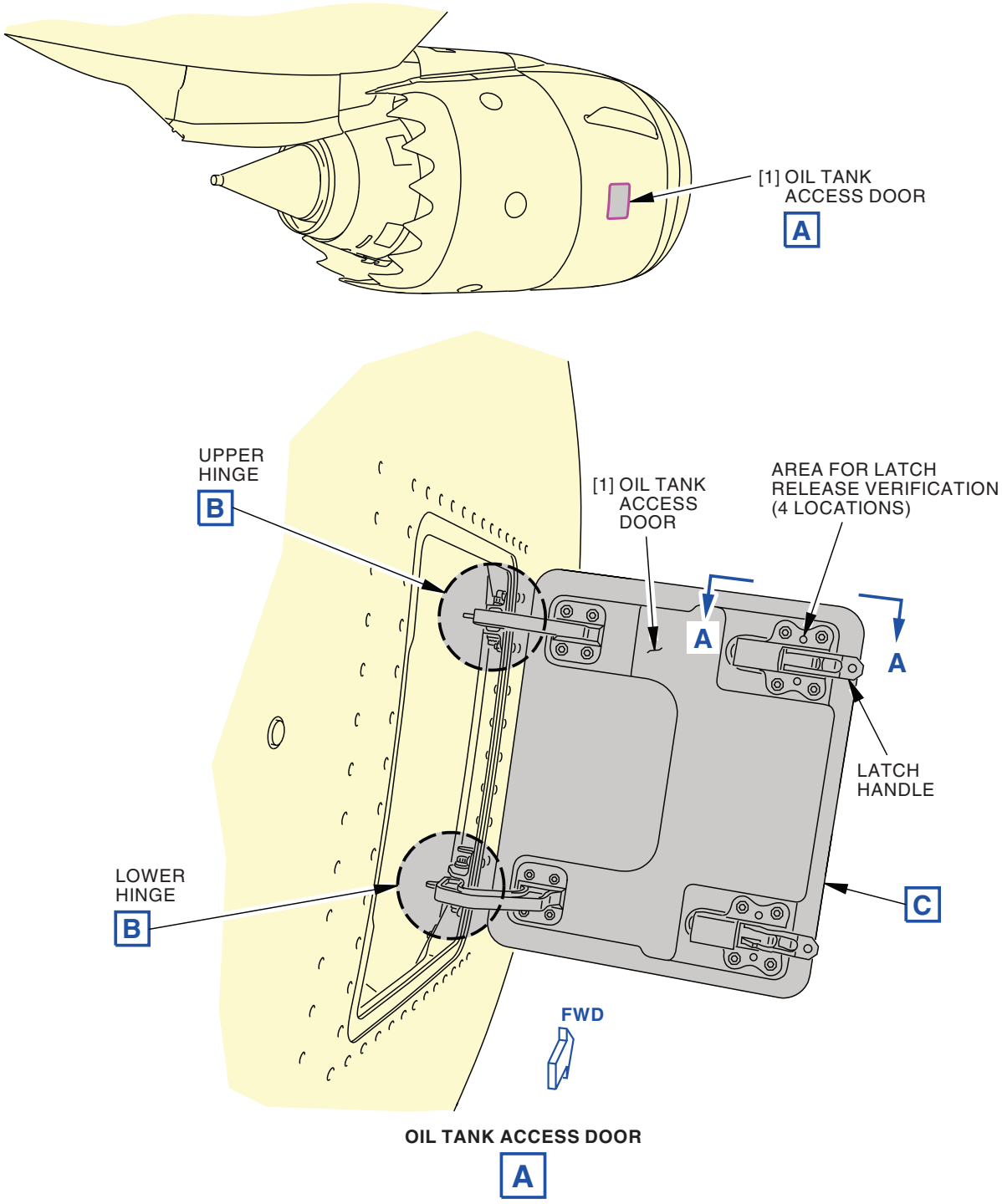
(c) Remove the nut [7], the washer [8], the washer [9], and the bolt [10] that attach the lower hinge fitting to the hinge clevis on the fan cowl.

(d) Remove the nut [7], the washer [8], the washer [9], and the bolt [10] that attach the upper hinge fitting to the hinge clevis on the fan cowl.

(e) Remove the access door [1] from the fan cowl panel.

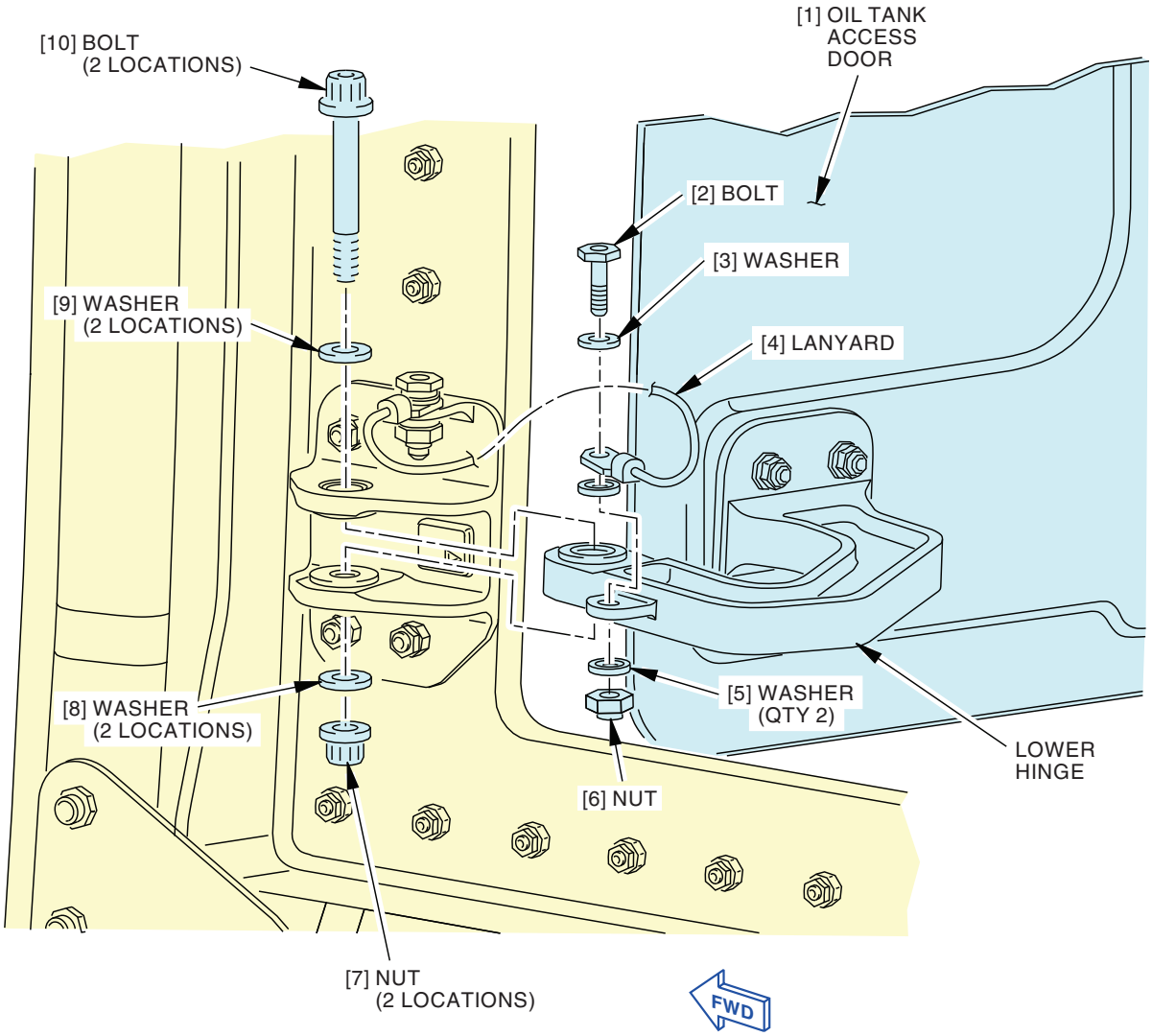
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



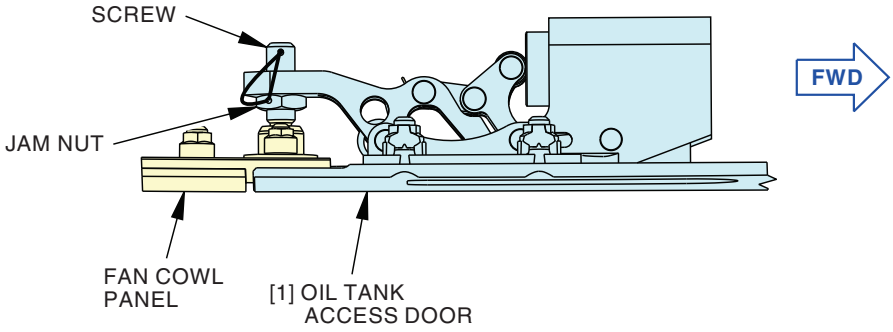
Engine Oil Tank Access Door Installation
Figure 401/71-11-10-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

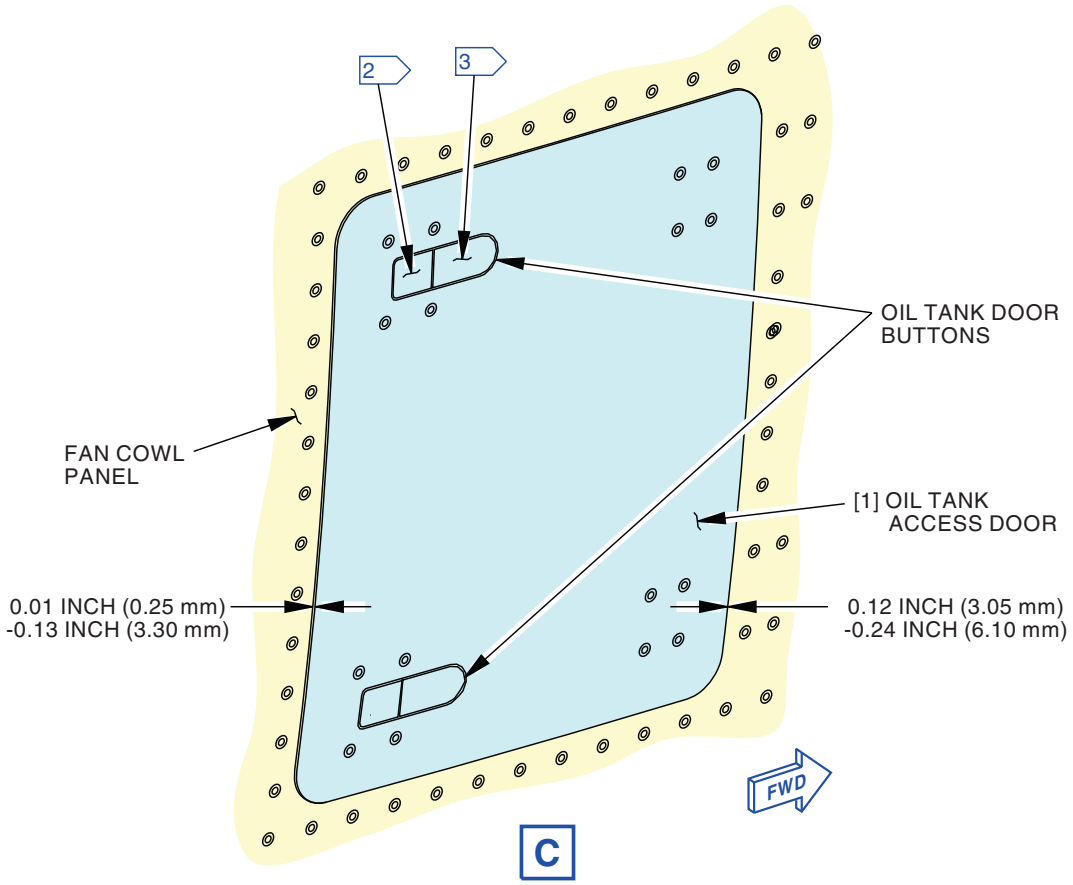


Engine Oil Tank Access Door Installation
Figure 401/71-11-10-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



2 PUSH CLOSE BUTTON
3 PUSH OPEN BUTTON

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Engine Oil Tank Access Door Installation
Figure 401/71-11-10-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-10-400-801-G00

3. Engine Oil Tank Access Door Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the engine oil tank access door.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-6371	Load Cell - 0 to 200 pound (91 kg) capability

D. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
G50667	Lockwire - MS20995N20, Inconel - 0.020 Inch (0.508 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Access door	71-11-10-01-115	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
414AR	Oil Tank Access / Pressure Relief Door, Engine 1
424	Right Fan Cowl, Engine 2
424AR	Oil Tank Access / Pressure Relief Door, Engine 2

H. Engine Oil Tank Access Door Installation

SUBTASK 71-11-10-420-001-G00

(1) Install the access door [1] as follows:

(a) Put the access door [1] in the open position on the fan cowl panel.

(b) Align the access door [1] hinge fitting bolt holes with the hinge clevis holes on the fan cowl.

(c) Install the bolt [10], the washer [9], the washer [8], and the nut [7] that attach the access door [1] upper hinge fitting to the hinge clevis on the fan cowl.

NOTE: Do not tighten the bolts at this time.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Install the bolt [10], the washer [9], the washer [8], and the nut [7] that attach the access door [1] lower hinge fitting to the hinge clevis on the fan cowl.

NOTE: Do not tighten the bolts at this time.

(e) Install the bolt [2], the washer [3], the washers [5], and the nut [6] that attach the lanyard [4] to the lower hinge fitting.

1) Tighten the nut to 33 in-lb (3.7 N·m) - 40 in-lb (4.5 N·m).

2) Fillet seal the ground stud and lanyard lug with sealant, A00767.

(f) Close and open the access door [1] to do a check on the clearance of the access door [1].

1) Make sure that the clearance between the access door [1] and the fan cowl panel at the aft edge is 0.01 in. (0.25 mm) - 0.13 in. (3.30 mm).

2) Make sure that the clearance between the access door [1] and the fan cowl panel at the forward edge is 0.12 in. (3.05 mm) - 0.24 in. (6.10 mm).

3) If it is necessary, adjust the hinge fittings.

4) Do a check on the clearance of the access door [1] again to make sure that the access door [1] operates smoothly.

(g) Tighten the nuts [7] to 82.0 in-lb (9.3 N·m) - 88.0 in-lb (9.9 N·m).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-10-410-001-G00

(1) Close the access door [1] as follows:

(a) Make sure that the latches are in the fully open position.

(b) Close these access panels:

Number	Name/Location
414AR	Oil Tank Access / Pressure Relief Door, Engine 1
424AR	Oil Tank Access / Pressure Relief Door, Engine 2

1) Push the access door [1] to close.

NOTE: If the latch jams in the fully open position and cannot close, push the forward “PUSH OPEN” button to reset the latch.

2) Use a load cell, STD-6371, to push each aft “PUSH CLOSE” button until it is flushed with the access door [1].

a) Measure the closing force.

b) Make sure that the closing force is 24.2 lb (11.0 kg) - 35.0 lb (15.9 kg).

<1> If it is necessary to adjust the closing force, do these steps:

<a> Adjust the latch screw.

<b> Install the MS20995N20 lockwire, G50667, between the latch screw and jam nut.

SUBTASK 71-11-10-010-003-G00

(2) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

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SUBTASK 71-11-10-750-001-G00

(3) Do a test of the release feature of each latch as follows:

(a) Release to open one latch of the oil tank door assembly.

(b) Use the load cell, STD-6371, to apply 85 lb (38.6 kg) - 103.5 lb (46.9 kg) evenly distributed on the two indicated areas of the latch that is closed.

NOTE: The indicated areas are marked with circular dots on the inside of the door on the latch.

(c) Make sure that the latch opens at the target applied force.

(d) Close the latch.

(e) Do the test of the release feature on the second latch.

SUBTASK 71-11-10-410-003-G00

(4) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-10-410-002-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HOLD-OPEN RODS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Fan cowl hold open rod removal

(2) Fan cowl hold open rod installation

(3) Fan cowl hold open rod inspection

(4) Fan cowl hold open rod cleaning.

TASK 71-11-11-000-801-G00

2. Fan Cowl Hold Open Rod Removal

(Figure 201)

A. General

(1) This task provides the instructions on how to remove the fan cowl hold open rod.

(2) Each fan cowl panel has a telescoped hold open rod. The fan cowl hold open rod is stowed in the vertical position on the aft end edge of the fan cowl panel when the fan cowl panel is closed.

(3) The removal task of the left side or the right side of the fan cowl hold open rod is the same.

B. References

Reference	Title
71-11-04-010-804-G00	Open the Fan Cowl Panels (Sling Method) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14546	Sling Eqpt - Fan Cowl, LEAP-1B Engine Part #: C71031-21 Supplier: 81205 Opt Part #: C71031-1 Supplier: 81205
STD-467	Hoist - 500 lb (226.8 kg)
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 71-11-11-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-11-010-005-G00

(2) On the applicable engine, open these access panels:

(TASK 71-11-04-010-804-G00)

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

NOTE: The fan cowl panels must continue to be held open using the hoist, STD-467, and fan cowl sling, SPL-14546, when the fan cowl hold open rod is removed.

SUBTASK 71-11-11-490-001-G00

(3) Put the engine and strut access platform, STD-653, inside the inside of the fan cowl panel.

G. Fan Cowl Hold Open Rod Removal

SUBTASK 71-11-11-020-001-G00

(1) Remove the fan cowl hold open rod [1] from the applicable fan cowl panel as follows:

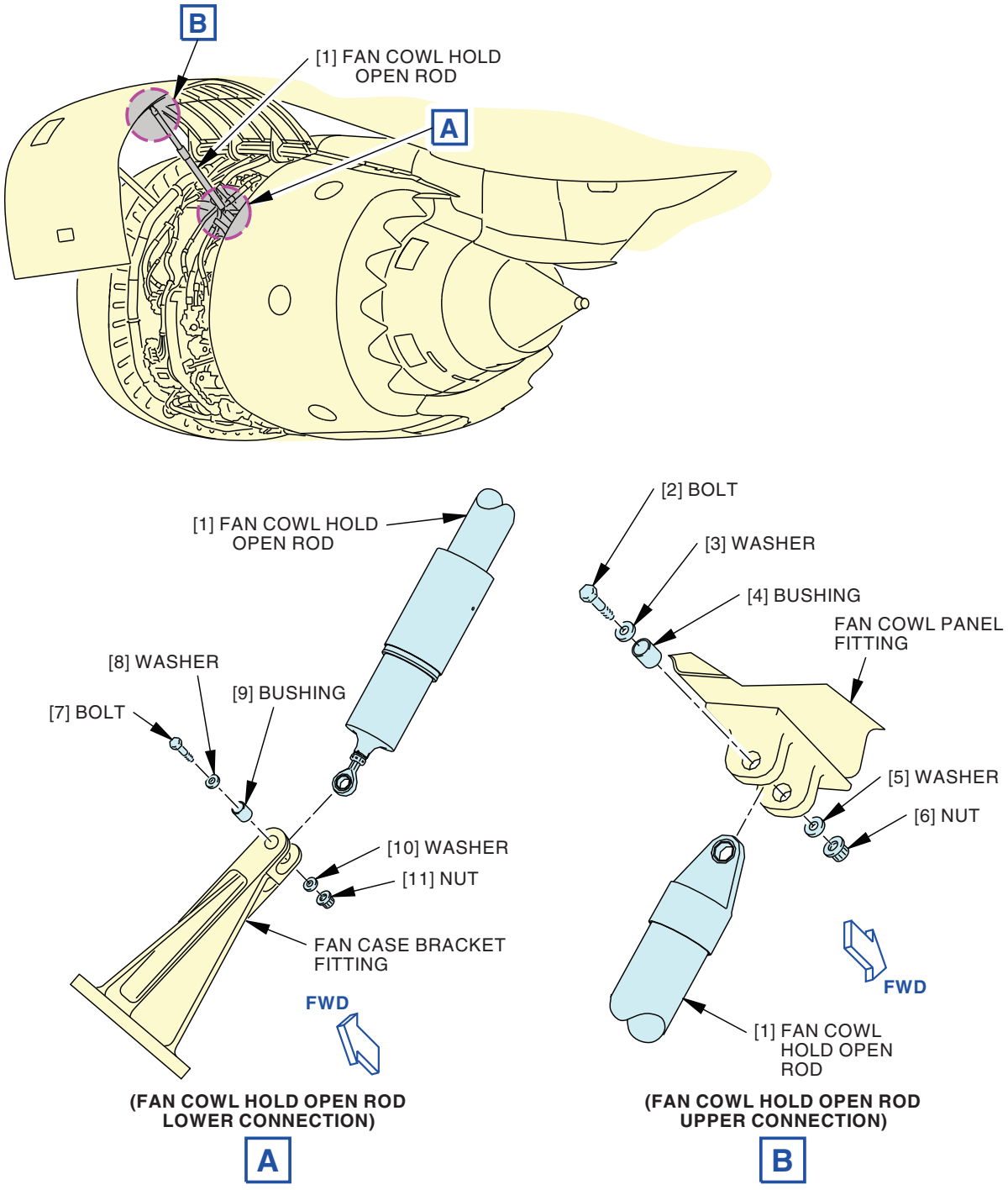
(a) Remove the nut [11], washer [10], bushing [9], washer [8], and bolt [7] that attach the fan cowl hold open rod [1] end to the fan case bracket fitting.

(b) Remove the nut [6], washer [5], bushing [4], washer [3], and bolt [2] that attach the fan cowl hold open rod [1] end to the fan cowl panel fitting.

(c) Remove the fan cowl hold open rod [1] from the fan cowl panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Cowl Hold Open Rod Installation
Figure 201/71-11-11-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-11-400-801-G00

3. Fan Cowl Hold Open Rod Installation

(Figure 201)

A. General

(1) This task provides the instructions on how to install the fan cowl hold open rod.

(2) The installation task of the left side or the right side of the fan cowl hold open rod is the same.

B. References

Reference	Title
71-11-04-410-804-G00	Close the Fan Cowl Panels (Sling Method) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fan cowl hold open rod	71-11-11-01-100	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Fan Cowl Hold Open Rod Installation

SUBTASK 71-11-11-420-001-G00

(1) Install the fan cowl hold open rod [1] on the applicable fan case panel as follows:

(a) Install the bolt [2], washer [3], bushing [4], washer [5], and nut [6] that attach the fan cowl hold open rod [1] end to the fan cowl panel fitting.

1) Tighten the nut [6] to 165 in-lb (18.6 N·m) - 175 in-lb (19.8 N·m).

(b) Install the bolt [7], washer [8], bushing [9], washer [10], and nut [11] that attach the fan cowl hold open rod [1] end to the fan case bracket fitting.

1) Tighten the nut [11] to 303 in-lb (34.2 N·m) - 347 in-lb (39.2 N·m).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-11-020-002-G00

(1) Move the engine and strut access platform, STD-653, away from the engine.

SUBTASK 71-11-11-410-008-G00

(2) On the applicable engine, close these access panels:

(TASK 71-11-04-410-804-G00)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-11-860-002-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 71-11-11-200-801-G00

4. Fan Cowl Hold-Open Rod Inspection

(Figure 202)

A. General

(1) This task provides the instructions on how to do a visual inspection of the fan cowl hold-open rod.

(2) You must open the fan cowl panels to do the inspection of the fan cowl hold-open rods.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-804-G00	Open the Fan Cowl Panels (Sling Method) (P/B 201)
71-11-04-410-804-G00	Close the Fan Cowl Panels (Sling Method) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14546	Sling Eqpt - Fan Cowl, LEAP-1B Engine Part #: C71031-21 Supplier: 81205 Opt Part #: C71031-1 Supplier: 81205
STD-467	Hoist - 500 lb (226.8 kg)

D. Location Zones

Zone	Area
411	Engine 1 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 71-11-11-869-001-G00

(1) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 71-11-11-040-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-11-010-006-G00

(3) Do this task: Open the Fan Cowl Panels (Sling Method), TASK 71-11-04-010-804-G00.

NOTE: The fan cowl panels must continue to be held open using the hoist, STD-467, and the fan cowl sling, SPL-14546, when the HOR is removed.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Fan cowl Hold-Open Rods Inspection

SUBTASK 71-11-11-210-001-G00

(1) Visually examine the applicable the fan cowl hold open rod [1] for damage as follows:

(a) Make sure that there are no cracks in the fan cowl hold open rod [1] end.

(b) Extend the fan cowl hold open rod [1] to make sure that they operate smoothly:

1) For the applicable fan cowl hold open rod [1], disconnect the bolt [7] at the bracket assembly on the fan case.

2) Manually extend and retract the fan cowl hold open rod [1]

a) Make sure that the fan cowl hold open rod [1] operates smoothly.

(c) Make sure that there is no contamination on the fan cowl hold open rod [1].

1) If there is contamination on the fan cowl hold open rod [1], clean the fan cowl hold-open rod (TASK 71-11-11-100-801-G00).

(d) Make sure that there is no corrosion on the fan cowl on the body and end of the fan cowl hold open rod [1].

SUBTASK 71-11-11-960-001-G00

(2) If you find damage, replace the fan cowl hold-open rod (TASK 71-11-11-000-801-G00 and TASK 71-11-11-400-801-G00).

SUBTASK 71-11-11-210-002-G00

(3) Visually examine the bracket assemblies on the fan cowl for damage.

(a) Make sure that there is no crack in the fan case bracket fittings and bushings of the fan cowl.

(b) Make sure that there is no corrosion on the brackets and the bushings of the fan cowl.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If you find damage to the brackets on the fan cowl, replace the brackets.

SUBTASK 71-11-11-210-003-G00

(4) Visually examine the fan case bracket for damage as follows:

(a) Make sure that there are no cracks on the fan case bracket.

(b) Make sure that there is no corrosion on the fan case bracket.

(c) If you find damage to the brackets on the fan cowl, replace the fan case brackets.

SUBTASK 71-11-11-410-002-G00

(5) Do this task: TASK 71-11-11-400-801-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-11-410-010-G00

(1) Do this task: Close the Fan Cowl Panels (Sling Method), TASK 71-11-04-410-804-G00.

(a) On the applicable engine, close these access panels:

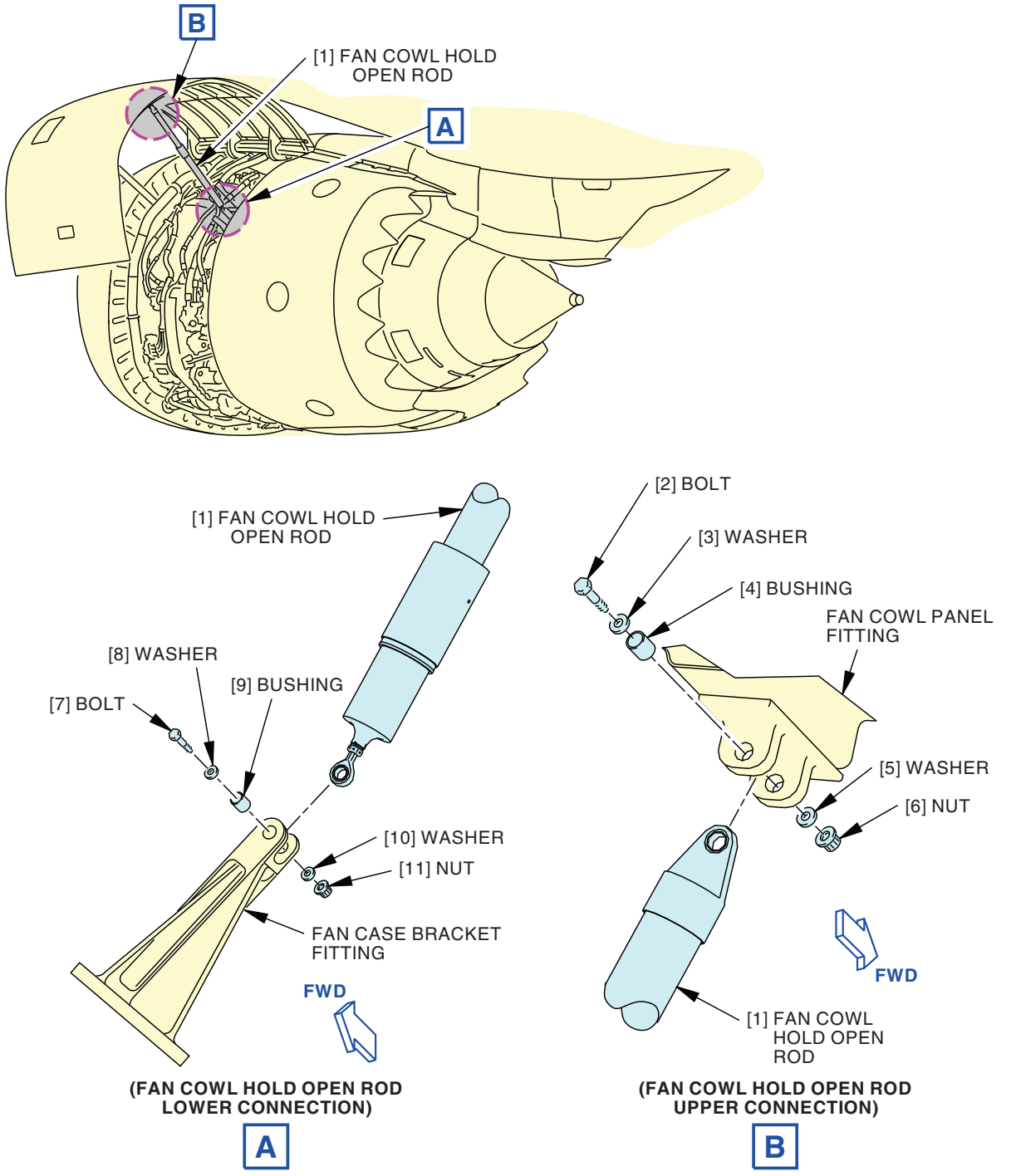
Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-11-440-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Cowl Hold-Open Rods Inspection
Figure 202/71-11-11-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-11-100-801-G00

5. Fan Cowl Hold-Open Rod Cleaning

(Figure 203)

A. General

(1) This task provides the instructions on how to clean the fan cowl hold open rod.

(2) You must open the fan cowl panels to clean the fan cowl hold-open rods.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-804-G00	Open the Fan Cowl Panels (Sling Method) (P/B 201)
71-11-04-410-804-G00	Close the Fan Cowl Panels (Sling Method) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14546	Sling Eqpt - Fan Cowl, LEAP-1B Engine Part #: C71031-21 Supplier: 81205 Opt Part #: C71031-1 Supplier: 81205
STD-467	Hoist - 500 lb (226.8 kg)

D. Consumable Materials

Reference	Description	Specification
B00090	Solvent - Inhibited Trichloroethane 1,1,1	MIL-T-81533
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Prepare for the Cleaning

SUBTASK 71-11-11-869-002-G00

(1) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-11-11-040-002-G00

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 71-11-11-010-007-G00

(3) Do this task: Open the Fan Cowl Panels (Sling Method), TASK 71-11-04-010-804-G00.

NOTE: The fan cowl panels must continue to be held open using the hoist, STD-467, and the fan cowl sling, SPL-14546, when the HOR is removed.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

H. Fan Cowl Hold-Open Rods Cleaning

SUBTASK 71-11-11-160-001-G00

(1) Clean the applicable fan cowl hold open rod [1] as follows:

(a) Make sure that the fan cowl hold open rod [1] is fully extended.

WARNING

DO NOT GET THE SOLVENT IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE IN THE FUMES FROM THE SOLVENT. MAKE SURE TO PUT ON PROTECTIVE SPLASH GOGGLES AND GLOVES WHEN USING THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAMES AND HEAT. THE SOLVENT IS POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY OR DAMAGE IF NOT HANDLED PROPERLY.

(b) Apply a large quantity of the solvent, B00090 to the fan cowl hold open rod [1].

(c) Extend and retract the fan cowl hold open rod [1] to flush the dirt, oil, or grease.

(d) Clean the bracket assemblies and the fittings.

(e) Dry the fan cowl hold open rod [1], the brackets, and the fittings with a clean dry cotton wiper, G00034.

(f) Extend and retract the fan cowl hold open rod [1] to make sure that it operates smoothly.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-11-410-009-G00

(1) Do this task: Close the Fan Cowl Panels (Sling Method), TASK 71-11-04-410-804-G00.

(a) On the applicable engine, close these access panels:

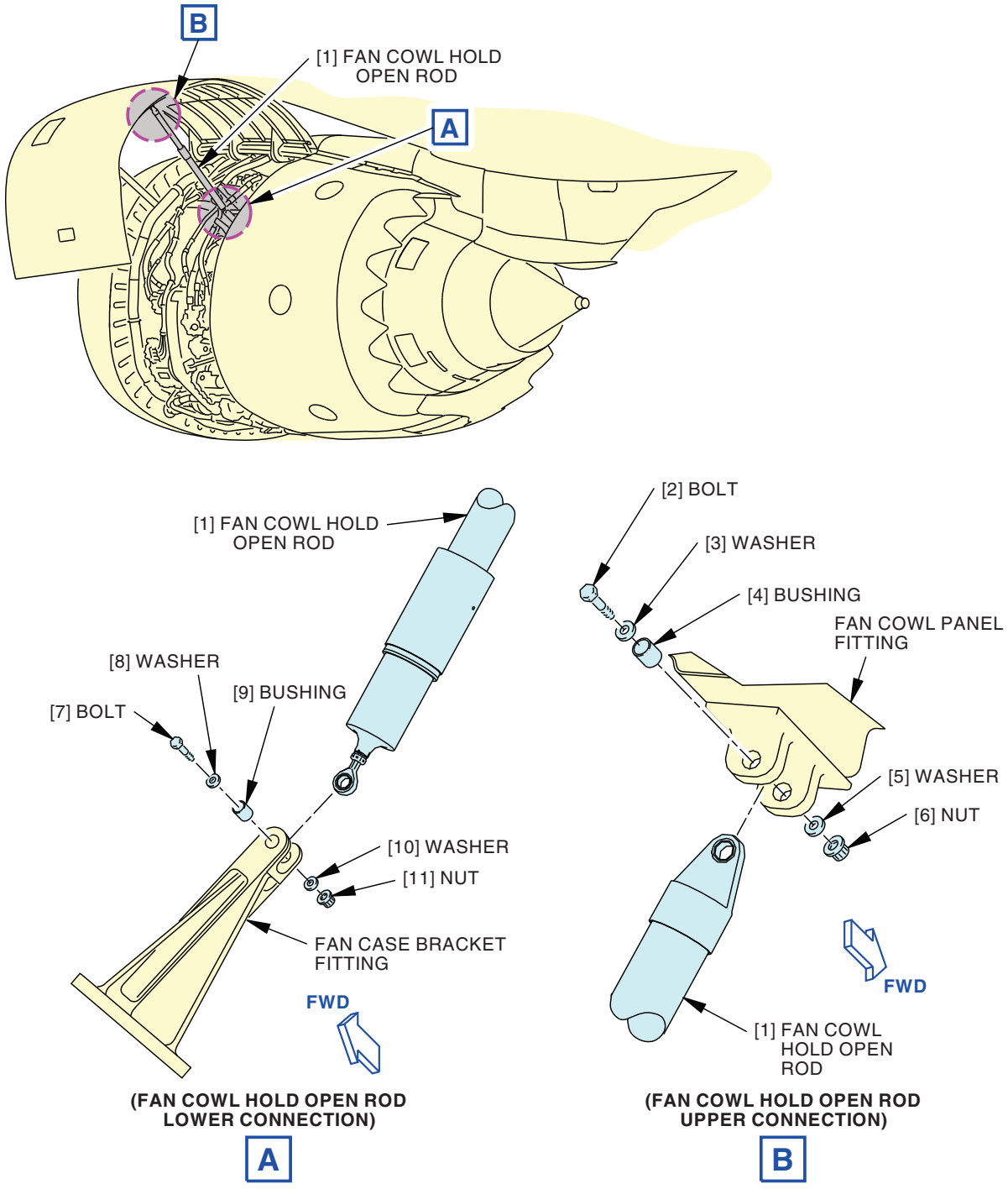
Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-11-440-002-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Cowl Hold Open Rod Cleaning
Figure 203/71-11-11-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL ALIGNMENT BUMPER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fan Cowl Alignment Bumper Removal

(2) Fan Cowl Alignment Bumper Installation.

TASK 71-11-12-000-801-G00

2. Fan Cowl Alignment Bumper Removal

(Figure 401 or Figure 402)

A. General

(1) This task gives the instructions to remove the fan cowl alignment bumper.

(2) The removal procedure of the left and the right fan cowl alignment bumper is the same.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right

E. Prepare for the Removal

SUBTASK 71-11-12-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-12-010-001-G00

(2) Open the applicable fan cowl panel (TASK 71-11-04-010-801-G00).

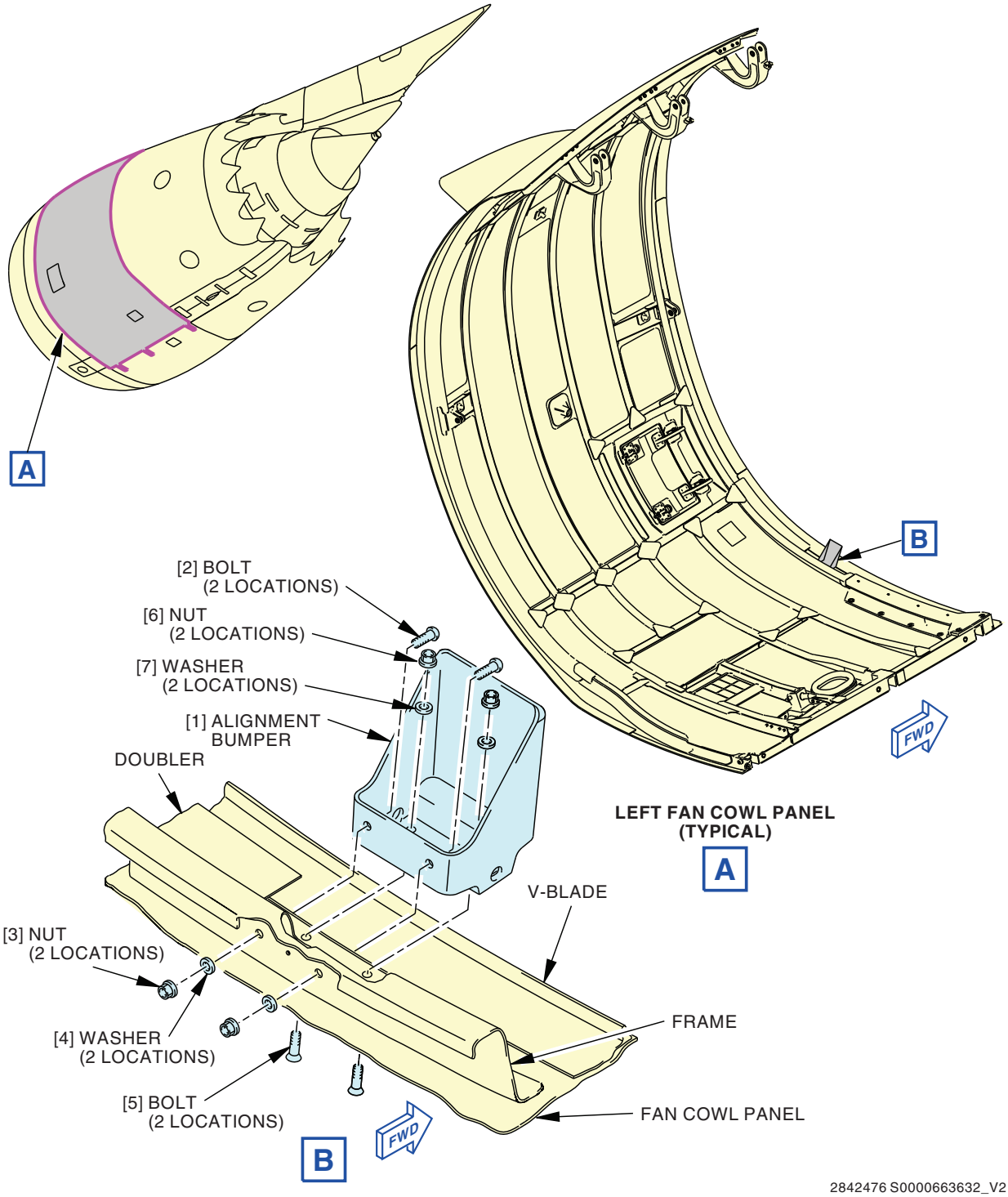
F. Fan Cowl Alignment Bumper Removal

SUBTASK 71-11-12-020-001-G00

(1) Remove the bolts [2], nuts [3], washers [4], bolts [5], nuts [6] and washers [7] that attach the alignment bumper [1] to the fan cowl panel.

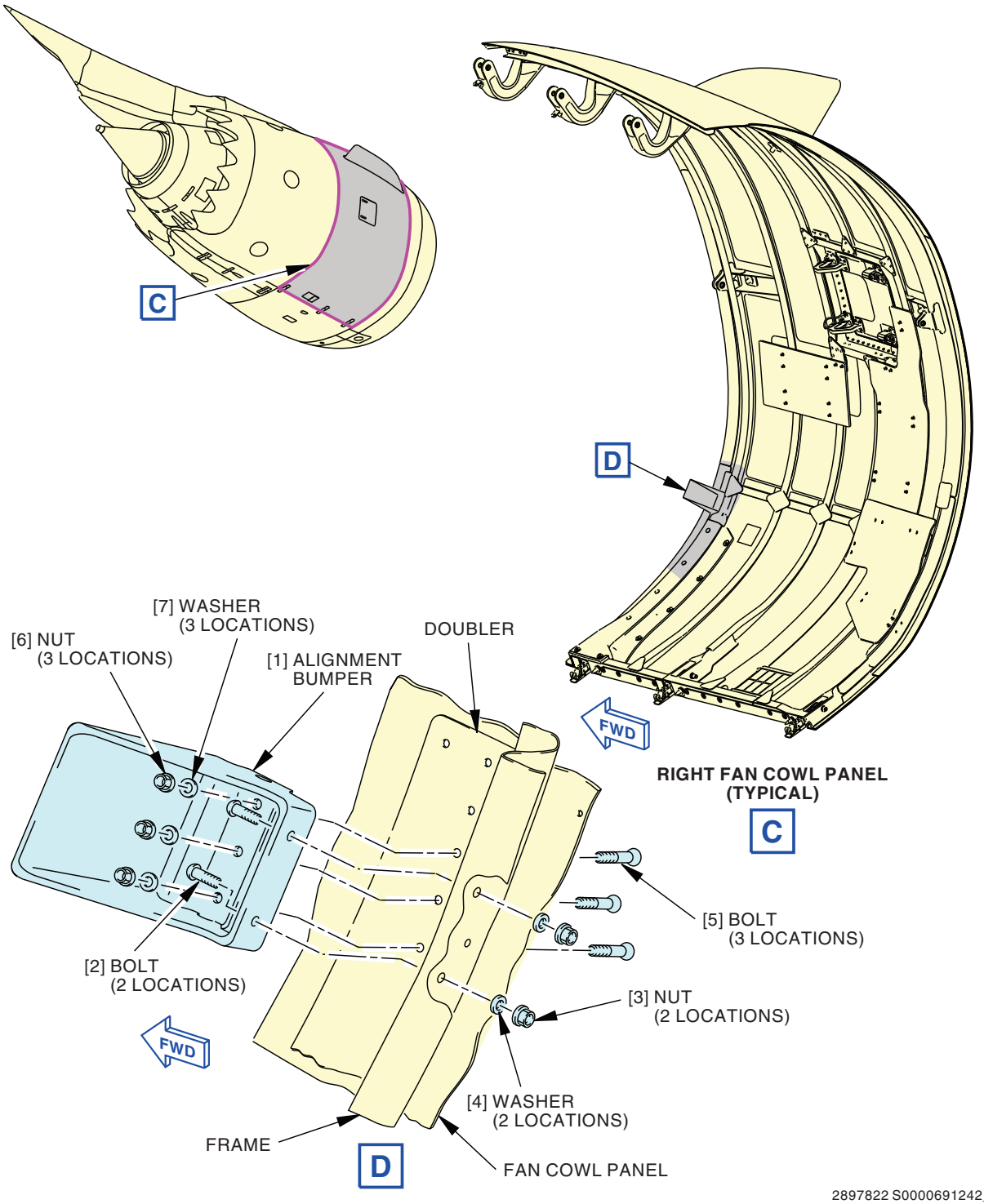
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left Fan Cowl - Alignment Bumper Installation
Figure 401/71-11-12-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right Fan Cowl - Alignment Bumper Installation
Figure 402/71-11-12-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-12-400-801-G00

3. Fan Cowl Alignment Bumper Installation

(Figure 401 or Figure 402)

A. General

(1) This task gives the instructions to install the fan cowl alignment bumper.

(2) The installation procedure of the left and the right fan cowl alignment bumper is the same.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50467	Sealant - Fuel Tank	BMS5-45, Class A, Grade 1

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Installation

SUBTASK 71-11-12-860-002-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

G. Fan Cowl Alignment Bumper Installation

SUBTASK 71-11-12-020-002-G00

(1) Install the alignment bumper [1] on the applicable fan cowl panel as follows:

(a) Put the alignment bumper [1] in position on the fan cowl panel.

(b) Use the sealant, A50467 to wet install the bolts [2] with the nuts [3] and washers [4] that attach the alignment bumper [1] to the fan cowl panel.

(c) Use the sealant, A50467 to wet install the bolts [5] with the nuts [6] and washers [7] that attach the alignment bumper [1] to the fan cowl panel.

(d) Tighten the bolts [2] and bolts [5] to 15.0 in-lb (1.7 N·m) - 17.0 in-lb (1.9 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-12-410-001-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-11-12-860-003-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE SIDE RADIAL RESTRAINT BRACKET - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the engine side radial restraint bracket

(2) An installation of the engine side radial restraint bracket.

TASK 71-11-13-000-801-G00

2. Engine Side Radial Restraint Bracket Removal

(Figure 401, Figure 402)

A. General

(1) This task includes the steps to remove the engine side radial restraint bracket.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 71-11-13-860-002-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-11-13-010-001-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Engine Side Radial Restraint Bracket Removal

SUBTASK 71-11-13-020-001-G00

(1) Remove the engine side radial restraint bracket [1] from the applicable engine as follows:

(a) Remove the bolt [3], washers [4], washer [5], and nut [6] at 2 locations.

(b) Remove the bolt [8], washer [9], washer [10] at 4 locations.

SUBTASK 71-11-13-020-002-G00

(2) Disassemble the engine side radial restraint bracket [1] from the engine side radial restraint flange [12] and radial restraint flange [20] as follows:

PTF ALL POST SB 737-71-2032

(a) Remove the nut [22], washer [23], and radial restraint pin [21].

PTF ALL PRE SB 737-71-2032

(b) Remove the bolt [16], washer [15] (if installed), washer [14] (if installed), and collar [13] or nut [13] at 4 locations.

PTF ALL POST SB 737-71-2032

(c) Remove the bolt [16], washer [15], washer [14], and nut [13] at 4 locations.

PTF ALL PRE SB 737-71-2032

(d) Remove the bolt [17], washer [15] (if installed), washer [14] (if installed), and collar [13] or nut [13] at 4 locations.

PTF ALL POST SB 737-71-2032

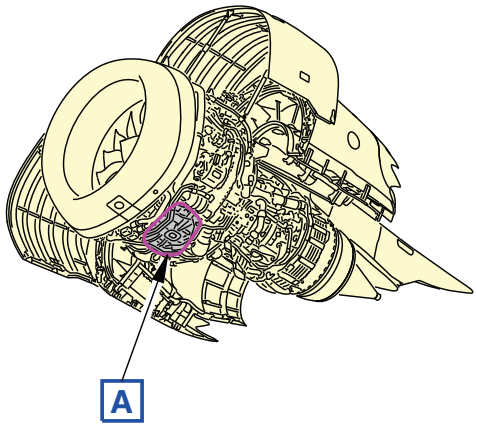
(e) Remove the bolt [17], washer [15], washer [14], and nut [13] at 4 locations.

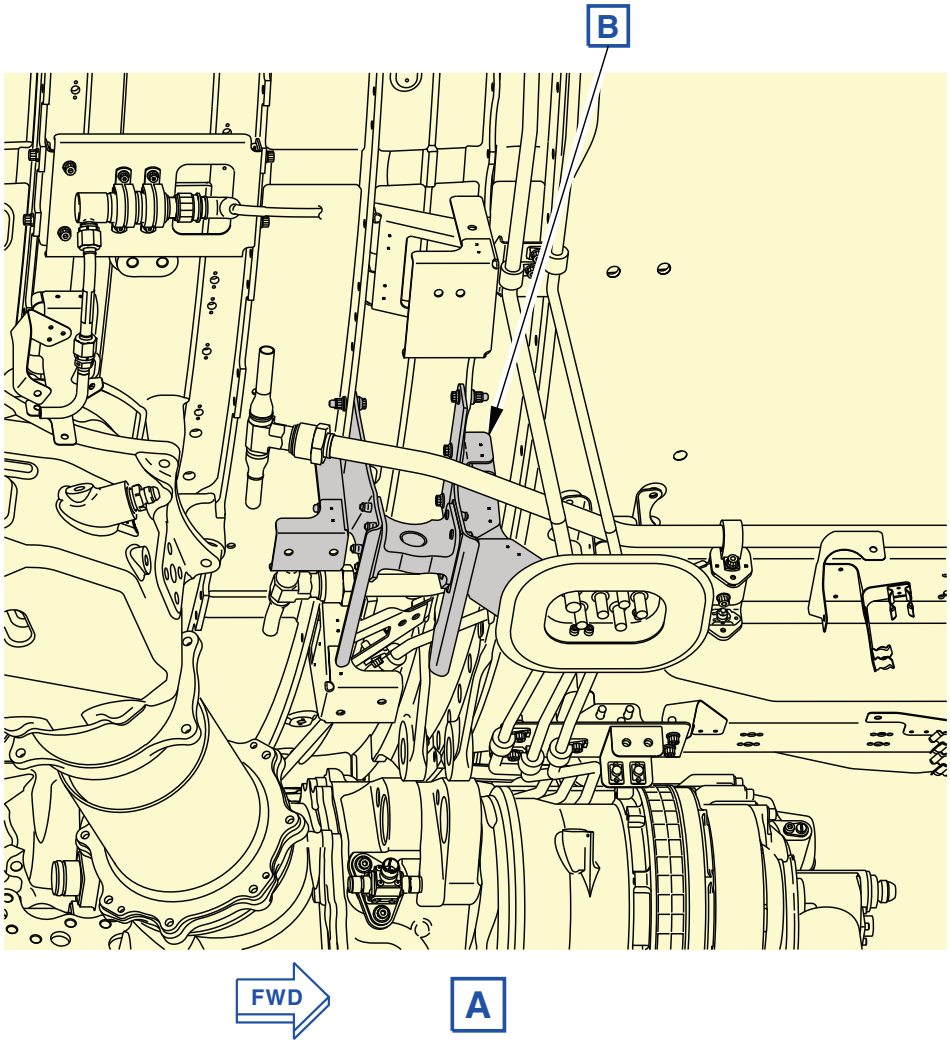
PTF ALL

(f) Remove the shim [18] and filler [19].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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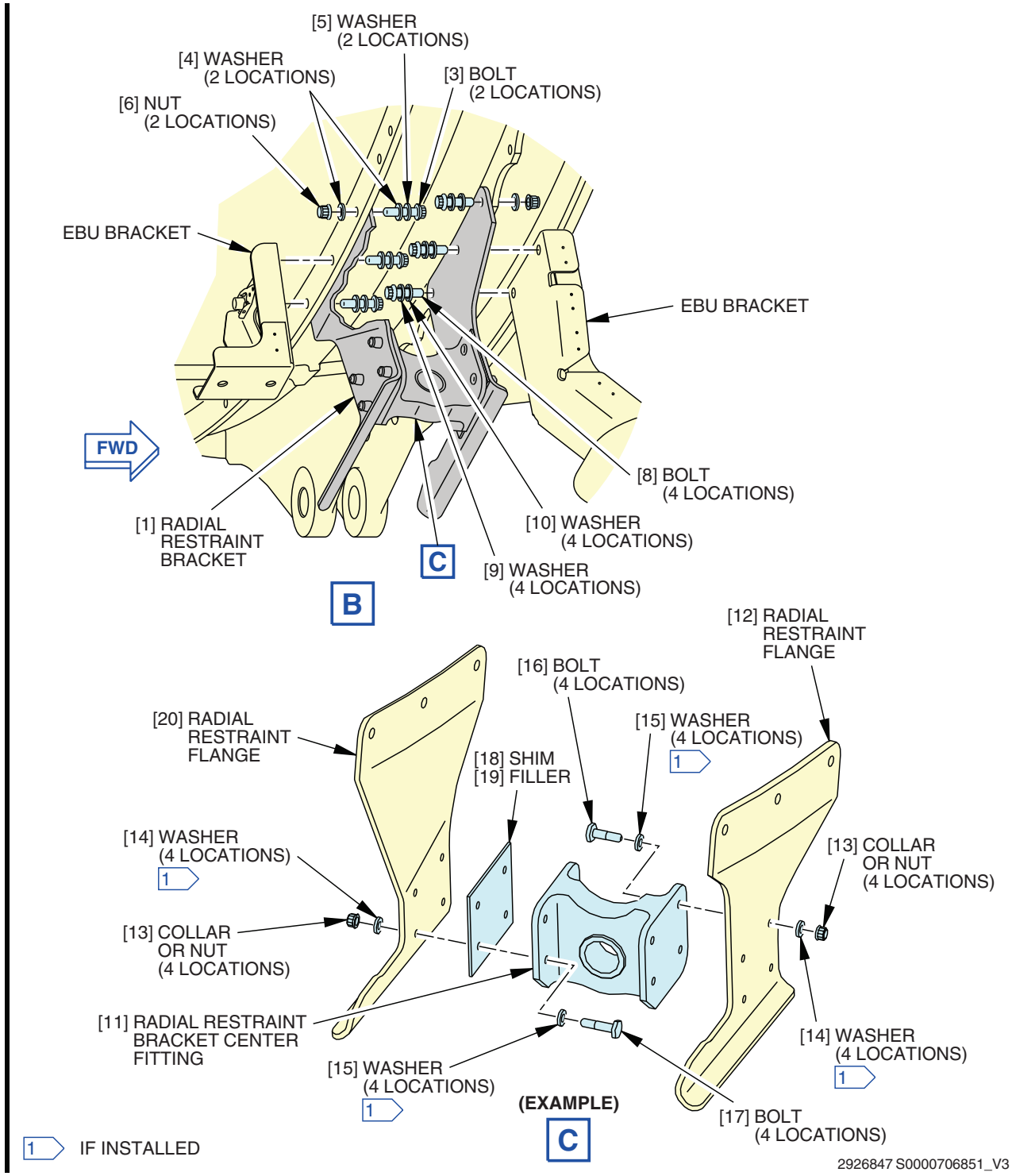
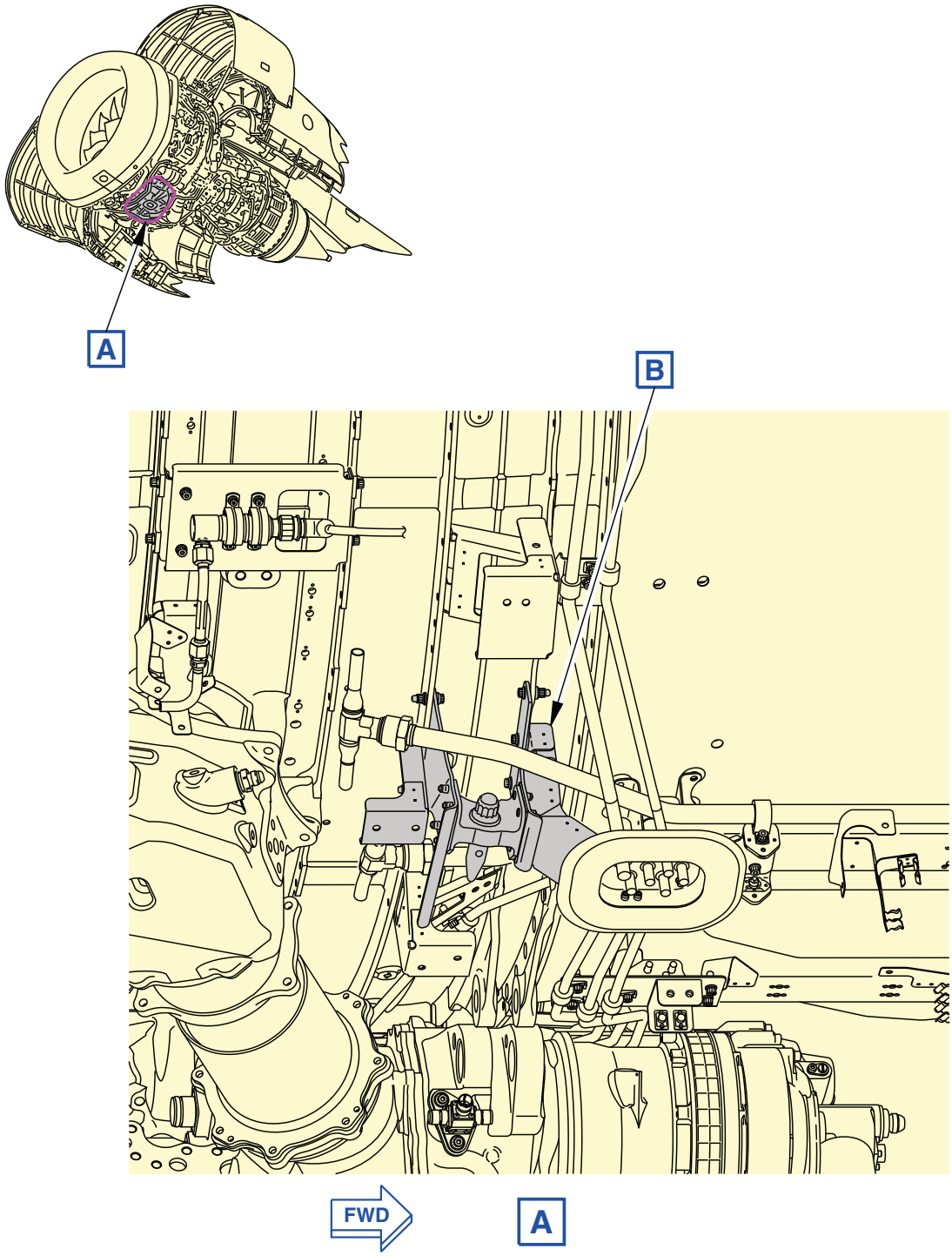
Engine Side Radial Restraint Bracket Installation
Figure 401/71-11-13-990-801-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



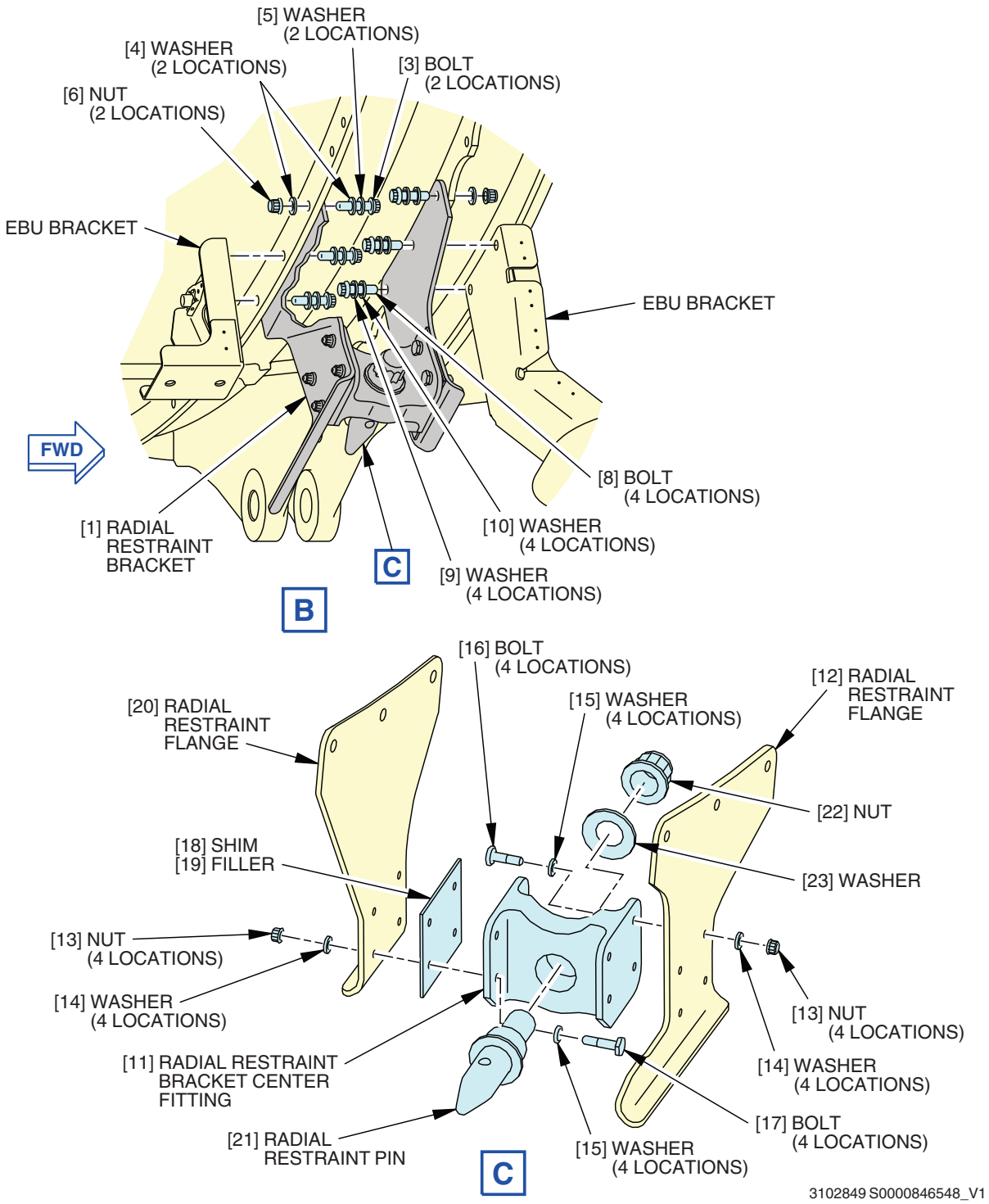
Engine Side Radial Restraint Bracket Installation
Figure 401/71-11-13-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



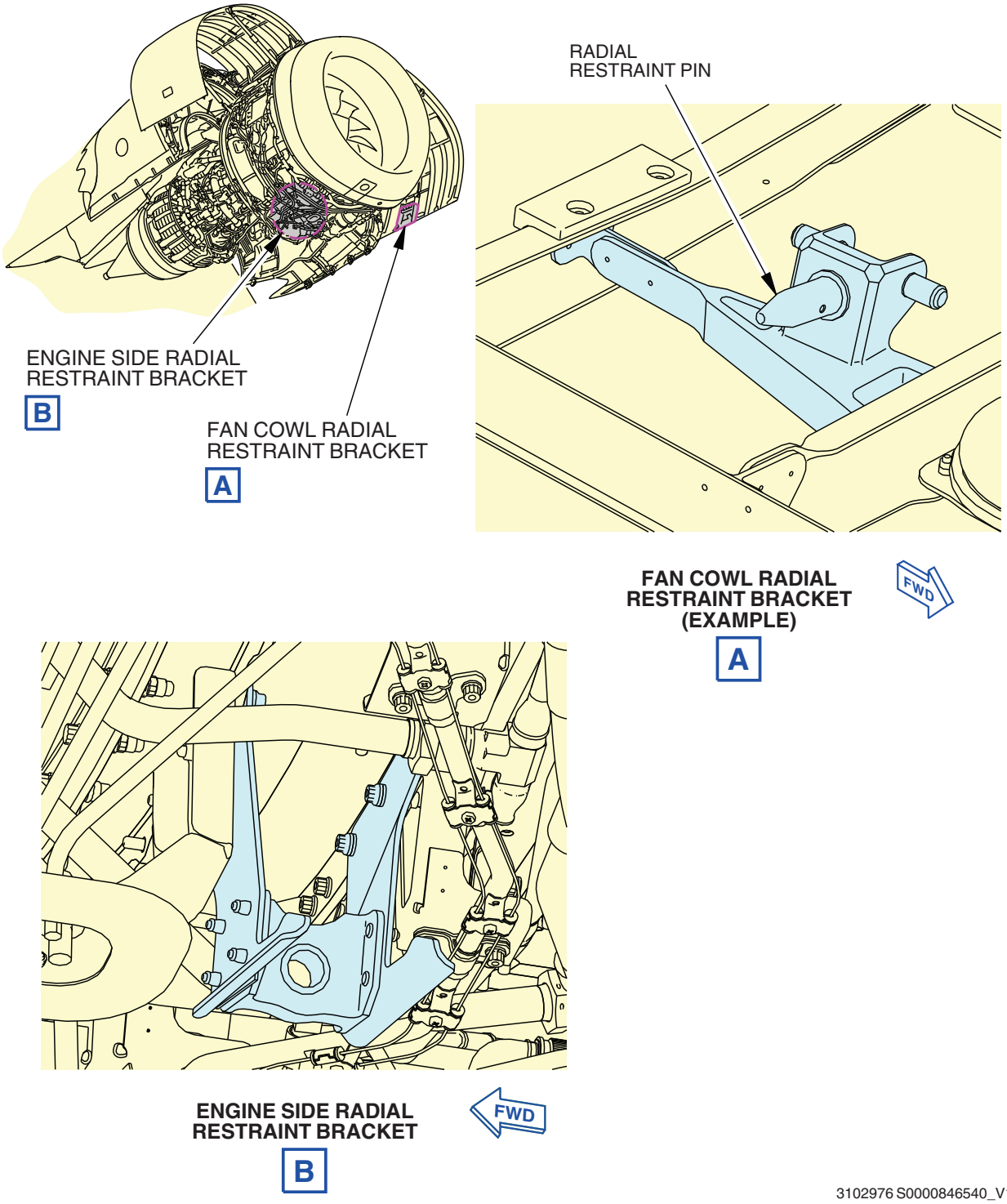
Engine Side Radial Restraint Bracket Installation
Figure 401/71-11-13-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



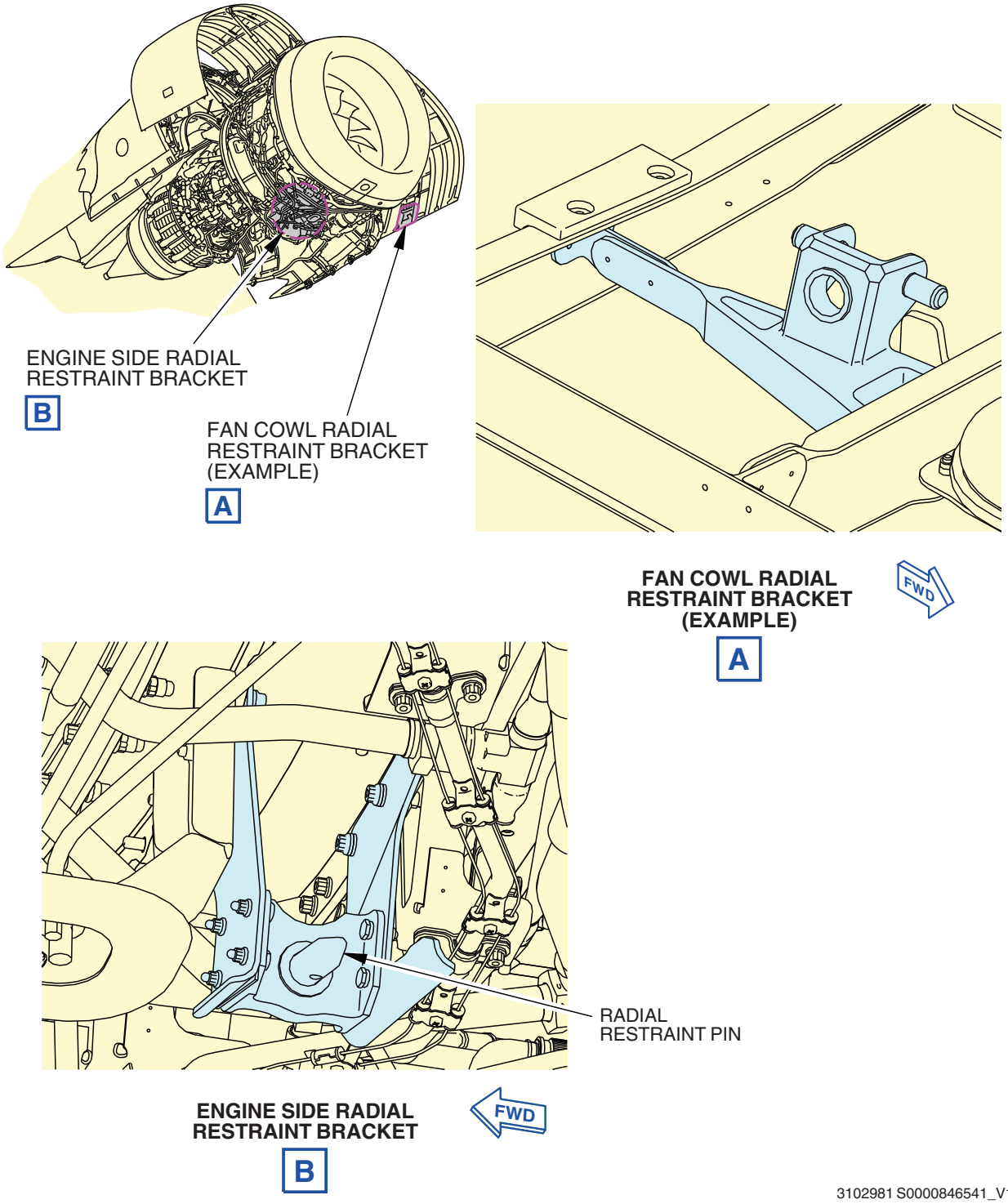
Engine Side Radial Restraint Bracket Installation
Figure 401/71-11-13-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl and Engine Radial Restraint Pin Configuration
Figure 402/71-11-13-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl and Engine Radial Restraint Pin Configuration
Figure 402/71-11-13-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-11-13-400-801-G00

3. Engine Side Radial Restraint Bracket Installation

(Figure 401, Figure 402)

A. General

(1) This task includes the steps to install the engine side radial restraint bracket.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

E. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Engine Side Radial Restraint Bracket Installation

SUBTASK 71-11-13-710-001-G00

WARNING MAKE SURE THAT THE FAN COWL RADIAL RESTRAINT BRACKET CONFIGURATION IS COMPATIBLE WITH THE ENGINE SIDE RADIAL RESTRAINT BRACKET CONFIGURATION. IF YOU DO NOT OBEY, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do a check to make sure that the fan cowl radial restraint bracket configuration is compatible with the engine radial restraint bracket configuration (Figure 402).

NOTE: A radial restraint pin must be installed in either bracket with a mating bushing installed in the opposite bracket.

SUBTASK 71-11-13-420-001-G00

(2) Assemble the engine side radial restraint bracket center fitting [11] to the engine side radial restraint flange [12] and radial restraint flange [20] as follows:

(a) Assemble the shim [18] and filler [19], all or in part to either side of the radial restraint bracket center fitting [11] to align hole with fan cowl door pin.

1) Use only one shim [18] and filler [19].

2) Make sure that the total thickness of the shim [18] and filler [19] assembled does not exceed 0.125 in. (3.175 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) If the shim [18] is damage or missing layers, replace the shim [18].

NOTE: It is possible to change the grip length of the bolt a maximum of ±3 sizes to fit the shim thickness provided that all standard installation requirements are met.

PTF ALL PRE SB 737-71-2032

(b) Install the bolt [16], washer [15] (if installed), washer [14] (if installed), and collar [13] or nut [13] at 4 locations.

PTF ALL POST SB 737-71-2032

(c) Install the bolt [16], washer [15], washer [14], and nut [13] at 4 locations.

PTF ALL PRE SB 737-71-2032

(d) Install the bolt [17], washer [15] (if installed), washer [14] (if installed), and collar [13] or nut [13] at 4 locations.

PTF ALL POST SB 737-71-2032

(e) Install the bolt [17], washer [15], washer [14], and nut [13] at 4 locations.

(f) Install the nut [22], washer [23], and radial restraint pin [21].

PTF ALL

SUBTASK 71-11-13-420-002-G00

(3) Install the engine side radial restraint bracket [1] to the applicable engine as follows:

(a) Apply Pure Nickel Special compound, D00006, to the shank and threads of the bolts [3] and bolts [8].

(b) Install the bolt [8], washer [9], and washer [10] at 4 locations.

1) Tighten the bolts [8] to 76 in-lb (8.6 N·m) - 84 in-lb (9.5 N·m).

(c) Install the bolt [3], washers [4], washer [5], and nut [6] at 2 locations.

1) Tighten the bolts [3] to 76 in-lb (8.6 N·m) - 84 in-lb (9.5 N·m).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-11-13-410-001-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 71-11-13-860-001-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL SDOS ACTUATORS - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A removal of the fan cowl SDOS actuators

(2) An installation of the fan cowl SDOS actuators.

TASK 71-12-01-000-801-G00

2. Fan Cowl SDOS Actuators Removal

(Figure 401, Figure 402 or Figure 403)

A. General

(1) This task gives the instructions to remove the fan cowl Spring Door Opening System (SDOS) actuators.

(2) Each fan cowl panel has one fan cowl SDOS actuator which is installed at the front end of the fan cowl panel bracket and the fan case fitting.

(3) The removal procedure of the left side or the right side of the fan cowl SDOS actuator is the same.

B. References

Reference	Title
71-11-04-010-804-G00	Open the Fan Cowl Panels (Sling Method) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14546	Sling Eqpt - Fan Cowl, LEAP-1B Engine Part #: C71031-21 Supplier: 81205 Opt Part #: C71031-1 Supplier: 81205
STD-467	Hoist - 500 lb (226.8 kg)
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G00440	Lockwire - MS20995C41, Corrosion Resistant Steel - 0.041 Inch (1.0414 mm) Diameter	NASM20995

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

G. Prepare for the Removal

SUBTASK 71-12-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-12-01-010-003-G00

(2) Do this task: Open the Fan Cowl Panels (Sling Method), TASK 71-11-04-010-804-G00.

NOTE: The fan cowl panels must continue to be held open using the hoist, STD-467, and fan cowl sling, SPL-14546, when the SDOS is removed.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-12-01-490-001-G00

(3) Put the engine and strut access platform, STD-653, inside the fan cowl panel.

SUBTASK 71-12-01-210-002-G00

WARNING YOU MUST OBEY ALL INSTRUCTIONS OF THE VISUAL INSPECTION FOR THE SPRING DOOR OPENING SYSTEM (SDOS) ACTUATOR. THE ACTUATOR CONTAINS DANGEROUS ENERGY FROM THE SPRING. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY TO PERSONNEL CAN OCCUR.

(4) Do a visual inspection of the fan cowl SDOS actuator [1].

(a) The fan cowl SDOS actuator [1] jammed or does not fully extend or retract (View A, Figure 402 or Figure 403).

NOTE: The forward part of the cowl will not fully open, or the black damper guide bushing is visible.

1) Not permitted.

(b) Separation of the tubes from the front support bush (View C, View A-A Figure 402 or Figure 403).

NOTE: There is a separation gap between the outer tube and front support bush.

1) Not permitted.

(c) Cracks on any part of the fan cowl SDOS actuator [1] (View A, View B, View C, View A-A Figure 402 or Figure 403).

1) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Bent, buckled (kinked) or cracks on any part of the fan cowl SDOS actuator [1].

1) Not permitted.

(e) One or more rivets missing, loose or protruding (View B, Figure 402 or Figure 403).

NOTE: The back end of the SDOS actuator is riveted with 6 rivets.

1) Not permitted.

(f) No damping action.

NOTE: Fan cowl closes rapidly. Leakage on the piston rod can be a sign of beginning damper oil loss.

1) Check the back end ventilation holes for oil accumulation.

a) Not permitted.

(g) Tabs missing from lock ring.

1) Not permitted.

(h) Nuts missing (View A, View B Figure 402 or Figure 403).

1) Not permitted.

PTF ALL; AIRPLANES WITH THE SDOS ACTUATORS P/N BOE-2001-901F

SUBTASK 71-12-01-210-004-G00

(5) Before you remove the SDOS actuators, secure the fan cowl SDOS actuator [1] with MS20995C41 lockwire, G00440 (View D, Figure 402 or Figure 403).

(a) Install a minimum of six independent lengths of lockwire as follows:

1) Install three independent lengths of lockwire on each side of the SDOS actuator.

2) Install the lockwires around the locking bracket pin and hinge point pin.

3) Cut out below the ball bearing of the back end.

4) Twist the ends of each length of lockwire together a minimum of ten times.

(b) Install one independent length of lockwire around the inner tube of the fan cowl SDOS actuator [1] as follows:

1) Wrap around the inner tube a minimum of three times over the six lockwires.

2) Twist the ends of the lockwire together a minimum of ten times.

PTF ALL; AIRPLANES WITH THE SDOS ACTUATORS P/N BOE-2001-901H AND ON

SUBTASK 71-12-01-210-003-G00

(6) If any of the above damage is seen, secure the fan cowl SDOS actuator [1] with MS20995C41 lockwire, G00440 (View D, Figure 402 or Figure 403).

(a) Install a minimum of six independent lengths of lockwire as follows:

1) Install three independent lengths of lockwire on each side of the SDOS actuator.

2) Install the lockwires around the hinge point pin.

3) Cut out below the ball bearing of the back end.

4) Twist the ends of each length of lockwire together a minimum of ten times.

(b) Install one independent length of lockwire around the inner tube of the fan cowl SDOS actuator [1] as follows:

1) Wrap around the inner tube a minimum of three times over the six lockwires.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH THE SDOS ACTUATORS P/N BOE-2001-901H AND ON (Continued)

2) Twist the ends of the lockwire together a minimum of ten times.

PTF ALL

H. Fan Cowl SDOS Actuators Removal

SUBTASK 71-12-01-870-001-G00

(1) Make sure that the applicable fan cowl panel is completely open.

SUBTASK 71-12-01-750-001-G00

(2) Make sure that you engage the lock mechanism on the hold-open rod and the SDOS.

SUBTASK 71-12-01-020-001-G00

WARNING MAKE SURE THAT THE SDOS ACTUATOR IS FULLY EXTENDED AND LOCKED. THE ACTUATOR CONTAINS STORED ENERGY. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY CAN OCCUR.

WARNING BE CAREFUL AFTER YOU REMOVE THE SDOS ACTUATOR. DO NOT POINT THE ACTUATOR END AT A PERSON. THE ACTUATOR CONTAINS ENERGY OF A SPRING. THIS CAN CAUSE INJURY OR DEATH.

(3) Remove the fan cowl SDOS actuator [1] from the applicable fan cowl panel:

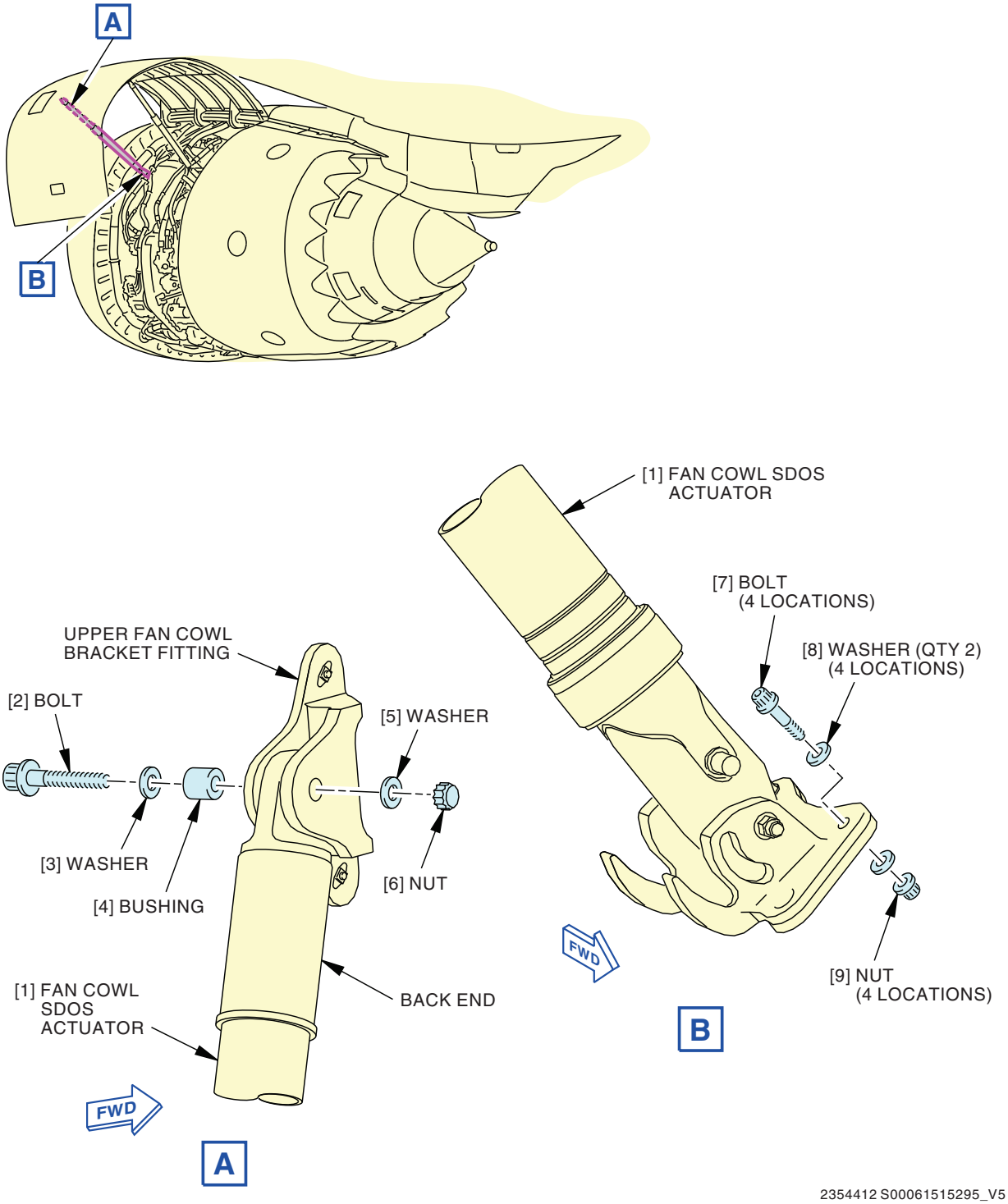
(a) Remove the bolts [7], washers [8], and nuts [9] that attach the fan cowl SDOS actuator [1] to the fan case bracket.

(b) Remove the bolt [2], washer [3], bushing [4], washer [5], and nut [6] that attach the fan cowl SDOS actuator [1] to the fan cowl panel bracket.

(c) Remove the fan cowl SDOS actuator [1] from the engine.

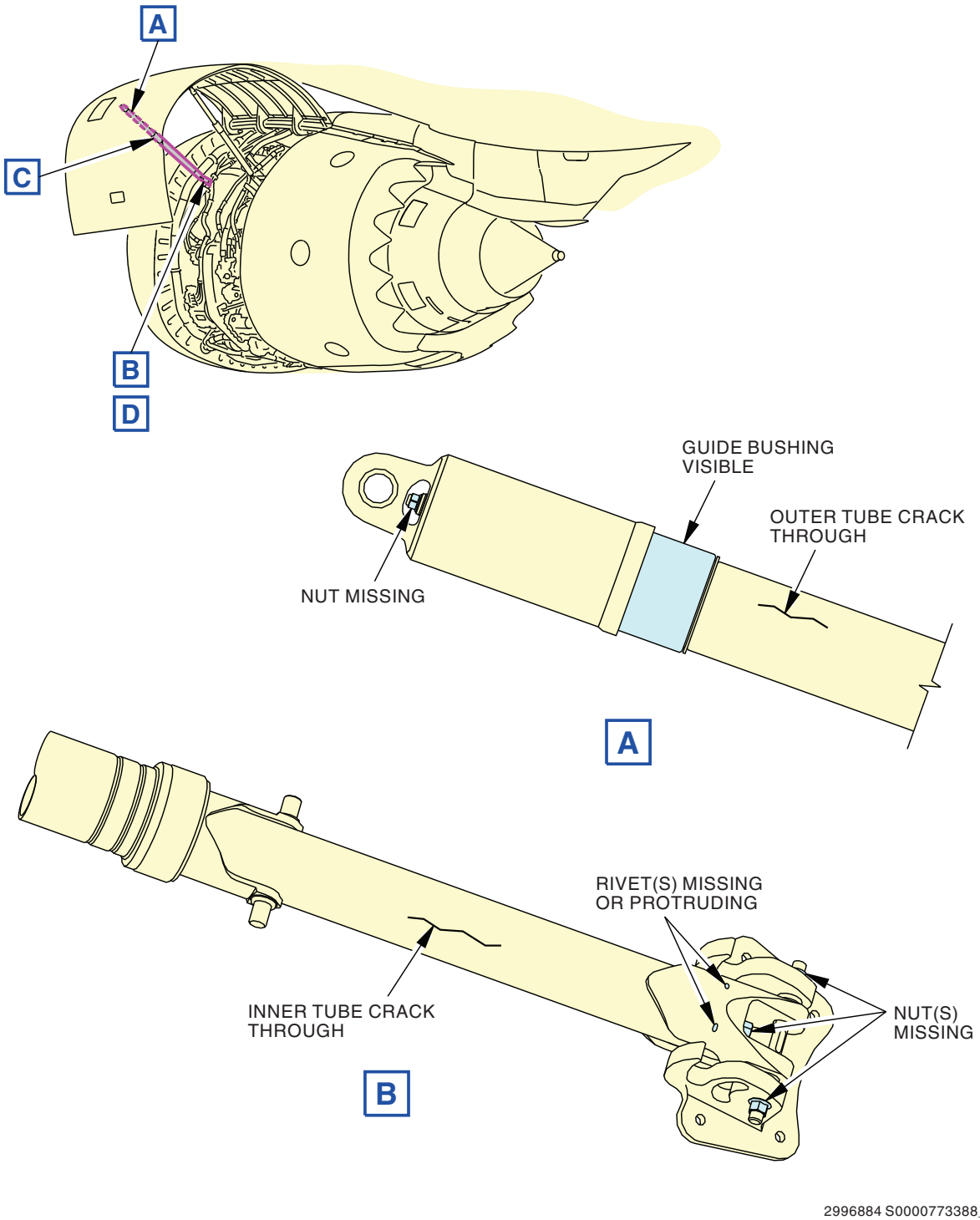
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



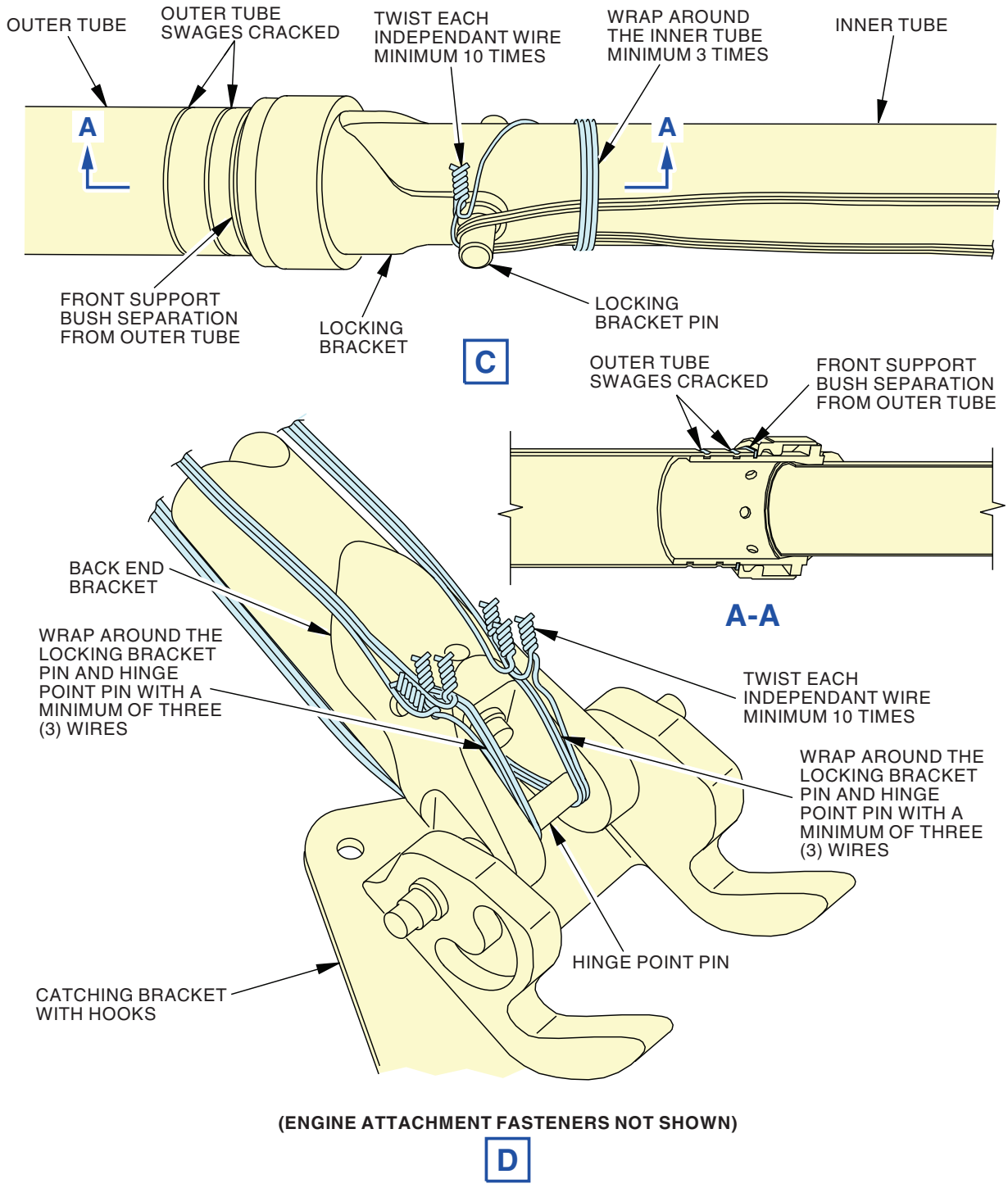
Fan Cowl SDOS Actuators Installation
Figure 401/71-12-01-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Cowl SDOS Actuators Visual Inspection
Figure 402/71-12-01-990-805-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Cowl SDOS Actuators Visual Inspection
Figure 402/71-12-01-990-805-G00 (Sheet 2 of 2)

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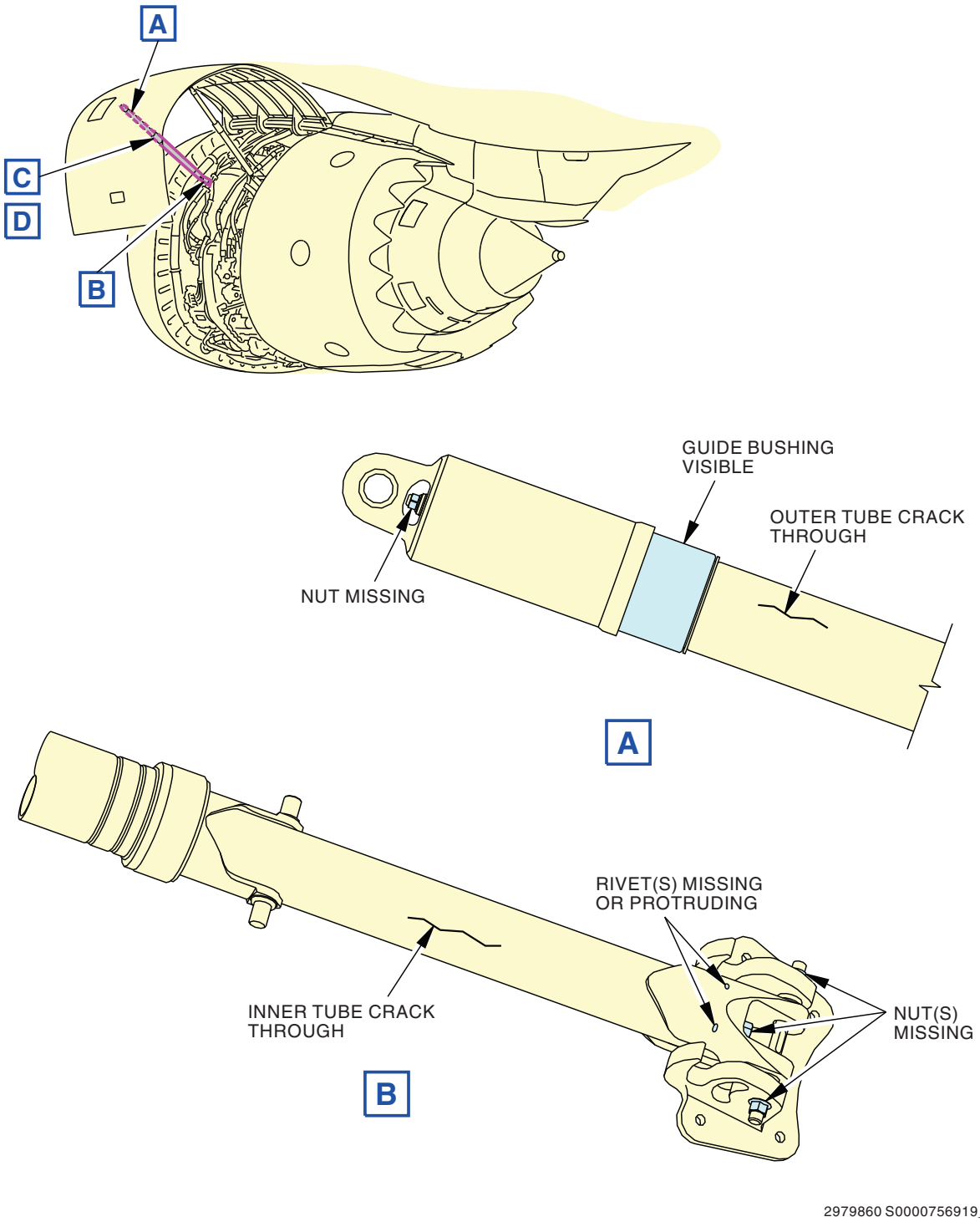

Fan Cowl SDOS Actuators Visual Inspection
Figure 403/71-12-01-990-804-G00 (Sheet 1 of 2)

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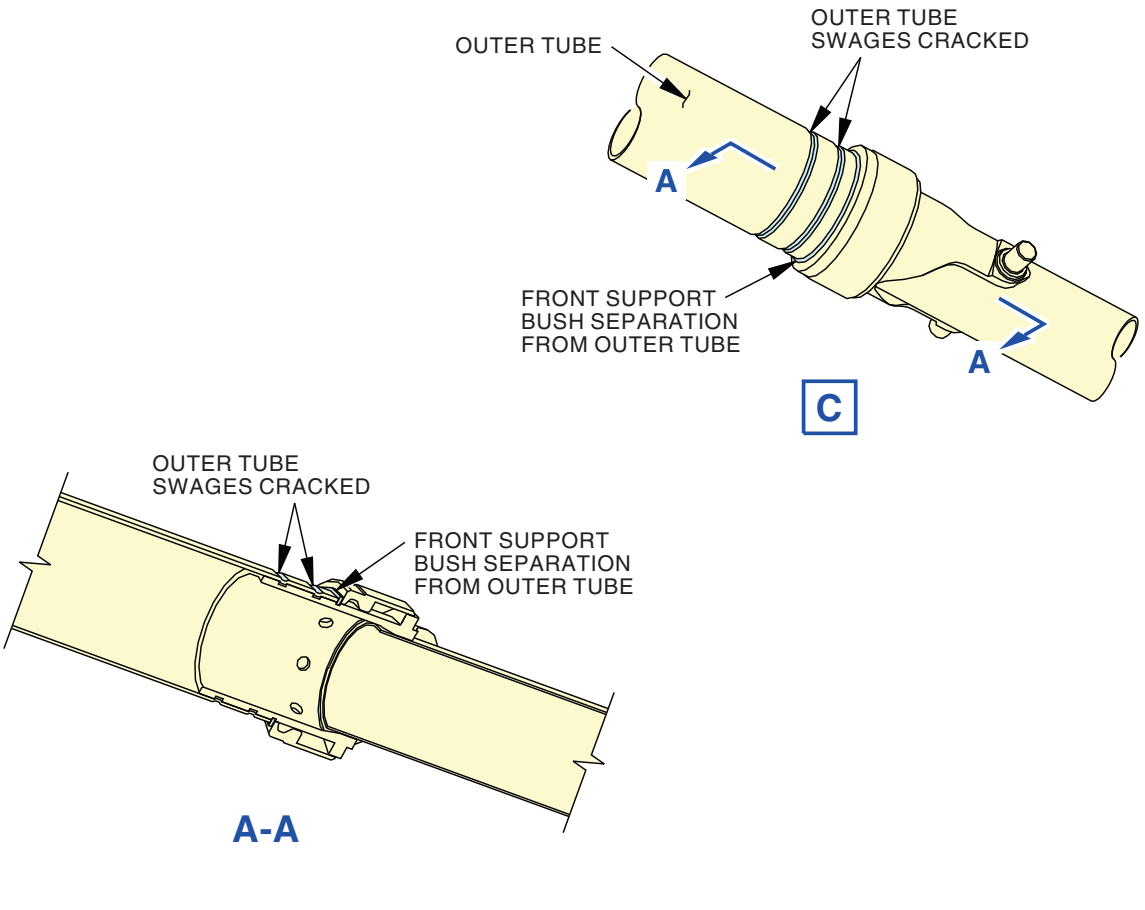

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Fan Cowl SDOS Actuators Visual Inspection
Figure 403/71-12-01-990-804-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-12-01-400-801-G00

3. Fan Cowl SDOS Actuators Installation

(Figure 401)

A. General

(1) This task gives the instructions for the installation of the fan cowl Spring Door Opening System (SDOS) actuators.

(2) The installation task of the left side or the right side of the fan cowl SDOS actuator is the same.

B. References

Reference	Title
71-11-04-410-804-G00	Close the Fan Cowl Panels (Sling Method) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fan cowl SDOS actuator	71-12-01-01-070	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Fan Cowl SDOS Actuators Installation

SUBTASK 71-12-01-420-002-G00

(1) Before installation, make sure that two safety markers p/n 12299, are installed on the outer tube of the fan cowl SDOS actuator [1] (cowl door side).

NOTE: Refer to Boeing Service Bulletin 737-71-1911 to ensure the P/N BOE-2001-901J SDOS actuator is in compliance with FAA AD 2021-13-18.

(a) If the safety markers p/n 12299, are not installed, do applicable corrective action until the safety markers are installed.

SUBTASK 71-12-01-420-001-G00

(2) Install the fan cowl SDOS actuator [1] on the applicable fan case panel:

(a) Install the fan cowl SDOS actuator [1] to the fan cowl panel bracket with the bolt [2], washer [5], bushing [4], washer [3], and nut [6].

1) Tighten the nut [6] to 310.0 in-lb (35.0 N·m) - 330.0 in-lb (37.3 N·m).

(b) Install the fan cowl SDOS actuator [1] to the fan case bracket with the bolts [7], washers [8], and nuts [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the fan cowl SDOS actuator [1] is installed on the fan case bracket in the correct orientation (Figure 401).

2) Tighten the bolts [7] to 97.0 in-lb (11.0 N·m) - 103.0 in-lb (11.6 N·m).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-12-01-010-002-G00

(1) Move the engine and strut access platform, STD-653, away from the engine.

SUBTASK 71-12-01-410-002-G00

(2) Do this task: Close the Fan Cowl Panels (Sling Method), TASK 71-11-04-410-804-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-12-01-710-001-G00

(3) Cycle the fan cowl panel partially open and closed three times to make sure that the fan cowl SDOS actuator [1] works smoothly.

SUBTASK 71-12-01-860-002-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN COWL SDOS ACTUATORS - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Fan Cowl SDOS Actuators Inspection.

TASK 71-12-01-200-801-G00

2. Fan Cowl SDOS Actuators Inspection

(Figure 601)

A. General

(1) This task gives the instructions to do the visual inspection of the fan cowl SDOS actuator.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
71-12-01-000-801-G00	Fan Cowl SDOS Actuators Removal (P/B 401)
71-12-01-400-801-G00	Fan Cowl SDOS Actuators Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 71-12-01-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-12-01-860-004-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 71-12-01-480-001-G00

(3) Put the engine and strut access platform, STD-653 into position.

G. Fan Cowl SDOS Actuators Inspection

SUBTASK 71-12-01-210-001-G00

(1) Examine the fan cowl SDOS actuator for damage with the limits that follow:

(a) Scratches

1) Permitted.

(b) Dusty sliding surfaces

1) Permitted.

(c) Unit is sliding, but with increased friction, following fretting marks on the tube

1) Permitted until the next convenient maintenance opportunity for these conditions:

a) The rubbed off area on the inner tube is not more than 0.5 in^2 (3.2 cm^2), and not deeper than 0.02 in. (0.51 mm).

b) The rubbed off area on the outer tube is not more than 1.6 in^2 (10.3 cm^2), and not deeper than 0.02 in. (0.51 mm).

(d) Dents on the inner tube

1) Permitted until the next convenient maintenance opportunity if the dents on the inner tube is not more than 0.5 in^2 (3.2 cm^2) and 0.02 in. (0.51 mm) in depth.

(e) Dents on the outer tube

1) Permitted until the next convenient maintenance opportunity if the dents on the outer tube is not more than 1.0 in^2 (6.5 cm^2) and 0.012 in. (0.3 mm) in depth.

(f) Worn out or loose bearing on the back end

1) Permitted until the next convenient maintenance opportunity.

(g) Bent tubes

1) Permitted until the next convenient maintenance opportunity if the eccentricity is not more than 1.0 in. (25 mm).

(h) Rusty deposits on ball bearings

1) Permitted until the next convenient maintenance opportunity.

(i) Minor leakage on the piston rod

1) Permitted until the next convenient maintenance opportunity.

2) Oil loss that results in decreased SDOS damping during fan cowl closing is not permitted.

NOTE: Loss of sufficient oil will cause SDOS damping to no longer control the fan cowl closing rate for part or all of the closing distance.

a) Replace the SDOS actuator (TASK 71-12-01-000-801-G00 and TASK 71-12-01-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Unit does not lock

1) Not permitted.

2) Replace the SDOS actuator (TASK 71-12-01-000-801-G00 and TASK 71-12-01-400-801-G00).

(k) Unit does not slide in or out anymore

1) Not permitted.

2) Replace the SDOS actuator (TASK 71-12-01-000-801-G00 and TASK 71-12-01-400-801-G00).

(l) Cracks on any part of the SDOS

1) Not permitted.

2) Replace the SDOS actuator (TASK 71-12-01-000-801-G00 and TASK 71-12-01-400-801-G00).

(m) Bent, buckled (kinked), or cracked tubes

1) Not permitted.

2) Replace the SDOS actuator (TASK 71-12-01-000-801-G00 and TASK 71-12-01-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-12-01-080-001-G00

(1) Remove the engine and strut access platform, STD-653.

NOTE: Make sure that the work area is clean of all tools and items used for the task.

SUBTASK 71-12-01-860-005-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 71-12-01-860-006-G00

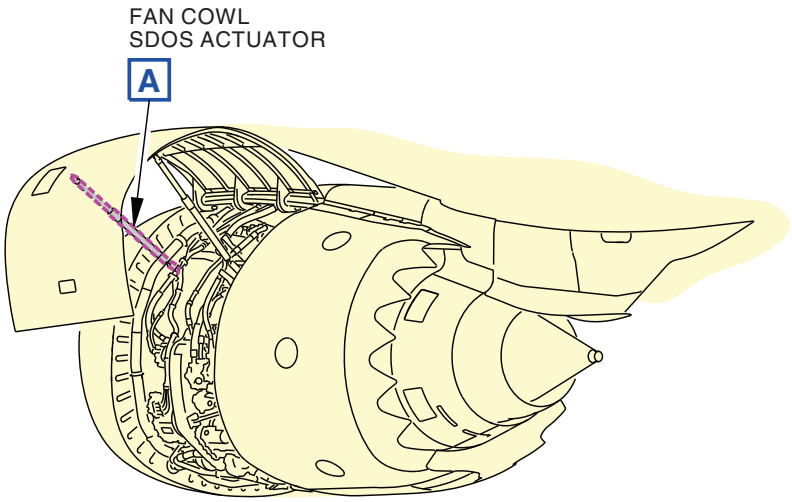
(3) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

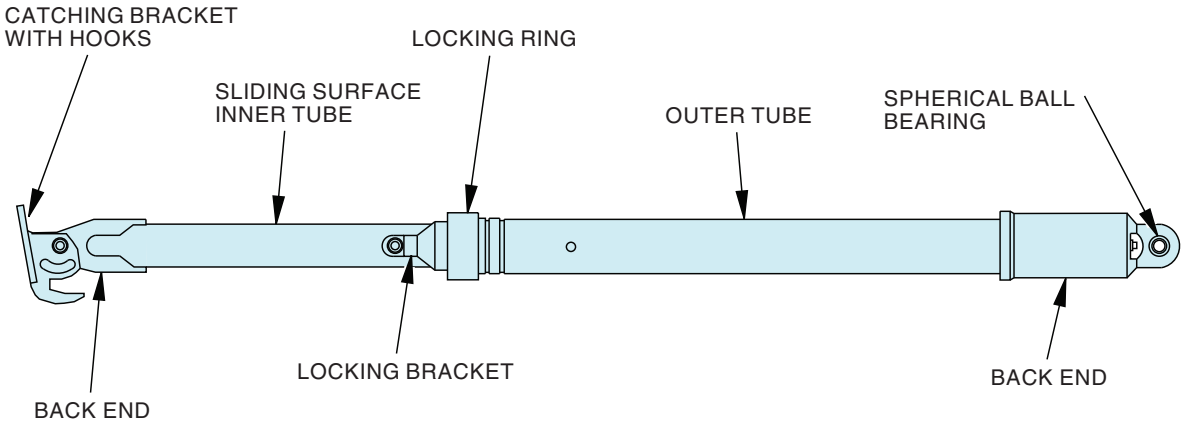
(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





FAN COWL SDOS ACTUATOR

A

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Fan Cowl SDOS Actuators Inspection
Figure 601/71-12-01-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE MOUNTS - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Forward engine mounts visual inspection

(2) Forward engine mounts detailed inspection

(3) Aft engine mount visual inspection

(4) Aft engine mount detailed inspection.

TASK 71-21-00-200-801-G00

2. Forward Engine Mount Visual Inspection

(Figure 601)

A. General

(1) This task provides the instructions on how to do a visual inspection of the forward engine mount.

(2) This task is used when the engine is off-wing.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-21-01-000-801-G00	Forward Engine Mount Removal (P/B 401)
71-21-01-400-801-G00	Forward Engine Mount Installation (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Inspection

SUBTASK 71-21-00-020-001-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

E. Forward Engine Mount Visual Inspection

SUBTASK 71-21-00-212-001-G00

(1) Examine the forward engine mount for these signs of damage.

(a) Corrosion is not permitted.

(b) Cracks are not permitted.

(c) Nicks are not permitted.

(d) Dents are not permitted.

(e) Scratches are not permitted.

(f) Bushing movement is not permitted.

(g) Damage on the tension bolt is not permitted.

(h) Damage to the side link assembly is not permitted.

(i) Damage to the shear pin is not permitted.

(j) Damage to the bearing is not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(k) Cracks to the bearing in the side link assembly is not permitted.

1) Cracks to the bearing ball are not permitted.

2) Seized bearing is not permitted.

3) Cracks in bearing are not permitted.

4) Bearing movement is not permitted.

5) All other types of damage to the bearing are not permitted.

(l) Fretting is not permitted.

NOTE: Fretting is caused when two adjacent pieces of metal rub one another.

(m) Pickup is not permitted.

NOTE: Pickup occurs when material is moved from one surface to a surface which it touches.

(n) High metal is not permitted.

NOTE: High metal is metal which is pushed above the surface adjacent to scratches or other damage.

SUBTASK 71-21-00-212-002-G00

(2) Examine the forward engine mount for worn areas.

(a) If you find signs of wear on the forward engine mount, do this task: TASK 71-21-00-211-801-G00.

SUBTASK 71-21-00-212-003-G00

(3) If damage is found on the forward engine mount, refer to the Structure Repair Manual for the allowable damage limits.

(a) Replace the forward engine mount, if it is more than the allowable damage limits.

1) Do this task: TASK 71-21-01-000-801-G00.

2) Do this task: TASK 71-21-01-400-801-G00.

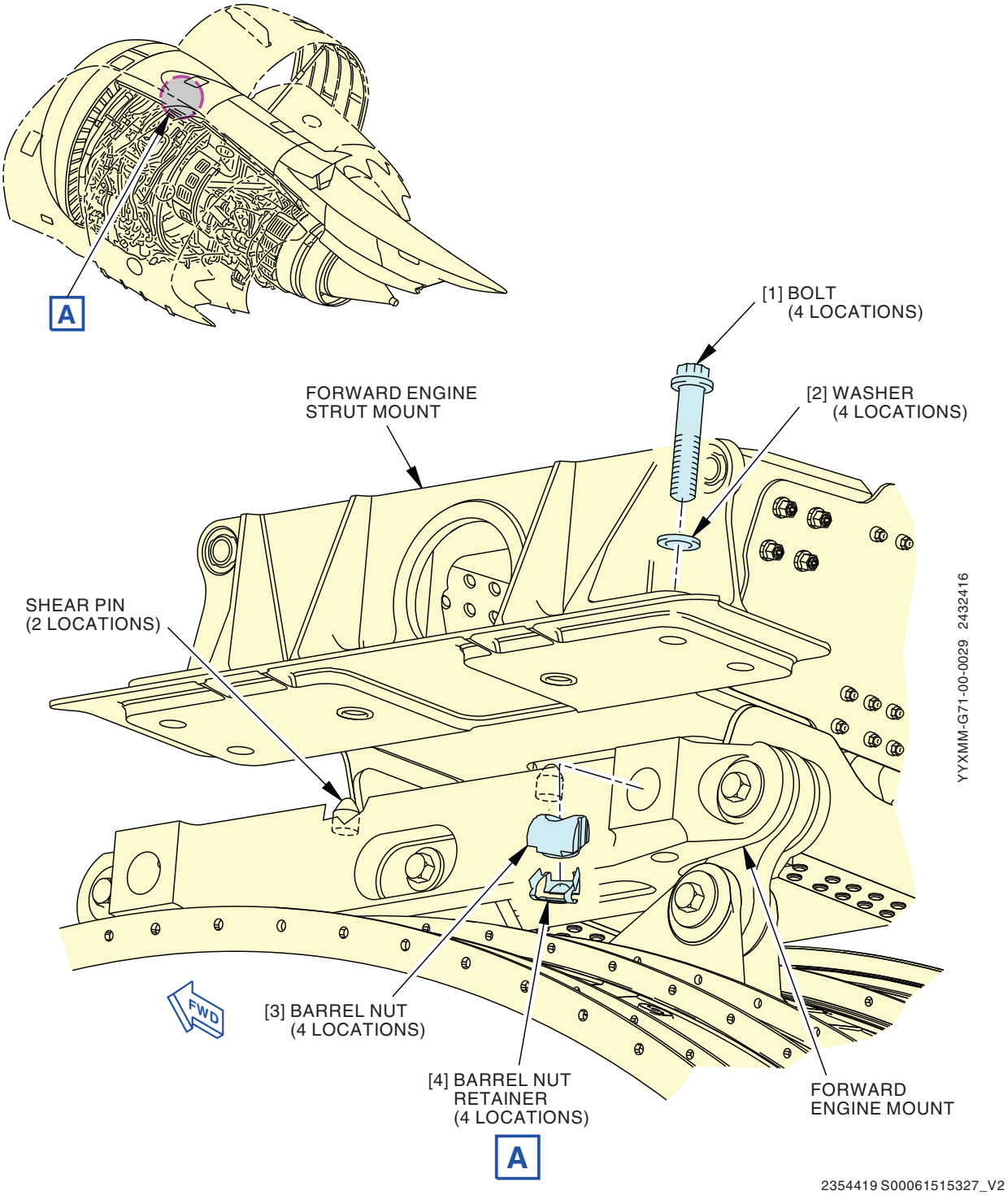
F. Put the Airplane Back to Its Usual Condition

SUBTASK 71-21-00-420-001-G00

(1) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Engine Mount Inspection
Figure 601/71-21-00-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-21-00-211-801-G00

3. Forward Engine Mounts Detailed Inspection

(Figure 602)

A. General

(1) This task provides the instructions on how to do a detailed inspection of the forward engine mount.

(2) This task is used when the engine is off-wing.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-21-01-000-801-G00	Forward Engine Mount Removal (P/B 401)
71-21-01-400-801-G00	Forward Engine Mount Installation (P/B 401)

C. Prepare for the Inspection

SUBTASK 71-21-00-020-002-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

D. Forward Engine Mounts Detailed Inspection

SUBTASK 71-21-00-211-001-G00

(1) Do a check of the component attach points for worn areas.

NOTE: If you find worn areas on the forward engine mount, refer to Component Maintenance Manual for wear limits and repair instructions.

(a) Examine the barrel nuts in the forward engine mount to make sure that they are free of damage.

(b) Examine the pin in the forward engine mount to make sure that they are free of damage.

(c) Examine the shear pins are free of damage.

(d) Examine the tension bolts are free of damage.

(e) Examine the forward engine mount left and right side links are free of damage.

(f) Make sure that they are free of damage and attached correctly.

(g) Examine the areas where the forward lower engine mount attaches to the engine fan case.

SUBTASK 71-21-00-211-002-G00

(2) If you find the component attach points are more than the wear limits, do these steps that follow:

(a) Replace the forward engine mount component, if it is necessary.

1) Do this task: TASK 71-21-01-000-801-G00.

2) Do this task: TASK 71-21-01-400-801-G00.

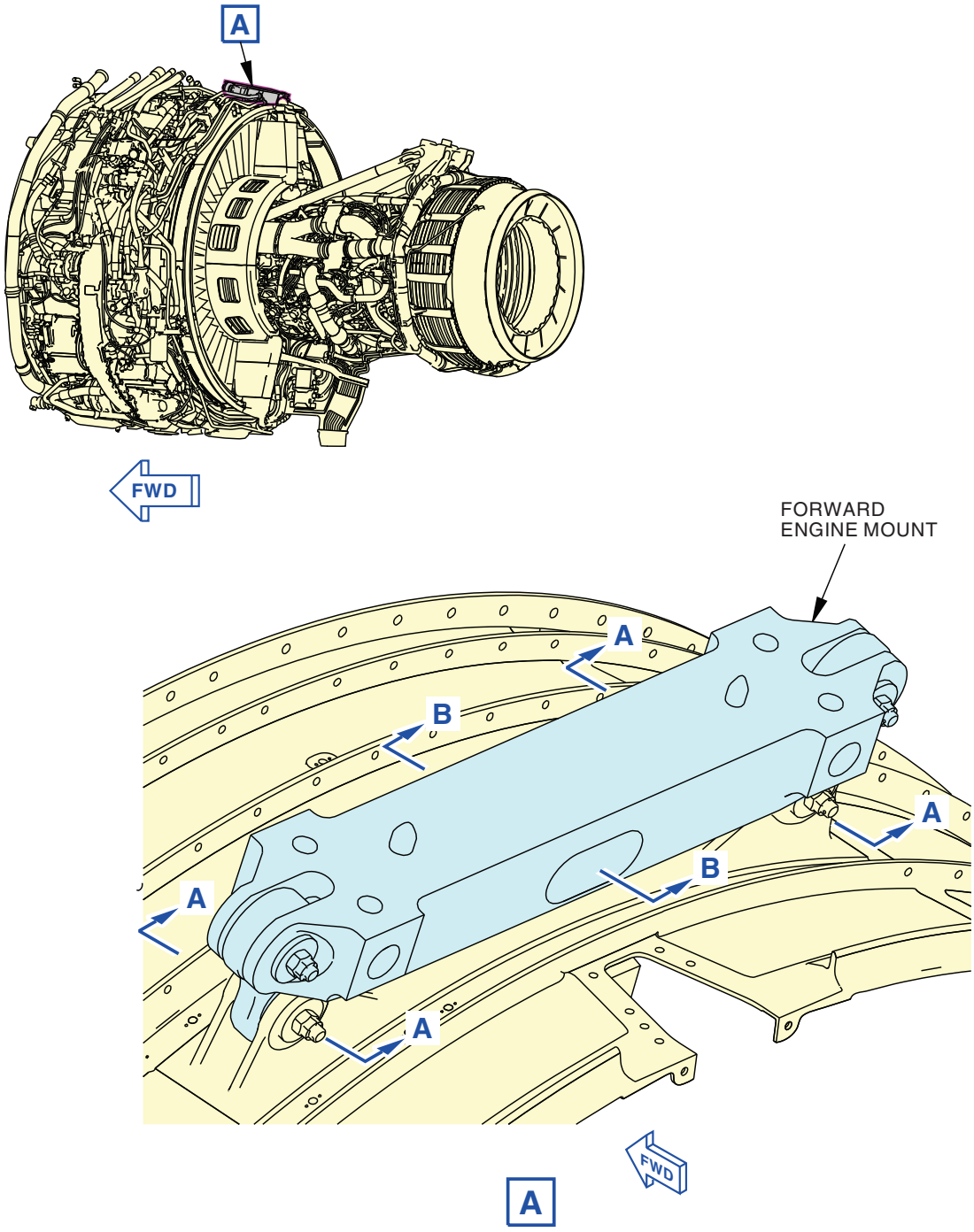
E. Put the Airplane Back to Its Usual Condition

SUBTASK 71-21-00-420-002-G00

(1) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

END OF TASK

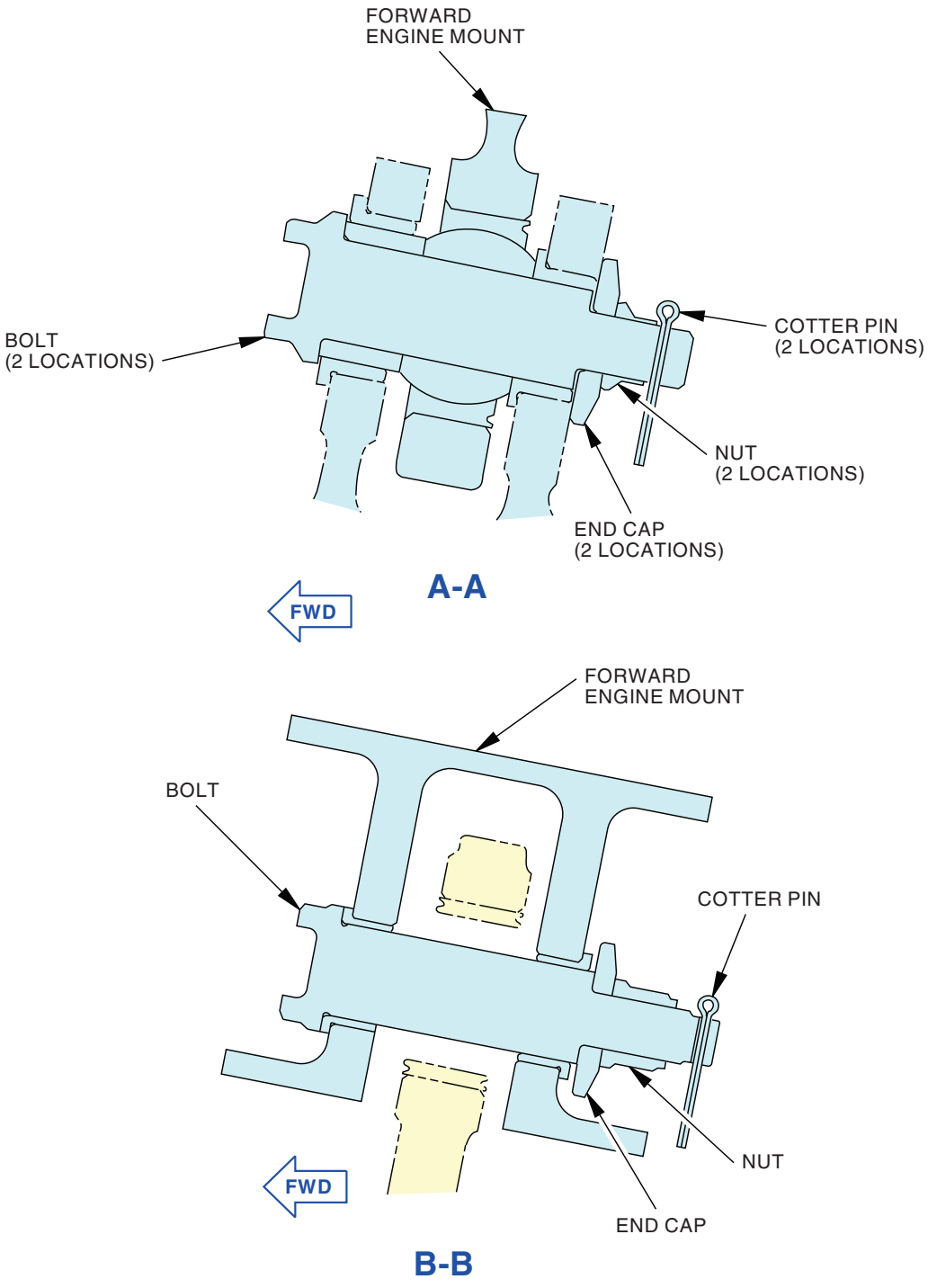
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354420 S00061515329_V2

Forward Engine Mount Inspection
Figure 602/71-21-00-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354421 S00061515330_V2

Forward Engine Mount Inspection
Figure 602/71-21-00-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-21-00-200-802-G00

4. Aft Engine Mount Visual Inspection

(Figure 603)

A. General

(1) This task provides the instructions on how to do a visual inspection of the aft engine mount.

(2) This task is used when the engine is off-wing.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-21-02-000-801-G00	Aft Engine Mount Removal (P/B 401)
71-21-02-400-801-G00	Aft Engine Mount Installation (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Inspection

SUBTASK 71-21-00-020-003-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

E. Aft Engine Mount Visual Inspection

SUBTASK 71-21-00-210-001-G00

(1) Examine the aft engine mount for these signs of damage:

(a) Corrosion is not permitted.

(b) Cracks are not permitted.

(c) Nicks are not permitted.

(d) Dents are not permitted.

(e) Scratches are not permitted.

(f) Bushing movement is not permitted.

(g) Damage on the tension bolt is not permitted.

(h) Damage to the side link assembly is not permitted.

(i) Damage to the shear pin is not permitted.

(j) Damage to the bearing is not permitted.

(k) Cracks to the bearing in the side link assembly is not permitted.

1) Cracks to the bearing ball are not permitted.

2) Seized bearing is not permitted.

3) Cracks in bearing are not permitted.

4) Bearing movement is not permitted.

5) All other types of damage to the bearing are not permitted.

(l) Fretting is not permitted.

NOTE: Fretting is caused when two adjacent pieces of metal rub one another.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) Pickup is not permitted.

NOTE: Pickup occurs when material is moved from one surface to a surface which it touches.

(n) High metal is not permitted.

NOTE: High metal is metal which is pushed above the surface adjacent to scratches or other damage.

SUBTASK 71-21-00-210-002-G00

(2) Examine the aft engine mount for worn areas.

(a) If you find signs of wear on the aft engine mount, do this task: TASK 71-21-00-211-802-G00.

SUBTASK 71-21-00-210-003-G00

(3) If damage is found on the aft engine mount, refer to the Structure Repair Manual for the allowable damage limits.

(a) Replace the aft engine mount, if it is more than the allowable damage limits.

1) Do this task: TASK 71-21-02-000-801-G00.

2) Do this task: TASK 71-21-02-400-801-G00.

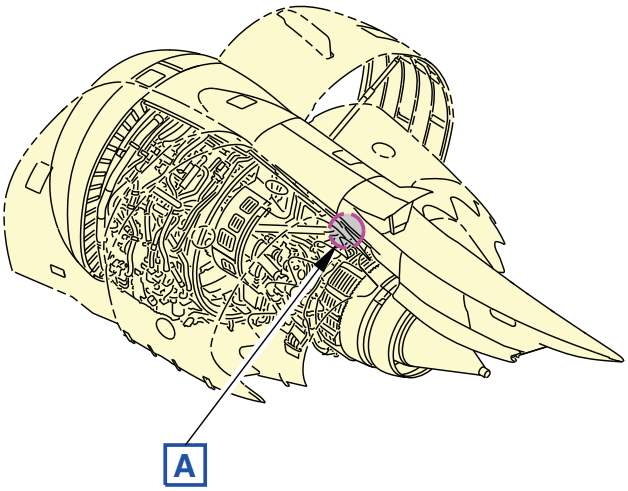
F. Put the Airplane Back to Its Usual Condition

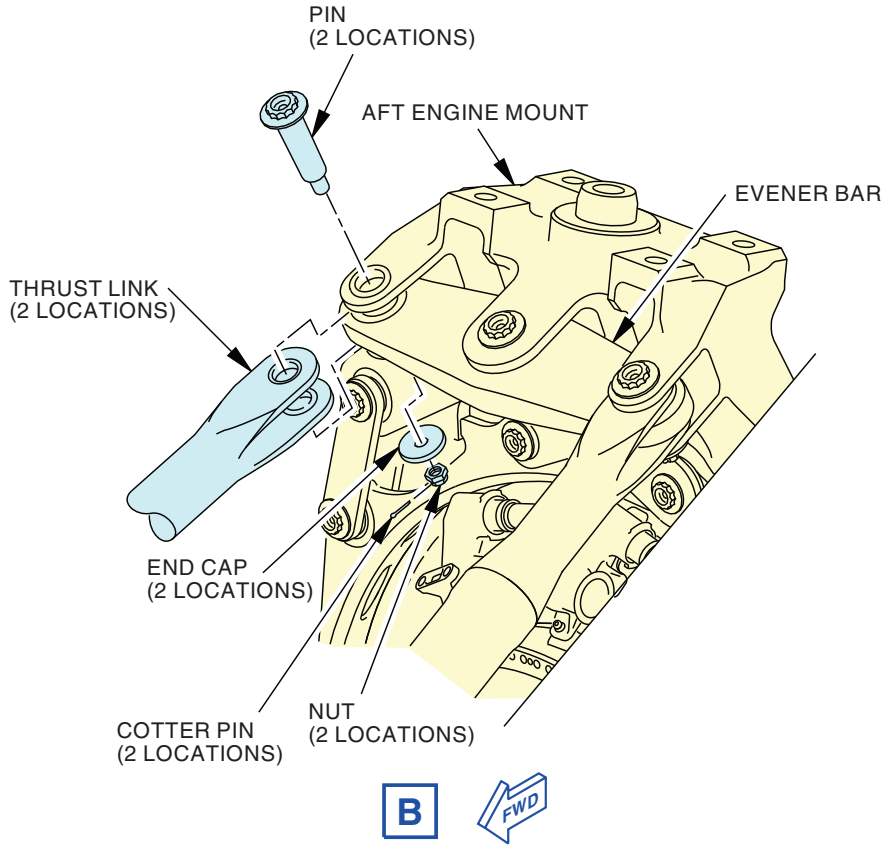
SUBTASK 71-21-00-420-003-G00

(1) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2354422 S00061515332_V2

Aft Engine Mount Inspection
Figure 603/71-21-00-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-21-00-211-802-G00

5. Aft Engine Mount Detailed Inspection

(Figure 604)

A. General

(1) This task gives the instructions to do a detailed inspection of the aft engine mount.

(2) This task is used when the engine is off-wing.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-21-02-000-801-G00	Aft Engine Mount Removal (P/B 401)
71-21-02-400-801-G00	Aft Engine Mount Installation (P/B 401)

C. Prepare for the Inspection

SUBTASK 71-21-00-020-004-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

D. Aft Engine Mounts Detailed Inspection

SUBTASK 71-21-00-211-003-G00

(1) Do a check of the component attach points for worn areas.

NOTE: If you find worn areas on the aft engine mount, refer to Component Maintenance Manual for wear limits and repair instructions.

(a) Examine the barrel nuts in the aft engine mount to make sure that they are free of damage.

(b) Examine the pin in the aft engine mount to make sure that they are free of damage.

(c) Examine the bearing and bearing balls are free of damage.

(d) Examine the bushings are free of damage.

(e) Examine the bolts are free of damage.

(f) Examine the shoulder bolts are free of damage.

(g) Examine the even bars are free of damage.

(h) Examine the aft engine mount side links are free of damage.

(i) Examine the fittings are free of damage.

(j) Make sure that they are free of damage and attached correctly.

(k) Examine the areas where the aft engine mount attaches to the engine exhaust outer case.

SUBTASK 71-21-00-211-004-G00

(2) If you find the component attach points are more than the wear limits, do these steps that follow:

(a) Replace the aft engine mount component, if it is necessary.

(b) Do this task: Aft Mount Removal (TASK 71-21-02-000-801-G00).

(c) Do this task: Aft Mount Installation (TASK 71-21-02-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

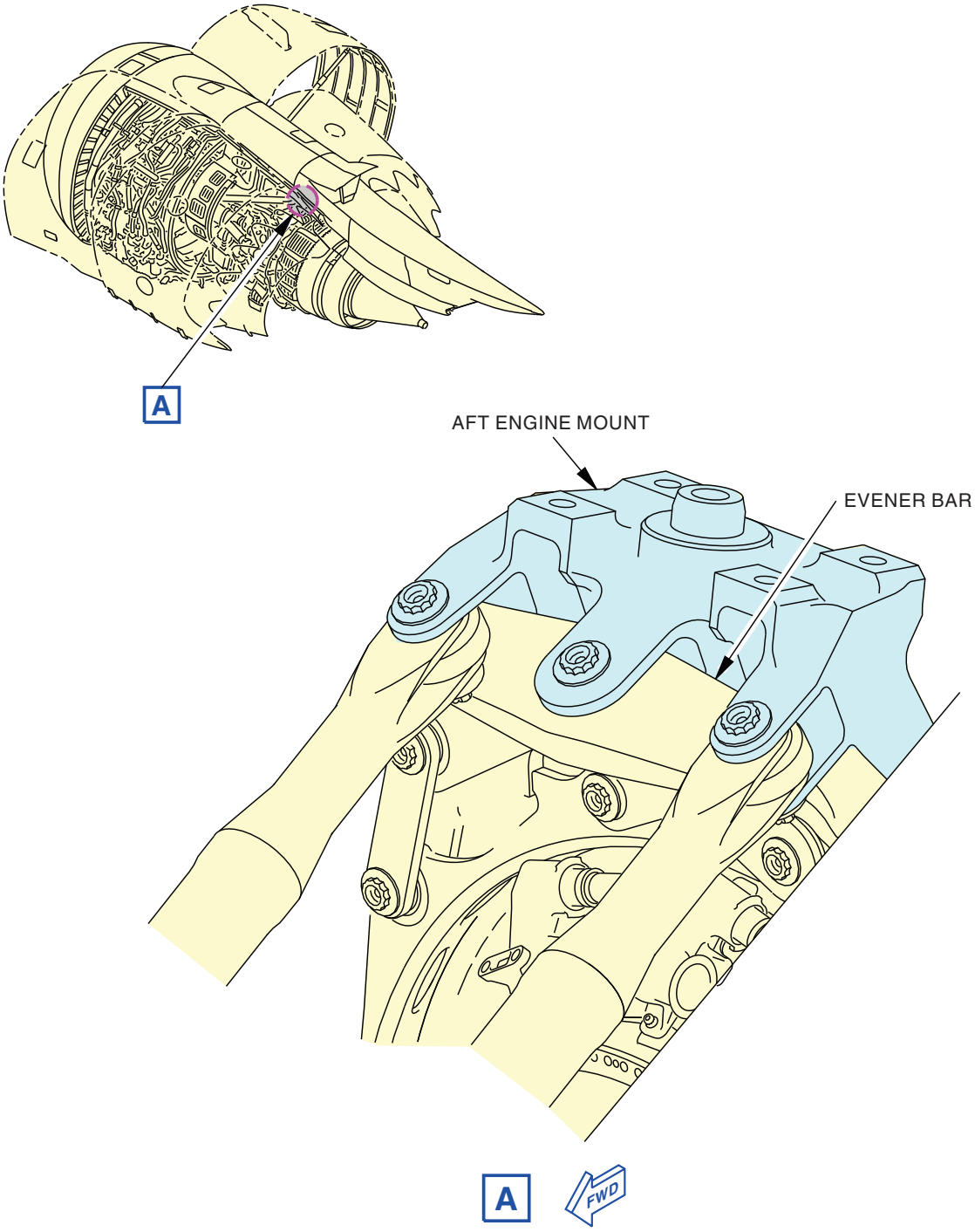
E. Put the Airplane Back to Its Usual Condition

SUBTASK 71-21-00-420-004-G00

(1) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

——— END OF TASK ———

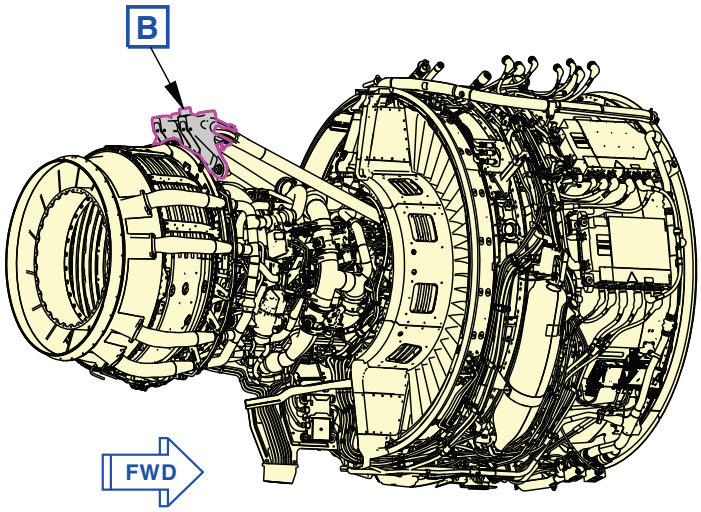
ECCN 9E991 BOEING PROPRIETARY - See title page for details



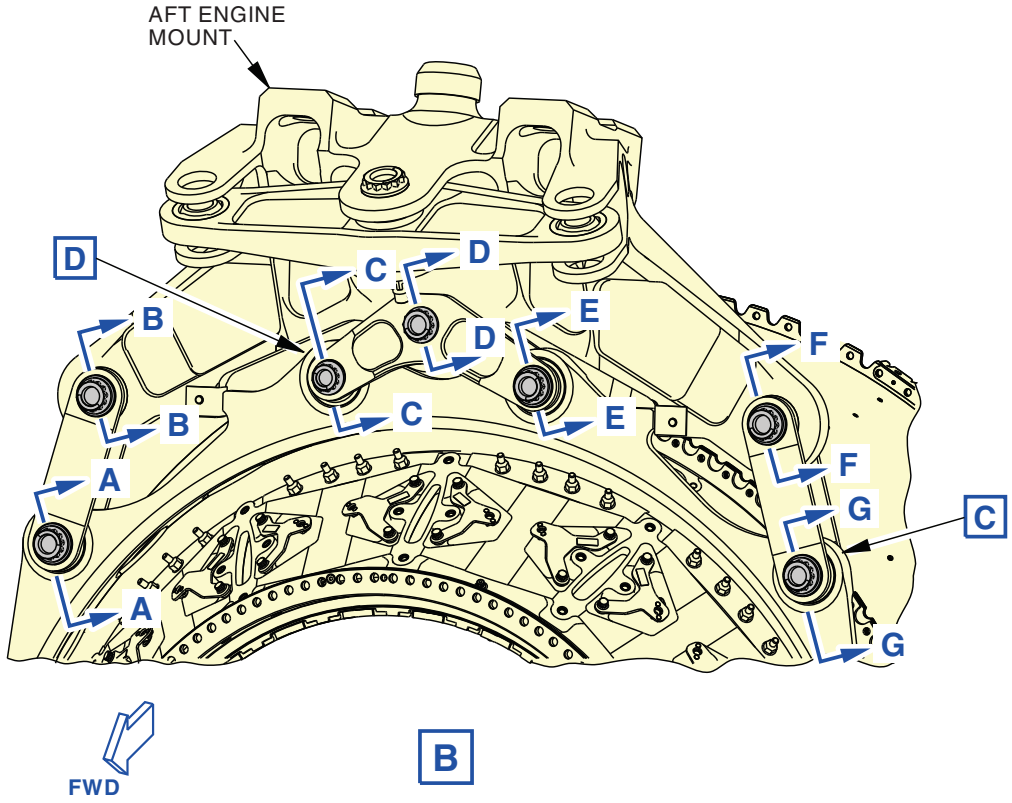
2354423 S00061515334_V1

Aft Engine Mount Inspection
Figure 604/71-21-00-990-804-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TYPICAL)

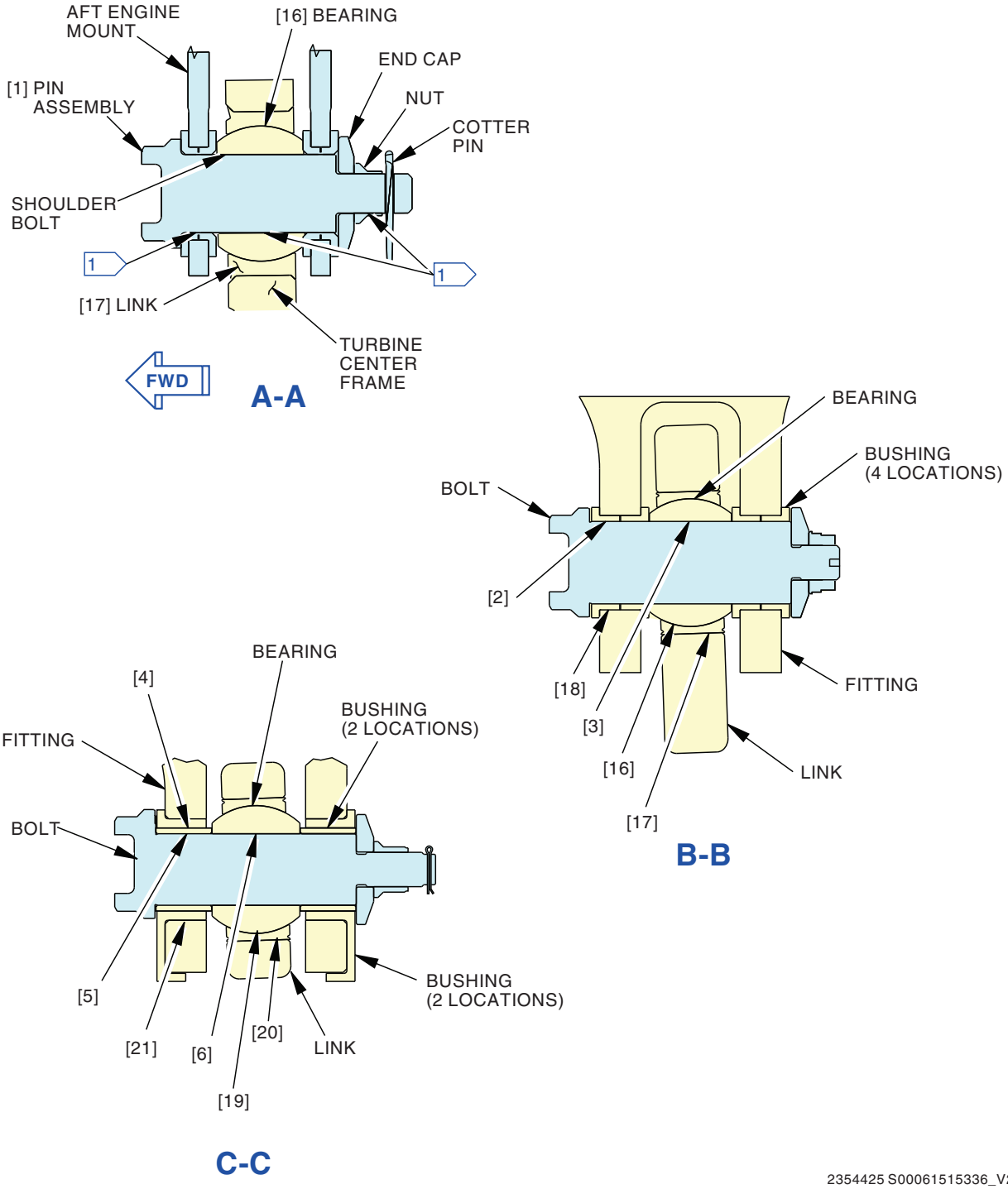


1 APPLY ANTI-SEIZE COMPOUND AS INDICATED.

2354424 S00061515335_V2

Aft Engine Mount Inspection
Figure 604/71-21-00-990-804-G00 (Sheet 2 of 6)

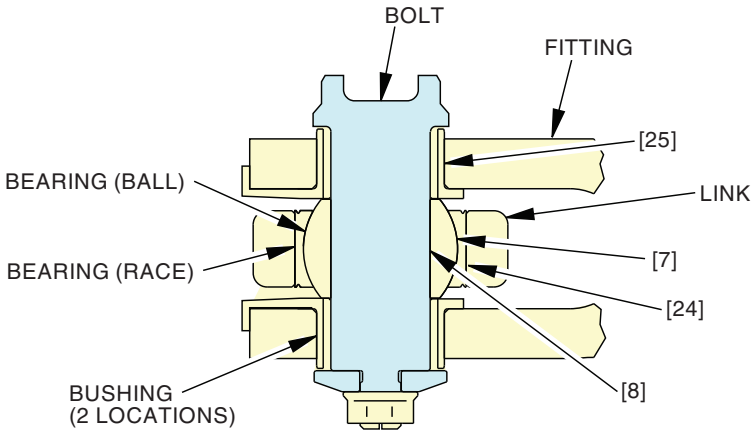
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354425 S00061515336_V2

Aft Engine Mount Inspection
Figure 604/71-21-00-990-804-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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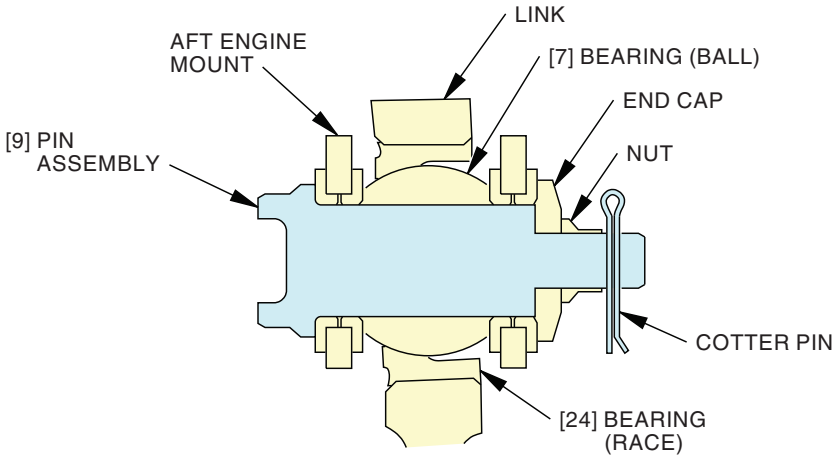
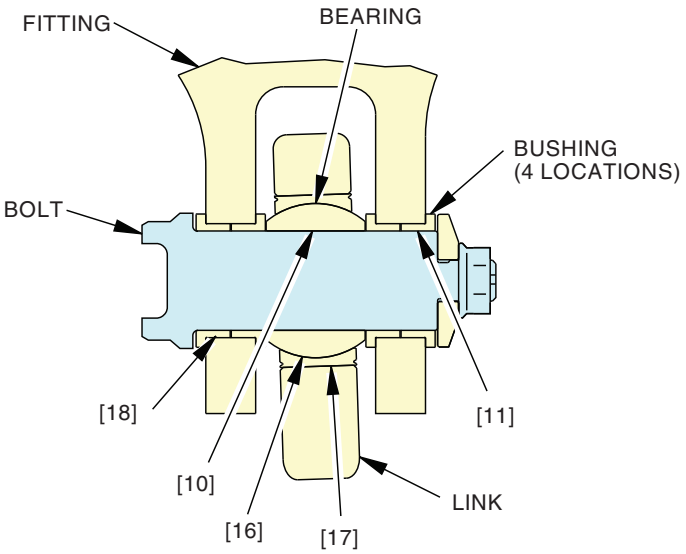


E-E

2354426 S00061515337_V1

Aft Engine Mount Inspection
Figure 604/71-21-00-990-804-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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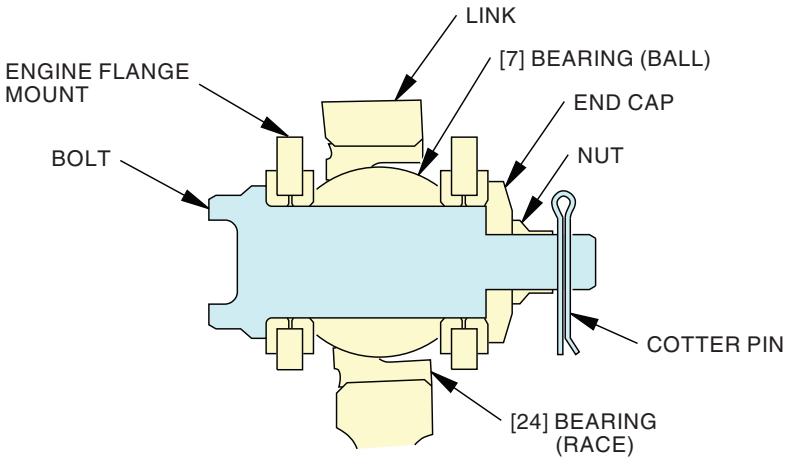
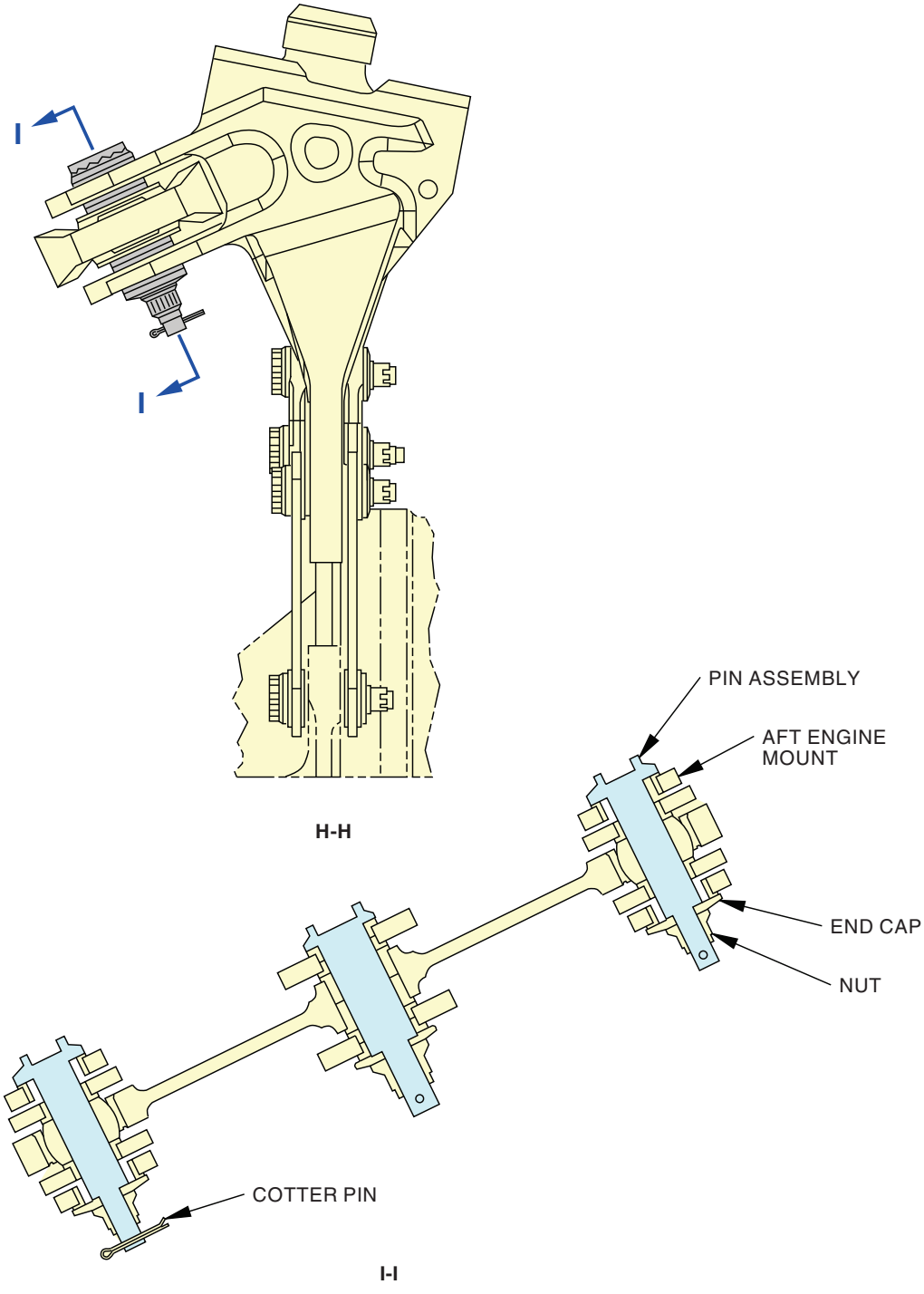


G-G

2354427 S00061515338_V1

Aft Engine Mount Inspection
Figure 604/71-21-00-990-804-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Engine Mount Inspection
Figure 604/71-21-00-990-804-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FORWARD ENGINE MOUNT - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Forward engine mount removal

(2) Forward engine mount installation.

TASK 71-21-01-000-801-G00

2. Forward Engine Mount Removal

(Figure 401)

A. General

(1) This task provides the instructions to remove the forward engine mount.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 71-21-01-010-001-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00..

SUBTASK 71-21-01-490-001-G00

(2) Move the engine and strut access platform, STD-653 into its position.

F. Forward Engine Mount Removal

SUBTASK 71-21-01-020-001-G00

(1) Remove the forward engine mount [1] as follows:

(a) Remove the left and right attachments of the forward engine mount [1] to the mount links:

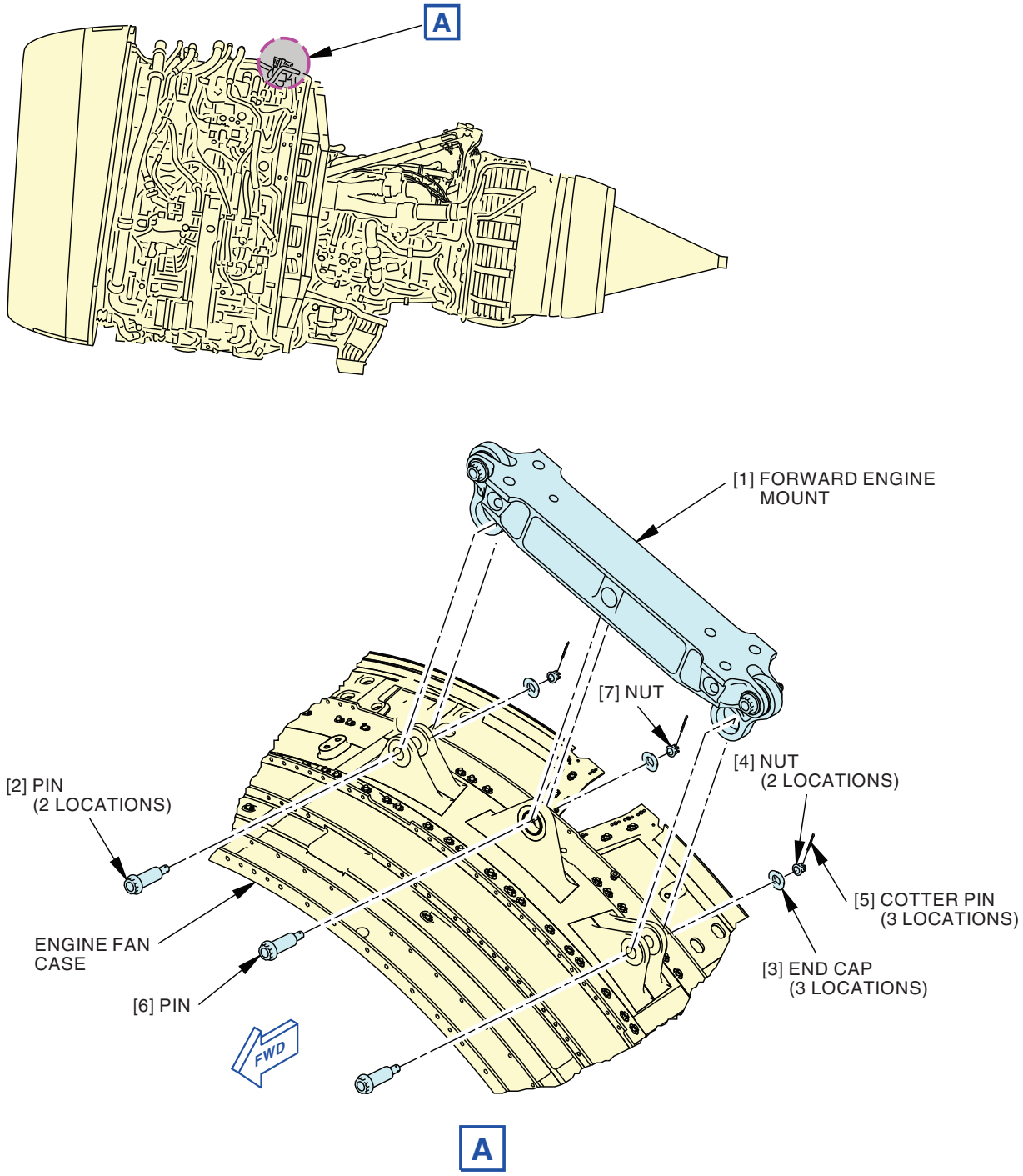
1) Remove the two cotter pins [5], the two nuts [4], the two end caps [3] and the two pins [2] that attach the forward engine mount [1] to the mount links.

(b) Remove the center attachment of the forward engine mount [1] to the fan case flange:

1) Remove the cotter pin [5], the nut [7], the end cap [3] and the pin [6] that attach the forward engine mount [1] to the fan case flange.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Engine Mount Installation
Figure 401/71-21-01-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-21-01-400-801-G00

3. Forward Engine Mount Installation

(Figure 401)

A. General

(1) This task provides the instructions to install the forward engine mount.

B. References

Reference	Title
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-653	Platform - Engine and Strut Access

D. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward engine mount	71-21-01-01-045	PTF ALL
5	Cotter pin	71-21-01-01-005	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Forward Engine Mount Installation

SUBTASK 71-21-01-420-001-G00

(1) Install the forward engine mount [1] as follows:

(a) Apply Pure Nickel Special compound, D00006 on these components:

1) The shanks and threads of the pins [2] and pin [6]

2) Below the pin heads of the pins [2] and pin [6]

3) The bore of the bushings on the fan case

4) The flange face of the bushings

5) The bore and mating surface of the spherical bearings.

(b) Install the two pins [2], two end caps [3], and the two nuts [4], that attach the forward engine mount [1] to the left and the right mount links.

1) Install the pins [2] with the head forward.

NOTE: Make sure that the flat side of the end cap is against the shoulder of the bolt.

(c) Install the pin [6], end cap [3], nut [7], that attach the forward engine mount [1] to the center fan case flange.

1) Install the pin [6] with the head forward.

NOTE: Make sure that the flat side of the end cap is against the shoulder of the bolt.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Tighten the nuts [4] and nut [7] to 170 in-lb (19 N·m) - 380 in-lb (43 N·m).

(e) Install the new cotter pins [5] in the nuts [4] and nut [7].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 71-21-01-090-001-G00

(1) Move the engine and strut access platform, STD-653 away from the engine.

SUBTASK 71-21-01-410-001-G00

(2) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AFT ENGINE MOUNT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aft engine mount

(2) An installation of the aft engine mount.

TASK 71-21-02-000-801-G00

2. Aft Engine Mount Removal

(Figure 401, Figure 402)

A. General

(1) This task provides the instructions on how to remove the aft engine mount.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-21-04-000-801-G00	Thrust Links Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2107	Fixture - Lift, Engine Aft Mount Part #: C71024-20 Supplier: 81205
STD-653	Platform - Engine and Strut Access
STD-1194	Crane - Lift, 250 Lb (113 kg) Capacity, Lift Range 20 to 60 Inches (508-1524 Millimeters)
STD-7220	Hoist - Portable, 200-lb Capacity

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 71-21-02-010-001-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

SUBTASK 71-21-02-490-001-G00

(2) Move the engine and strut access platform, STD-653, into its position.

SUBTASK 71-21-02-010-002-G00

(3) Do this task: Thrust Links Removal, TASK 71-21-04-000-801-G00.

F. Aft Engine Mount Removal

SUBTASK 71-21-02-490-002-G00

(1) Attach the fixture, SPL-2107, to the top of the aft engine mount [1] as follows:

(a) Install the "SHORT LEG" to the front hole on the left side as marked on the strap.

(b) Install the "LONG LEG" to the aft hole on the right side as marked on the strap.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Keep the straps loose.

SUBTASK 71-21-02-080-002-G00

(2) Support the aft engine mount [1], do the following steps:

(a) Connect the portable hoist, STD-7220, to the straps.

(b) Attach the portable hoist, STD-7220, to the 250 lb (113 kg) capacity crane, STD-1194.

(c) Adjust the portable hoist, STD-7220, to hold the weight of the aft engine mount [1].

SUBTASK 71-21-02-480-001-G00

CAUTION

INSTALL THE PROTECTIVE COVERS TO ALL TOP OPENINGS OF THE TURBINE CENTER FRAME BELOW THE AFT ENGINE MOUNT. AFT ENGINE MOUNT PARTS (PINS, CAPS, NUTS) CAN FALL INTO THE ENGINE. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

(3) Install the protective covers to all top openings of the turbine center frame.

SUBTASK 71-21-02-020-005-G00

WARNING

BE CAREFUL WHEN YOU MOVE THE COMPONENT. THE COMPONENT IS HEAVY. INJURIES TO PERSONS CAN OCCUR.

(4) Remove the aft engine mount [1] as follows:

(a) Remove the two cotter pins [4], two end caps [7], two nuts [8], and two pins [9] (View B, Figure 401).

(b) Remove the cotter pin [4], end cap [5], nuts [6], and pin [10] (View B, Figure 401).

(c) Remove the cotter pin [4], end cap [2], nut [3], and pin [11] (View B, Figure 401).

SUBTASK 71-21-02-080-004-G00

(5) Remove the protective covers from the openings of the turbine center frame.

SUBTASK 71-21-02-080-003-G00

WARNING

BE CAREFUL WHEN YOU MOVE THE COMPONENT. THE COMPONENT IS HEAVY. INJURIES TO PERSONS CAN OCCUR.

(6) Lift the aft engine mount [1] slowly up and away of the engine.

SUBTASK 71-21-02-020-003-G00

(7) Remove the fixture, SPL-2107, from the aft engine mount [1].

(a) Lower the portable hoist, STD-7220, to loosen the straps.

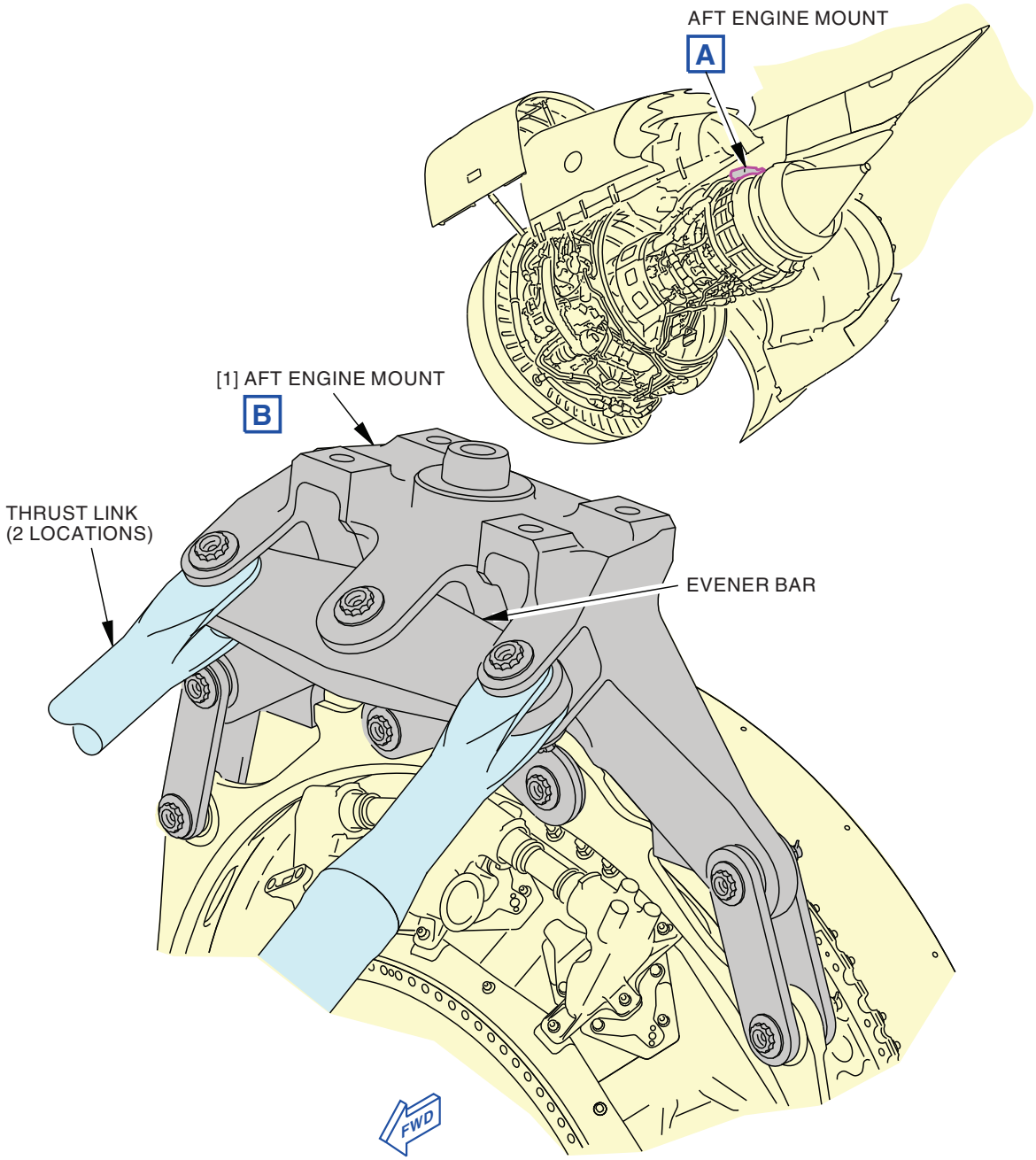
(b) Remove the "SHORT LEG" from the front hole on the left side as marked on the strap.

(c) Remove the "LONG LEG" from the aft hole on the right side as marked on the strap.

(d) Move the fixture, SPL-2107, away from the engine.

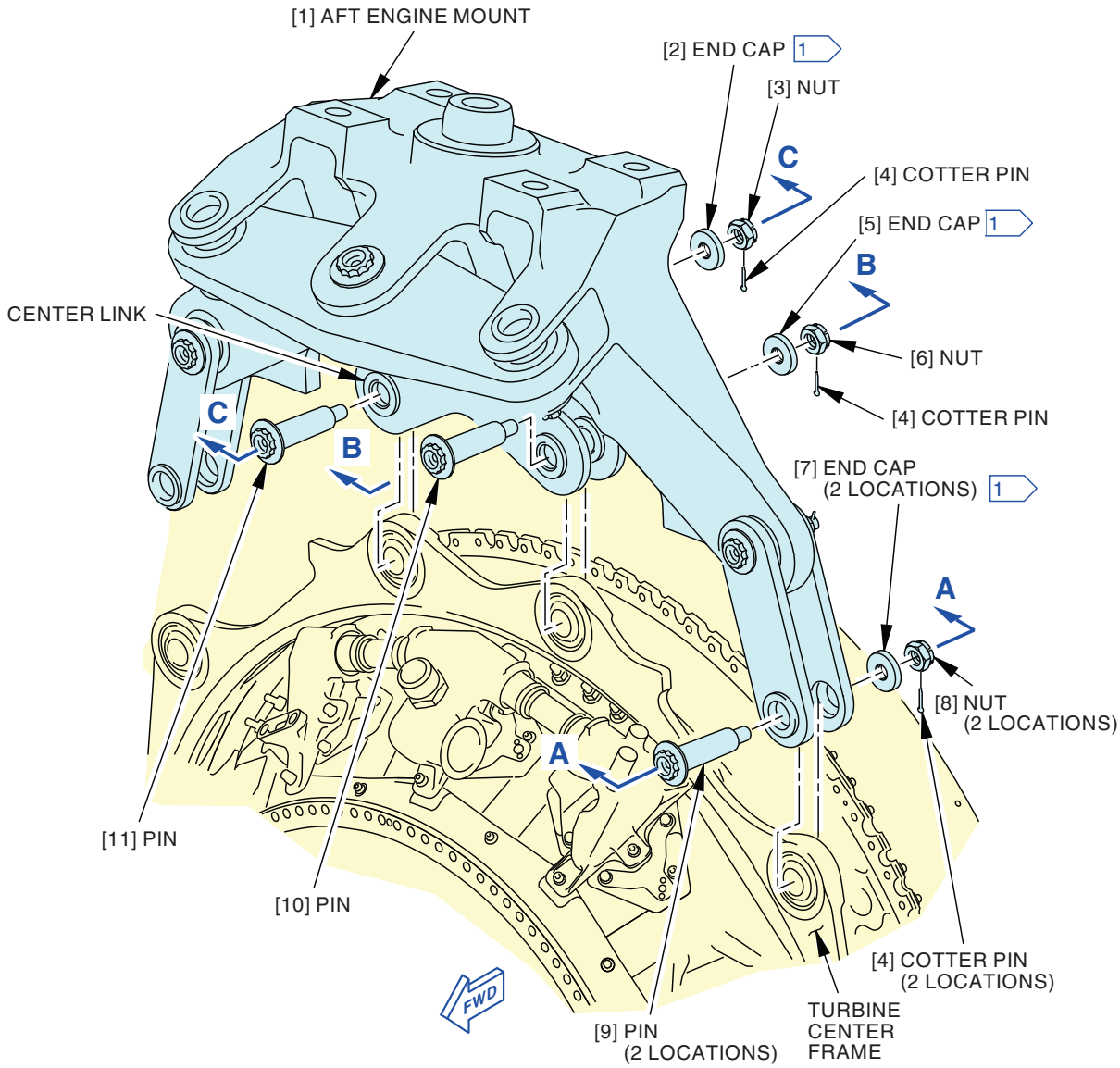
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Engine Mount Installation
Figure 401/71-21-02-990-802-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AFT ENGINE MOUNT

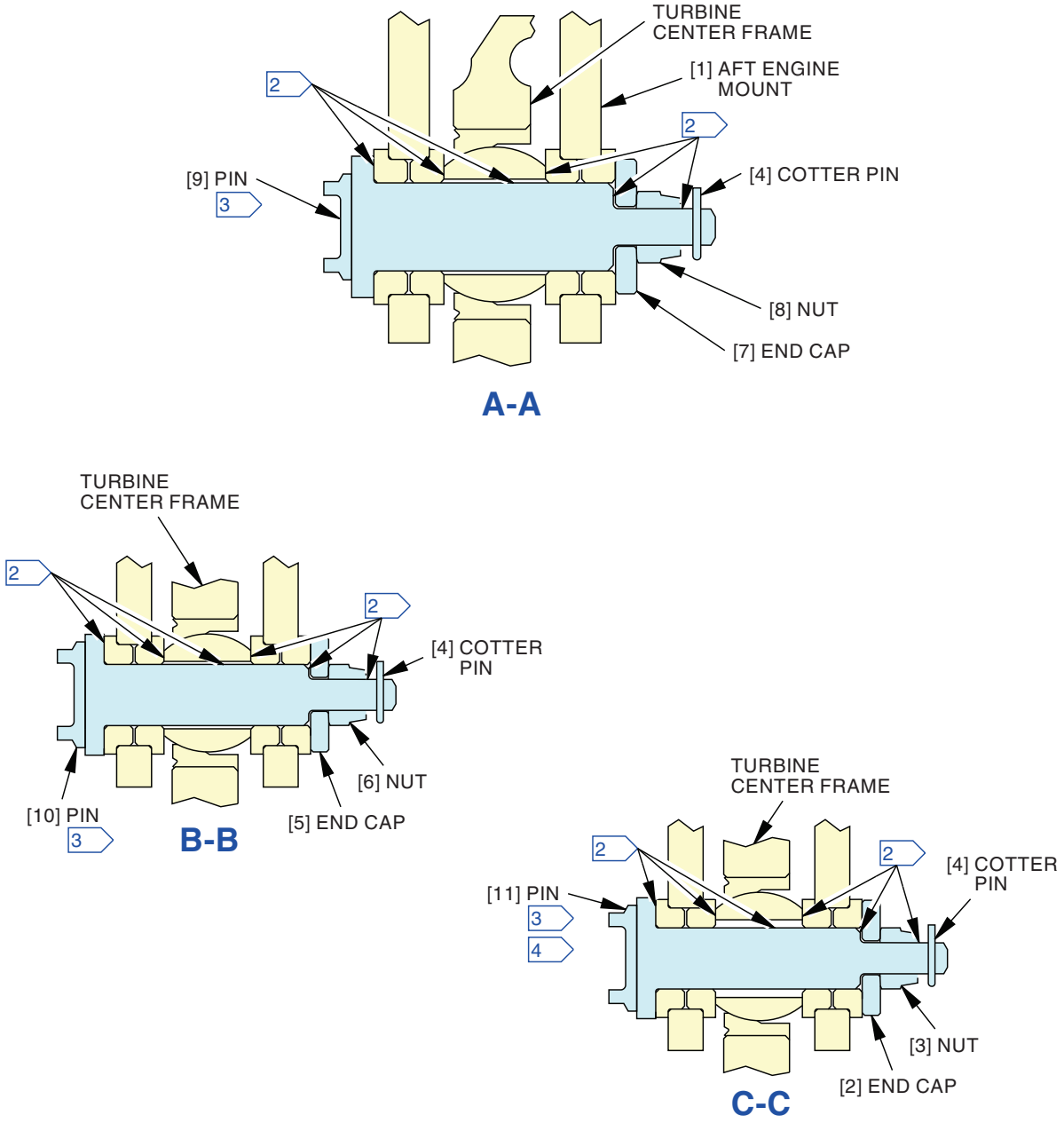
B

1 INSTALL THE END CAP WITH THE FLAT SIDE AGAINST THE SHOULDER OF THE PIN.

2427961 S0000561682_V7

Aft Engine Mount Installation
Figure 401/71-21-02-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

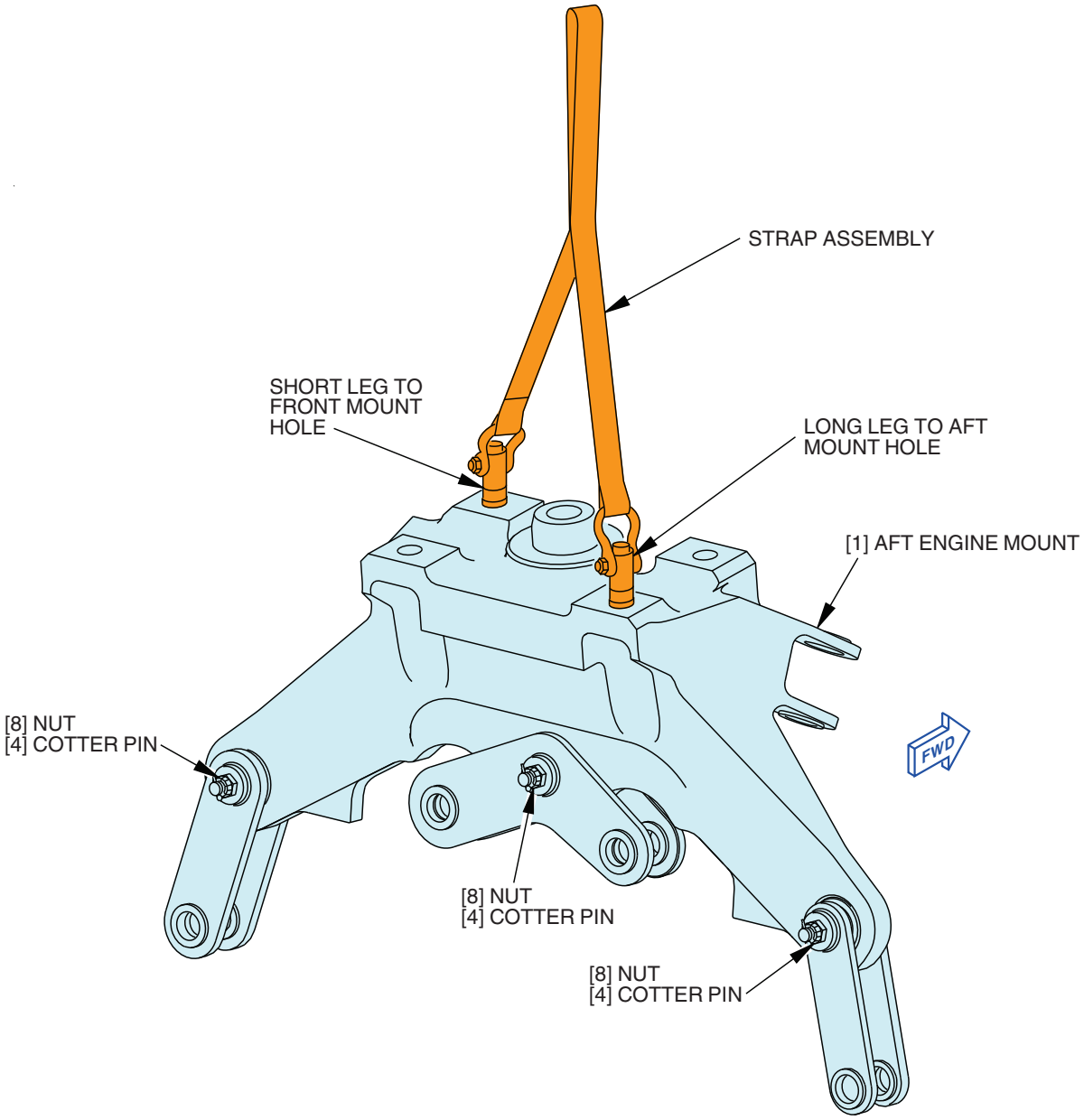


2 APPLY ANTI-SEIZE COMPOUND AS INDICATED.
3 PIN CAN TURN WITH HAND PRESSURE AFTER INSTALLATION.
4 CATCHER BEARING BALL IS OVERSIZED AND NOT CO-AXIAL TO PIN AT ROOM TEMPERATURE.

2427431 S0000558936_V7

Aft Engine Mount Installation
Figure 401/71-21-02-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Engine Mount Installation Tool
Figure 402/71-21-02-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-21-02-400-801-G00

3. Aft Engine Mount Installation

(Figure 401, Figure 402)

A. General

(1) This task provides the instructions on how to install the aft engine mount.

B. References

Reference	Title
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-21-04-400-801-G00	Thrust Links Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2107	Fixture - Lift, Engine Aft Mount Part #: C71024-20 Supplier: 81205
STD-653	Platform - Engine and Strut Access
STD-1194	Crane - Lift, 250 Lb (113 kg) Capacity, Lift Range 20 to 60 Inches (508-1524 Millimeters)
STD-7220	Hoist - Portable, 200-lb Capacity

D. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Aft engine mount	71-21-02-01-035	PTF ALL
4	Cotter pin	71-21-02-01-005	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Prepare for the Installation

SUBTASK 71-21-02-210-001-G00

(1) Do a check of the aft engine mount assembly for the correct installation of the evener bar.

(a) Make sure that the evener bar is installed with the slotted entry sides of the bearings facing up.

SUBTASK 71-21-02-490-003-G00

(2) Attach the fixture, SPL-2107, to the top of the aft engine mount [1] as follows:

(a) Install the "SHORT LEG" to the front hole on the left side as marked on the strap.

(b) Install the "LONG LEG" to the aft hole on the right side as marked on the strap.

(c) Connect the hoist to the straps.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Attach the portable hoist, STD-7220, to the 250 lb (113 kg) capacity crane, STD-1194.

(e) Adjust the portable hoist, STD-7220, to hold the weight of the aft engine mount [1].

(f) Lift the aft engine mount [1] slowly to the top of the turbine center frame.

H. Aft Engine Mount Installation

SUBTASK 71-21-02-420-001-G00

WARNING BE CAREFUL WHEN YOU MOVE THE COMPONENT. THE COMPONENT IS HEAVY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Install the aft engine mount [1] on the turbine center frame as follows:

(a) Apply Pure Nickel Special compound, D00006, to these components:

1) The shanks of the pins
2) Below the pin heads of the pins
3) The bore of the bushings
4) The flange face of the bushings
5) The bore of the spherical bearing balls
6) The flat surface of the spherical bearing balls.

CAUTION INSTALL THE PROTECTIVE COVERS TO ALL TOP OPENINGS OF THE TURBINE CENTER FRAME BELOW THE AFT ENGINE MOUNT. AFT ENGINE MOUNT PARTS (PINS, CAPS, NUTS) CAN FALL INTO THE ENGINE. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

(b) Install the protective covers to all top openings of the turbine center frame.

(c) Align the aft engine mount [1] outer connections to the turbine center frame flanges.

(d) Install the two pins [9], two end caps [7], and two nuts [8] at the two outer locations.

1) Make sure that the end cap [7] with the flat side are installed against the shoulder of the pins.

(e) Align the aft engine mount [1] inner connections to the turbine center frame flanges.

(f) Install the pin [10], end cap [5], and nut [6] from the aft engine mount [1] at the left inner location (view looks forward).

1) Make sure that the end cap [5] with the flat side is installed against the shoulder of the pin.

(g) Install the pin [11], end cap [2], and nut [3] from the aft engine mount [1] at the right inner location (view looking forward).

1) Make sure that the end cap [2] with the flat side is installed against the shoulder of the pin.

(h) Tighten the nut [6] to 170 in-lb (19.2 N·m) - 380 in-lb (42.9 N·m) at the inner location (view looks forward).

1) Install the new cotter pin [4] on the nut [6].

(i) Tighten the nut [3] to 529 in-lb (60 N·m) - 621 in-lb (70.2 N·m).

1) Install the new cotter pin [4] on the nut [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Tighten the two outer nuts [8] to 170 in-lb (19.2 N·m) - 380 in-lb (42.9 N·m).

1) Install the new cotter pins [4] on the nuts [8].

SUBTASK 71-21-02-220-001-G00

(2) Check the upper fasteners of the center and outer links of the aft engine mount [1] as follows (Figure 402):

(a) Remove and discard the three cotter pins [4].

(b) Make sure that the three nuts [8] are tightened to 170 in-lb (19.2 N·m) - 380 in-lb (42.9 N·m).

(c) Install the three new cotter pins [4] on the three nuts [8].

SUBTASK 71-21-02-080-005-G00

(3) Remove the protective covers from the openings of the turbine center frame.

SUBTASK 71-21-02-080-001-G00

(4) Disconnect the fixture, SPL-2107, from the aft engine mount [1] as follows:

(a) Lower the portable hoist, STD-7220, to loosen the straps.

(b) Remove the "SHORT LEG" from the front hole on the left side as marked on the strap.

(c) Remove the "LONG LEG" from the aft hole on the right side as marked on the strap.

(d) Move the fixture, SPL-2107, away from the engine.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 71-21-02-090-001-G00

(1) Remove the fixture, SPL-2107, from the portable hoist, STD-7220.

SUBTASK 71-21-02-410-002-G00

(2) Do this task: Thrust Links Installation, TASK 71-21-04-400-801-G00.

SUBTASK 71-21-02-090-002-G00

(3) Remove the engine and strut access platform, STD-653.

SUBTASK 71-21-02-410-001-G00

(4) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE MOUNT BOLTS AND NUTS - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Engine mount bolts and nuts inspection.

TASK 71-21-03-200-801-G00

2. Engine Mount Bolts and Nuts Inspection

(Figure 601)

A. General

(1) This task provides the instructions on how to do an inspection of the forward and aft engine mount bolts and barrel nuts.

(2) This task is used when the engine is off-wing.

(3) You must remove the engine mount bolts and barrel nuts to do the inspection.

(4) The inspection of the forward and aft engine mount bolts are the same.

(5) You must have a certification in fluorescent dye penetrant inspection techniques to do this task.

(6) Use this task, or an equivalent inspection procedure, to do a fluorescent dye penetrant system inspection to a sensitivity level of Level 3 - High Sensitivity, according to the classifications given by Aerospace Material Specification (AMS) 2644.

NOTE: To do the inspection of the bolts and nuts, you must comply with AMS 2644, Inspection Material - Penetrant, and Boeing Specification Support Standard (BSS) 7039, Liquid Penetrant Inspection.

B. Tools/Equipment

Reference	Description
STD-122	Brush
STD-442	Gun - Heat, 180 deg. F (82 deg. C) Maximum Output Temperature
STD-569	Light - Black, Ultraviolet

C. Consumable Materials

Reference	Description	Specification
B00074	Solvent - Degreasing	MIL-PRF-680 (Supersedes P-D-680)
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02036	Penetrant - Zyglo ZL-27A Fluorescent Post Emulsifiable Penetrant	AMS2644 Type I (Level 3 - High Sensitivity)
G50189 [CP3102]	Developer - Developer in Powder Form - Zyglo ZP 4B	
G50704 [C05-051]	Emulsifier - Zyglo ZE-4B Penetrant Emulsifier	AMS 2644

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Engine Mount Bolts and Barrel Nuts Inspection

SUBTASK 71-21-03-211-001-G00

(1) Do a visual check of the engine mount bolts [1] and bolts [5] for damage as follows:

(a) Use the solvent, B00074, and a brush, STD-122, to clean the bolts [1], the bolts [5], the barrel nuts [3], and the barrel nuts [7].

(b) Use forced air to dry the engine mount bolts and barrel nuts.

(c) Surface damage is not permitted.

(d) Cracks are not permitted.

(e) Damage threads are not permitted.

(f) Corrosion is not permitted.

SUBTASK 71-21-03-211-002-G00

(2) Examine the engine mount bolts [1], the bolts [5], the barrel nuts [3], and the barrel nuts [7] with the dye penetrant for damage as follows:

(a) Use the solvent, B00074, and a brush, STD-122, to clean the bolts and nuts.

(b) Use forced air to dry the engine mount bolts and barrel nuts.

NOTE: The engine mount bolts and barrel nuts must be fully dry before you apply the penetrant.

1) Use a 180 deg. F (82 deg. C) maximum output temperature heat gun, STD-442 to apply 140.0°F (60.0°C) air to dry the engine mount bolts and barrel nuts.

(c) Apply the Zyglo ZL-27A penetrant, G02036 to the engine mount bolts and barrel nuts.

NOTE: Make sure that all the surfaces of the engine mount bolts and barrel nuts are covered with the Zyglo ZL-27A penetrant, G02036.

(d) Let the Zyglo ZL-27A penetrant, G02036, stay on the surface of the engine mount bolts and barrel nuts for a minimum of 10 minutes.

(e) Apply the Zyglo ZE-4B emulsifier, G50704 [C05-051] to the engine mount bolts and barrel nuts.

(f) Remove the engine mount bolts and barrel nuts from the Zyglo ZE-4B emulsifier, G50704 [C05-051].

(g) Let the Zyglo ZE-4B emulsifier, G50704 [C05-051] drain from the engine mount bolts and barrel nuts for a minimum of 2 minutes.

(h) Flush the Zyglo ZE-4B emulsifier, G50704 [C05-051] and Zyglo ZL-27A penetrant, G02036 from the engine mount bolts and barrel nuts with water.

(i) Dry the engine mount bolts and barrel nuts.

NOTE: The engine mount bolts and barrel nuts must be fully dry before you apply the developer.

1) Use a 180 deg. F (82 deg. C) maximum output temperature heat gun, STD-442 to apply 140.0°F (60.0°C) air to dry the engine mount bolts and barrel nuts.

(j) Apply the powder developer - Zyglo ZP 4B, G50189 [CP3102] to the engine mount bolts and barrel nuts.

(k) Let the powder developer - Zyglo ZP 4B, G50189 [CP3102] stay on the engine mount bolts and barrel nuts for 10 minutes.

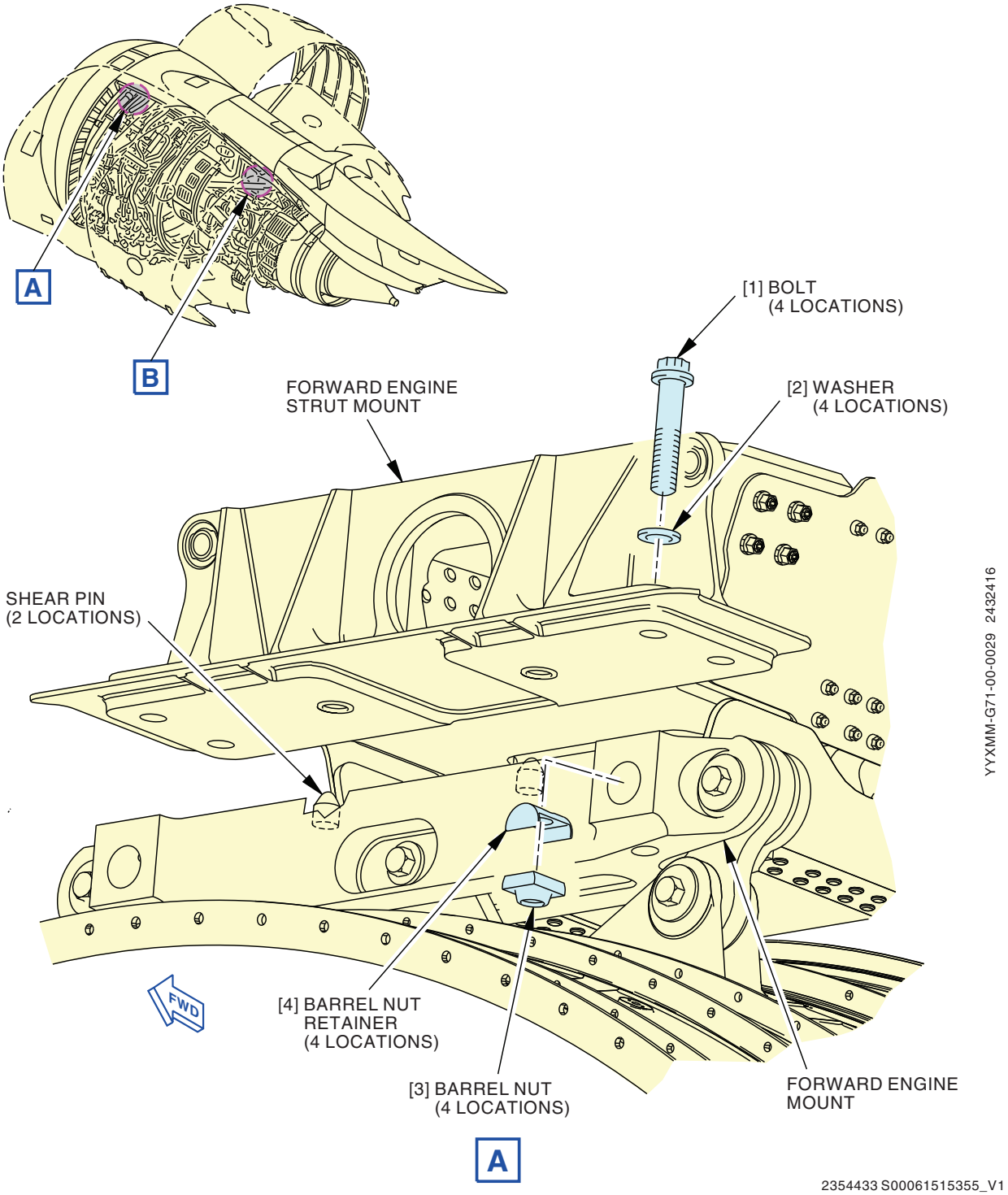
(l) Use a ultraviolet black light, STD-569 to examine the engine mount bolts and barrel nuts for cracks or other defects.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Cracks are not permitted.
2) If you do not find crack on the engine mount bolts and barrel nuts, clean the bolts and barrel nuts with clean water and brush, STD-122 for 2 - 3 minutes.
3) Use forced air or a clean dry cotton wiper, G00034 to dry the bolts and barrel nuts.

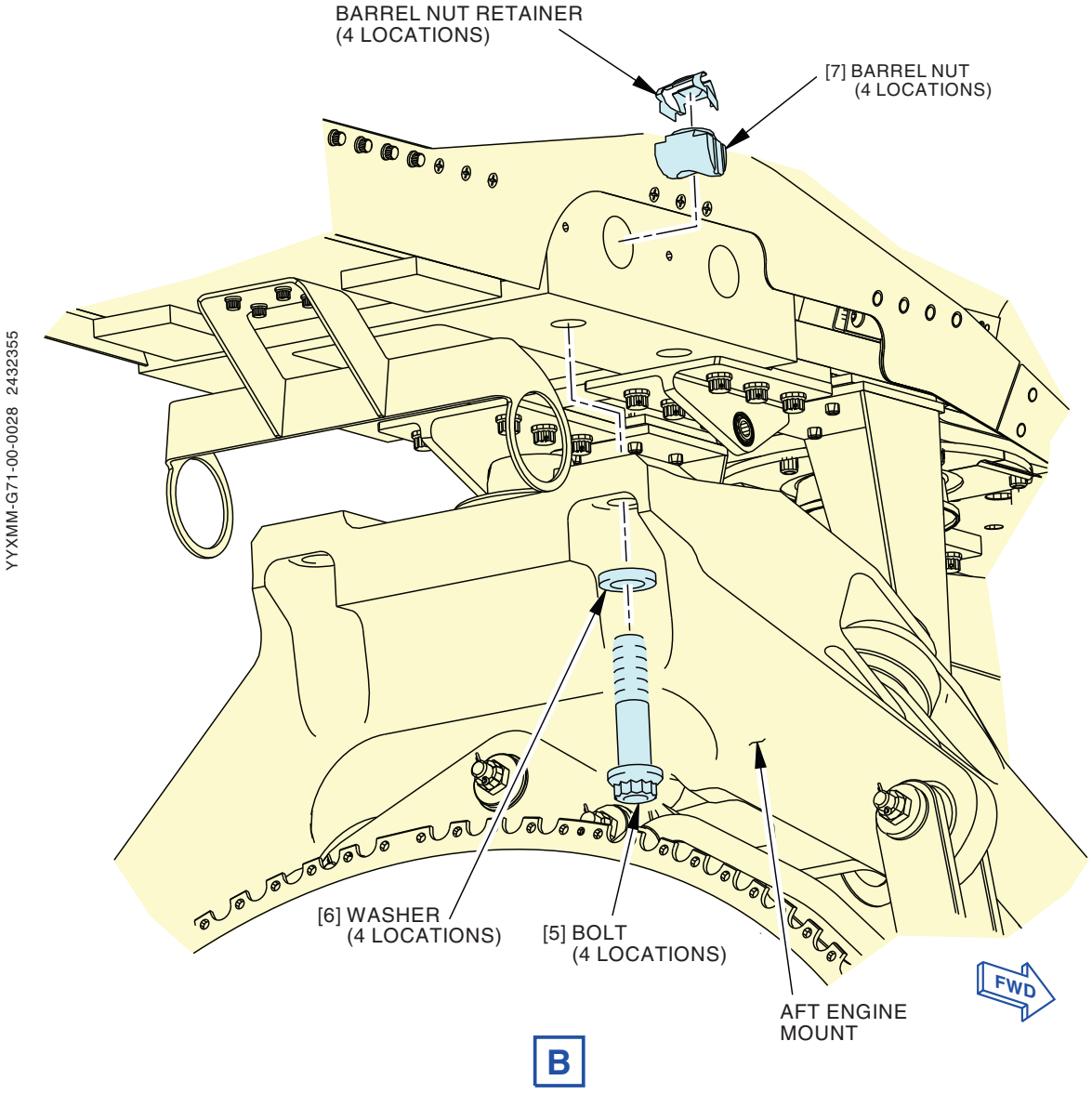
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Mount Bolts and Nuts Inspection
Figure 601/71-21-03-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Mount Bolts and Nuts Inspection
Figure 601/71-21-03-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST LINKS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Thrust Links Removal

(2) Thrust Links Installation.

TASK 71-21-04-000-801-G00

2. Thrust Links Removal

(Figure 401, Figure 402, and Figure 403)

A. General

(1) This task gives the instructions to remove the thrust link.

(2) The thrust links are installed between the fan frame and the aft engine mount.

(3) You can only remove one thrust link at a time when the power plant is installed.

(4) The removal procedures of the left and the right thrust links are the same.

(5) The upper thread protector, C71038-5, SPL-14641, lower thread protector, C71038-4, SPL-20763, and the link adjuster assembly, C71038-2, SPL-20764 are components of the thrust link equipment, SPL-23486.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
73-22-01-000-801-G00	Electronic Engine Control (EEC) Electrical Harness (HJ3A) Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14641	Upper Thread Protector, C71038-5 (included in C71038 eqpt) Part #: C71038-1 Supplier: 81205
SPL-20763	Lower Thread Protector, C71038-4 (included in C71038 eqpt) Part #: C71038-1 Supplier: 81205
SPL-20764	Link Adjuster Assy, C71038-2 (included in C71038-1 eqpt) Part #: C71038-1 Supplier: 81205
SPL-23486	Alignment/Thread Protector Equipment - Thrust Link Part #: C71038-1 Supplier: 81205
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 71-21-04-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-21-04-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

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WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 71-21-04-490-001-G00

(3) Put an engine and strut access platform, STD-653, in its position to get access to the thrust links.

SUBTASK 71-21-04-010-002-G00

(4) Disconnect the LPT/HPTACC hose from the VBV inlet.

(a) Loosen the saddle clamp.

SUBTASK 71-21-04-020-004-G00

(5) Remove the applicable shroud segment to get access to the thrust link [1] (Figure 403).

(a) For the right thrust link [1], remove the shroud 1 segment and the shroud 2 segment at the 1:00 and 3:00 o'clock positions (TASK 72-23-04-000-801-G00).

NOTE: You must remove the shroud 2 segment first before you remove the shroud 1 segment.

(b) For the left thrust link [1], remove the shroud 5 segment at the 9-12:00 o'clock position (TASK 72-23-04-000-801-G00).

(c) Remove the applicable VBV duct.

1) Do not remove the VBV inlet from the shroud.

SUBTASK 71-21-04-010-003-G00

(6) Disconnect the applicable wire bundles from the shroud segments (TASK 73-22-01-000-801-G00).

(a) Remove brackets from the shrouds.

G. Thrust Links Removal

SUBTASK 71-21-04-470-001-G00

(1) Install a link adjuster assembly, C71038-2, SPL-20764, as follows (Figure 402):

(a) Install a link adjuster assembly, C71038-2, SPL-20764, on the side of the aft engine mount opposite the thrust link [1] you will remove.

1) Tighten a link adjuster assembly, C71038-2, SPL-20764, until you can reduce the compressive force on the thrust link [1] and remove the thrust link [1].

NOTE: This will keep the aft engine mount angle in a correct position while remove the thrust link.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-21-04-020-001-G00

WARNING

BE CAREFUL WHEN YOU MOVE THE COMPONENT. THE COMPONENT IS HEAVY. INJURIES TO PERSONS CAN OCCUR.

(2) Remove the applicable thrust link [1] as follows (Figure 401):

(a) Disconnect the thrust links [1] from the aft engine mount as follows:

1) Remove the cotter pins [4] from the nuts [3].
2) Remove the pins [5], end caps [2], and nuts [3] at the end of the thrust links [1].
3) Install a upper thread protector, C71038-5, SPL-14641, on the pins [5] at the end of the thrust links [1].

(b) Disconnect the thrust links [1] from the fan frame thrust link fitting as follows:

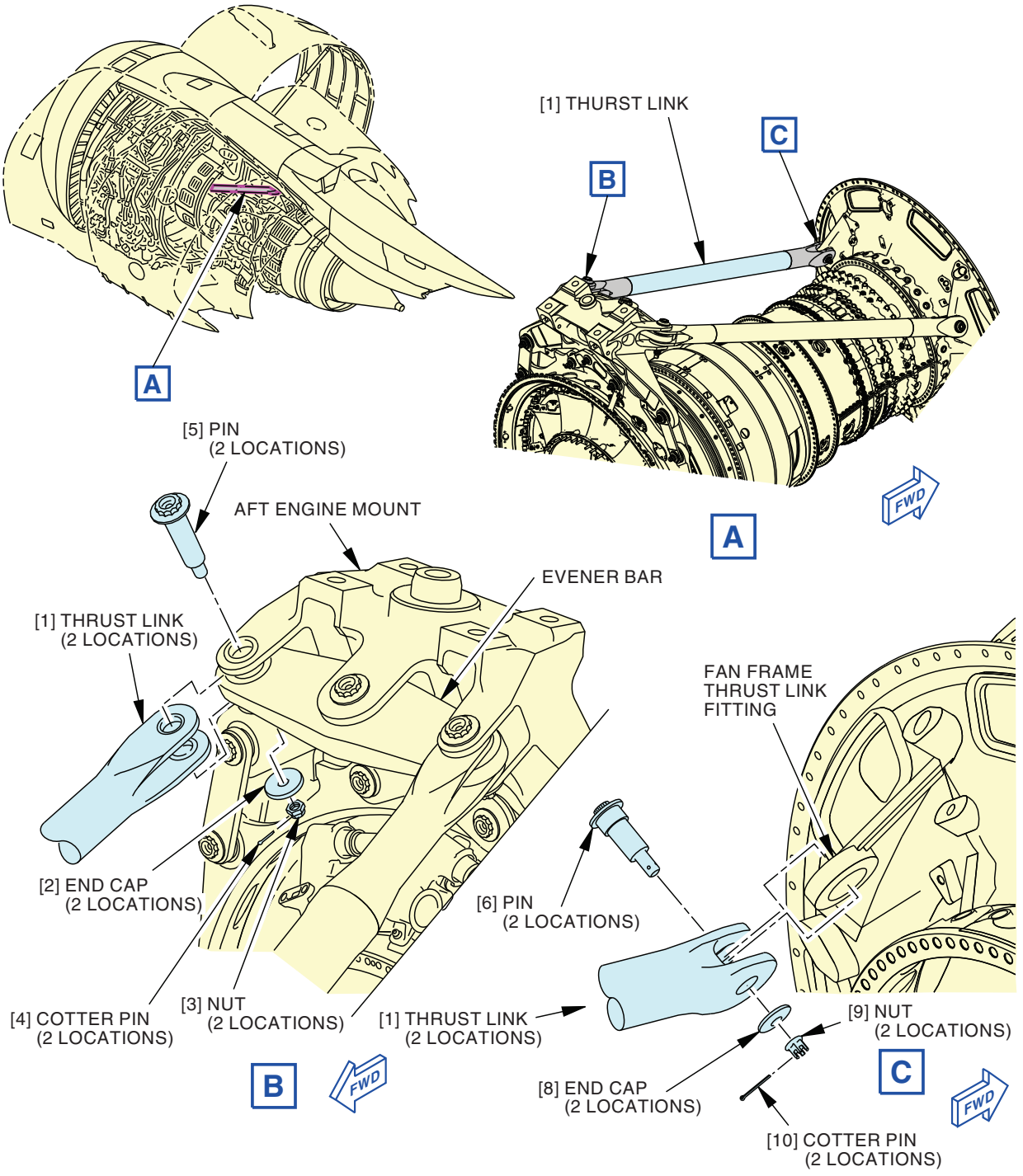
1) Remove the cotter pins [10] from the nuts [9].
2) Remove the pin [6], end cap [8], and nut [9] at the end of the thrust link [1].
3) Install a lower thread protector, C71038-4, SPL-20763, on the pin [6] at the end of the thrust link [1].

(c) Remove the thrust link [1] in the down direction.

NOTE: You will need to slide the thrust link aft until the forward end clears the cables.

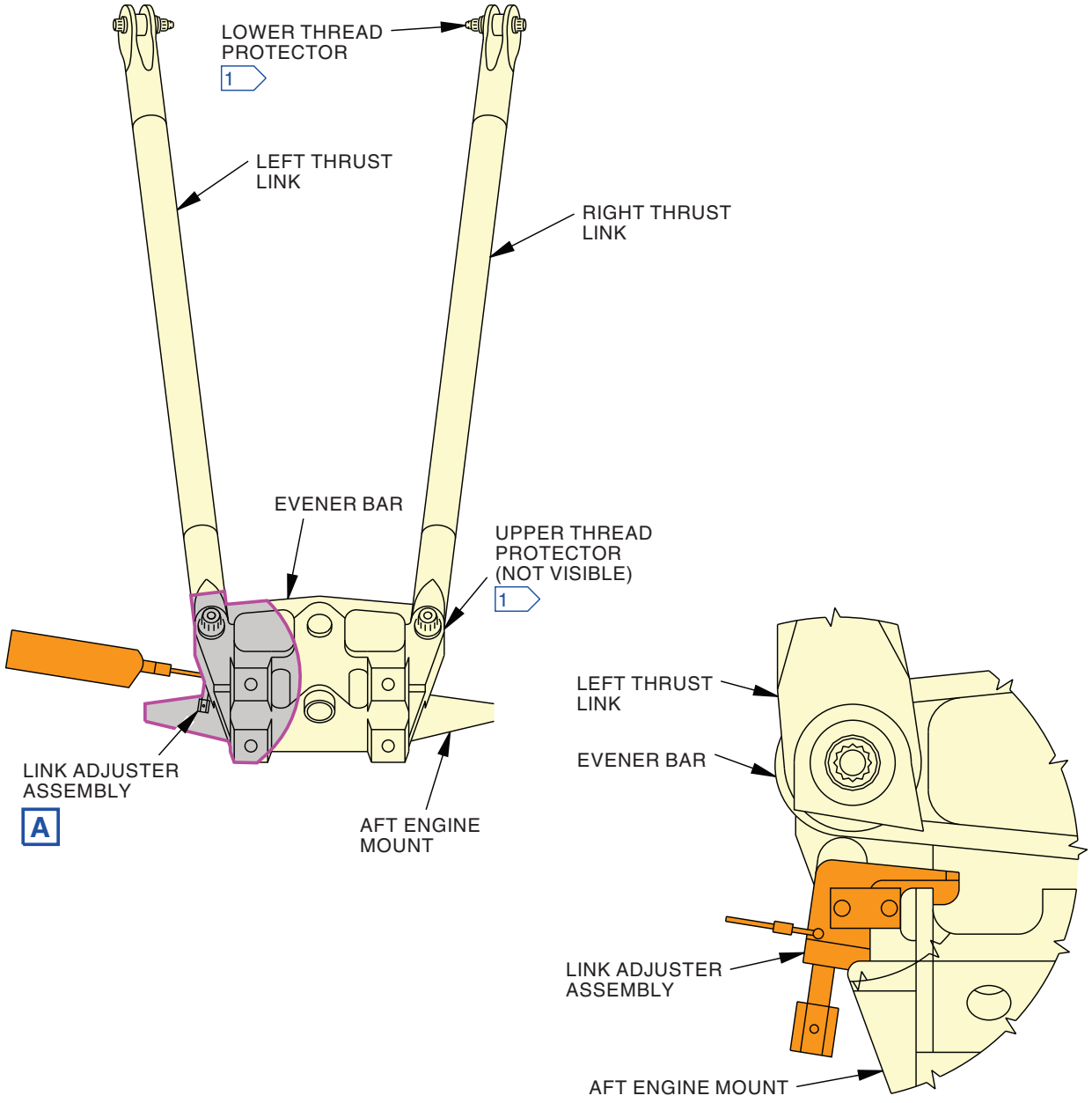
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Link Installation
Figure 401/71-21-04-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 THREAD PROTECTOR IS USED FOR ALIGNMENT PURPOSE AND PROTECT THREADS DAMAGE.

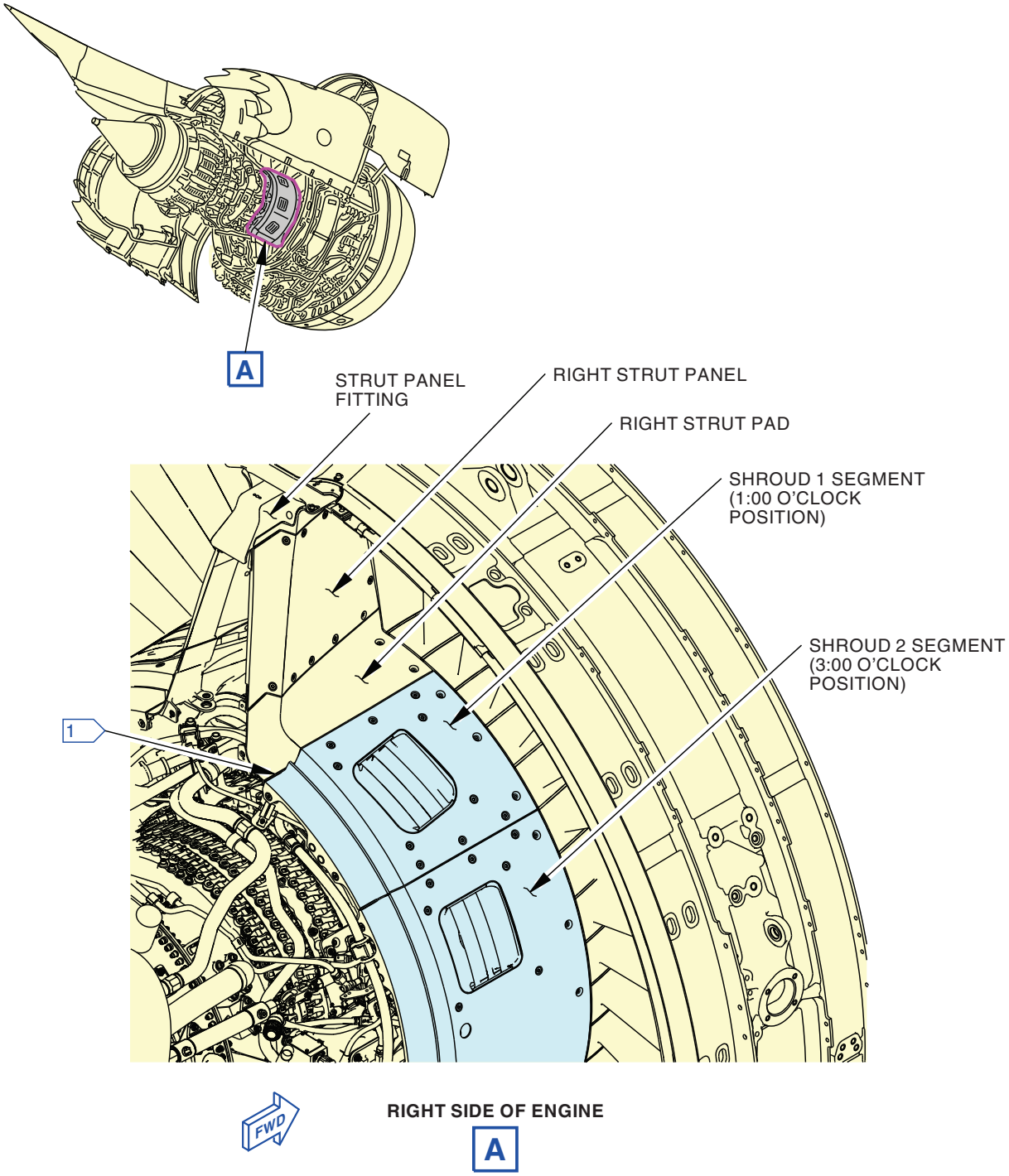
LINK ADJUSTER ASSEMBLY

A

2354435 S00061515360_V2

Thrust Link Installation Tool
Figure 402/71-21-04-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

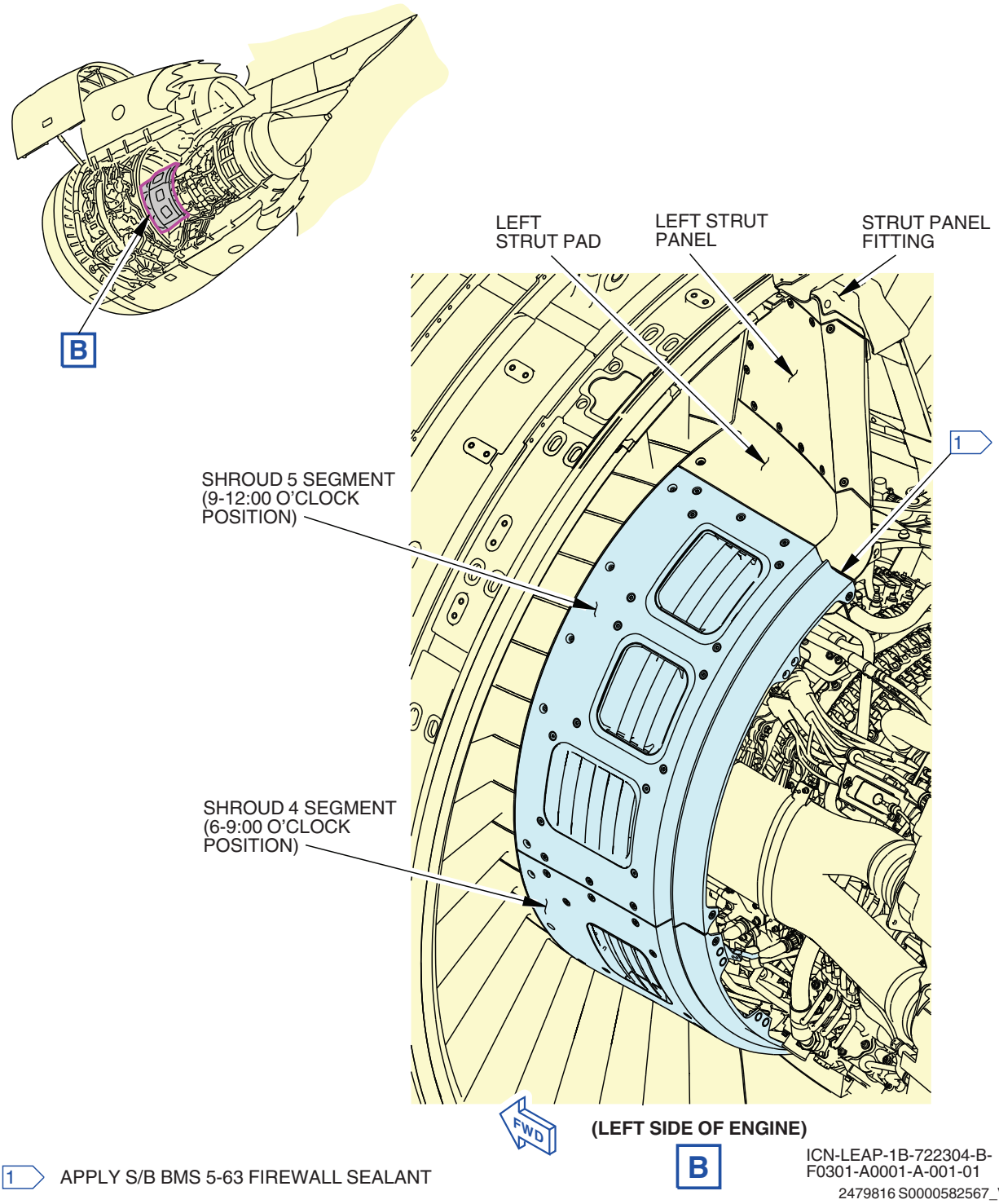


1 APPLY S/B BMS 5-63 FIREWALL SEALANT

2479761 S0000582566_V4

Strut Panels and Shroud Segments Firewall Sealant Installation
Figure 403/71-21-04-990-806-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Panels and Shroud Segments Firewall Sealant Installation
Figure 403/71-21-04-990-806-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-21-04-400-801-G00

3. Thrust Links Installation

(Figure 401, Figure 402, and Figure 403)

A. General

(1) This task gives the instructions to install the thrust link.

(2) You can only install one thrust link at a time when the power plant is installed.

(3) You must use the thrust link alignment tool to install the thrust link on the engine.

(4) The installation procedures of the left and the right thrust links are the same.

(5) You must apply the firewall sealant at the strut panel and the shroud segment edge.

(6) The upper thread protector, C71038-5, SPL-14641, lower thread protector, C71038-4, SPL-20763, and the link adjuster assembly, C71038-2, SPL-20764, are components of the thrust link equipment, SPL-23486.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
73-22-01-400-801-G00	Electronic Engine Control (EEC) Electrical Harness (HJ3A) Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14641	Upper Thread Protector, C71038-5 (included in C71038 eqpt) Part #: C71038-1 Supplier: 81205
SPL-20763	Lower Thread Protector, C71038-4 (included in C71038 eqpt) Part #: C71038-1 Supplier: 81205
SPL-20764	Link Adjuster Assy, C71038-2 (included in C71038-1 eqpt) Part #: C71038-1 Supplier: 81205
SPL-23486	Alignment/Thread Protector Equipment - Thrust Link Part #: C71038-1 Supplier: 81205
STD-653	Platform - Engine and Strut Access
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
B00130	Alcohol - Isopropyl	TT-I-735
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

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(Continued)

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G51249	Agent - Parting, Peelable	BAC5000

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Thrust link	71-21-04-01-055	PTF ALL
		71-21-04-01-060	PTF ALL
4	Cotter pin	71-21-04-01-005	PTF ALL
10	Cotter pin	71-21-04-01-010	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Thrust Links Installation

SUBTASK 71-21-04-640-001-G00

(1) Apply Pure Nickel Special compound, D00006, to the threads, shank and below the head of the pin [5] and pin [6] to the bushings and bores.

SUBTASK 71-21-04-480-001-G00

(2) Make sure that you install a upper thread protector, C71038-5, SPL-14641, and lower thread protector, C71038-4, SPL-20763, on the applicable pin to protect the pin from alignment damage.

SUBTASK 71-21-04-420-001-G00

(3) Install the applicable thrust link [1].

(a) Put the thrust link [1] between the engine fitting and the aft engine mount.

(b) Connect the thrust link [1] to the aft engine mount as follows:

1) Align the aft end of the thrust link [1] with the aft engine mount and align the forward end with the thrust link fitting.

2) Install the pin [5] at the aft end of the thrust link [1].

a) Remove a upper thread protector, C71038-5, SPL-14641, from the pin [5].

b) Apply Pure Nickel Special compound, D00006, to the threads of the pin [5].

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3) Install the end cap [2] and nut [3].

a) Make sure that the flat side of the end cap [2] is against the shoulder of the pin.

(c) Connect the thrust link [1] to the fan frame thrust link fitting.

1) Install the pin [6] at the forward end of the thrust link [1].

a) Remove a lower thread protector, C71038-4, SPL-20763, from the pin [6].

b) Install the end cap [8] and nut [9].

c) Apply Pure Nickel Special compound, D00006, to the threads of the nut [9].

<1> Make sure that Pure Nickel Special compound, D00006, is not applied onto the Teflon liner of the spherical bearing ball.

2) Install the end cap [8] and nut [9].

a) Make sure that the flat side of the end cap [8] is against the shoulder of the pin.

(d) Tighten the two nuts [3] or the head of the pins to 450.0 in-lb (50.8 N·m) - 650.0 in-lb (73.4 N·m).

NOTE: You can apply the torque to the nut or the head of the pin.

1) Install the new cotter pin [4] on the nut [3].

(e) Tighten the two nut [9] or the head of the pins to 170.0 in-lb (19.2 N·m) - 380.0 in-lb (42.9 N·m).

NOTE: You can apply the torque to the nut or the head of the pin.

1) Install the new cotter pin [10] on the nut [9].

SUBTASK 71-21-04-470-002-G00

(4) Remove a link adjuster assembly, C71038-2, SPL-20764, as follows:

(a) Loosen a link adjuster assembly, C71038-2, SPL-20764, and remove from the side of the aft engine mount.

SUBTASK 71-21-04-940-001-G00

(5) Remove the protective cover around the thrust link.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 71-21-04-420-004-G00

(1) For the left thrust link [1], install the shroud 5 segment at the 9-12:00 o'clock position (TASK 72-23-04-400-801-G00).

SUBTASK 71-21-04-420-005-G00

(2) For the right thrust link [1], install the shroud 1 segment and the shroud 2 segment at the 1:00 and 3:00 o'clock positions (TASK 72-23-04-400-801-G00).

NOTE: You must install the shroud 1 segment before you install the shroud 2 segment.

SUBTASK 71-21-04-410-002-G00

(3) Connect the applicable wire bundle to the applicable shroud segments (TASK 73-22-01-400-801-G00).

NOTE: It is necessary to install brackets on the applicable shrouds.

SUBTASK 71-21-04-410-003-G00

(4) Connect the Low Pressure Turbine (LPT)/High Pressure Turbine Active Clearance Control (HPTACC) hose to the Variable Bleed Valve (VBV) inlet.

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(a) Tighten the saddle clamp to 52.25 in-lb (5.9 N·m) - 57.75 in-lb (6.5 N·m).

SUBTASK 71-21-04-860-003-G00

(5) Apply the firewall sealant (Figure 403).

(a) Clean the surfaces of the aft edge of the left strut pad, right strut pad, shroud 1 segment, and shroud 5 segment with alcohol, B00130, and cotton wiper, G00034.

(b) Apply peelable parting agent, G51249, to the shroud 1 segment and shroud 5 segment surfaces prior to sealant, A00160, application.

(c) For the left thrust link [1], apply a minimum 1.30 in. (33.02 mm) bead of sealant, A00160, to the aft edge of the left strut pad, to fill the gap between the left strut pad and shroud 5 segment.

NOTE: The sealant, A00160, is to be flush between adjacent surfaces.

(d) For the right thrust link [1], apply a minimum 1.30 in. (33.02 mm) bead of sealant, A00160, to the aft edge of the right strut pad, to fill the gap between the right strut pad and shroud 1 segment.

NOTE: The sealant, A00160, is to be flush between adjacent surfaces.

SUBTASK 71-21-04-090-001-G00

(6) Move an engine and strut access platform, STD-653, away from the engine.

SUBTASK 71-21-04-410-001-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 71-21-04-860-002-G00

(8) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

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(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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THRUST LINKS - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Thrust Links Visual Inspection

(2) Thrust Links Detailed Inspection.

TASK 71-21-04-210-801-G00

2. Thrust Links Visual Inspection

(Figure 601)

A. General

(1) This task gives the instructions to do a visual inspection of the thrust links.

(2) The thrust links are installed between the inner fan frame and the aft engine mount.

(3) This task is used when the engine is off-wing.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-21-04-000-801-G00	Thrust Links Removal (P/B 401)
71-21-04-400-801-G00	Thrust Links Installation (P/B 401)

C. Prepare for the Inspection

SUBTASK 71-21-04-020-002-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

D. Thrust Links Visual Inspection

SUBTASK 71-21-04-210-001-G00

(1) Examine the thrust links for damage:

NOTE: If damage is found on the thrust link, refer to the Structure Repair Manual for the allowable damage limits.

(a) Cracks are not permitted.

(b) Nicks are not permitted.

(c) Dents are not permitted.

(d) Scratches are not permitted.

(e) Pitting is not permitted.

(f) Fretting is not permitted.

NOTE: Fretting is caused when two adjacent pieces of metal rub one another.

(g) Pickup is not permitted.

NOTE: Pickup occurs when material is moved from one surface to a surface which it touches.

(h) High metal is not permitted.

NOTE: High metal is metal which is pushed above the surface adjacent to scratches or other damage.

(i) Bushing movement is not permitted.

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SUBTASK 71-21-04-210-002-G00

(2) Examine the thrust link for worn areas.

(a) If you find signs of wear on the thrust links, do this task: TASK 71-21-04-211-801-G00.

SUBTASK 71-21-04-210-003-G00

(3) If damage is found on the thrust link, refer to the Structure Repair Manual for the allowable damage limits.

(a) Replace the thrust link, if it is more than the allowable damage limits.

1) Do this task: TASK 71-21-04-000-801-G00.

2) Do this task: TASK 71-21-04-400-801-G00.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 71-21-04-420-002-G00

(1) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

——— END OF TASK ———

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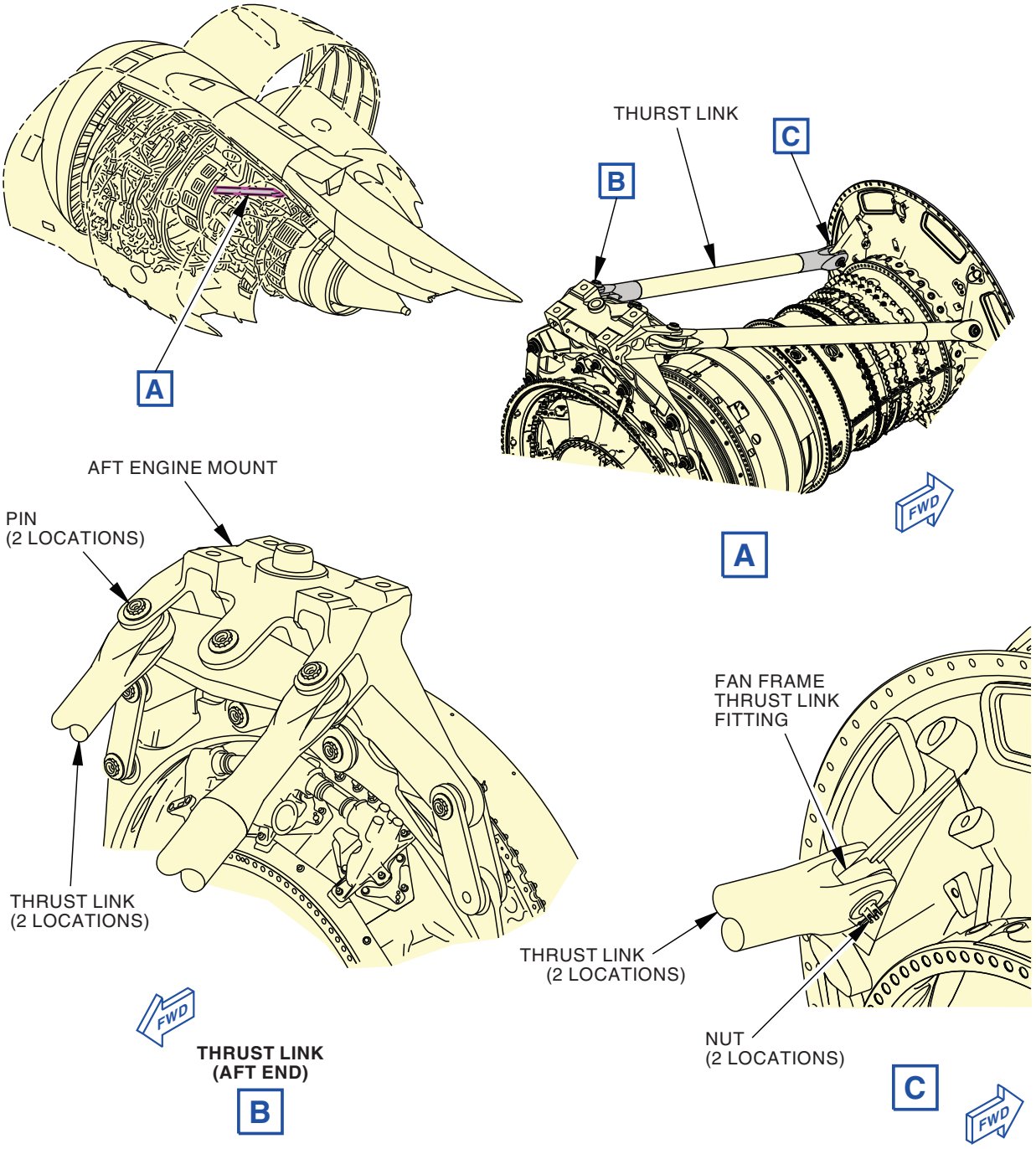

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Thrust Link Inspection
Figure 601/71-21-04-990-803-G00

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TASK 71-21-04-211-801-G00

3. Thrust Links Detailed Inspection

(Figure 602)

A. General

(1) This task gives the instructions to do a detailed inspection of the thrust link.

(2) You do this task after you remove the thrust link.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-21-04-000-801-G00	Thrust Links Removal (P/B 401)
71-21-04-400-801-G00	Thrust Links Installation (P/B 401)

C. Prepare for the Inspection

SUBTASK 71-21-04-020-003-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

D. Thrust Links Detailed Inspection

SUBTASK 71-21-04-211-001-G00

(1) Do a check of the thrust link for worn areas as follows:

NOTE: If you find worn areas on the thrust link, refer to Component Maintenance Manual for wear limits and repair instructions.

(a) Examine the areas where the thrust links attach points to the aft and forward engine mount on the engine compressor case.

1) Examine the bearing and bearing balls are free of damage.

2) Examine the bushings are free of damage.

3) Examine the bolts are free of damage.

4) Examine the shoulder bolts are free of damage.

5) Examine the fittings are free of damage.

6) Make sure that they are free of damage and attached correctly.

(b) If you find the component attach points are more than the wear limits, do the steps that follows (Table 601).

1) Replace the worn component of the thrust links, if it is necessary (TASK 71-21-04-000-801-G00 and TASK 71-21-04-400-801-G00).

Table 601/71-21-04-993-801-G00 Thrust Link Wear Limits

			DIMENSION		WEAR LIMITS (INCHES)
INDEX NO.	PART NAME	DIM.	MIN	MAX	MAXIMUM CLEARANCE
1	BUSHING	ID	-	0.8755	0.0030
	PIN	OD	0.8720	-	

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Table 601/71-21-04-993-801-G00 Thrust Link Wear Limits (Continued)

			DIMENSION		WEAR LIMITS (INCHES)
INDEX NO.	PART NAME	DIM.	MIN	MAX	MAXIMUM CLEARANCE
2	BUSHING	ID	-	1.1904	0.0052
	PIN BUSHING	OD	1.1826	-	
3	BUSHING	ID	-	1.0025	0.0040
	PIN	OD	0.9965	-	

E. Put the Airplane Back to Its Usual Condition

SUBTASK 71-21-04-420-003-G00

(1) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

——— END OF TASK ———

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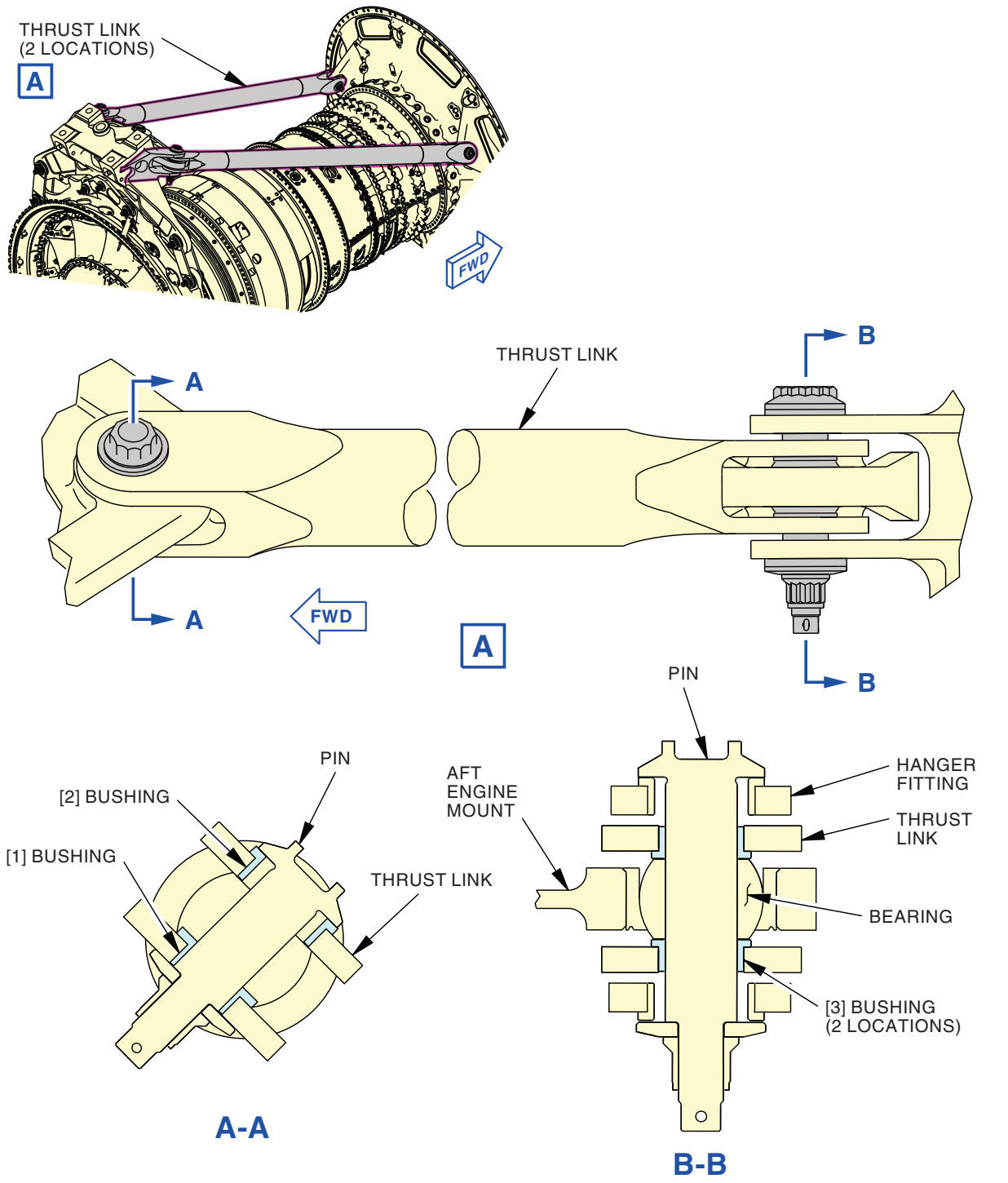

Thrust Link Inspection
Figure 602/71-21-04-990-804-G00

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ENGINE VENTS AND DRAINS - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Engine Vents and Drains Inspection.

TASK 71-71-00-200-801-G00

2. Engine Vents and Drains Inspection

(Figure 601 or Figure 602)

A. General

(1) This task gives the instructions to do the inspection of all engine vents and drains for leakage.

(2) This task contains data that lets you get an estimate of the condition of the Line Replaceable Unit (LRU) when you look at the rate of leakage from the engine vent and drain system.

(3) The leakage rate for the different drains is determined under the conditions as follows:

(a) Operation of the engine at minimum idle to monitor the fuel or oil leakage from the vents and drains.

(b) Operation of the engine at part power to isolate and monitor the oil leakage of specific accessory gearbox seals.

B. References

Reference	Title
24-11-00-700-801	IDG Oil System Static Leak Check (P/B 201)
24-11-11-000-801	Integrated Drive Generator (IDG) Removal (P/B 401)
24-11-11-400-801	Integrated Drive Generator (IDG) Installation (P/B 401)
24-11-13-210-801	IDG Air/Oil Heat Exchanger - Inspection (P/B 601)
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
70-00-04-000-801-G00	Preformed Packing Seal and Packing Retainer Removal (P/B 201)
70-00-04-400-801-G00	Preformed Packing Seal and Packing Retainer Installation (P/B 201)
71-00-00-710-801-G00	Test No. 5 - Part Power Leak Check (P/B 501)
71-00-00-730-802-G00	Test No. 15 - Actuator Operational Test (P/B 501)
71-00-00-790-803-G00	Test No. 3 - Wet Motor Leak Test (P/B 501)
71-00-00-790-804-G00	Test No. 4 - Idle Leak Test (P/B 501)
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-64-01-000-801-G00	Accessory Gearbox Rotating Seals Removal (P/B 401)
72-64-01-400-801-G00	Accessory Gearbox Rotating-Seals Installation (P/B 401)
73-11-01-000-801-G00	Main Fuel Pump Removal (P/B 401)
73-11-01-400-801-G00	Main Fuel Pump Installation (P/B 401)
75-23-02-000-801-G00	Left Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)
75-23-02-000-802-G00	Right Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)

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(Continued)

Reference	Title
75-23-02-400-801-G00	Left Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
75-23-02-400-802-G00	Right Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
75-24-01-000-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal (P/B 401)
75-24-01-400-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation (P/B 401)
75-24-04-000-801-G00	High Pressure Turbine Active Clearance Control (HPTACC) Valve Removal (P/B 401)
75-24-04-400-801-G00	High Pressure Turbine Active Clearance Control (HPTACC) Valve Installation (P/B 401)
75-31-02-000-801-G00	Left Variable Stator Vane (VSV) Actuator Removal (P/B 401)
75-31-02-000-802-G00	Right Variable Stator Vane (VSV) Actuator Removal (P/B 401)
75-31-02-400-802-G00	Left Variable Stator Vane (VSV) Actuator Installation (P/B 401)
75-31-02-400-803-G00	Right Variable Stator Vane (VSV) Actuator Installation (P/B 401)
75-32-01-000-801-G00	Left Variable Bleed Valve Actuator Removal (P/B 401)
75-32-01-000-802-G00	Right Variable Bleed Valve Actuator Removal (P/B 401)
75-32-01-400-802-G00	Left Variable Bleed Valve Actuator Installation (P/B 401)
75-32-01-400-803-G00	Right Variable Bleed Valve Actuator Installation (P/B 401)
75-33-01-000-801-G00	Transient Bleed Valve Removal (P/B 401)
75-33-01-400-801-G00	Transient Bleed Valve Installation (P/B 401)
75-34-03-000-801-G00	Reverse Bleed Valve Removal (P/B 401)
75-34-03-400-801-G00	Reverse Bleed Valve Installation (P/B 401)
78-11-03-210-801-G00	Center Vent Tube Extension (CVTE) Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
79-00-00-200-804-G00	Drain Tubes Inspection and Cleaning (P/B 601)
79-11-03-000-801-G00	Oil Tank Filler Cap Removal (P/B 401)
79-11-03-400-801-G00	Oil Tank Filler Cap Installation (P/B 401)
79-21-04-211-801-G00	Scavenge Screen Plug Inspection (P/B 201)
79-21-07-200-801-G00	Oil Filter Cartridge Element Inspection (P/B 601)
79-21-17-000-801-G00	Oil Eductor Valve Removal (P/B 401)
79-21-17-400-801-G00	Oil Eductor Valve Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-20664	Fixture - Leak Check Part #: 956A3624G01 Supplier: 58828
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)

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(Continued)

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 71-71-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

G. Engine Vents and Drains Inspection

SUBTASK 71-71-00-210-003-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

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(WARNING PRECEDES)

WARNING DO NOT GET FUEL OR OIL ON YOUR SKIN. REMOVE THE FUEL OR OIL FROM YOUR SKIN IMMEDIATELY. IF YOU LET FUEL OR OIL STAY ON YOUR SKIN, IT CAN CAUSE INJURIES TO YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Do a general visual inspection of the following engine drains and vents:

NOTE: You must identify the origin of the leak and do the corrective action for leakage which is not water.

(a) Inlet exhaust and inlet drain holes (Figure 601 (Sheet 1)).

1) Serviceable limit: 0

2) Corrective action:

a) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

(b) Fan cowl drain holes and starter exhaust vents (Figure 601 (Sheet 2)).

1) Serviceable limit: 0

a) The fan cowl drain holes and the starter exhaust vents do not have limits. Any flammable fluids need to be addressed in these location as there are ignition sources present in the fan compartment. Fuel leaks in particular must be inspected to ensure the integrity of the fuel supply system is maintained. Suction fuel feed capability must not be compromised (no fuel leakage is allowed from the fuel supply line).

2) Corrective action:

a) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

(c) Lower bifurcation drain scuppers and thrust reverser fan sleeve static ground drain (Figure 601 (Sheet 3)).

1) Serviceable limit: 0

2) Corrective action:

a) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

(d) Main core compartment drains (Figure 601 (Sheet 3)).

1) Serviceable limit: 0

a) The main core compartment drain holes do not have explicit leakage limits. Flammable fluids need to be addressed in this location as there are ignition sources present in the core compartment.

2) Corrective action:

a) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

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(e) Thrust reverser translating sleeve cowl drain No. 1 forward latch (Figure 601 (Sheet 3)).

NOTE: Water and hydraulic fluid may drain from the No. 1 forward latch drain.

1) Serviceable limit: 0

2) Corrective action:

a) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

(f) Dry bay condensate drains (Figure 601 (Sheet 4)).

1) Serviceable limit: 0

2) Corrective action:

a) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

(g) Aft fairing compartment drain and aft fairing drain tube (Figure 601 (Sheet 5)).

1) Serviceable limits:

a) The aft fairing compartment drain directs fluid from the aft fairing compartment and out through a gap between the trailing edge fitting and the drain tube, which are concentric. Sources of flammable fluids coming out of the aft fairing compartment drains should be inspected. Use leakage limits for said components, for example, fuel tank and hydraulic tubing leaks.

b) The aft fairing drain tube drains fluid from the aft upper strut (strut to wing leakage zone) as well as the wing leading edge (adjacent to the engine). Any fuel leaking from the aft fairing drain tube must be inspected to ensure the integrity of the fuel supply system is maintained. Suction fuel feed capability must not be compromised (no fuel leakage is allowed from the fuel supply line).

2) Corrective action:

a) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

(h) Turbine case drain tubes (Figure 601 (Sheet 5)).

1) Serviceable limits:

a) The turbine case drains collect and drain flammable fluids from the engine core compartment and the engine core. Fuel leakage at the turbine case drains is expected following a wet motor procedure or if there is a no-light event where residual fuel from the engine core will drain through Turbine Rear Vane (TRV) drains into turbine case drains. If the engine has not experienced a recent wet motor maintenance task or aborted no-light off event, inspect for fluid leaks in engine core compartment.

2) Corrective action:

a) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

SUBTASK 71-71-00-210-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

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(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT GET FUEL OR OIL ON YOUR SKIN. REMOVE THE FUEL OR OIL FROM YOUR SKIN IMMEDIATELY. IF YOU LET FUEL OR OIL STAY ON YOUR SKIN, IT CAN CAUSE INJURIES TO YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Do the visual inspection of the engine drain masts as follows:

(a) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(b) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(c) Do a check of the fan drain mast [2] for signs of leakage (Figure 602 (Sheet 5) and Figure 602 (Sheet 6)).

(d) Do a check of the core drain mast [1] for sign of leakage (Figure 602 (Sheet 1), Figure 602 (Sheet 2) and Figure 602 (Sheet 3)).

1) When the engine starts, oil leakage at the core drain mast may be observed (very cold weather, engine not operated recently, or recently installed). These leaks are permitted if they stop after five minutes of engine idle run.

2) At engine shutdown, oil leakage from the sumps at the core drain mast may be observed.

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a) The leaks at engine shutdown are permitted as long as they stop before 15 minutes and the total volume of oil is not more than 20 cc (20 ml), cumulated on all oil core drains, in case of multiple leakage sources, refer to the serviceable limits for A-Sump, B-Sump, C-Sump, and Aux, in topic H of this task.

<1> If the leak continues for more than 15 minutes after engine shutdown or if the total volume of oil exceeds 20 cc (20 ml), refer to the steps to identify if there is a sign of leakage from the core drain mast in topic G of this task.

NOTE: Limits given in this paragraph are applicable only at engine shutdown.

b) It is permitted to continue operation for 5 flight cycles prior assessing the duration and volume of the oil leakage provided the three following conditions are met:

- oil leak is coming only from C-Sump drain
- oil consumption is below the troubleshooting limit of 0.36 qt (0.34 l) per flight hour (TASK 71-00-00-800-802-G00)
- there is no active oil leak or fumes or fogs from core drain mast at idle.

3) The following leakage rate limits apply to engines that are not operated and engines that are operated at all power settings.

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NOTE: The markings on the drain mast are as follow:

- A-SUMP 1R: drain line from the A-Sump for the N°1 Roller Bearing
- A-SUMP 2R: drain line from the A-Sump for the N°3 Roller Bearing
- LRU CORE LEFT: drain line for the LRUs located on the left hand side of the engine
- LRU CORE RIGHT: drain line for the LRUs located on the right hand side of the engine
- AUX: drain line from the Turbine Center Frame (TCF)
- B-SUMP: drain line from the B-Sump
- C-SUMP: drain line from the C-Sump

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0180

NOTE: The markings on the drain mast are as follow:

- A-S1: drain line from the A-Sump for the N°1 Roller Bearing
- A-S3: drain line from the A-Sump for the N°3 Roller Bearing
- LRU-L: drain line for the LRUs located on the left hand side of the engine
- LRU-R: drain line for the LRUs located on the right hand side of the engine
- AUX: drain line from the TCF
- B-S: drain line from the B-Sump
- C-S: drain line from the C-Sump.

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(e) If there are no signs of leakage, the engine is serviceable.

(f) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the threshold and serviceable limits.

1) For fuel leaks:

a) If the total leakage rate from the drains is less than the threshold limit, you can continue to operate the engine.

NOTE: Maintenance action is not necessary.

b) If the total leakage rate from the drains is more than the threshold limit, but less than the serviceable limit, corrective action must be done before 25 flight cycles.

c) If the total leakage rate from the drains is more than the serviceable limit, you must do the corrective action.

2) For oil leaks:

a) If the leakage rate is less than the serviceable limit, you can continue to operate the engine.

b) If the leakage rate is more than the serviceable limit, maintenance must be done before you operate the engine again.

<1> Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” below for the serviceable limit and corrective action.

3) For hydraulic leaks:

a) If the total leakage rate from the strut drain is more than the serviceable limit, you must do the corrective action.

(g) If there is a sign of leakage from the fan drain mast [2], identify the origin of the leak as follows:

NOTE: These steps are not necessary if the fuel leakage rate is less than the threshold limit for the fan drain mast.

1) Put the oil resistant container, STD-203, below the fan drain mast [2].

2) Install an oil and fuel resistant bottle to each drain line at the fan drain mast [2].

a) Make sure that the bottle is sealed.

3) Do this task: Test No. 4 - Idle Leak Test, TASK 71-00-00-790-804-G00.

a) Record the time of the engine start.

b) Run the engine at idle power for 10 minutes.

4) Monitor the drain for drops of oil or fuel.

5) Calculate the total engine operation time.

6) Identify leaks origin with the placard installed on oil tank (Figure 602 (Sheet 6) and Figure 602 (Sheet 7)).

7) Record the number of drops.

8) Change the number of drops per minute to cc’s per hour.

NOTE: 20 drops equal to approximately one cc.

a) For example: 20 drops per minute x 60 = 1200 drops per hour / 20 = 60 cc's per hour.

9) Refer to the topic “Engine Vents and Drains Leakage Limits and Corrective Action” for the permitted leakage rate for the component.

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(h) If there is a sign of leakage from the Integrated Drive Generator (IDG) drain pad (Figure 602 (Sheet 6)), identify the source of the leak as follows:

1) Do this task: IDG Oil System Static Leak Check, TASK 24-11-00-700-801.

NOTE: If there is a static leak, the leakage is from the IDG input shaft seal. If there is no static leak, the leakage is from the Accessory Gear Box (AGB) pad seal.

(i) If there is a sign of leakage from the core drain mast [1], identify the origin of the leak as follows:

NOTE: These steps are not necessary if the fuel leakage rate is less than the threshold limit for the core drain mast.

1) Install the leak check fixture, SPL-20664, as an option to collect oil in case of leaks.

a) Put the test tube holder on a work table and put all seven test tubes in each of the grommets of the rack.

b) Install the tube holder in the rack on the top of the test tubes and tightly attach it with knobs.

c) Identify the drain master and each of the collecting tubes with the diameter in table below:

Table 601/71-71-00-993-802-G00 Collecting Tubes Diameters

Drain Mast Tube	Diameter	Collecting Tube
A-SUMP 1R	0.3125 in. (7.9 mm)	4
A-SUMP 3R	0.3125 in. (7.9 mm)	4
LRU CORE LEFT	0.3125 in. (7.9 mm)	4
LRU CORE RIGHT	0.3125 in. (7.9 mm)	4
AUX.	0.41 in. (10.4 mm)	4
B-SUMP	0.41 in. (10.4 mm)	5
C-SUMP	0.44 in. (11.2 mm)	5

CAUTION MAKE SURE THAT THE O-RINGS DO NOT HAVE DAMAGES AND INSTALLATION IS CORRECT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

d) Put each collecting tube into the related drain master tube until they touch the core drain mast [1].

e) Put the rack below the core drain mast [1].

<1> Make sure that the rack points in the FWD direction.

f) Put the hoses into the holder in the related test tube.

<1> Make sure that the marks on the core drain mast [1] match with the marks on the holder.

2) Install an oil and fuel resistant bottle to each drain line or at the core drain mast [1].

a) Make sure that the bottle is sealed.

3) Do this task: Test No. 4 - Idle Leak Test, TASK 71-00-00-790-804-G00.

a) Record the time of the engine start.

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b) Run the engine at idle power for 10 minutes.

4) Monitor the drain for drops of oil or fuel.

5) If the leak stops or is within acceptable limits, the engine is serviceable.

6) Calculate the total engine operation time.

7) Identify leaks origin with the number written and identification on drain mast (Figure 602 (Sheet 8) or Figure 602 (Sheet 10)).

8) Record the number of drops or volume inside tooling tubes.

9) Change the number of drops per minute to cc's per hour.

NOTE: 20 drops equal to approximately one cc.

a) For example: 20 drops per minute x 60 = 1200 drops per hour / 20 = 60 cc's per hour.

10) Refer to the topic "Engine Vents and Drains Leakage Limits and Corrective Action" for the permitted leakage rate for the component.

11) If the leakage rate is more than the permitted limits:

a) If the leakage comes from the Engine Fuel equipment Left side or Right side, do these steps to identify leakage origin:

<1> Do this task: Test No. 15 - Actuator Operational Test, TASK 71-00-00-730-802-G00.

<2> Do an engine run at idle for 5 minutes.

<3> If the leak stops or is within acceptable limits, the engine is serviceable.

<4> If leaks does not stop and stay above the serviceable limit, do these steps:

<a> Open the left and right thrust reversers (TASK 78-31-00-010-801-G00).

<b> Refer to (Figure 602 (Sheet 12)) for the left LRU fuel drain leakage.

<c> Refer to Figure 602 (Sheet 14) for the right LRU fuel drain leakage.

H. Engine Vents, Drains, and Drain Masts Leakage Limits and Corrective Action

SUBTASK 71-71-00-220-001-G00

(1) Fuel drainage from core drain mast [1].

(a) Fluid: Fuel.

(b) Threshold limit: 180 cc/hr (60 drops per minute).

1) If the leakage is less than the threshold limit, no maintenance action is necessary.

2) If the leakage is more than the threshold limit and less than the serviceable limit, you can continue the engine in service for not more than 25 flight cycles before you replace the applicable component.

(c) Serviceable limit: 270 cc/hr (90 drops per minute).

(d) Replace the applicable component:

1) For the left Variable Stator Vane Actuator (VSVA), these are the tasks:

Left Variable Stator Vane (VSV) Actuator Removal, TASK 75-31-02-000-801-G00,

Left Variable Stator Vane (VSV) Actuator Installation, TASK 75-31-02-400-802-G00.

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2) For the right VSVA, these are the tasks:

Right Variable Stator Vane (VSV) Actuator Removal, TASK 75-31-02-000-802-G00,

Right Variable Stator Vane (VSV) Actuator Installation, TASK 75-31-02-400-803-G00.

3) For the left Variable Bleed Valve (VBV) actuator, these are the tasks:

Left Variable Bleed Valve Actuator Removal, TASK 75-32-01-000-801-G00,

Left Variable Bleed Valve Actuator Installation, TASK 75-32-01-400-802-G00.

4) For the right VBV actuator, these are the tasks:

Right Variable Bleed Valve Actuator Removal, TASK 75-32-01-000-802-G00,

Right Variable Bleed Valve Actuator Installation, TASK 75-32-01-400-803-G00.

5) For the Transient Bleed Valve (TBV), these are the tasks:

Transient Bleed Valve Removal, TASK 75-33-01-000-801-G00,

Transient Bleed Valve Installation, TASK 75-33-01-400-801-G00.

6) For the left Modulated Turbine Cooling (MTC) valve, these are the tasks:

Left Modulated Turbine Cooling (MTC) Valve Removal, TASK 75-23-02-000-801-G00,

Left Modulated Turbine Cooling (MTC) Valve Installation, TASK 75-23-02-400-801-G00.

7) For the right MTC valve, these are the tasks:

Right Modulated Turbine Cooling (MTC) Valve Removal, TASK 75-23-02-000-802-G00,

Right Modulated Turbine Cooling (MTC) Valve Installation, TASK 75-23-02-400-802-G00.

8) For the High Pressure Turbine Active Clearance Control (HPTACC) valve, these are the tasks:

High Pressure Turbine Active Clearance Control (HPTACC) Valve Removal, TASK 75-24-04-000-801-G00,

High Pressure Turbine Active Clearance Control (HPTACC) Valve Installation, TASK 75-24-04-400-801-G00.

9) For the Low Pressure Turbine Active Clearance Control (LPTACC) valve, these are the tasks:

Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal, TASK 75-24-01-000-801-G00,

Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation, TASK 75-24-01-400-801-G00.

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10) For the Reverse Bleed Valve (RBV), these are the tasks:

Reverse Bleed Valve Removal, TASK 75-34-03-000-801-G00,

Reverse Bleed Valve Installation, TASK 75-34-03-400-801-G00.

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SUBTASK 71-71-00-220-002-G00

(2) Engine oil drain system fan drain mast [2].

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(a) Fluid: Oil.

(b) Serviceable limit: 20 cc/hr (7 drops per minute).

(c) Replace the applicable component:

1) Do these steps to replace the AGB rotating seal of the applicable pad for the fuel pump, hydraulic pump, hand cranking pad, or IDG:

Accessory Gearbox Rotating Seals Removal, TASK 72-64-01-000-801-G00,

Accessory Gearbox Rotating-Seals Installation, TASK 72-64-01-400-801-G00.

SUBTASK 71-71-00-220-003-G00

(3) Hydraulic drain system.

(a) Fluid: Hydraulic fluid.

(b) For the limits of the hydraulic engine-driven pump (TASK 29-00-00-790-801).

SUBTASK 71-71-00-220-004-G00

(4) IDG input shaft seal.

(a) Fluid: Oil.

(b) Serviceable Limit: 1 cc/hr.

(c) Corrective action: replace the IDG (TASK 24-11-11-000-801 and TASK 24-11-11-400-801).

SUBTASK 71-71-00-220-012-G00

(5) If it is necessary, repair or replace the IDG air/oil heat exchanger (TASK 24-11-13-210-801).

SUBTASK 71-71-00-220-006-G00

(6) Oil Tank Scupper.

(a) Fluid: Oil.

(b) Serviceable limit: Oil that flows from the oil tank servicing only.

(c) Corrective action:

1) Make sure that the oil tank cap is installed correctly.

2) Make sure that there is no oil leakage when you service the oil tank.

3) Make sure that there is no damage to the oil tank cap O-ring.

a) If there is damage to the O-ring, replace the oil tank cap O-ring (TASK 70-00-04-000-801-G00 and TASK 70-00-04-400-801-G00).

4) Make sure that there is no damage to the oil tank cap.

a) If there is damage to the oil tank cap, replace the oil tank cap (TASK 79-11-03-000-801-G00 and TASK 79-11-03-400-801-G00).

SUBTASK 71-71-00-220-007-G00

(7) A-Sump.

NOTE: Limits given in this paragraph are applicable only at idle and high power run.

(a) Fluid: Oil.

(b) Serviceable limit: 40 cc/h for the A-Sump 1R and A-Sump 3R together.

(c) Continue-In-Service (CIS) limit: 540 cc/h for each A-Sump 1R and A-Sump 3R.

1) If the leak is less than the serviceable limit, no maintenance action is necessary.

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2) If the leak at IDLE is more than the serviceable limit and less than the CIS limit with all the following conditions, then you can continue the engine in service for 150 cycles:

a) If cumulative oil leak (from A-Sump, B-Sump, C-Sump and Aux drains) is below 540 cc/h at IDLE.

b) If oil consumption is within serviceable limit (TASK 71-00-00-800-802-G00).

c) If there is no leak above 40 cc/h during stabilized high power (operate the engine at minimum of 70 percent N1 for 10 minutes (TASK 71-00-00-710-801-G00)).

<1> Perform the following maintenance inspections within these 150 cycles:

<a> Oil filter cartridge inspection (TASK 79-21-07-200-801-G00).

<b> Drain tubes inspection (TASK 79-00-00-200-804-G00).

<c> Scavenge Screen Plug Inspection, TASK 79-21-04-211-801-G00.

<d> Center Vent Tube Extension (CVTE) Inspection, TASK 78-11-03-210-801-G00.

<e> Do a check of the eductor air valve pop-out window to verify the position of the valve.

NOTE: If the pop-out is closed and you cannot read OPEN on the pop-out, it is necessary to replace the oil eductor valve (TASK 79-21-17-000-801-G00 and TASK 79-21-17-400-801-G00).

<2> Within these 150 cycles, measure the leak again.

<3> If the leak at IDLE is more than the serviceable limit and less than the CIS limit with all the following conditions, then you can continue the engine in service for 150 cycles:

<a> If cumulative oil leak (from A-Sump, B-Sump, C-Sump and Aux drains) is below 540 cc/h at IDLE.

<b> If oil consumption is within serviceable limit (TASK 71-00-00-800-802-G00).

<c> If there is no leak above 40 cc/h during stabilized high power (operate the engine at minimum of 70 percent N1 for 10 minutes (TASK 71-00-00-710-801-G00)).

<4> Perform the following maintenance inspections within these 150 cycles:

<a> Drain tubes inspection (TASK 79-00-00-200-804-G00).

<b> Center Vent Tube Extension (CVTE) Inspection, TASK 78-11-03-210-801-G00.

<5> Within these 150 cycles, measure the leak again and apply the recommendation in steps <3> to <5> above.

3) If the leak is more than the CIS limit, contact CFM.

(d) Corrective action:

1) Replace the engine (TASK 71-00-02-000-801-G00 and TASK 71-00-02-400-801-G00).

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SUBTASK 71-71-00-220-008-G00

(8) B-Sump, C-Sump and Aux.

NOTE: Limits given in this paragraph are applicable only at idle and high power run.

(a) Fluid: Oil.

(b) Serviceable limit : 80 cc/h for the B-Sump, C-Sump and Aux together.

(c) CIS limit: 540 cc/h for each B-Sump, C-Sump, and Aux.

1) If the leak is less than the serviceable limit, no maintenance action is necessary.

2) If the leak at IDLE is more than the serviceable limit and less than the CIS limit with all the following conditions, then you can continue the engine in service for 150 cycles:

a) If cumulative oil leak (from A-Sump, B-Sump, C-Sump and Aux drains) is below 540 cc/h at IDLE.

b) If oil consumption is within serviceable limit (TASK 71-00-00-800-802-G00).

c) If there is no leak above 80 cc/h during stabilized high power (operate the engine at minimum of 70 percent N1 for 10 minutes (TASK 71-00-00-710-801-G00)).

<1> Perform the following maintenance inspections within these 150 cycles:

<a> Oil filter cartridge inspection (TASK 79-21-07-200-801-G00).

<b> Drain tubes inspection (TASK 79-00-00-200-804-G00).

<c> Scavenge Screen Plug Inspection, TASK 79-21-04-211-801-G00.

<d> Center Vent Tube Extension (CVTE) Inspection, TASK 78-11-03-210-801-G00.

<e> Do a check of the eductor air valve pop-out window to verify the position of the valve.

NOTE: If the pop-out is closed and you cannot read OPEN on the pop-out, it is necessary to replace the oil eductor valve (TASK 79-21-17-000-801-G00 and TASK 79-21-17-400-801-G00).

<2> Within these 150 cycles, measure the leak again.

<3> If the leak at IDLE is more than the serviceable limit and less than the CIS limit with all the following conditions, then you can continue the engine in service for 150 cycles:

<a> If cumulative oil leak (from A-Sump, B-Sump, C-Sump and Aux drains) is below 540 cc/h at IDLE.

<b> If oil consumption is within serviceable limit (TASK 71-00-00-800-802-G00).

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<c> If there is no leak above 80 cc/h during stabilized high power (operate the engine at minimum of 70 percent N1 for 10 minutes (TASK 71-00-00-710-801-G00)).

NOTE: If there is a leak above 80 cc/h during stabilized high power or if leak rate increases at IDLE after stabilized high power, compared to first measurement at IDLE, refer to the steps to identify if there is a sign of leakage from the core drain mast in topic G of this task. Contact CFM.

<4> Perform the following maintenance inspections within these 150 cycles:

<a> Drain tubes inspection (TASK 79-00-00-200-804-G00).

<b> Center Vent Tube Extension (CVTE) Inspection, TASK 78-11-03-210-801-G00.

<5> Within these 150 cycles, measure the leak again and apply the recommendation in steps <3> to <5> above.

3) If the leak is more than the CIS limit, contact CFM.

(d) Corrective action:

1) Replace the engine (TASK 71-00-02-000-801-G00 and TASK 71-00-02-400-801-G00).

SUBTASK 71-71-00-220-009-G00

(9) Service Disconnect (Strut) Drain.

(a) Fluid: Hydraulic fluid.

1) Serviceable limit: 30 cc/hr (10 drops per minute).

2) Corrective action:

a) Find and correct the hydraulic fluid leak, do this task: Hydraulic System External Leakage Check, TASK 29-00-00-790-801.

(b) Fluid: Fuel.

1) Threshold limit: 0 cc/hr.

2) Serviceable limit: 0 cc/hr.

3) Corrective action:

a) Find and correct the fuel leak, do this task: Test No. 3 - Wet Motor Leak Test, TASK 71-00-00-790-803-G00.

SUBTASK 71-71-00-220-010-G00

(10) Aft fairing drain.

(a) Fluid: Hydraulic fluid.

1) Serviceable limit: 30 cc/hr (10 drops per minute).

2) Corrective action:

a) Find and correct the hydraulic fluid leak, do this task: Hydraulic System External Leakage Check, TASK 29-00-00-790-801.

(b) Fluid: Fuel.

1) Threshold limit: 180 cc/hr (60 drops per minute).

a) If the leakage is less than the threshold limit, no maintenance action is necessary.

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b) If the leakage is more than the threshold limit and less than the serviceable limit, you can continue the engine in service for not more than 25 flight cycles before you replace the applicable component.

NOTE: If fuel is coming from the strut/fuel supply line, you may not dispatch and the leak must be corrected.

2) Serviceable limit: 270 cc/ hr (90 drops per minute).

3) Corrective action:

a) Find and correct the fuel leak, do this task: Test No. 3 - Wet Motor Leak Test, TASK 71-00-00-790-803-G00.

SUBTASK 71-71-00-220-011-G00

(11) Fuel pump pad drain.

NOTE: Leakage from the drain of the Main Fuel Pump can be fuel or oil.

(a) Fluid: Oil.

1) Serviceable limit: 20 cc/hr (7 drops per minute).

2) Replace the applicable component:

a) Do these steps to replace the AGB rotating seal of the applicable pad for the fuel pump, hydraulic pump, hand cranking pad, or IDG:

Accessory Gearbox Rotating Seals Removal, TASK 72-64-01-000-801-G00,

Accessory Gearbox Rotating-Seals Installation, TASK 72-64-01-400-801-G00.

(b) Fluid: Fuel.

1) Threshold limit: 180 cc/hr (60 drops per minute).

a) If the leakage is less than the threshold limit, no maintenance action is necessary.

b) If the leakage is more than the threshold limit and less than the serviceable limit, you can continue the engine in service for not more than 25 flight cycles before you replace the applicable component.

2) Serviceable limit: 270 cc/ hr (90 drops per minute).

3) Do these steps to replace the main fuel pump:

Main Fuel Pump Removal, TASK 73-11-01-000-801-G00,

Main Fuel Pump Installation, TASK 73-11-01-400-801-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 71-71-00-840-001-G00

(1) Remove the leak check fixture, SPL-20664.

(a) Remove the hoses from the holder in the test tubes.

(b) Remove collecting tubes from the drain master tubes.

(c) Remove tubes holder from the rack.

SUBTASK 71-71-00-420-001-G00

(2) If it is necessary, re-install and tighten the witness tee-nut(s) to 145 in-lb (16 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 71-71-00-860-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 71-71-00-860-004-G00

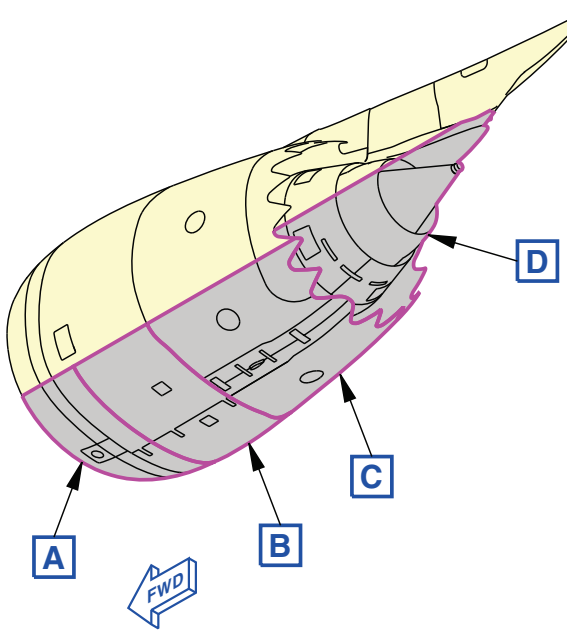
(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

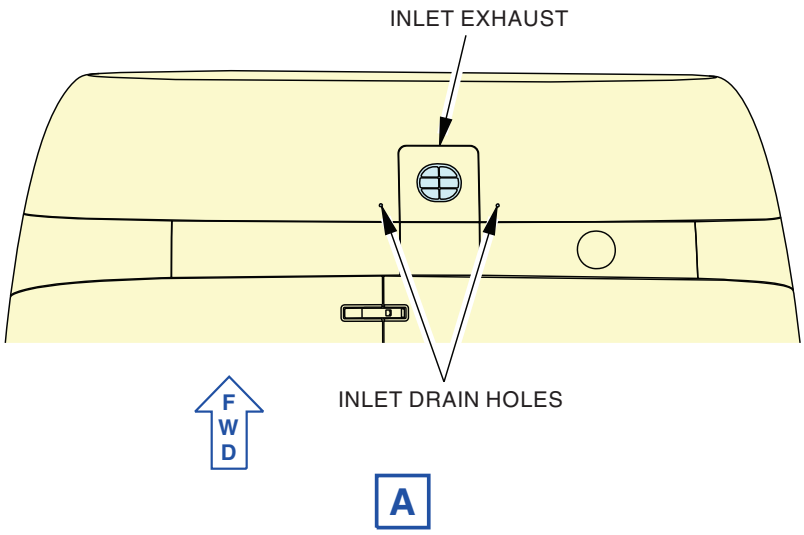
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

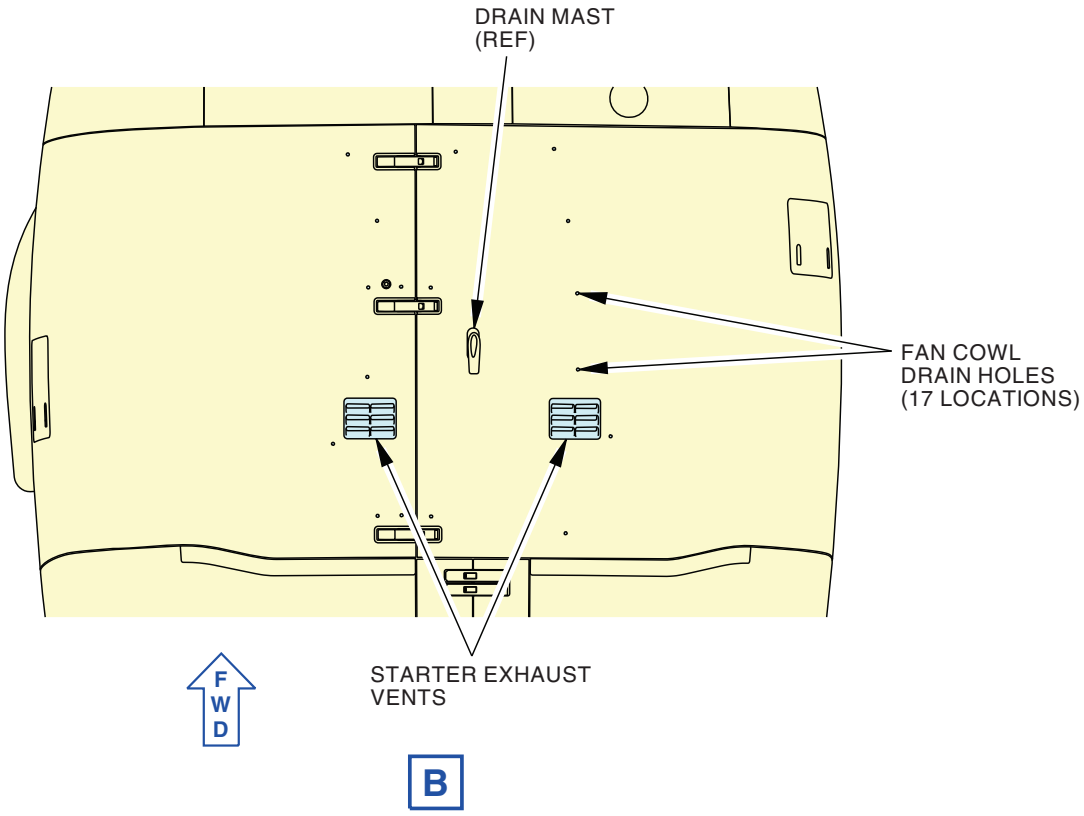




2955627 S0000732783_V1

Engine Drains and Vents Inspection
Figure 601/71-71-00-990-804-G00 (Sheet 1 of 5)

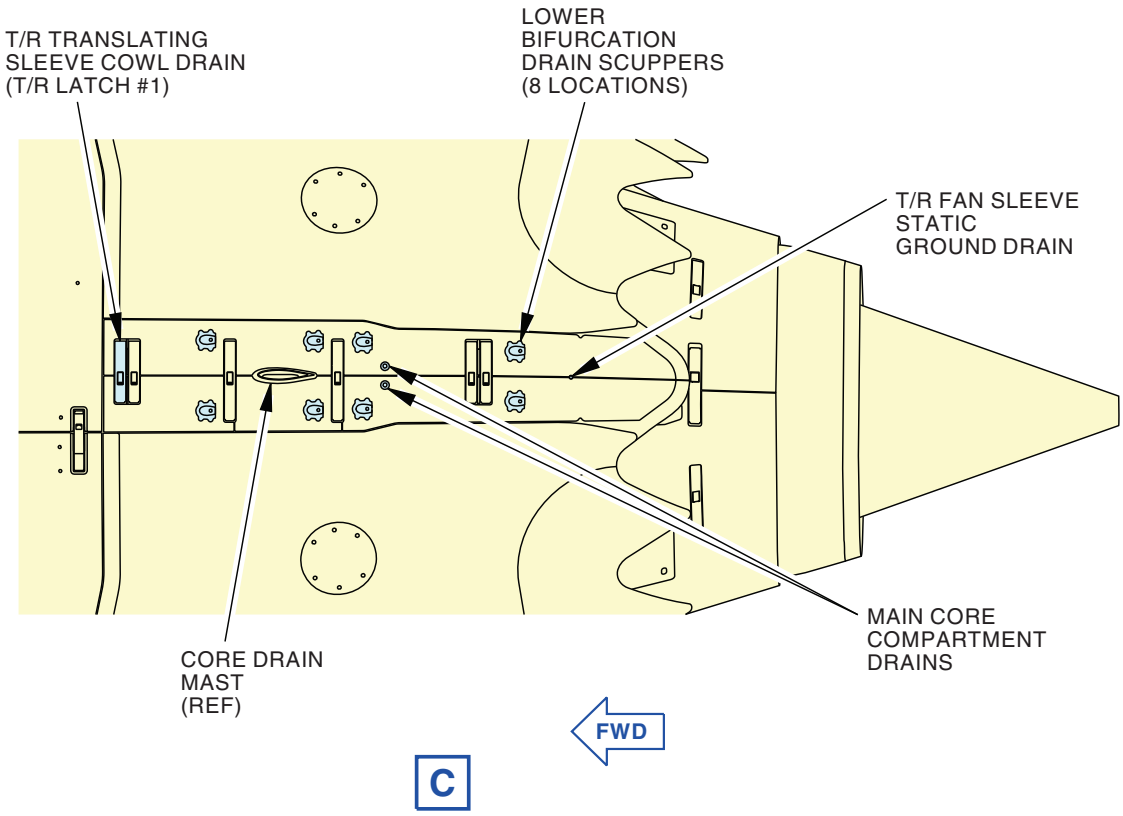
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2955628 S0000732791_V1

Engine Drains and Vents Inspection
Figure 601/71-71-00-990-804-G00 (Sheet 2 of 5)

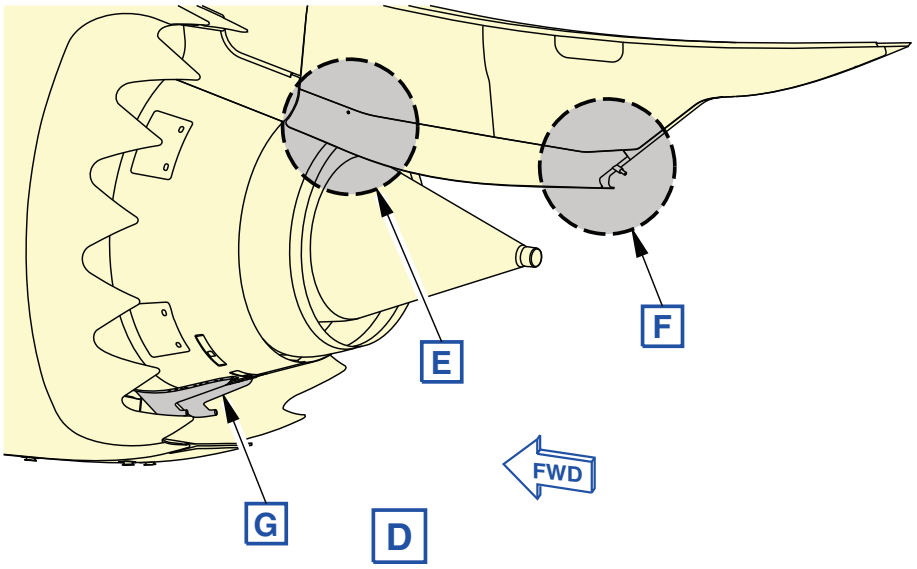
ECCN 9E991 BOEING PROPRIETARY - See title page for details

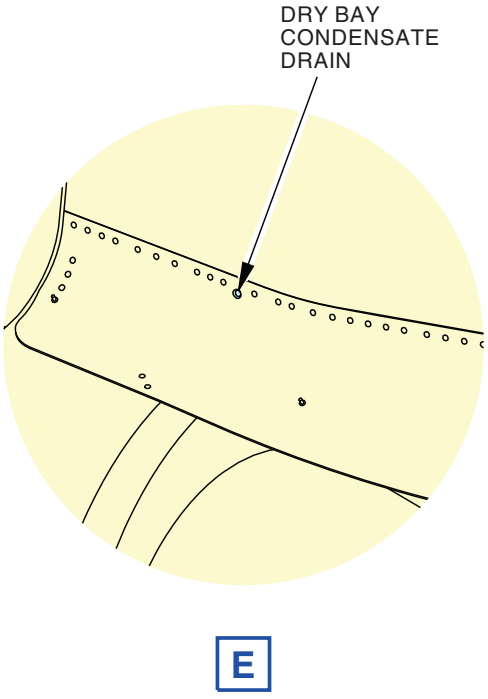


2955629 S0000732794_V1

Engine Drains and Vents Inspection
Figure 601/71-71-00-990-804-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

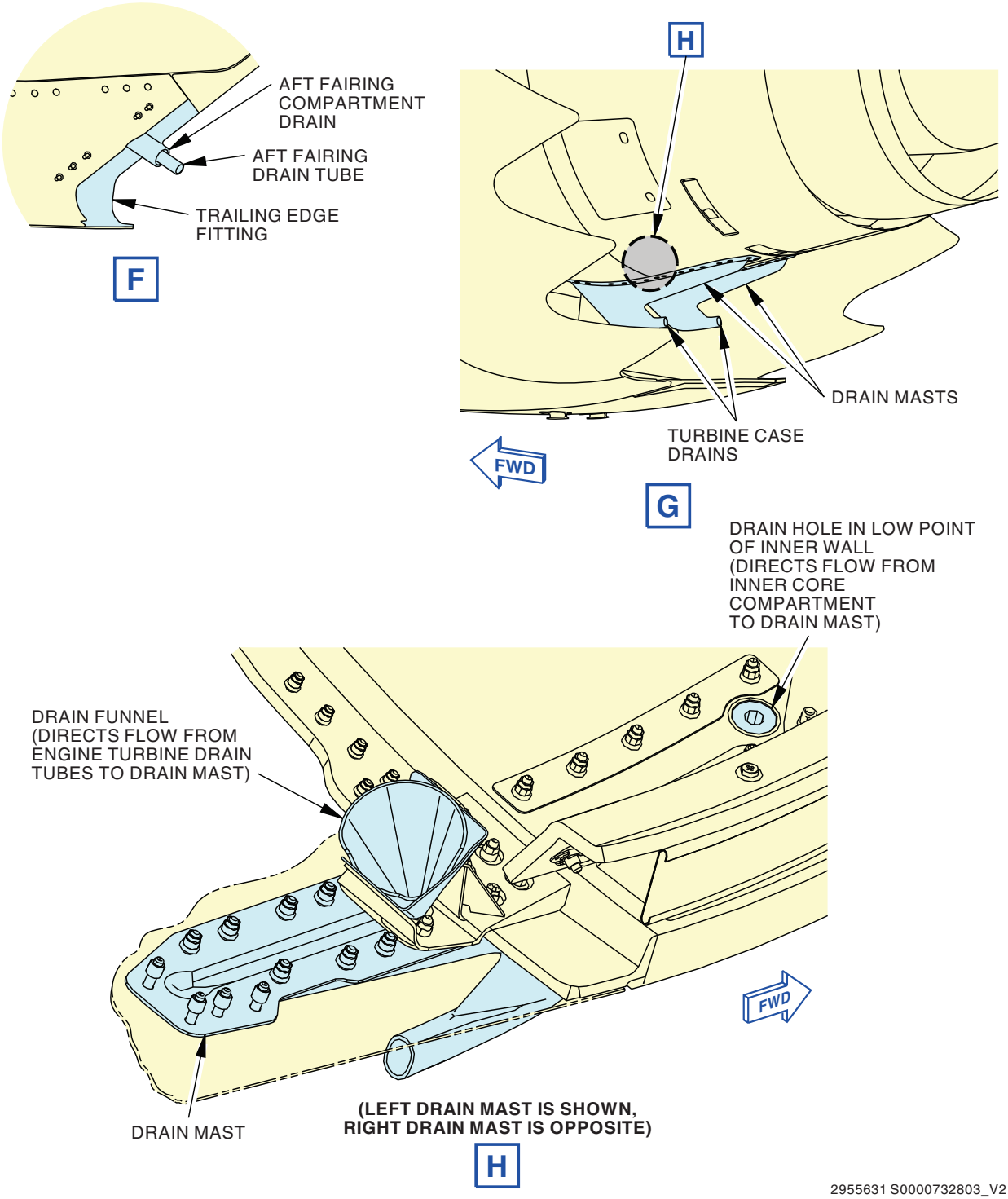




2955630 S0000732797_V1

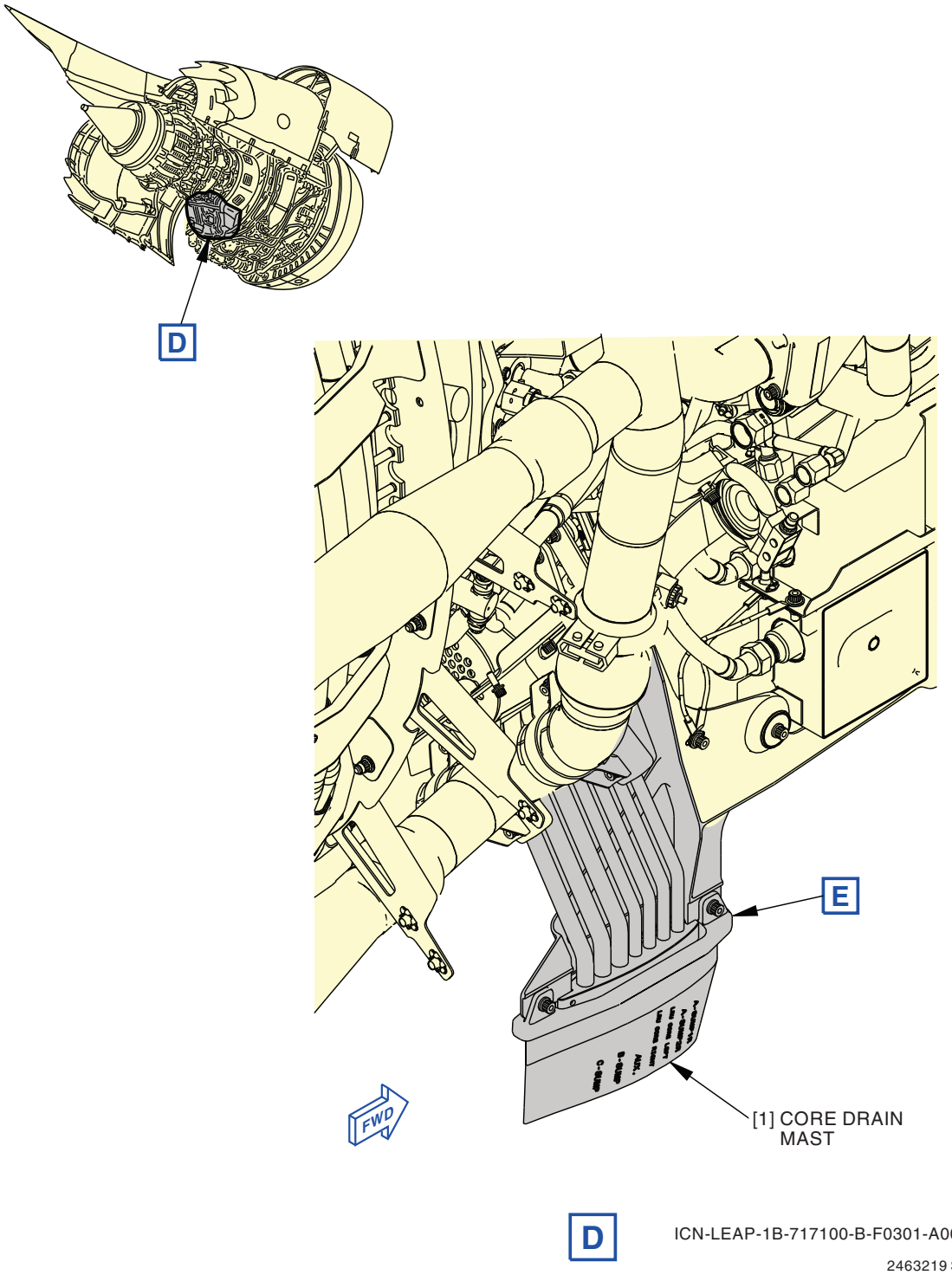
Engine Drains and Vents Inspection
Figure 601/71-71-00-990-804-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



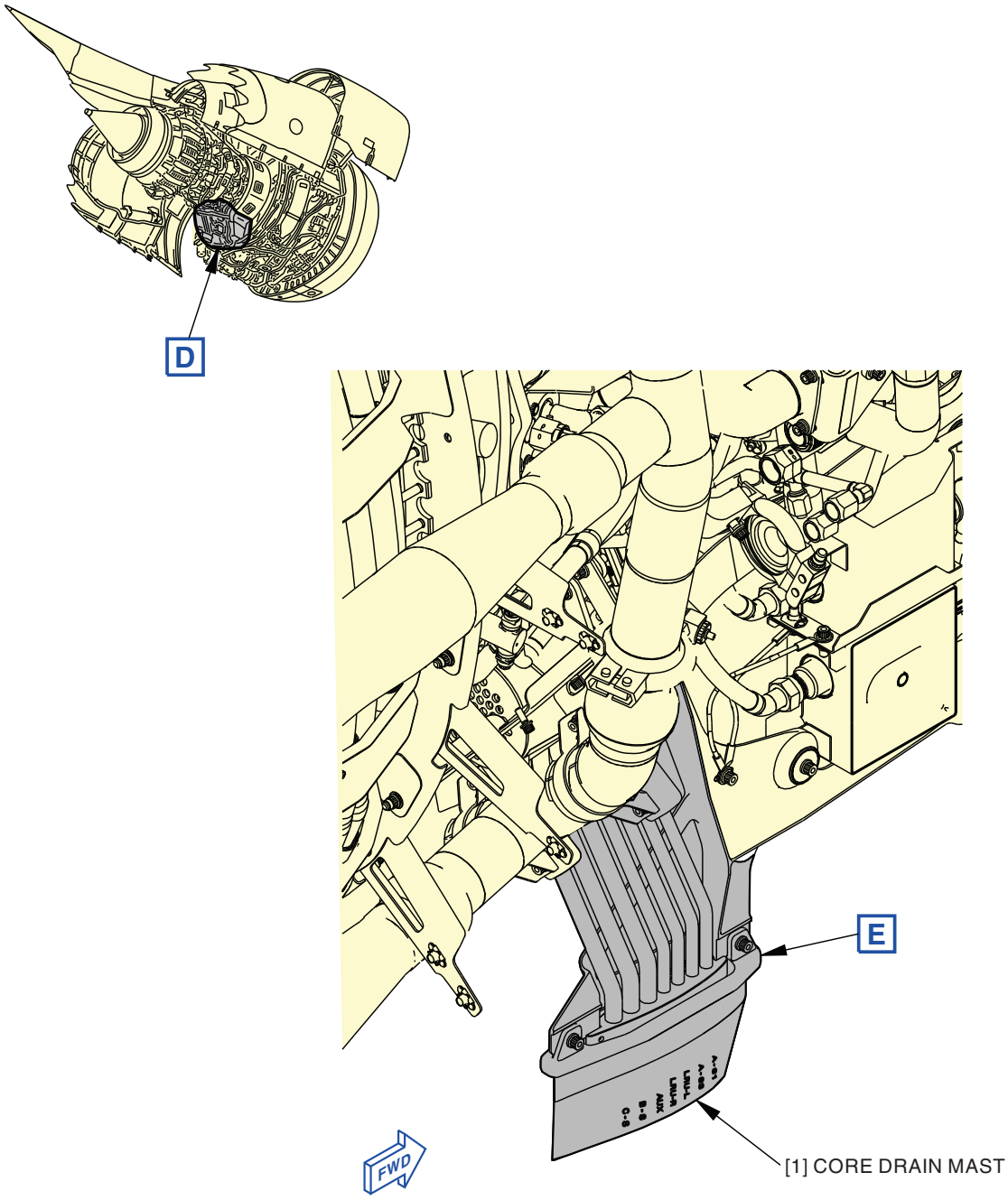
Engine Drains and Vents Inspection
Figure 601/71-71-00-990-804-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



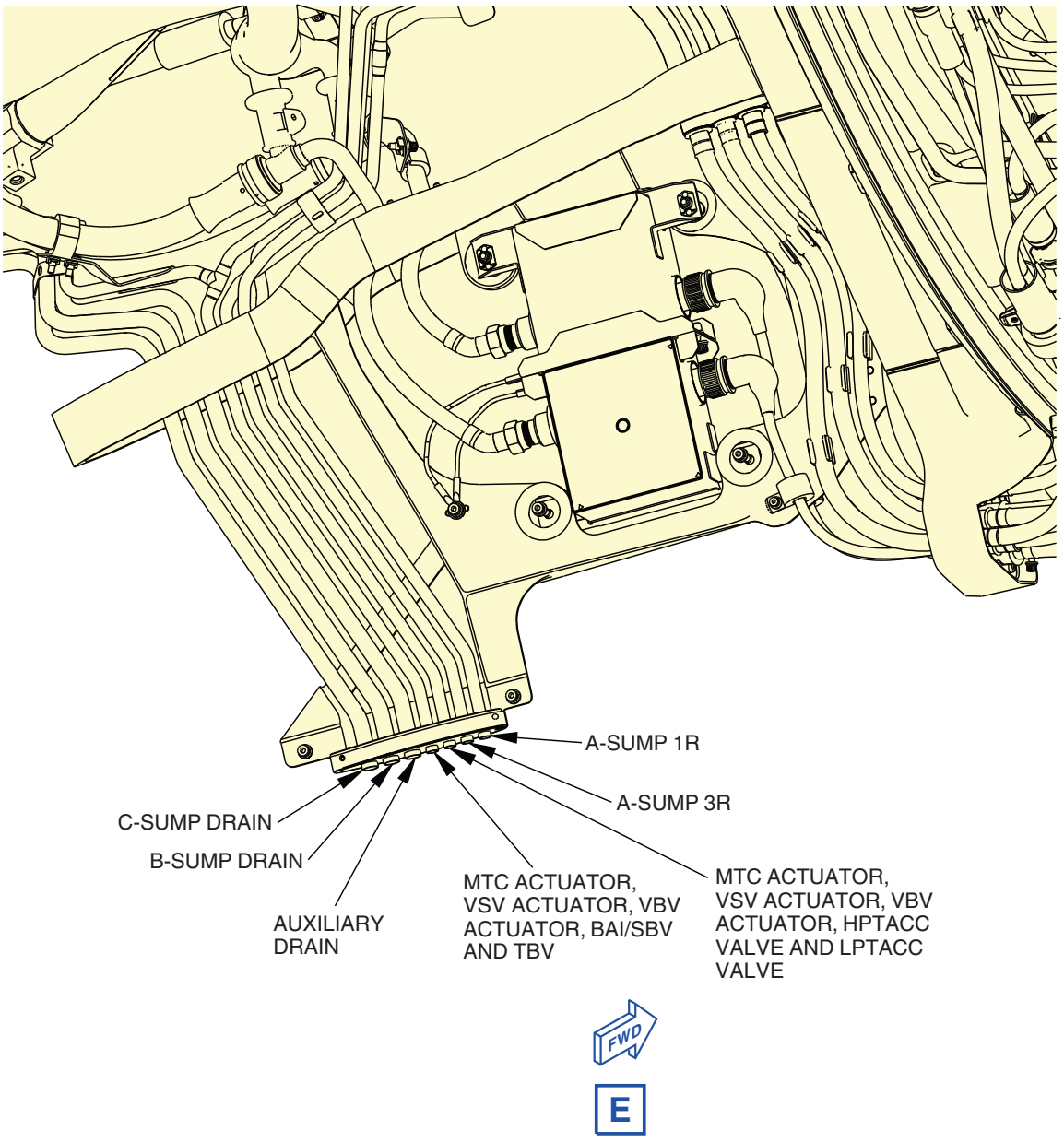
Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 1 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 2 of 16)

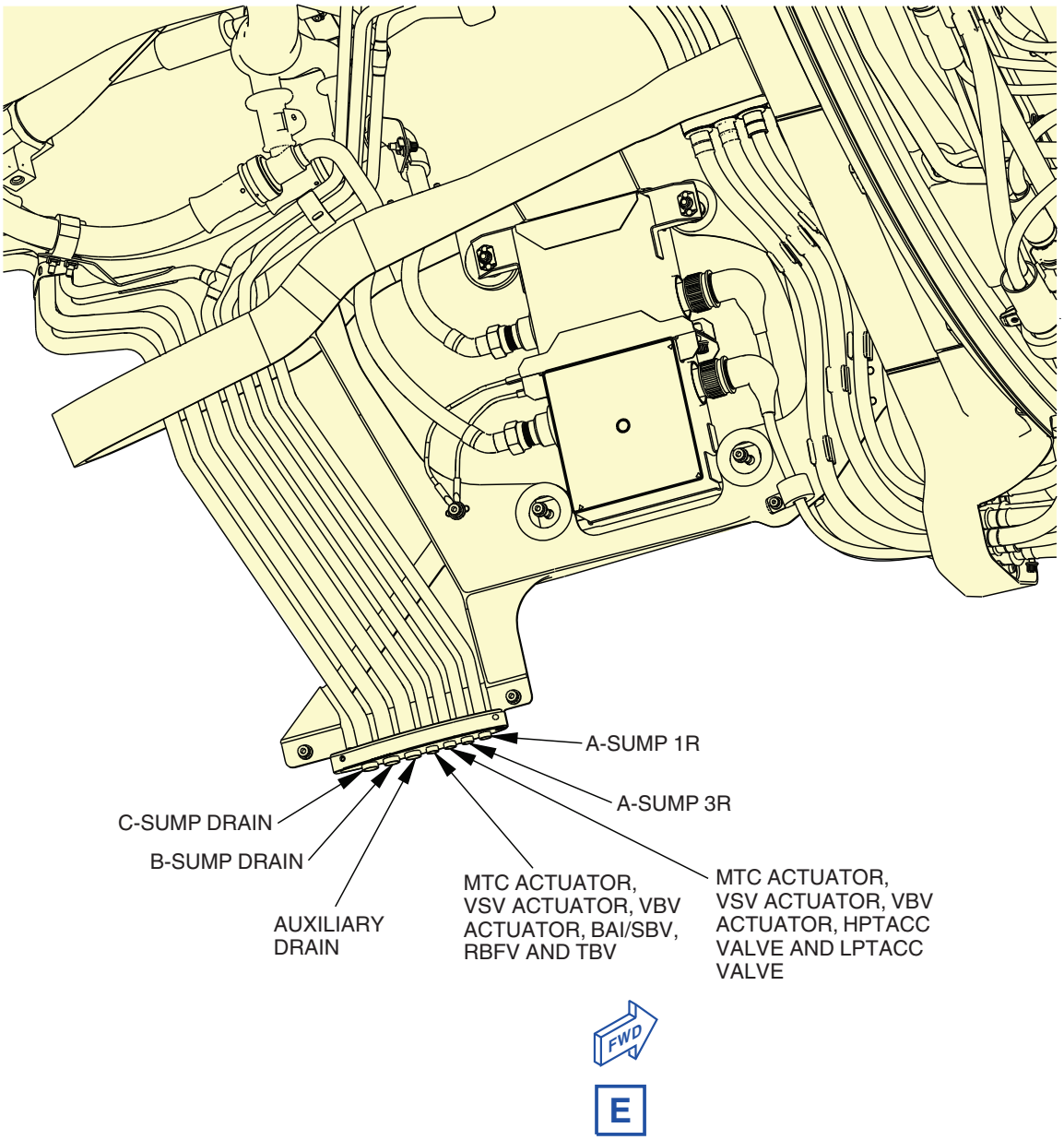
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-717100-B-F0301-A0005-A-001-01
2463221 S0000573545_V3

Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 3 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-717100-B-F0301-A0005-A-001-01
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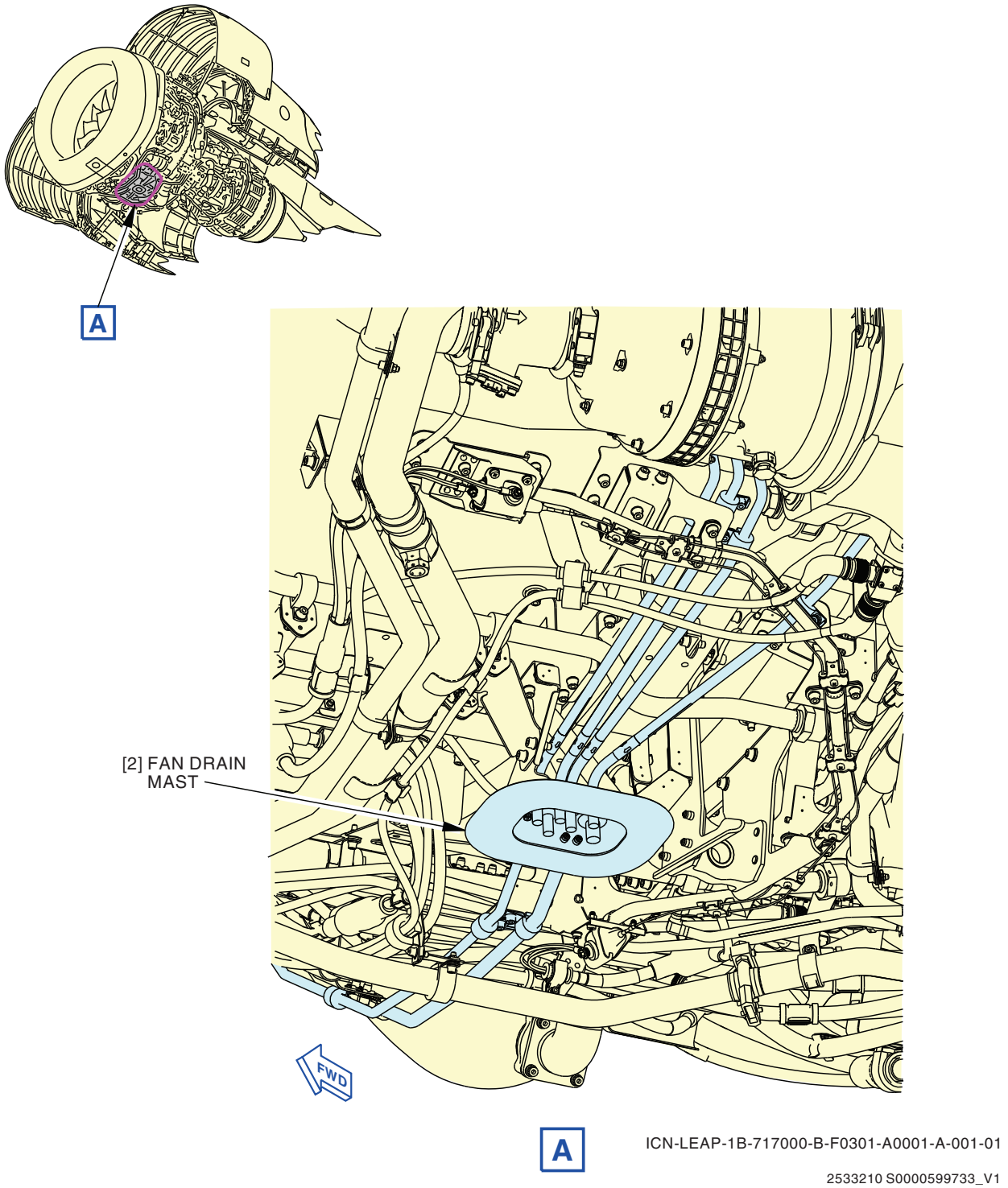
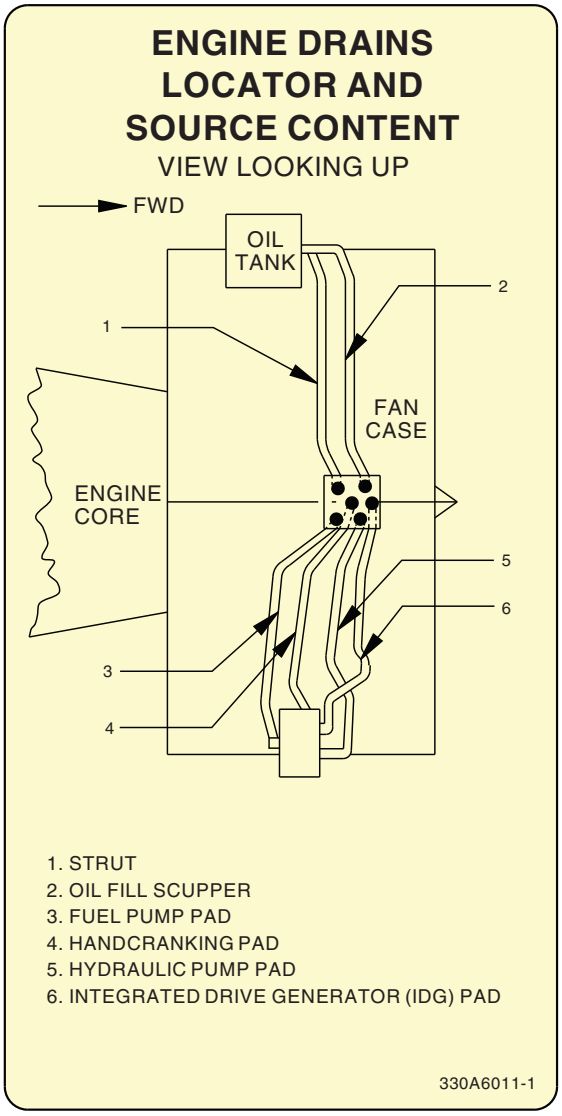
Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 4 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 5 of 16)

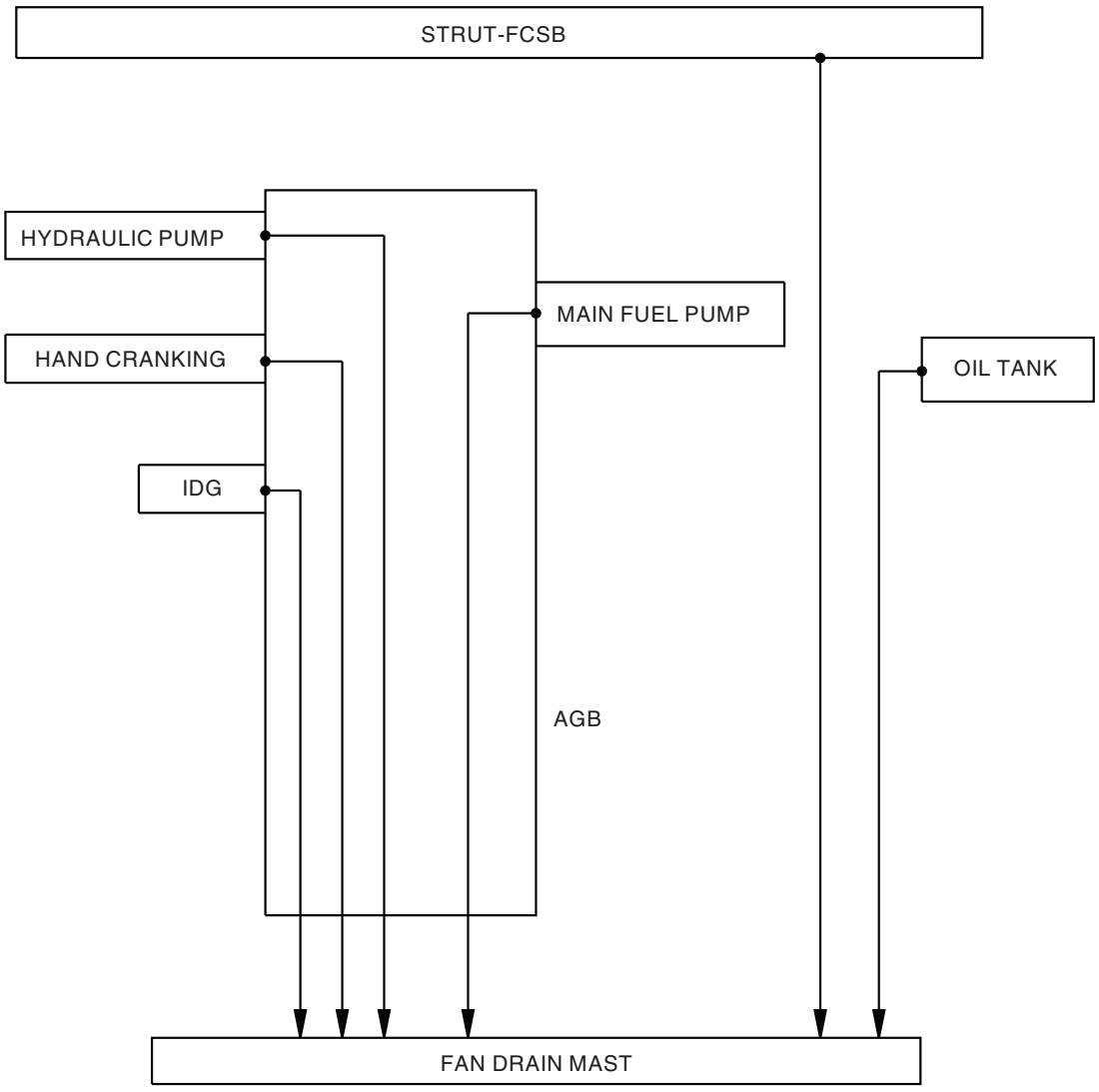
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2533187 S0000599698_V1

Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 6 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

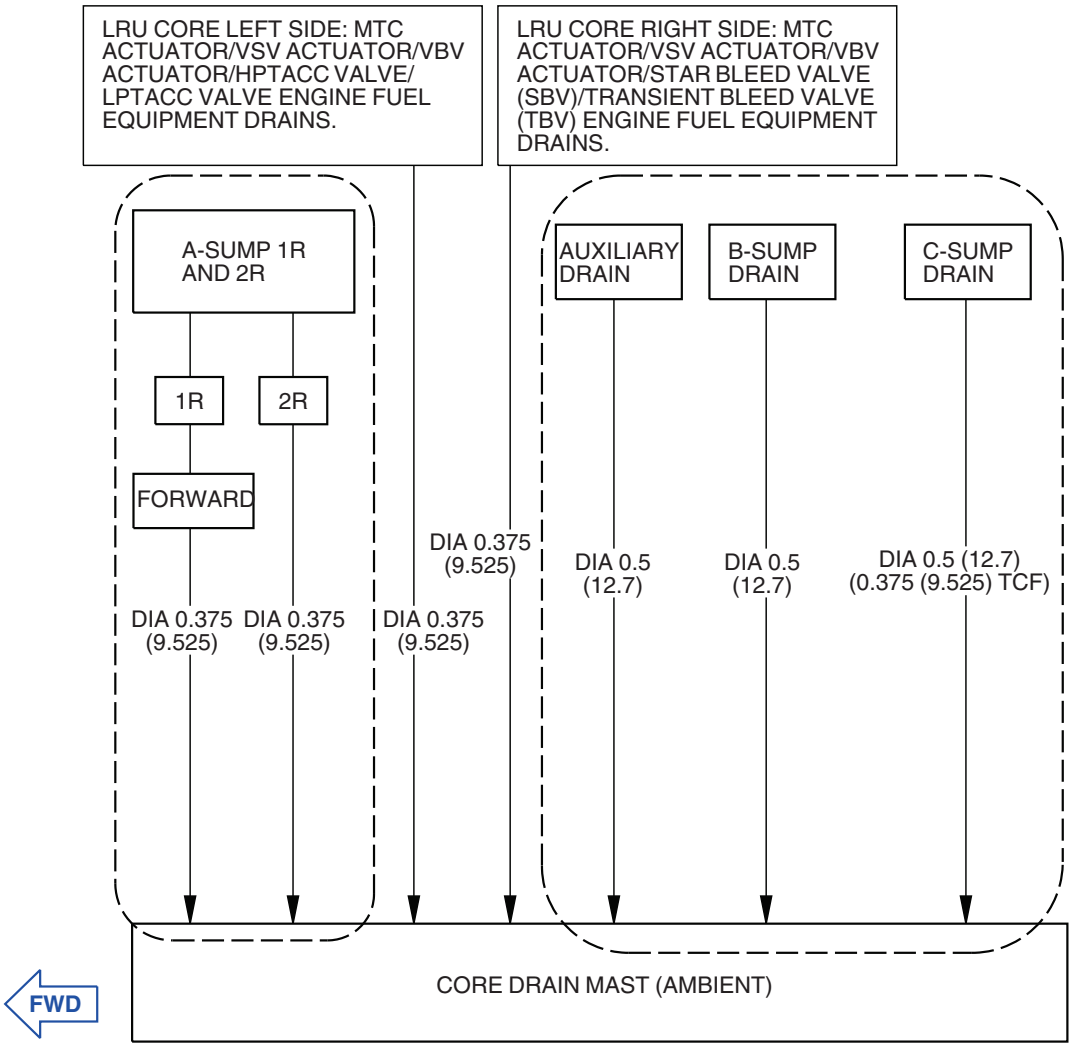


ICN-LEAP-1B-717100-B-F0301-A0008-A-001-01

2533207 S0000599699_V2

Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 7 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

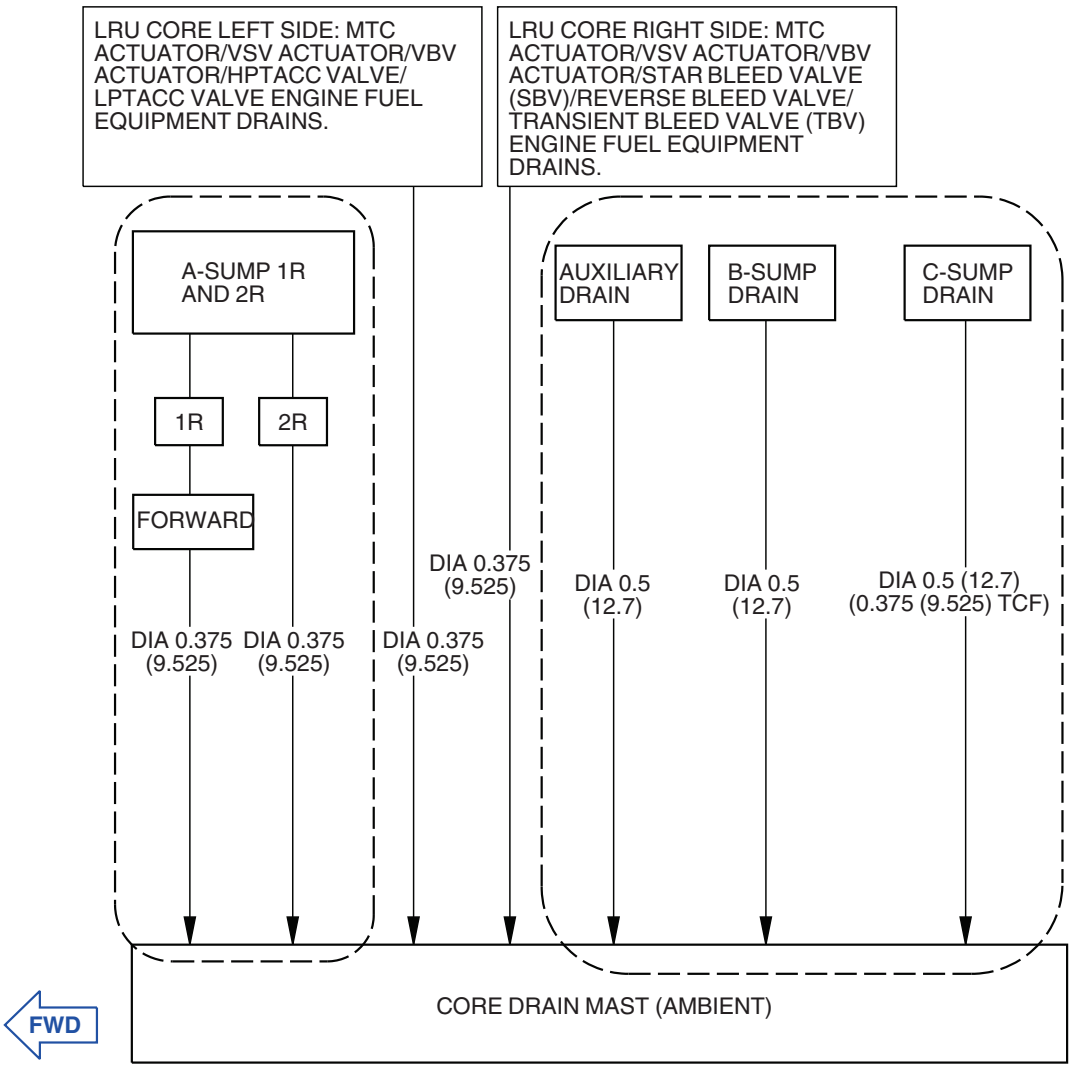
ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

ICN-LEAP-1B-717100-B-F0301-A0009-A-001-01

2354440 S00061515373_V5

Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 8 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

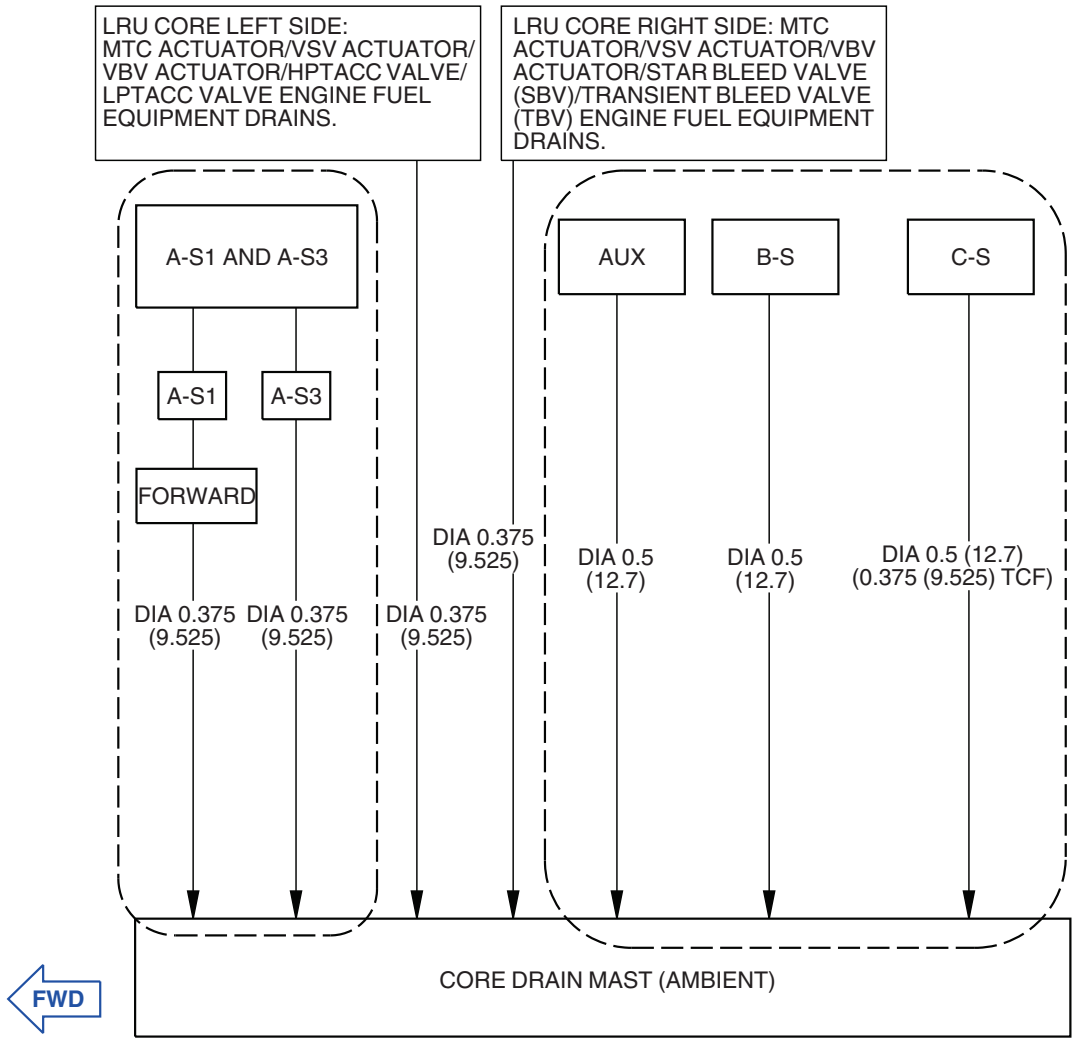
ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

ICN-LEAP-1B-717100-B-F0301-A0009-A-001-01

3129703 S0000861265_V1

Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 9 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



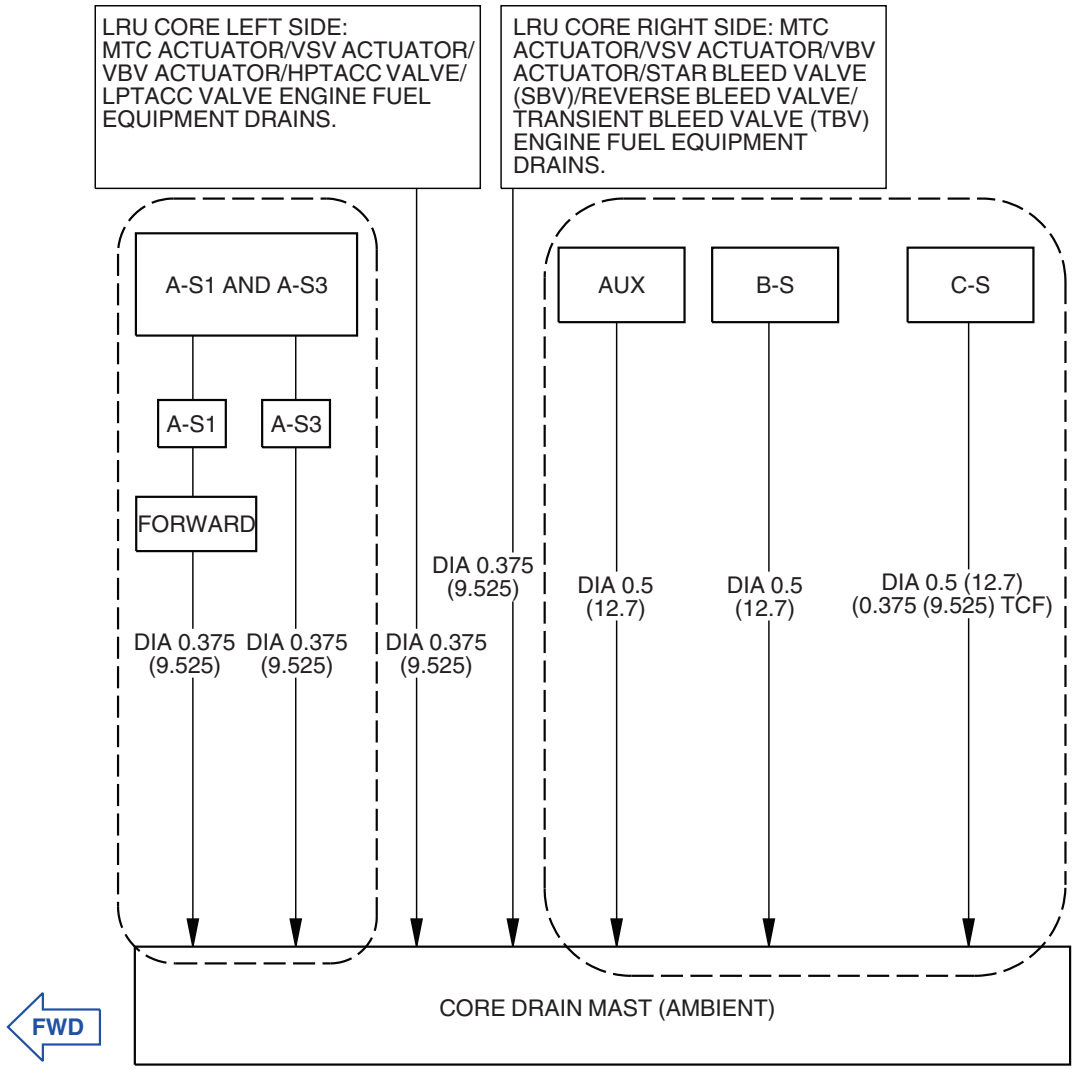
NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

ICN-LEAP-1B-717100-D-58828-00001-A-001-01
2836144 S0000660388_V2

Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 10 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



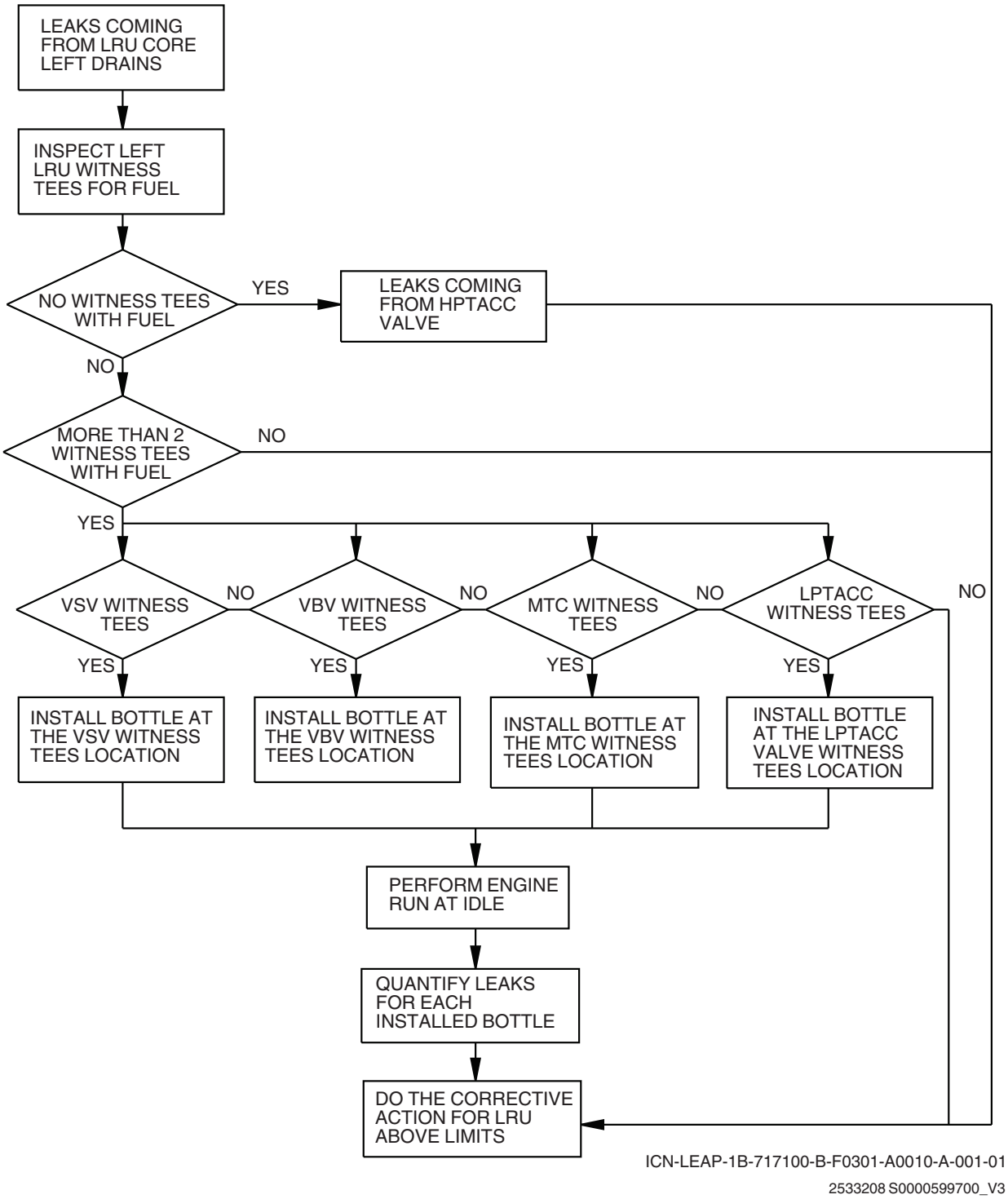
NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

ICN-LEAP-1B-717100-D-58828-00001-A-001-01
3129704 S0000861266_V1

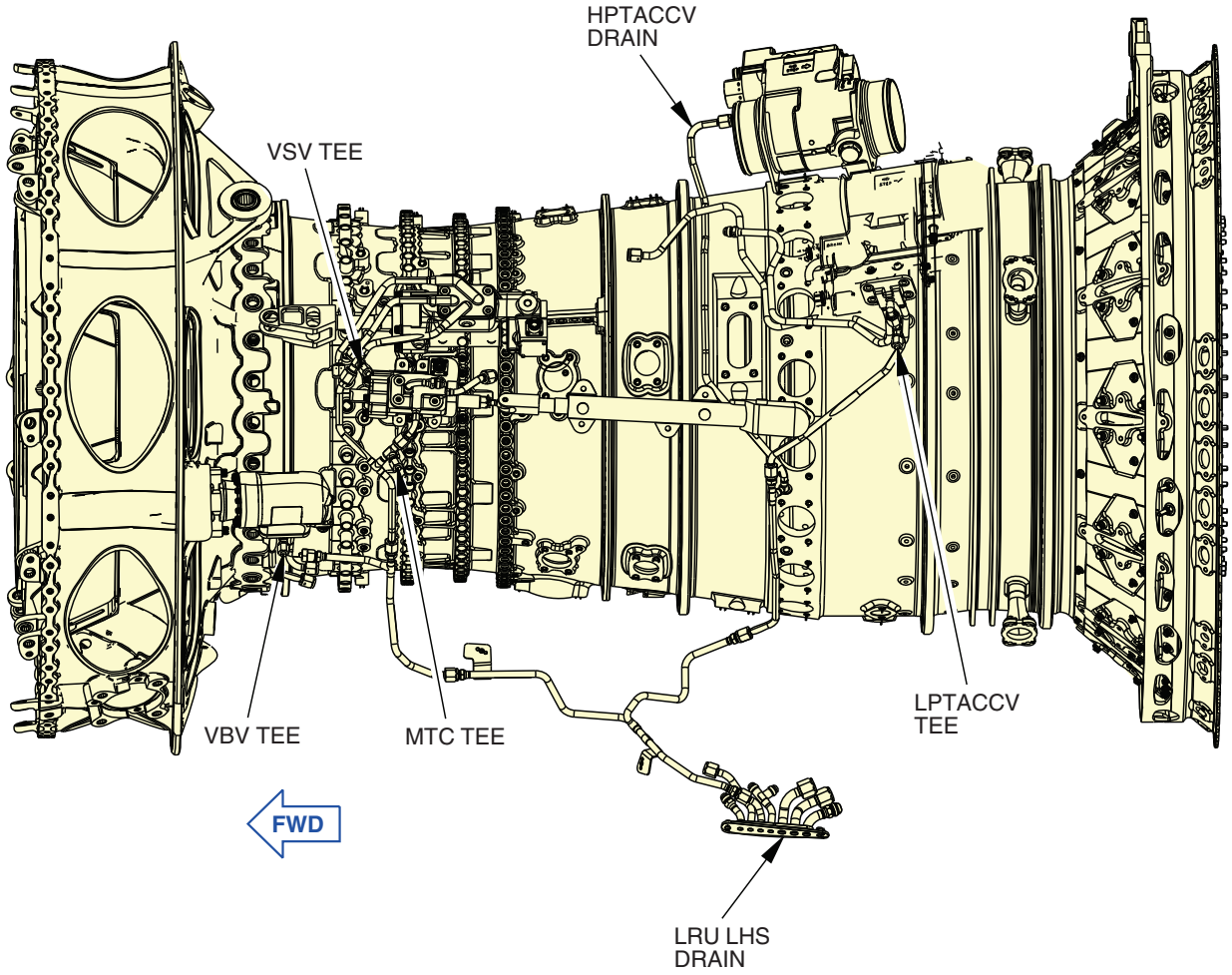
Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 11 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 12 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

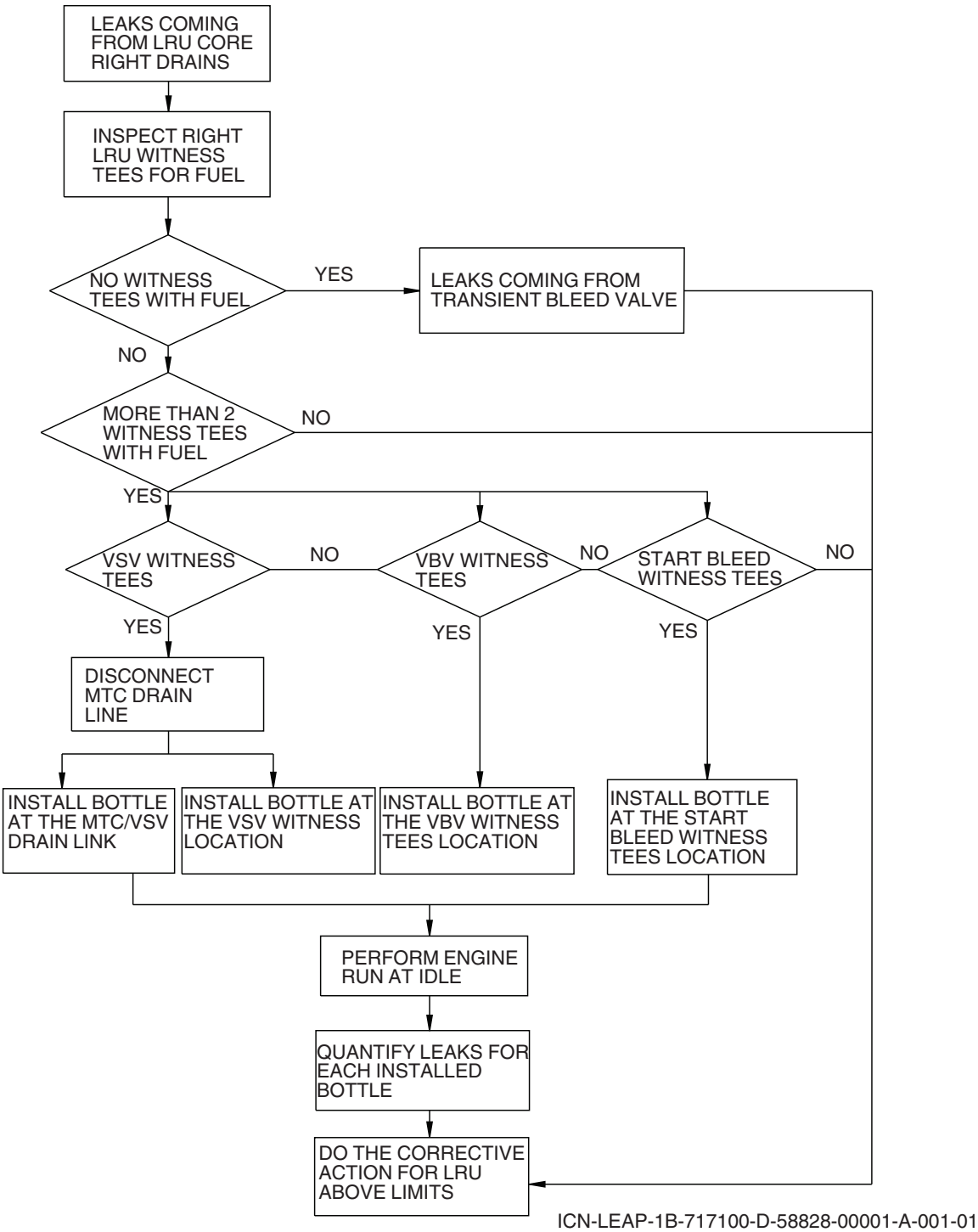


6056786-00

3119714 S0000855001_V1

Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 13 of 16)

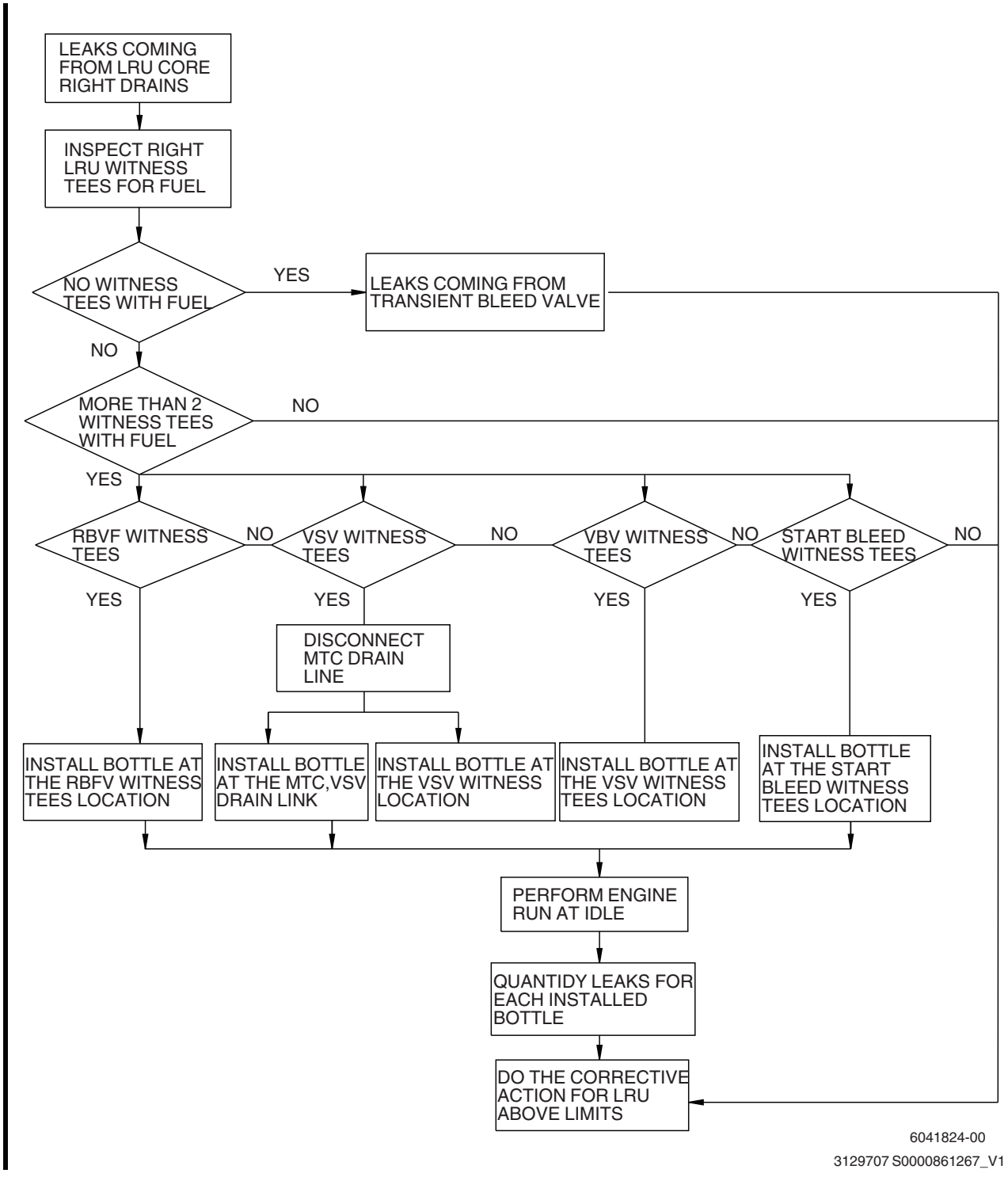
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-717100-D-58828-00001-A-001-01
2533209 S0000599701_V3

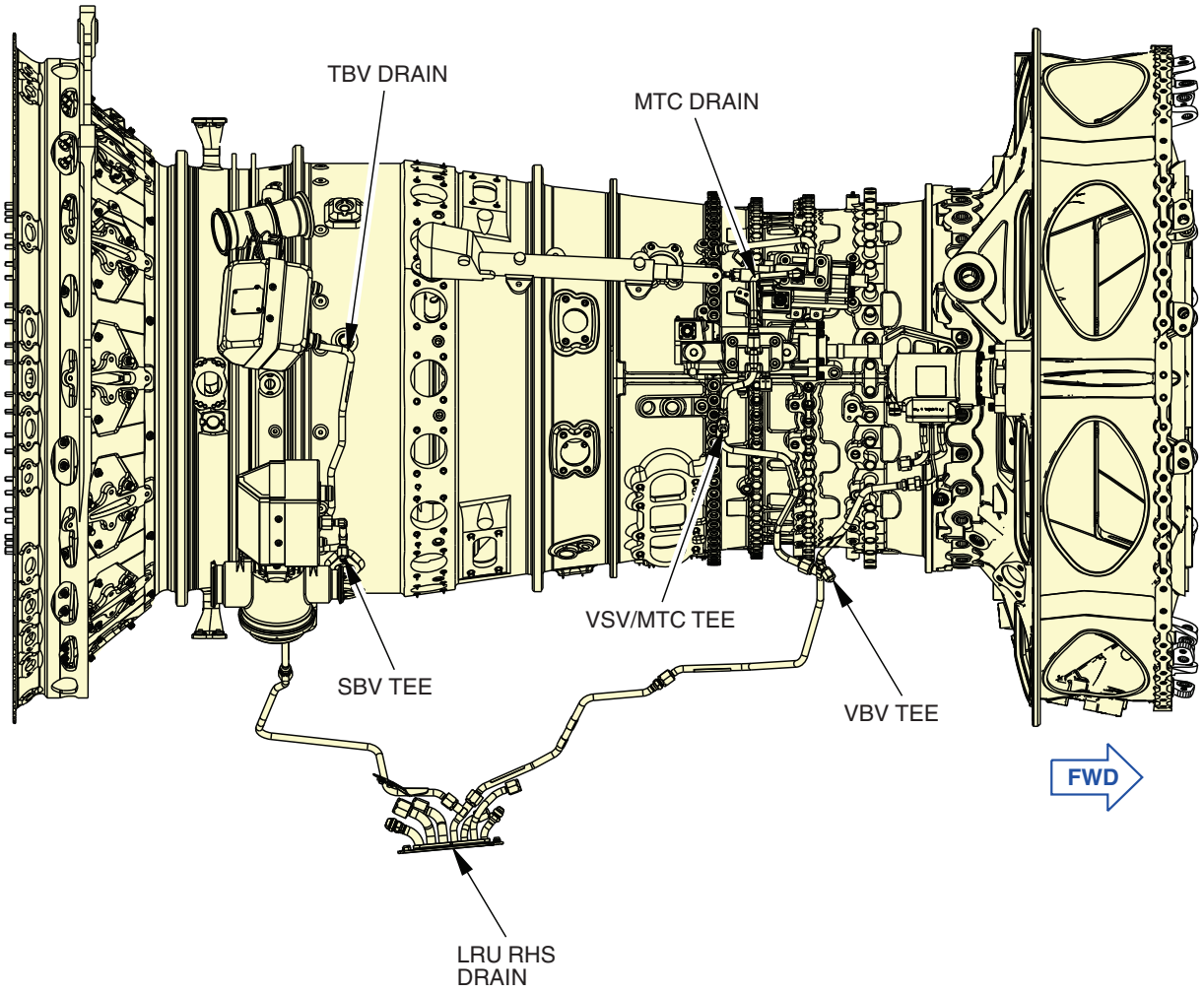
Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 14 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 15 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



6056787-00

3119718 S0000855002_V1

Engine Drain Masts Inspection
Figure 602/71-71-00-990-802-G00 (Sheet 16 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DRAIN MAST - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Drain mast removal

(2) Drain mast installation.

TASK 71-71-01-000-801-G00

2. Drain Mast Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the drain mast.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 71-71-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 71-71-01-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Drain Mast Removal

SUBTASK 71-71-01-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the drain mast [1] as follows:

(a) Remove the two bolts [2] and the two nuts [3].

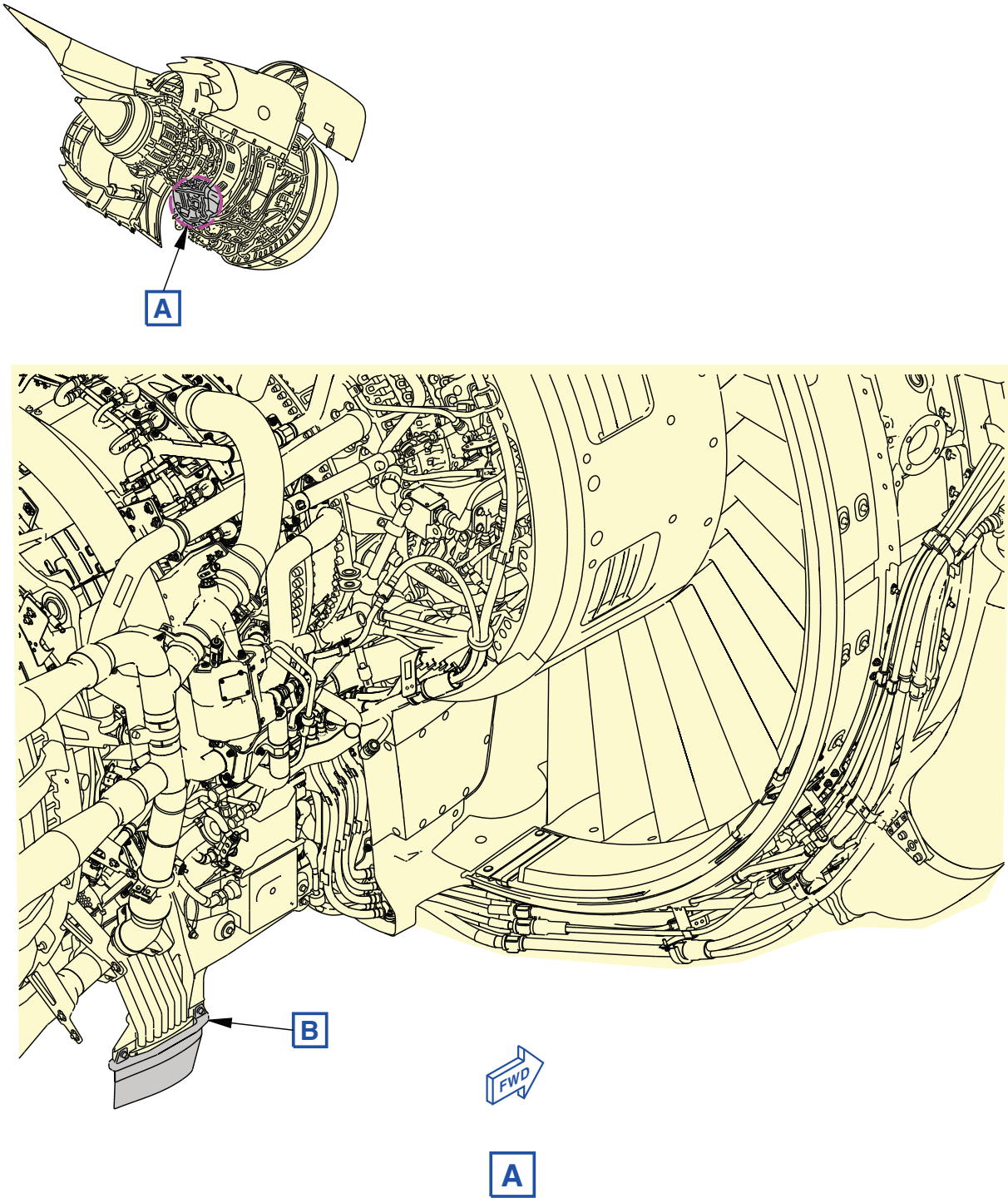
(b) Remove the drain mast [1] from the 6:00 o'clock bifurcation.

NOTE: It may require the dowel pin replacement for the installation of a new drain mast, refer to CFM SB LEAP-1B-72-00-0180.

(c) Install the protective covers on the drain mast [1] and the bifurcation.

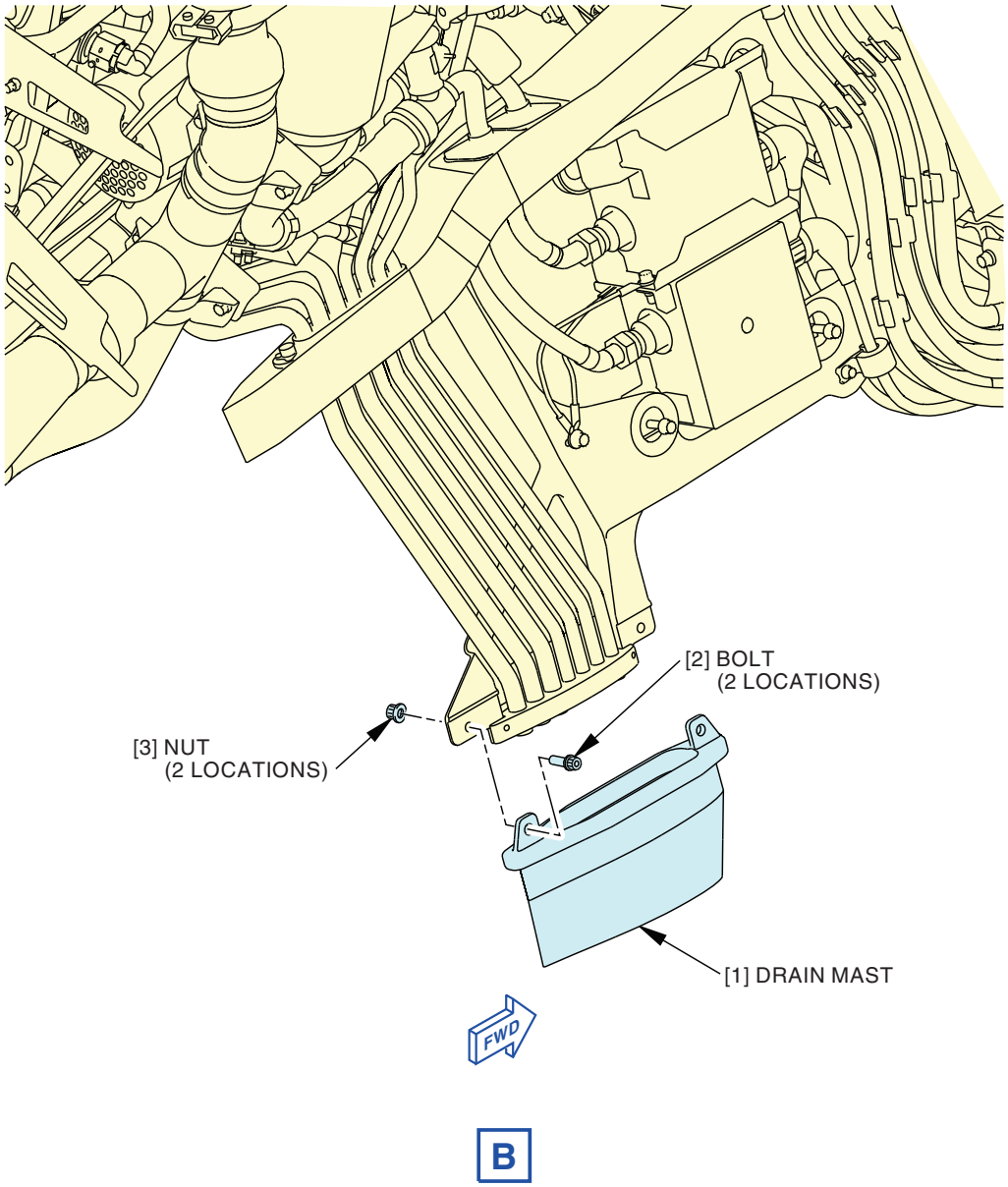
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Drain Mast Installation
Figure 401/71-71-01-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Drain Mast Installation
Figure 401/71-71-01-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 71-71-01-400-801-G00

3. Drain Mast Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the drain mast.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Drain mast	72-00-00-21-410	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Drain Mast Installation

SUBTASK 71-71-01-420-001-G00

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the drain mast [1] as follows:

(a) Remove the protective covers from the drain mast [1] and the bifurcation.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Install the drain mast [1] in a correct position on the engine at the 6:00 o'clock bifurcation.

NOTE: It may require the dowel pin replacement for the installation of a new drain mast, refer to CFM SB LEAP-1B-72-00-0180.

(d) Lubricate the threads of the two bolts [2] with the graphite mineral vaseline, D00601 [CP2101].

(e) Install the two bolts [2] and the two nuts [3] on the drain mast [1].

(f) Tighten the nuts [3] to 109.25 in-lb (12.34 N·m) to - 120.75 in-lb (13.64 N·m).

(g) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 71-71-01-860-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 71-71-01-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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